Franzen Complete Lessons Binder

November 8, 2026 • 458 pages • 72 public topics • 37 unique papers

Begin with mathematics, then follow the lessons through the Earth and sky, exploration, weather, language and history. Every available public topic is reproduced in full, followed by complete companion papers and clearly labeled earlier mathematics editions.

Coverage and known gaps

Maurolico and Snell: no standalone lessons were located in the collected sources. The scientific references to Tycho Brahe and Kepler are included, especially Kepler’s Mars triangulation; separate Tycho childhood, duel/nose and Stjerneborg stories, and Kepler family biographies, were not found. This binder does not reconstruct missing lessons.

How to use this collection

Click any contents entry or use the PDF bookmarks. The page number in the added bottom band is the binder page. Numbers printed inside a reproduced paper belong to its original edition. Complete HTML lessons and standalone paper editions can overlap; byte-identical PDF downloads appear only once. Archival mathematics versions are in Appendix B.

Sources: the public Faria learning log and the existing lesson-paper collection. Wind (topic 71) and expanded longitude (topic 70) reflect the published November 8 revisions. Only lesson content is reproduced.

Contents

Every topic and paper • click a title to begin PAGE

1. Mathematics foundations 6

Master mathematics: algebra, geometry, trigonometry, calculus and growth rates 7

Trigonometry: the ladder, sine, cosine, tangent and Galileo's Moon 29

Growth rates compared 35

Flight calculus terms 38

2. Earth, sky and navigation 42

Topic 18 — Eratosthenes measures the Earth 43

Topic 19 — The shadow stick and why the lines meet at Earth's center 46

Topic 20 — Why the sun's rays count as parallel (Aristarchus) 49

Topic 21 — The flat-Earth myth and Columbus 50

Topic 22 — Copernicus, 1543 51

Topic 23 — Bruno, Galileo and “Eppur si muove” 52

Topic 45 — Looking Up: Polaris, Planets, Ptolemy and Copernicus 54

Topic 46 — Galileo 58

Topic 47 — Map Surprises: Latitude and Longitude Tricks 69

Topic 48 — Polaris: the North Star, explained 73

Topic 49 — Nathaniel Bowditch: the self-taught navigator 88

Topic 50 — The sidereal day and latitude sailing 100

Topic 51 — The Harvard Computers: women who measured the stars 108

Topic 52 — From a Venus transit to the Cepheids 124

Topic 62 — Aristarchus: a moving Earth and a measurable Moon 145

Topic 63 — Hipparchus: the Moon changes place when we move 148

Topic 64 — Galileo's moons: new centers and lasting names 150

Topic 65 — Romer and Io: light needs time 152

Topic 66 — Kepler's Mars triangles, and an honest scale check 154

Topic 69 — Thumb to Moon: calculating the Moon's diameter 157

Topic 70 — Time to longitude: converting hours, minutes, and seconds to degrees 161

3. Voyages, surveying and empire 169

Topic 16 — The Suez Canal and the route to India and Australia 170

Contents • continued

Every topic and paper • click a title to begin PAGE

Topic 17 — The Red Sea rift and Bab-el-Mandeb 173

Topic 53 — Captain James Cook: precision, encounter, and empire 174

Topic 54 — Joseph Banks: the botanist who built a network 176

Topic 55 — The Venus transits: a worldwide ruler 178

Topic 56 — Charles Mason: an astronomer on difficult ground 183

Topic 57 — Jeremiah Dixon: the surveyor in the partnership 185

Topic 58 — The Mason-Dixon survey: how a boundary became real 189

Topic 59 — Ships of exploration: four craft, four different jobs 191

Topic 60 — Bounty: mutiny, survival, and the people of Pitcairn 193

Topic 61 — Breadfruit and empire: moving a living crop 195

4. Weather, medicine and communication 197

Topic 06 — RADAR is an acronym 198

Topic 07 — The telegraph and the first storm warnings 199

Topic 08 — Father Benito Viñes, Padre Huracán 200

Topic 09 — Galveston 1900 and the dismissed Cuban warnings 202

Topic 10 — How a Cuban warning reached Texas 204

Topic 11 — The transatlantic cables of 1858 and 1866 205

Topic 12 — The Great Eastern and Isambard Kingdom Brunel 207

Topic 27 — Hurricane, typhoon, cyclone, and the circle of learning 209

Topic 28 — Quarantine, the lazaretto and Yellow Jack 211

Topic 29 — Walter Reed, Carlos Finlay and yellow fever, Cuba 1900 213

Topic 35 — Tambora, the Year Without a Summer and Frankenstein 215

Topic 37 — Typhoon Scales vs Saffir–Simpson 217

Topic 71 — Hurricane Wind Force: Force Grows with Velocity Squared 221

Topic 72 — A Feather and an Anvil Falling Through Air 228

5. Words, books, memory and history 231

Topic 01 — Disaster means “bad star” 232

Topic 02 — Astronaut, nautical and the sailor root 233

Topic 03 — King Lear: Edmund's star speech and the scientific method 234

Contents • continued

Every topic and paper • click a title to begin PAGE

Topic 04 — King Lear: the whole play 236

Topic 05 — King Lear: the great lines, and how to say Gloucester 238

Topic 13 — Fulton's Nautilus, a submarine for Napoleon 240

Topic 14 — The Count of Monte Cristo and Napoleon 242

Topic 15 — Dumas's father, the general who defied Napoleon 244

Topic 24 — Jules Verne's A Floating City 245

Topic 25 — Twenty Thousand Leagues Under the Seas 246

Topic 26 — Mark Twain's The Innocents Abroad 248

Topic 30 — Clue, the Labyrinth and the Minotaur 249

Topic 31 — Shakespeare's Sonnet 18: Shall I compare thee 250

Topic 32 — T. S. Eliot and The Waste Land 252

Topic 33 — The Love Song of J. Alfred Prufrock 254

Topic 34 — The method of loci and a memory palace 256

Topic 36 — Shelley's Ozymandias 258

Topic 38 — Milton's Paradise Lost 260

Topic 39 — Plutarch and the Parallel Lives 265

Topic 40 — Delphi and the Oracle 273

Topic 41 — Ancient Greek Religion 286

Topic 42 — Alexander and Caesar 289

Topic 43 — Lycurgus and Numa 293

Topic 44 — The Real Abbé Faria 298

Topic 67 — Jos. A. Bank: an American clothier, not Joseph Banks 303

Topic 68 — Hartz and Hart Schaffner and Marx: names are not a pedigree 305

Appendix A. Complete companion papers 307

Aristarchus: heliocentrism, lunar eclipses and distance 308

Bounty: Bligh, Tofua and Pitcairn’s families 311

Breadfruit: botany, transport and Caribbean plantation slavery 313

Captain James Cook: precision, encounter and empire 315

Charles Mason: astronomer, transit observer and surveyor 317

Chronometer longitude in plain language 319

Contents • continued

Every topic and paper • click a title to begin PAGE

A Feather and an Anvil Falling Through Air 320

Galileo’s moons: Medicean stars, Marius and Kepler 321

Hartz versus Hart Schaffner & Marx: evidence and family claims 323

The Harvard Computers: women who measured the stars 325

Hipparchus: lunar parallax, eclipse reports and uncertainty 332

Jeremiah Dixon: surveyor, observer and partner 334

Jos. A. Bank: American clothier, Baltimore and ownership 338

Joseph Banks: botanist, collections and empire 340

Kepler’s Mars triangle method, illustrated 342

Kepler’s Mars triangulation: true scale and exaggeration 344

The Mason–Dixon survey: geometry, corrections and Indigenous limits 346

The Moon’s diameter: a four-city U.S. map 348

Nathaniel Bowditch: the self-taught navigator 349

Picard, 1669: triangulation map 356

Polaris: the North Star, with Shakespeare’s Sonnet 116 358

Rømer, Io eclipses, finite light and Huygens’s speed 365

Ships of exploration: Endeavour, Bounty, Seahorse and Bligh’s launch 367

The sidereal day and latitude sailing 369

Thumb to Moon: calculating the Moon’s diameter 375

Time to longitude: hours, minutes and seconds to degrees 377

Venus transit parallax: the triangle 378

From Venus transit to Cepheid distances 380

Venus transits, 1761 and 1769: parallax and expedition risks 398

Hurricane Wind Force: Force Grows with Velocity Squared 403

Appendix B. Earlier mathematics editions 404

Original 17-page mathematics paper — before coffee cooling 405

Original 18-page mathematics paper — before the two-point derivative explanation 422

Original 19-page mathematics paper — before growth rates 440

1. Mathematics foundations

The current mathematics paper comes first. The standalone ladder, growth-rate and flight glossaries follow in their original complete editions. Overlap is retained deliberately.

4 complete papers • begins on binder page 7

A MASTER PAPER / BEGIN WHEREVER YOU FEEL COMFORTABLE

Mathematics, one idea at a time

Jeff, this is not a test. Think of it as a walk from a shelter room to a ladder, then out to the Moon and the moving planets. We will name each idea before asking it to do any work.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: **say what you know, say what you want, and connect the two.**

CONTENTS

1. Algebra 2

Solving an unknown • shelter rooms and a truck

2. Geometry 4

Euclid • right triangles • Pythagoras • equal sides

3. Trigonometry 7

One ladder • sine, cosine and tangent • Galileo

4. Calculus 13

Newton and Leibniz • change • accumulation • orbits

4.5–4.6 Coffee cooling and its derivative 17–18

5. Growth rates 19

Odometer • garden • concrete • bacteria • digits • seating

Sources & further reading 22

The history and science behind the stories

How to read: stop whenever an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation.

The contents and colored page references are clickable. Click any page footer to return here. Examples simplify real situations; they are not operating limits or engineering advice.

1.1 / ALGEBRA STARTS WITH AN EMPTY PLACE

How many people in the shelter?

Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An **unknown** is simply the answer we have not found yet. We can leave an empty box for it, or give it a letter as a label.

Here, let **C** stand for the total number of people. The letter is not a special object. It is just the name on the empty place we want to fill.

The sign **×** means “times,” or “multiplied by.” The sign **=** means “has the same value as.” It is not an instruction to guess.

THREE GROUPS OF FORTY PEOPLE

C = 3 × 40 = 120 people

Check the answer: 40 people in the first room, 40 in the second and 40 in the third make 120. In symbols, **+** means “add”: **40 + 40 + 40 = 120**.

We did not make the problem harder by writing C. We made the question visible: “What total belongs here?”

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements and local rules.

1.2 / WORK BACKWARD, THEN CHECK

How many pallets in the truck?

Now a truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let **P** be the number of pallets. Before we know P, we can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

NAME THE UNKNOWN

P × 500 pounds = 3,000 pounds

“How many groups of 500 fit into 3,000?” is a division question. The sign ÷ means “divided by.” Division undoes multiplication.

FILL THE EMPTY PLACE

P = 3,000 ÷ 500 = 6 pallets

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting our answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves **P = 6**.

Name the unknown, use the relationship, then check. We will do it again with the ladder in Section 3 (page 7).

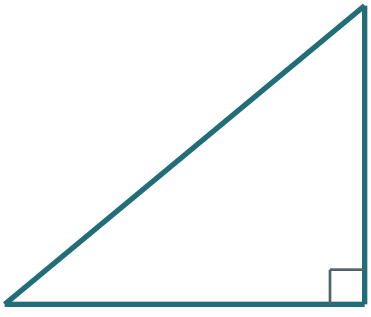
“lb” on the boxes means pounds. This is weight arithmetic only—not a loading plan. Real loads must meet vehicle, axle, space, balance and securement limits.

2.1 / GEOMETRY GIVES THE PROBLEM A SHAPE

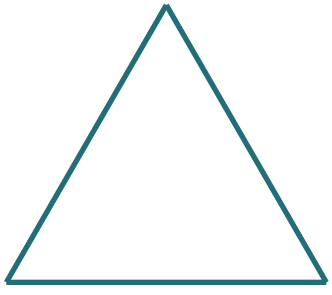
Euclid’s world of lines and corners

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE—about 2,300 years ago—he organized much earlier knowledge into a book called Elements.

Geometry studies shapes, sizes and positions. Euclid’s lasting example was to begin with clear assumptions and build an explanation, step by step. A **proof** is reasoning that shows why a claim must be true.



Right triangle



Equilateral triangle

An **angle** describes how widely two lines open apart. We measure angles in **degrees**, written with the little sign °. A full turn is 360 degrees; a square corner is 90 degrees.

A **right triangle** has one square corner. “Right” here means 90 degrees, not the right-hand side of the page. Its **hypotenuse** is the side across from that corner, and is always the longest side.

An **equilateral triangle** has three equal sides. On flat paper, each of its three angles is 60 degrees. It does **not** have a square corner.

These are different triangles. We can split the equal-sided one to make a right triangle; that connection will explain the ladder’s one-half relationship on the next diagram (page 6).

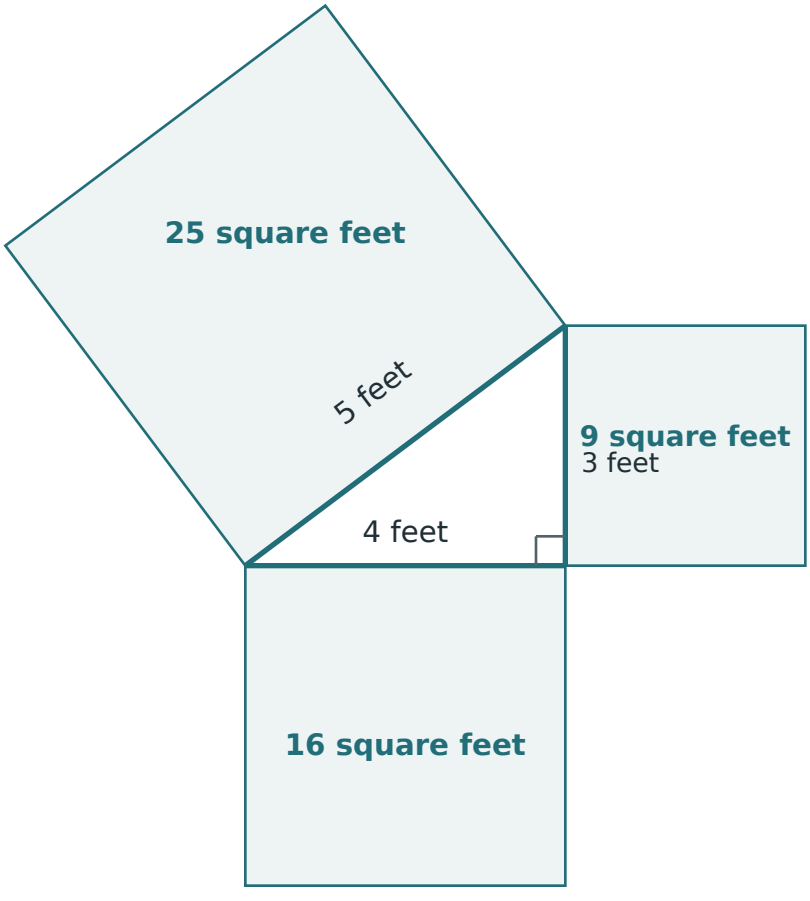
BCE means “before the common era.” Geometry is much older than any one mathematician. History sources are collected on page 22.

2.2 / PYTHAGORAS CONNECTS THREE LENGTHS

The 3–4–5 triangle

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A **theorem** is a statement established by proof.



Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A **square foot** is a square one foot on each side.

So our picture says: **3 × 3 + 4 × 4 = 5 × 5**, or **9 + 16 = 25**. If the long side were unknown, we would ask: "What length times itself gives 25?" The answer is 5.

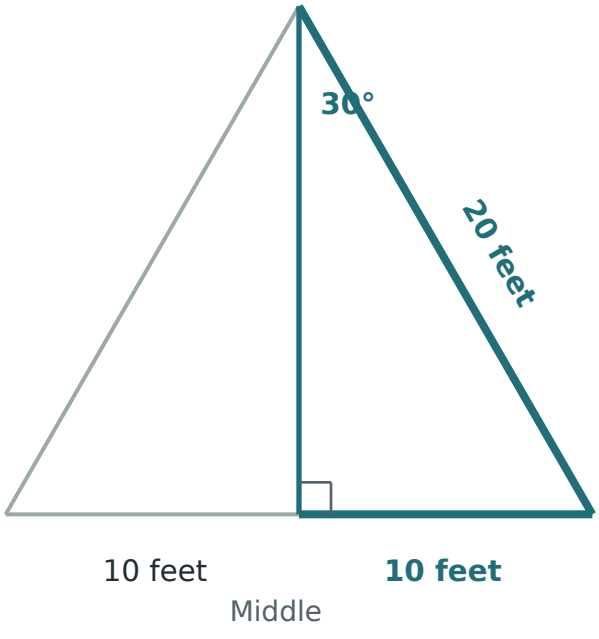
The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning (page 12).

2.3 / A BRIDGE FROM GEOMETRY TO TRIGONOMETRY

Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.



In either half, the side **opposite the 30-degree angle** is 10 feet long. The sloping side—the hypotenuse—is 20 feet long. So the opposite side is **half** the hypotenuse.

Turn that half-triangle around on the page: the lengths and angles do not change. It is the same shape as our 20-foot ladder reaching 10 feet up the wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not a lucky measurement or a number invented by a calculator.

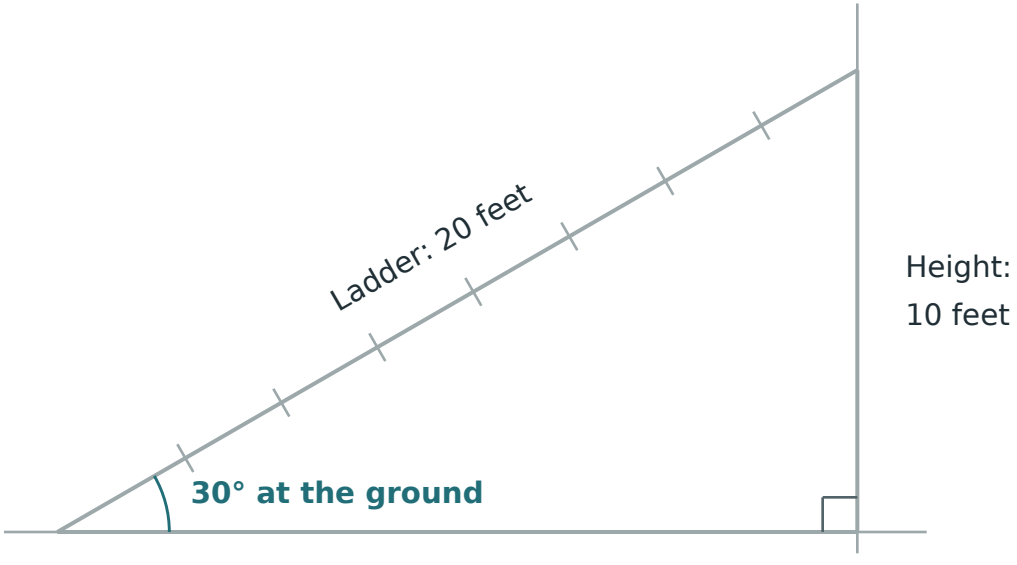
This ratio is called **sine** (Section 3, page 8). Geometry supplies the shape; trigonometry uses its proportions.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Return to the right triangle from Section 2 (page 4). A ladder leaning against a wall makes that shape. We will keep the **same ladder** throughout this section.

Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown (Section 1, page 2). You do not need calculus yet.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means "degrees." A square corner is 90 degrees.

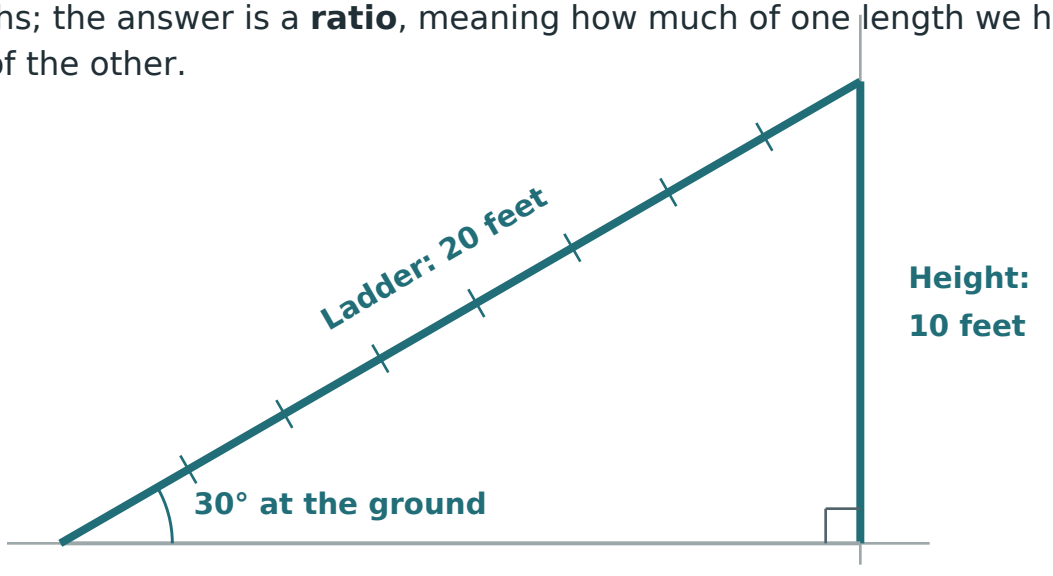
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. The equal-sided triangle on page 6 explains why; a calculator or table supplies the same ratio.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

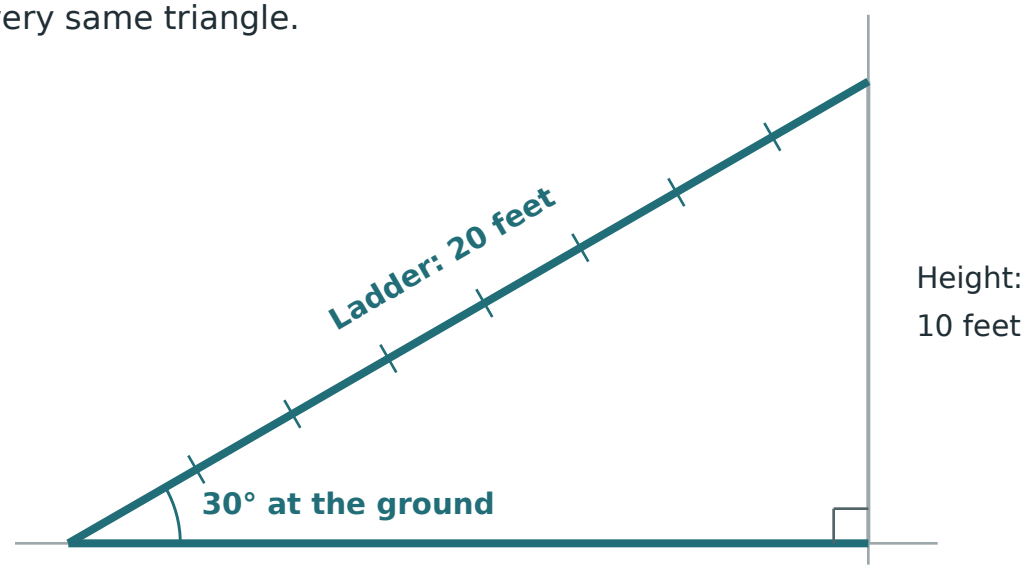
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



Base: about 17.32 feet

For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

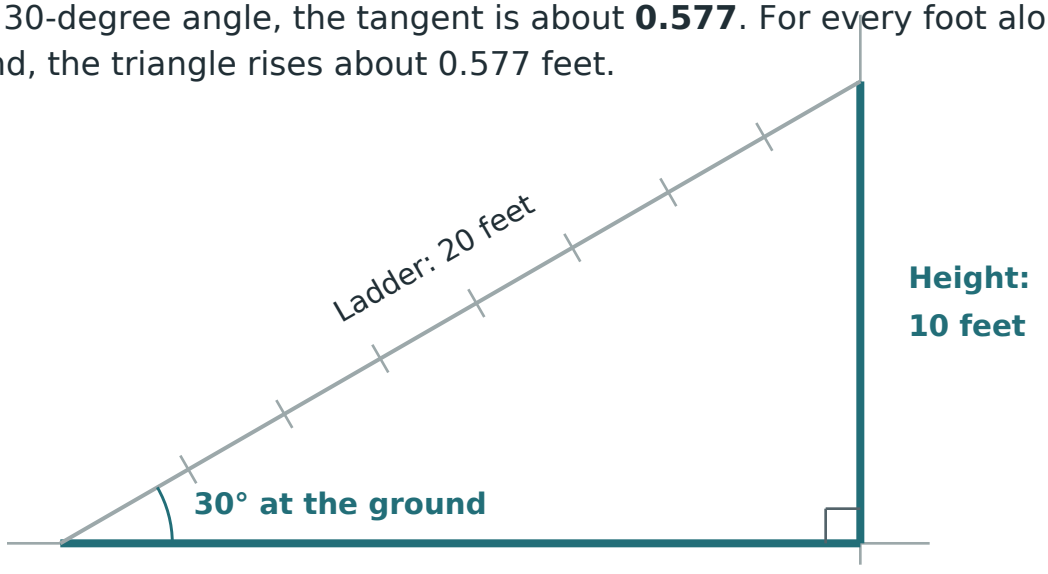
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

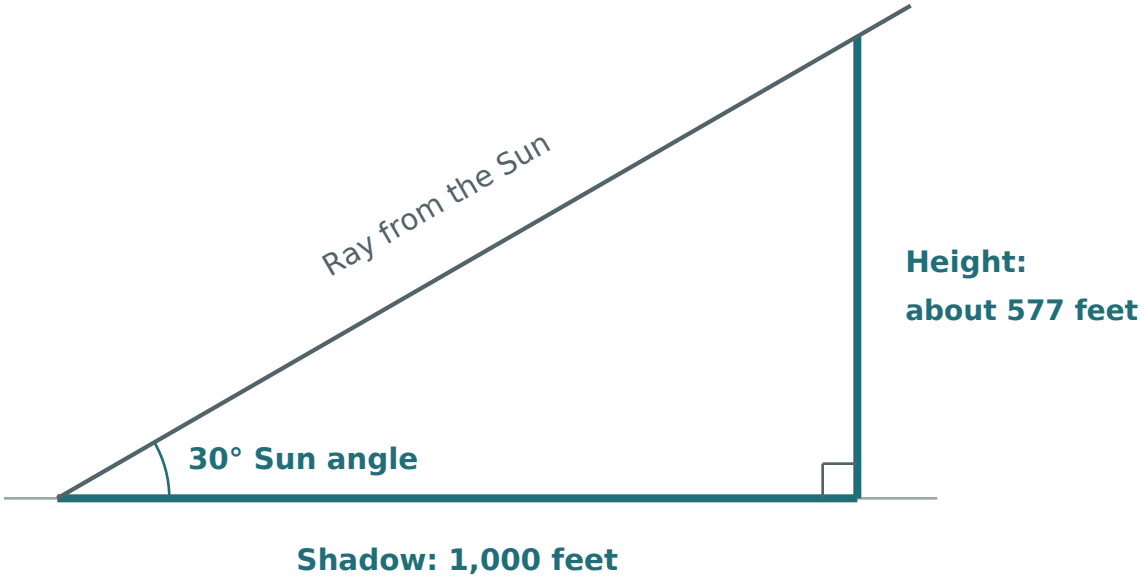


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule** (Section 2, page 5), the relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Where the story goes next

Geometry gives us a shape; trigonometry compares its sides. What if the shape or position changes from moment to moment? That is our next subject: Section 4, calculus (page 13).

Galileo's original text and the related shadow method are linked in the sources on page 22. The shadow example on page 11 is invented, not Galileo's data.

4.1 / NEWTON, LEIBNIZ AND THE MOVING WORLD

Now let something change

The shelter total, triangle and ladder all held still while we examined them. But a truck speeds up, a planet changes direction, and a tide rises. **Calculus** helps us describe continuous change and what builds up from it.

In the late 1600s, Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this new mathematics closely with motion. It was not one person's invention out of nothing.

How fast, right now?

A rate of change

How much altogether?

Accumulation

A **rate of change** compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If we know the truck's speed throughout a journey, we can add up the small distances it covers to find the whole trip.

These two questions are partners: from the journey's distance record, find the changing speed; from the changing speed, recover the distance traveled.

The earlier sections are foundations, not a claim that Pythagoras invented calculus. Geometry describes positions and lengths; algebra expresses relationships. Calculus asks how those relationships behave as things change.

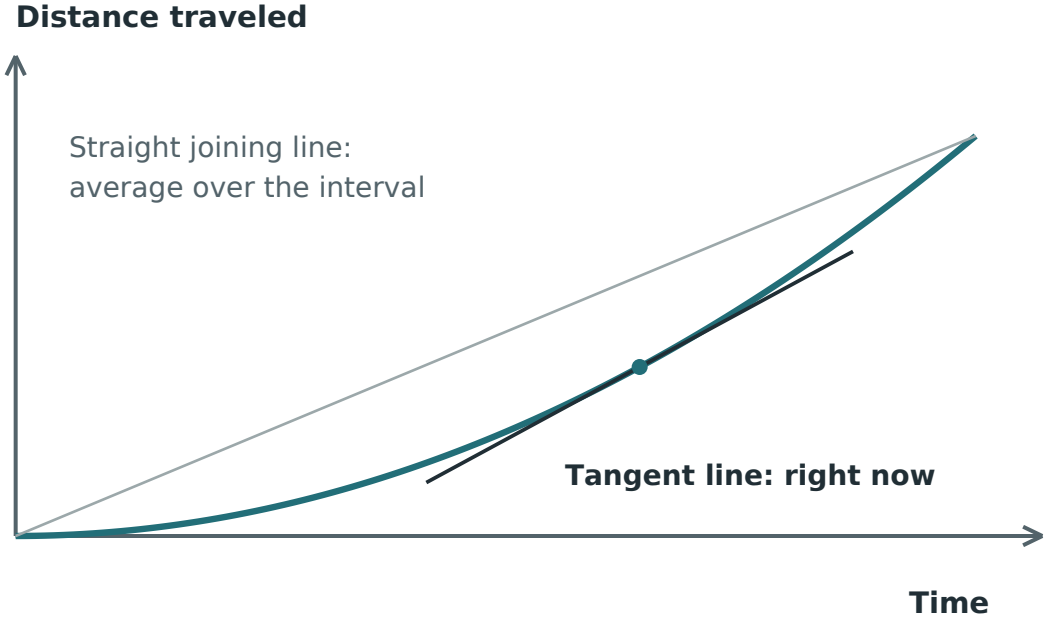
You can understand the central ideas without learning the formal symbols first. We will stay with the truck before returning to the sky.

4.2 / AVERAGE IS NOT THE SAME AS RIGHT NOW

A speedometer and a curve

Suppose the truck travels 60 miles in two hours. Its **average speed** is the total distance divided by the total time: **60 ÷ 2 = 30 miles per hour**. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment—the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.



This picture is a **graph**: time runs across; distance runs upward. The curve records a journey. Its **slope** means rise compared with horizontal run—here, distance change compared with time change.

A **tangent line** follows the curve's direction at one point. Its slope gives the rate there. Finding that local rate is called **differentiation**; the rate we find is called a **derivative**.

This is not quite the same use of "tangent" as on page 10. There, **tan** is a triangle's height-to-base ratio. Here, a **tangent line** is a line beside a curve. Their connection is slope: rise divided by run.

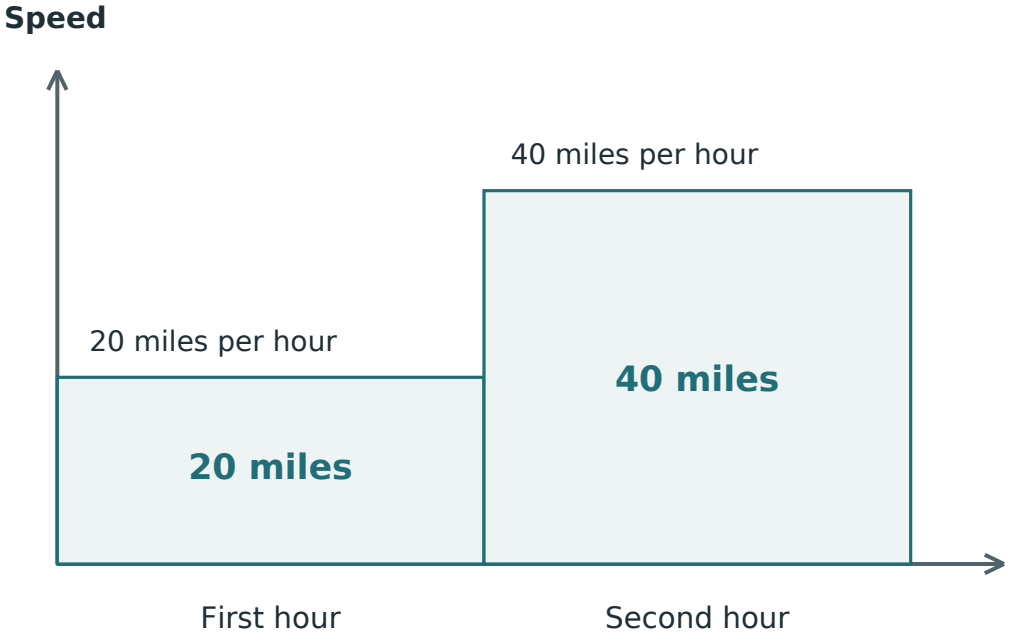
The graph is conceptual, not the numerical record of the 60-mile trip above. No formula for the curve is needed to read the idea.

4.3 / ADD THE LITTLE PIECES

From speed back to distance

Now turn the question around. Suppose we know how fast the truck moved. If it held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

So the whole distance is **20 + 40 = 60 miles**. We found a total by adding the contributions of different parts of the journey.



In a speed-versus-time graph, a rectangle’s height is the speed and its width is the time. Multiplying them gives the distance during that time. The **area under the speed graph** therefore represents distance.

When speed changes smoothly, divide the journey into short time pieces. Use each piece’s speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called **integration**; the accumulated result is an **integral**.

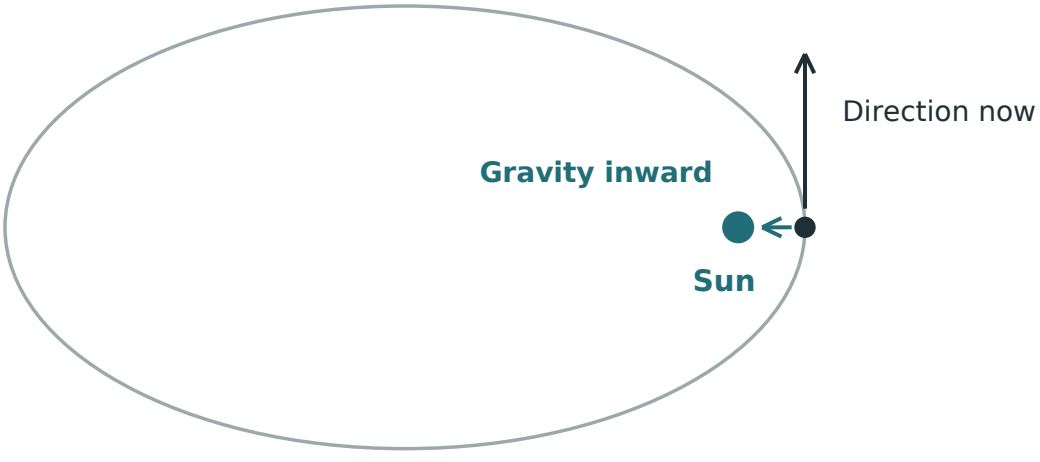
Differentiation asks for the local rate; integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 / FROM MOVING TRUCKS TO MOVING PLANETS

Orbits, tides and changing direction

For Newton, motion raised a deeper question: how does a pull change an object’s journey? **Gravity** is the attraction between masses—between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. **Acceleration** means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity and acceleration from moment to moment.



Schematic ellipse; sizes and distances are not to scale.

Forward motion and the Sun’s inward pull make the planet’s path curve. Without gravity it would continue straight in its current direction. Orbits need not be circles; many are **ellipses**—oval-shaped paths.

The Moon pulls harder on Earth’s near side than on its center, and less on the far side. This **difference across Earth** helps stretch the oceans and creates tides. The Sun contributes too.

Newton’s gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape and other effects. Calculus describes change; it does not make geography disappear.

Geometry describes shapes; calculus describes how they change.

4.5 / DIFFERENTIAL EQUATIONS, IN EVERYDAY LIFE

Why hot coffee cools more slowly

A fresh cup of coffee cools quickly at first, then more slowly as it approaches room temperature. A **differential equation** connects a changing quantity with its **rate of change**—how quickly it changes. Here is a simple model, called Newton's law of cooling.

First, name the pieces. **T** is the coffee's temperature, in degrees Fahrenheit. **t** is elapsed time, in minutes. **T_room** is the steady room temperature; "room" is a label, not multiplication. The **gap** is coffee temperature minus room temperature: **T − T_room**. The sign **−** means subtract.

dT/dt reads "rate of temperature change": degrees Fahrenheit per minute. Think of **dT** as a tiny temperature change and **dt** as a tiny time change; the derivative describes the limiting rate. **k** is a positive **cooling coefficient**—a multiplier telling how quickly the gap shrinks. Its units are **per minute**, or inverse time.

The equals sign means both sides have the same value. Parentheses group the gap. Writing k beside parentheses means multiply. Now we can read the equation:

dT/dt = −k(T − T_room)

In words: **the size of the cooling rate is k times the gap**. The minus sign before k makes the temperature-change rate negative: hot coffee's temperature is falling. A smaller gap gives slower cooling.

The **time constant** is one characteristic cooling time. Its name is **tau**, written **τ**. The rule is **τ = 1/k**: one divided by k. Because k is per minute, tau is in minutes. The actual time depends on the cup and conditions; we have not chosen a number of minutes.

The sign **%** means "out of a hundred." After **one** time constant, about **37% of the original gap remains**: 63% has closed. After **three**, about **5% remains**: 95% has closed. These percentages describe the **gap**, not the room temperature.

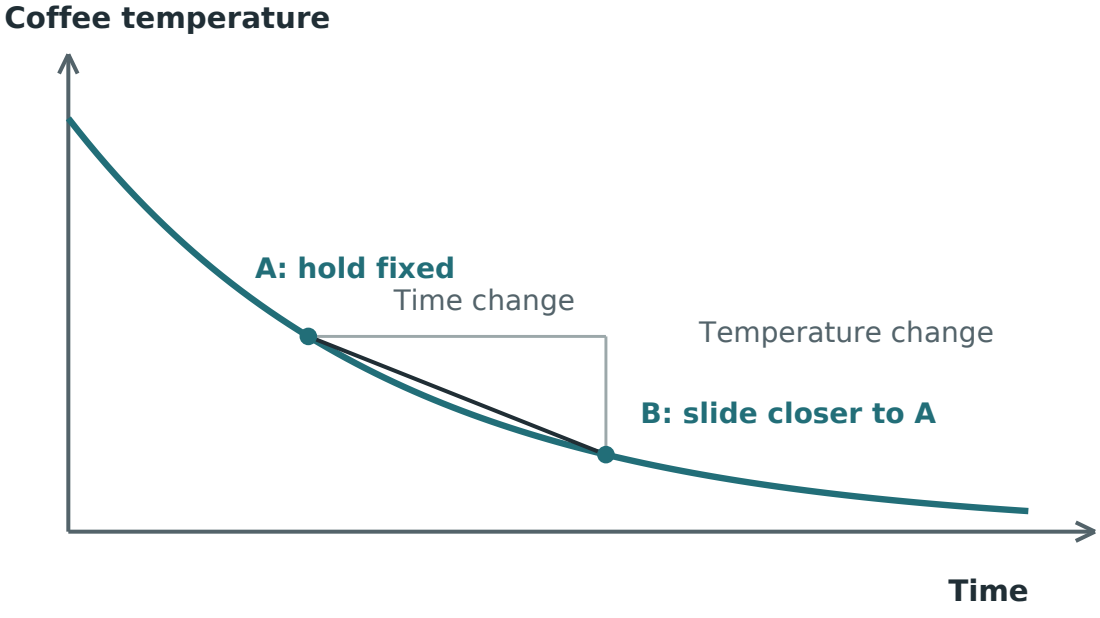
Example: coffee starts at 170°F in a 70°F room: a 100-degree gap. "°F" means degrees Fahrenheit. After one time constant it is about **107°F**; after three, about **75°F**.

The model assumes steady room temperature and approximately constant k. It only approximates real coffee. It approaches room temperature without reaching it exactly in finite time: page 14's local-rate idea at work.

4.6 / THE DERIVATIVE, ONE INSTANT AT A TIME

Two points, then one instant

On a **coffee-temperature graph**, time runs horizontally and temperature runs vertically. Choose an instant and mark its temperature as **point A**. Choose a nearby later instant and mark its temperature as **point B**. The letters are only labels.



Schematic curve, not observed measurements. The dark line joins A and B.

Join the points. The line’s slope is the **vertical temperature change divided by the horizontal time change**—rise divided by run. The run is positive; the rise is negative because coffee cools. The ratio is degrees Fahrenheit per minute.

Hold A fixed and slide B closer along the curve. Each two-point ratio estimates the slope at A. The value these ratios approach as the time gap shrinks is the **limit**: the derivative at A. We never divide by zero.

Every instant has its own slope. In this cooling model the slopes become less negative as the curve flattens. The limit is exact for the ideal smooth model; real measurements only estimate it.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: steady and square

5.1 Linear — add the same amount

Linear growth adds the same amount for every equal step in the input.

Example: An odometer records 10 more miles each hour while a vehicle holds 10 miles per hour.

Driving time (hours)	Trip distance (miles)
1	10
2	20
3	30
4	40
5	50
6	60
7	70
8	80

5.2 Quadratic — the square pattern

Quadratic growth follows the area of a square as its side gets longer.

Example: A square garden bed 2 feet on each side covers 4 square feet; a square foot is a 1-foot square.

Garden bed’s side (feet)	Garden area (square feet)
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

“Input” is the number we start with; “output” is what the rule gives back. These tables use different units, so compare their patterns—not miles directly with square feet.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: cubes and doubling

5.3 Cubic — the cube pattern

Cubic growth follows the volume of a cube as its side gets longer.

Example: A concrete cube 2 feet on each side has 8 cubic feet of volume; a cubic foot is a 1-foot cube.

Cube's side (feet)	Cube's volume (cubic feet)
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512

5.4 Exponential — multiply each time

Exponential growth multiplies by the same factor at each equal step.

Example: Imagine bacteria doubling each hour, starting with one at hour zero.

Time elapsed (hours)	Bacteria in the ideal model
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Bacteria cannot double forever: this model assumes unlimited resources. Doubling may start below a square pattern before eventually passing it.

5 / SIX PATTERNS, WITH EVERYDAY EXAMPLES

Growth rates: orders and digits

5.5 Factorial — count possible orders

Factorial growth counts the different ways to place distinct people in a row of seats.

Example: Three different people can occupy a row of three seats in six different ways.

n! (“n factorial”) multiplies whole numbers 1 through group size n; ! is not excitement.

People to seat (n)	Seating arrangements (factorial)
1	1
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320

5.6 Logarithmic — count multiplication steps

Logarithmic growth adds a fixed amount whenever the input grows by a fixed factor.

Example: log10 counts tenfold steps from 1: each step multiplies by ten.

Inputs are powers of ten, not 1–8; “digits” counts written figures (100 has three).

Input number	Tenfold steps (log10)	Written digits
1	0	1
10	1	2
100	2	3
1,000	3	4
10,000	4	5
100,000	5	6
1,000,000	6	7
10,000,000	7	8

Digit counts change in whole-number steps, not along a smooth log curve. “Log10 plus one” gives the digit counts shown **for these powers of ten**.

For these fixed rules, factorial eventually grows fastest; small-input rankings differ. Compare patterns, not quantities with different units.

READ ONLY IF YOU WANT TO FOLLOW THE THREADS

Sources, and what to remember

The historical stories are real; the shelter, truck and shadow numbers are teaching examples. The master paper reworks the standalone ladder lesson into Section 3. Original PDFs are left unchanged.

Geometry and its history
University of St Andrews, MacTutor: Euclid and Pythagoras. Euclid organized earlier knowledge; the right-triangle relationship predates Pythagoras.

[1] Galileo's actual mountain calculation
Galileo, Sidereus Nuncius (1610), E. S. Carlos translation (1880), pp. 28–30: Project Gutenberg ebook 46036. Sunlit peaks beyond the terminator; right-triangle calculation; "more than 4 Italian miles."

[2] The related shadow method
Mary McIntyre, BBC Sky at Night Magazine: "How to measure craters on the Moon". Height equals shadow length times the tangent of the local Sun angle, with simplifying assumptions.

Calculus and its development
University of St Andrews, MacTutor: "A history of the calculus". Earlier work, Newton, Leibniz, rates and accumulation. Orbits: NASA, "Gravity & Mechanics".

Why tides happen
NOAA: "Tides" and "What Causes Tides?". Uneven gravitational pull, the Moon and Sun, and the influence of local geography.

Four questions, one growing toolkit
Algebra: What number is missing?
Geometry: What shape and lengths do we have?
Trigonometry: What do the angles tell us about the sides?
Calculus: How is it changing, and what does it add up to?

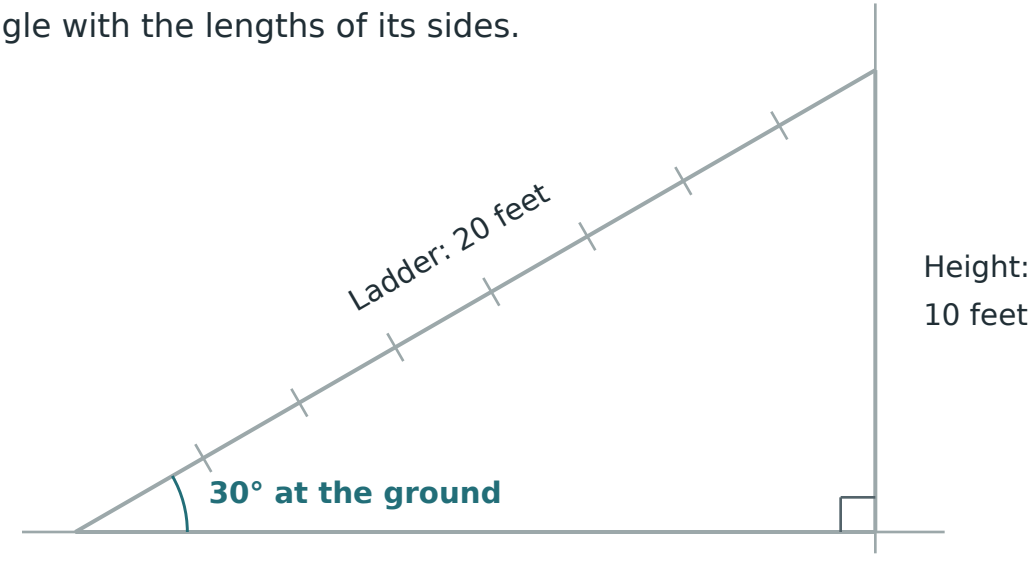
You do not have to memorize every word. Recognizing the question is already a beginning.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Imagine a ladder leaning against a straight wall on level ground. The ladder, wall and ground make a triangle. We will keep this **same triangle** throughout the lesson.

You do not need calculus. **Trigonometry** simply helps us connect the angles of a triangle with the lengths of its sides.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means “degrees.” A square corner is 90 degrees.

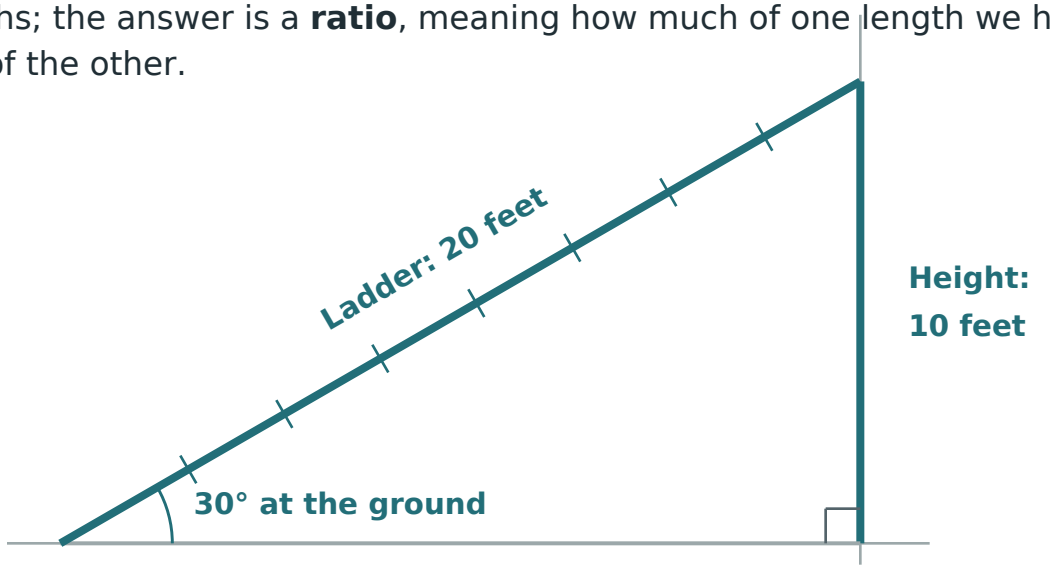
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. A calculator or a table of triangle ratios tells us this—not another measurement.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

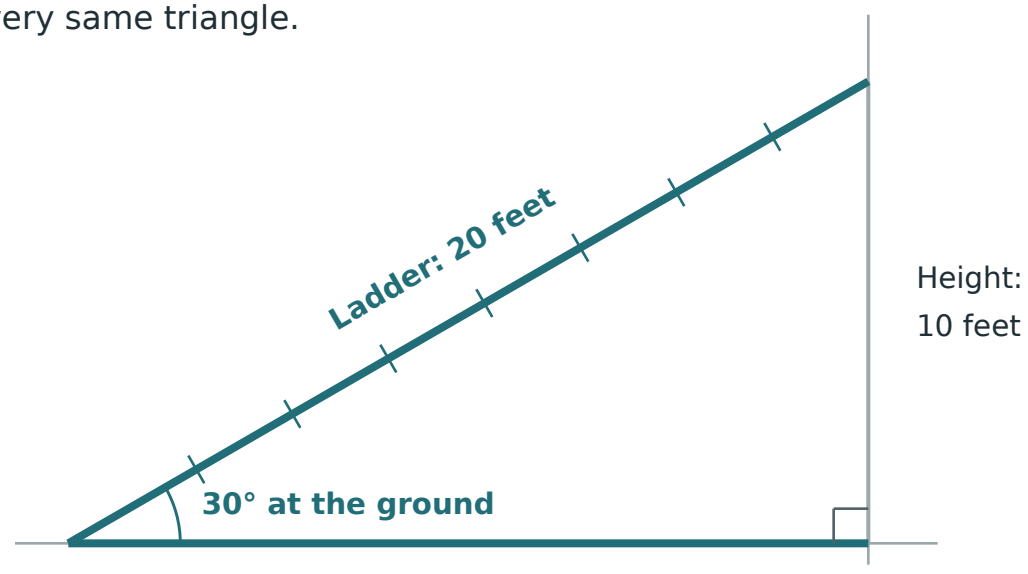
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

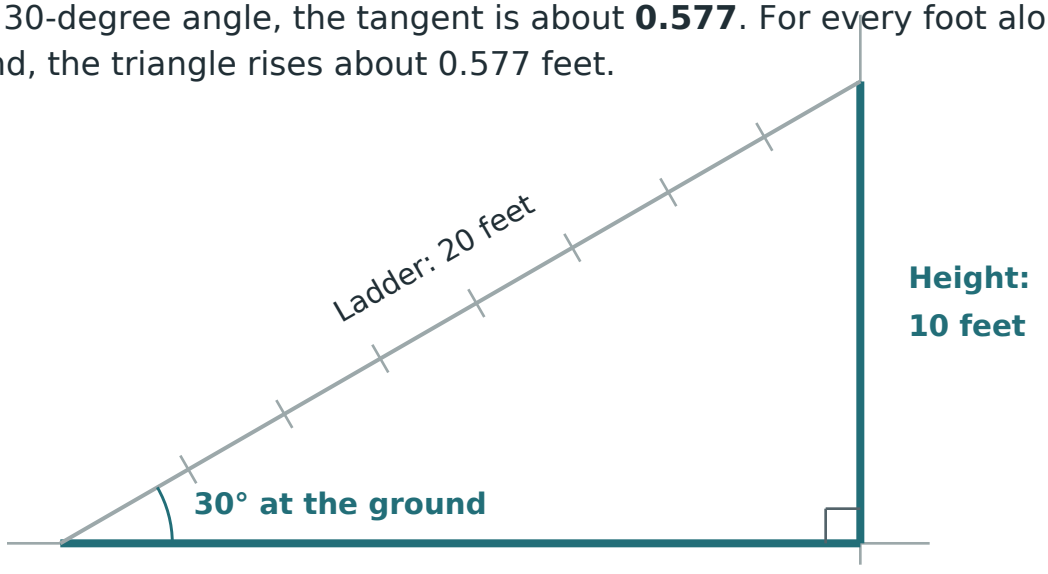
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

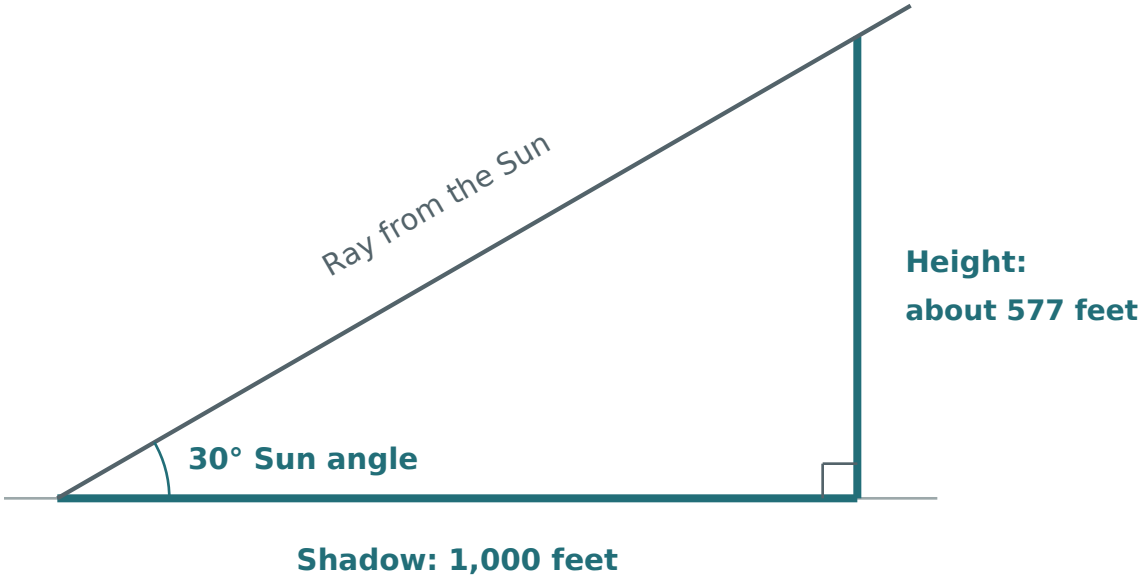


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule**, another relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Sources & further reading

[1] Galileo, Sidereus Nuncius (1610), lunar mountains; E. S. Carlos's English translation (1880), pp. 28–30. Project Gutenberg, ebook 46036.

[2] BBC Sky at Night Magazine, "How to measure craters on the Moon" (shadow method and its limits). The source titles are clickable in this PDF.

Growth rates compared

Begin with the question: what changes when the input gets bigger?

1. Linear — add the same amount

Linear growth adds the same amount for every equal step in the input.

Example: An odometer records 10 more miles each hour while a vehicle holds 10 miles per hour.

Driving time (hours)	Trip distance (miles)
1	10
2	20
3	30
4	40
5	50
6	60
7	70
8	80

2. Quadratic — the square pattern

Quadratic growth follows the area of a square as its side gets longer.

Example: A square garden bed 2 feet on each side covers 4 square feet; a square foot is a 1-foot square.

Garden bed’s side (feet)	Garden area (square feet)
1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

“Input” is the number we start with; “output” is what the rule gives back. These tables use different units, so compare their patterns—not miles directly with square feet.

Growth rates compared

Two more patterns: grow in three directions, or keep doubling.

3. Cubic — the cube pattern

Cubic growth follows the volume of a cube as its side gets longer.

Example: A concrete cube 2 feet on each side has 8 cubic feet of volume; a cubic foot is a 1-foot cube.

Cube’s side (feet)	Cube’s volume (cubic feet)
1	1
2	8
3	27
4	64
5	125
6	216
7	343
8	512

4. Exponential — multiply each time

Exponential growth multiplies by the same factor at each equal step.

Example: Imagine bacteria doubling each hour, starting with one at hour zero.

Time elapsed (hours)	Bacteria in the ideal model
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Bacteria cannot double forever: this model assumes unlimited resources. Doubling may start below a square pattern before eventually passing it.

Growth rates compared

One pattern grows especially fast; the other compresses large numbers.

5. Factorial — count possible orders

Factorial growth counts the different ways to place distinct people in a row of seats.

Example: Three different people can occupy a row of three seats in six different ways.

n! (“n factorial”) multiplies whole numbers 1 through group size n; ! is not excitement.

People to seat (n)	Seating arrangements (factorial)
1	1
2	2
3	6
4	24
5	120
6	720
7	5,040
8	40,320

6. Logarithmic — count multiplication steps

Logarithmic growth adds a fixed amount whenever the input grows by a fixed factor.

Example: log10 counts tenfold steps from 1: each step multiplies by ten.

Inputs are powers of ten, not 1–8; “digits” counts written figures (100 has three).

Input number	Tenfold steps (log10)	Written digits
1	0	1
10	1	2
100	2	3
1,000	3	4
10,000	4	5
100,000	5	6
1,000,000	6	7
10,000,000	7	8

Digit counts change in whole-number steps, not along a smooth log curve. “Log10 plus one” gives the digit counts shown **for these powers of ten**.

For these fixed rules, factorial eventually grows fastest; small-input rankings differ. Compare patterns, not quantities with different units.

Flight calculus, in plain language

Hypothetical commercial flight: Los Angeles to New York.

1 / Name the quantities and their changes

Calculus connects how fast something is changing with how much has built up.

“Elapsed time” means time since takeoff; no actual flight data are used here.

Variable

A variable is a quantity whose value can change.

During our flight, elapsed time, altitude, fuel remaining and speed are variables.

Function

A function is a rule that gives one output for each input.

For our chosen flight, entering an elapsed time into the altitude record gives one altitude.

Independent and dependent variables

An independent variable is a chosen input, while a dependent variable is an output we follow as that input changes.

We choose elapsed time as input and follow altitude or fuel remaining as output.

Rate of change

A rate of change says how much one quantity changes compared with another.

Fuel-burn rate tells us how much fuel is used per unit of elapsed time.

Slope

Slope compares a vertical change on a graph with a horizontal change.

On an altitude-versus-elapsed-time graph, slope is vertical speed, not how quickly the airplane travels across the ground.

Flight calculus, in plain language

Hypothetical commercial flight: Los Angeles to New York.

2 / Find what is happening at one instant

Instantaneous versus average

An instantaneous rate describes one moment, while an average rate describes a whole interval.

Our average ground speed is total traveled path distance divided by elapsed time, while an instantaneous reading estimates ground speed at one moment.

Limit

A limit is the value a quantity approaches as we narrow in on a chosen situation.

Using shorter time intervals around one flight moment lets us approach its rate without ever dividing by a zero-length interval.

Derivative

A derivative is the exact instantaneous rate in an ideal smooth mathematical model.

The altitude derivative is zero in level cruise even though the airplane may be moving quickly across the ground.

Differentiation

Differentiation is the process of finding a derivative.

Differentiating our modeled altitude using elapsed time gives modeled vertical speed.

Tangent line

A tangent line matches a smooth curve’s direction at one chosen point.

Its slope on our altitude-time graph gives modeled vertical speed at that instant, not the triangle ratio called tangent.

Flight calculus, in plain language

Hypothetical commercial flight: Los Angeles to New York.

3 / Add up what happens along the way

Integral

An integral adds a continuously varying quantity over an interval to find a total.

The integral of our fuel-burn rate over elapsed time gives total fuel used, not fuel remaining.

Integration

Integration is the process of finding an integral.

Integrating ground speed over elapsed time gives distance along the flight’s traveled path, not the straight-line distance between its endpoints.

Area under a curve

For a positive rate-versus-time graph, area under the curve represents the accumulated total.

On our fuel-burn-rate graph the area means fuel used, whereas area under an altitude-time graph does not mean distance traveled.

Initial condition

An initial condition is a known starting value used to anchor a model.

Our model could start with a known fuel amount and subtract accumulated fuel use to find what remains.

Differential equation

A differential equation connects a changing quantity with one or more of its rates of change.

A simple flight-fuel model says that the rate of change of fuel remaining is the negative of the fuel-burn rate.

Flight calculus, in plain language

Hypothetical commercial flight: Los Angeles to New York.

4 / Describe motion and make sensible choices

Velocity

Velocity describes speed together with direction.

Our airplane’s velocity includes both its ground speed and its direction of travel.

Acceleration

Acceleration is a change in velocity, whether speed, direction or both change.

Our airplane accelerates during a turn even if its speed stays constant.

Maximum and minimum

A maximum is a highest value and a minimum is a lowest value within the range being considered.

For this flight, we could ask for the maximum altitude reached or the minimum fuel remaining over the modeled journey.

Optimization

Optimization means choosing the best available result while respecting stated limits.

A fuel-saving flight plan must still respect safety, weather and arrival requirements rather than simply seek the fastest trip.

Model

A model is a simplified description that helps us reason about a real situation.

Our smooth flight curves are models, while real instrument readings are measurements with uncertainty.

A conversation starter: “The derivative tells me the rate right now; the integral tells me how much has added up.”

2. Earth, sky and navigation

Measurement grows from shadows and angles into planets, stars, clocks and distance. Public topic numbers remain visible so that the printed collection and the learning log can be used together.

21 complete topics • begins on binder page 43

Saturday, October 31, 2026

Eratosthenes measures the Earth

About 240 BC, one angle (7.2°) and one distance gave the size of the whole planet.

Pronunciation

Eratosthenes: **air-uh-TOSS-thuh-neez**. Stress on the third syllable, TOSS.

Euclid: **YOO-klid**.

Bematist: **BEE-muh-tist** (from Greek bēma, a step).

Who he was

Eratosthenes of Cyrene (about 276–194 BC) was the chief librarian of the great **Library of Alexandria** in Egypt. He coined the word **geography** (geo, earth + graphia, writing or drawing), which is why he is called the father of geography. Rivals nicknamed him "Beta," second best at everything, because he worked in so many fields.

The clue

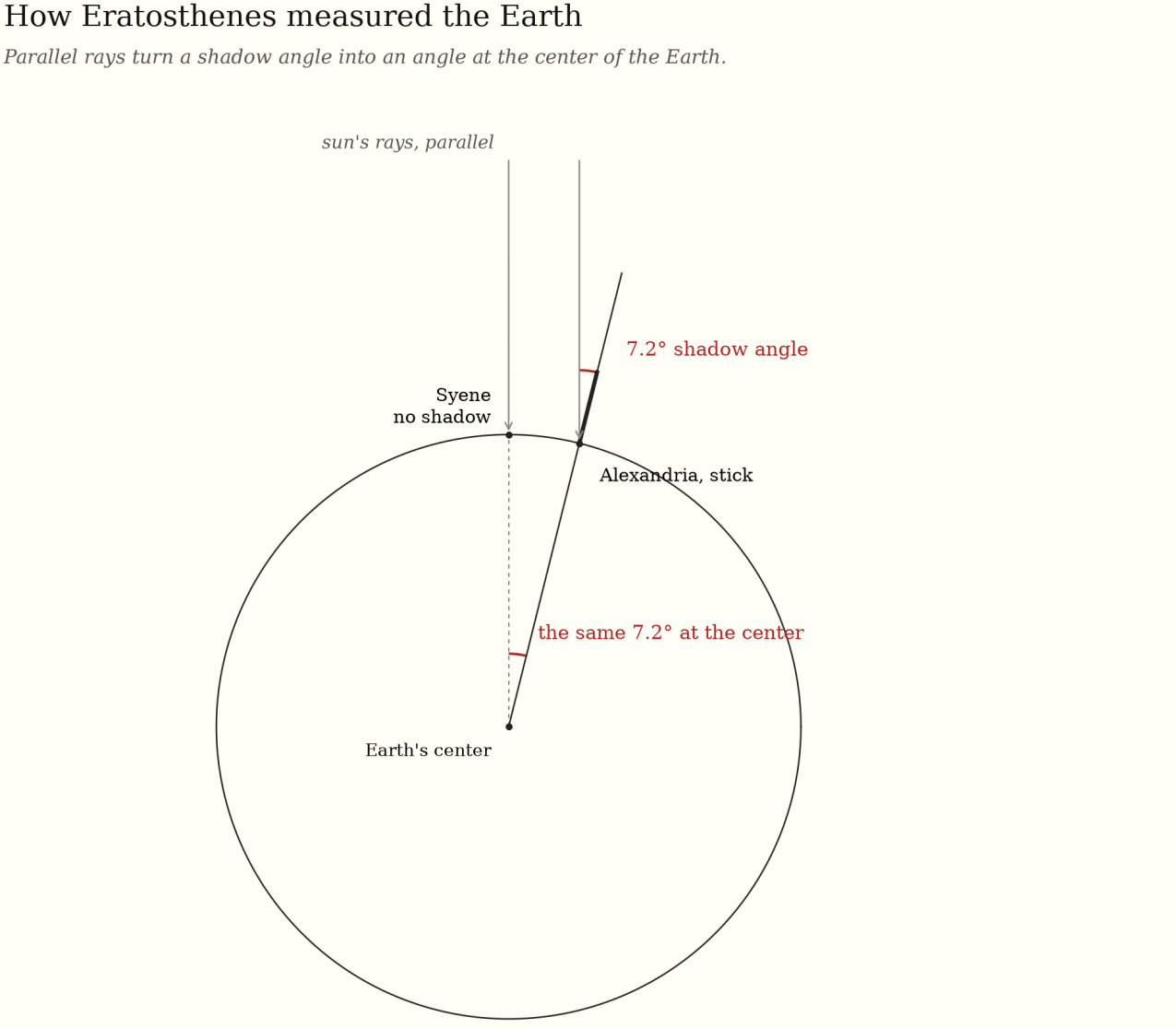
He heard that at **Syene** (modern Aswan), far up the Nile, at noon on the summer solstice the sun shone straight down a deep well, lighting the water at the bottom. An upright stick cast no shadow. The sun was directly overhead.

On the same day at noon in **Alexandria**, to the north, an upright stick did cast a shadow. The sun was about **7.2°** off overhead.

The reasoning: Euclid's alternate angles

If the Earth were flat, the sun's parallel rays would make the same shadow everywhere. A different shadow means the ground is curved.

Treat the sun's rays as **parallel lines** (see why sun rays count as parallel). Extend the Syene stick and the Alexandria stick down into the Earth; both point to the center. The Alexandria stick's line is a **transversal** cutting across two parallel lines (the ray at Alexandria and the ray going down the Syene well). Euclid proved that when a line crosses two parallel lines, the **alternate angles are equal**. So the 7.2° angle at the top of the Alexandria stick equals the angle between the two cities at the center of the Earth.



Parallel sun rays at Syene and Alexandria; the 7.2° angle at the stick equals the 7.2° angle at Earth's center (angle exaggerated so it's visible)

The arithmetic

7.2° is **1/50 of a full circle** (360 ÷ 7.2 = 50). So the distance from Syene to Alexandria is 1/50 of the way around the Earth.

That distance was about **5,000 stadia**. 5,000 × 50 = **250,000 stadia** (he later used 252,000). Depending on which length of stadion he used, that is roughly 24,000 to 29,000 miles. The true figure is about **24,901 miles**. With a stick and a shadow, he was very close.

The bematists

How did he know the distance? Egypt employed **bematists**, professional pacers trained to walk with exactly even steps and count them, to survey land and roads. Their measurements, along with camel caravan travel times and royal surveys, gave the Syene–Alexandria distance. (Some of it was approximate: Syene is not exactly due south of Alexandria, and it isn't exactly on the Tropic of Cancer. The errors partly cancel.)

Key facts

- Eratosthenes (air-uh-TOSS-thuh-neez), father of geography: measured Earth with one 7.2° angle and one distance
- Euclid: a line crossing parallel lines makes equal alternate angles
- Bematists: Greek pacers who measured distance by counting steps (bēma = step)
- 7.2° is 1/50 of a circle; 5,000 stadia × 50 = 250,000 stadia
- True circumference about 24,901 miles; his figure was close

Saturday, October 31, 2026

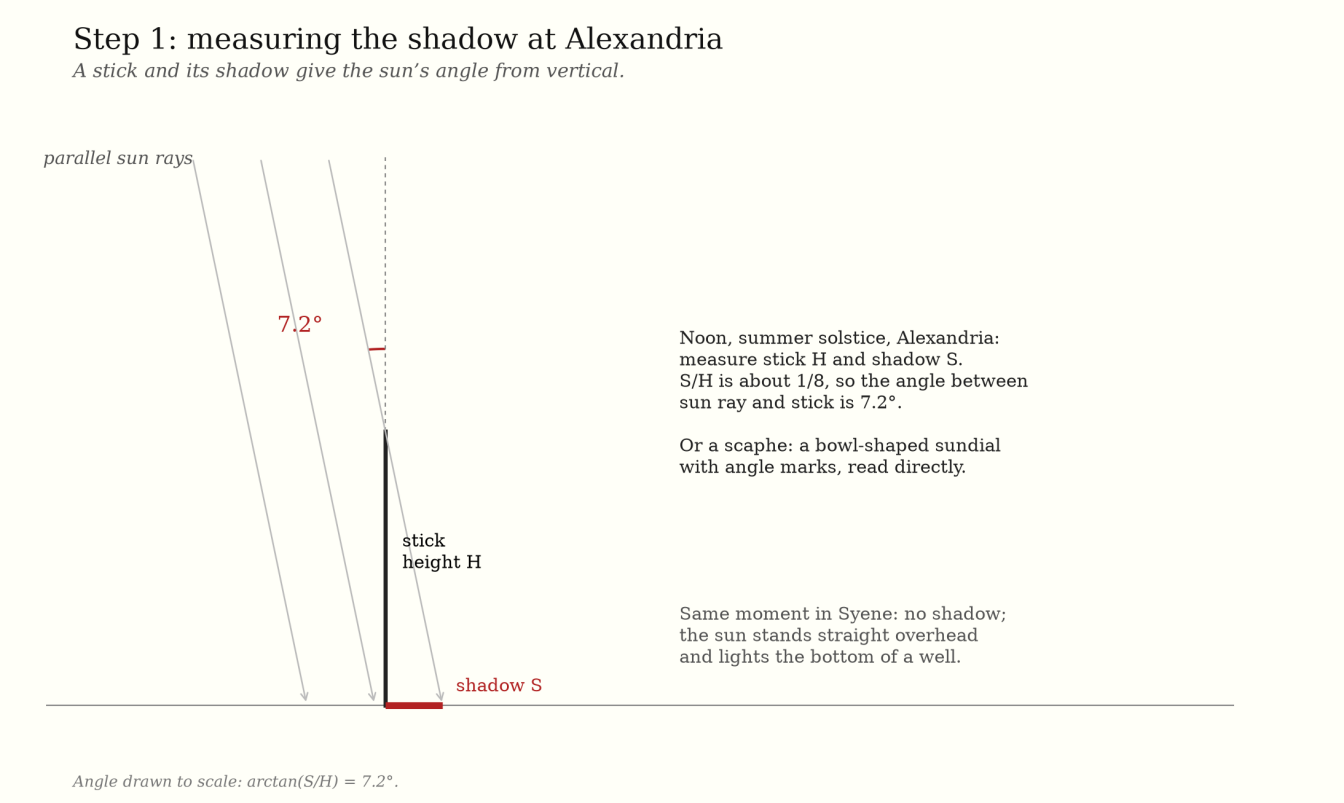
The shadow stick and why the lines meet at Earth's center

How the 7.2° was measured at Alexandria, and why a plumb line points to the center of the Earth.

Measuring the shadow at Alexandria

At noon on the summer solstice he stood a stick (a gnomon) perfectly upright and compared the length of its shadow with its height. The shadow came out at about **1/8 of the stick's height**. A right triangle with those proportions has an angle at the top of about **7.2°**.

He may also have used a **scaphe**, a bowl-shaped sundial with a pointer at the bottom and angle marks scratched on the inside, so the shadow's tip shows the angle directly.



At Alexandria the upright stick's shadow is about 1/8 of its height, giving 7.2°

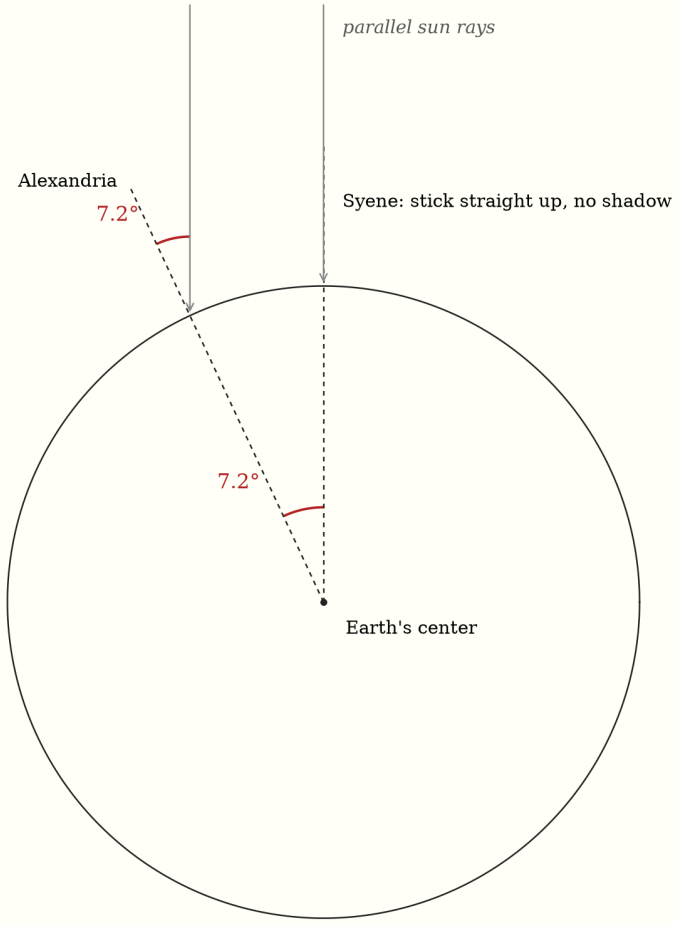
How do we know the lines meet at the center?

Honestly, this was an **assumption**, but a sound one.

You make a stick "upright" with a **plumb line**: a weight hanging on a string. The string hangs the way gravity pulls. On a round Earth, gravity pulls toward the center, so every plumb line, everywhere, points at the center of the Earth. Extend the Syene stick and the Alexandria stick downward and they cross there.

Step 2: why the sticks point to Earth's center

Plumb lines follow gravity, and gravity points to the center.



A stick set upright with a plumb line (a weight on a string) points along gravity, which on a round Earth runs straight toward the center. Extend both sticks downward and they cross there. The sun's rays are parallel, so the Alexandria stick crosses two parallel lines and the alternate angles are equal (Euclid). 7.2° is 1/50 of 360°, so Earth is 50 × 5,000 = 250,000 stadia around.

Angle exaggerated (drawn at 25°) so it is visible.

Plumb lines at Syene and Alexandria both point to Earth's center, where they meet at 7.2°

Then Euclid does the rest: the Alexandria stick crosses two parallel lines (its own sun ray and the Syene stick's line, which runs along the sun's ray), so the angle at the top

equals the angle at the center. 7.2° is 1/50 of a circle, so the Earth is **50 times** the Syene–Alexandria distance around.

Key facts

- At Alexandria the shadow was about 1/8 of the stick's height, about 7.2°
- A scaphe is a bowl sundial with angle marks
- A plumb line hangs toward Earth's center, so extended sticks meet there

Saturday, October 31, 2026

Why the sun's rays count as parallel (Aristarchus)

The Greeks already knew the sun was very far away, so rays reaching two cities 500 miles apart are effectively parallel.

The worry

Eratosthenes' method needs the sun's rays at Syene and Alexandria to be parallel. If the sun were close, the rays would spread out like a lamp's and the angle would mean something else. How did he avoid that mistake?

The answer: the sun was already known to be very far away

Around **260 BC**, **Aristarchus of Samos** (air-iss-TAR-kus) estimated that the sun was about **20 times farther than the Moon**. He did it with geometry: at half moon, the Sun, Moon and Earth form a right angle at the Moon, and the angle he measured at Earth gave the ratio.

That is far too small; the real figure is about **400 times**. But even 20 times was enough to show that two cities 500 miles apart see the sun from almost exactly the same direction. Over that tiny baseline, the rays are parallel for all practical purposes.

Eclipses helped too

During a lunar eclipse, the Earth's curved shadow falls across the Moon. Its size and shape showed the Earth is round and the sun is huge and distant. Aristarchus even concluded that because the sun is so big, it, not the Earth, should be at the center, and the Earth should go around it, some 1,800 years before Copernicus.

Key facts

- Aristarchus (about 260 BC) put the sun about 20 times farther than the Moon; really about 400
- Even that showed the sun's rays are effectively parallel over 500 miles
- Aristarchus proposed a sun-centered system 1,800 years before Copernicus

Saturday, October 31, 2026

The flat-Earth myth and Columbus

Educated Europeans never forgot the Earth was round. The argument with Columbus was about its size, and he was wrong.

"Then we got stupid for hundreds of years"?

Partly true, but not about the shape of the Earth. After Eratosthenes, educated people in Europe and the Islamic world **never lost** the knowledge that the Earth is a sphere. Medieval scholars like the Venerable Bede (700s) and Thomas Aquinas took it for granted, and the standard medieval astronomy textbook was literally called On the Sphere (Sacrobosco, around 1230). Kings held an orb, a globe, as a symbol of their rule.

Columbus

The famous story that Columbus had to prove the world was round against people who thought he'd sail off the edge is a **myth**. It was popularized by Washington Irving's romantic 1828 biography of Columbus, and later spread in the 1800s.

The real argument in 1492 was about **size**. Columbus claimed Asia was only about 2,400 miles west of the Canary Islands. He did this by choosing the smallest available estimate of the Earth and the largest estimate of how far Asia stretched east. The experts, who knew Eratosthenes' figure, said the distance was far greater and no ship could carry enough food and water. **They were right.** Columbus survived only because an unknown continent was in the way. He died insisting he had reached Asia.

What did get worse

What medieval Europe did lose for centuries was a lot of Greek science in general, which was preserved and extended by scholars in the Islamic world and came back through Arabic translations. And on the question of whether the **Earth moves**, the Church's resistance was real. See Copernicus and Bruno and Galileo.

Key facts

- Educated medieval Europeans knew the Earth was round
- The Columbus flat-Earth story was popularized by Washington Irving, 1828
- The real 1492 dispute was the Earth's size; Columbus underestimated it and the experts were right

Saturday, October 31, 2026

Copernicus, 1543

Nicolaus Copernicus put the sun at the center and published as he was dying.

The book

Nicolaus Copernicus (1473–1543), a Polish astronomer and church canon, worked out a system with the **sun at the center** and the Earth as one planet circling it, spinning on its axis daily. He held back publishing for decades, partly fearing ridicule.

De revolutionibus orbium coelestium (On the Revolutions of the Heavenly Spheres) was finally printed in **1543**. The story goes that the first printed copy was put in his hands on his deathbed, on May 24, 1543.

Why it didn't cause an uproar at first

An unsigned preface, added by the Lutheran minister Andreas Osiander who oversaw the printing, said the system was just a mathematical tool for calculating, not a claim about reality. Copernicus was dead and couldn't object. The book was dense mathematics, read mostly by astronomers. The real fight came about 70 years later. See Bruno and Galileo.

Key facts

- Copernicus 1543: Earth moves around the sun, published as he was dying
- De revolutionibus orbium coelestium; Osiander's preface called it only a calculating tool

Saturday, October 31, 2026

Bruno, Galileo and "Eppur si muove

Were people put to death for saying the Earth isn't the center? Bruno was burned in 1600, mostly for theology; Galileo was convicted in 1633.

Giordano Bruno, burned in 1600

Giordano Bruno, an Italian friar and philosopher, went beyond Copernicus: he taught that the universe is infinite, the stars are other suns, and they may have their own worlds with life. The Roman Inquisition burned him at the stake in Rome's **Campo de' Fiori** on **February 17, 1600**.

But it is not quite true that he was killed for astronomy. The main charges were theological: denying the Trinity, the divinity of Christ and other core doctrines. His cosmology was part of the picture, not the whole of it. A statue of him stands where he was burned.

Galileo

Galileo Galilei (1564–1642) turned a telescope on the sky in 1609–10 and saw moons circling Jupiter, the phases of Venus (which only make sense if Venus orbits the sun) and mountains on the Moon. Strong evidence for Copernicus.

- **1616**: the Church declared the sun-centered idea contrary to scripture, put Copernicus's book on the banned list until corrected, and warned Galileo not to hold or defend it.
- **1632**: he published the Dialogue Concerning the Two Chief World Systems, which made the case anyway and put the Pope's own argument in the mouth of a character named Simplicio.
- **1633**: tried by the Inquisition, found "vehemently suspect of heresy," forced to kneel and **abjure** (formally renounce) the idea. Sentenced to prison, commuted to **house arrest** for life at his villa at Arcetri near Florence. He died there in 1642, the year Newton was born.
- **1992**: Pope John Paul II formally acknowledged the Church had erred.

"Eppur si muove" (And yet it moves)

Legend says that after abjuring, Galileo muttered **"Eppur si muove"** (EP-poor see MWOH-veh), "And yet it moves," meaning the Earth. It is **probably a legend**: it first appears in print in 1757, more than a century later (Giuseppe Baretti), though a painting from the 1640s may show the words. Saying it aloud in front of the Inquisition

would have been suicidal. But it captures the truth: you can force a man to deny something, but the Earth goes on moving.

Key facts

- Bruno burned February 17, 1600, Campo de' Fiori; mainly for theological heresy
- Galileo 1633: convicted, house arrest; "Eppur si muove" (And yet it moves) is likely legend
- Galileo saw Jupiter's moons and Venus's phases (1610)
- Church acknowledged its error in 1992

Wednesday, November 4, 2026

Looking Up: Polaris, Planets, Ptolemy and Copernicus

The turning sky, the star that tells your latitude, the wanderers that loop backward, and the 1,400-year model that predicted well and was still wrong.

The sky turns around a point

Watch the night sky for an hour and every star wheels around one point, the **celestial pole**, the spot Earth's axis points to. In the north, a moderately bright star sits almost on it: **Polaris**, the North Star (today it's within about **0.7°** of the true pole, so it barely moves while everything else circles it).

Polaris tells your latitude

The height of Polaris above the horizon equals your latitude.

- At the **North Pole** (90°N) it is **straight overhead**.
- At the **equator** it sits on the horizon.
- In **Florida** (about 25°–31°N) it stands roughly **27°** up, about three fists held at arm's length.

That's why sailors could always find latitude on a clear night. Longitude was another matter (see Galileo). For the full story of Polaris itself — the triple star, Cepheids, precession, every name, and Shakespeare's Sonnet 116 — see Polaris: the North Star, explained.

The wanderers

The stars keep their patterns; five bright lights don't. The Greeks called them **planētai**, **"wanderers,"** which gave us **planet**.

Mars's retrograde motion. Mars normally drifts eastward against the stars night after night. But every two years or so it **slows, stops, runs backward (westward) for a couple of months, then turns forward again**, tracing a loop or zigzag.

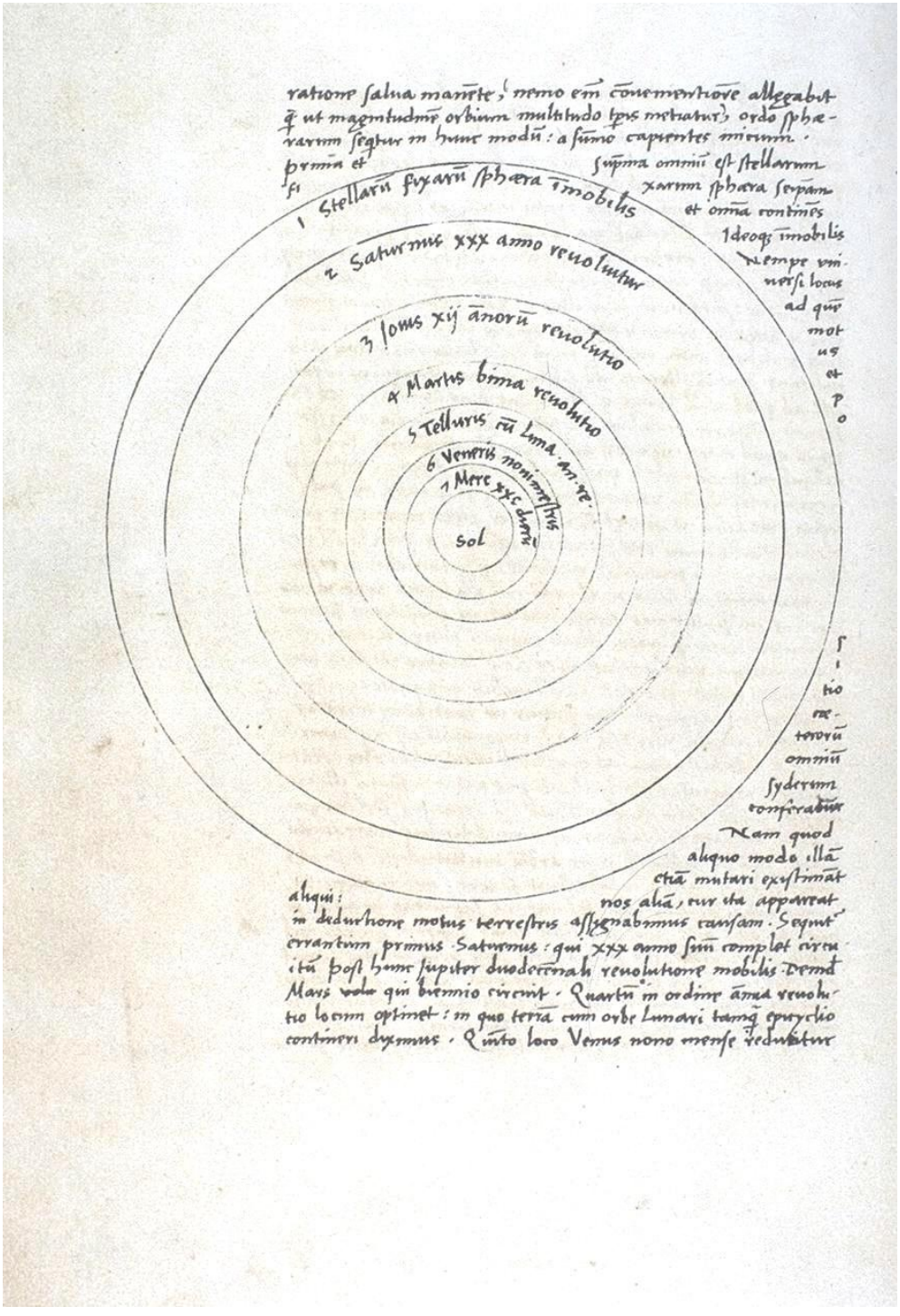
Ptolemy's answer: circles on circles

Claudius Ptolemy of Alexandria (about **AD 150**) kept **Earth at the center** and explained the loops with **epicycles**: each planet rides a small circle whose center rides a big circle around Earth. When the small circle carries the planet backward faster than the big one carries it forward, the planet appears to reverse. His great book, the

Almagest (Arabic al-majisti, "the greatest"), predicted planetary positions well enough to **dominate for about 1,400 years**. It is **the universe of Dante and Milton**: nested spheres around a central Earth.

Copernicus: the Sun in the middle

In **1543**, the year he died, **Nicolaus Copernicus** published **De revolutionibus orbium coelestium** with the **Sun at the center**. Retrograde motion became simple: **Earth, on a faster inner orbit, overtakes Mars**. It's like **passing a slower car on the highway**: for a moment the other car seems to slide backward against the distant hills, though both are moving forward.



Copernicus's own diagram of the heliocentric order, from the manuscript of De revolutionibus

Copernicus's own diagram of the heliocentric order, from the manuscript of De revolutionibus. Nicolaus Copernicus, Public domain, via Wikimedia Commons.

Star words

- **Disaster**: from Italian disastro, from dis- ("bad, ill") + astro (Latin astrum, "star"): **a "bad star,"** an ill-starred event.

- **Consider**: from Latin considerare, **possibly** con- + sidus, sideris ("star, constellation"), "to examine the stars" closely. The etymology is plausible but not certain.
- **Influenza**: Italian for **"influence"**, the **influence of the stars**, once blamed for epidemics.

"A correct wrong model is bad news"

The Socratic question: why is a wrong model that gives correct answers dangerous? Jeff's answer:

A wrong model that predicts well hides its own error. As long as it gets the answers right, nobody has a reason to question the structure underneath. It's the **forecasting** problem: a model that verifies well for the wrong physical reasons will fail exactly when conditions change, and you won't see it coming.

Ptolemy fell not because his predictions were bad, but through three blows:

1. **Copernicus's simplicity** (1543): the same motions with fewer contrivances.
2. **Kepler's ellipses** (1609): orbits are ellipses, not circles, and the predictions became far more accurate.
3. **Galileo's evidence** (1610): Jupiter's moons and the phases of Venus, which Ptolemy's arrangement could not produce.

Key facts

- Polaris's altitude = your latitude (about 27° in Florida; 90° at the pole).
- Planētai = "wanderers."
- Ptolemy, c. AD 150: Earth-centered, epicycles, the Almagest; dominant ~1,400 years.
- Copernicus, 1543: Sun-centered; retrograde = Earth overtaking Mars.
- Disaster = "bad star"; influenza = influence of the stars.

Wednesday, November 4, 2026

Galileo

The telescope, the five discoveries that broke Ptolemy's universe, the trial, Milton's visit, a mnemonic for Jupiter's four moons, and how those moons became a clock for longitude.

Portrait of Galileo Galilei (1636)

https://faria.jbf.com/topics/46-galileo/

Portrait of Galileo Galilei (1636). Justus Sustermans, Public domain, via Wikimedia Commons.

A life between giants

Galileo Galilei, 1564–1642. He was born in Pisa on **15 February 1564**, the year **Shakespeare** was born and three days before **Michelangelo died**. He died at Arcetri on **8 January 1642**; **Isaac Newton** was born on Christmas Day that year (by the English calendar then in use).

The telescope, 1609

In 1608 Dutch spectacle-makers produced a **spyglass** magnifying about 3 times. Hearing of it in **1609**, Galileo worked out the optics and ground his own lenses, reaching about **8×, then 20×, and later about 30×**. He first called it a **perspicillum**. The name **telescope** (Greek tele, "far," + skopein, "to look") was proposed by **Giovanni Demisiani** at a **banquet in Rome in April 1611** held by the Accademia dei Lincei in Galileo's honor.

The five discoveries

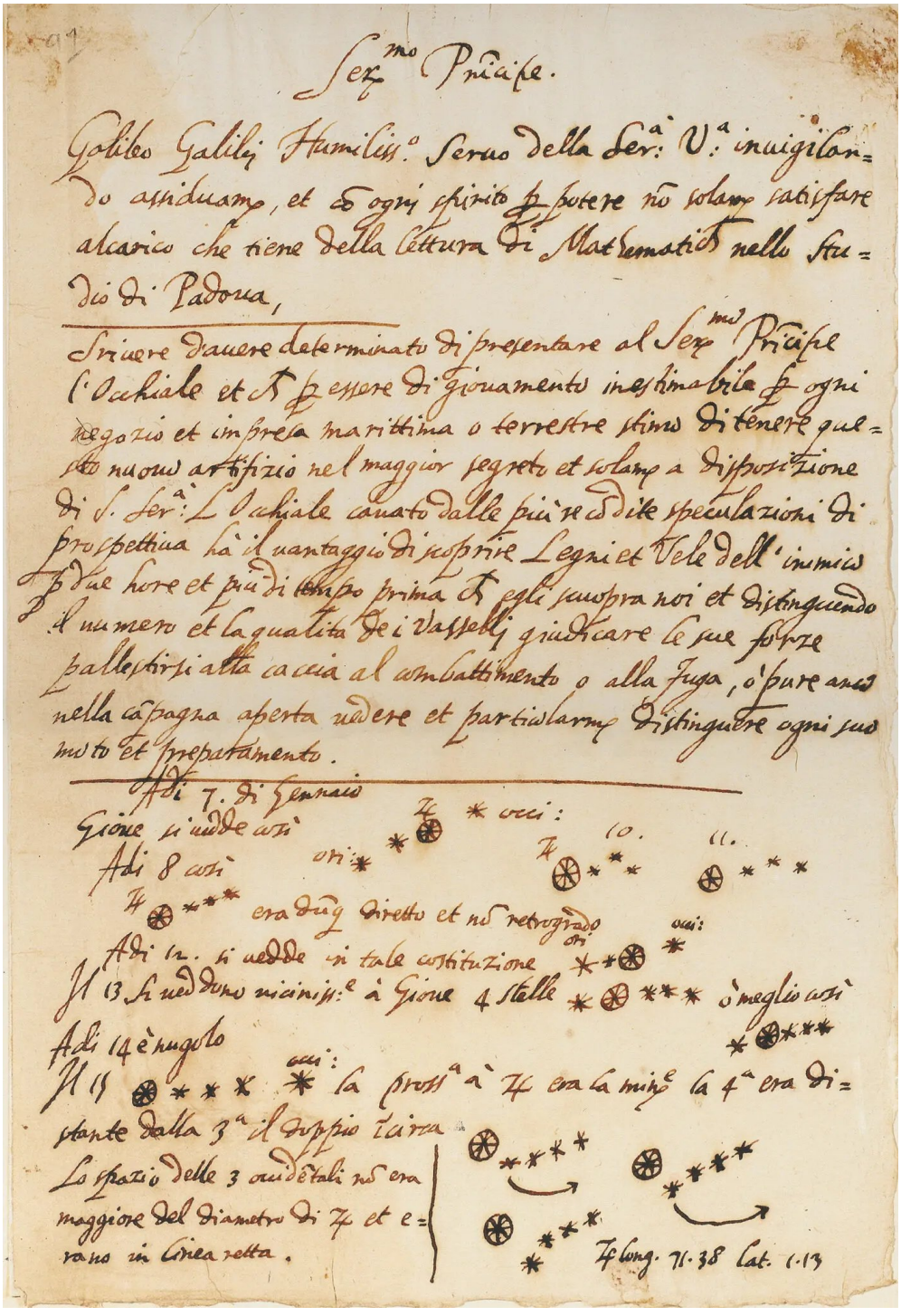
1. Mountains on the Moon (late 1609). Near the **terminator**, the line between lunar day and night, he saw **points of light in the dark part**: mountaintops catching sunrise before the plains below. From how far beyond the terminator they lay, he calculated heights of about **four miles**. He recorded what he saw in **ink-wash drawings**. The Moon was not a perfect crystal sphere; it was a world like Earth.

Galileo's ink-wash drawings of the Moon, 1609

Galileo's ink-wash drawings of the Moon, 1609. Galileo Galilei, Public domain, via Wikimedia Commons.

2. Jupiter's moons (7 January 1610). He saw **three, then four, "stars"** strung in a line beside Jupiter, changing position night to night: moons circling Jupiter. He named them the **"Medicean Stars"** for **Cosimo II de' Medici**, Grand Duke of Tuscany, and his brothers, and won the post of **court mathematician and philosopher** in

Florence. They answered a key objection to Copernicus, that **a moving Earth would leave the Moon behind**: Jupiter moves and keeps four.


Sermo Prīcipe.

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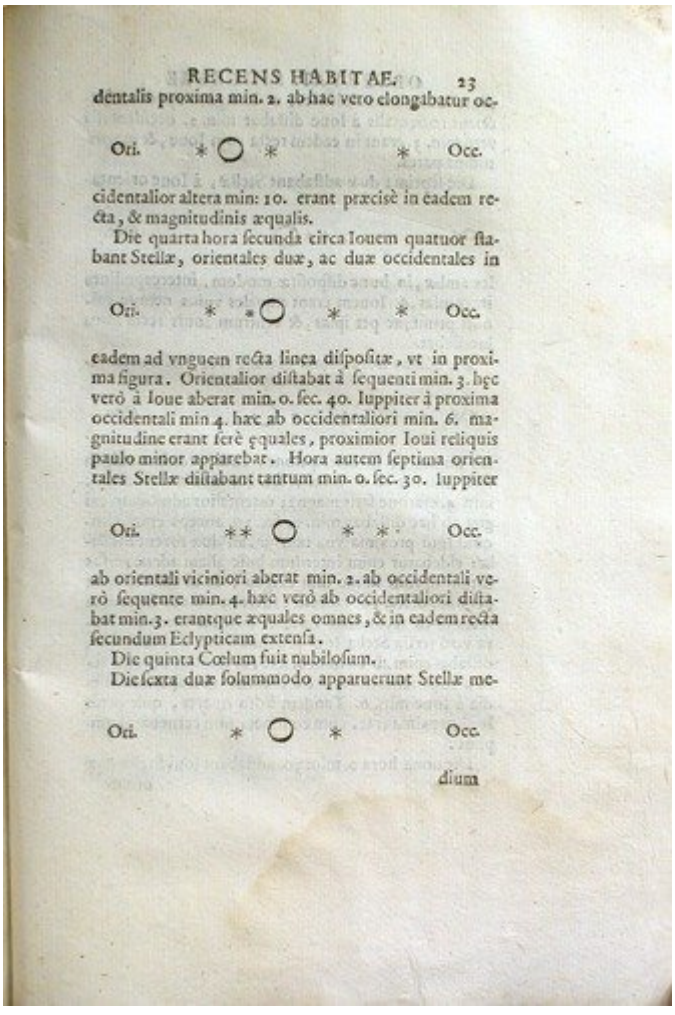
Galileo's draft letter of January 1610, with his sketches of Jupiter's moons at the bottom

Galileo's draft letter of January 1610, with his sketches of Jupiter's moons at the bottom. Galileo Galilei, Public domain, via Wikimedia Commons.



RECENS HABITAE. 23

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Ori. * ○ * * Occ.

cidentalior altera min: 10. erant præcisè in eadem re-
cta, & magnitudinis æqualis.

Die quarta hora secunda circa Iouem quatuor sta-
bant Stellæ, orientales duæ, ac duæ occidentales in

Ori. * * ○ * * Occ.

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ma figura. Orientalior distabat à sequenti min. 3. hæc
verò à Ioue aberat min. o. sec. 40. Iuppiter à proxima
occidentali min 4. hæc ab occidentaliori min. 6. ma-
gnitudine erant ferè æquales, proximior Ioui reliquis
paulo minor apparebat. Hora autem septima orien-
tales Stellæ distabant tantum min. o. sec. 30. Iuppiter

Ori. ** ○ * * Occ.

ab orientali viciniori aberat min. 2. ab occidentali ve-
rò sequente min. 4. hæc verò ab occidentaliori dista-
bat min. 3. erantque æquales omnes, & in eadem recta
secundum Eclypticam extensa.

Die quinta Cœlum fuit nubilosum.

Die sexta duæ solummodo apparuerunt Stellæ me-

Ori. * ○ * Occ.

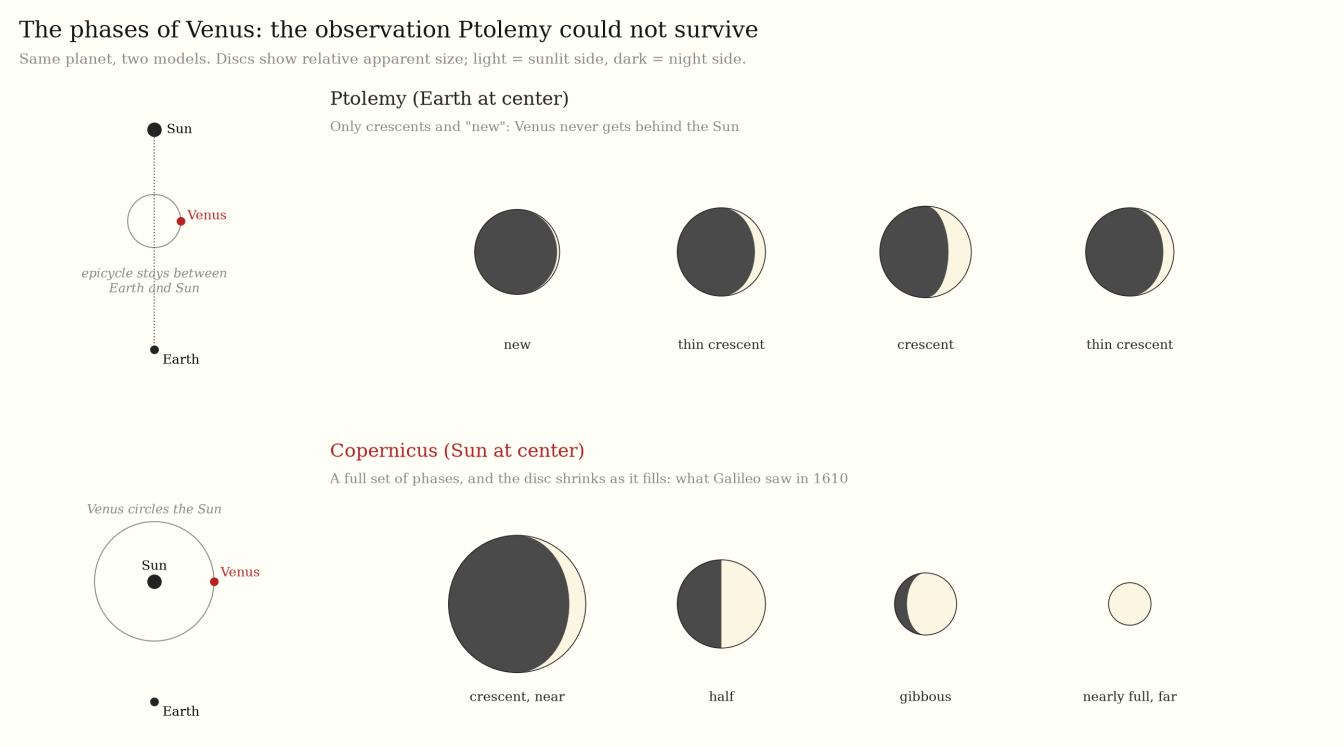
dium

Observations of the Medicean Stars printed in Sidereus Nuncius (1610)

Observations of the Medicean Stars printed in Sidereus Nuncius (1610). Galileo Galilei; scan by History of Science Collections, University of Oklahoma Libraries, Public domain, via Wikimedia Commons.

3. The phases of Venus (late 1610). In Ptolemy's arrangement Venus always lies between Earth and the Sun, so it could only ever show **crescents**. Galileo saw a **full cycle of phases**, crescent to gibbous to nearly full, and the planet's **size changing**: big as a crescent, small when nearly full. Only a **Venus orbiting the Sun** does that. It killed Ptolemy's arrangement, though **Tycho Brahe's** hybrid system (planets circle the Sun, the Sun circles Earth) still fit. To claim priority before he was sure, he sent a scrambled Latin anagram, later revealed as:

> **"Cynthiae figuras aemulatur mater amorum."** "The mother of loves [Venus] imitates the shapes of Cynthia [the Moon]."



The phases of Venus in the Ptolemaic and heliocentric models

Diagram drawn for this site.

4. Sunspots (1611–12). By **projecting** the Sun's image onto paper, he tracked dark spots crossing the disc and showed they lay **on the Sun's surface**, which therefore **rotates, about once a month** (about 27 days as seen from Earth). The Jesuit **Christoph Scheiner** argued they were small bodies orbiting the Sun; the priority dispute made a dangerous enemy. Galileo went **blind in 1638**, but likely from **cataracts and glaucoma**, not, as legend says, from staring at the Sun.

5. The Milky Way and countless stars. The Milky Way resolved into **innumerable stars**. Around **Orion's belt and sword**, where the eye sees about **nine** stars, he plotted **over 80** more; in the **Pleiades**, where the eye sees **six or seven**, he drew **36** (he wrote "more than forty"). And **stars stayed points** in the telescope while **planets showed disks**: the stars were vastly farther away.

The Starry Messenger and the trial

- **Sidereus Nuncius** (The Starry Messenger), **March 1610**, announced the Moon, the stars and Jupiter's moons in 24 pages.
- **1616:** the Church's Congregation of the Index **suspended Copernicus's book** pending correction and condemned heliocentrism as contrary to Scripture; **Cardinal Bellarmine** formally warned Galileo **not to hold or defend** it (the exact terms of the injunction were later disputed).

- **1632: Dialogue Concerning the Two Chief World Systems** staged a debate in which the Ptolemaic side is voiced by **Simplicio**, and Simplicio is given **Pope Urban VIII's own favorite argument** (that God could produce any effect in ways we can't know). The Pope, once his friend, took it as mockery.
- **1633:** tried by the Inquisition, Galileo knelt and swore that he would **"abjure, curse and detest"** his errors. Sentenced to prison, commuted to **house arrest** at Arcetri for the rest of his life.
- **"Eppur si muove"** ("And yet it moves") is a **legend**: the first record is **Giuseppe Baretti's in 1757**, over a century later.

Milton visits Galileo

In **1638**, the young **John Milton**, on his Italian tour, visited the old man. He wrote in **Areopagitica** (1644):

> **"There it was that I found and visited the famous Galileo, grown old, a prisoner to the Inquisition, for thinking in astronomy otherwise than the Franciscan and Dominican licensers thought."**

Galileo is the only contemporary named in **Paradise Lost**, and he appears first in Book I as the "Tuscan artist":

> "the moon, whose orb / Through optic glass the Tuscan artist views / At evening from the top of Fesole, / Or in Valdarno, to descry new lands, / Rivers or mountains in her spotty globe."

Milton was not blind then. He was **29**. His sight failed in the **mid-1640s**, he was **totally blind by about 1652**, and he *dictated Paradise Lost in the 1660s*. His sonnet on his blindness:

> **"When I consider how my light is spent"** ... ending, **"They also serve who only stand and wait."**

Two great minds met in 1638: one going blind, the other who would.

Afterward

- **1758:** the general ban on books teaching heliocentrism was dropped from the Index (Galileo's Dialogue itself stayed listed until **1835**).
- **1992: Pope John Paul II** accepted a commission's finding that the Church's judges had erred.

Memorize the four moons: I Eat Green Carrots

> ## **I E**at **G**reen **C**arrots
> **Io · Europa · Ganymede · Callisto**, innermost to outermost

Each is a lover of **Zeus (Jupiter)**, so the names come with stories:

- **Io**: a priestess Zeus turned into a **heifer** to hide her from Hera.
- **Europa**: carried off by Zeus disguised as a **white bull**; she gave her name to **Europe**.
- **Ganymede**: the beautiful Trojan youth carried to Olympus to be the gods' **cupbearer**. Ganymede is the **largest moon in the solar system, larger than the planet Mercury** (in diameter).
- **Callisto**: turned into a **bear** and set in the sky as the **Great Bear**, Ursa Major.

(The names were proposed by Simon Marius in 1614; Galileo called them simply I to IV.)

Latitude is easy, longitude needs a clock

Latitude comes straight from the sky: the height of Polaris (or the noon Sun) gives it (see Looking Up and the full Polaris lesson, including Shakespeare's "height be taken").

Longitude is a matter of time. Earth turns **360° in 24 hours = 15° per hour**. If you know the time **here** and the time at a **reference place** at the same instant, the difference is your longitude.

Galileo's idea: Jupiter's moons as a clock in the sky. Io disappears into Jupiter's shadow **about every 42.5 hours**, and that eclipse is **seen at the same instant everywhere on Earth**. Make tables of eclipse times at a reference city, then observe one wherever you are.

Worked example. The tables say Io vanishes at **midnight in Florence**. You see it vanish at **10:00 p.m. local time**. Difference **2 hours × 15° = 30° west of Florence.**

At sea it failed: you can't hold a long telescope steady on a pitching deck. **On land it worked: Giovanni Domenico Cassini's** Jupiter-moon tables (1668) let French astronomers **remap France in the 1670s–80s**. The coast came out well inland of where maps had it, and **Louis XIV reportedly** complained that his astronomers had cost him more territory than his enemies.

Sea longitude was solved by a clock instead: **John Harrison's** marine chronometer (H4, tested in the **1760s**).

The GIS link: every coordinate on every map started as an angle measured against the sky and a clock. GPS still works the same way, by timing: satellites carry atomic clocks.

Longitude by Jupiter's clock: worked examples

Latitude is easy; longitude needs time

Latitude is easy: the height of Polaris above the horizon gives it directly. **Longitude** needs **time**. The Earth turns **15° an hour**, so:

> **(local time − home time) × 15 = your degrees east (+) or west (−) of home.**

The method, step by step

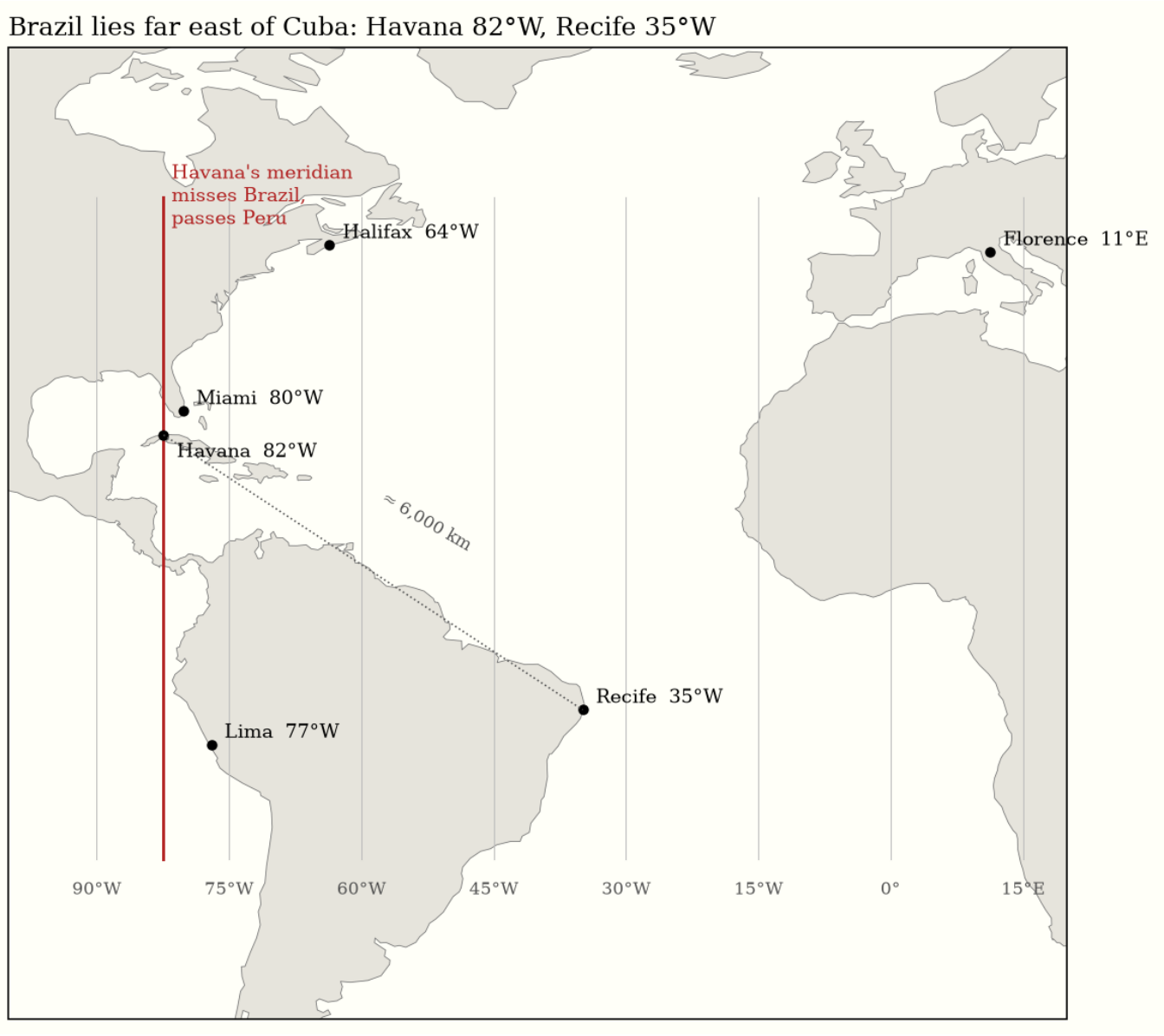
1. **Find local noon** from the **shortest shadow** of a vertical stick, and set your clock to 12:00.
2. **Read home time** from Galileo's printed table of **Io eclipses**, for example: "Io vanishes at 9:14 p.m. Florence time." The eclipse happens **at one instant for every observer on Earth**, like a **starter's pistol in the sky**. Everyone sees it at the same moment; only their **local clocks** read differently.
3. **Subtract and multiply by 15.**

Worked examples (home: Florence, about 11°E)

Your clock reads 10:14 p.m. That's **1 hour later** than Florence, so **15° east**: about **26°E**, the **Aegean**, between **Athens (24°E)** and **Istanbul (29°E)**. Jeff guessed Turkey or Afghanistan: **western Turkey fits**, but **Afghanistan (~65°E)** would be about **3.6 hours ahead** of Florence.

Your clock reads 6:14 p.m. That's **3 hours behind**, so **45° west**: about **34°W**, just off **Recife, Brazil (35°W)**. Jeff guessed near Cuba, but **Havana is at about 82°W**, about **6 hours behind** Florence.

The surprise: South America lies far east



Havana's meridian (82°W) runs south and misses South America; Recife lies at 35°W

Havana's meridian (82°W) runs south and misses South America; Recife lies at 35°W. Tufte-style map, Abbé Faria, 2026.

South America sits much farther east than most people picture. **Recife is east of Halifax, Nova Scotia (64°W).** Fly **due south from Havana** and you clip **western Panama**, then pass over **open Pacific just off Ecuador and Peru**: you miss South America entirely. Even **Lima (77°W) is east of Havana**. **Havana to Recife is about 6,000 km** (about 6,200 km great-circle), roughly **New York to Paris**.

Jeff:

> "I think that would trick 99.99% of the world."

More map tricks like this are collected in Map Surprises: Latitude and Longitude Tricks.

Why it didn't work at sea

On a **rolling deck**, keeping a long telescope on Jupiter's tiny moons was impractical. **On land** it worked beautifully: **Cassini** used the Io tables to **remap France**, which **shrank** it as the coastlines moved inland. At sea the problem was solved by **John Harrison's marine chronometer in the 1760s**: carry home time in a box instead of reading it from the sky. Until then sailors often **ran down the latitude** — hold a parallel they could measure from Polaris or the noon Sun — because longitude was the unsolved half; see The sidereal day and latitude sailing and Bowditch.

Rømer, 1676: the clock that ran late

While compiling Io tables, **Ole Rømer** noticed the eclipses ran **up to about 22 minutes late** when **Earth was far from Jupiter**, and early when near. The cause: light takes time to cross the extra distance. It was **the first measurement of the speed of light** (the modern figure for the delay across Earth's orbit is about **16.5 minutes**). The eclipse is still **simultaneous for all observers on Earth**; the delay is between Jupiter and us, not among us, so only the **local clock readings** differ.

Question to think about

Was Galileo a **martyr to truth**, or **done in by his own arrogance**, mocking a pope who had been his friend?

Key facts

- Galileo 1564–1642; telescope 1609; "telescope" named 1611.
- Five discoveries: Moon mountains; Jupiter's moons (7 Jan 1610); Venus's phases; sunspots; countless stars.
- Sidereus Nuncius 1610; warning 1616; Dialogue 1632; trial and abjuration 1633.
- "Eppur si muove": first recorded 1757.
- Milton visited 1638, aged 29, not yet blind; blind by ~1652.
- **I Eat Green Carrots**: Io, Europa, Ganymede, Callisto.
- 15° of longitude = 1 hour; Io eclipses every ~42.5 h.

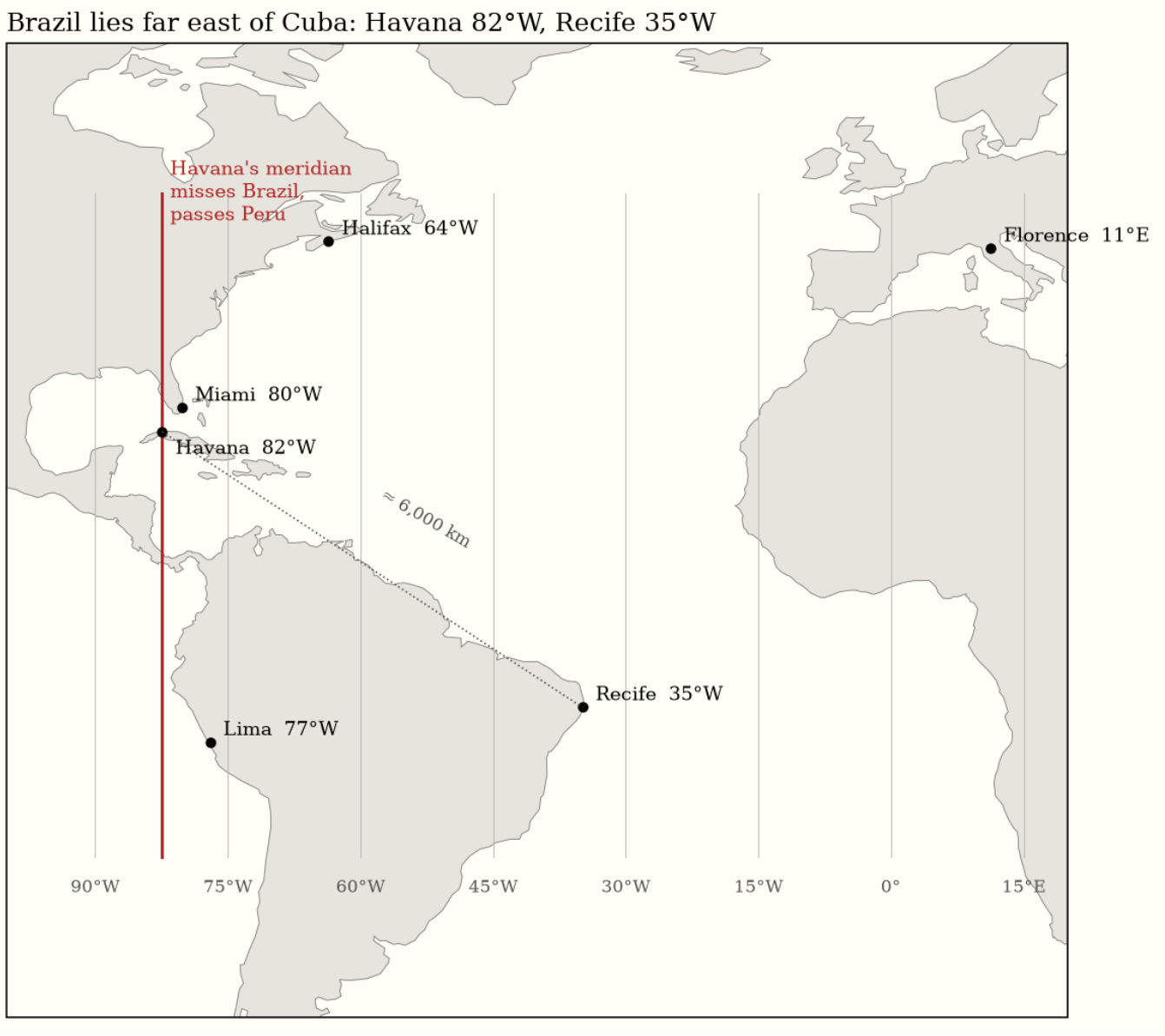
Wednesday, November 4, 2026

Map Surprises: Latitude and Longitude Tricks

A growing collection of map anomalies, places that sit much farther east, west, north or south than our mental maps say. More will be added.

This is a growing collection: new map surprises will be added as we find them. It grew out of the longitude work on Galileo (see "Longitude by Jupiter's clock: worked examples").

1. Due south from Havana misses South America



Havana's meridian (82°W) runs south and misses South America; Recife lies at 35°W

Havana's meridian (82°W) runs south and misses South America; Recife lies at 35°W. Tufte-style map, Abbé Faria, 2026.

Due south from Havana on a flat map and on a globe

The same line on a flat map and on a globe: due south from Havana (82.4°W) passes just west of Punta Pariñas, Peru (81.3°W), the westernmost point of South America, and reaches Antarctica. Jeff: "I like showing both versions." Tufte-style, Abbé Faria, 2026.

The globe proves the flat map isn't lying: a meridian is a straight north–south line on both, and the miss is only about 1°, roughly 120 km.

Most mental maps put South America roughly below North America. It actually sits **far to the east**.

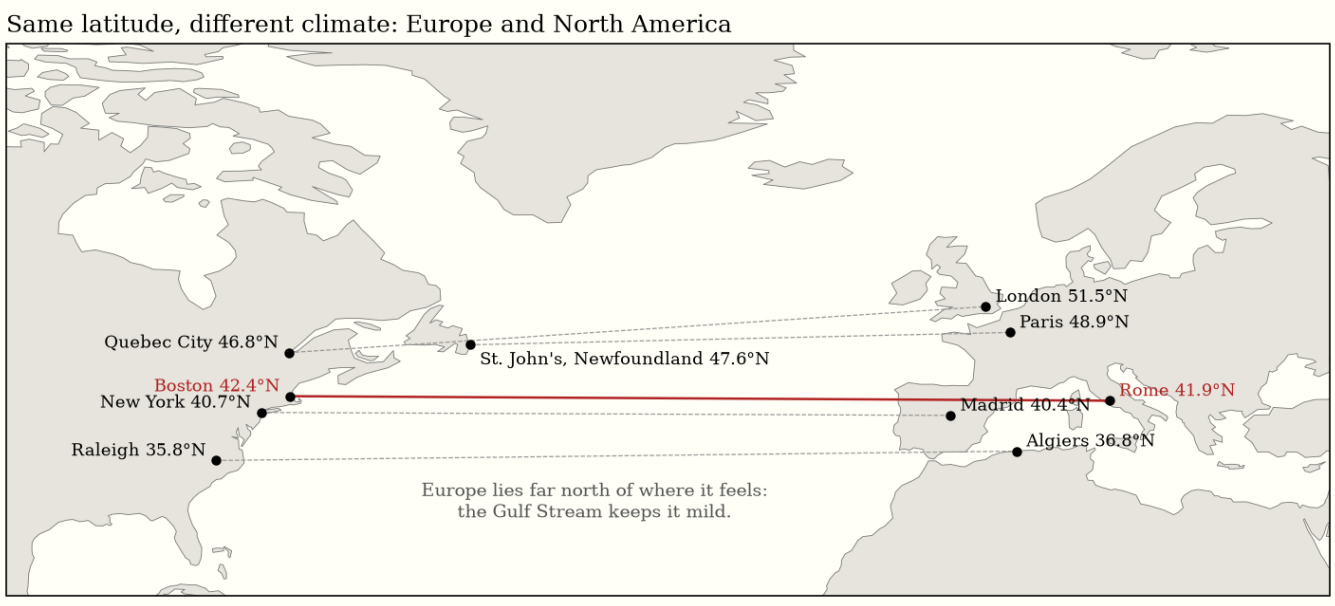
- **Havana: 82°W**
- **Lima: 77°W**, so even Lima, on South America's west coast, is **east of Havana**
- **Halifax, Nova Scotia: 64°W**
- **Recife, Brazil: 35°W**, so Recife is **east of Halifax**

Fly **due south from Havana** and you **clip western Panama**, then pass over **open Pacific just off Ecuador and Peru**. You never cross South America at all.

Jeff:

> "I think that would trick 99.99% of the world."

2. Europe sits far north



European cities and the North American cities at the same latitude

European and North American cities at matching latitudes. Tufte-style map, Abbé Faria, 2026.

Europe's great cities lie much farther north than their mild weather suggests. Match them by latitude:

Europe	Latitude	North America	Latitude
London	51.5°N	Quebec City (London is farther north)	46.8°N
Paris	48.9°N	St. John's, Newfoundland	47.6°N
Rome	41.9°N	Boston	42.4°N
Madrid	40.4°N	New York	40.7°N
Algiers	36.8°N	Raleigh	35.8°N

- **London is north of Quebec City.**
- **Paris** is level with **St. John's, Newfoundland**.
- **Rome** is level with **Boston**. Jeff guessed **Atlanta**, but Atlanta is at **33.7°N**, level with **Morocco** and **Beirut**.
- **Madrid** is level with **New York**.
- **Algiers**, in Africa, is level with **Raleigh, North Carolina**.

Why Europe is so mild: the **Gulf Stream** and its extension, the **North Atlantic Drift**, carry warm water from the tropics toward Europe, and the **prevailing**

westerlies blow that ocean-warmed air onto the continent. Eastern North America, at the same latitudes, gets its winter winds from the frozen interior.

Latitude itself is read from the sky: the altitude of Polaris equals your latitude (Shakespeare's "height be taken"), corrected with Bowditch tables because of the sidereal-day wobble. Longitude needs a clock — see Galileo.

Key facts

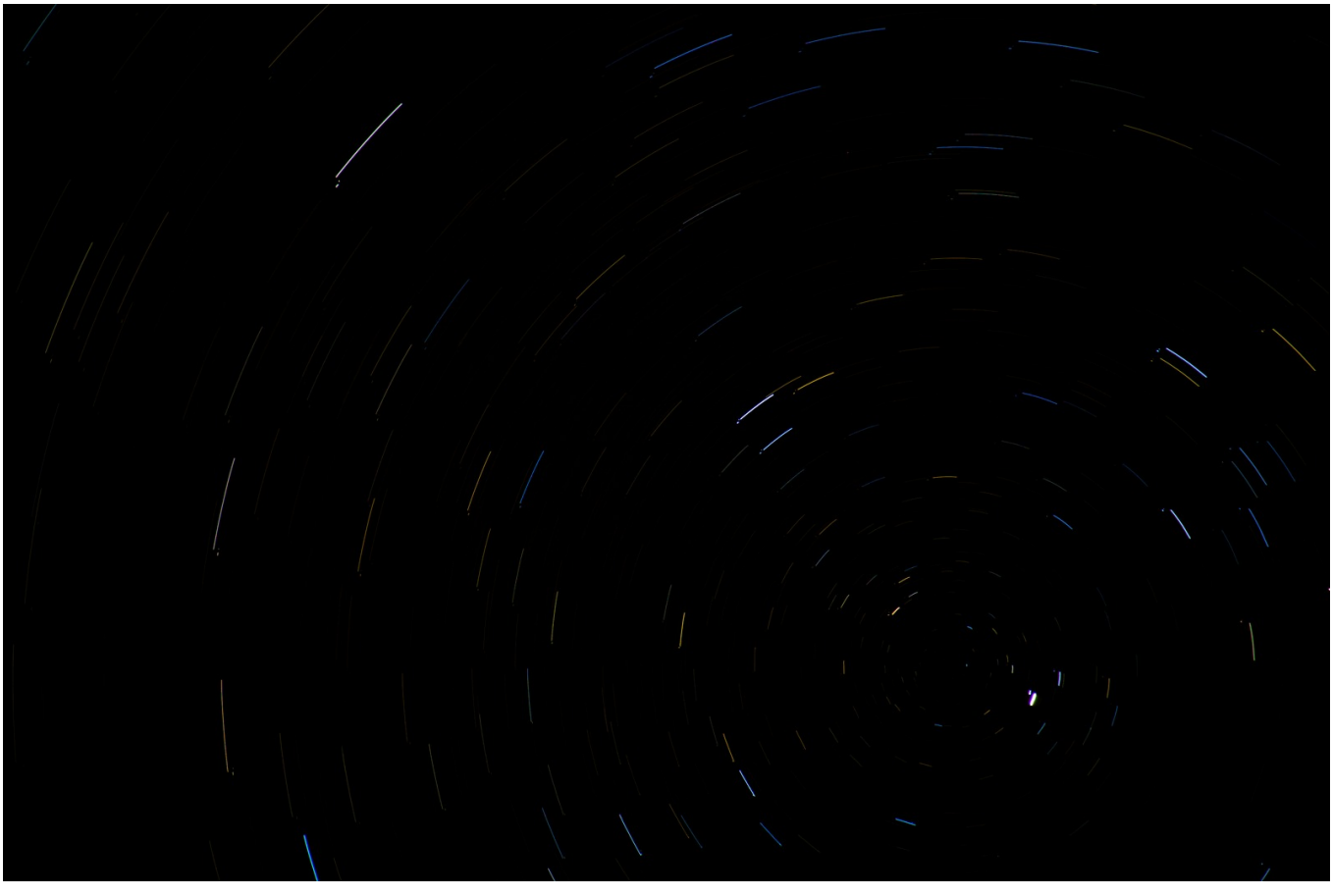
- Havana 82°W, Lima 77°W, Halifax 64°W, Recife 35°W: due south from Havana misses South America.
- London (51.5°N) is north of Quebec City (46.8°N); Rome ≈ Boston; Madrid ≈ New York.
- The Gulf Stream, North Atlantic Drift and westerlies keep western Europe mild.

Friday, November 6, 2026

Polaris: the North Star, explained

The whole Wikipedia entry on Polaris, made clear: a triple star, a Cepheid yardstick, the precession of the pole, every name from cynosure to Dhruva, and Shakespeare's Sonnet 116 line by line.

Jeff read the Wikipedia / SkyView entry on Polaris and wanted to understand **all of it**. This page is that walkthrough: the triple star, why it stands still, the 26,000-year wobble of Earth's axis, Cepheids and Henrietta Leavitt, the fight over how far away it is, every name, and Shakespeare's Sonnet 116 as the centerpiece. It grows out of Looking Up (altitude = latitude) and pairs with Galileo (longitude needs a clock), Map Surprises, Bowditch (the correction tables), The sidereal day and latitude sailing (why the correction is needed), and The Harvard Computers (what "computer" meant, and Leavitt's full story).



Circumpolar star trails with Polaris near the center

Long-exposure star trails around the north celestial pole; Polaris sits near the still center. Photograph by Daniel Seßer (Seßn), CC BY-SA 3.0, via Wikimedia Commons.

1. Not one star: a multiple system

To the naked eye Polaris is a single point of light. It is not.

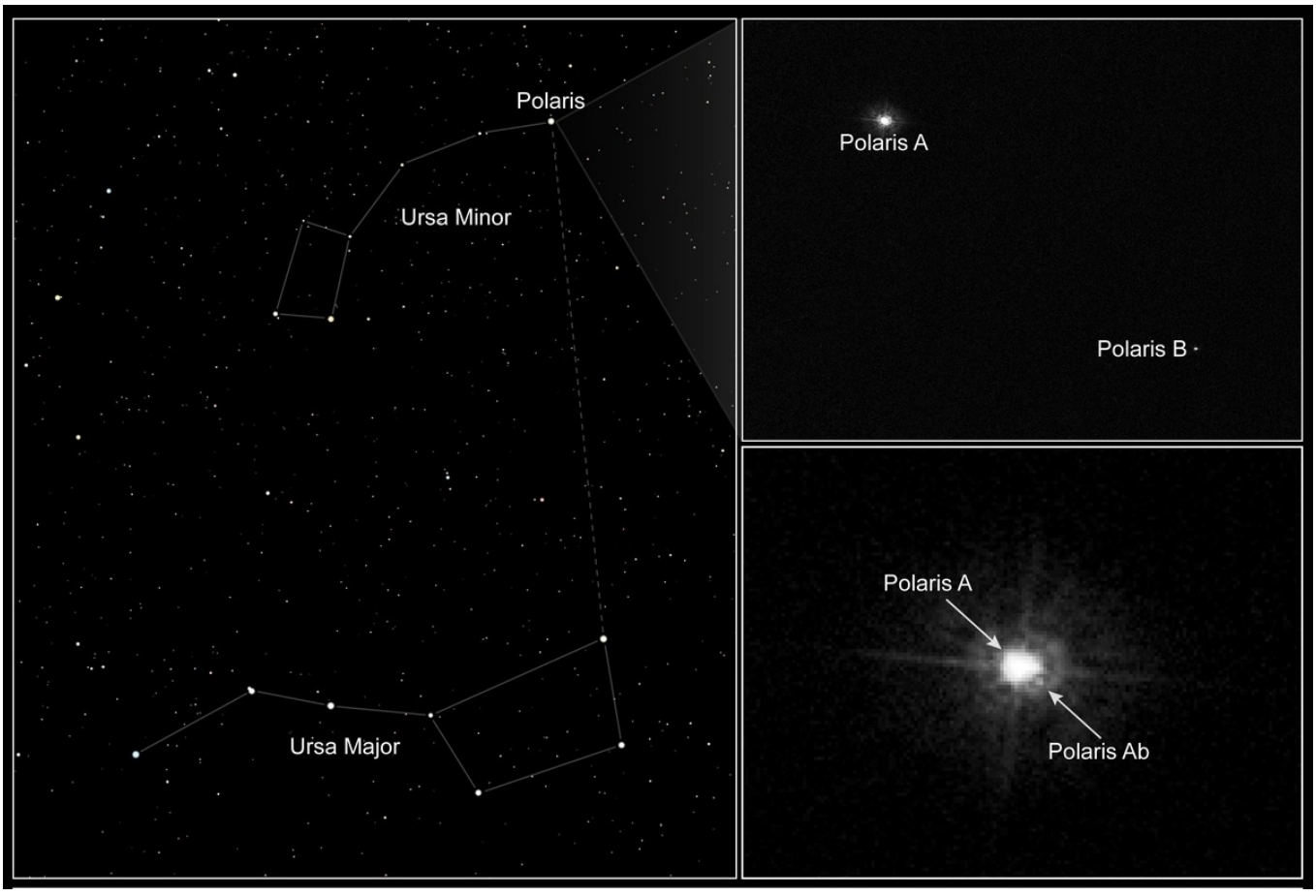
α Ursae Minoris (Alpha Ursae Minoris, abbreviated **α UMi**) means "the brightest star of Ursa Minor," the Little Bear / Little Dipper. Greek letters were assigned by **Johann Bayer** in 1603: α = brightest, β = next, and so on. "Ursae Minoris" is the Latin genitive, "of the Lesser Bear."

What you see as Polaris is really a **triple system** (two more distant stars once thought to belong have been shown **not** to be bound to it):

Component	What it is	Notes
Polaris Aa	Yellow **supergiant**, spectral type **F7Ib**	The star you see. Mass about **5.13 ± 0.28** solar masses (Evans et al. 2024, CHARA + Hubble + Gaia); radius about **46** solar radii; thousands of times the Sun's luminosity.
Polaris Ab	Close dwarf, roughly **F6V**	Orbits Aa in about **29.4 years**. Angular separation only ~0.17″; at the Gaia distance the closest approach is about **6 AU**. Hubble first imaged it in **January 2006**. A's glare had hidden it for centuries.
Polaris B	Main-sequence dwarf, spectral type **F3V**	Much farther out, about **2,400 AU** from A. **William Herschel** discovered it in **August 1779** with a reflecting telescope he built himself. Visible in a modest amateur telescope.

Spectral classes F7 and F3. The letters O B A F G K M run from hottest to coolest. **F** stars are yellow-white, a little hotter than our G-type Sun. The number (0–9) is a finer grade: F3 is hotter than F7. The suffix **Ib** means a luminous supergiant; **V** means a normal main-sequence ("dwarf") star like the Sun.

Polaris C and D were once listed as distant companions. Modern proper-motion and distance work shows they are **not physically associated** with the Polaris system: background (or foreground) stars that happen to lie near it on the sky.



Hubble image of the Polaris system

Polaris as seen by the Hubble Space Telescope, resolving the close companion. NASA / ESA / HST, Public domain, via Wikimedia Commons.

Artist's concept of the Polaris triple system

Artist's concept: supergiant Polaris Aa, close dwarf Ab, and distant companion B. NASA / ESA / G. Bacon (STScI), Public domain, via Wikimedia Commons.jpg).

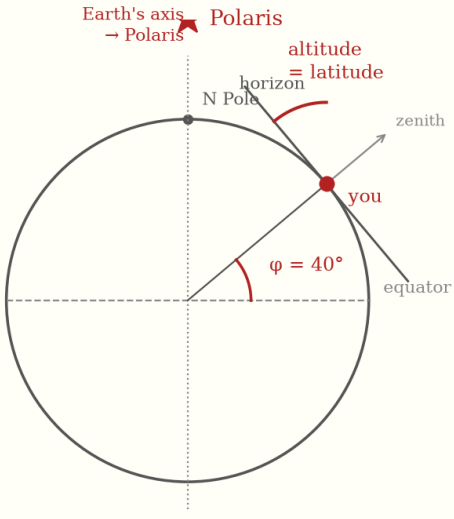
2. Why it stands still

Earth spins on an axis. Extend that axis into the sky and you hit the **north celestial pole**. Polaris sits almost on that point, so as Earth turns, every northern star wheels around Polaris while Polaris itself barely moves.

How close? In 2018 it was about **0.66°** (about **40 arcminutes**) from the true pole: roughly **1.4 times the width of the full Moon**. It therefore traces a tiny daily circle about **1.3°** across. Navigators have always corrected for that.

Finding true north with the Big Dipper. The two stars at the "bowl" end of the Big Dipper, **Dubhe** and **Merak** (the "pointers"), point toward Polaris. Because Polaris itself is a little off the pole, its azimuth is true north only twice each sidereal day; the rest of the time you correct east or west. A classic rule of thumb treats the leading edge of the Dipper like a **clock face** and reads the correction from Dubhe–Merak. Nathaniel *Bowditch's American Practical Navigator **(first edition** 1802*) lists Polaris among the navigational stars and has been teaching American sailors this geometry for more than two centuries — full story in Bowditch; why the daily correction exists in The sidereal day.

Altitude = latitude. The height of Polaris above your northern horizon **is** your latitude (to within that ~0.7° offset, which tables correct). At the North Pole it is overhead; at the equator it sits on the horizon; in Florida (~28°N) it stands about 28° up. That is the GIS / navigation fact already on Looking Up and the reason latitude was easy for sailors while longitude was hard (see Galileo).



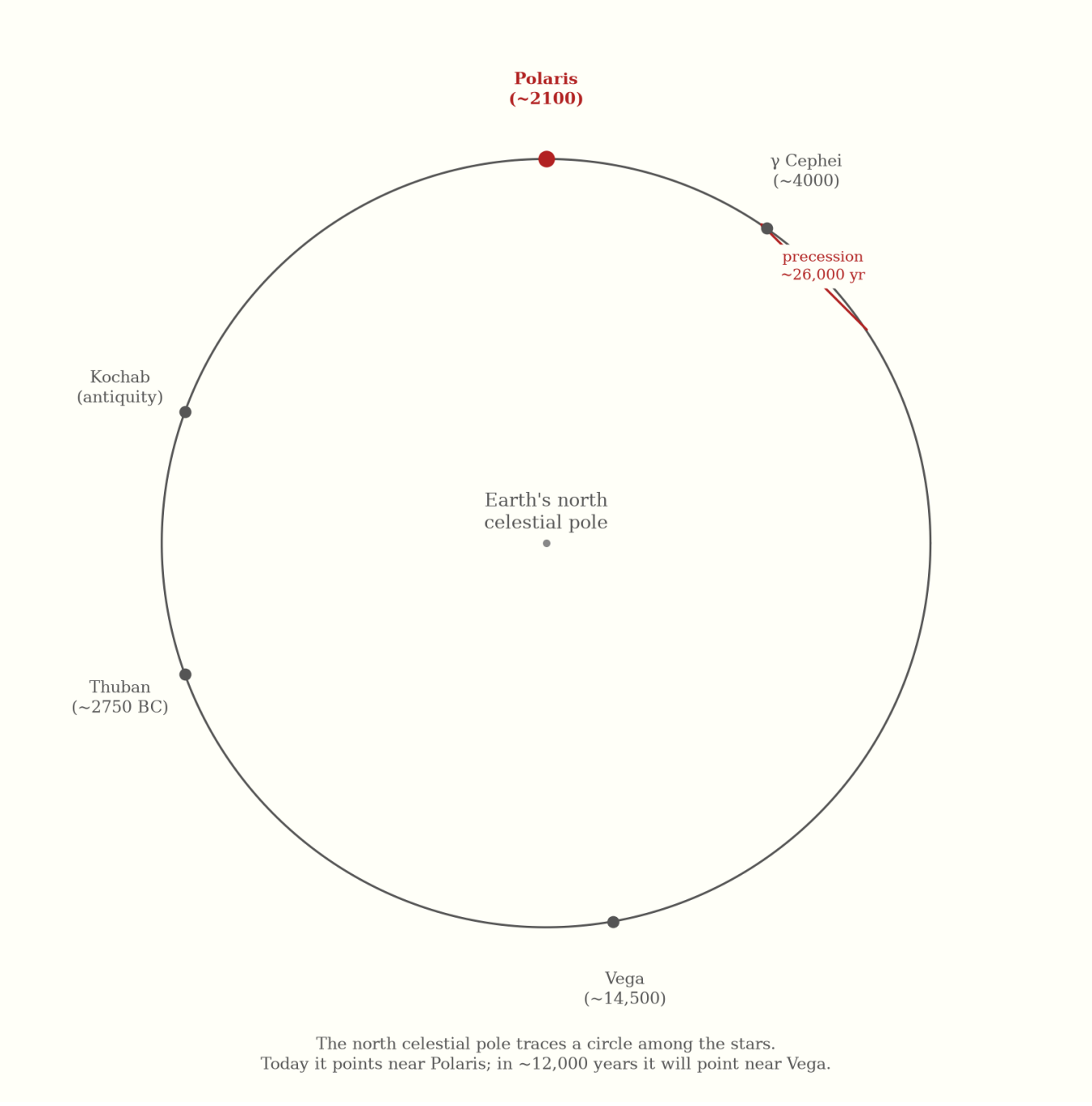
Because Polaris lies almost on Earth's axis, its height above your northern horizon equals your latitude. At the North Pole it is overhead; at the equator it sits on the horizon; in Florida (~28°N) it stands ~28° up.

Polaris altitude equals latitude

Because Polaris lies almost on Earth's axis, its altitude above the northern horizon equals the observer's latitude. Diagram drawn for this site.

3. It will not always be the pole star: precession

Earth's axis does not point forever at the same spot. It **wobbles**, like a slowing top, tracing a circle on the sky once every roughly **26,000 years**. This is the **precession of the equinoxes**. The north celestial pole therefore drifts among the stars, and different stars take turns as "the" North Star.



Precession circle of pole stars

The north celestial pole traces a ~26,000-year circle among the stars. Diagram drawn for this site.

Key stops on that circle:

When	Pole star (or none)
~2750 BC	**Thuban** (α Draconis), in Draco; near the pole when the pyramids were rising
Classical antiquity	Closer to **Kochab** (β UMi) than to Polaris, though still ~10° from either
~320 BC	The Greek navigator **Pytheas** described the pole as **devoid of stars**
5th century AD	**Stobaeus** called Polaris ἀεί φανής (aei phanēs), **"always visible"**
~2100 AD	Polaris's closest approach (~0.45° / 27′)
~4000 AD	**Gamma Cephei** (Errai)
~10,000 AD	Near **Deneb**
~14,500 AD	Near **Vega**
+26,000 years	Back to Polaris

Shakespeare's Caesar boasts of being "as constant as the northern star" (Julius Caesar, ~1599). In Caesar's own century there was **no** bright star sitting on the pole; Polaris was still several degrees off. The line is poetry, not astronomy.

4. Cepheids: a pulsing yardstick

Pronounce it SEF-ee-id.

A **Cepheid** is a star that **pulses**: it swells and shrinks on a steady beat, brightening and dimming like a slow heartbeat. The class is named for **Delta Cephei**, the prototype studied in the late 18th and 19th centuries.

Henrietta Swan Leavitt (1868–1921), working as a "computer" at Harvard College Observatory (full story: The Harvard Computers), measured variable stars on photographic plates of the Magellanic Clouds. In **1912** she published the result that made cosmology possible: a Cepheid's **period tells you its true (intrinsic) brightness**. The slower the pulse, the more luminous the star. Compare that true brightness with how bright the star looks, and you know how far away it is. Cepheids became **standard candles**.

Edwin Hubble used Leavitt's period–luminosity law on Cepheids in the Andromeda Nebula (the 1920s) to show that Andromeda is a separate galaxy, outside the Milky Way, and that the universe is expanding.

Henrietta Swan Leavitt

Henrietta Swan Leavitt (1868–1921). Public domain, via Wikimedia Commons.

Polaris is the nearest classical Cepheid. Its period is about **3.97 days** (roughly four days). Two odd facts make astronomers stare:

- **The amplitude is shrinking (and misbehaving).** Before 1963 the brightness swing was over 0.1 magnitude and slowly decreasing; after **1966** it dropped

rapidly below 0.05 magnitude, then wandered. Reports since then suggest the amplitude may be growing again, a reversal not seen in other Cepheids.

- **The period has been lengthening** by around **4.5 seconds per year** for much of the modern record (with a pause around 1963–65). Recent work (Torres 2023) finds the period change has **reversed** and been decreasing since roughly 2010.
- **It may be brightening over centuries.** Research reported in Science (2004) suggests Polaris is about **2.5 times brighter** now than when **Ptolemy** recorded it ~1,900 years ago (third magnitude then, second now). Astronomer **Edward Guinan**: if real, "these changes are **100 times larger** than [those] predicted by current theories of stellar evolution."

Polaris is overluminous compared with the best Cepheid evolution models, and its age is hard to reconcile with its companions. It remains a puzzle even while it is a yardstick.

5. How far away? Parallax, and why it matters

Parallax, the finger-blink analogy. Hold a finger at arm's length and blink each eye in turn: the finger jumps against the background. The closer the finger, the bigger the jump. Astronomers do the same with Earth's orbit. Look at a star in January, then again in July (Earth has moved ~2 AU), and measure how far the star shifts against distant background stars. That angle is the **stellar parallax**. One **parsec** is the distance at which the parallax is one arcsecond; a light-year is about 0.3066 parsecs.

The fight over Polaris's distance:

Method	Distance	Notes
Hipparcos satellite (observations 1989–93; revised reduction 2007)	about **432** light-years (133 pc)	Direct trigonometric parallax of the primary
Turner et al. (spectral / photometric estimates, 2004–2013)	about **323** light-years (≈99 pc)	"Short" distance from spectroscopy and main-sequence fitting of B
Gaia (DR3, via companion B)	about **446.5** light-years (136.9 pc)	Current best geometric estimate for the system, assuming A and B share a distance

Why binaries spoil the measurement. Polaris Ab tugs Aa around a 29-year orbit. That orbital wobble mixes with the tiny parallax wobble and with the Cepheid pulsation itself. Hipparcos had to disentangle all three. Close companions and variability are exactly what make bright, famous stars harder to measure, not easier.

Why it matters: the cosmic distance ladder. Cepheids are the rung that reaches from our neighborhood out to nearby galaxies. Every rung above them (Type Ia supernovae, Hubble's constant, the age and expansion rate of the universe) rests on how well we know Cepheid luminosities. Polaris is the **nearest** classical Cepheid, so an

error in its distance calibrates an error into everything farther out. That is why grown astronomers argue in public about 323 vs 434 vs 446 light-years. Polaris is a prized check because it is the nearest classical Cepheid. It is a difficult one (a companion, a small amplitude), so the modern zero point uses many Galactic Cepheids, not this star alone. How the rungs under that yardstick were built, from Kepler's relative orbits through the Venus transits and stellar parallax to the calibration Leavitt's slope still needed, is From a Venus transit to the Cepheids.

6. Names: a word-lover's cabinet

Jeff loves words. Polaris has collected them for millennia.

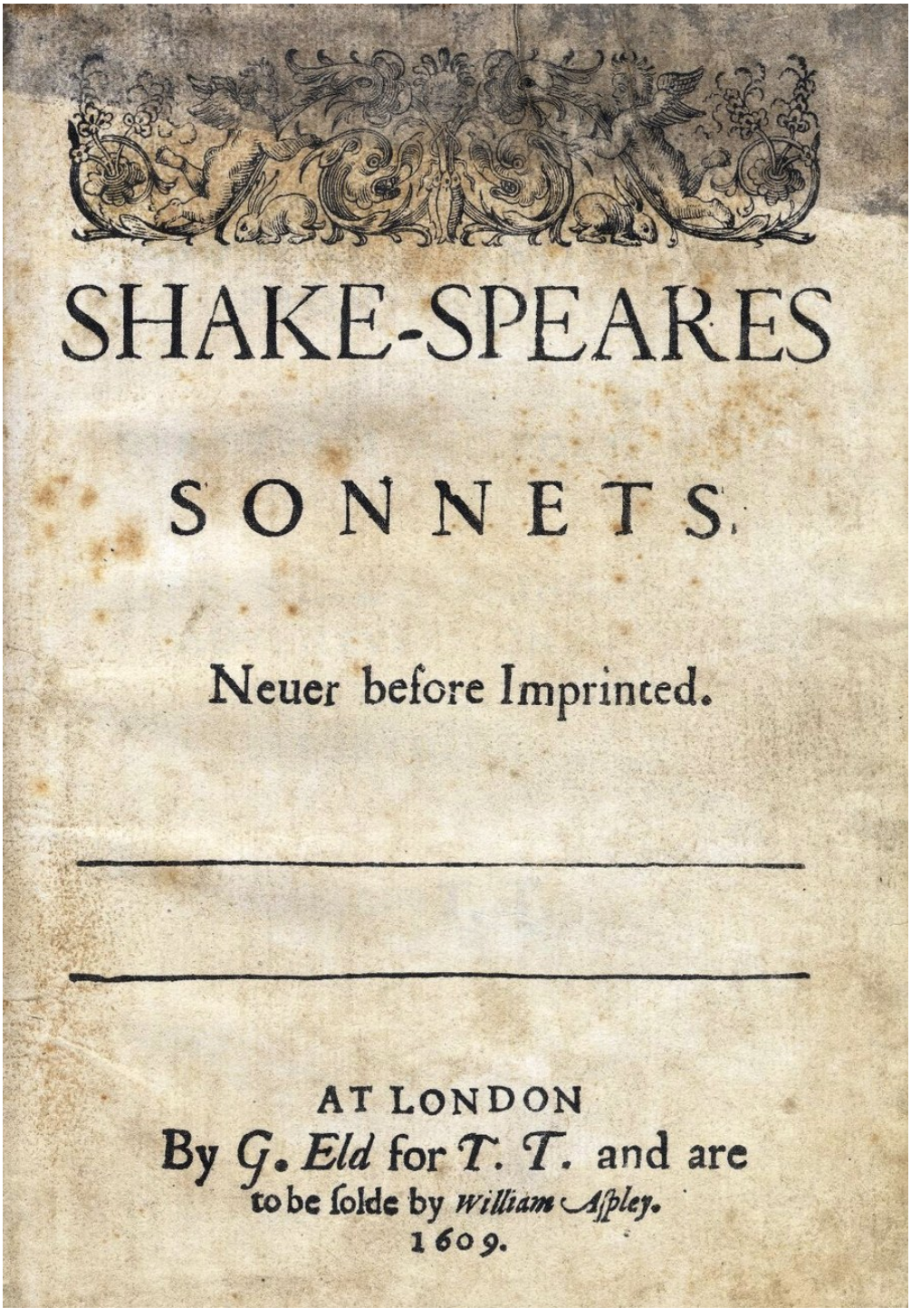
- **Cynosure.** From Greek **κυνόσουρα** (kynosoura), **"dog's tail."** The Little Bear was once pictured as a dog; Polaris is the tip of its tail. Because everyone stared at it, cynosure came to mean **"the center of attention"** ("she was the cynosure of all eyes"). Pronounce **SY-nuh-shoor** (or SY-nuh-syoor).
- **Lodestar.** Middle English: the **guiding star**. Related to lode (a way or course) and to **lodestone**, the magnetic rock that also points the way.
- **Spenser's "steadfast star."** In The Faerie Queene I.ii.1: "By this the northern wagoner had set / His sevenfold team behind the **steadfast star**." The "northern wagoner" is the Big Dipper; the steadfast star is Polaris.
- **Stella polaris / Polaris.** Neo-Latin **"polar star,"** in print by the Alfonsine Tables of **1492**; Gemma Frisius used stella polaris in 1547. The IAU made **Polaris** the official proper name of α UMi Aa in 2016.
- **Stella Maris**, **"star of the sea,"** a medieval Marian title (Bartholomaeus Anglicus, c. 1270s) transferred to the pole star; the hymn **Ave Maris Stella** ("Hail, Star of the Sea") carries the same image.
- **Sanskrit Dhruva (Dhruva tara).** The word ध्रुव means "fixed, constant, immovable." In the Puranas the devotee Dhruva is set in the sky as the pole star.
- **Arabic.** Pre-Islamic / medieval astronomy called it **al-Jady** ("the kid," a young goat). Star-name dictionaries also record **al-quṭb al-shamālī** (**"the northern axle / pole"**) and **mismār** (**"needle"**): the pin on which the sky seems to turn.
- **Old English.** Scip-steorra, **"ship-star."** And in the **Old English Rune Poem**, the T-rune **Tir**:

> **Tir biþ tacna sum, healdeð trywa wel**
> **wiþ æþelingas; a biþ on færylde**
> **ofer nihta genipu, næfre swiceþ.**

> "Tir is a certain sign (a guiding star); well does it keep faith with princes; it is ever on its course over the mists of night and never fails."

(Dickins and others read this as a circumpolar / guiding star; the original rune-name was likely the god **Tiw**, with tir "glory" a later or alternate reading.)

7. Shakespeare, Sonnet 116: the centerpiece



Title page of Shakespeare's Sonnets, 1609

Title page of the 1609 Quarto of Shake-speares Sonnets, published by Thomas Thorpe. Public domain, via Wikimedia Commons.

The full sonnet (public domain; 1609 Quarto, modernized spelling)

> Let me not to the marriage of true minds
> Admit impediments. Love is not love
> Which alters when it alteration finds,
> Or bends with the remover to remove:
> O, no! it is an ever-fixed mark
> That looks on tempests and is never shaken;
> It is the star to every wandering bark,
> Whose worth's unknown, although his height be taken.
> Love's not Time's fool, though rosy lips and cheeks
> Within his bending sickle's compass come;
> Love alters not with his brief hours and weeks,
> But bears it out even to the edge of doom.
> If this be error and upon me proved,
> I never writ, nor no man ever loved.

Line-by-line close reading

"Let me not to the marriage of true minds / Admit impediments." Echoes the Book of Common Prayer marriage service ("if either of you know any impediment..."). Shakespeare refuses to allow any obstacle into a union of minds, not just of bodies or estates.

"Love is not love / Which alters when it alteration finds, / Or bends with the remover to remove." If it changes when circumstances change, or yields when someone tries to take it away, it was never love. The stacked alter / alteration and remover / remove are deliberate: love that mirrors change is only change wearing love's name.

"O, no! it is an ever-fixed mark / That looks on tempests and is never shaken." A **sea-mark**: a beacon, landmark, or seamark fixed on shore (or a shoal) by which sailors steer. Tempests howl; the mark does not move. "Ever-fixed" is the same steadfastness Spenser gave the pole star.

"It is the star to every wandering bark, / Whose worth's unknown, although his height be taken." This is the GIS line.

- **Bark** = a **ship** (poetic for any sailing vessel; from Old French barque). A "wandering bark" is a ship off on the open sea.
- **"The star"** = the pole star, Polaris: the one star every northern sailor steered by.
- **"His height be taken"** = a sailor **measuring the star's altitude** above the horizon with an **astrolabe**, **quadrant**, or **cross-staff** (later a sextant). That altitude **is the ship's latitude**. This is exactly the rule on your GIS Month talk and on Looking Up: **altitude of Polaris = latitude**.

- **"Whose worth's unknown"** = even though you can measure the star (take its height), you cannot know its **true nature or value**. In 1609 nobody knew Polaris was a triple Cepheid 446 light-years away; Shakespeare means something deeper: the guide can be used without being understood. Love, likewise, can be relied on without being fathomed.

So the couplet says: love is to the lover what Polaris is to the sailor. You can take its height (measure it, depend on it for your position) and still not know its worth.

"Love's not Time's fool, though rosy lips and cheeks / Within his bending sickle's compass come." Time is a reaper with a curved sickle; beauty falls inside its sweep. Love is not Time's toy or victim.

"Love alters not with his brief hours and weeks, / But bears it out even to the edge of doom." Doom = Judgment Day. Love lasts to the end of the world.

"If this be error and upon me proved, / I never writ, nor no man ever loved." The poet stakes his entire life's work and the reality of human love on the claim. If he is wrong, no poem was ever written and no one ever loved: a deliberate impossibility, sealing the argument.

Pronunciation notes

Word	Say it	Meaning
Polaris	puh-LAIR-iss (US) / puh-LAH-riss (UK)	the pole star
Cepheid	**SEF-ee-id**	pulsing variable star
cynosure	**SY-nuh-shoor**	dog's tail; center of attention
precession	pree-SESH-un	the 26,000-year wobble of Earth's axis
parallax	PAIR-uh-lacks	the apparent shift used to measure distance

8. Coordinates: latitude and longitude for the sky

Polaris (J2000):

- **Declination: +89° 15′ 50.8″**
- **Right ascension: 02h 31m 49.09s**

Declination is **latitude on the sky**: 0° at the celestial equator, +90° at the north celestial pole. Polaris at +89° 16′ is almost at the top of the sky's globe.

Right ascension (RA) is **longitude on the sky**, but measured in **hours, minutes, and seconds** (0h to 24h), not degrees. Why hours? Because the sky turns **360° in 24 hours = 15° per hour**, the same arithmetic as terrestrial longitude (see Galileo). RA 2h 31m is about 37.8° east of the vernal equinox, the sky's Greenwich.

Because Polaris sits so near the pole, its right ascension changes rapidly as precession moves the pole: from about 2.5h in AD 2000 toward 6h by AD 2100.

Review questions

Try these before peeking at the answers at the bottom.

1. If you stood at the **North Pole**, where would Polaris be in your sky, and what latitude would that report?
2. What did **Henrietta Leavitt** discover in 1912, and why did that let Hubble prove Andromeda is a separate galaxy?
3. In Sonnet 116, what does **"wandering bark"** mean, and what does it mean to **"take the height"** of the star?
4. Why will Polaris **not** be the North Star in the year 14,500?
5. Why does a **wrong distance** to Polaris ripple into measurements of the whole universe?

Key facts

- Polaris = α Ursae Minoris, a triple: Aa (F7Ib Cepheid ~5.13 M☉), Ab (close F6V, ~29.4 y orbit), B (F3V at ~2400 AU, found by Herschel 1779).
- Altitude of Polaris ≈ latitude; ~0.66° from the true pole (about 1.4 Moon-widths).
- Precession ~26,000 y: Thuban ~2750 BC; Pytheas (c. 320 BC) saw a pole with no star; Polaris closest ~2100; Vega ~14,500.
- Cepheid (SEF-ee-id): pulsating star; Leavitt 1912 period–luminosity law; Polaris period ~3.97 d.
- Distance debated: Hipparcos ~432 ly; Turner ~323 ly; Gaia (via B) ~446.5 ly; calibrates the cosmic distance ladder.
- Cynosure = Greek "dog's tail" → "center of attention"; lodestar; Stella Maris; Dhruva; OE Tir "never fails."
- Sonnet 116: love = "the star to every wandering bark, / Whose worth's unknown, although his height be taken" — bark = ship; height taken = measuring altitude for latitude.
- Sky coordinates: dec +89° 15′ 51″, RA 02h 31m 49s (RA in hours because 15° = 1 h).

Answers

1. **Straight overhead** (at the zenith). That is latitude **90°N**.

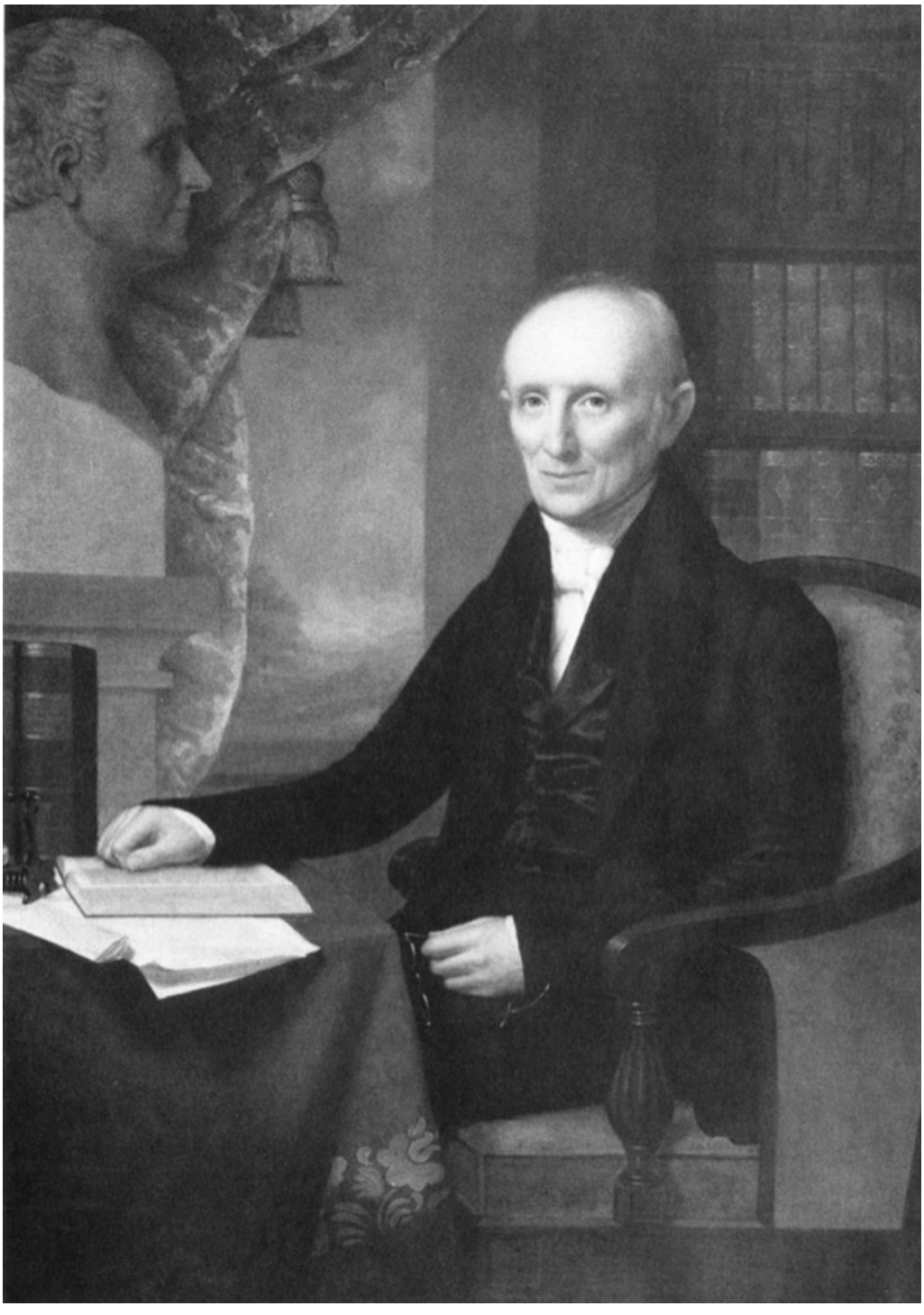
2. That a Cepheid's **pulse period reveals its true brightness**. Measure period → know luminosity → compare with apparent brightness → get distance. Hubble found Cepheids in Andromeda and showed it lies far outside the Milky Way.
3. **Bark** = ship. **Taking the height** = measuring the star's altitude above the horizon (with astrolabe, quadrant, or cross-staff) to find the ship's **latitude**.
4. Because of **precession**: Earth's axis will then point near **Vega**, not Polaris.
5. Polaris is the **nearest classical Cepheid**. Cepheids calibrate distances to galaxies; those calibrate supernovae and Hubble's constant. An error in Polaris's distance feeds every rung above it on the cosmic distance ladder.

Friday, November 6, 2026

Nathaniel Bowditch: the self-taught navigator

Salem boy who left school at ten, found ~8,000 errors in the standard British navigator, wrote the book American sailors still call 'Bowditch,' taught a whole crew lunars, translated Laplace, and drew the curves on Jeff's oscilloscope.

Jeff asked who **Bowditch** was after we talked about Polaris correction tables. This page is the full answer. It pairs with Polaris (the star whose tiny offset Bowditch's tables correct), The sidereal day and latitude sailing (why the correction is needed, and Jeff's "oscilloscope wave" insight), Galileo (longitude as time, before chronometers), Map Surprises, and The Harvard Computers (human computers of the Nautical Almanac were Bowditch's cousins; Leavitt was a later kind).



Portrait of Nathaniel Bowditch by Charles Osgood

Nathaniel Bowditch (1773–1838), portrait by **Charles Osgood**. Public domain, via Wikimedia Commons (Peabody Essex Museum).

1. Salem childhood: school ends at ten

Nathaniel Bowditch was born in **Salem, Massachusetts**, on **26 March 1773**, the fourth of seven children of **Habakkuk Bowditch**, a cooper (and sometime sailor who quit the sea after his ship went aground in 1775), and **Mary Ingersoll Bowditch**.

Family money ran short. At about **age ten** his formal schooling stopped and he went to work in his father's cooperage. At **twelve** he was **indentured for nine years** as a bookkeeping apprentice to a **ship chandler** — a dealer in rope, canvas, tar, and the other stores a ship needs. He lived and worked in the chandlery and learned accounts.

That "left school at ten" is the fact Jeff asked about. It is confirmed in the standard biographies (MacTutor / Biographical Encyclopedia of Astronomers; Wikipedia following Henry Ingersoll Bowditch's memoir of his father).

2. Self-taught: algebra, calculus, Latin, French, Newton's Principia

In the chandlery Bowditch taught himself.

- **1786** (age 14): algebra.
- **1788**: calculus.
- **1790**: **Latin**, specifically so he could read Isaac Newton's Philosophiae Naturalis Principia Mathematica.
- **1792**: **French**, for the European mathematical literature.

At eighteen he copied out all the mathematical papers in the Philosophical Transactions of the Royal Society. In **1793** he found an error in Newton's Principia.

Serendipity helped. In **1781** the Beverly privateer Pilgrim captured a ship carrying the scientific library of the Irish chemist **Richard Kirwan** — Newton, the Bernoullis, Chambers's Cyclopedia, and more. The books were bought for Salem and eventually housed with the Rev. **John Prince**, who in **1791** let the eighteen-year-old Bowditch use what had become the best scientific library north of Philadelphia. Without that captured library, the self-education would have been far harder.

3. Five voyages, 1795–1803

Bowditch went to sea:

Voyage	Role	Notes
1st	Ship's clerk / second mate	Began finding errors in Moore's tables; conceived a clearer method of **lunars**
2nd–4th	Supercargo	Continued corrections; taught crew navigation
5th	**Master and part owner**	Last voyage; returned to Salem **1803**

Between the first and second voyages, in **March 1798**, he married **Elizabeth Boardman**; she died seven months later while he was at sea. In **1800** he married his cousin **Mary "Polly" Ingersoll**; they had eight children.

After 1803 he left the sea for mathematics, insurance, and the book that made his name.

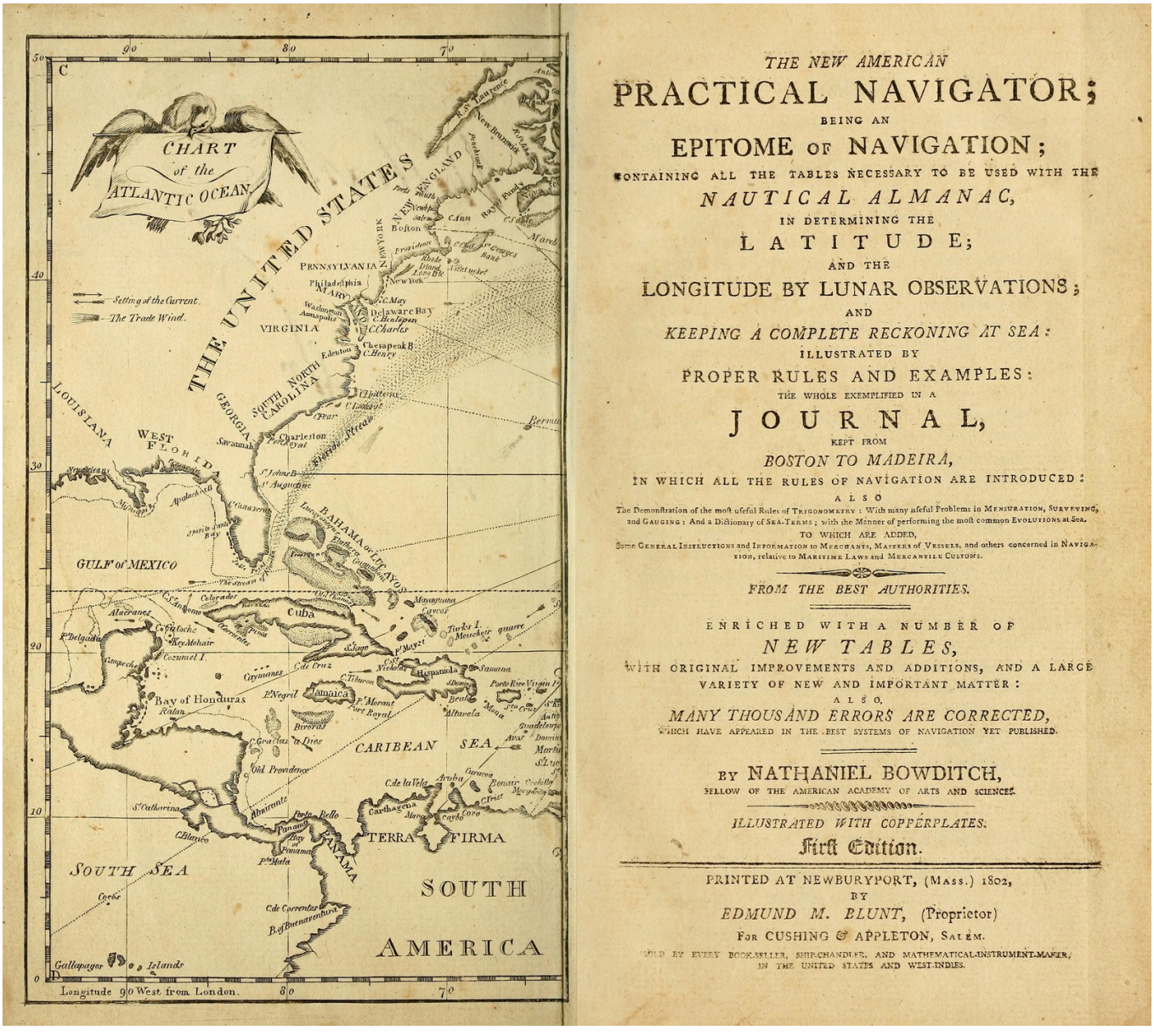
4. ~8,000 errors in Moore's Practical Navigator

The standard English handbook was **John Hamilton Moore**'s The Practical Navigator (London). Its tables were known to be faulty. On his first voyage Bowditch began checking them. By the time he was done he had compiled a list of **more than 8,000 errors**.

A careful caveat (from a modern study of Moore and Bowditch in the Journal of Navigation): most of those errors were **small** — last-decimal-place mistakes in traverse tables, proportional logarithms, and log-trig functions. Saying "8,000 errors" can make Moore sound like a fraud; he was not. But at sea a wrong last digit can still put you on a reef. Bowditch recomputed everything.

He first sent corrections to the American publisher **Edmund March Blunt**. Blunt issued corrected American editions of Moore in **1799** and **1800**. By the third round the book had changed so much that Blunt published it under Bowditch's own name.

5. The New American Practical Navigator (1802)



Title page of the 1802 first edition of Bowditch's Navigator

Title page / opening of the **1802** first edition of The New American Practical Navigator. Public domain, via Wikimedia Commons (U.S. Naval Observatory copy).

In **1802** Blunt published **Nathaniel Bowditch**, The New American Practical Navigator, printed at Newburyport, Massachusetts. Bowditch's rule for what went in:

> **"put down in the book nothing I can't teach the crew."**

The book rearranged Moore, added tables, and included Bowditch's clearer solutions for the **spherical triangle** and for **lunar distances** — finding longitude by measuring the Moon's angular distance from the Sun or a star, then comparing with an almanac. Accurate marine chronometers (Harrison's) existed but were still too expensive for

most merchant captains; lunars were the poor man's clock in the sky. (See Galileo for the older dream of a Jupiter-moon clock.)

Still in print. In **1867–68** the U.S. Hydrographic Office bought the copyright. The book has been in continuous government publication ever since. Today it is **NGA Pub. No. 9**, The American Practical Navigator, issued free by the National Geospatial-Intelligence Agency in two volumes (2024 edition as of this writing). Every commissioned U.S. Navy ship still carries it.

Sailors simply call it **"Bowditch"** — "the bible of the sea." Midshipmen before the Naval Academy existed called it "the immaculate Bowditch."

6. Teaching the crew — including the cook

Bowditch's teaching story is the one Jeff heard.

On the **fourth voyage**, to **Manila** in the ship Astræa, Captain **Henry Prince** was asked in Manila how an American ship had found its way so well against the northeast monsoon. Prince answered (as recorded in Henry Ingersoll Bowditch's memoir of his father) that he had a crew of **twelve men**, every one of whom could

> "take and work a lunar observation as well, for all practical purposes, as **Sir Isaac Newton** himself were he alive."

Bowditch sat through the conversation "as modest as a maid," slate pencil in his mouth. Another merchant added that there was "more knowledge of navigation on board that American ship than there has been in all the ships that ever came into Manila Bay."

Later tellings (including Baron Franz von Zach and many navy histories) add that **even the ship's cook** became proficient in lunar distances. The cook detail is traditional and widely repeated; the core claim — an entire twelve-man crew able to work lunars — is anchored in Captain Prince's contemporary boast. Bowditch's own standard for the book was the same spirit: write nothing you cannot teach the forecastle.

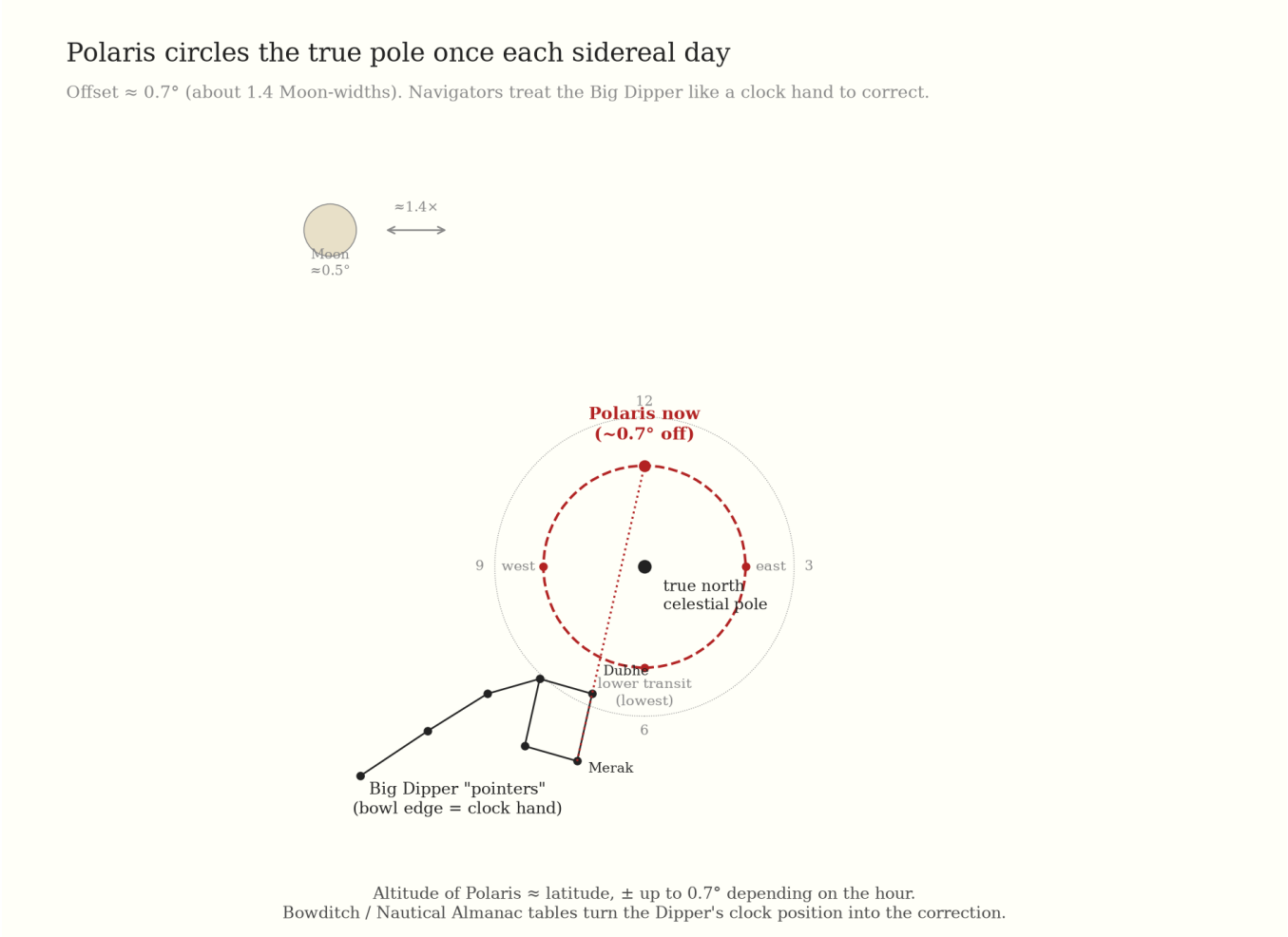
7. How Bowditch corrects Polaris

This is the bridge to Polaris and to Jeff's insight on the sidereal day page.

Polaris is not exactly on the north celestial pole. It sits about **0.7°** off (roughly **1.4 Moon-widths**), and therefore **circles the true pole once each sidereal day**. If you take the raw altitude of Polaris as your latitude, you can be wrong by up to **0.7°**, about **42 nautical miles** (since 1° of latitude = 60 nmi).

Navigators fix that with a **Polaris correction**. The classic rule of thumb treats the **Big Dipper's pointer stars** (**Dubhe** and **Merak**, the outer edge of the bowl) like a **clock hand** around the pole. From the "hour" on that clock you look up a correction in the

tables — in Bowditch historically, and today in the Nautical Almanac (corrections a0, a1, a2) — and add or subtract it. The wave flattens. The altitude of Polaris, corrected, **is** your latitude.



Polaris circles the pole; Big Dipper as clock

Polaris traces a ~0.7° circle around the true pole once per sidereal day. The Big Dipper's pointers act as a clock hand for the correction. Diagram drawn for this site.

8. Laplace's Mécanique céleste

Bowditch taught himself French partly to read **Pierre-Simon Laplace**'s monumental Traité de mécanique céleste. He finished translating the first four volumes by about **1818**, then spent years adding commentaries that roughly **doubled** the length: supplying steps Laplace had skipped, inserting later results, and giving credits Laplace had omitted. Publication (at Bowditch's own expense) ran **1829–1839**. The translation became a foundation of American physical astronomy in the nineteenth century. Benjamin Peirce helped with the later volumes.

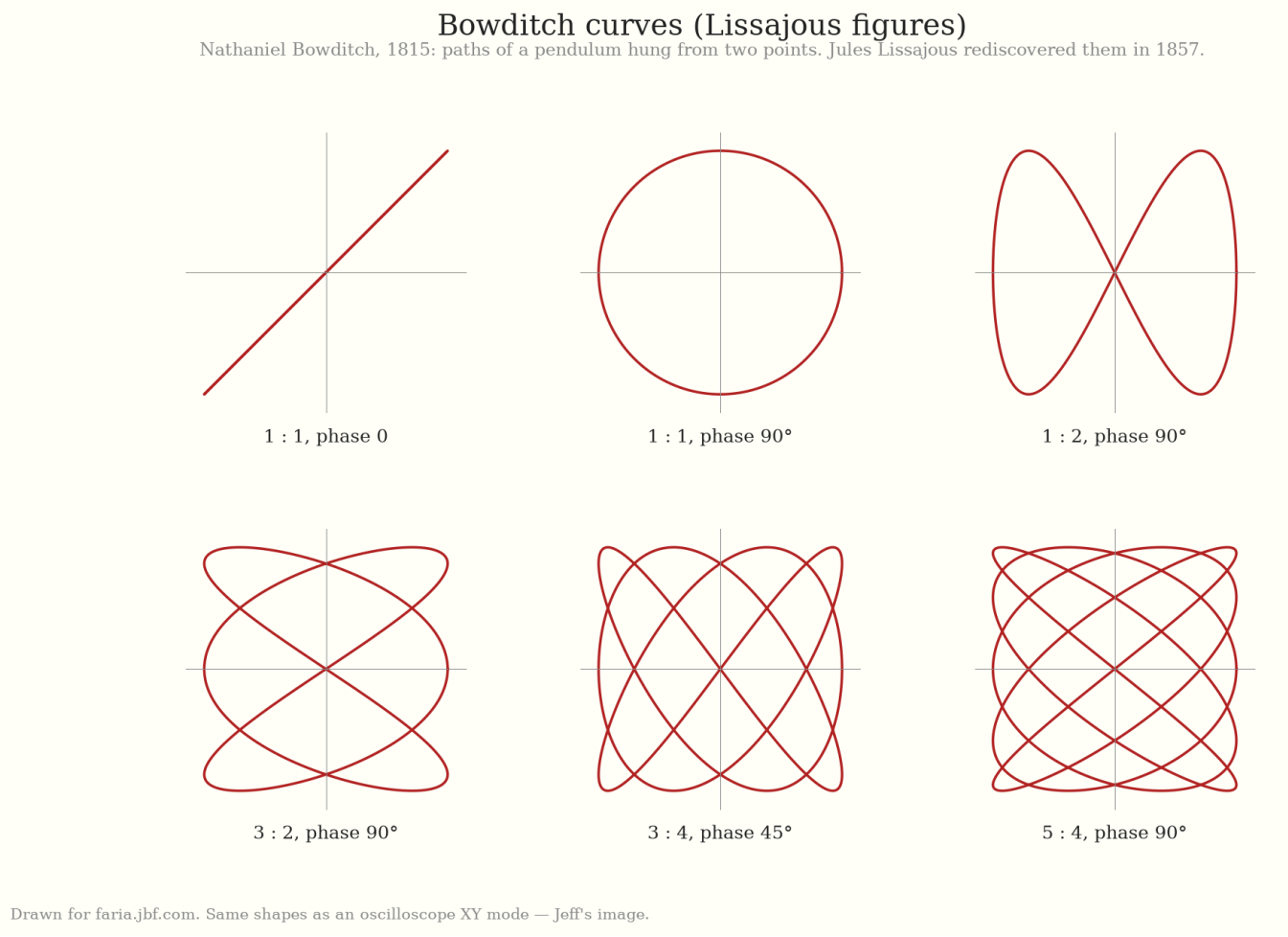
Harvard offered him the chair of mathematics and physics in **1806**; West Point and the University of Virginia also offered chairs. He turned them all down — the insurance salary was higher, and he preferred the work.

9. Actuary and "money manager"

In **1804** Bowditch became president of the **Essex Fire and Marine Insurance Company** in Salem — often called America's first insurance actuary in the modern sense. The company prospered through the War of 1812. In **1823** he moved to Boston as actuary for the **Massachusetts Hospital Life Insurance Company**, investing the fortunes of sea-made wealth into manufacturing towns such as Lowell. When he moved, he took more than **2,500 books**, **100 maps**, and **29 volumes** of his own manuscripts.

He was elected to the American Academy of Arts and Sciences (**1799**), the American Philosophical Society (**1809**), and later the Royal Societies of London and Edinburgh and the Royal Irish Academy. In **1829** he became the first American foreign associate of the Royal Astronomical Society. Harvard gave him an honorary degree in **1802** and an LL.D. in **1816**.

10. Bowditch curves = Lissajous figures = Jeff's oscilloscope



Bowditch / Lissajous curves

Bowditch curves (Lissajous figures): the paths of a pendulum suspended from two points. Drawn for this site. Same family of shapes as an oscilloscope in XY mode — the image Jeff had in mind.

In **1815** Bowditch published "On the Motion of a Pendulum Suspended from Two Points" in the Memoirs of the American Academy of Arts and Sciences. A pendulum hung from two points (a Y-shaped or Blackburn suspension) can swing at different frequencies in two perpendicular directions. The bob traces the curves

> $x = a \sin(n\,t + c), \quad y = b \sin(t)$

Jules Antoine Lissajous studied them independently and more famously in **1857**, using light reflected from tuning forks; his name stuck. Because Bowditch was first, mathematicians also call them **Bowditch curves**.

Jeff's connection. When Jeff pictured an uncorrected Polaris track as "cycles on an oscilloscope," he reached for exactly this family of shapes. An oscilloscope in **XY mode** draws Lissajous / Bowditch figures. The navigator who wrote the bible of the sea also

drew the curves on the screen Jeff was imagining. See The sidereal day and latitude sailing for the wavy track itself.

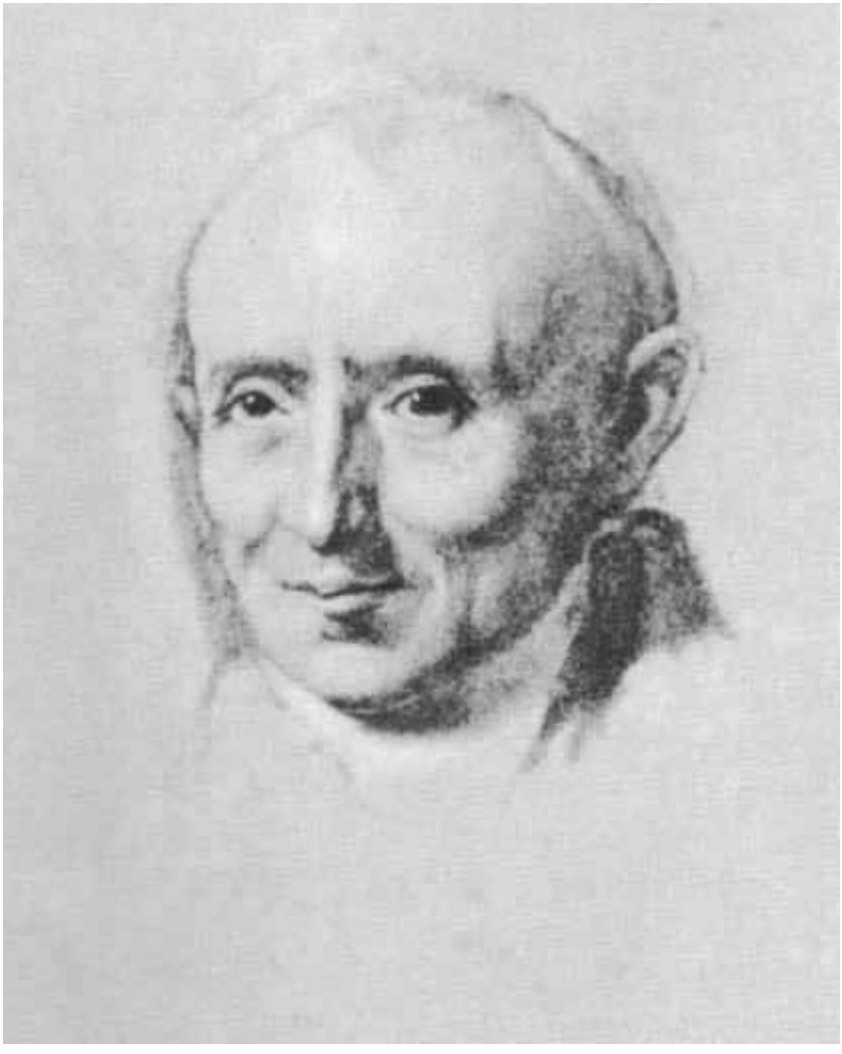
11. Death, eulogy, and legacy

Bowditch died in Boston on **16 March 1838** of stomach cancer. He is buried in **Mount Auburn Cemetery**, Cambridge; the memorial statue by **Robert Ball Hughes** was the first life-size bronze cast in America.

The **Salem Marine Society** eulogized him:

> "In his death a public, a national, a human benefactor has departed. Not this community nor our country only, but the whole world has reason to do honor to his memory. When the voice of eulogy shall cease to flow, no monument will be needed to keep alive his memory among men; but as long as ships shall sail, the needle point to the north, and the stars go through their wonted courses in the heavens, the name of Dr. Bowditch will be revered as of one who has helped his fellowmen in time of need, who was and is a guide to them over the pathless oceans, and one who forwarded the great interests of mankind."

Named for him: the oceanographic survey ship **USNS Bowditch**; a lunar crater; a Salem–Boston ferry; a Maine windjammer; schools; the Department of Marine Transportation building at the U.S. Merchant Marine Academy; and Jean Lee Latham's 1955 Newbery Medal novel Carry On, Mr. Bowditch.



Unfinished Gilbert Stuart portrait of Bowditch, 1828

Unfinished portrait by **Gilbert Stuart**, 1828 — Bowditch's family considered it the best likeness. Public domain, via Wikimedia Commons.

Review questions

1. Why did Bowditch teach himself **Latin**?
2. What was his rule for what belonged in the Navigator?
3. Why does Polaris need a **correction table**, and what "clock" do sailors use?
4. What do **Bowditch curves** have to do with an oscilloscope?

Key facts

- Nathaniel Bowditch, Salem, 26 Mar 1773 – Boston, 16 Mar 1838.
- Left school ~age 10; indentured at 12 to a ship chandler for nine years; self-taught algebra, calculus, Latin (for Newton), French (for Laplace).

- Five voyages 1795–1803; found **more than 8,000** errors in J. H. Moore's Practical Navigator (mostly small tabular mistakes).
- The New American Practical Navigator, 1802; still published as **NGA Pub. No. 9**; sailors call it **"Bowditch."**
- Rule: "put down in the book nothing I can't teach the crew"; on the Manila voyage the whole twelve-man crew (cook included, by tradition) could work lunars.
- Polaris correction: star is ~0.7° off the pole; Big Dipper pointers as clock; tables remove up to ~42 nmi of latitude error.
- Translated and annotated Laplace's Mécanique céleste (vols. I–IV, pub. 1829–39); declined Harvard / West Point / Virginia chairs.
- America's early insurance actuary (Essex Fire & Marine, then Mass. Hospital Life).
- **Bowditch curves** (1815) = Lissajous figures (1857) = oscilloscope XY shapes.
- USNS Bowditch; Mount Auburn bronze; Salem Marine Society eulogy: "as long as ships shall sail…"

Answers

1. So he could read Newton's Principia in the original Latin.
2. **"Put down in the book nothing I can't teach the crew."**
3. Polaris is ~0.7° from the true pole and circles it once per sidereal day; raw altitude can be wrong by up to ~42 nmi. Sailors use the **Big Dipper pointers** as a clock hand and look up the correction in Bowditch / the Almanac.
4. They are the same family of curves: Bowditch studied them in 1815 as pendulum paths; an oscilloscope in XY mode draws them live.

Friday, November 6, 2026

The sidereal day and latitude sailing

Why a star-day is 23h 56m, why the sky shifts 4 minutes a night, how Polaris's 0.7° wobble weaves a north-south wave like Jeff's oscilloscope, and how sailors 'ran down the latitude' — with Columbus as the cautionary case.

Jeff asked what **sidereal** means and why a day measured by the stars is only **23 hours 56 minutes**. Then he pictured what happens if you trust Polaris raw while sailing west: the track would "wiggle… like cycles on an oscilloscope." This page is both answers, plus the older sailor's trick of **latitude sailing**. It grows straight out of Polaris and Bowditch, and sits beside Galileo (longitude as time), Map Surprises, and The Harvard Computers.

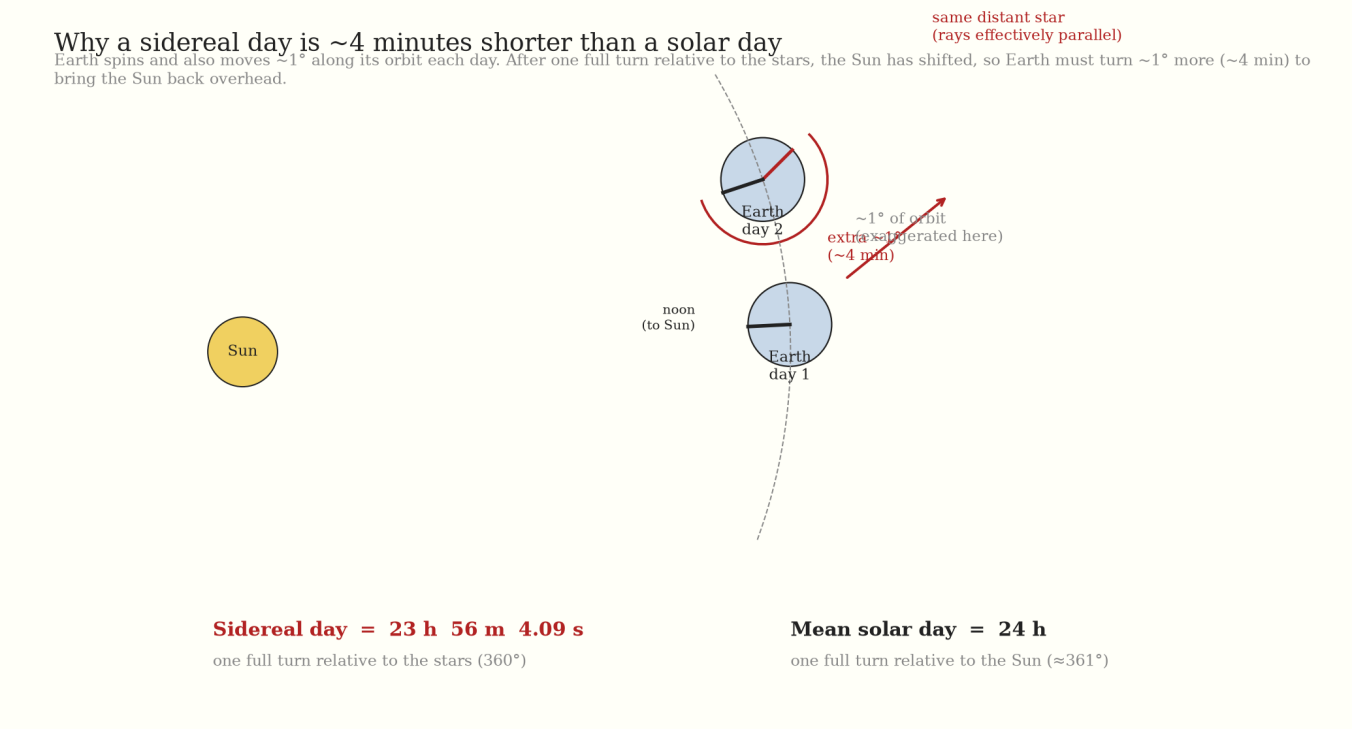
1. The word: sidereal

Sidereal — pronounce **sy-DEER-ee-ul**.

From Latin **sidus**, genitive **sideris**, "star" (also "constellation, heavenly body"). Related English words: sidereal, consider (literally "with the stars," from an old sense of consulting the heavens), desire (from de-sidus, "down from a star," via a folk etymology of longing), and the astronomical sidereal time, sidereal year.

A **sidereal day** is a day measured against the **stars**, not against the Sun.

2. 23 h 56 m 4.09 s versus 24 h — and why



Why a sidereal day is shorter than a solar day

Earth must turn ~1° extra each day to bring the Sun back overhead, because it has also moved ~1° along its orbit. Diagram drawn for this site (angle exaggerated).

Two different "days":

Kind of day	Length	What returns overhead
Sidereal day	**23 h 56 m 4.091 s** of mean solar time	A distant star (really: the vernal equinox)
Mean solar day	**24 h** exactly (by definition of civil time)	The mean Sun

Why the difference? While Earth spins, it also moves along its orbit around the Sun — about **360° / 365.25 ≈ 0.986° per day**, "about a degree." After Earth has turned a full **360°** relative to the stars, the Sun has shifted by that ~1° in the sky. Earth must turn about **one more degree** to bring the Sun back to the same meridian. One degree of rotation takes **4 minutes** (because 360° × 4 min = 1440 min = 24 h). So the solar day is about **four minutes longer** than the sidereal day.

Numbers Jeff asked about:

- "Why only 56 minutes?" — because 24 h minus ~4 minutes is **23 h 56 m**. The precise figure is **23 h 56 m 4.091 s**.

- "366 sidereal days per year?" — yes, almost: in one **mean solar year** (~365.2422 mean solar days) Earth turns once extra relative to the stars, so there are about **366.2422 sidereal days** in a year. Roughly: **365 Sun-days, 366 star-days**.

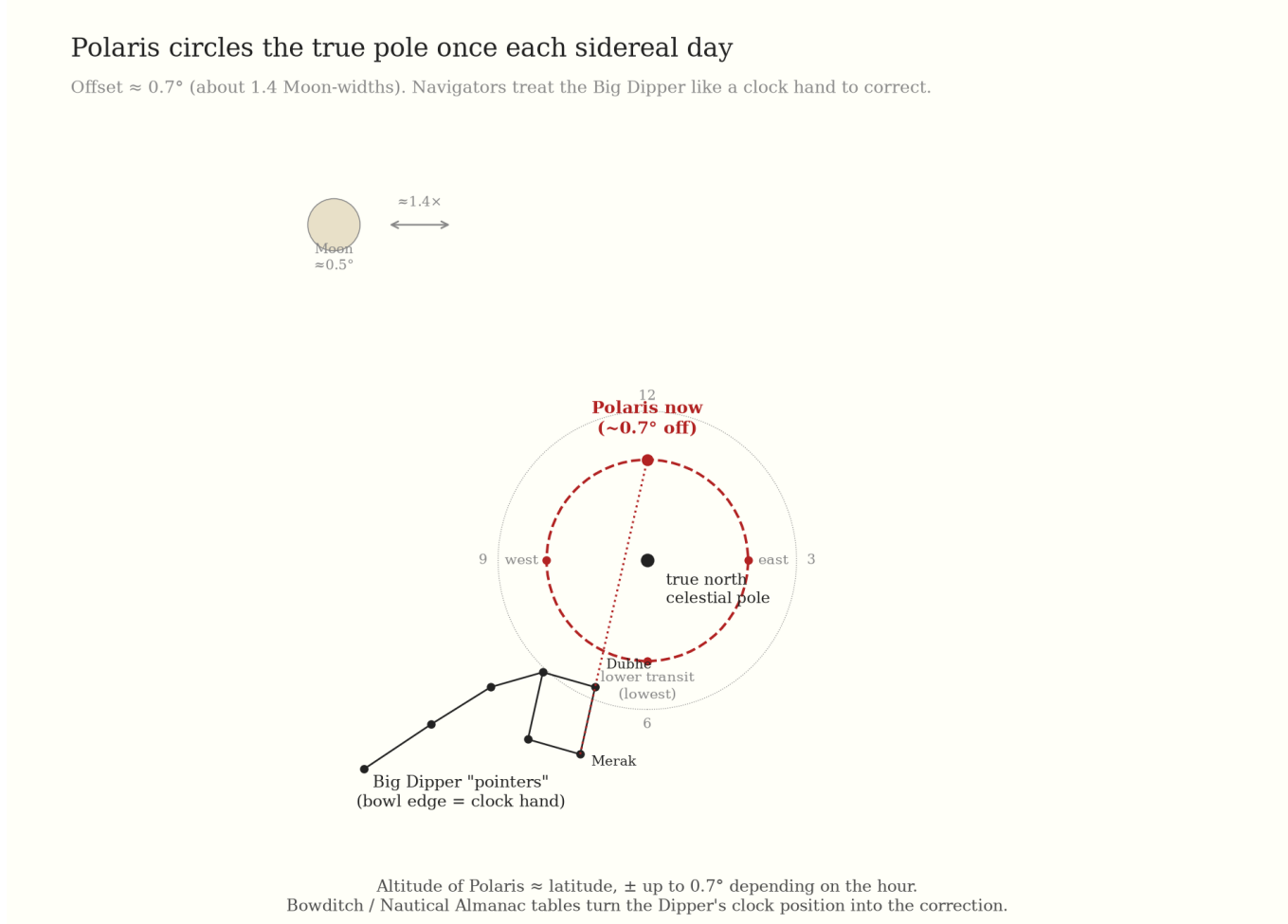
3. Stars rise ~4 minutes earlier each night

Because the sidereal day is shorter, the starry sky runs **about 4 minutes ahead** of civil clocks each night. Consequences Jeff already uses without naming them:

- **~2 hours per month.** 4 min × 30 ≈ 2 hours. The same constellation rises two hours earlier after a month.
- **Seasonal constellations.** Over a year those daily 4-minute slips add up to a full 24 hours: winter evenings show Orion; summer evenings show Scorpius. Same sky, different clock time.
- **Right ascension in hours.** Astronomers measure sky-longitude in hours precisely because the sky turns 15° per hour — the same arithmetic as terrestrial longitude (Galileo, Polaris §8).

4. Polaris's tiny circle — once per sidereal day

Polaris is about **0.7°** from the true north celestial pole (epoch ~2010: ~0.69°; ~0.66° in more recent figures; closest approach ~0.45° around **2100**). It therefore traces a small circle around the pole **once each sidereal day** (23 h 56 m), not once each solar day.



Polaris circle and Big Dipper clock

Polaris's ~0.7° circle, with the Big Dipper pointers as a clock hand. Diagram drawn for this site.

Only at **upper and lower transit** (when Polaris is due above or below the pole on the meridian) is its azimuth true north. The rest of the time its altitude is a little high or a little low compared with the observer's true latitude. Maximum error ≈ **0.7°**.

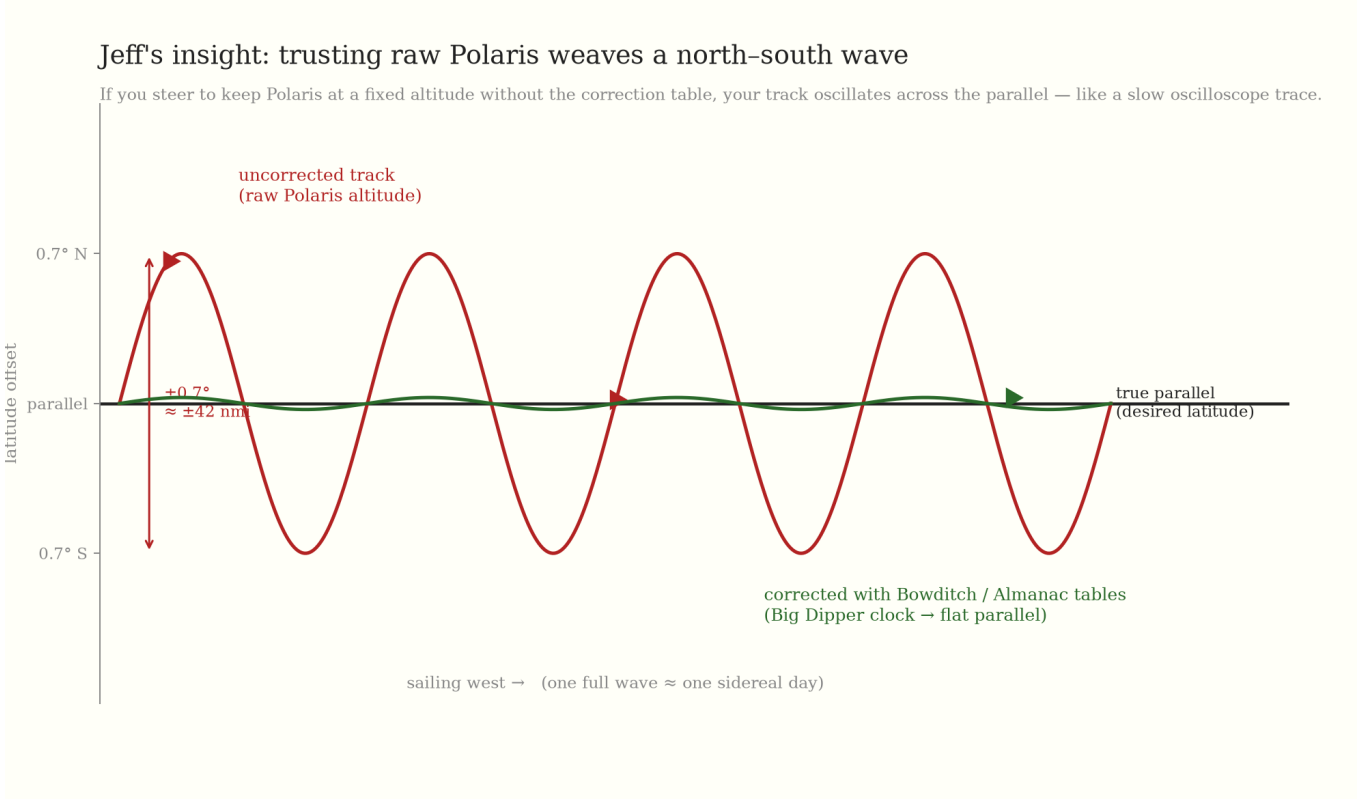
How far is 0.7° on Earth? One degree of latitude is **60 nautical miles** (that is the definition that makes the nautical mile useful). So **0.7 × 60 = 42 nmi** — about **48 statute miles**, or roughly the distance from Washington, D.C. to Baltimore. Trust raw Polaris and you can be almost a Baltimore-run off your parallel.

5. Jeff's insight: the oscilloscope wave

Jeff, thinking about sailing west while keeping Polaris at a fixed altitude:

> "You would wiggle your line… like cycles on an oscilloscope. You just kind of go a little left and right."

Almost perfect. The wiggle is **north–south**, not left–right (east–west), because altitude error is a **latitude** error. As Polaris rides its daily circle — high, then low — a helmsman who steers to "correct" the raw altitude weaves **above and below** the parallel. One full wave takes one **sidereal day**. Amplitude up to **±0.7° ≈ ±42 nmi**.



Uncorrected Polaris track weaves like an oscilloscope

Uncorrected (red): a north–south sine wave across the parallel. Corrected with tables (green): nearly flat. Diagram drawn for this site from Jeff's insight.

The fix is exactly what Bowditch teaches: read the **Big Dipper** like a clock, look up the Polaris correction, apply it. The wave flattens into a straight parallel. Modern Nautical Almanac tables (a0, a1, a2) do the same job.

And the oscilloscope image is not a stretch: the curves an oscilloscope draws in XY mode are **Lissajous figures**, also called **Bowditch curves**, first studied by Nathaniel Bowditch in **1815**. Jeff reached for the navigator's own mathematics without knowing it.

6. Latitude sailing: "running down the latitude"

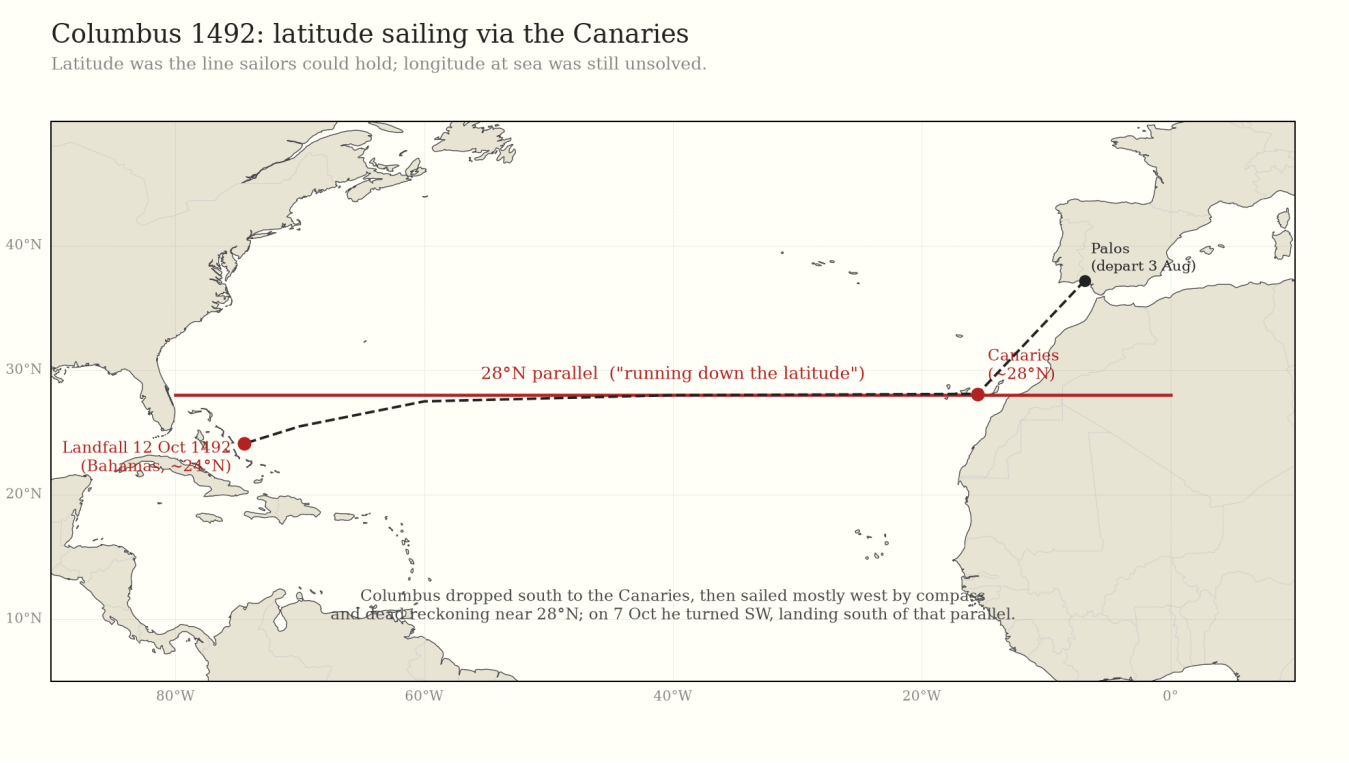
Before sailors could find **longitude** at sea, they could still find **latitude** — from the Sun at noon, or from Polaris at night. So a standard tactic was:

1. Sail north or south until you reach the **latitude of your destination**.
2. Turn east or west and **hold that latitude** (keep Polaris, or the noon Sun, at the right height) until land appears.

This is called **latitude sailing**, **parallel sailing**, or **"running down the latitude."** You follow a **rhumb line** along a parallel: constant compass heading, constant latitude. It is not the shortest path on a globe (that is a **great circle**), but it is the path you can find again tomorrow with only a quadrant and a compass.

Why it mattered. Longitude at sea was the unsolved problem until **John Harrison's** marine chronometers in the 1760s (and, for the poor, workable lunar-distance methods that Bowditch helped teach). See Galileo for the earlier dream of timing Io's eclipses. Latitude was the only line you could trust. So you aimed for the right parallel and ran along it.

7. Columbus 1492: mostly that idea, with a caveat



Columbus 1492 route via the Canaries along ~28°N

Columbus dropped south to the Canaries (~28°N), then sailed mostly west near that parallel; on 7 October he turned southwest and made landfall in the Bahamas (~24°N). Map drawn for this site.

Christopher Columbus, first voyage, **1492**:

- Left **Palos**, Spain, **3 August**.
- Dropped south to the **Canary Islands** at about **28°N** — a deliberate move into the easterly trade winds and onto a useful parallel.
- From the Canaries he sailed **mostly west**, by **compass** and **dead reckoning**, near that latitude. He was not yet a Polaris-table navigator in the Bowditch sense; the star method became routine later. But the logic is latitude sailing: get on the parallel, hold west.
- On **7 October**, seeing flocks of birds, he turned **southwest**.
- Landfall **12 October** in the **Bahamas**, around **24°N** — south of the 28°N parallel he had been roughly holding. (Which island is still argued: traditionally San Salvador / Watling.)

So: Columbus is the textbook illustration of "drop to the latitude, then run west," with the honest footnote that he steered more by compass and log than by nightly Polaris altitudes, and that his final days left the parallel on purpose.

8. Rhumb line versus great circle

A quick GIS distinction for the talk:

Path	What it is	On a Mercator chart
Rhumb line (loxodrome)	Constant compass bearing	A **straight** line
Great circle	Shortest path on the sphere	A **curve** (unless it is the equator or a meridian)

Latitude sailing along a parallel is a special rhumb line (bearing due east or due west). Airlines and great-circle sailing prefer the shorter curved path; sixteenth- and seventeenth-century ocean sailors preferred the rhumb they could hold with a compass. That is why Mercator charts were invented: so a rhumb looks straight.

Review questions

1. Pronounce **sidereal** and give its Latin root.
2. Why must Earth turn ~1° extra each solar day?
3. If Polaris is 0.7° off the pole, what is the maximum latitude error in **nautical miles**, and why?
4. Jeff said the uncorrected track would wiggle "left and right." What is the one-word correction to that picture?

5. What is **"running down the latitude,"** and why did sailors use it before chronometers?

Key facts

- **Sidereal** (sy-DEER-ee-ul) ← Latin sidus, sideris "star."
- Sidereal day = **23 h 56 m 4.091 s**; mean solar day = 24 h; difference ≈ 4 min because of ~1°/day orbital motion.
- ≈ **366.24 sidereal days** per mean solar year (one extra turn relative to the stars).
- Stars rise ~4 min earlier each night → ~2 h/month → seasonal constellations.
- Polaris ~0.7° from the pole; circles it once per sidereal day; raw altitude error up to **±42 nmi**.
- Jeff's insight: uncorrected track = north–south **oscilloscope wave**; Bowditch / Almanac tables flatten it. Those curves are themselves Bowditch/Lissajous figures.
- **Latitude sailing / parallel sailing / "running down the latitude"**: reach the destination's latitude, then hold east/west on that parallel (a rhumb). Used because longitude at sea was unsolved before chronometers (and practical lunars).
- Columbus 1492: Canaries ~28°N, then mostly west; turned SW 7 Oct; Bahamas landfall ~24°N.
- Rhumb (constant bearing, straight on Mercator) ≠ great circle (shortest, curved on Mercator).

Answers

1. **sy-DEER-ee-ul**; Latin sidus / sideris, "star."
2. Because Earth also advanced ~1° along its orbit; after a 360° spin relative to the stars, the Sun has moved, so Earth needs ~1° more rotation (~4 min) to bring the Sun back overhead.
3. **42 nmi**, because 1° of latitude = 60 nmi, and 0.7 × 60 = 42.
4. **North–south** (a latitude error), not east–west.
5. Sail to the destination's latitude, then hold east or west along that parallel until land appears. Latitude could be measured from the sky; longitude at sea could not (until chronometers / lunars).

Friday, November 6, 2026

The Harvard Computers: women who measured the stars

Before 'computer' meant a machine, it meant a person who computed. At Harvard College Observatory, Edward Pickering hired women — Fleming, Cannon, Leavitt, Maury, and later Payne — who turned half a million glass plates into the map of the stars, Cepheid yardsticks, and the proof that stars are mostly hydrogen.

Jeff, reading about Cepheids on the Polaris page, stopped at **"Henrietta Swan Leavitt (1868–1921), a Harvard 'computer'"** and said he could not be left hanging on the word. This page is the full answer: what computer meant for three centuries, who the Harvard Computers were, what they found on the glass plates, and how their work underwrites everything from spectral class **F7** on Polaris to the size of the universe. It sits beside Bowditch (human computers of the Nautical Almanac) and The sidereal day.

The Harvard Computers at work, circa 1891

Women astronomical computers at the Harvard College Observatory, circa **1891**, often nicknamed "Pickering's Harem." **Williamina Fleming** stands near the center; **Antonia Maury** (with magnifying glass) is third from the left. Henrietta Leavitt and Annie Jump Cannon are not in this particular photograph. Harvard College Observatory / Public domain, via Wikimedia Commons.

1. The word: computer meant a person

Computer is **compute** + **-er**: a person who computes.

- Earliest recorded English use: **Richard Brathwaite** (also spelled Braithwaite), The Yong Mans Gleanings (London, **1613**):[1]

> "I haue read the truest **computer** of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number."

- The Oxford English Dictionary sense that matters here (chiefly historical): "A person who makes calculations or computations… specifically a person employed to make calculations in an observatory, in surveying, etc."
- Machine sense arrives much later: calculating machines mid-19th century; programmable electronic computers mid-20th century (OED's early electronic citations cluster around the **1940s**). Until then, computer almost always meant a human being.

Gloss. In Brathwaite the "computer" is **God**, the reckoner of our days ("The daies of Man are threescore and ten"); the word then moved from a reckoner of time, to human calculators like the Harvard women, to machines.

Human computers before Harvard. The British Nautical Almanac (founded **1767**, tables from **1767** onward under Nevil Maskelyne) hired dozens of human computers to produce the ephemerides sailors needed for lunars and for finding longitude. From **1765 to 1811** about **35** computers worked on it; one of them, **Mary Edwards**, was a woman who computed for a living after her husband's death. That world is the direct cousin of Nathaniel Bowditch's world: tables of the Moon, the Sun, and the stars, checked and rechecked by hand, then bound into the book a navigator trusted with a ship.

So when Jeff hit the word computer on a Polaris page and felt something was off, his ear was right. The word had not yet been stolen by the machine.

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[1] **Footnote — Brathwaite 1613.** Exact wording as cited by the Oxford English Dictionary (earliest quotation for computer): Richard Brathwaite ("R. B."), The Yong Mans Gleanings (London, 1613), p. 1: "I haue read the truest computer of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number." Original spelling preserved (haue, euer, dayes). The passage continues: "The

daies of Man are threescore and ten" (Psalm 90:10) — so the "computer of Times" is God counting out a human life, not a clerk with an abacus. Same quotation in Wikipedia's Human computer etymology note.

2. Edward Pickering and the factory observatory

Edward Charles Pickering (1846–1919) became director of the **Harvard College Observatory** in **1877**. Photography and spectroscopy were turning astronomy into a flood of data. Glass photographic plates could freeze the whole sky; objective-prism spectra could record thousands of stellar fingerprints on a single exposure. Someone had to measure them.

Pickering specialized the labor. Night observers (mostly men) took the plates; daytime staff measured positions, magnitudes, and spectra. Starting in the early **1880s** (the program is often dated from **1881**, when Williamina Fleming moved from the household staff into the computing room; by **1883** half a dozen women were already at work, and by the late 1880s about fourteen), he hired **women** for that daytime analysis. Over the decades the group grew to roughly **80** women in total. They were called **computers**. The observing itself stayed mostly closed to them.

Why women? Skilled, careful, and — crucially for the budget — cheap. Typical pay was about **25 cents an hour** (Sobel and AIP sources; some cite 25–30¢). Men doing similar work earned substantially more (AIP: nearly five times as much in one comparison; other accounts say about 40% more). Fleming, once she became curator of astronomical photographs (**1899**), earned about **50 cents an hour**. The mocking contemporary nickname for the group was **"Pickering's Harem."** Modern writers drop the sneer and say **the Harvard Computers**.

The glass. Over decades the Observatory (Cambridge, Massachusetts, plus a southern station at **Arequipa, Peru**) amassed on the order of **500,000** glass photographic plates — still one of the great sky archives on Earth. The plates were funded in large part by **Anna Palmer Draper**, widow of the pioneer astrophotographer **Henry Draper**. She underwrote the **Henry Draper Memorial** that paid for the spectral survey and the catalogues that still carry his name: the **Henry Draper Catalogue** (HD).

Harvard Computers examining plates, circa 1890

Computers at work over astronomical photographs, circa **1890**. Public domain, via Wikimedia Commons.

3. Williamina Fleming: from housemaid to curator

Williamina Paton Stevens Fleming (1857–1911) was a Scottish schoolteacher who emigrated to Boston, was abandoned by her husband while pregnant, and took work as a **maid / housekeeper** in Pickering's household around **1879**. In **1881** he moved her onto the Observatory payroll as a computer. The story that he declared his housemaid could do better than his (male) assistants is the popular version of a real promotion.

What she did:

- Helped invent and run the early **Harvard** spectral classification (letters for types of spectra).

- Discovered the **Horsehead Nebula** on a plate taken **1888** February 6 by William Henry Pickering (Edward's brother); she noted the dark nebula on **1888** May 19 while examining the plate. (Credit for the discovery is hers; the plate was his.)
- Found hundreds of variable stars, dozens of novae, and the first recognized **white dwarfs** (spectral oddities that later got that name).
- In **1899** became Harvard's first **Curator of Astronomical Photographs** — the first woman given a Corporation title at Harvard.

Williamina Fleming, circa 1890s

Williamina Paton Stevens Fleming, circa 1890s. Public domain, via Wikimedia Commons.

4. Antonia Maury: a finer classification

Antonia Maury (1866–1952), a niece of Henry Draper, built a more refined spectral scheme that distinguished stars of the same letter by the width and appearance of their lines — in modern terms, separating **giants** from **dwarfs**. Pickering preferred Fleming's simpler letter system for the big catalogue; Maury's subtler scheme was vindicated later when Hertzsprung and Russell made luminosity classes central to stellar evolution. She left and returned more than once; the tension between speed and depth is part of the story of the plates.

Oh Be A Fine Girl, Kiss Me

How Annie Jump Cannon sorted the stars.

Annie Jump Cannon (1863–1941) studied at Wellesley, lost much of her hearing after scarlet fever, and joined the Observatory in **1896**. She took Fleming's letters and **reordered** them into the temperature sequence still used worldwide:

> **O – B – A – F – G – K – M**

That mnemonic — the one Cannon's students and every later textbook used — is the heading of this section. It looks alphabetical. It is not. It is a **temperature** sequence: **O** hottest (blue, ≥30,000 K), **M** coolest (red, a few thousand K). Our Sun is a **G2** star. Polaris Aa, on the Polaris page, is **F7Ib** — an F-type yellow-white supergiant. That "F7" is Cannon's language.

Cannon's spectral classes O through M

The Harvard spectral sequence, with rough temperatures and familiar examples. Diagram drawn for this site. (Temperatures are approximate modern ranges; the sequence itself is Cannon's.)

How many stars? For the nine volumes of the **Henry Draper Catalogue** (1918–1924) she classified about **225,300** spectra. Across her lifetime, counting the Henry Draper Extension and later lists, the usual figure is about **350,000** (some archival tallies push toward **~400,000**). She could classify several stars a minute by eye. The International Astronomical Union adopted her system in **1922**. She became Curator of Astronomical Photographs after Fleming, received the **Henry Draper Medal** (1931) — the first woman so honored by the National Academy of Sciences — and an honorary doctorate from Oxford, but was not made a Harvard professor until **1938**, three years before her death.

Annie Jump Cannon, 1922

Annie Jump Cannon, 1922 portrait. Public domain, via Wikimedia Commons.

6. Henrietta Swan Leavitt: the period–luminosity law

Henrietta Swan Leavitt (4 July **1868**, Lancaster, Massachusetts – 12 December **1921**, Cambridge) graduated from what became **Radcliffe College**, volunteered at the Observatory in **1895**, and took a permanent staff post in the early **1900s** (often dated **1902**). She was nearly **deaf**. Her assignment was **variable stars** — stars that

brighten and fade — measured by comparing glass plates of the same field taken on different nights.

1908. In Annals of Harvard College Observatory 60: "1777 Variables in the Magellanic Clouds," she listed periods for a handful of variables in the Small Magellanic Cloud and wrote the sentence that starts everything:

> **"It is worthy of notice that the brighter variables have the longer periods."**

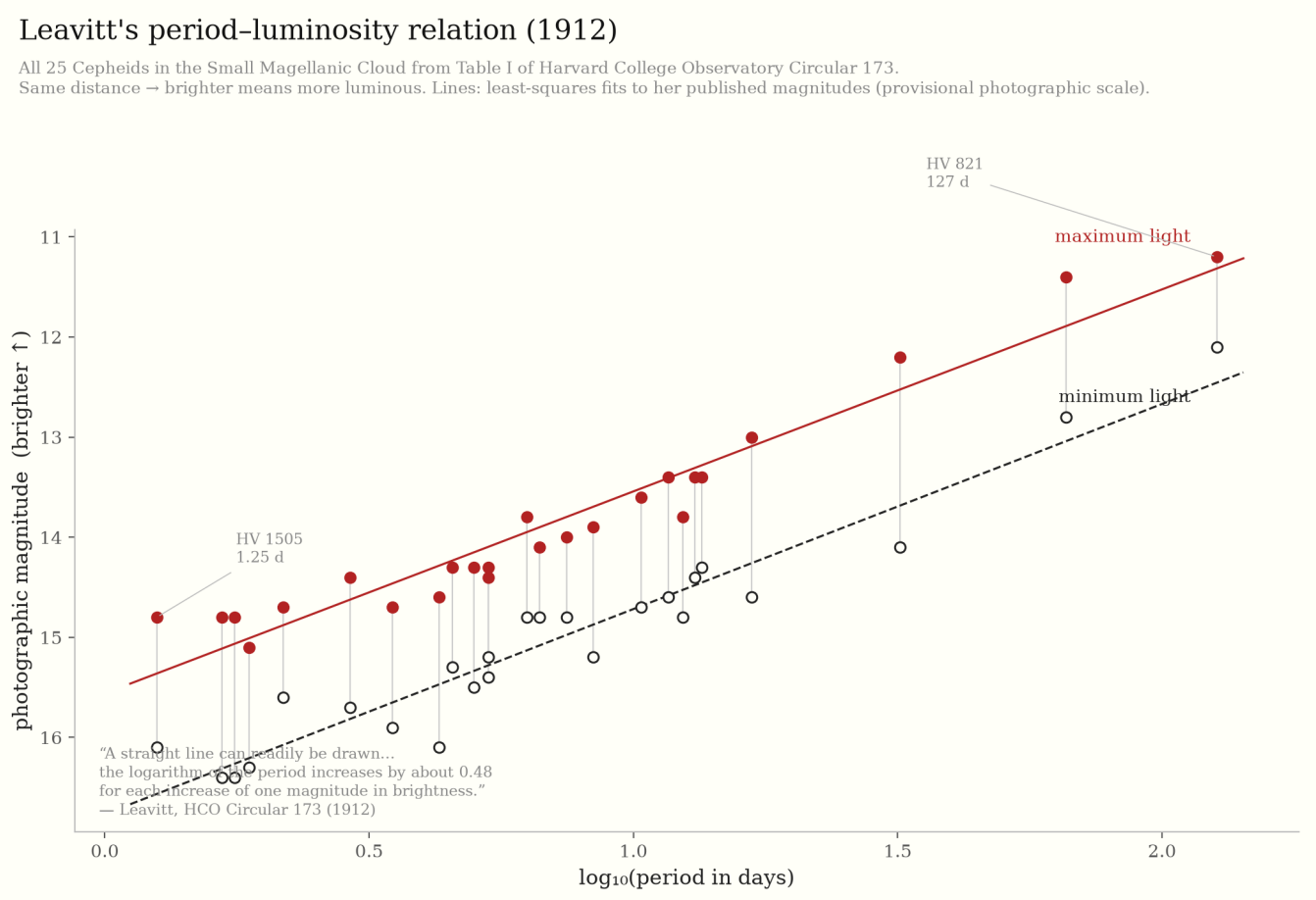
She cautioned that the sample was still small.

1912. Harvard College Observatory **Circular 173** (3 March 1912), "Periods of 25 Variable Stars in the Small Magellanic Cloud," opens:

> "The following statement regarding the periods of 25 variable stars in the Small Magellanic Cloud has been prepared by **Miss Leavitt**."

The circular carries **Pickering's** signature as director — the institutional form of the day — but the first sentence names her. With **25** Cepheids, all at roughly the same distance in the Small Magellanic Cloud, she plotted magnitude against the **logarithm of the period** and wrote:

> "A straight line can readily be drawn among each of the two series of points corresponding to maxima and minima, thus showing that there is a simple relation between the brightness of the variables and their periods. **The logarithm of the period increases by about 0.48 for each increase of one magnitude in brightness.**"



Leavitt 1912 period-luminosity data

All 25 stars from Table I of HCO Circular 173 (1912), replotted for this site from her published periods and provisional photographic magnitudes. Brighter is upward. The lines are least-squares fits to her table — the same relation she described as "a straight line."

Why this changes the universe. Apparent brightness alone cannot tell distance (a faint star may be dim, or far, or both). But if period tells you **true** brightness, then comparing true with apparent brightness yields distance. Cepheids become **standard candles**. **Ejnar Hertzsprung** (1913) and later **Harlow Shapley** calibrated the zero point; **Edwin Hubble**, using Cepheids in the Andromeda "nebula" in **1923–1924** (announced **1924–25**), showed Andromeda lies far outside the Milky Way. The universe got vastly larger overnight. That ladder still starts with Leavitt's relation — now often called **Leavitt's Law**. Circular 173 fixed the slope, because the Cloud stars share a distance. It did not fix the zero point in watts or in parsecs; she said the parallaxes still had to be measured. The rungs under and above that slope, and the difference between a solar parallax and a stellar one, are From a Venus transit to the Cepheids.

Deafness, pay, death. Leavitt's hearing had been damaged by illness; she worked through it. She was paid a little more than the base computer rate (one institutional

account says **30¢/hour**). She died of cancer in **1921**, age 53, before cosmology finished catching up to her.

The Nobel letter that came too late. On **23 February 1925**, the Swedish mathematician **Gösta Mittag-Leffler** wrote to her at Harvard, unaware she had been dead for three years:

> "Honoured Miss Leavitt, What my friend and colleague Professor von Zeipel of Uppsala has told me about your admirable discovery of the empirical law touching the connection between magnitude and period-length for the S Cepheid-variables of the Little Magellan's cloud, has impressed me so deeply that I feel seriously inclined to nominate you to the Nobel prize in physics for 1926…"

Shapley, by then director, replied that she was dead. Nobel Prizes are not awarded posthumously. She never received the nomination.

Henrietta Swan Leavitt

Henrietta Swan Leavitt (1868–1921). Public domain, via Wikimedia Commons.

Cannon and Leavitt, 1913

Annie Jump Cannon (left) and Henrietta Swan Leavitt (right), 1913. Public domain, via Wikimedia Commons.

7. Cecilia Payne: stars are mostly hydrogen

Cecilia Payne (later **Payne-Gaposchkin**, 1900–1979) came from Cambridge, England, to Harvard on a fellowship in **1923**, under Shapley. Her **1925** Ph.D. thesis — often called the most brilliant in astronomy's history — used Cannon's spectral classes

and the new quantum theory of ionization (Saha) to show that stars are composed overwhelmingly of **hydrogen** and **helium**, not of terrestrial rock-forming elements. A senior reviewer (Henry Norris Russell) urged her to soft-pedal the conclusion; she did in the printed thesis, then lived to see it become orthodoxy. In **1956** she became Harvard's first woman professor of astronomy. She is the bridge from the Computers' plates to modern astrophysics.

8. Later recognition

- **Dava Sobel**, The Glass Universe (2016): the narrative history of the plates and the women who read them.
- **DASCH** (Digital Access to a Sky Century at Harvard) and **Project PHaEDRA**: ongoing efforts to digitize the ~500,000 plates and the computers' notebooks, so a century of sky is searchable again.
- A crater on the Moon is named **Leavitt**; others honor Cannon and Fleming.
- The period–luminosity relation is increasingly called **Leavitt's Law** (a 2008 AAS resolution urged the name).

The unpaid and underpaid work that looked like "just computing" was, in Jeff's own phrase about not being left hanging, the entry into modern cosmology. Without Fleming's plates, Cannon's letters, and Leavitt's straight line, there is no Cepheid rung — and the Polaris page's yardstick has nothing to stand on.

Review questions

1. What did the English word **computer** mean in 1613, and who used it?
2. Roughly what was a Harvard computer paid per hour, and what was the mocking nickname for the group?
3. Who discovered the Horsehead Nebula, on whose plate, and in what year?
4. Recite Cannon's spectral sequence and its mnemonic. Which letter is the Sun? Which is Polaris Aa?
5. What did Leavitt notice in 1908, and what did Circular 173 (1912) prove with 25 stars?
6. Why does a shared distance (the Small Magellanic Cloud) make a period–luminosity relation possible?
7. What did Mittag-Leffler write in 1925, and why did Leavitt never get a Nobel Prize?
8. What did Cecilia Payne's 1925 thesis conclude about the composition of stars?

Key facts

- **Computer** = person who computes (Brathwaite 1613; OED historical sense); first cite is God as "computer of Times"; then human calculators; machine sense mid-20th century.
- Human computers also made the British Nautical Almanac (Mary Edwards among them) — same world as Bowditch.
- Edward C. Pickering, Harvard College Observatory director from 1877; women computers from early 1880s; ~80 women over the decades; ~**500,000** glass plates; Henry Draper Memorial funded by **Anna Palmer Draper**.
- Pay ~**25¢/hour** (Fleming as curator ~50¢; Leavitt reportedly ~30¢); nickname **"Pickering's Harem."**
- **Williamina Fleming**: housemaid → computer (1881) → Curator (1899); Horsehead Nebula **1888**; variables, novae, white dwarfs.
- **Antonia Maury**: finer spectral scheme distinguishing giants/dwarfs.
- **Annie Jump Cannon**: **O B A F G K M** (temperature); mnemonic Oh Be A Fine Girl, Kiss Me; ~**225,300** stars in the Henry Draper Catalogue, ~**350,000** in her lifetime; IAU 1922.
- **Henrietta Swan Leavitt** (1868–1921): nearly deaf; 1908 "brighter variables have the longer periods"; 1912 Circular 173, 25 SMC Cepheids, log P vs magnitude; foundation for Hubble's 1924–25 Andromeda distances; Mittag-Leffler Nobel letter **1925**, too late.
- **Cecilia Payne(-Gaposchkin)**, 1925 thesis: stars mostly **hydrogen** (and helium).
- Later: Sobel's The Glass Universe; DASCH / PHaEDRA plate digitization; Leavitt lunar crater; "Leavitt's Law."

Answers

1. A **person who calculates**; **Richard Brathwaite**, Yong Mans Gleanings (1613), "the truest computer of Times."
2. About **25 cents an hour**; **"Pickering's Harem."**
3. **Williamina Fleming**, on a plate taken by **W. H. Pickering**, **1888**.
4. **O B A F G K M** — "Oh Be A Fine Girl, Kiss Me." Sun = **G** (G2). Polaris Aa = **F** (F7Ib).
5. 1908: brighter Magellanic variables have longer periods. 1912: a **straight-line** relation between log(period) and brightness for 25 Cepheids — period tells intrinsic luminosity.
6. If the stars are at nearly the same distance, differences in apparent brightness are differences in true brightness; period can then be matched to luminosity without knowing the absolute distance.

7. He wrote that he felt "seriously inclined" to nominate her for the **1926 Nobel Prize in Physics**; she had died in **1921**, and Nobels are not awarded posthumously.
8. That stars are composed overwhelmingly of **hydrogen** and **helium**, not Earth-like heavy elements.

Friday, November 6, 2026

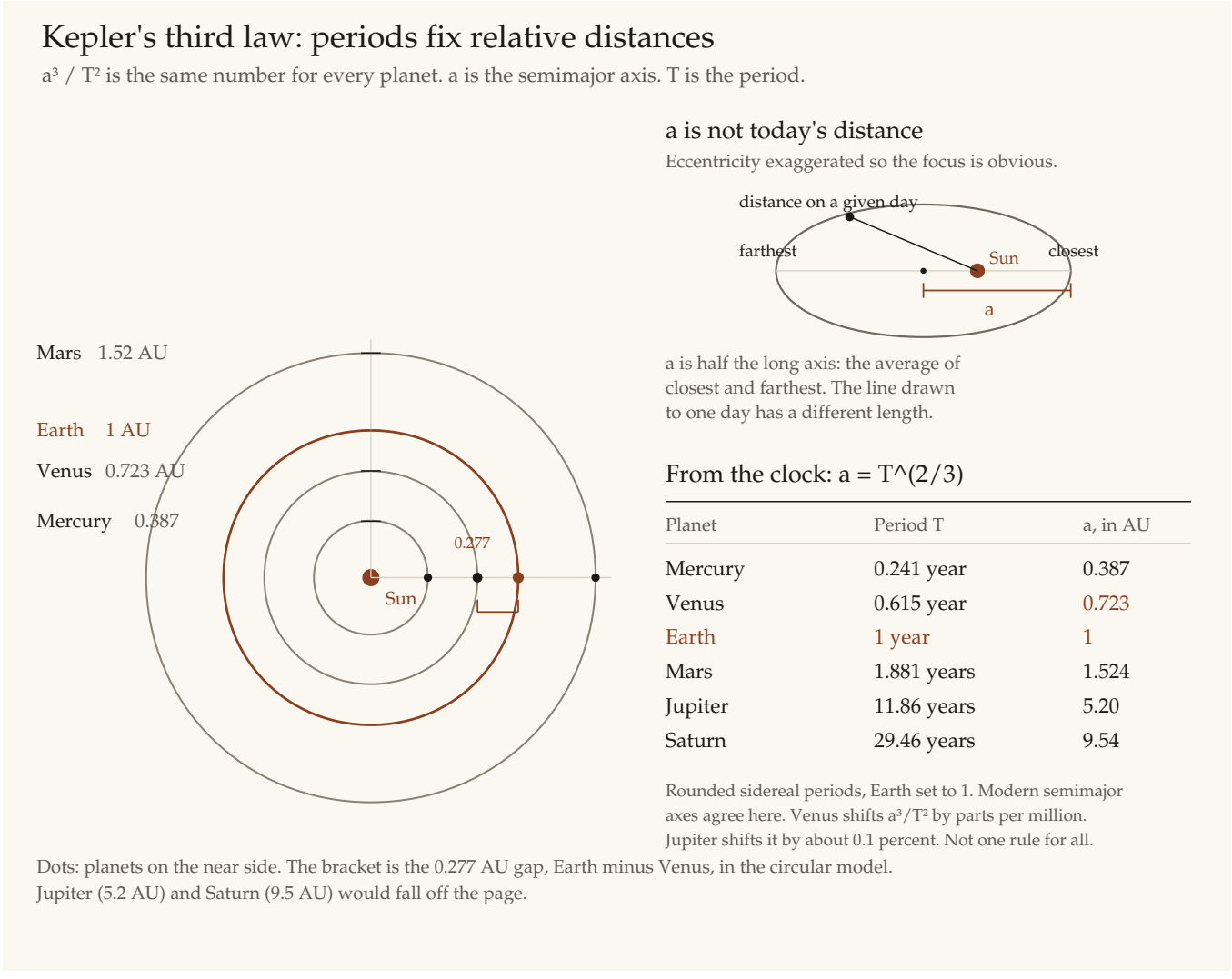
From a Venus transit to the Cepheids

Kepler's clock gives the shape of the solar system, the Venus transits of 1761 and 1769 turn that shape into a length, Earth's orbit becomes the baseline for stellar parallax, and a calibrated Cepheid period-luminosity relation carries distances across the Milky Way and beyond.

Jeff, the Polaris page called a Cepheid a yardstick, and the Harvard Computers page showed where Henrietta Leavitt's straight line came from. This page is the road under that yardstick. Four rungs, in order:

1. **Kepler's third law** gives planetary distances relative to Earth's orbit. It does not give miles.
2. **Transits of Venus in 1761 and 1769**, timed from widely separated places, turn that relative map into a length. The thing measured is a difference of parallaxes, read from durations or from contact times, not one angle taken at one station and not a latitude by itself.
3. Once the astronomical unit is a known length, **Earth's orbit is a baseline**. A nearby star's annual shift against the background is its parallax. January and July are a teaching pair, not a rule for every star. The first success, Bessel in 1838, used a heliometer, not a photograph.
4. **Leavitt's period-luminosity relation** tells which Cepheid is intrinsically brighter than which other one. It does not, by itself, say how many watts a ten-day Cepheid emits. Later calibrations, anchored on parallax and on other rungs, pin that zero point. Then Cepheids measure distances across the Milky Way and out into other galaxies.

A printed copy of this whole lesson, with the same four diagrams, is Venus transit to Cepheid distances (PDF).



Orbits of Mercury, Venus, Earth, and Mars to scale, the 0.277 AU gap marked, and an ellipse showing that the semimajor axis is not one day's distance

Kepler's third law drawn for this lesson. The orbits are circles of the right semimajor axes. The ellipse at right has its eccentricity exaggerated so the focus is obvious. Jupiter and Saturn are in the table and off the drawing.

1. The clock, and what it measures

In 1619 Johannes Kepler published Harmonices Mundi (The Harmony of the World). In Book 5 he stated what we still call his third law. In his words, the ratio of the periodic times of any two planets is the **sesquialterate** proportion of their mean distances: the 3/2 power. If Earth is the unit,

> **a cubed, over T squared, is the same number for every planet.**

T is the sidereal period in Earth years. a is the **semimajor axis** in astronomical units. Set Earth's year and Earth's semimajor axis both to 1, and the shared number is 1. Then

> **a = T to the power 2/3.**

"Mean distance," in Kepler's language, is this a. It is half the long axis of the ellipse. It equals the average of the closest distance and the farthest distance. It is **not** the distance on a Tuesday.

Look at the ellipse on the diagram. The Sun sits at a focus, not at the center. The long axis is 2a. The segment marked a runs from the center to either end. The line from the Sun to "distance on a given day" has a different length. For Venus the difference is small, because the orbit is nearly round (eccentricity about 0.0068). For a comet it is the whole story. The law never said "the distance right now."

A deeper step, if you want it. Newton showed why the number is shared. For a small planet around a heavy sun,

> a^3 / T^2 = G (M + m) / (4 pi squared),

where M is the Sun's mass and m is the planet's. The constant grows by the factor (M + m) / M, which is 1 + m/M. That factor is not the same for every planet. Venus is a few parts per million of the solar mass, so for Venus the shift in a^3/T^2 really does sit in the sixth decimal. Jupiter is about 1/1047 of the Sun, and the constant is larger by about **0.1 percent**. Saturn is about 0.03 percent. The rounded table on the diagram agrees at the printed decimals because the periods were rounded. It does not mean every planet's mass is negligible.

Worked arithmetic for Venus. NASA's Venus fact sheet gives a sidereal period of **224.701 days**. Earth's sidereal year is **365.256 days**.

- 224.701 / 365.256 = **0.61519** year.
- Square it: 0.61519 squared = **0.3785**.
- The semimajor axis is the cube root of 0.3785. Try 0.72: 0.72 cubed = 0.373. Try 0.723: 0.723 squared = 0.5227, times 0.723 = **0.3779**. Close enough: **a = 0.723 AU**.

The fact sheet's own semimajor axis is **0.723332 AU**. The clock and the modern orbit agree. The same rule, with rounded sidereal periods, fills the table on the diagram: Mercury 0.387, Mars 1.524, Jupiter 5.20, Saturn 9.54.

The law was not the only road to a ratio. Venus never strays much past about **46 degrees** from the Sun. In a circular picture, the sine of that greatest elongation is the size of Venus's orbit in AU. sin 46 degrees = **0.719**, and the modern a is 0.723. Copernicus already had that triangle. What the third law adds is a single rule that ties every planet's period to the same scale, including Mars, Jupiter, and Saturn, for which there is no "greatest elongation" of the Venus kind. The transit calculators of the 1760s used that relative model. They were not inventing the shape. They were trying to put a mile on it.

2. The simplified gap, and why it is simplified

Put the orbits in a flat circle and line the planets up on the near side of the Sun, as the dots on the diagram do. Earth's orbit is 1 AU. Venus's is 0.723 AU. The gap is

> **1 - 0.723 = 0.277 AU.**

That is the Earth-Venus separation at inferior conjunction **in this model**: circular, coplanar, and using the semimajor axes as if they were the distances that day.

Three real complications, none of which wreck the lesson:

- The orbits are ellipses. Venus's Sun-distance runs from about 0.718 to 0.728 AU, Earth's from about 0.983 to 1.017. The radial gap at a real conjunction is near 0.277 AU and is not identical to it.
- Venus's orbit is tilted about **3.4 degrees** to Earth's. The true separation is a little larger than the difference of the two solar distances, except near the nodes, which is exactly where a transit can happen.
- "Conjunction" means the same ecliptic longitude, not "the smallest possible 3-D gap on an average Tuesday."

So 0.277 AU is the right number to think with. A reduction of a real transit uses the orbital elements for that week. Nobody in 1769 subtracted 0.723 from 1.000 on a slate and called the result the measured distance.

3. A ratio is not a mile

An astronomical unit, in the eighteenth century, meant "Earth's semimajor axis, whatever that is in feet." Kepler hands you Venus at 0.723 of that unit. He does not hand you the unit.

The transit measures an angle: the **solar parallax**. The horizontal solar parallax is the angle subtended by Earth's equatorial radius as seen from a distance of 1 AU. Once you have that angle, and once you know Earth's radius in miles from a survey,

> **AU = Earth's radius / the parallax, in radians.**

The survey is the old problem on the Eratosthenes page, done much more carefully by 1769. Earth's size was already known far better than the Sun's distance. The blur in the transit era was the angle, not the radius. Halley, in 1716, still had to remind his readers how bad the old guesses were: Ptolemy at 1,200 Earth radii, Kepler at almost 3,500, and his own working hypothesis of 12.5 arcseconds, which is about 16,500 Earth radii. The modern figure is about **23,500** Earth radii. The expeditions were trying to settle a number that learned men still had wrong by a large factor.

4. Three different parallaxes

This is the step that gets flattened in a hurry, so it is worth going slowly.

Solar parallax, written here as pi-sun, is Earth's radius seen from the Sun. The modern value is about **8.794 arcseconds**. It is no longer an independent mystery: the astronomical unit is a defined length (below), Earth's equatorial radius is about **6,378.137 km**, and the angle follows.

> 6,378.137 / 149,597,870.7 = 0.000042635, in radians.
> Times 206,265 arcseconds in a radian = **8.794 arcseconds**.

Venus's parallax is the same baseline seen from Venus. In the circular model Venus is 0.277 AU away at inferior conjunction, so

> pi-Venus is about pi-sun / 0.277, which is about **3.61 times** the solar parallax.

The differential parallax is what two observers actually see as a disagreement in where Venus sits on the Sun. Both bodies shift when you move your eye. The nearer one shifts more. The relative shift is the difference:

> pi-Venus minus pi-sun is about **(3.61 - 1) times pi-sun = 2.61 times the solar parallax**.

Check the arithmetic the long way. 1 / 0.277 = 3.610. Subtract 1. That is 2.610. If you insert the modern solar parallax only to see the size of the effect, 2.61 times 8.79 arcseconds is about **23 arcseconds**.

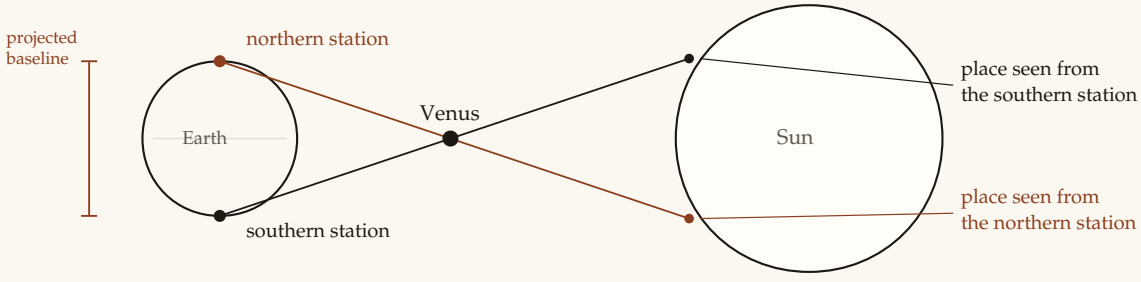
Read that 23 carefully. It is the track separation for one stated baseline: a projected separation of **one Earth radius**, square across the line of sight. That is the same length hidden in the definition of horizontal parallax. It is an illustration, not a number from a logbook, and it is **not a ceiling**. The straight-line distance between two observers can be longer than one radius. It can be as much as about an Earth **diameter**, two radii, when the stations stand near opposite ends of the sunlit hemisphere and can both still see the Sun. A side-on diameter would about double the shift, to something near 46 arcseconds in this same circular model. Real stations fall somewhere in that range, and the baseline is tilted, so the separation on the disk is whatever that projected length produces. The diagram labels 23 arcseconds as the one-radius case.

Halley himself, in 1716, did the same kind of arithmetic with a trial parallax, and he said so. He supposed the solar parallax was 12.5 arcseconds "till its precise quantity be made to appear more certain by the trial I propose." On that supposition he put Venus's parallax at 43 arcseconds, "almost 4 times" the Sun's (43 / 12.5 = 3.44, so he was using an Earth-Venus distance near 0.29 AU, close to our 0.277 and not identical). Subtract, and "there will remain half a minute at least" for the parallax of Venus from

the Sun. Half a minute is 30 arcseconds. That is his hypothesis. It is not a measurement, and it is not the modern 8.8 arcseconds.

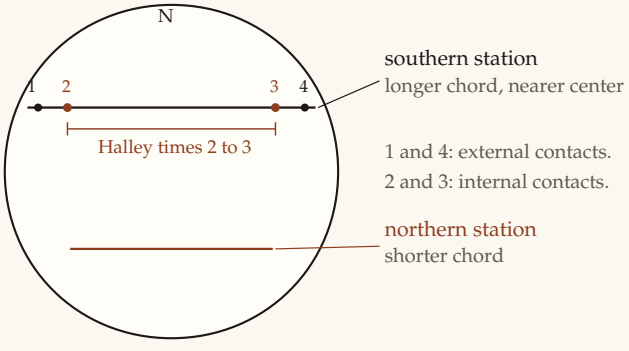
What two stations actually compare

Schematic. Sizes and angles are exaggerated so the shift can be seen. Nothing here is traced from a 1769 logbook.

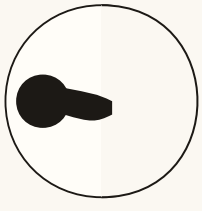


Raise your head and a near finger jumps down against a far wall.
A northern station sees Venus shifted south on the Sun. The Sun shifts too.
Venus, being nearer, shifts more. The measurement is that difference.

Over the hours, each station sees a chord across the Sun



Why contact 2 smears



Black drop. A dark neck seems to tie Venus to the outside sky, so "contact" is a judgment, not a click.

Circular model: Earth to Venus at inferior conjunction is about 0.277 AU.
Differential parallax is about (1/0.277 − 1) times the solar parallax, or 2.61 times it.

Illustrative, not a measurement. One Earth radius, side-on, separates the tracks by about 23 arcseconds if the solar parallax is the modern 8.79. That is one radius, not a ceiling.
Stations can be farther apart, up to about a diameter, and the baseline is usually tilted.

Delisle times one contact in absolute time and needs longitude. Halley's duration uses one local clock. Latitude by itself is not the baseline.

Mercury shows a black drop too. The neck is mostly blur and the Sun's dim edge, not only Venus's air.

Two stations see Venus shifted differently against the Sun; the tracks are two chords whose durations Halley compared; a black drop blurs internal contact

Schematic, drawn for this lesson. Earth, Venus, and the angles are much too large. The 23 arcseconds in the note is the one-radius illustration above, not a ceiling and not a 1769 observation. North is up on the solar disk: a northern station sees the nearer planet shifted south.

5. Halley's clock, and Delisle's clock

Edmond Halley watched a transit of Mercury from St Helena in 1677. Mercury is too far, at a transit, for its parallax to stand well clear of the Sun's. Venus at inferior conjunction is the useful one: nearer, larger differential, rare. He published the method in the Philosophical Transactions in **1716** (Methodus singularis, volume 29, number 348, page 454). He would be dead before 1761. The paper is a set of instructions to the next generation.

What he asked them to time. Not a tiny angle on a divided circle. He had already said that single seconds of arc, even ten seconds, defeat the instruments of the day. He asked for the **duration**: the local time from internal ingress (contact 2, when Venus has just come fully onto the disk) to internal egress (contact 3, when it is about to leave). A clock that runs truly between those two moments is enough for the duration. An error in the absolute longitude mostly cancels, because you subtract two readings on the same clock.

Why a duration is a parallax: the stations see Venus travel different **chords** of the Sun. A chord nearer the center is longer, so the transit lasts longer. The difference of durations, the known speed of Venus across the disk, and the known direction of the path turn back into an angle: the separation of the chords. That separation is the **differential** parallax for the **projected baseline** between the stations. Divide by the geometric factor (about 2.61 in the simple model; the real factor uses that month's distances) and you have the solar parallax. Then the AU, in Earth radii, is the reciprocal of that angle in radians.

Halley's own trial numbers show the proportion, and only the proportion. He reckoned that if the solar parallax were 12.5 arcseconds, the duration at Hudson Bay would exceed the duration in the East Indies by about **17 minutes**. A larger observed difference would mean a larger parallax, in nearly the same ratio. He thought two seconds of time would pin the parallax to a small fraction of an arcsecond, and the distance to one part in five hundred. Hold that hope lightly. He also assumed a path that Venus did not take. In 1761 she passed much nearer the Sun's center than his 4 minutes of latitude, his favored stations were wrong for the whole duration, and a later note on the translation records an error in the angle of the path. The 17 minutes is a worked example under stated assumptions. It is not the 1769 result, and it is not a number to paste onto Cook's journal.

Delisle's method is the other half of the campaign. Joseph-Nicolas Delisle, who had visited Halley, proposed timing **one** internal contact at two stations in absolute time. You no longer need both ingress and egress, so a station that sees only the entrance, or only the exit, still counts. The price is the longitude. Both clocks have to be reduced to one meridian, and the east-west part of the baseline matters as much as the north-south part. Halley's duration is forgiving of a bad absolute clock and hungry for a complete transit. Delisle's instant is forgiving of a cloud in the middle of the afternoon

and hungry for a known longitude. Eighteenth-century longitude was the harder measurement. Neither method is "stand on a latitude and measure one angle."

The projected baseline is the component that actually separates the chords or shifts the contact time: the station-to-station line as seen from the Sun, square to the line of sight, and then taken along the direction that matters for the path. Two towns at the same latitude, far apart in longitude, are useless for a pure north-south chord shift and very useful for a timing method. Two towns on the same meridian, far apart in latitude, are Halley's classic pair. The map coordinate you print is not the baseline. The geometry of that day is.

6. Who went, and what "Siberia" means

Transits of Venus come in pairs eight years apart, then not again for more than a century. Jeremiah Horrocks and William Crabtree saw the transit of **1639** from England. There was no distant second station, so Halley's comparison could not be made. The coordinated campaigns are **1761** and **1769**.

1761, 6 June. The French Academy sent Jean-Baptiste Chappe d'Auteroche overland to **Tobolsk**, in western Siberia. He got there, timed a lunar eclipse to fix the longitude, and, in clear weather and with armed protection (local feeling blamed the strange instruments for a flood), watched the whole transit. He published the observations; the journey became Voyage en Sibérie (1768). Charles Mason and Jeremiah Dixon, sent toward Bencoolen in Sumatra and diverted by war and weather, observed from the **Cape of Good Hope**. Dozens of other stations in Europe took part. The published solar parallaxes from this transit scattered by several tenths of an arcsecond.

Mikhail Lomonosov, watching from St Petersburg, reported a luminous ring at the limb and argued that Venus has an atmosphere. The atmosphere is real. How much of what he saw was the atmosphere, and how much was the optics of a bright edge, is still argued. Keep that ring separate from the black drop below. One is a bright aureole. The other is a dark neck that spoils a clock reading.

1769, 3 June at Tahiti and 4 June at stations far to the east, the same transit on two local dates. The Royal Society sent James Cook in the Endeavour, with the astronomer Charles Green, formerly Maskelyne's assistant at Greenwich. They anchored in Matavai Bay on **13 April 1769**, built Fort Venus on the point still called Point Venus, and had a clear day. Cook, Green, and Daniel Solander observed. Cook's journal records the gift and the trouble in one breath: not a cloud, and yet

> "we very distinctly saw an Atmosphere or dusky shade round the body of the Planet which very much disturbed the times of the Contacts particularly the two internal ones."

He adds that he, Green, and Solander "differ'd from one another in observeing the times of the Contacts much more than could be expected." The times and the drawings

of that ligament were published in the Philosophical Transactions for **1771** (volume 61, pages 397-421), Green named with Cook. Green did not live to finish the voyage home.

The black drop. Near internal contact, Venus seems to stay tied to the Sun's limb by a dark neck, so the moment of "internal contact" is a judgment spread over many seconds. It is tempting to blame Venus's air, and Cook's "dusky shade" points that way. Mercury, which has almost no atmosphere, shows a black drop too. The neck is mostly the telescope's blur acting on the Sun's darkened limb. The atmosphere produces a different, real, bright rim. The practical result in 1769 was the same either way: Halley's hoped-for single second of time was not what the observers could agree on.

Russia in 1769 is a network, not one Siberian dot. Stepan Rumovsky, director of the Academy observatory, organized it under Catherine II, with observers from Russia, Germany, Sweden, and Geneva. The stations, from the IMCCE chronological table and the Siberian local-history accounts, were:

Station	Who	Where it actually is
Yakutsk	Ivan Isleniev; the Siberian encyclopedia also names F. Chernyi, while the IMCCE table says an unnamed companion	On the Lena, deep in Siberia. The Siberian station.
Orsk	Christoph Euler	Southern Urals, the edge of Europe and Asia, not deep Siberia
Orenburg	Wolfgang Ludwig Krafft	Same edge
Guriev	Petr Inokhodtsev and Georg Moritz Lowitz	Caspian shore, now Atyrau, Kazakhstan
Kola	Rumovsky	Kola Peninsula, European Arctic
Ponoi	Jacques Andre Mallet	Kola Peninsula
Umba	Jean-Louis Pictet	Kola Peninsula
St Petersburg	Anders Lexell and colleagues	The capital

Isleniev's party was in the Yakutsk region from June 1768 to August 1769. The published work includes Venus leaving the Sun, Jupiter's satellites for the longitude, and coordinates that later improved the Academy's maps. Do not fold every Russian station into the word Siberia, and do not move Chappe's **1761** Tobolsk into 1769.

Other 1769 stations that matter for the geometry, and that are not Siberia: Maximilian Hell and Janos Sajnovics at **Vardo** in northern Norway; William Wales and Joseph Dymond at Prince of Wales Fort on **Hudson Bay**, the northern station Halley had

named in 1716; and Chappe again, this time at **San Jose del Cabo** in Baja California. He observed the transit on 3 June and died in the epidemic there on 1 August 1769. The observations were carried back.

No single station, and no single transit, printed the modern astronomical unit. The 1761 reductions disagreed among themselves. The 1769 campaign was better and still left a spread. Johann Franz Encke's careful combination of both transits, published in **1824**, gave **8.5776 arcseconds**, with an uncertainty he put near 0.037. That is about 0.22 arcsecond below the modern 8.794, which makes the Sun about **2.5 percent too far** (8.794 / 8.5776 is about 1.025). A revision in 1835, after a dispute over Hell's times, moved it only to about 8.571. The transits of **1874 and 1882** brought photography and another generation of reductions. Simon Newcomb's later synthesis sat near **8.79 to 8.80 arcseconds**, close to what we use, and still not a definition frozen on transit day. Radar echoes from Venus, in the twentieth century, measured a distance in kilometres by the travel time of a signal. The transit had become history. The geometry it taught had not.

7. The number we actually use

On 31 August 2012 the International Astronomical Union, in Resolution B2, redefined the astronomical unit as a conventional length:

> **exactly 149,597,870,700 metres.**

The symbol they asked for is **au**. The old definition, tied to the Gaussian gravitational constant and to the Earth's orbit, was retired. The Earth's real Sun-distance still changes through the year, from about 0.983 au at perihelion to about 1.017 au at aphelion. "One au" is no longer a synonym for "however far the Sun is today."

In international miles, 1,609.344 metres to the mile:

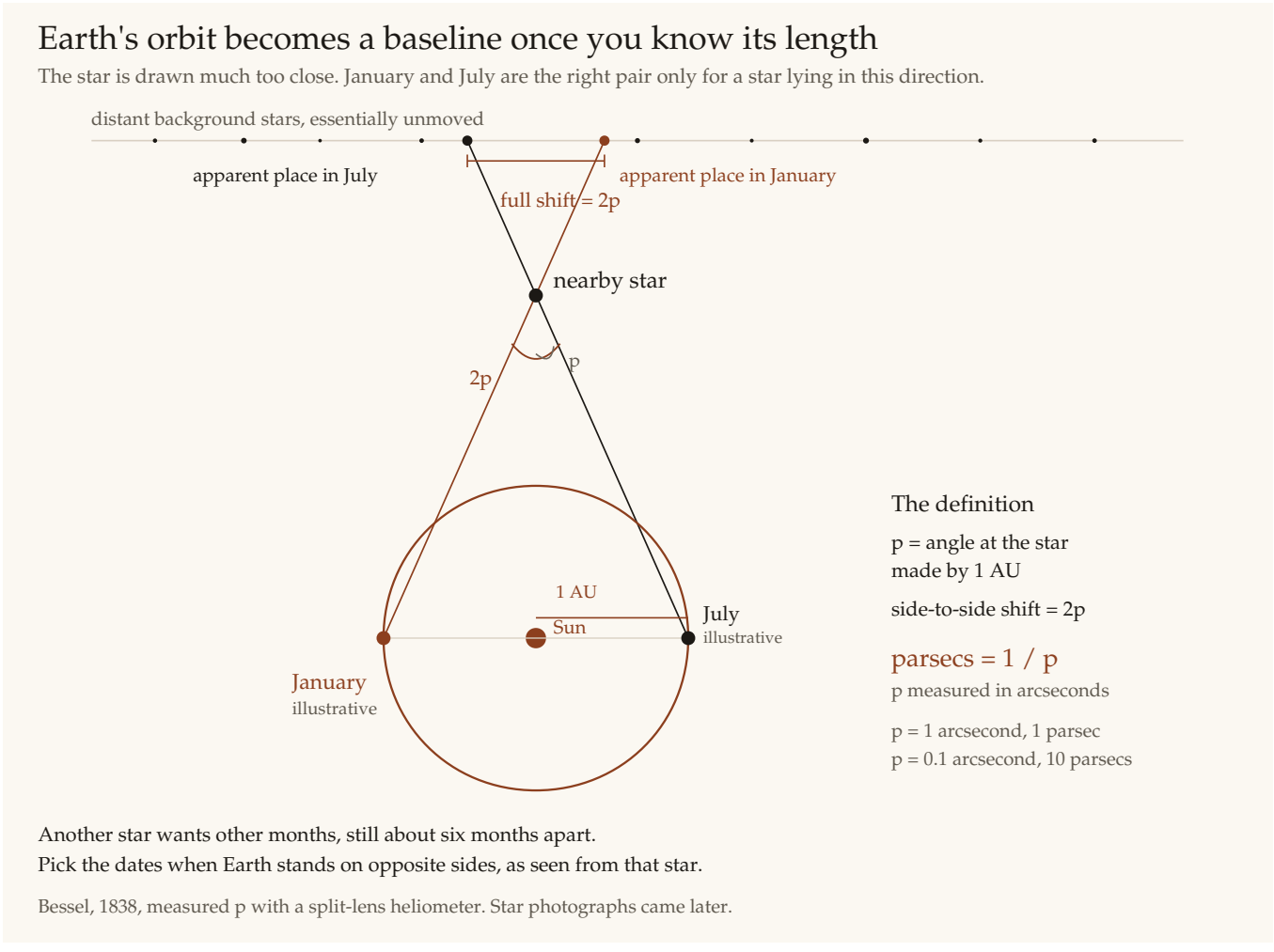
> 149,597,870,700 / 1,609.344 = **92,955,807 miles**,

which is **92.96 million miles** if you round to the hundredth of a million. That is the figure to remember. It is a definition, inherited from the best radar-era estimates, not a stopwatch reading from Tahiti.

8. The orbit as a baseline

You now have a stick. It is one au long, and you know that length in metres. Look at a nearby star in two seasons, from opposite sides of the stick, and the star shifts against the distant background. That shift is the **annual parallax**.

Hold a finger at arm's length and blink each eye. The finger jumps. The nearer it is, the larger the jump. The distant wall barely moves. Earth's orbit is the gap between the two eyes. The star is the finger.



Earth on opposite sides of its orbit, January and July illustrative, the full background shift marked 2p and the one-au angle marked p

Drawn much too close, or the angle would be invisible. January and July are the right pair for a star in this direction. Another star wants other months. Bessel's 1838 measurement was a heliometer angle, not a pair of photographs.

p and 2p. The parallax angle p is the angle at the star subtended by **1 AU**, not by the full width of the orbit. The star's apparent jump from one side of the orbit to the other is **2p**. Textbooks that say "the parallax is the shift from January to July" are describing 2p and naming p. The definition that makes the distance formula clean is the half-shift:

> **distance in parsecs = 1 / p**, with p in arcseconds.

One parsec is the distance at which 1 AU subtends one arcsecond. A tenth of an arcsecond means ten parsecs. A parsec is about 3.26 light-years, or about 206,265 AU.

January and July are a drawing, not a calendar rule. They are opposite, and they are the right pair when the star lies square to the January-July line, as on the diagram. A star somewhere else on the ecliptic wants the two dates when Earth is on opposite sides of the Sun as seen from that star. Those dates slide around the year with the star's ecliptic longitude. A star near the ecliptic pole traces a small circle, and any

opposite pair will do. What you must not do is observe in January and February and call the tiny piece of the orbit a baseline of 2 AU.

The first success was not a photograph. Friedrich Wilhelm Bessel, at Konigsberg, used a Fraunhofer **heliometer**: a telescope whose objective lens is cut in half, so the two halves can be slid until a nearby star and a comparison star coincide. He chose **61 Cygni** for its large proper motion, a hint that it is near, and because the pair can be measured cleanly. In 1838, in Astronomische Nachrichten, he published

> **p = 0.3136 arcsecond, give or take 0.0202.**

He turned that into 657,700 sun-distances and a light-time of **10.3 years**. The modern parallax is about **0.286 arcsecond**, near **11.4 light-years**. He was close, and he was first to publish a result that the errors could not swallow.

Two others were on the same problem. Thomas Henderson had measured Alpha Centauri from the Cape in 1832-33 and published in 1839, after Bessel. Wilhelm Struve measured Vega; a preliminary notice is 1837 and the fuller result 1840, and it was less accurate. "Bessel, 1838" is the right first success. It is not a claim that nobody else was looking.

Photography cannot be the method, because it was not ready. Daguerre's process was announced in 1839. The first successful photograph of a star, Vega, was made at Harvard by John Adams Whipple and William Cranch Bond on 17 July 1850. A pair of plates, January against July, is a clear way to explain the geometry, and it is how later parallax programs worked. It is not what Bessel did. The Polaris page uses the January/July picture for the finger-blink. That picture is fair teaching. The instrument in 1838 was the split lens.

Polaris itself is a warning about easy parallaxes. It is the nearest classical Cepheid, and the distance fight on that page (Hipparcos near 432 light-years, a spectroscopic "short" distance near 323, Gaia near 446) exists because a companion and a pulsation sit on top of a tiny angle. The nearest famous star is not the easiest calibrator.

9. Brightness is not distance

A star can look faint because it is dim, or because it is far, or because dust is in the way. **Apparent** brightness is what the plate records. **Intrinsic** brightness, the luminosity, is what the star emits.

Light spreads over a sphere. Double the distance and the same light covers four times the area, so the flux falls to a quarter. That is the **inverse-square law**. In magnitudes, with distance d in parsecs and no dust,

> **m - M = 5 log10(d) - 5.**

m is the apparent magnitude. M is the absolute magnitude, the apparent magnitude the star would have at 10 parsecs. If dust dims the star by A magnitudes, add A to the right-hand side. Ignore the dust and you will set the star too far.

A purely illustrative sum, not a real star. Suppose a calibrated relation says some Cepheid has absolute magnitude -4.0, and you measure its apparent magnitude as 16.0, and you ignore dust. The difference m - M is 20. Then

> d = 10 to the power of (20 + 5) / 5 = 10 to the 5th = **100,000 parsecs**.

The arithmetic is the whole point. The -4.0 was supplied. Nothing in the period, by itself, wrote that -4.0 down.

10. What Leavitt measured, and what she did not

The Harvard Computers page has her life, the plates, and the plot of all 25 stars. The facts that matter for the ladder are these.

The Small Magellanic Cloud is a cloud of stars at essentially **one** distance. (The depth of the cloud is real and small beside the distance.) Stars in that cloud can be faint because they are faint, not because one of them is vastly farther. Difference in apparent brightness is difference in luminosity.

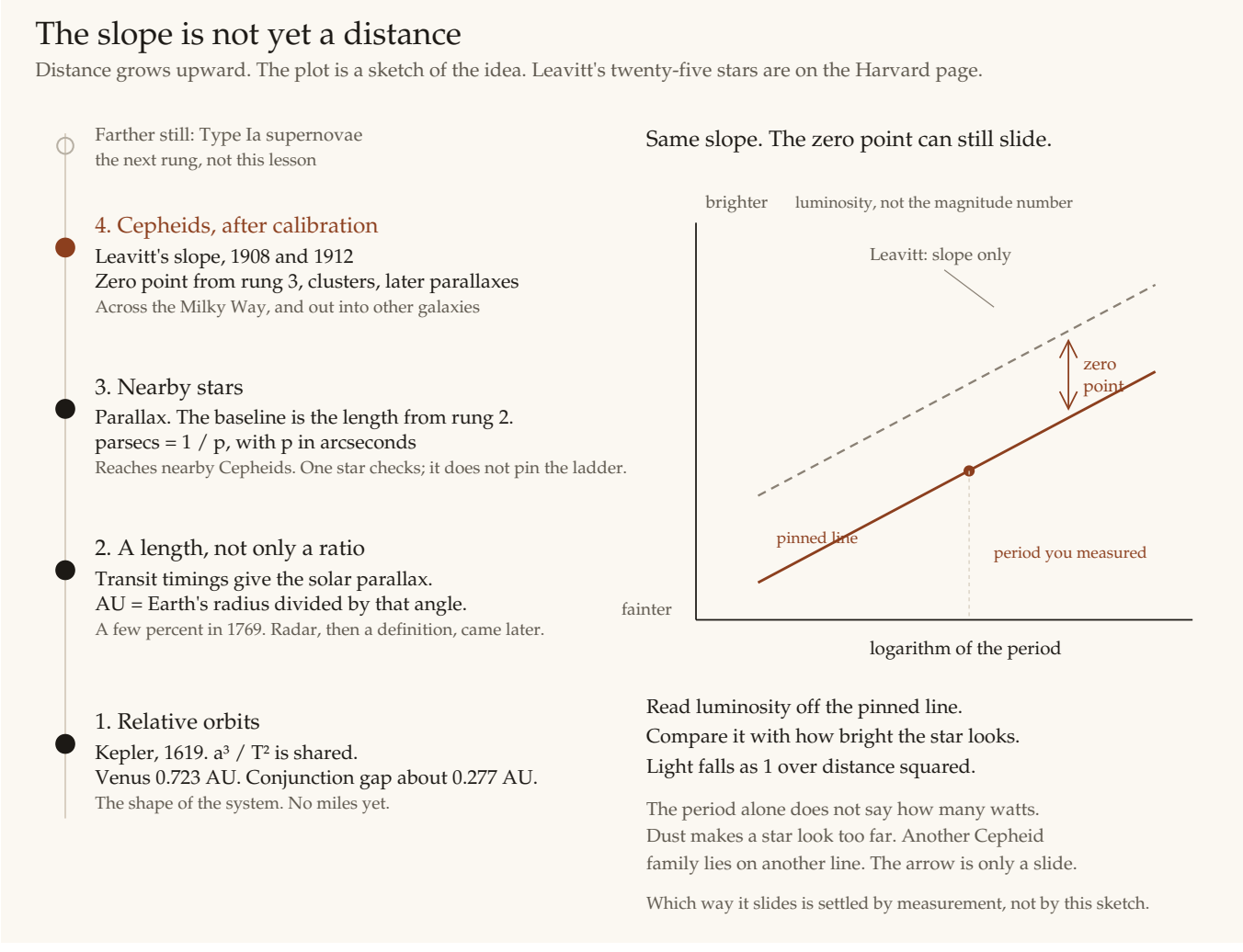
In 1908, in Annals of the Harvard College Observatory, volume 60, Leavitt wrote that it was worthy of notice that **the brighter variables have the longer periods**, and that the sample was still small. In **Harvard College Observatory Circular 173**, 3 March 1912, "prepared by Miss Leavitt" and issued over Pickering's name, she had 25 periods. On a plot of magnitude against the logarithm of the period, "a straight line can readily be drawn." She gave the slope in her own words: the logarithm of the period increases by about **0.48** for each increase of one magnitude in brightness. The average swing from maximum to minimum was about 1.2 magnitudes. And, because they share a distance, "their periods are apparently associated with their actual emission of light."

That is the **slope**. A longer period means a more luminous star, by a definite step on that line.

It is not the **zero point**. The circular is careful. The magnitudes are on a provisional photographic scale. When she says "absolute scale of magnitudes," she means tying those faint comparison stars to a standard sequence, the North Polar Sequence. She does not mean the modern M, which already requires a distance. The closing hope is explicit:

> "It is to be hoped, also, that the parallaxes of some variables of this type may be measured."

In 1912 those parallaxes were out of reach. A classical Cepheid's parallax is a few thousandths of an arcsecond. The period told her which star was brighter than which other star in the Cloud. It did not tell her how far the Cloud was, in parsecs, until someone pinned the line to a luminosity.



Four rungs of the distance ladder, and a sketch of a period-luminosity line whose slope is known while its zero point can still slide

The ladder drawn for this lesson. Distance increases upward. On the graph, brighter means more luminous, upward, so a longer period stands higher. That axis is not a magnitude scale, on which the numbers run the other way. The graph is a sketch of a sliding zero point, not a replot of Leavitt's 25 stars; those are on the Harvard Computers page. The arrow can slide either way. Measurement decides which way.

11. Pinning the line

Ejnar Hertzsprung, in 1913, made the first rough zero point from the motions of galactic Cepheids: a statistical parallax, not a clean trigonometric one. **Harlow Shapley, around 1918**, revised that calibration and used it to size the Milky Way and

to put the Sun off center. The revision was bold and partly wrong. He mixed classical Cepheids, the bright young pulsators of Leavitt's Cloud, with fainter stars of similar light curves in globular clusters (Type II Cepheids, and the RR Lyrae stars then called cluster variables). Interstellar dust was not yet in the sum. The zero point in use came out too faint by about **1.5 magnitudes**.

Edwin Hubble, on plates of 1923, found a Cepheid in the Andromeda "nebula." The result read to the astronomers in 1925 placed Andromeda far outside the Milky Way, on Shapley's scale. The island-universe question was answered. The miles were still short.

Walter Baade, in 1952, separated the two families. A classical, Population I Cepheid is about 1.5 magnitudes brighter than the Population II star of the same period that had been sitting on the same line. A magnitude difference of 1.5 is a distance factor of 10 to the power 1.5/5, which is 10 to the 0.3, which is **about 2**. The galaxies jumped outward. That was not a new slope from the Cloud. It was a new pin.

Dust, if you want the extra care: grains dim a star and redden it. Leave them out and the inverse-square distance is too large. A color excess estimates the dimming. Metallicity is a smaller, real shift: Cloud Cepheids are poorer in heavy elements than Cepheids near the Sun, and the luminosity at a given period depends slightly on that composition. Overtone pulsators, which complete a cycle in a shorter period than the fundamental tone, must not be dropped onto the fundamental line. Polaris is a nearby check and an awkward one. The pin is a crowd of Cepheids with geometric distances, not one star.

The modern pin is geometric. Parallaxes of Milky Way Cepheids, and of Cepheids that belong to open clusters whose distances are also parallaxes, put luminosities on the line directly. Gaia's data releases are the current basis of that work; one 2023 cluster study (Cruz Reyes and Anderson, Astronomy & Astrophysics) is an example of a zero point at the one-percent level, and of the remaining argument over a tiny parallax offset. Hubble Space Telescope parallaxes of Galactic Cepheids did the same job a generation earlier, with fewer stars. The top of the ladder, where Cepheids calibrate Type Ia supernovae and those supernovae measure the expansion, is still debated. The debate is about a few percent and about the next rung. It is not a return to "we have no idea how far."

Cepheids are bright: thousands of times the Sun, and the slow ones brighter still. They are visible across the Milky Way, in the Magellanic Clouds where Leavitt found them, in Andromeda, and, with a telescope in orbit, in galaxies tens of millions of light-years away that also hosted a supernova. The rung is not "inside our galaxy and then a wall." The wall is the next method.

12. The four rungs, said once

Rung	What you use	What you get	What you still do not have
Kepler, 1619	Periods, and a^3 / T^2	a in AU. Venus 0.723. Gap about 0.277 in the circular model	Miles
Transits, 1761 and 1769	Durations or contact times, a known baseline, Earth's radius	The AU as a length, to a few percent; refined for two centuries after	A single transit's exact modern metre
Parallax	The AU as a baseline	parsecs = 1 / p. Nearby stars, including nearby Cepheids	A luminosity for a Cepheid you cannot resolve
Cepheids, calibrated	Leavitt's slope, plus a zero point from parallax and later rungs	Distances across the Milky Way and into other galaxies	The supernova rung beyond them

Today the second rung is a definition plus radar history. The shape of the argument is unchanged. A ratio, then a length, then an angle against the stars, then a clock inside a star whose brightness that angle has taught you.

Try the arithmetic again

1. Venus takes 0.615 year. Square it. You should be near 0.378. What cube root, to three decimals, brings you back?
2. In the circular model, what is 1 minus that cube root?
3. 1 / 0.277 is about 3.61. What is the differential factor, and what angle is 2.61 times 8.79 arcseconds for a side-on baseline of one Earth radius? Is that a ceiling?
4. A star has p = 0.25 arcsecond. How many parsecs? About how many light-years, at 3.26 light-years to the parsec?
5. Two Cepheids have the same apparent brightness. One has twice the period of the other. What else do you need before you can say which is farther?

Review questions

1. What does a shared a^3 / T^2 give you, and what does it refuse to give you?
2. Why is the semimajor axis not the distance today?
3. Where does the 0.277 AU come from, and what three facts make it a model rather than a transit-day measurement?
4. Name the solar parallax, Venus's parallax, and the differential parallax. Which one moves the black spot differently for two observers?

5. What did Halley tell observers to time, and why is that not one angle at one place?
6. Why does Delisle's method need longitude, while a duration mostly cancels a bad absolute clock?
7. Who observed at Tahiti on 3 June 1769, and which 1769 station is the one deep in Siberia? Where was Chappe in 1761, and where was he in 1769?
8. What did Encke publish in 1824, and why is "the transit gave us 92.96 million miles" false?
9. In the parallax drawing, what is p, what is 2p, and why are January and July only examples?
10. What instrument did Bessel use in 1838, and why is a photograph the wrong picture of that measurement?
11. What did the Small Magellanic Cloud do for Leavitt that a scatter of Milky Way Cepheids could not, and what did she still ask for?
12. Why did Baade's 1.5 magnitudes roughly double the distances to galaxies?

Glossary

- **Semimajor axis (a).** Half the long axis of an elliptical orbit. The average of the nearest and farthest distances from the focus. Kepler's "mean distance."
- **Astronomical unit (au).** Since 2012, exactly 149,597,870,700 metres. Historically, the Earth's semimajor axis.
- **Inferior conjunction.** An inner planet between the Earth and the Sun, on the near side. A transit is an inferior conjunction that also crosses the Sun's disk.
- **Solar parallax.** The angle subtended by Earth's equatorial radius at a distance of 1 au. About 8.794 arcseconds.
- **Differential parallax.** Venus's parallax minus the Sun's, the shift of Venus relative to the solar disk. About 2.61 times the solar parallax in the circular model.
- **Projected baseline.** The station-to-station separation as it appears from the Sun, in the direction that changes the chord or the contact time. Not the latitude printed on a map.
- **Internal contacts.** Contact 2, Venus fully onto the disk, and contact 3, about to leave. Halley's duration runs from 2 to 3.
- **Black drop.** The dark neck that makes internal contact uncertain. Mostly optical. Not the same thing as Venus's bright atmospheric rim.
- **Parallax angle p.** The angle at a star subtended by 1 au. The full side-to-side shift is 2p. Distance in parsecs is 1/p if p is in arcseconds.
- **Parsec.** The distance at which 1 au subtends 1 arcsecond. About 3.26 light-years.

- **Heliometer.** A telescope with a split objective, used to measure small angles. Bessel's instrument.
- **Apparent magnitude m.** How bright a star looks.
- **Absolute magnitude M.** How bright it would look from 10 parsecs. A measure of luminosity.
- **Period-luminosity relation.** Longer period, higher luminosity, for a given family of Cepheids. Leavitt found the slope. The zero point is the pin that turns the slope into a distance.
- **Zero point.** Where the line sits vertically: the actual luminosity at a chosen period. Sliding it changes every distance and leaves the slope alone.
- **Standard candle.** An object whose luminosity you know, so apparent brightness gives distance by the inverse-square law.

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- Encke's 1824 solar parallax 8.5776 arcseconds, and the 1835 revision near 8.571, as summarized by Andreas Verdun, Archives des Sciences 57 (2004), and by the historical survey arXiv:1402.1945. Newcomb's later value near 8.79-8.80 arcseconds is from his re-reductions for the American Ephemeris papers.
- IAU Resolution B2 (2012), Beijing: the astronomical unit is exactly 149,597,870,700 metres.
- Bessel, "Bestimmung der Entfernung des 61sten Sterns des Schwans," Astronomische Nachrichten 16 (1838); English in Monthly Notices of the Royal Astronomical Society 4 (1838), 152-161. The 0.3136 and 0.0202 arcseconds, and the 10.3 years, are his. The heliometer is described in the AAS "This Month in Astronomical History," November 2021.
- Leavitt, Annals of the Harvard College Observatory 60 (1908), and Harvard College Observatory Circular 173 (3 March 1912). Quotations from the circular as transcribed by the AIP Center for History of Physics. The 25-star figure is on the Harvard Computers page.
- Walter Baade's 1952 separation of the two Cepheid populations, as told by Allan Sandage, "The First 50 Years at Palomar," and the standard factor: 1.5 magnitudes, about twice the distance. Hubble's Andromeda Cepheid is the 1923-25 result.
- Cruz Reyes and Anderson, "A 0.9% calibration of the Galactic Cepheid luminosity scale based on Gaia DR3," Astronomy & Astrophysics (2023), as one modern geometric pin, not as the last word.

Key facts

- Kepler, 1619: a^3 / T^2 is shared. a is the semimajor axis, the average of nearest and farthest, not the distance today.
- Venus: period 224.701 days = 0.615 year, so a = 0.723 AU. Circular coplanar gap at inferior conjunction: 1 - 0.723 = 0.277 AU. Real conjunctions are near that, not identical to it.
- Greatest elongation near 46 degrees already gave sin 46 = 0.72 by geometry. The third law extends one scale to every planet.

- The transit measures a differential parallax, about 2.61 times the solar parallax in the simple model, using a projected baseline. Halley compared durations of internal contacts. Delisle compared one contact in absolute time and needed longitude. Not one angle, and not latitude alone.
- Halley's 12.5 arcseconds, 43 arcseconds, and 17 minutes are his 1716 trial, not results. About 23 arcseconds is the illustrative shift for a side-on baseline of one Earth radius and the modern 8.79 arcsecond solar parallax. A baseline can be longer, up to about a diameter. It is not a logbook entry and not a ceiling.
- 1761: Chappe at Tobolsk, Siberia, 6 June. Mason and Dixon at the Cape. 1769: Cook, Green, and Solander at Point Venus, Tahiti, 3 June; Isleniev at Yakutsk, the deep-Siberian station, in Rumovsky's wider Russian network. Hell was at Vardo, not Siberia. Chappe's 1769 station was Baja California.
- The black drop spoiled contact times. It is mostly telescope blur and the Sun's dim limb. Venus's atmosphere is a separate bright rim. Mercury shows a black drop too.
- Encke 1824: 8.5776 arcseconds, about 2.5 percent too small, so the Sun too far. No one transit delivered the modern au.
- Since IAU 2012 Resolution B2, 1 au = 149,597,870,700 m exactly, about 92.96 million miles. Earth's actual distance still varies from about 0.983 to 1.017 au.
- Stellar parallax: full opposite-orbit shift is 2p; the defined p is the angle of 1 AU; parsecs = 1/p in arcseconds. January and July are illustrative. Bessel, 1838, 61 Cygni, heliometer, 0.3136 arcsecond. Photography of stars begins later (Vega, 1850).
- Leavitt 1908 and Circular 173 (1912) fixed the slope from SMC stars at one distance. She asked for parallaxes. She did not set the zero point. Hertzsprung and Shapley tried; Baade in 1952 separated two Cepheid families and roughly doubled the extragalactic distances. Gaia-era parallaxes pin the luminosity now. Cepheids reach across the Milky Way and into other galaxies.

Answers

Arithmetic. (1) 0.723 AU. (2) 0.277 AU. (3) The factor is 2.61. 2.61 times 8.79 is about 23 arcseconds for a side-on baseline of one Earth radius. That is an illustration, not a measurement, and not a ceiling: stations can be up to about a diameter apart. (4) 4 parsecs, about 13 light-years. (5) The calibrated period-luminosity relation, so you know which period is the more luminous. Equal apparent brightness plus a known luminosity ratio gives the distance ratio. The period ratio alone does not.

Review. (1) Relative distances in AU. Not miles, and not the distance on a given day. (2) It is half the long axis, the average of closest and farthest. A point on the ellipse lies at some other distance from the focus. (3) From 1 minus 0.723 in a circular, flat model. Real orbits are ellipses, tilted by about 3.4 degrees, and a conjunction is a shared

longitude, so the gap that week is computed from the elements. (4) Solar parallax: Earth's radius at 1 au, about 8.794 arcseconds. Venus's parallax: the same baseline at Venus's distance, about 3.61 times larger in the model. Differential: the difference, about 2.61 times the solar parallax. The differential is what shifts Venus on the disk. (5) The duration from internal contact 2 to internal contact 3. The difference of durations is a difference of chords, which is a differential parallax across a known projected baseline. (6) One contact has to be timed on a shared absolute clock, so the longitudes must be known. A duration subtracts two times on one local clock, so an overall clock error mostly cancels. The places still have to be known. (7) Tahiti: Cook, Green, and Solander, 3 June 1769. Deep Siberia in 1769: Isleniev at Yakutsk. Chappe was at Tobolsk in 1761 and at San Jose del Cabo in 1769. (8) 8.5776 arcseconds. It is about 2.5 percent smaller than the modern solar parallax, and the modern mile-length is a 2012 definition, 149,597,870,700 metres, after radar. The expeditions got the scale to a few percent and did not finish it. (9) p is the angle subtended by 1 AU. The full shift from one side of the orbit to the other is 2p. January and July fit the star on the diagram. Another ecliptic longitude wants another pair, still about six months apart. (10) A Fraunhofer heliometer at Konigsberg. The first stellar photograph, Vega, is 1850. Plates are a later way to explain the same angle. (11) A common distance turns apparent-brightness differences into luminosity differences, so period can be matched to luminosity without already knowing the miles. She still asked that parallaxes of brighter variables of the type be measured. That request is the missing zero point. (12) 1.5 magnitudes is a factor of 10 to the power 0.3, about 2, in distance. The galaxies were about twice as far as the old pin had said.

Saturday, November 7, 2026

Aristarchus: a moving Earth and a measurable Moon

Archimedes says Aristarchus put the Earth in orbit. The surviving book is a different job: a quarter-moon triangle and a lunar eclipse, neither of which prints the modern distance to the Moon.

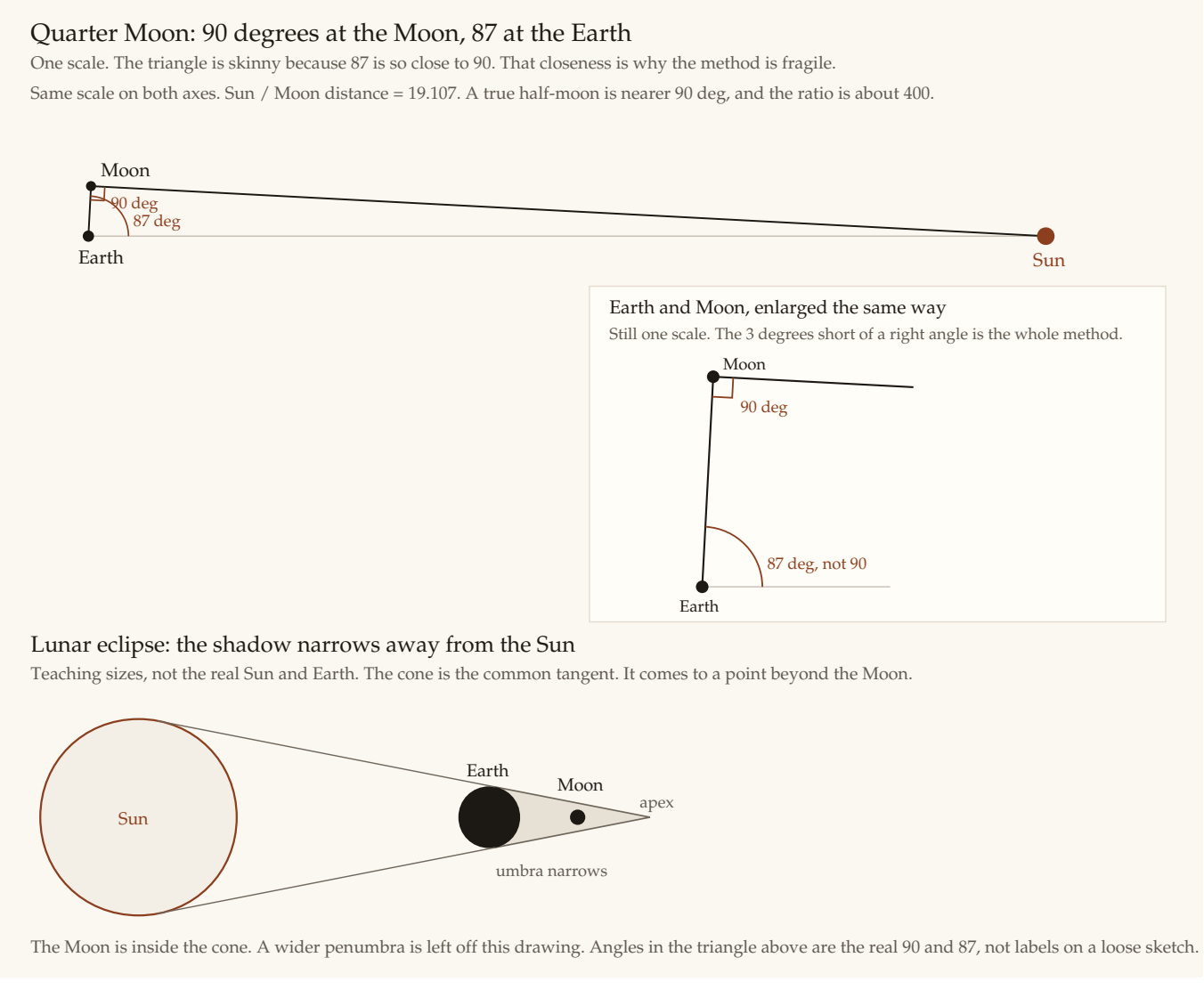
Aristarchus of Samos, about 310 to 230 BCE, is the first Greek we can name who set the Earth moving around the Sun, and turning once a day, with the stars so distant that the year shows no shift. We know that proposal because Archimedes reports it. The book itself is lost. The book we still have, On the Sizes and Distances of the Sun and Moon, does geometry. It is not the heliocentric treatise, and it should not be asked to prove one.

Three claims get fused in retellings. They are different.

First, the heliocentric idea, attested, not surviving in his own diagrams.

Second, the quarter Moon. At half phase, the angle at the Moon, between Sun, Moon, and Earth, is a right angle. Measure how far the Moon stands from the Sun in the sky, and the triangle gives the Sun's distance in units of the Moon's. Aristarchus took that elongation as 87 degrees and found a Sun about 18 to 20 times farther than the Moon. The real ratio is about 400. Half-moon is so close to 90 degrees from the Sun that a small mistake in "exactly half" wrecks the answer. The diagram draws 87 degrees so you can see the right angle. The sky is less obliging.

Third, a lunar eclipse. The Moon crosses Earth's shadow. That shadow is a cone that tapers, because the Sun is a wide disk, not a point of light. You need the Moon's angular size and the Sun's geometry. The width of the dark patch, alone, is not a distance.



The quarter-moon triangle and a tapering eclipse shadow

The triangle is one scale: 90 degrees at the Moon, 87 at the Earth, Sun 19.107 times the Moon's distance. The eclipse uses teaching sizes. The true half-moon is nearer 90 degrees, and the ratio is about 400.

Do not hand Aristarchus the modern "about 60 Earth radii" as if his book printed it and got it right. Reconstructions of the angles he adopted are shaky. What is solid is the distinction he forces on us: a physical width, an angle, and a distance are three different things, and a triangle can connect them. No calculus is required. Hipparchus, next, uses a baseline on the Earth and other people's eclipse reports. A thumb at arm's length, much later in this day's lessons, is the same triangle with worse numbers and an assumed distance.

The illustrated lesson is download the PDF.

Sources

- MacTutor, biography of Aristarchus: https://mathshistory.st-andrews.ac.uk/Biographies/Aristarchus/
- MacTutor, on the sizes and distances: https://mathshistory.st-andrews.ac.uk/BEA/aristachus_bea.pdf

Key facts

- The heliocentric proposal is known from Archimedes. It is not the surviving Sizes and Distances.
- The quarter-moon triangle, with an assumed 87-degree elongation, gave a Sun only about 18 to 20 times farther than the Moon. The real ratio is about 400.
- A lunar eclipse needs the taper of Earth's shadow and the Moon's angular size. Shadow width alone is not the distance.
- The modern Moon, near 60 Earth radii, is not a figure to attribute to Aristarchus as a successful original result.

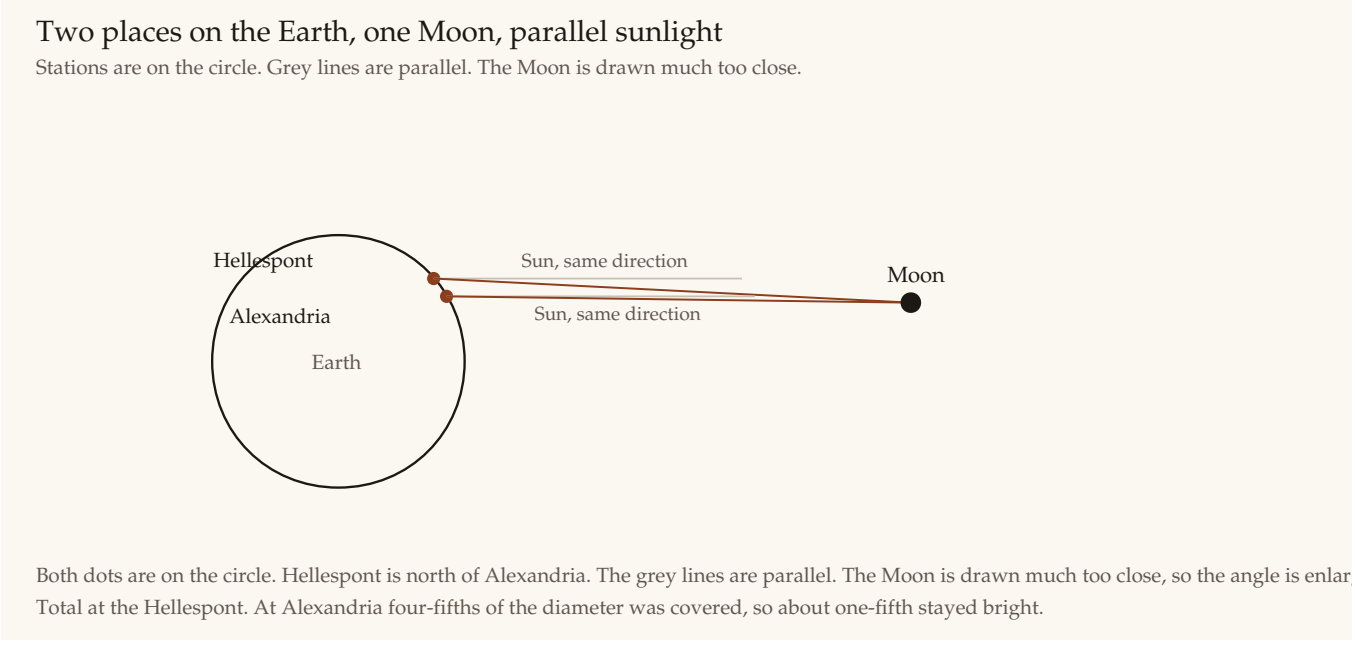
Saturday, November 7, 2026

Hipparchus: the Moon changes place when we move

Hipparchus used eclipse reports from two latitudes, not a double journey of his own. The Moon's parallax against a shared solar direction gave a distance with wide edges, near the modern value and not a lucky single number.

Hold a finger at arm's length and look with one eye, then the other. The finger jumps against the far wall. The gap between your eyes is the baseline. The angle of the jump, with that baseline, is a distance. That is parallax. Hipparchus, in the second century BCE, used it on the Moon. Most of his books are lost. Later astronomers rebuilt the argument from what they still had.

One famous piece uses a solar eclipse. Near the Hellespont the Sun's disk was totally obscured. At Alexandria four-fifths of that disk was obscured, so only about one-fifth of the Sun's diameter stayed uncovered. The whole shift between the two places was that fifth of the diameter. It is a fraction of the diameter, not of the area: four-fifths of the area would be a different, larger bite. A date often proposed is 14 March 190 BCE. The identification is not secure, and this lesson will not pretend it is. Hipparchus used reports. He did not stand in both cities.



Two stations, one Moon, a shared solar direction

The angle is drawn far too large. The Sun's own parallax is treated as negligible beside the Moon's. Drawn for this lesson.

The baseline is the north-south distance on the Earth between those latitudes. The sightlines to the Moon differ. The Sun is so far that its direction can be treated as the shared wall. G. J. Toomer's reading of the reconstruction gives one lower bound near 71 Earth radii. Other ways of treating the shadow and the Sun's distance give a band roughly 59 to 67 Earth radii. That band is not a modern confidence interval. The Moon really averages near 60 Earth radii, and the distance changes a little through the month. An ancient model can land close to the truth and still be a model with wide edges. Nearness is not the same thing as a tight measurement.

Compare the sticks. The Venus stations of 1761 and 1769 stand on the Earth's surface, a baseline of hundreds or thousands of miles. Kepler's Mars triangles, later in this day's lessons, use a baseline the size of Earth's orbit. Hipparchus is the same idea with the cities of two reports. Aristarchus, just before this page, was comparing sizes inside one triangle. Here the observer moves, and the Moon moves on the sky.

The illustrated lesson is download the PDF.

Sources

- MacTutor, biography of Hipparchus: https://mathshistory.st-andrews.ac.uk/Biographies/Hipparchus/
- MacTutor, Dictionary of Scientific Biography excerpt: https://mathshistory.st-andrews.ac.uk/DSB/Hipparchus.pdf

Key facts

- Parallax is a known baseline and an angle. Hipparchus's own books are mostly lost and were rebuilt by later writers.
- The eclipse was total at the Hellespont. At Alexandria four-fifths of the Sun's diameter was covered, so about one-fifth of the diameter stayed bright. That fifth is the parallax shift. It is not four-fifths of the area. He used reports. The 190 BCE date is uncertain.
- Reconstructions give a lower bound near 71 Earth radii in one reading, and roughly 59 to 67 in others. That is not one confidence interval.
- The modern mean is near 60 Earth radii and varies. Venus transits use a surface baseline. Kepler's Mars work uses the orbit.

Saturday, November 7, 2026

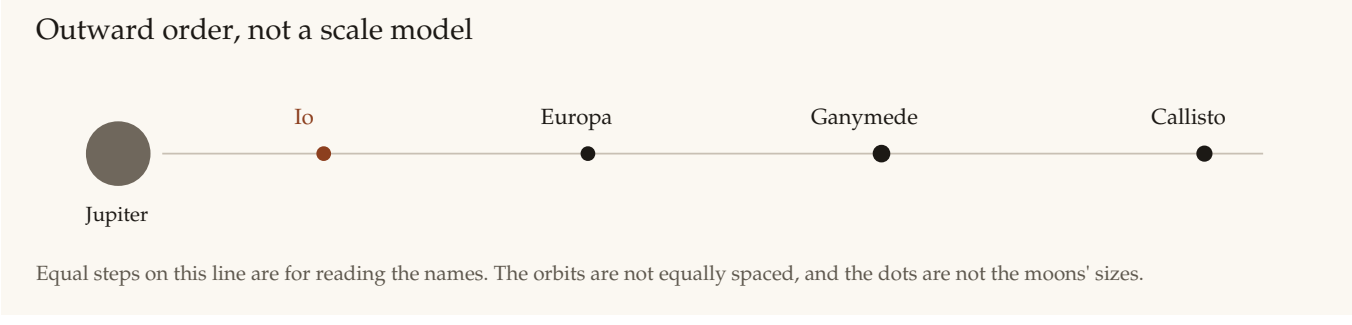
Galileo's moons: new centers and lasting names

In January 1610 Galileo found four bodies circling Jupiter and called them the Medicean Stars. The names Io, Europa, Ganymede, and Callisto come from Marius in 1614, with a calendar trap in the priority story.

On 7 January 1610 Galileo saw points of light beside Jupiter, lined up east and west and changing place from night to night. He came to understand four bodies going around Jupiter. Sidereus Nuncius, in March 1610, told Europe. The fuller story of his telescope and his trial is the Galileo page. This page is only the moons, and the names.

They matter because not every moving body has to orbit the Earth. One old objection to Copernicus was that a traveling Earth would leave its Moon behind. Jupiter travels, in any of the systems then argued, and it keeps four companions. That weakens the objection. It does not finish the case. Tycho Brahe's system, Sun around the Earth and the other planets around the Sun, can hold moons around Jupiter just as well. New centers of motion are real. They are not, by themselves, a proof of a moving Earth.

Galileo named the four, together, the Medicean Stars, for the Medici who paid the patronage. He did not give them the names on a modern chart. Those names, Io, Europa, Ganymede, and Callisto, in that order outward, are Simon Marius's, in Mundus Iovialis of 1614. Marius said Kepler suggested them.



The four moons in outward order, not to scale

A reading order, drawn for this lesson. The gaps are equal so the names can be read. The orbits are not equally spaced, and the dots are not the moons' sizes.

Marius also said he first saw the moons on 29 December 1609. That date is Julian. On the Gregorian calendar it is 8 January 1610, the day after Galileo's 7 January. Do not treat the two calendars as if they were the same day, and do not call Marius a fraud because the numbers sit close. Who published a dated book first is one question. Who

looked, independently, is another. The quarrel is real. A collapsed calendar is not evidence.

Io, the inner one, later became a clock. When it enters Jupiter's shadow the reflected sunlight stops, and the timetable of those eclipses taught Rømer that light takes time. That is the next lesson.

The illustrated lesson is download the PDF.

Sources

- Museo Galileo, on Jupiter's moons: https://catalogue.museogalileo.it/indepth/JupitersMoonsSatellites.html
- Museo Galileo, Sidereus Nuncius (1610): https://www2.museogalileo.it/it/biblioteca-e-istituto-di-ricerca/progetti/incontra-galileo/31-opere/454-sidereus-nuncius-1610.html
- Pasachoff, "Simon Marius's Mundus Iovialis and the discovery of the moons of Jupiter," IAU proceedings, 2018: https://ui.adsabs.harvard.edu/abs/2018iau3.book...38P/abstract

Key facts

- Galileo recorded moons of Jupiter from 7 January 1610 and published them in March 1610 as the Medicean Stars.
- Moons around Jupiter answer one objection to a moving Earth. Tycho's system can include them too. They are not a complete proof of Copernicus.
- Io, Europa, Ganymede, and Callisto are Marius's 1614 names, which he credited to a suggestion from Kepler.
- Marius's 29 December 1609 is Julian. Gregorian, it is 8 January 1610. Do not equate the calendars or assume fraud from the numbers.

Saturday, November 7, 2026

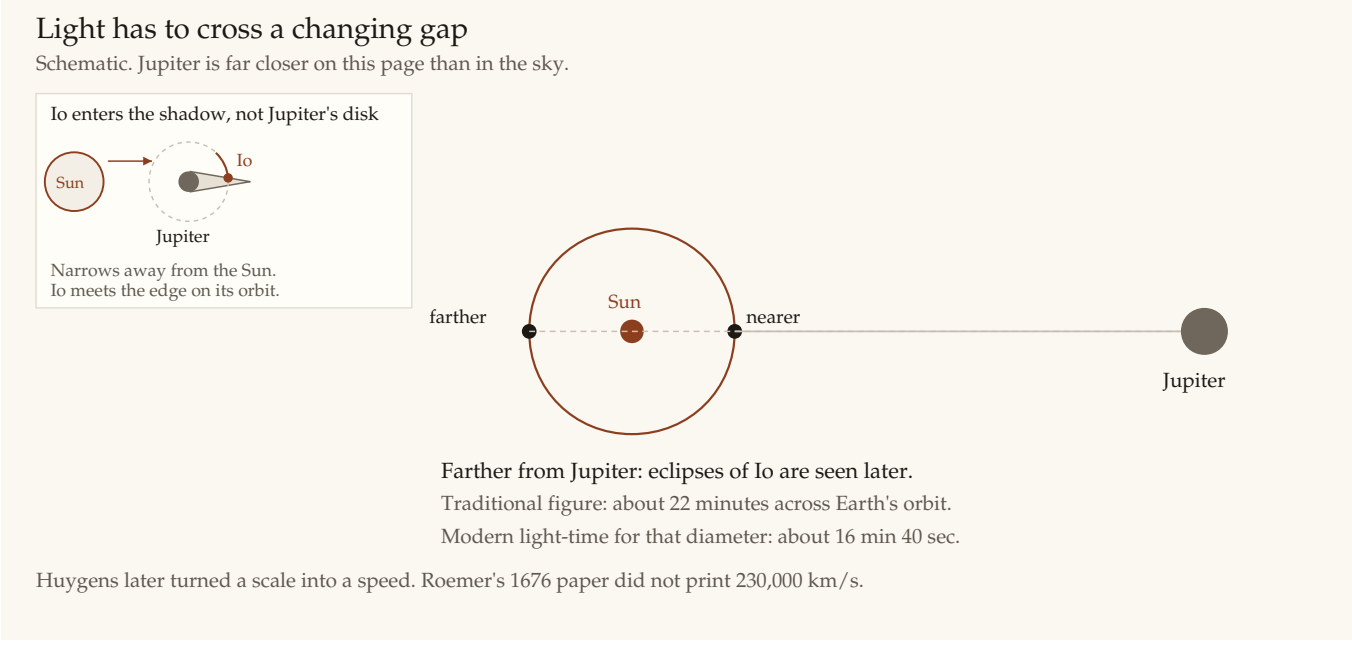
Romer and Io: light needs time

In 1676 Romer argued that Io's eclipses run late or early because light takes time to cross a changing distance. He did not publish 230,000 kilometers a second. Huygens made a speed later, from an orbital scale.

Io goes around Jupiter in about 42.5 hours. We see it by sunlight reflected from its surface. When Io enters Jupiter's shadow, that sunlight is cut off. The moon has not switched off a lamp of its own. The disappearances and returns are eclipses, and they can be timed. The names and the order of the four moons are the previous lesson.

Giovanni Cassini had been building tables of those times. Ole Romer, working with that program in Paris, argued in 1676 that the tables made sense if light travels at a finite speed. Earth moves around the Sun once a year. When Earth is traveling away from Jupiter, each eclipse is seen a little later, because the light has a longer trip. When Earth is traveling toward Jupiter, the events come early. The extra minutes are the light-time of the extra distance.

Jupiter moves too, and Io's orbit is not a perfect clock. Many nights are required before the leftover can be blamed on light rather than on the tables. The traditional round number is about 22 minutes for light to cross the diameter of Earth's orbit. The modern light-time for that same diameter is about 16 minutes 40 seconds.



Io's shadow eclipse, and Earth near and far from Jupiter

A schematic, drawn for this lesson. Jupiter is much farther, and it moves. The inset is the local event: reflected light cut off by the shadow.

Romer did not publish a speed of 230,000 kilometers a second. Christiaan Huygens, in the Treatise on Light of 1690, converted an orbital scale into a speed that comes out near that figure in modern units. The defined speed of light is now 299,792.458 kilometers a second. A conversion is not a sentence Romer printed.

You also cannot pick one opposition and one conjunction and call them a clean pair. Near conjunction, Jupiter sits in the Sun's glare and Io's eclipses cannot be watched. The argument is a stack of dates. The drawing shows the idea, not the observing calendar.

The illustrated lesson is download the PDF.

Sources

- Paris Observatory, "340 years ago, Romer showed that light has a finite speed": https://observatoiredeparis.psl.eu/340-years-ago-romer-showed.html
- Observatoire de Paris, CLEA / CLIC notes on Romer: https://clic.obspm.fr/

Key facts

- Io's eclipses are the cutting off of reflected sunlight in Jupiter's shadow, on a clock of about 42.5 hours. Io does not emit the light.
- Romer in 1676 explained early and late eclipses by a finite speed of light as Earth moved nearer or farther.
- The traditional 22 minutes across Earth's orbit compares with a modern 16 minutes 40 seconds. Jupiter's motion and Io's orbit also affect the times.
- Huygens, not Romer, produced the speed later quoted near 230,000 km/s. The defined value is 299,792.458 km/s.

Saturday, November 7, 2026

Kepler's Mars triangles, and an honest scale check

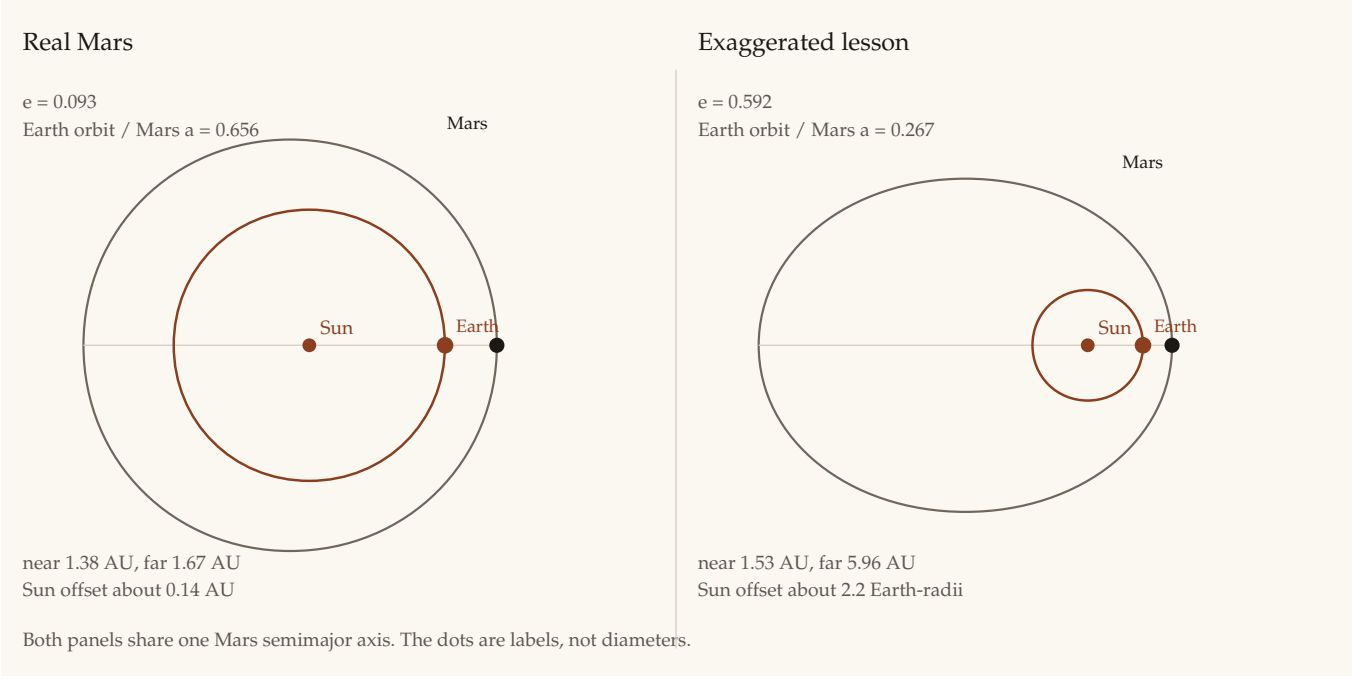
Kepler placed Mars by crossing lines of sight from a moving Earth, many times, until the path was an ellipse. An early teaching drawing of that ellipse was exaggerated on purpose. This page shows the real shape beside it.

Kepler did not guess Mars's orbit from one night. Mars returns to the same place in its path about every 687 days. Earth has moved. If you know where Earth is, two lines of sight to that same Martian place cross, and you have a point in space. Angles measured against the stars give the directions. One angle plus a guessed radius is not enough. Kepler also had to repair Earth's own orbit before the crossings would sit still. Many pairs, not one clever triangle, drew the curve. It is an ellipse, with the Sun at a focus.

Opposition and conjunction are easy to mix up with a measurement. At opposition the order is Sun, Earth, Mars, with Earth in the middle. Mars is up all night and near its brightest. At conjunction the order is Earth, Sun, Mars. Mars is lost in the glare. Nothing is eclipsed. A straight alignment is not, by itself, a distance. Hipparchus's lesson uses a short baseline on the Earth. Here the baseline is Earth's whole orbit, the same stick that, once it has a length in miles, carries the transit lesson outward.

An early illustrated sheet of this ellipse, made to be obvious in a small picture, stretched the truth and said so in a caption. It was not a printer's accident. On that sheet Mars's semiaxes were 247 and 199 drawing units, and Earth's orbital radius was 66. The ratio is about 3.74, against the real ratio of semimajor axes, about 1.524. The drawing is about 2.46 times too stretched. Its eccentricity is about 0.59, against Mars's real 0.093, roughly six times too oval. The Sun sits about 2.2 of those Earth-radii from the center. In the sky the offset is about 0.14 AU. If you treat the drawn Earth radius as 1 AU, the sheet's nearest and farthest distances are about 1.53 and 5.96 AU. The real ones are about 1.38 and 1.67 AU.

Earth's orbital radius ought to be about 66 percent of Mars's semimajor axis, not about 27 percent. That is why Earth looks tiny on the old panel. The tininess is the exaggeration. It is the wrong reason for the eye to think Earth is small. The planet dots on that sheet were 14 and 12 units across, a ratio near 1.17. Mars's diameter is about 6,792 kilometers and Earth's about 12,756, a ratio near 0.53. Dots are names, not worlds.



Real Mars beside the exaggerated teaching ellipse

Both panels use the same Mars semimajor axis, and both put the Sun at the focus. The right-hand ellipse is the exaggerated lesson, labeled as such. The dots are not the planets' physical sizes.

The illustrated lesson is download the PDF.

Sources

- Steven Austin, notes on Kepler's Mars orbit: https://faculty.uca.edu/saustin/3110/mars.pdf
- Physics Today, on Kepler and Mars: https://physicstoday.aip.org/features/the-great-martian-catastrophe-and-how-kepler-fixed-it
- NASA Mars fact sheet: https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html
- NASA, Mars at opposition: https://science.nasa.gov/asset/hubble/close-encounter-mars-at-opposition/
- NASA, geometry of a Mars conjunction: https://science.nasa.gov/resource/geometry-of-mars-solar-conjunction/

Key facts

- A repeated heliocentric place of Mars, seen from two known positions of Earth, fixes a point. Many points make the ellipse. One angle is not enough.
- Opposition puts Earth between the Sun and Mars. Conjunction hides Mars in the Sun's glare. Neither alignment is a distance by itself.

- The exaggerated sheet has orbital ratio about 3.74 instead of 1.52, and eccentricity about 0.59 instead of 0.093. Earth's orbit should be about 66 percent of Mars's semimajor axis, not 27 percent. The dots are not sizes: Mars is about half Earth's diameter.

Saturday, November 7, 2026

Thumb to Moon: calculating the Moon's diameter

A thumb can hide the Moon because apparent size depends on distance. Half a degree at an assumed 240,000 miles is about 2,100 miles if you divide the whole circle by 720, close to the Moon's real 2,160-mile diameter.

A little window on a large Moon

Let us put the arithmetic aside for a moment. Imagine the Moon as a distant coin. It has a real width, measured in miles, and an apparent width: how much of your view it occupies. A nearby thumb can hide that enormous object because apparent width depends on distance as well as size.

Close one eye and hold your thumb at arm’s length. For a rough picture, think of a thumb about two centimeters—roughly three-quarters of an inch—wide, held about sixty centimeters, or two feet, from your eye. It covers about two degrees. Your own thumb and arm may differ; this is a picture, not a precision instrument. Never use this exercise to look at the Sun.

The Moon spans about half a degree. That is roughly one-quarter of our two-degree thumb width. So a small object covering the Moon at a distance of two feet would be only about a fifth of an inch wide: approximately 0.21 inches, or 5.3 millimeters, at 24 inches.

A correction worth keeping

A TWO-INCH-wide thumb held 24 inches away would cover about 4.77 degrees, not two degrees. Centimeters and inches cannot be exchanged without changing the answer. The rough two-degree example above uses two centimeters, not two inches.

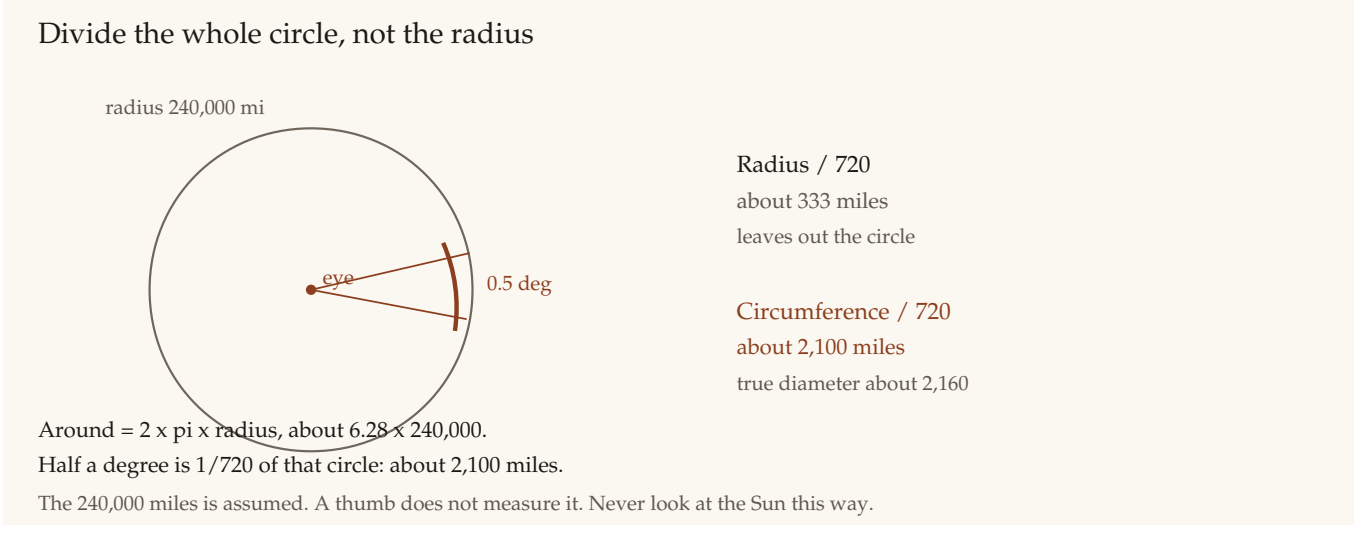
The rule behind the picture

For the SAME apparent width, an object twice as far away must be twice as wide. Imagine sliding a small coin farther from your eye while enlarging it just enough to keep its edges on the same two sightlines. The triangle grows; its shape stays the same. Moving your thumb does not change the Moon. It changes only how much of your view the thumb covers.

From your arm to the Moon

We must already know the distance before we can calculate the Moon’s physical width. Take a rounded distance of 240,000 miles. A mile contains 5,280 feet, so that is 1,267,200,000 feet. Divide by our two-foot arm length: the Moon is about 633,600,000 arm lengths away.

Enlarge the 0.21-inch Moon-covering width by that same factor. The units look cumbersome, but the thought is simple: the same narrow view reaches a much wider object farther away. Converting the enlarged width back into miles gives about 2,100 miles.



A measuring circle: half a degree is one part in 720 of the circumference

Teaching circle centered on the eye, not the Moon's orbit. Drawn for this page from the lesson's own arithmetic. The distance is assumed.

An easier route: go around a circle

Imagine an enormous circle centered on your eye, with a radius reaching to the Moon: 240,000 miles. This is an imaginary measuring circle, not a claim about the Moon’s orbit. A full turn contains 360 degrees; half a degree is one part in 720.

We must divide the WHOLE circumference by 720—not the radius. The distance around a circle is about 6.28 times its radius. That factor is twice pi; pi is approximately 3.14. Our circle is therefore about 1,508,000 miles around.

Diameter ≈ 240,000 × 6.28 ÷ 720 ≈ 2,100 miles.

The little arc covering half a degree is almost the same length as the straight width across it. At this tiny angle, the difference is negligible for our rough inputs. Dividing the distance alone by 720 would give about 333 miles and omit the essential distance-around-the-circle step.

Putting the Moon's width on a US map

The rounded 2,100 miles is a diameter. On the Earth that is a circle 1,050 miles in radius. The map sets that circle on the United States as a best fit to four real cities: Chicago, Boston, Miami, and Denver. The cities stay put. At this diameter the circle cannot pass through all four. Chicago sits about 272 miles inside, Boston about 327 miles outside, Miami about 267 miles inside, and Denver about 89 miles inside.

This is a best fit, not an exact match, and not a literal footprint of the Moon pressed onto the curved Earth. The lesson's 2,100 miles is the rounded figure. The Moon's real mean diameter is about 2,160 miles. The city dots are place markers from Natural Earth, which is in the public domain. They are not survey monuments.

Best-fit circle 2100 miles across on a US map, with Chicago, Boston, Miami, and Denver

A 1,050-mile geodesic circle, drawn on an azimuthal projection centered so distances from the middle stay true. Chicago, Miami, and Denver fall inside. Boston falls outside. The fit is the best one of this size, not a circle through every city.

The map itself, already reviewed, is The Moon's width on a US map (PDF).

Does the answer make sense?

Yes. The Moon’s known mean diameter is about 2,160 miles, or 3,475 kilometers. Our rounded calculation gives about 2,100 miles, roughly 3,370 kilometers: a good approximation, not an exact measurement. Apparent size changes with lunar distance, and our thumb and arm are rough guides.

We have not discovered the distance with a thumb. We assumed it. Angular width PLUS distance gives physical width. Ancient eclipse geometry offers a different route to size and distance relationships; that is a separate lesson.

SOURCE

NASA Science, Moon Facts: mean distance 238,855 miles (384,400 km); rounded radius 1,080 miles (1,740 km). Geometry and illustrative calculations in this lesson are modern derivations, not claims about an exact ancient measurement. https://science.nasa.gov/moon/facts/

The printed lesson, reviewed and kept byte for byte, is Thumb to Moon (PDF).

Key facts

- A thumb about 2 cm wide at 60 cm covers about 2 degrees. A 2-inch thumb at 24 inches covers about 4.77 degrees. The units are not interchangeable.
- Half a degree at an assumed 240,000 miles is about 2,100 miles if you divide the circumference by 720. Dividing the distance alone by 720 gives about 333 miles and drops the factor of about 6.28.
- The Moon's mean diameter is about 2,160 miles. The thumb does not measure the distance. It uses a distance you already assumed.
- On a US map, a 2,100-mile circle fitted to Chicago, Boston, Miami, and Denver does not pass through all four. Chicago is about 272 miles inside, Boston about 327 outside, Miami about 267 inside, Denver about 89 inside.
- NASA's figures: mean distance 238,855 miles, radius about 1,080 miles. https://science.nasa.gov/moon/facts/

Saturday, November 7, 2026

Time to longitude: converting hours, minutes, and seconds to degrees

Earth turns once in twenty-four hours of solar time, which is 360 degrees. Two hours and thirty-five minutes behind Greenwich is 38 degrees 45 arcminutes west, or 38.75 degrees, and -38.75 degrees in the common signed convention.

Earth turns once relative to the mean Sun in twenty-four hours of solar time. That full turn is three hundred sixty degrees of longitude. Divide evenly and each hour of solar-time difference corresponds to fifteen degrees of longitude: two hours, thirty degrees. We work with solar time here, not with the slightly shorter sidereal day of a pure spin against the stars.

Keep the two unit systems apart. Time uses hours, minutes, and seconds. Angle uses degrees, arcminutes, and arcseconds. One degree equals sixty arcminutes. One arcminute equals sixty arcseconds. One hour of time equals sixty minutes. One minute equals sixty seconds. The bridge between them is simple. One hour of time equals fifteen degrees of angle. One minute of time equals fifteen arcminutes. One second of time equals fifteen arcseconds of angle.

Longitude difference in degrees is 15 times (hours + minutes/60 + seconds/3600).

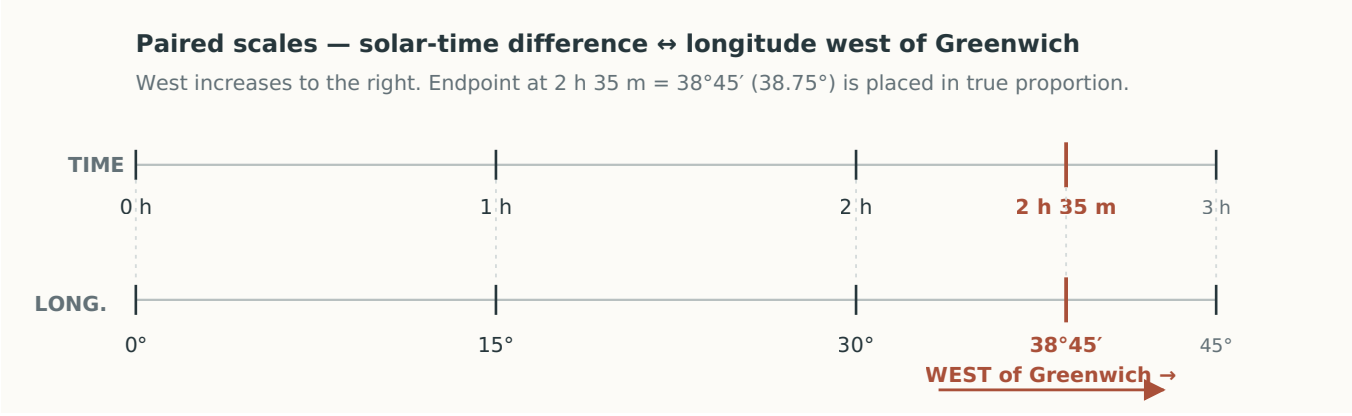
Suppose local solar time is two hours thirty-five minutes behind the time at Greenwich, compared at the same instant.

Two hours of time are 2 times 15 degrees, which is 30 degrees. Thirty-five minutes of time are 35 times 15 arcminutes, which is 525 arcminutes. Divide by 60: 8 degrees, with 45 arcminutes left over, so 8 degrees 45 arcminutes. Add them. 30 degrees plus 8 degrees 45 arcminutes is 38 degrees 45 arcminutes 00 arcseconds west of Greenwich. As a decimal that is 38.75 degrees west. In the common signed convention, east positive and west negative, it is -38.75 degrees.

If local solar time is behind Greenwich at the same instant, you are west. If local solar time is ahead, you are east. Picture Greenwich at noon while a western place still shows 9:25. That is 2 hours 35 minutes behind, hence 38 degrees 45 arcminutes west. Always name the reference meridian. Today Greenwich is usual. Chappe's published result, for example, used Paris, which is a different zero and is told in the Venus lesson.

This is a solar-time conversion, not a way to read exact longitude from today's time-zone label. Civil and political zones, and daylight-saving shifts, are administrative clocks, not precise meridians.

Coordinates may be written both ways. Computers commonly work with decimal degrees. People often read degrees, minutes, and seconds. GPS and mapping apps can display either. Neither form belongs only to the machine or only to the reader.



Paired scales: solar-time difference and longitude west of Greenwich

Time difference and longitude march together. 0 hours is 0 degrees, 1 hour is 15 degrees, 2 hours is 30 degrees, and 2 hours 35 minutes is 38 degrees 45 arcminutes west, placed in true proportion. The reference meridian is Greenwich.

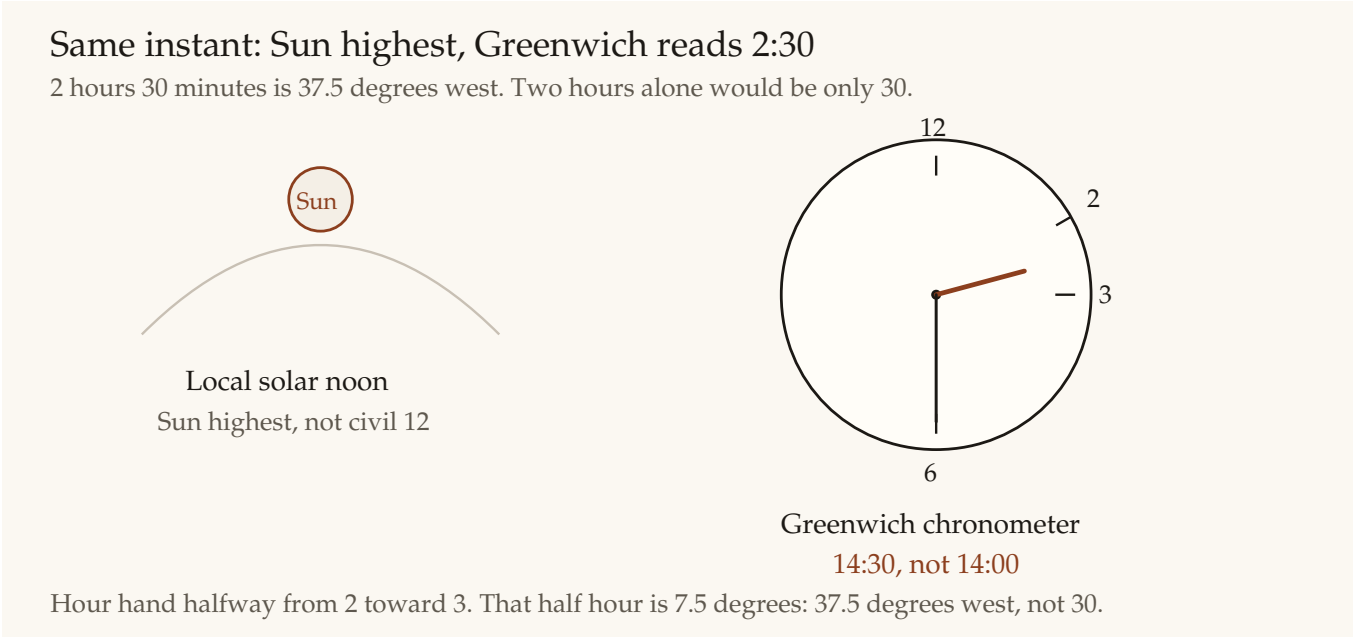
If local solar time is one hour behind Greenwich, the longitude that suggests is fifteen degrees west of Greenwich.

The illustrated lesson is download the PDF.

Longitude with a clock, in plain language

Before you sail, set an accurate chronometer to Greenwich time and do not reset it at sea. It carries Greenwich with you. It does not count the hours you have been under way.

At sea, watch the Sun. Local noon is when the Sun stands highest, not when a civil clock says 12. Read the chronometer then.



Same instant: the Sun at its highest, and a Greenwich chronometer at 14:30 (2:30 p.m.)

Local noon is the Sun's highest point. The hour hand sits between the 2 and the 3. The minute hand points at the 6.

If it reads 14:30 (2:30 p.m.), Greenwich is 2 hours 30 minutes ahead, so you are west. Each hour is 15 degrees. Two hours are 30 degrees. The half hour adds 7.5 degrees. So 2.5 times 15 is 37.5 degrees west, 37 1/2 degrees, not 30. That angle is your longitude. A reading of 10:00 would put you 30 degrees east. The 2 hours 35 minutes case higher on this page is different again: 38 degrees 45 arcminutes west.

The 15 degrees per hour does not depend on latitude. The miles do. Parallels shrink toward the poles, so the same angle covers less ground. The clock method is not limited to the tropics, and it still works at every latitude where you can see the Sun.

Near the poles the hard part is the Sun, not the clock. The highest point is flatter, and in polar winter the Sun may not rise for weeks. At the pole every meridian meets, so longitude has no meaning.

The example skips the equation of time, the seasonal gap between the sundial and the clock, up to about 16 minutes. An almanac supplies that correction.

What if we had started with tens?

We could divide a day into 10 hours, each hour into 100 minutes, and each minute into 100 seconds. A circle could have 400 divisions, called gradians, rather than 360 ordinary degrees. Revolutionary France actually tried decimal time. Tens fit our ten fingers and the way we write numbers.

Sixty has a useful advantage: it divides evenly by 2, 3, 4, 5, 6, 10, and 12. Ten does not. Decimal notation and modern calculation make that bookkeeping advantage less important, though one-third still cannot be written as a finite decimal.

A 400-day year would look tidy beside a 400-part circle, but nature does not oblige. The seasonal year is about 365.24 ordinary days. Changing the units on a clock, or the number of parts in a circle, changes neither the length of the day nor the length of the year. It would not remove leap-day corrections, and it would not make them fewer.

The plain-language sheet is Longitude with a clock (PDF).

From a carried clock to a broadcast clock, and then to GPS

The sheet above tells you how to turn a time difference into an angle, once you already have Greenwich time in your hand. It does not say where that carried clock came from, and it does not say what replaced the need to trust one watch for a whole voyage. Those are the next sentences. None of them were on this page before.

The watch was made in 1759 and tried at sea in 1761

John Harrison's fourth marine timekeeper, H4, is dated 1759 by the Royal Museums Greenwich. Work on it had started earlier, in 1755, after a smaller watch made to his ideas had behaved well. "Finished in 1761" is the wrong year for the building of it. 1761 is the year of the first sea trial. The Commissioners allowed his son William to take H4 toward Jamaica. The trial looked promising, including a landfall called early at Madeira. The Board of Longitude did not treat that voyage as the end of the test. A further trial, to Barbados, followed in 1764. In February 1765 the Board agreed that H4 had kept time within the strictest limit of the 1714 Longitude Act, and it asked Parliament to pay 10,000 pounds once Harrison explained the watch, and the rest, less sums already paid, only after other makers could build the same kind of thing. Harrison thought the full reward was already due, and that the rules had been moved. That argument became the Longitude Act of 10 May 1765. It was not a handshake and a single purse.

Later testing at Greenwich went badly, in Maskelyne's report, and Harrison disputed the report. In 1773 he went to Parliament rather than back to the Board. The Act of that year, in its own title, pays him a further reward for the invention and for disclosing how it was made. It is not worded as the Board handing over the lump prize of 20,000 pounds. Accounts of that Act put the further sum at 8,750 pounds. Royal Museums Greenwich says he was rewarded, but not by the Board, and not with everything he believed he was owed. Earlier payments, later payments, and this one are a long argument. They are not one sentence that says "he won."

The watches also stayed expensive. Wider use waited on later makers who could build them more simply, into the early 1800s. A prize on paper is not a chronometer in every chart room.

A clock finds the ship. A transit finds the Sun's distance

Keep the two jobs apart. The chronometer answers a question about the ship: how far east or west of a named meridian am I, at the moment the Sun is highest? The Venus transits of 1761 and 1769 answered a different question: how far away is the Sun, in Earth-radii or in miles? The Venus-to-Cepheid lesson is that second job. This page is the first.

A picture that sets London at one end of the Earth and Cape Town at the other is the wrong picture. On a modern map those two cities are about 87 degrees apart along a great circle. Points that are truly opposite are 180 degrees apart. The straight line through the Earth between London and Cape Town is a chord. A chord is a cut across the ball, shorter than a diameter. A diameter would join opposite points. And even the chord is not the baseline a transit uses. The useful baseline is the projected one: the separation of the stations as seen from the Sun, square to the line of sight, taken along the direction that actually shifts the contact. The same lesson already says the map coordinate you print is not that baseline.

A second mix-up lives in a small angle. A solar parallax near 8.5 arcseconds is the angle filled by the Earth's radius if you could look back from the Sun. The modern value of that angle is about 8.8 arcseconds, and the eighteenth-century arguments about it are on the Venus lesson, including Encke's 8.5776. That angle is not the width of Venus on the Sun, and it is not the angle between London and the Cape, which is tens of degrees, not seconds of arc. The chronometer never measured it. The transit campaign was trying to.

A radio tick can check the clock. It did not retire it

A time signal on the radio is still a reference clock. The difference is that the reference no longer has to survive the whole voyage inside one box. A ship can compare its chronometer with a tick that left a known station at a known moment, and then set or rate the watch. That is a check. It is not the same act as throwing the watch overboard.

WWV, the station of the National Bureau of Standards, now NIST, is easy to misdate. NIST's own history says the call was in use by 1920, and that a regular service of standard frequencies, for calibrating other radios, was underway in 1923. Time pulses on WWV came later. A NIST history of time-signal stations puts the first seconds pulses in 1937, and says the United States Navy was still the nation's official radio time broadcaster then. Voice announcements of the time came in 1950. Earlier stations, including time signals from the Eiffel Tower referenced to the clocks in Paris, were already being used so that ships could correct marine chronometers. Correct, not replace. Chronometers stayed in the chart room for decades after the first ticks. A broadcast lets you audit a carried clock. It does not, by itself, end the need for one.

GPS times the journey of the signal

GPS is still a time measurement, and it is not the noon-sun formula. NASA describes the satellites as sending their position and their time. The receiver works out how long each signal took to arrive. Distance, here, is that travel time multiplied by the speed of the signal. Several satellites are required, because the receiver is solving for where it is and for how wrong its own clock is, all at once. Four are the usual minimum for a position and a clock correction. Nothing in that sum asks when the Sun stood highest, and nothing in it is "one hour of solar time is 15 degrees" applied to a wristwatch. The longitude that comes out is still an angle on the Earth. The clock that produced it is a set of precise signals in flight, not a chronometer read at local noon.

The grid is older than a measurement you could trust at sea

Claudius Ptolemy, in the second century, wrote a Geography that placed thousands of towns by latitude and longitude in degrees. NASA's note on that book says the coordinates let later mapmakers reconstruct his world, and that his own drawn maps do not survive. Latitude from the sky was already a practical art. Longitude in his lists was a coordinate he could write down from reports, travelers, and earlier compilers. Writing a number is not the same as recovering it from the deck of a ship. The sky method and the carried clock, centuries later, were attempts to earn a figure his grid had already given a place to hold. The astronomical Ptolemy, circles on circles, is the Looking Up lesson. The longitude clock, in its first sea trials, is the one on this page.

Sources

- The factor of 15 is 360 degrees divided by the 24-hour mean solar day. The Greenwich numbers on this page are that arithmetic.
- Jean-Baptiste Chappe d'Auteroche, Voyage en Californie, Cassini's reductions, Paris, 1772, pp. 88-92, where the published longitude is measured west of Paris rather than Greenwich: https://archive.org/download/voyageencalifforn00chap/voyageencalifforn00chap_djvu.txt
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- U.S. Naval Observatory, "The Equation of Time": https://aa.usno.navy.mil/faq/eqtime
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- Royal Museums Greenwich, H4, object record, date made 1759: https://www.rmg.co.uk/collections/objects/rmgc-object-79142

- Royal Museums Greenwich, "Longitude found: the story of Harrison's timekeepers," on the 1761 Jamaica trial, the 1764 Barbados trial, the 1765 Act, and a reward that did not come from the Board: https://www.rmg.co.uk/stories/time/harrisons-clocks-longitude-problem
- Act of Parliament, 1773, 13 Geo. 3, c. 77, whose title calls the payment a further reward for the timekeeper and for the disclosure of its principles: https://www.legislation.gov.uk/apgb/Geo3/13/77/enacted
- NIST, "History of Radio Station WWV," on standard-frequency broadcasts from 1923: https://www.nist.gov/pml/time-and-frequency-division/time-distribution/radio-station-wwv/history-radio-station-wwv
- NIST, "Time Signal Stations," on seconds pulses from WWV in 1937 and on earlier signals used to correct marine chronometers: https://tf.nist.gov/general/pdf/2131.pdf
- NASA, "GPS," satellites sending position and time, and a receiver timing the signals' arrival: https://www.nasa.gov/directorates/somd/space-communications-navigation-program/gps/
- NASA, on Ptolemy's Geography and its latitude and longitude for thousands of places: https://science.nasa.gov/mission/landsat/outreach/geographia/

Key facts

- Twenty-four hours of solar time equal 360 degrees. One hour of time is 15 degrees. One minute of time is 15 arcminutes. One second of time is 15 arcseconds.
- Time is hours, minutes, and seconds. Angle is degrees, arcminutes, and arcseconds. One degree is 60 arcminutes, and one arcminute is 60 arcseconds.
- Longitude in degrees is 15 times (hours + minutes/60 + seconds/3600). Name the meridian. Behind it is west. Ahead of it is east.
- Two hours 35 minutes behind Greenwich is 38 degrees 45 arcminutes west, which is 38.75 degrees west, and -38.75 degrees if east is positive.
- A civil time zone or daylight saving is an administrative clock, not this angle. Decimal degrees and degrees-minutes-seconds are two displays. GPS and mapping software can show either.
- The worked example uses Greenwich. Chappe's published longitude used Paris.
- A chronometer kept on Greenwich time, read when the Sun is highest, gives longitude. A reading of 14:30 (2:30 p.m.) is 2.5 hours, which is 37.5 degrees west, not 30. The hours do not depend on latitude. Miles along a parallel do. Near the poles the hard part is seeing solar noon. The equation of time, up to about 16 minutes, needs an almanac.

- A decimal day of 10 hours, and a circle of 400 gradians, were tried in Revolutionary France. Sixty divides evenly by 2, 3, 4, 5, 6, 10, and 12. The seasonal year is about 365.24 ordinary days. New clock units would not shorten the year or reduce the need for leap days.
- Harrison's H4 is dated 1759. Its first sea trial was in 1761. The 1773 Act paid a further reward, not a one-line award of the whole Board prize. Radio time signals let a ship check a chronometer. They did not retire it at once. GPS times the travel of signals from several satellites. It is not the local-noon formula.
- London and Cape Town are not opposite ends of the Earth. The transit baseline is a projected separation, not a diameter. A solar parallax near 8.5 arcseconds is the Earth's radius seen from the Sun, not an angle of Venus and not the angle between those cities.

3. Voyages, surveying and empire

Routes, instruments, ships and people: the public lessons connect measurement with exploration and its consequences.

11 complete topics • begins on binder page 170

Saturday, October 31, 2026

The Suez Canal and the route to India and Australia

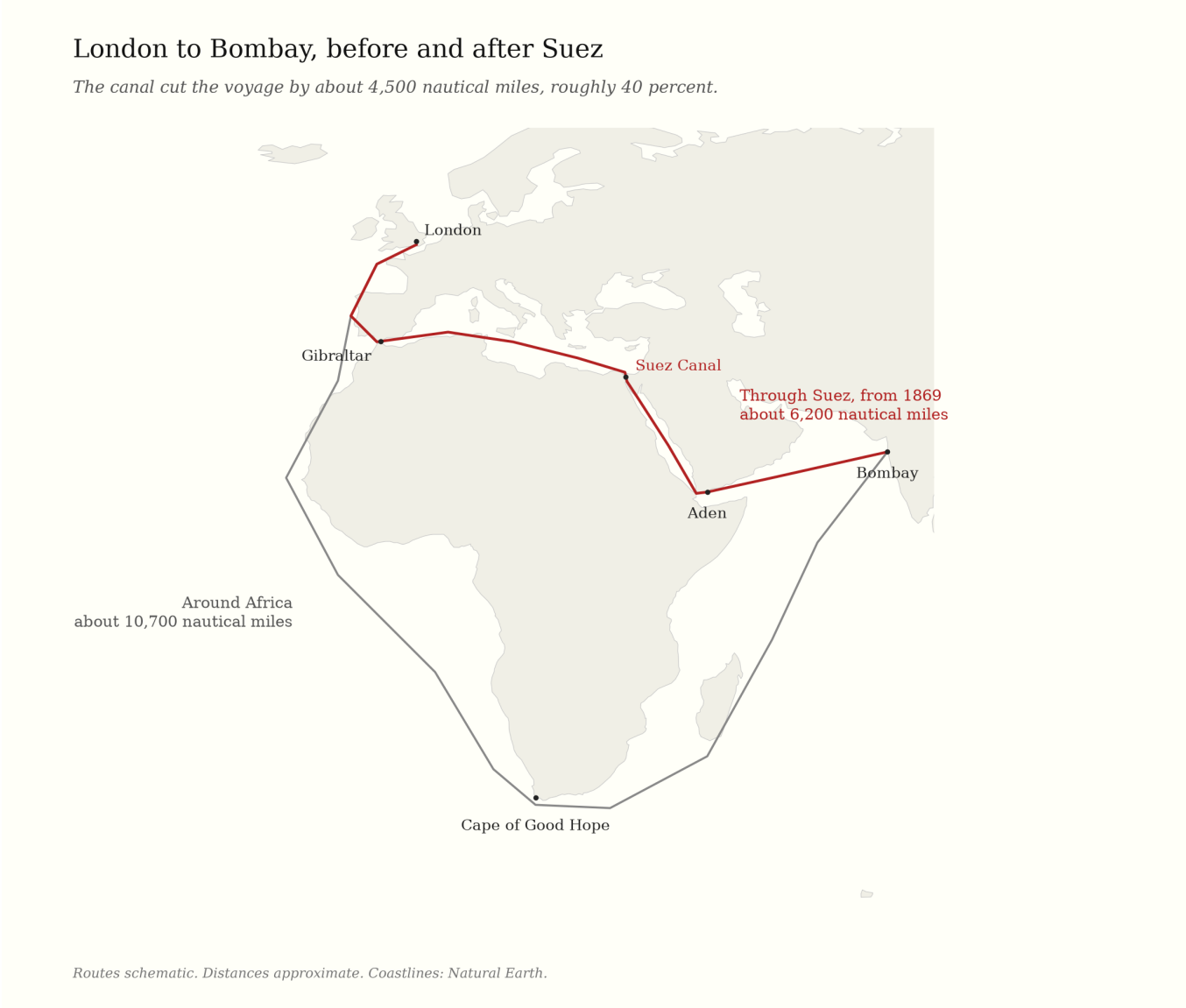
Opened in 1869, the canal cut London to Bombay from about 10,700 to about 6,200 nautical miles.

What it is

The **Suez Canal** runs about 120 miles across Egypt, from Port Said on the Mediterranean to Suez on the Red Sea. It is a sea-level canal with no locks. It was built by Ferdinand de Lesseps's French-led company and opened on **November 17, 1869**.

Why it matters

Before it, a ship from Britain to India or Australia had to sail down the whole west coast of Africa, around the **Cape of Good Hope** and back up the east side. The canal cut straight from the Mediterranean into the Red Sea, then out past Arabia into the Indian Ocean.



London to Bombay: around Africa (about 10,700 nautical miles) against through Suez (about 6,200)

- **London to Bombay**: about 10,700 nautical miles around the Cape, about 6,200 through Suez, roughly 40% shorter.
- **To Australia**: the saving is smaller, because Australia lies farther south and east, but ships still saved weeks, and coaling stations along the Suez route (Port Said, Aden) let ordinary steamers make the trip.

That was the end of the Great Eastern's reason to exist: she had been built to carry coal for the long way around, and she was too big to fit through the canal. See the Great Eastern.

In the novels

Jules Verne has the Nautilus pass from the Red Sea to the Mediterranean through a secret undersea tunnel, the "Arabian Tunnel," beneath the isthmus, before the canal

was even finished. Phileas Fogg passes through Suez in Around the World in Eighty Days (1872).

Key facts

- Suez Canal opened November 17, 1869 (Ferdinand de Lesseps); about 120 miles, no locks
- London to Bombay: about 10,700 nautical miles around Africa, about 6,200 through Suez
- Suez made the Great Eastern's design obsolete

Saturday, October 31, 2026

The Red Sea rift and Bab-el-Mandeb

The long narrow Red Sea is a crack where Africa and Arabia are pulling apart, closed at the south by the Gate of Grief.

Why is there a long, narrow sea there?

It looks odd on the map: a thin finger of water about 1,400 miles long and at most about 220 miles wide, running between Africa and the Arabian Peninsula. That is because the Red Sea is a **rift**, a crack in the Earth's crust. The **Arabian plate** is pulling away from the **African plate**, about a centimeter a year, roughly as fast as fingernails grow.

Starting some 25–30 million years ago, Arabia began tearing away from Africa. The gap sank and filled with sea. Hot new crust rises along the middle, so in many millions of years the Red Sea may widen into an ocean, the way the Atlantic once began.

At its southern end, in the **Afar** region of Ethiopia and Djibouti, three rifts meet (the Red Sea, the Gulf of Aden and the East African Rift), one of the few places on land where you can watch a new ocean being born, with volcanoes and earthquakes.

Bab-el-Mandeb

The southern mouth of the Red Sea is the strait of **Bab-el-Mandeb**, Arabic for "**Gate of Grief**" (or Gate of Tears), named for how dangerous it was to sail. It is about 20 miles wide at its narrowest, between Yemen and Djibouti/Eritrea. Every ship using the Suez Canal to or from Asia must pass through it, which is why it matters so much for trade and security today.

Key facts

- The Red Sea is a rift where the Arabian plate pulls away from Africa, about 1 cm a year
- Afar, in Ethiopia and Djibouti, is where three rifts meet
- Bab-el-Mandeb means Gate of Grief; about 20 miles wide

Saturday, November 7, 2026

Captain James Cook: precision, encounter, and empire

Cook's three Pacific voyages were feats of surveying, and meetings with people who already lived on the coasts he charted. He was killed at Kealakekua Bay in 1779. The third voyage itself lasted until the ships came home in 1780.

James Cook was born in Yorkshire in 1728. He learned the sea in the coal trade, then joined the Royal Navy in 1755. Before any Pacific voyage he surveyed Newfoundland. The Navy already trusted him to draw a coast so that the next ship could use the drawing.

In 1768 the Admiralty gave him Endeavour. The announced task was the transit of Venus, 3 June 1769, at Tahiti, with the astronomer Charles Green. Joseph Banks and Daniel Solander sailed to collect and classify. Cook commanded, and he surveyed. After Tahiti he opened sealed instructions to sail farther and to chart what he found. In 1769 and 1770 he charted New Zealand and the east coast of Australia. Those shores were not empty, and they were not unknown to the people who lived there. A European chart is a new record for London. It is not the first knowledge of a place.

Cook's three Pacific voyages on a timeline

Three commissions, drawn for this lesson. The first includes the Tahiti transit and the charts of New Zealand and eastern Australia. The second crosses the Antarctic Circle. The third runs from 1776 to 1780. Cook died in 1779. The ships came home the next year.

The second voyage, 1772 to 1775, in Resolution and Adventure, crossed the Antarctic Circle. Cook put an end to the hope of a green southern continent sitting in the temperate ocean. He did not claim to have seen the mainland of Antarctica.

The third voyage, 1776 to 1780, in Resolution and Discovery, searched for a passage in the north. In 1778 the ships reached Hawaii. On 14 February 1779, at Kealakekua Bay,

Cook was killed as a relationship that had already turned violent broke apart. His death did not end the expedition. Charles Clerke took command, and the ships reached England in 1780. The charts stayed in the Navy's hands. So did the harder consequence: disease, weapons, and later colonial power traveled along the same routes.

Hold both parts of the man. The angles and the coastlines were careful. The encounters were not a blank page, and they were not a moral slogan in either direction. The transit itself, and the risk of getting the observers there, is the next lesson in this day's chain. Banks, who collected beside him and did not sail the second voyage, is the lesson after that.

The illustrated lesson is download the PDF.

Sources

- Royal Museums Greenwich, "Captain James Cook: timeline," https://www.rmg.co.uk/stories/maritime-history/captain-james-cook-timeline
- Whipple Museum, Cambridge, on the quadrant stolen at Tahiti and recovered: https://www.whipplemuseum.cam.ac.uk/explore-whipple-collections/astronomy-and-empire/stolen-quadrant

Key facts

- Cook, born 1728 in Yorkshire, joined the Royal Navy in 1755 and had already surveyed Newfoundland.
- Endeavour, 1768-1771: Tahiti transit on 3 June 1769 with Green; then charts of New Zealand and Australia's east coast, lands already inhabited.
- Second voyage, 1772-1775, crossed the Antarctic Circle and did not claim a sighting of the Antarctic mainland.
- Third voyage, 1776-1780, in Resolution and Discovery. Cook was killed at Kealakekua Bay on 14 February 1779. The ships came home in 1780.

Saturday, November 7, 2026

Joseph Banks: the botanist who built a network

Joseph Banks paid for collecting, sailed with Cook, presided over the Royal Society, and organized the breadfruit missions. He was not Kew's director, not the Bounty's captain, and not the American clothier.

Joseph Banks (1743-1820) was a British naturalist with money enough to turn curiosity into cargo. In 1766 he collected in Newfoundland and Labrador. In 1768 he joined Cook on Endeavour. Daniel Solander classified. Sydney Parkinson drew. A dried plant, a label, and a drawing are a scientific record. They are a European record. People who already grew or gathered those plants had not been waiting for a London name.

What traveled home was not a conquest of the plant. A pressed specimen keeps a shape. Parkinson's drawing keeps a color the dried sheet loses. Solander's name puts the plant in a European cabinet. None of those three objects moves the people who already cooked the leaves or fruit. The cabinet is a record of a visit. It is not a discovery in the sense of finding an empty world.

Banks came home in 1771. He did not go on Cook's second voyage. In 1772 he sailed to Iceland instead. From 1778 until he died in 1820 he was president of the Royal Society. He advised King George III and had a long influence on the gardens at Kew. Influence is the right word. He was not the formal director of Kew, and this lesson will not give him that title.

He is also the organizer of the breadfruit voyages, not their captain. The plan was to move a living Pacific crop to the Caribbean as cheap food for enslaved plantation workers. The packing of living trees was ingenious. The purpose sat inside an empire that ran on forced labor. Both sentences belong in the account. The voyage itself, and the mutiny that stopped the first attempt, are told with the Bounty. The plant is told in the breadfruit lesson.

One warning about the name. Jos. A. Bank, the Baltimore clothier, is a different man and a different century. The botanist has a final s. The firm usually does not. Nothing in Banks's career is a clothing chain, and this page will not invent one.

The link back to Cook is the ship they shared in 1768. The link forward is the crate of breadfruit trees Banks wanted sent, and the American firm that only looks like his name. A label in a cabinet is not a deed to the plant, and a presidency is not a captain's log.

The illustrated lesson is download the PDF.

Sources

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- Royal Botanic Gardens, Kew, history of the gardens: https://www.kew.org/about-us/history-of-kew
- Royal Society note on Banks: https://royalsociety.org/science-events-and-lectures/2017/09/joseph-banks/

Key facts

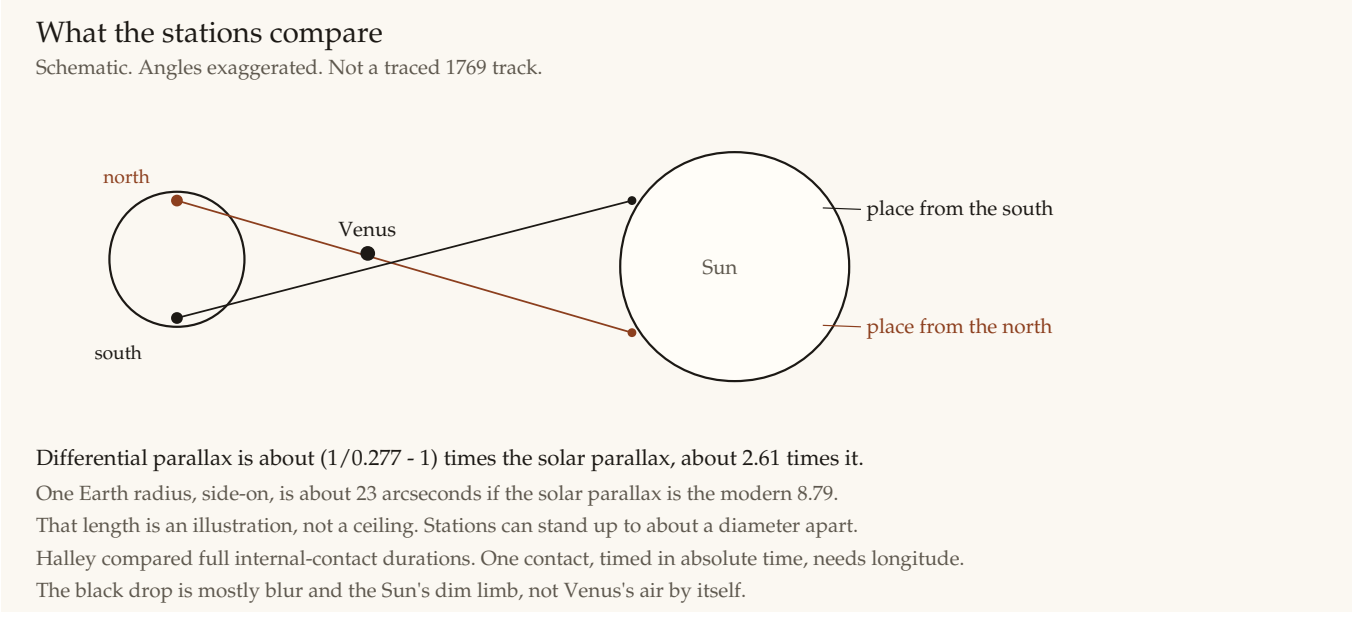
- Banks (1743-1820) collected in Newfoundland and Labrador in 1766 and sailed on Endeavour in 1768 with Solander and Parkinson.
- He returned in 1771, did not join Cook's second voyage, and went to Iceland in 1772.
- President of the Royal Society 1778-1820. He advised the king and influenced Kew. He was not Kew's formal director.
- He organized the breadfruit missions. He was not captain of the Bounty. He is not the clothier Jos. A. Bank.

Saturday, November 7, 2026

The Venus transits: a worldwide ruler

The 1761 and 1769 transits turned a relative solar system into miles, badly at first and better later. No single station, and no single reduction, was the modern astronomical unit.

Kepler's clock gives the planets in units of Earth's orbit. A transit of Venus can turn that ratio into a length, if two stations far apart time the same event. What they compare is a differential parallax: Venus, being nearer, shifts on the Sun more than the Sun itself shifts. The full story of the ladder, including why 23 arcseconds is a one-radius illustration and not a ceiling, is the Venus-to-Cepheid lesson. This page is the human campaign.



Two stations and the differential shift of Venus on the Sun

Teaching geometry, not a 1769 tracing. Halley compared durations of the internal contacts. A single contact timed on an absolute clock needs the longitude. The black drop is mostly blur and limb darkening.

Jeremiah Horrocks and William Crabtree, observing separately, saw the transit of 1639. Horrocks argued from Venus's small apparent disk, and from a model of sizes, that the solar parallax was not above 14 arcseconds, and the Sun at least about 15,000 Earth radii. That is not a two-station parallax. The book came out after his death, in 1662. Converted with modern constants, 14 arcseconds is about 58.4 million miles and 15,000 Earth radii about 59.4 million miles. Those mile figures are our conversion, not his words.

In 1761 Mason and Dixon were sent toward Bencoolen. Their ship, the Seahorse, was attacked in January. Mason's letter counts eleven killed and thirty-seven wounded. After repairs they observed from the Cape, and only the end of the transit, after sunrise on 6 June. Cloud early, then a clearing for the contacts. Do not promote that into a cloudless day. James Short compared the Cape with fifteen European stations. He was not their exclusive partner. His 8.47 and 8.52 arcseconds are for the day of the transit, not the mean distance. The mean-distance figure near 8.65 converts to about 94.5 million miles, roughly 1.7 percent high. A later Short value, 8.69, converts to about 94.1 million miles. Again, the miles are a modern conversion.

There was no one official answer. Pingre's 1765 reduction from Rodrigues, 10.25 arcseconds, converts to about 79.8 million miles, about 14 percent low. Hornsby, using 1769 and publishing in 1771-72, got a mean near 8.78 arcseconds, about 93.1 million miles, roughly 0.16 percent above today's 92.956 million miles. The astronomical unit is now defined as exactly 149,597,870,700 metres. Encke in 1824 combined both transits and got 8.5776 arcseconds, about 95.3 million miles, 2.5 percent high. That combination is not the 1769 result alone.

Some observers got little or nothing. Le Gentil, kept from Pondicherry by war in 1761, watched from a moving ship and the record was unusable. In 1769 a cloud hid Pondicherry again. Maskelyne at St Helena could not send a complete observation. Pingre at Rodrigues saw the Sun rise in thick cloud. At Greenwich, Bliss and Green missed the start for cloud.

In 1769 Cook and Green had a clear 3 June at Tahiti, after the quadrant was stolen on 2 May and recovered damaged but repairable. Green died of dysentery on the way home, on 29 January 1771. Wales and Dymond wintered at Hudson Bay and did observe. Their story is not a stopped clock. Mason's share of the Cape work is the next lesson. Dixon's sky in Norway, and whether anyone "won" against Chappe, is Dixon's lesson. This page will not rank them.

Chappe in Baja California, 1769

Pedro Espinoza Meléndez, writing in Meyibo, follows Chappe to San Jose del Cabo. The copy used here does not show the article's year or its pages. It says the party reached California on 19 May 1769, two weeks before the transit, after a month crossing the gulf. Chappe's diary, quoted there, has them camped on the shore and determined to reach San Jose the next day. Nineteen May is the landing the article dates. The walk into the town may be the day after. The article does not decide.

He set up in a granary, half the roof removed on the south side. Before 3 June he set the pendulum clock, checked the telescopes, and made repeated observations of the Sun. The transit was 3 June. On 5 June the Spaniards with the party fell ill, and so did the people of San Jose. Chappe nursed the sick and then became ill himself. The article does not date each of those days of care. On 18 June, fifteen days after the transit, he

observed a lunar eclipse while very weak. Cassini, printing the register in 1772, says Chappe was then too weak to take the Moon at the meridian. That night was not the last line in the book. The same tables still enter later emergences of Jupiter's moons as Chappe's, including 20 June and 16 July. He did not leave Baja. He died on 1 August 1769. Counted on the calendar, that is fifty-nine days after the transit and seventy-four days after the 19 May landing. Those counts are arithmetic, not a sentence in either source. No date was found here for when the survivors sailed. The Meyibo article says it is not clear whether the disease was typhus or yellow fever. This page says epidemic.

The article quotes Chappe's own instruments: a quadrant of three-foot radius by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. Dixon's weather, and the comparison of the two kits, is his lesson.

What the 18 June eclipse did not settle

The eclipse was a check on the longitude of San Jose. It did not become the longitude the book adopts, and it did not leave a measured improvement.

In Voyage en Californie (Paris, 1772) Cassini works the place in the astronomical observations, pages 88 to 92. Five emergences of Jupiter's moons, which he attributes to Chappe, average 7 hours 28 minutes 53 seconds west of Paris. The five include 20 June and 16 July, after the eclipse, so the average is not a baseline from before it. Cassini gives more weight to the transit. From the transit he has 7 hours 28 minutes 6 seconds, and he adopts 7 hours 28 minutes 10 seconds, which is 112 degrees 2 minutes 30 seconds west of Paris. Turning those hours into degrees is the time-to-longitude lesson. The worked example there uses Greenwich. This one uses Paris.

The eclipse, set only against a time calculated for Paris, gives 7 hours 29 minutes 7 seconds at the start and 7 hours 29 minutes 40 seconds at the end. Cassini says he knew of no observation of that eclipse at another known place. The two results stand 57 seconds and 90 seconds of time away from the figure he adopted. In angle, at fifteen seconds of arc to one second of time, that is 14 minutes 15 seconds and 22 minutes 30 seconds of longitude. The conversion is arithmetic done here. It is a disagreement. It is not a measured error, and it is not a measured improvement. No distance in miles is taken from it.

His remark that Jupiter's moons, without enough matched observations, should not be trusted closer than about 20 or 30 seconds of time is a caution about the method. It is not a personal error for Chappe. Nothing on these pages shows the eclipse tightening the longitude. The figure he asks the reader to keep is the one from the transit. A later ESO note says he stayed in order to fix the longitude. Staying made the check possible. It did not write the adopted value.

The illustrated lesson is download the PDF.

Sources

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- Jean-Baptiste Chappe d'Auteroche, Voyage en Californie, Cassini's reductions, Paris, 1772, pp. 88-92: https://archive.org/download/voyageencaliforn00chap/voyageencaliforn00chap_djvu.txt
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Key facts

- Horrocks in 1639 set an upper bound near 14 arcseconds from Venus's apparent size, not from a second station. Publication was posthumous, 1662.
- Mason and Dixon reached the Cape after the Seahorse fight (11 dead, 37 wounded) and timed the end of the 6 June 1761 transit.
- Short, Hornsby, Pingre, and Encke published different parallaxes. Converted miles on this page are modern arithmetic, not their sentences.
- The modern au is exactly 149,597,870,700 metres, about 92.956 million miles. No 1761 or 1769 paper printed that definition.
- Chappe's party reached the Baja shore on 19 May 1769. The diary leaves open whether the town was the next day. Transit on 3 June, lunar eclipse on 18 June while he was very weak, death there on 1 August, of an epidemic. Later Jupiter-moon timings are still entered as his.

- Cassini, Voyage en Californie (1772), pp. 88-92, adopts 7 hours 28 minutes 10 seconds west of Paris from the transit. The eclipse was a check against a calculated Paris time. No measured improvement is given.

Saturday, November 7, 2026

Charles Mason: an astronomer on difficult ground

Mason went from Bradley's assistant at Greenwich to the Cape transit of 1761 and then to the boundary survey. The place of the observer had to be known before the sky, or the line, could be trusted.

Charles Mason was born in Gloucestershire in 1728, a baker's or miller's son. In 1756 James Bradley brought him to Greenwich as an assistant. Not in 1750. The five years sometimes counted backward from 1761 are a way of saying he had already been there a while. The appointment year is 1756.

In September 1760 the Royal Society named him for the transit expedition toward Bencoolen, in Sumatra, with Jeremiah Dixon as his assistant. John Bird, the instrument maker, is the man usually credited with putting Dixon forward. That recommendation is the sturdy tradition. A document of their first handshake is not. This page will not invent the meeting.

The plan did not survive the voyage. After the attack on the Seahorse, told in the transit lesson, they observed from the Cape of Good Hope on 6 June 1761, and only the ending of the transit. Mason's times then entered other people's reductions. James Short compared the Cape with a set of European stations. Mason was an observer in a network, not the owner of the solar parallax.

Pingre later worried that the Cape longitude might be off, and Short had to answer him in print. The argument is a reminder, not a verdict against Mason. A clock time from the Cape is only as good as the place you assign to the Cape. Observers in Europe could not repair a bad longitude by wishing. They could only compare, reject, and say so.

In 1763 the same pair took the Penn-Calvert boundary. A zenith sector watches stars nearly overhead and yields a latitude. Chains and offsets turn that latitude into a mark on the ground. The legal line is geometry. It is useless if you do not know where the instrument stood. That is the transit rule brought ashore: the observer's place is part of the observation.

The survey lesson tells how the tangent was tried, found 22.51 chains off an older point, and corrected without a theatrical restart. Dixon's lesson tells the partnership from his side. Mason's job here is the simpler moral of the instruments. A number in the sky, or a post in the woods, means something only after you can say where you were standing when you took it.

The illustrated lesson is download the PDF.

Sources

- Edwin Danson, on Mason and Dixon's astronomy, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf
- Todd Babcock, on the boundary survey, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Mason and Dixon's 1767 letter on the line, from the survey record: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

- Mason was born in 1728 and became Bradley's assistant at Greenwich in 1756, not 1750.
- He led the 1761 transit party, with Dixon as assistant. Bird is generally credited with recommending Dixon. Their first meeting is not documented.
- After the Seahorse attack they timed the end of the transit from the Cape. Short and others reduced those times with European stations.
- On the boundary, latitude from a zenith sector still required a known place on the ground.

Saturday, November 7, 2026

Jeremiah Dixon: the surveyor in the partnership

Dixon was a Durham surveyor and amateur astronomer, recommended through John Bird's circle. On the Cape and on the boundary he was Mason's partner. In 1769, cloud at Hammerfest kept the transit from being satisfactory. That is weather, not a verdict that Chappe was the better astronomer.

Jeremiah Dixon was born in 1733 at Cockfield, in County Durham, into a Quaker family with means. He trained as a surveyor and worked as an amateur astronomer. When the Royal Society needed a second man for the 1761 transit, John Bird's recommendation is the account to trust, and to trust only as far as it goes. Bird made instruments and knew both worlds, the workshop and the observatory. A family story about an examination at Woolwich is thinner: the Ordnance minutes that would confirm it have not been produced. Say "recommended through Bird's network," not "we watched them meet."

A Quaker surveyor in the north of England already knew how a chain lies, how a slope cheats a distance, and how a neighbor disputes a hedge. That craft is not astronomy, and astronomy is not that craft. The Royal Society needed both on one road, which is why the pairing matters more than which man stood an inch to the left.

Dixon was not a spare hand hired to hold Mason's chain. At the Cape he observed. On the boundary, from 1763, surveying was his profession and astronomy was Mason's daily habit. The work needed both. A chain of standard length, a slope corrected, a trial line checked by a second measurement: those are the surveyor's corrections, and they are legitimate. They are not a confession that the first number was a scandal.

In October 1767 the Six Nations escort stopped at the war path. Their leader told the party that this was as far as his commission from the chiefs ran, and that he would not go one step farther west. That sentence is about a limit set by the people who sent him. It is not a license to invent a speech about fear, and it is not a permission slip written by the geometry. Some of the hired men, separately, had already refused to cross a river for fear of the Shawnee and the Delaware. Two facts. Do not melt them into one slogan.

Dixon died in Britain in 1779. The line he helped lay is the next lesson. Mason's lesson has the Greenwich years and the rule that a place must be known before a sight is interpreted. The pair only works if you can see both men.

Norway in 1769: Dixon and Chappe compared

Who observed the transit of 3 June 1769 better, Jeremiah Dixon in Norway or Jean-Baptiste Chappe d'Auteroche in Baja California? Weather, instruments, timing, and later praise all belong in the question. They do not add up to a winner.

Dixon was not at Vardo with Maximilian Hell, and he was not beside William Bayly. Nils Voje Johansen's account puts Bayly at Honningsvag, near the North Cape, and Dixon at Rypeklubben, just outside Hammerfest. Cloud held Dixon until 20 May, his first clear observations. On 3 June the weather was cloud and gales. Johansen writes that both men "failed to observe the transit satisfactorily." That sentence is about the transit. Dixon and Bayly still published the positions of the observatories, altitudes of the Sun, meridian transits, the solar eclipse that followed, and notes on the weather. A missed contact is not an empty expedition.

Dixon's own instrument list is not known. Johansen reconstructs Bayly's from the papers they printed: a quadrant of one-foot radius by Bird, a two-foot reflector by Dollond, a four-foot transit instrument by Bird with a Dollond lens, and clocks by Shelton, including an astronomical clock with a gridiron pendulum. Bayly said Dixon's instruments were similar, except that Dixon had no dipping needle. That is a reconstruction, not a case opened and listed. A contact time to the second, for Dixon, was not established here. Bayly wrote down a hazy contact of his own. Those seconds stay his.

Chappe's list is his own, quoted by Espinoza Meléndez: a three-foot-radius quadrant by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. He had checked the clock and the telescopes, with repeated observations of the Sun, before 3 June. The landing, the illness, and the death on 1 August are on the Venus lesson. So is the longitude. Cassini's adopted figure in the 1772 volume comes from the transit, not from an improvement measured by the eclipse of 18 June. A seconds-level accuracy for Chappe himself was not established. Cassini's 20 or 30 seconds of time is a caution about the Jupiter-moon method, not a personal error bar.

Later pages praise the Baja work. A Smithsonian exhibition describes the conditions as ideal. An American Scientist article calls the observations among the best made of either transit, and the position work unusually accurate for hard fieldwork. That is later praise, from those pages, not a jury sitting in 1769. No formal ranking of Dixon against Chappe was found, and nothing found here says Dixon was the less skillful man. The sky at Hammerfest failed him. No weighting by Encke that picks a winner was found.

The illustrated lesson is download the PDF.

Sources

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- Todd Babcock, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- The 1767 letter, Swarthmore transcription: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html
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Key facts

- Dixon was born in 1733 at Cockfield, County Durham, and worked as a surveyor and amateur astronomer.
- Bird's recommendation brought him to Mason. The literal first meeting is unknown, and the Woolwich exam story is unverified.
- He observed at the Cape in 1761 and surveyed the boundary from 1763. Planned corrections are part of the method.
- In October 1767 the escort stopped at the commission's geographic limit. Dixon died in Britain in 1779.
- In 1769 Dixon was at Rypeklubben, near Hammerfest, and Bayly at Honningsvag. Hell at Vardo was a separate station. Cloud until 20 May, and cloud and gales on 3 June. Johansen's words are that both "failed to observe the transit satisfactorily." They still published the place and other observations.
- Dixon's exact instruments are not known. Johansen's list reconstructs Bayly's kit, which Bayly called similar except for the dipping needle. Chappe's own list names the Canivet and English quadrants, a transit instrument, a parallactic machine, two Dollond telescopes, and a Berthoud clock.

- Later histories praise Chappe's Baja observations. No contemporary ranking, and no Encke weighting, was found. The adopted longitude in the 1772 volume is the transit's, not an eclipse improvement. The weather, not a judgment of skill, is what failed Dixon.

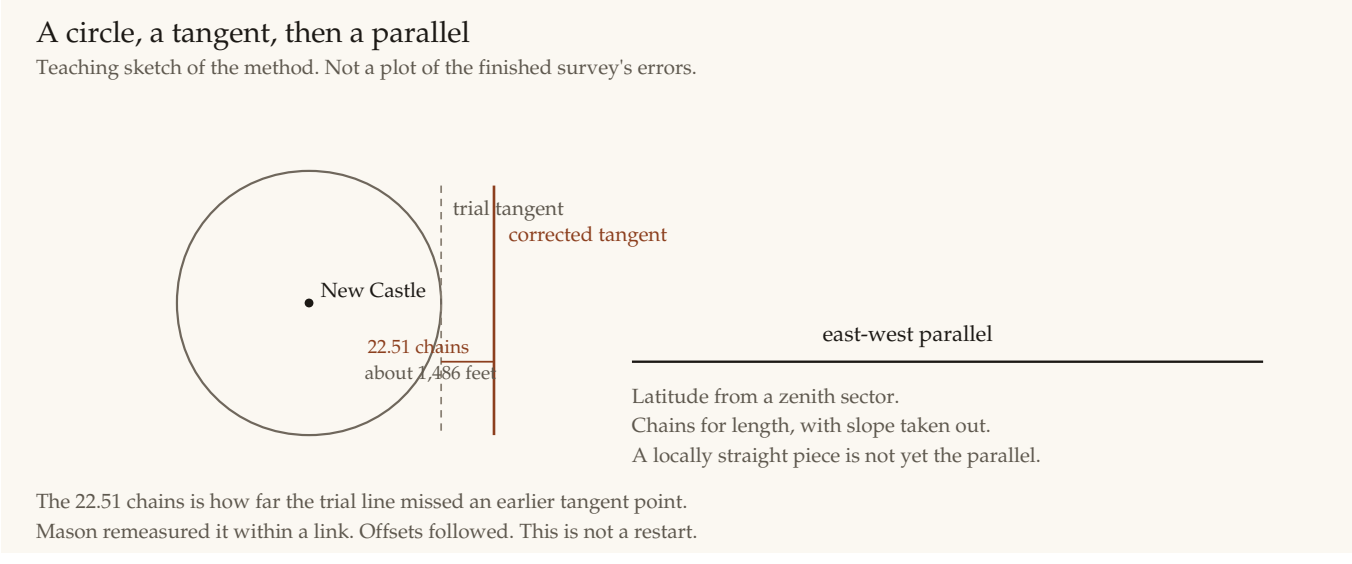
Saturday, November 7, 2026

The Mason-Dixon survey: how a boundary became real

The famous line is a circle, a tangent, and a parallel, laid with a zenith sector and chains. A 1764 offset of 22.51 chains was corrected in place. In 1767 the Six Nations escort stopped at the limit of its commission.

The Mason-Dixon line is not one endless straight stroke. The Penn and Calvert grant needed a circle around New Castle, a tangent across the Delmarva peninsula, and then an east-west parallel. Each piece answers a clause in a charter. None of them is "draw west until the paper runs out."

Latitude came from a zenith sector, a telescope pointed near the zenith so the star's height gives the parallel. Length came from chains of a known size, with slope leveled out. On a round Earth a short piece that looks straight on the ground is not yet a parallel of latitude. You compute the parallel, then you step the offsets that keep you on it.



The New Castle circle, a trial tangent, and the parallel

A sketch of the method, drawn for this lesson. It is not a graph of leftover error along the finished line.

In August 1764 a trial tangent about 81 miles long missed an earlier colonial tangent point by 22.51 chains, about 1,486 feet. Read that carefully. It is the offset of a planned trial from an older mark, not the collapse of a finished survey. Mason took a man and remeasured the gap himself. He came within a link, about eight inches. Through the autumn they laid proportional offsets and declared a true tangent. They did not burn the work and begin again. No document found shows one single gross mistake that

forced a full restart. That sentence does not prove the party never erred. Later resurveys, in the nineteenth and twentieth centuries, are different instruments and different legal questions.

On 17 September 1767 a falling tree killed the pack-horse drivers William Baker and John Carpenter. The names come from a careful later account of the survey; they are not written in Mason's own journal, so the lesson keeps that distinction. On 9 October the leader of the escort, who had joined in July with Mohawk and Onondaga men, said the war path was as far as the chiefs of the Six Nations had commissioned him to go. He would not proceed one step farther west. Earlier, twenty-six hired men had refused a river for fear of the Shawnee and the Delaware. The journal states the commission's limit. It does not, in its own words, explain why the chiefs drew it there. Later markers that say hostile nations "ordered them to halt" are an interpretation, not the 1767 sentence.

The line later became a symbol of North and South. That symbol is a nineteenth-century use of an eighteenth-century colonial boundary. Dixon's lesson has the man who helped lay it. The geometry here is only the geometry.

The illustrated lesson is download the PDF.

Sources

- Todd Babcock, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Edwin Danson, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf
- Mason and Dixon, 1767 letter: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

- The boundary is a New Castle circle, a Delmarva tangent, and an east-west parallel, not one long line.
- The 1764 trial tangent missed an older point by 22.51 chains. Mason remeasured within a link, and offsets corrected the line. It was not a documented full restart.
- On 9 October 1767 the escort stopped at the war path, the limit of the Six Nations commission.
- Baker and Carpenter were killed by a falling tree on 17 September 1767 in later accounts of the work, not in Mason's journal itself.

Saturday, November 7, 2026

Ships of exploration: four craft, four different jobs

Endeavour, Seahorse, Bounty, and Bounty's launch were built or bought for different work. Mixing them up mixes up Cook, Mason and Dixon, and Bligh.

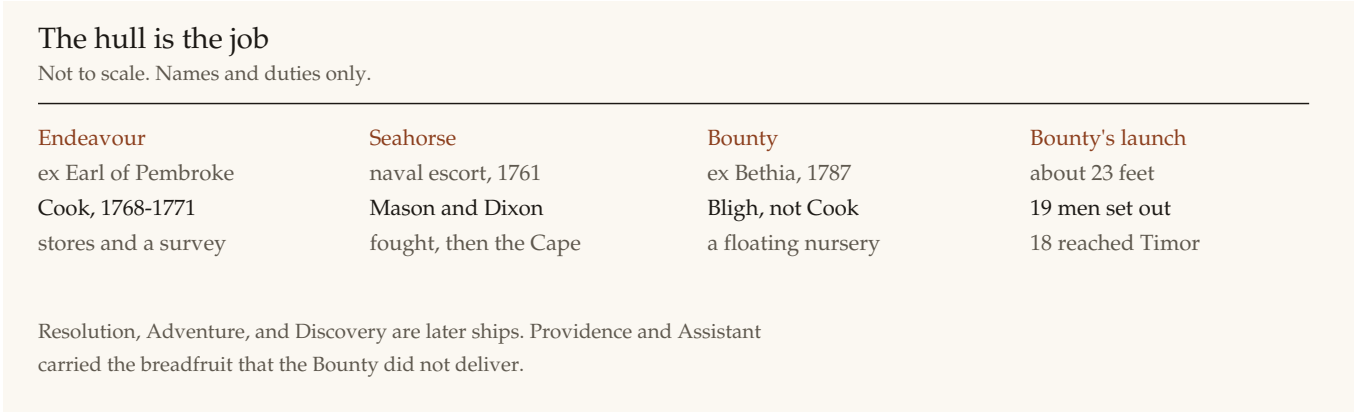
A ship's name is a job description if you let it be. Four vessels from this day's stories are easy to tangle, and they should stay apart.

Endeavour had been the collier Earl of Pembroke. The Navy bought a cargo ship because Cook knew that kind of hull and because a survey voyage needs room for stores, not a frigate's gun deck. She carried the first voyage, 1768 to 1771: Tahiti, New Zealand, the Australian east coast. Resolution, Adventure, and Discovery came later, for the second and third voyages. They are not nicknames for Endeavour.

Seahorse, in 1761, was the naval ship that took Charles Mason and Jeremiah Dixon toward Sumatra. In January she fought a French frigate. Eleven men were killed and thirty-seven wounded, by Mason's letter. The transit party went on to the Cape. Seahorse did not draw the American boundary, and she had nothing to do with the Bounty.

Bounty had been the merchant Bethia. The Navy took her up in 1787 and rebuilt her to carry living breadfruit, a nursery where a great cabin had been. William Bligh commanded. Cook had been dead for eight years. Joseph Banks organized the plan and stayed in England. Bounty is a breadfruit ship that became a mutiny. She is not Cook's surveyor.

Her launch was a working boat about 23 feet long, carried on board like any ship's boat. After the mutiny, nineteen men were in it. At Tofua, John Norton was killed. The remaining eighteen reached Timor. The distance usually given is about 3,618 nautical miles, not statute miles. How you count the days depends on which day's entry you start from, so this page will not pretend one duration is the only honest one. A later pair, Providence and Assistant, in 1791 to 1793, did deliver breadfruit. The boat's story is the mutiny lesson. The plant's story is the breadfruit lesson. Cook's collier is the Cook lesson.



Four vessels and the job each one actually did

Roles, not a scale model. Drawn for this lesson.

The illustrated lesson is download the PDF.

Sources

- Royal Museums Greenwich, Cook timeline: https://www.rmg.co.uk/stories/maritime-history/captain-james-cook-timeline
- Royal Museums Greenwich, mutiny on the Bounty: https://www.rmg.co.uk/stories/maritime-history/mutiny-on-bounty
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- Royal Museums Greenwich, the Bounty launch model: https://www.rmg.co.uk/collections/objects/rmgc-object-68956

Key facts

- Endeavour, ex Earl of Pembroke, was Cook's first-voyage collier. Later voyages used other ships.
- Seahorse carried Mason and Dixon in 1761 and was attacked. She is not the boundary survey and not the Bounty.
- Bounty, ex Bethia, was fitted in 1787 as a breadfruit nursery under Bligh, not under Cook or Banks.
- The launch of about 23 feet took 19 men. After Norton's death at Tofua, 18 reached Timor, some 3,618 nautical miles, not statute miles.

Saturday, November 7, 2026

Bounty: mutiny, survival, and the people of Pitcairn

Christian took the Bounty on 28 April 1789. Bligh's launch reached Timor after Norton was killed at Tofua. On Pitcairn the Polynesian women and children, not Adams alone, made the community that lasted.

On 28 April 1789 Fletcher Christian took HMS Bounty from William Bligh. Bligh and eighteen others were put in the launch. Not everyone left aboard had chosen the mutiny in the same way, and not every man in the boat was a loyal volunteer in a simple story. The ship's jobs, nursery and boat, are the ships lesson. This page is what people did.

Bligh could navigate. He was also harsh to his company. A court later acquitted him of losing the ship. An acquittal says the Navy did not convict him. It does not wash the temper out of the record.

At Tofua on 2 May his own narrative is the one we have. He had told his men to agree on a cover: the ship had sunk on 1 May. Islanders tried to haul the boat ashore, and Bligh brandished a cutlass. When he refused to stay the night, a chief threatened them. John Norton ran up the beach to cast off the stern line, though he was called back. As the crew boarded, the attack began. Norton was struck down. Bligh cut the rope with a pocket knife. Bligh inferred that the visitors meant to take the boat and the stores, and that they had learned the party had no firearms. That is his inference. There is no account here from Tofua, and there is no proven single provocation that rests on Norton alone.

They reached Timor. Not every man of the launch lived to see England again.

In January 1790 nine British mutineers, six Polynesian men, twelve women, and an infant came to Pitcairn. Infant counts vary, and this page will not pretend they do not. People had lived on the island before. It was empty then. The Bounty was burned on 23 January. The Pitcairn government describes the Polynesian men as treated more like slaves than like partners. Land, weapons, and the women's choices were not shared. In September 1793 five of the mutineers were killed: Christian, Mills, Martin, Williams, and Brown. By 1794 all six Polynesian men were dead. Later tellings disagree about who struck whom. McCoy killed himself in 1798. Young and Adams killed Quintal in 1799. Young died in 1800. Adams lived until 1829, the last of the mutineers, not the author of a peaceful idyll. The women and the children are the center of what continued.

The American ship Topaz, under Mayhew Folger, found the settlement in 1808. In 1831 a removal to Tahiti brought disease and deaths, then a return. In 1856 the community

went to Norfolk Island. Some families came back to Pitcairn later. Head counts differ by source, so this lesson will not pick a false exact number. Descendants live on both islands. The breadfruit those ships were built to move is the next lesson in the chain.

The illustrated lesson is download the PDF.

Sources

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Key facts

- The mutiny was 28 April 1789. Bligh and 18 others took the launch. Not everyone remaining aboard was a willing mutineer in the same sense.
- At Tofua on 2 May, Norton was killed while casting off the stern line. Bligh's book is the account we have. It infers motive. It is not a Tongan testimony.
- Pitcairn, January 1790: nine British mutineers, six Polynesian men, twelve women, and an infant. The ship was burned on 23 January.
- By 1794 the Polynesian men were dead and five mutineers had been killed. Adams was the last surviving mutineer, not the whole community.

Saturday, November 7, 2026

Breadfruit and empire: moving a living crop

Breadfruit was a Pacific staple long before Banks. The Bounty was meant to carry living plants to feed enslaved Caribbean workers. Providence and Assistant finished that delivery in 1793.

Breadfruit, Artocarpus altilis, is a starchy staple of the Pacific. People there cultivated it long before a British ship asked for cuttings. Many of the kinds grown for food are seedless. You do not ship them as a bag of seed. You ship a living plant: stem, root, and the water and shade those roots need. That is why a warship's cabin was rebuilt as a nursery. The botany is real. It is also the reason the voyage was so easy to ruin.

A seedless fruit has no seed to dry and pocket. Gardeners cut a shoot, keep the cut damp, and wait for roots. At sea the same shoot needs fresh water, air that is not a hold's fug, and space so one mold does not take the whole crate. Those are practical facts. They are also why the first cargo was fragile, and why a mutiny did not merely delay a sack of grain. It ended a greenhouse.

Joseph Banks organized the plan. He did not captain it. The stated use was cheap food for enslaved workers on Caribbean plantations, not a clean gift from one people to another. Scientific care and an oppressive purpose traveled in the same pots. The Banks lesson has the organizer. The Bounty lesson has the mutiny that threw the first cargo away in 1789.

A second try worked. HMS Providence and Assistant, under Bligh again, sailed in 1791 and in 1793 landed breadfruit at St Vincent and Jamaica, first into botanical gardens and then out toward plantations. The trees that grow in the Caribbean now are part of the food of those islands. Genetic studies tie modern cultivars back to Pacific stock moved in that era. A plant can become ordinary in a kitchen and still have a harsh reason for the crossing.

Nothing in that success rewrites the first plan. Ingenuity in keeping a tree alive at sea is not the same thing as a kind motive. The gardeners who received the pots in the Caribbean were doing botany. The people the fruit was meant to feed were not free to refuse the plantation. If you want the ship that failed, read the mutiny. If you want the man who ordered the trees, read Banks. This page is only the crop, and the fact that it was sent as provisions for people who were not free.

The illustrated lesson is download the PDF.

Sources

- Natural History Museum, on Banks: https://www.nhm.ac.uk/discover/joseph-banks-scientist-explorer-botanist.html
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- Royal Museums Greenwich, William Bligh: https://www.rmg.co.uk/stories/maritime-history/william-bligh
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Key facts

- Artocarpus altilis was cultivated in the Pacific long before European voyages. Seedless kinds travel as living plants, not as seed.
- Banks's plan for the Bounty was cheap food for enslaved Caribbean workers, not a purely humanitarian gift.
- The Bounty never delivered. Providence and Assistant brought plants to St Vincent and Jamaica in 1793.
- Modern Caribbean trees are tied by genetics and by history to that transfer. The meal does not erase the purpose.

4. Weather, medicine and communication

Warnings and their journeys, hurricane language, disease and the new wind-force comparison. The published November 8 wind and expanded longitude lessons are included in full.

14 complete topics • begins on binder page 198

Saturday, October 31, 2026

RADAR is an acronym

RAdio Detection And Ranging, a US Navy coinage from 1940, and how it changed storm watching.

The letters

RADAR = **RA**dio **D**etection **A**nd **R**anging. The US Navy coined it in 1940. It's an acronym that became an ordinary word, and it happens to be a palindrome: it reads the same backwards, fitting for a signal that goes out and bounces back.

How it works

Send out a pulse of radio waves. When it hits something (a plane, a ship, raindrops), part of it bounces back. The time it takes to return tells you how far away the object is (the "ranging"). The direction the antenna was pointing tells you where.

Why it matters for storms

Radar was built for war, to spot aircraft and ships, but World War II operators noticed rain cluttering their screens. After the war that "clutter" became the point: weather radar lets forecasters see a hurricane's rain bands and eye as it approaches the coast. Before radar, and long before satellites, forecasters had only what people could see and measure on the ground and the reports the telegraph carried. See the telegraph and storm warnings.

Key facts

- RADAR: RAdio Detection And Ranging (US Navy, 1940)
- Radar measures distance by the echo's return time
- Rain "clutter" on wartime radar became weather radar

Saturday, October 31, 2026

The telegraph and the first storm warnings

For the first time, news of a storm could travel faster than the storm itself.

Before the telegraph

Before about 1850, news moved only as fast as a horse or a ship. A hurricane moves at roughly 10–20 miles an hour, and usually nobody ahead of it knew it was coming until the sky and sea turned. Each town met it alone.

What changed

Samuel Morse's electric telegraph (first public line 1844, "What hath God wrought") sent messages over wire almost instantly. So **reports could outrun the storm**. If an observer down the coast saw the barometer falling and the wind swinging, a message could reach the next city hours or days ahead.

That made weather a network science. You can't see a storm's shape from one place, but with many simultaneous reports you can draw a map of pressure and wind and watch the system move. The US Army Signal Corps began national telegraphed weather reports in 1870; that service became the US Weather Bureau in 1891.

The catch

A warning is only as good as the people who decide whether to pass it on. The telegraph made the Cuban hurricane warnings possible, and human pride kept them from being heeded in 1900. See Benito Viñes and Galveston 1900.

Key facts

- Telegraph changed storm warnings: reports could outrun the storm
- Morse's first public line: 1844, "What hath God wrought"
- US national weather reports by telegraph began 1870 (Army Signal Corps); Weather Bureau 1891

Saturday, October 31, 2026

Father Benito Viñes, Padre Huracán

A Jesuit priest in Havana who read the sky so well he issued the first scientific hurricane warning in 1875.

Who he was

Father Benito Viñes (BEN-ee-toh VEEN-yes), a Spanish Jesuit priest, became director of the observatory at **Belén College** (Colegio de Belén) in Havana, Cuba, in 1870. Cubans called him **Padre Huracán**, Father Hurricane.

How he and his team became experts

Nobody trained them; there was no such field yet. They became experts the way science always works: patient, careful observation, many times a day, for years.

- **Constant measurements.** Barometer, wind direction and speed, temperature and humidity, recorded hour after hour.
- **The clouds.** Viñes studied the high, thin **cirrus** clouds that fan out from a hurricane's top. He worked out that they spread out ahead of the storm, often a day or two before it arrives, and their streaks point back toward its center.
- **Wind shifts and the barometer.** The way the wind swings and the pressure drops told him which side of the storm a place was on and whether it was approaching.
- **A network.** Viñes set up a network of observers across Cuba and the Caribbean who sent their readings to Havana by telegraph. Combined, they showed the storm's track.
- **Writing it down.** He published his rules for reading storms (Apuntes relativos a los huracanes de las Antillas, 1877), which ship captains used across the Caribbean.

His first public, scientific hurricane warning went out in **September 1875**, and it saved lives.

Could he "see" the spin?

No, and this is an important correction. Nobody could see a hurricane's spiral from the ground; that wasn't possible until satellites in the 1960s. What Viñes did was read **indirect signs**: which way the cirrus clouds streamed, how the barometer fell, how the wind shifted. From those he reasoned out where the center was and where it was heading. It was inference, not sight, which is what makes it so impressive.

After Viñes

Viñes died in 1893. The Belén Observatory carried on his methods, and in September 1900 Cuban forecasters warned of the storm that destroyed Galveston. See Galveston 1900.

Key facts

- Father Benito Viñes (Belén College, Havana): cirrus clouds fan out ahead of a hurricane and point to its center
- Viñes issued the first scientific hurricane warning in 1875, using an observer network linked by telegraph
- Nicknamed Padre Huracán; nobody could see the spiral until satellites

Saturday, October 31, 2026

Galveston 1900 and the dismissed Cuban warnings

The deadliest natural disaster in US history, and the Weather Bureau's refusal to listen to Havana.

The storm

On **September 8, 1900**, a hurricane struck **Galveston, Texas**, a booming port city on a low barrier island, its highest point under 9 feet above sea level. A storm surge of about 15 feet swept over the island. Somewhere between **6,000 and 12,000 people** died, most often cited as about 8,000. It remains the deadliest natural disaster in US history.

The people

- **Willis Moore**, chief of the US Weather Bureau in Washington. He distrusted the Cuban forecasters and wanted the Bureau to be the only authority. The Bureau had already moved to stop Cuban storm warnings from going out over the telegraph lines it controlled, and in 1900 it dismissed, and effectively banned, the Cuban warnings about this storm.
- **Isaac Cline**, the Weather Bureau's meteorologist in charge at Galveston. In 1891 he had written in the Galveston News that the idea of a hurricane seriously damaging Galveston was "an absurd delusion." The city's confidence helped ensure no seawall was built.
- **Joseph Cline**, Isaac's younger brother, also a Weather Bureau man at the Galveston office.

What the Cubans said, and what Washington said

The storm crossed Cuba first. Forecasters there, working in Viñes's tradition, judged that it would carry on into the Gulf of Mexico toward Texas. The Weather Bureau in Washington decided instead that it would curve northeast toward Florida and the Atlantic coast. Washington was wrong.

The day

Isaac Cline later said he drove along the beach warning people to move to higher ground; Joseph said that he himself raised the hurricane warning flags. Historians debate how much warning was actually given and who did what. Either way, by the

time the danger was plain, the water was rising, the bridges to the mainland were going under, and there was nowhere to go.

The Clines' own house, built to be strong, collapsed in the surge. Isaac's wife, Cora, died. Isaac, Joseph and Isaac's daughters survived by clinging to wreckage through the night.

Why didn't people act when they saw the sky?

They had the signs, the long swells, the strange high clouds, the falling barometer, but without a forecast they trusted, those signs weren't read as "leave now." Many people even went down to the beach to watch the big waves. Viñes-style reading of the sky needed training and a network; Galveston had neither listening to Havana. And there were no evacuation plans or routes off the island.

Afterward

Galveston built a 17-foot **seawall** and then did something astonishing: it raised the entire city, jacking up more than 2,000 buildings and pumping sand underneath. Houston, inland and safer, took over as Texas's great port.

The book on this is Erik Larson's Isaac's Storm (1999).

Key facts

- Father Benito Viñes warned of hurricanes; Galveston 1900 dismissed the Cuban warnings
- Galveston, Sept 8, 1900: about 6,000 to 12,000 dead, the deadliest US natural disaster
- Willis Moore, Weather Bureau chief, dismissed and blocked the Cuban warnings
- Isaac Cline, Galveston meteorologist, had called a destructive hurricane "an absurd delusion" (1891); brother Joseph Cline worked with him
- Washington forecast the storm would curve toward Florida; Cuba said Texas
- Galveston built a seawall and raised the whole city

Saturday, October 31, 2026

How a Cuban warning reached Texas

The undersea telegraph cable from Havana to Key West, laid around 1867, tied Cuba into the US wire network.

The question

How could a warning from Havana get to Galveston in 1900, with no radio and no telephone across the sea?

The answer: a wire under the sea

Around **1867**, an undersea telegraph cable was laid across the Florida Straits from **Havana to Key West**, about 90 miles. From Key West, lines ran up the Florida Keys and into the mainland US telegraph network, and from there to Washington, New Orleans and Galveston.

So a message tapped out in Morse code in Havana could arrive in Texas in minutes, relayed station to station. Undersea cables were cutting-edge then: the first lasting transatlantic cable had worked only since 1866. See the transatlantic cables.

Who controlled the wire mattered

Because the wires went through Weather Bureau and US government channels, Washington could decide which warnings were passed along. That is how the Cuban forecasts could be blocked in 1900. See Galveston 1900.

Key facts

- Havana to Key West undersea telegraph cable: about 1867, about 90 miles
- Cuban warnings reached the US over that cable, then the mainland telegraph

Saturday, October 31, 2026

The transatlantic cables of 1858 and 1866

Cyrus Field's first cable carried Queen Victoria's message in 16 hours and died in three weeks; the Great Eastern laid the one that lasted.

The dream

Before the cable, news between Europe and America took about ten days by fast steamship. **Cyrus Field**, an American businessman, set out in the 1850s to run a telegraph wire along the floor of the Atlantic, about 2,000 miles from **Valentia Island, Ireland**, to **Newfoundland**.

1858: the first cable

After failed attempts (the cable snapped and was lost), two ships, the British HMS Agamemnon and the American USS Niagara, met in mid-ocean, spliced their halves together and sailed in opposite directions, paying out cable. In August 1858 it worked.

Queen Victoria sent a message of congratulations to President James Buchanan. It was about 98 words long and took **about 16 hours** to transmit, because the signal was so faint and smeared. There were fireworks and parades in New York (part of City Hall caught fire from the celebrations).

Then, after **about three weeks**, the cable died. The electrician Wildman Whitehouse had tried to push the weak signal through with very high voltage, which burned out the insulation. Public celebration turned to suspicion that it had all been a hoax.

1866: the cable that lasted

The Civil War delayed everything. Then Field had the perfect ship: Brunel's giant **Great Eastern**, the only vessel big enough to carry the whole cable, about 2,300 nautical miles of it, in one load. A 1865 attempt failed when the cable broke two-thirds of the way across. In **July 1866** the Great Eastern landed a working cable at Heart's Content, Newfoundland. Then she went back, grappled the lost 1865 cable from the sea floor two miles down, spliced it and completed a second line.

Messages that had taken ten days now took minutes. The physics that made it work came partly from William Thomson (later Lord Kelvin), who worked out how signals travel through long cables.

Key facts

- 1858 first transatlantic cable (Cyrus Field): Victoria's message took 16 hours, died in 3 weeks
- 1866 lasting transatlantic cable laid by the Great Eastern; Havana to Key West cable about 1867
- The cable ran from Valentia Island, Ireland, to Newfoundland
- Too much voltage killed the 1858 cable

Saturday, October 31, 2026

The Great Eastern and Isambard Kingdom Brunel

Why anyone built a ship five times too big, and the brilliant engineer it killed.

The man

Isambard Kingdom Brunel (1806–1859), British engineer, is one of the great builders of the Victorian age. The name itself is wonderful: Isambard from his French father, **Marc Isambard Brunel**, an engineer who fled the French Revolution; Kingdom from his English mother, Sophia Kingdom.

What he built:

- The **Thames Tunnel** with his father, the first tunnel under a river in a city (opened 1843). Young Isambard nearly drowned when it flooded.
- The **Great Western Railway**, London to Bristol, with its viaducts, tunnels and stations.
- The **Clifton Suspension Bridge** in Bristol (designed by him, finished after his death).
- Three ships, each the biggest in the world in its time: the **Great Western** (1838), a wooden paddle steamer built to extend his railway across the Atlantic to New York; the **Great Britain** (1843), the first big iron, screw-propeller ocean liner; and the **Great Eastern**.

The famous photograph of Brunel, cigar in mouth, hat on, in front of the huge launching chains, was taken at the Great Eastern's yard.

The ship

The **Great Eastern** was about **692 feet** long, nearly six times the size of any ship afloat by tonnage, designed for **4,000 passengers**. She had both paddle wheels and a screw propeller, plus six masts for sails. No ship would beat her length until 1899.

Why build a ship that big? Who thought that was a good idea?

It was actually a logical idea, built on one problem: **coal**. In the 1850s the route from Britain to India and Australia went all the way around Africa, and there were few places to refuel. Steamers had to stop often or carry so much coal there was little room for cargo. Brunel's answer: a ship big enough to carry all the coal for a round trip to Australia without refueling, plus thousands of passengers. The size was the solution.

Then the world changed under her:

- **The Suez Canal (1869)** cut the route to India by thousands of miles, and coaling stations spread along it. A ship built for the long way around was suddenly unnecessary. And she was too big to fit through the canal anyway. See the Suez Canal.
- **The launch was a disaster.** She was built sideways to the Thames, because the river was too narrow to launch her lengthwise. In November 1857 she stuck on the slipway; it took about three months of hydraulic jacks and chains to push her in, in January 1858. The cost ruined the company.
- **Brunel died with her.** He collapsed with a stroke on her deck in September 1859, just before her first trial voyage. Days later an explosion on board blew off a funnel, and Brunel died on September 15, 1859, aged 53.
- She rolled badly, never carried full passenger loads and lost money on the Atlantic run.

Her real calling: laying cable

She found her purpose by accident. Only the Great Eastern could carry an entire transatlantic cable in one load. She laid the lasting **1866** cable (see the transatlantic cables) and more cables after that. She was scrapped in 1889–90; it took nearly two years to break her up.

Jules Verne sailed on her

In 1867 Jules Verne crossed the Atlantic on the Great Eastern to America, with his brother Paul. He turned the trip into a novel, A Floating City. See A Floating City.

Key facts

- Isambard Kingdom Brunel (1806–1859): Thames Tunnel, Great Western Railway, Clifton Suspension Bridge, three giant ships
- Great Eastern: about 692 feet, 4,000 passengers, paddles plus screw plus sails
- Built huge to carry coal for a nonstop trip to Australia
- The Suez Canal (1869) made her design obsolete, and she was too big for it
- Launched sideways, stuck for months (1857–58); Brunel died in 1859 just after her trials
- Found her calling laying the 1866 transatlantic cable

Sunday, November 1, 2026

Hurricane, typhoon, cyclone, and the circle of learning

A Taíno storm god, a Chinese great wind and a Greek circle, which also hides inside bicycle and encyclopedia.

Hurricane: a storm god

Hurricane comes from **Hurakán** (**hoo-rah-KAHN**), the storm god of the **Taíno**, the people Columbus met in the Caribbean. Spanish sailors took it home as huracán after Columbus's voyages, and English borrowed it from Spanish in the 1500s.

Farther west, the **Maya** had a creator god named **Huracan**, "one-legged," the heart of the sky. In the Maya creation book, the **Popol Vuh**, Huracan sends a great flood to wipe out an earlier race of failed people. Whether the Taíno and Maya names share a source is debated, but both tie the word to divine wind and flood.

Typhoon: a great wind and a Greek monster

Typhoon probably comes from Chinese **tai fung**, "great wind," as sailors heard it in the South China Sea. In English it blended with **Typhon**, the monstrous Greek storm giant, father of the winds, whom Zeus defeated and buried under Mount Etna. Two origins, one word.

Cyclone: a circle

Cyclone comes from Greek **kyklos**, "circle" or "wheel." It was coined in **1848** by **Henry Piddington**, a British sea captain in **Calcutta** who studied the storms of the Bay of Bengal and wanted one word for winds that whirl in a circle around a center.

Today they are all the same kind of storm: a **hurricane** in the Atlantic and northeast Pacific, a **typhoon** in the northwest Pacific, a **cyclone** in the Indian Ocean and South Pacific.

The same circle in other words

- **Bicycle**: two (bi-) circles, two wheels
- **Cycle**, **recycle**: going around again
- **Encyclopedia**: from Greek **enkyklios paideia**, "education in the circle," the well-rounded circle of subjects a free person should know: en (in) + kyklos (circle) + **paideia** (education, child-rearing)

Paideia: the child root

Paideia comes from **pais, paidos**, "child." The same root gives:

- **Pediatrician**: paidos + iatros (healer), a children's doctor
- **Pedagogy**: from **paidagōgos**, literally "child-leader": in ancient Greece, the slave who walked a boy to school, carried his things and watched his manners. The word for the person who led the child became the word for the art of teaching.

Key facts

- Hurricane: from Taíno storm god Hurakán (hoo-rah-KAHN), via Spanish huracán
- Maya creator god Huracan, "one-legged," sends the flood in the Popol Vuh
- Typhoon: probably Chinese tai fung, great wind, blended with the Greek monster Typhon
- Cyclone: Greek kyklos, circle; coined 1848 by Henry Piddington in Calcutta
- Encyclopedia: enkyklios paideia, education in the circle
- Pedagogy: from paidagogos, the slave who walked a boy to school

Sunday, November 1, 2026

Quarantine, the lazaretto and Yellow Jack

Forty days in Venice, thirty in Ragusa, an island named for Lazarus, and the sailors' flag for yellow fever.

Forty days

Quarantine comes from Venetian **quaranta giorni**, "forty days," the time a ship arriving from a plague port had to wait at anchor before anyone could come ashore.

Ragusa came first, with thirty

The first such law was passed in **1377** by **Ragusa** (today **Dubrovnik**, Croatia), a trading republic on the Adriatic. It required **30 days** of isolation, a trentino. **Venice** later stretched it to **40**.

Why forty? The number was more **biblical than medical**: Jesus's forty days in the desert, Noah's forty days of rain, the forty days of Lent, Moses's forty days on the mountain. It just happened to be longer than the plague's incubation period, so it worked.

The lazaretto

Venice isolated the sick and suspect on an island hospital, the **lazaretto**, named after **Lazarus**: the beggar covered in sores in Jesus's parable (and associated with Lazarus raised from the dead). Lazar became an old word for a leper or diseased beggar.

Yellow Jack

Yellow Jack was sailors' slang for **yellow fever**, named for the **jaundice**, the yellowing of the skin and eyes as the disease attacks the liver. A **jack** is a small flag flown at the bow of a ship, so "yellow jack" was both the disease and the flag.

Yellow fever struck the Americas over and over. In **Philadelphia in 1793**, then the US capital, an epidemic killed about **a tenth of the city's people**, and the government fled.

The Q flag today

Today the plain yellow signal flag stands for the letter **Q** in the international code. Flown alone by an arriving ship, it means the opposite of the old warning: "**my ship is healthy, and I request free pratique**," clearance to come ashore.

One correction to a popular story: the ships under Ragusa's 1377 law did **not** fly a yellow flag. The yellow quarantine flag came much later.

Key facts

- Quarantine: Venetian quaranta giorni, 40 days
- Ragusa (Dubrovnik) passed the first law in 1377, for 30 days; Venice stretched it to 40
- Forty was biblical (desert, Noah, Lent), but longer than plague's incubation
- The lazaretto was named after Lazarus
- Yellow Jack: sailors' slang for yellow fever, named for jaundice; a jack is a small bow flag
- Philadelphia 1793 lost about a tenth of its people to yellow fever
- The yellow Q flag today means "healthy, requesting clearance"

Sunday, November 1, 2026

Walter Reed, Carlos Finlay and yellow fever, Cuba 1900

A Cuban doctor named the mosquito in 1881 and was mocked for 19 years; the US Army proved him right the same month Galveston drowned.

The setting

After the Spanish-American War of **1898**, the **US Army occupied Cuba**. Yellow fever was killing more American soldiers than Spanish bullets had. Most doctors still believed in **miasma**, bad air, or that the disease spread through contact with the sick and their soiled clothes and bedding ("fomites").

Carlos Finlay, the mosquito man

Dr. Carlos Finlay, a Cuban physician in Havana, proposed in **1881** that yellow fever was carried by a mosquito, the one now called **Aedes aegypti**. He presented it to scientific meetings and ran experiments for years, but for **19 years** he was dismissed and mocked as "**the mosquito man**."

The Yellow Fever Commission

In 1900 the Army sent the **Yellow Fever Commission**, led by Major **Walter Reed**, with **James Carroll**, **Jesse Lazear** and **Aristides Agramonte**. They met Finlay, who gave them his mosquito eggs.

- **Jesse Lazear** let infected mosquitoes bite himself and others. In **September 1900** he died of yellow fever after a bite, the **same month as the Galveston hurricane**.
- At **Camp Lazear**, named for him, volunteers (American soldiers and Spanish immigrants, who signed consent forms, among the first in medical history) were tested two ways. Some slept for weeks in a closed hut, in the **soiled bedding and clothing** of yellow fever victims; they **stayed healthy**. Others, in a clean, screened hut, were bitten by infected mosquitoes; they got sick.
- That **disproved the miasma and contact theories** and proved the mosquito.

Gorgas clears Havana, then Panama

Army doctor **William Gorgas** wiped out mosquito breeding places in Havana: screens, draining standing water, oil on cisterns. By **1902** yellow fever was gone from the city. He then did the same in **Panama**, where yellow fever and malaria had helped defeat

the French canal effort of the 1880s. That made the American **Panama Canal** possible (opened 1914).

The irony of 1900

In the same year, the United States **ignored Cuba's hurricane warnings**, and Galveston drowned, while it **used Cuba's mosquito theory**, and **Walter Reed got the credit**. See Galveston 1900 and Benito Viñes.

Latin America remembers Finlay: **December 3**, his birthday, is **Latin American Medicine Day** (Día de la Medicina Latinoamericana).

After Reed

Walter Reed died in **1902** of a burst appendix (peritonitis). **Walter Reed General Hospital** opened in Washington in **1909**. In **2011** it merged with the naval hospital at **Bethesda**, Maryland, as the **Walter Reed National Military Medical Center**.

Key facts

- Carlos Finlay proposed the mosquito (Aedes aegypti) in 1881; mocked for 19 years as "the mosquito man"
- Jesse Lazear died after an infected mosquito bit him, September 1900, the same month as Galveston
- At Camp Lazear, volunteers in soiled bedding stayed healthy: miasma disproved
- Gorgas cleared Havana of yellow fever by 1902, then Panama
- 1900 irony: the US ignored Cuba's hurricane warnings but used its mosquito theory, and Reed got the credit
- December 3, Finlay's birthday, is Latin American Medicine Day
- Reed died 1902 (burst appendix); Walter Reed hospital opened 1909, merged with Bethesda in 2011

Sunday, November 1, 2026

Tambora, the Year Without a Summer and Frankenstein

A volcano in Indonesia darkened the skies of 1816, trapping Byron and the Shelleys indoors, where Mary Shelley began Frankenstein.

The eruption

Mount Tambora, on the island of Sumbawa in Indonesia, erupted in **April 1815**, peaking on April 10–11. It is the **largest eruption in recorded history**. Tens of thousands of people died on the islands from the blast, ash and the famine that followed.

1816, the Year Without a Summer

Tambora threw so much sulfur gas into the upper atmosphere that it formed a haze that reflected sunlight around the world. Temperatures fell. In **1816**:

- **Snow fell in June in New England**, and frosts killed crops through the summer. New Englanders called it "eighteen hundred and froze to death."
- Harvests failed across Europe; there was **famine**, food riots and disease.
- Nobody then knew the cause; the link to Tambora was made long afterward.

It's a disaster that reached around the world: a "bad star" in the oldest sense. See Disaster means bad star.

The Villa Diodati

In June 1816 **Lord Byron** rented the **Villa Diodati** on **Lake Geneva** in Switzerland. Nearby stayed **Percy Bysshe Shelley**, **Mary Godwin** (soon to be Mary Shelley), her stepsister Claire Clairmont, and Byron's doctor, **John Polidori**. The cold, stormy weather kept them indoors. Byron challenged everyone to write a **ghost story**.

- **Mary Shelley**, only **18**, after a waking nightmare of a scientist kneeling beside the thing he had put together, began **Frankenstein; or, The Modern Prometheus**, published in **1818**. Prometheus stole fire from the gods for mankind and was punished forever: the scientist who seizes the power of creation.
- **Polidori** wrote "**The Vampyre**" (1819), the first modern vampire story in English, ancestor of Dracula.

Frankenstein is the scientist

Victor Frankenstein is the young scientist. **The creature has no name.** Calling the monster "Frankenstein" is a later mix-up.

The creature learns

Abandoned by his maker, the creature hides in a hovel beside a cottage and secretly watches the **De Lacey** family. Through a chink in the wall he learns to speak and read as they teach their foreign guest. He finds three books: Milton's **Paradise Lost**, **Plutarch's Lives** and Goethe's **The Sorrows of Young Werther**. From them he learns about history, love, and God and creation.

When he finally faces Victor, he says:

> Remember that I am thy creature; I ought to be thy Adam, but I am rather the fallen angel, whom thou drivest from joy for no misdeed.

Like Dantès and the Abbé Faria, he is educated in secret, in hiding, by a teacher on the other side of a wall. Both are rejected by the world and turn their learning toward revenge.

The grunting monster

The flat-headed, bolt-necked, grunting monster most people picture comes from the **1931** film with **Boris Karloff**. In the novel, the creature is eloquent and quotes Milton.

Key facts

- Tambora erupted April 1815, the largest eruption in recorded history
- 1816, the Year Without a Summer: June snow in New England, famine in Europe
- At the Villa Diodati, Byron's ghost-story challenge led to Frankenstein and The Vampyre
- Mary Shelley was 18; Frankenstein; or, The Modern Prometheus was published in 1818
- Frankenstein is the scientist; the creature has no name
- The creature learns from the De Lacey family and Paradise Lost, Plutarch's Lives and Werther
- "I ought to be thy Adam, but I am rather the fallen angel"
- The grunting monster comes from the 1931 Boris Karloff film

Monday, November 2, 2026

Typhoon Scales vs Saffir–Simpson

Why one storm can be a "Violent Typhoon" in Tokyo and a "Category 5 super typhoon" to the Americans, and how every basin keeps its own scale.

The question

Looking at Wikipedia's RSMC Tokyo typhoon scale, Jeff asked: does Saffir–Simpson not apply to typhoons? The answer: it does, but not officially.

RSMC Tokyo: the official scale

The **Japan Meteorological Agency (JMA)**, acting as **RSMC Tokyo**, is the World Meteorological Organization's official warning center for the western North Pacific. Its scale uses **10-minute sustained winds**:

- **Tropical Depression**: ≤33 kt (≤61 km/h)
- **Tropical Storm**: 34–47 kt (62–88 km/h)
- **Severe Tropical Storm**: 48–63 kt (89–117 km/h)
- **Typhoon**: 64–84 kt (118–156 km/h)
- **Very Strong Typhoon**: 85–104 kt (157–193 km/h)
- **Violent Typhoon**: ≥105 kt (≥194 km/h)

The American view: JTWC

The US **Joint Typhoon Warning Center (JTWC)** measures **1-minute sustained winds**, which read about **12% higher** than 10-minute averages for the same storm. JTWC unofficially applies the **Saffir–Simpson** categories to typhoons, and calls any storm of **≥130 kt** (1-minute) a **super typhoon**, roughly a strong Category 4 to a Category 5.

One storm, two labels

Typhoon Haiyan (called **Yolanda** in the Philippines), 2013, was a **Violent Typhoon** to JMA and a **Category 5 super typhoon** to JTWC. Same storm, different averaging window, different vocabulary.

Every basin has its own scale

- **India (IMD)** ranks North Indian Ocean storms in steps up to **super cyclonic storm**.

- **Australia (Bureau of Meteorology)** has its own **Category 1–5** scale based on **wind gusts**, not sustained winds, so an Australian Category 3 is not the same as a US Category 3.

Comparison table: Tokyo vs Saffir–Simpson

Rule of thumb for converting: **1-minute ≈ 10-minute × 1.14**. The matches below are approximate.

Tokyo (JMA) class	10-min winds	≈ 1-min equivalent	Saffir–Simpson (1-min)
Tropical Depression	≤33 kt	≤38 kt	Tropical Depression (≤33 kt)
Tropical Storm	34–47 kt	39–54 kt	Tropical Storm (34–63 kt)
Severe Tropical Storm	48–63 kt	55–72 kt	Strong Tropical Storm to low Cat 1
Typhoon	64–84 kt	73–96 kt	Cat 1 to Cat 2 (64–95 kt)
Very Strong Typhoon	85–104 kt	97–119 kt	Cat 3 to low Cat 4
Violent Typhoon	≥105 kt	≥120 kt	Strong Cat 4 to Cat 5 (≥137 kt)

Saffir–Simpson reference ranges (1-minute winds):

Category	Winds
Cat 1	64–82 kt
Cat 2	83–95 kt
Cat 3	96–112 kt
Cat 4	113–136 kt
Cat 5	≥137 kt

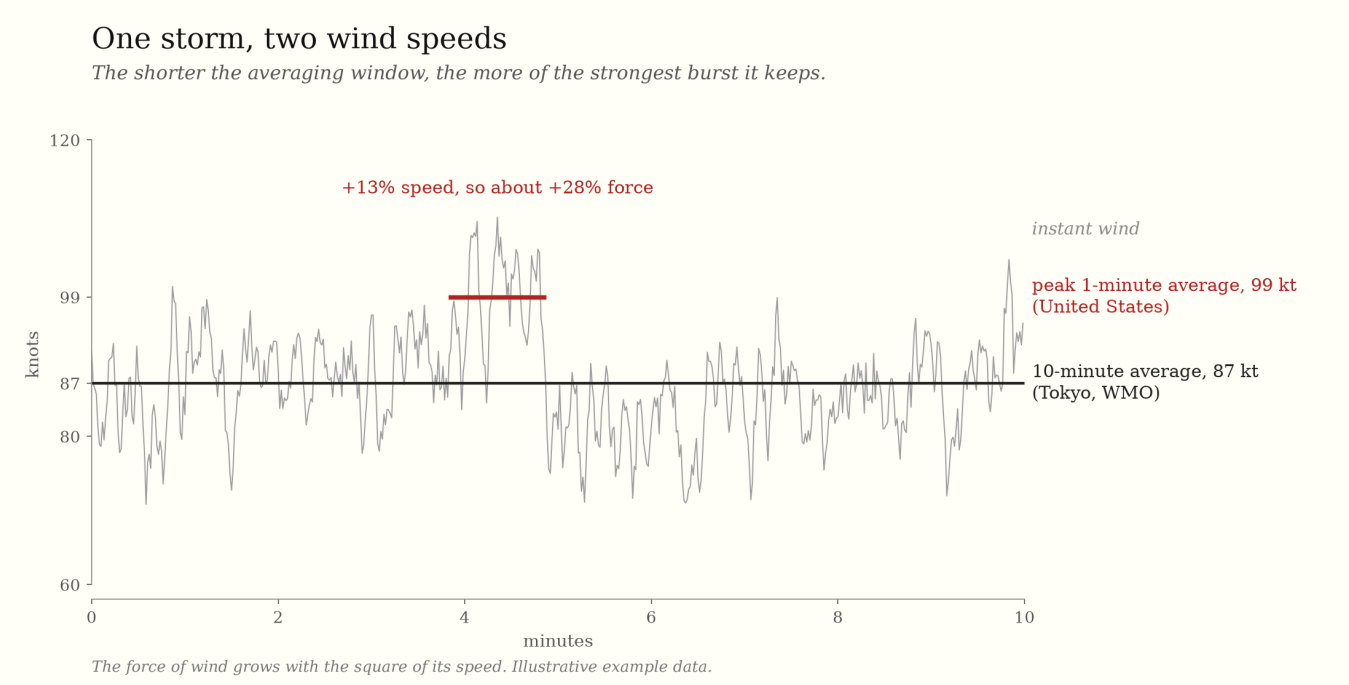
The US **super typhoon** threshold (≥130 kt 1-minute) falls inside Tokyo's **Violent Typhoon** class.

Why 1-minute vs 10-minute matters

Wind is never steady: it comes in **gusts and lulls**. Averaging over a longer window dilutes the gusts with the lulls, so a 10-minute average always gives a **lower number** than a 1-minute average. The 1-minute figure reads about **12% higher**. The US uses 1-minute winds; the **WMO standard is 10-minute**.

> **The force of wind grows with the square of its speed.**

12% more speed ≈ 25% more force; double the speed = four times the force.



One storm, two wind speeds. Grey: instant wind. Black: 10-minute average (Tokyo/WMO). Red: peak 1-minute average (US). Illustrative example data.

- **Same storm, different numbers:** 100 kt (10-min) ≈ 114 kt (1-min). That is **Cat 3 vs Cat 4** for the same wind.
- **Damage:** structures fail in the strongest bursts, and wind force scales with **speed squared**, so ~12% more speed ≈ **~25% more force** ($1.12^2 \approx 1.25$).
- **Records:** always check the averaging period before comparing storms from different agencies or basins.

Connection

This picks up the Nov 1 lesson on the words themselves: **hurricane** from the Taíno god **Hurakán**, and **typhoon** from the Chinese **tai fung**, "great wind." The words differ by ocean, and so do the rulers used to measure the storms.

Key facts

- The force of wind grows with the square of its speed: 12% more speed ≈ 25% more force; double the speed = four times the force.
- Saffir–Simpson applies to typhoons only unofficially; RSMC Tokyo (JMA) is the WMO's official western North Pacific center.
- JMA uses 10-minute sustained winds; Saffir–Simpson and JTWC use 1-minute winds, about 12% higher.
- JMA scale: TD ≤33 kt, TS 34–47, STS 48–63, Typhoon 64–84, Very Strong 85–104, Violent ≥105 kt.

- JTWC "super typhoon" = ≥130 kt 1-minute winds, roughly strong Cat 4 to Cat 5.
- Haiyan (Yolanda) 2013: Violent Typhoon to JMA, Category 5 super typhoon to JTWC.
- India's IMD tops out at "super cyclonic storm."
- Australia's BoM Categories 1–5 are based on gusts and don't match US categories.
- Conversion rule of thumb: 1-minute wind ≈ 10-minute wind × 1.14.
- Saffir–Simpson (1-min): Cat 1 64–82 kt, Cat 2 83–95, Cat 3 96–112, Cat 4 113–136, Cat 5 ≥137.
- Tokyo's Typhoon ≈ Cat 1–2; Very Strong Typhoon ≈ Cat 3 to low Cat 4; Violent Typhoon ≈ strong Cat 4 to Cat 5.
- The US super typhoon threshold (≥130 kt 1-min) falls inside Tokyo's Violent Typhoon class.
- Longer averaging dilutes gusts with lulls, so 10-minute winds read lower; the WMO standard is 10-minute, the US standard 1-minute.
- 100 kt 10-min ≈ 114 kt 1-min: Cat 3 vs Cat 4 for the same storm.
- Wind force scales with speed squared, so ~12% more speed ≈ ~25% more force.
- Check the averaging period before comparing storm records.

Sunday, November 8, 2026

Hurricane Wind Force: Force Grows with Velocity Squared

On a comparable wall, in comparable air, the force of the wind follows the square of the speed. Equal steps in miles per hour add larger and larger pieces of force, but they do not multiply the force by one fixed factor. That fixed factor is the exponential pattern, the one in compound interest, and it is not the wind's force. The power flowing in the wind is a third story, and it follows the cube.

A hurricane wind field in a GIS is a map of speed. The number on the pixel is miles per hour. The load on a wall, a sign, or the edge of a roof is not that number, and it is not the word people reach for when the damage looks as if it ran away. They say the force grows exponentially. For the push itself, under conditions we can name, the honest word is quadratic. Quadratic means the speed is used twice: multiplied by itself.

Force here means the push, a shove you could imagine weighing. Pressure means that push spread over an area. Dynamic pressure is the pressure that moving air brings with it because the air has mass and speed. Density is how much mass sits in a given volume of air. A drag coefficient is the plain number that carries the shape: a flat sign and a round tank do not bargain with the wind in the same way. Area is how much surface the wind meets. Hold the density, the area, and the shape still, and the speed is the only thing left to change.

NASA's Glenn Research Center puts the aerodynamic force in one sentence of that kind. The force equals a constant times the density times the speed times the speed. The dynamic pressure of the flow is one half of the density times the speed squared, and the force is directly proportional to that dynamic pressure. Doubling the speed quadruples the force. The constant is where the area and the shape sit. A National Weather Service technical note makes the density part vivid from the other direction: the same speed pushes less where the air is thinner, higher up, because there is less mass in each volume. Our comparison is the sea-level kind of comparison, comparable air, the same object.

So the working rule, once the words are in place, is small. Dynamic pressure is one half times density times speed times speed. Force is that pressure times area times the drag coefficient. Cross out the things we are holding still, and force stands in proportion to speed times speed.

A fixed baseline of 100 miles per hour

Take 100 miles per hour as the baseline, and ask only about the ratio. A ratio is one speed divided by the other. 120 divided by 100 is 1.20. The miles per hour are in the

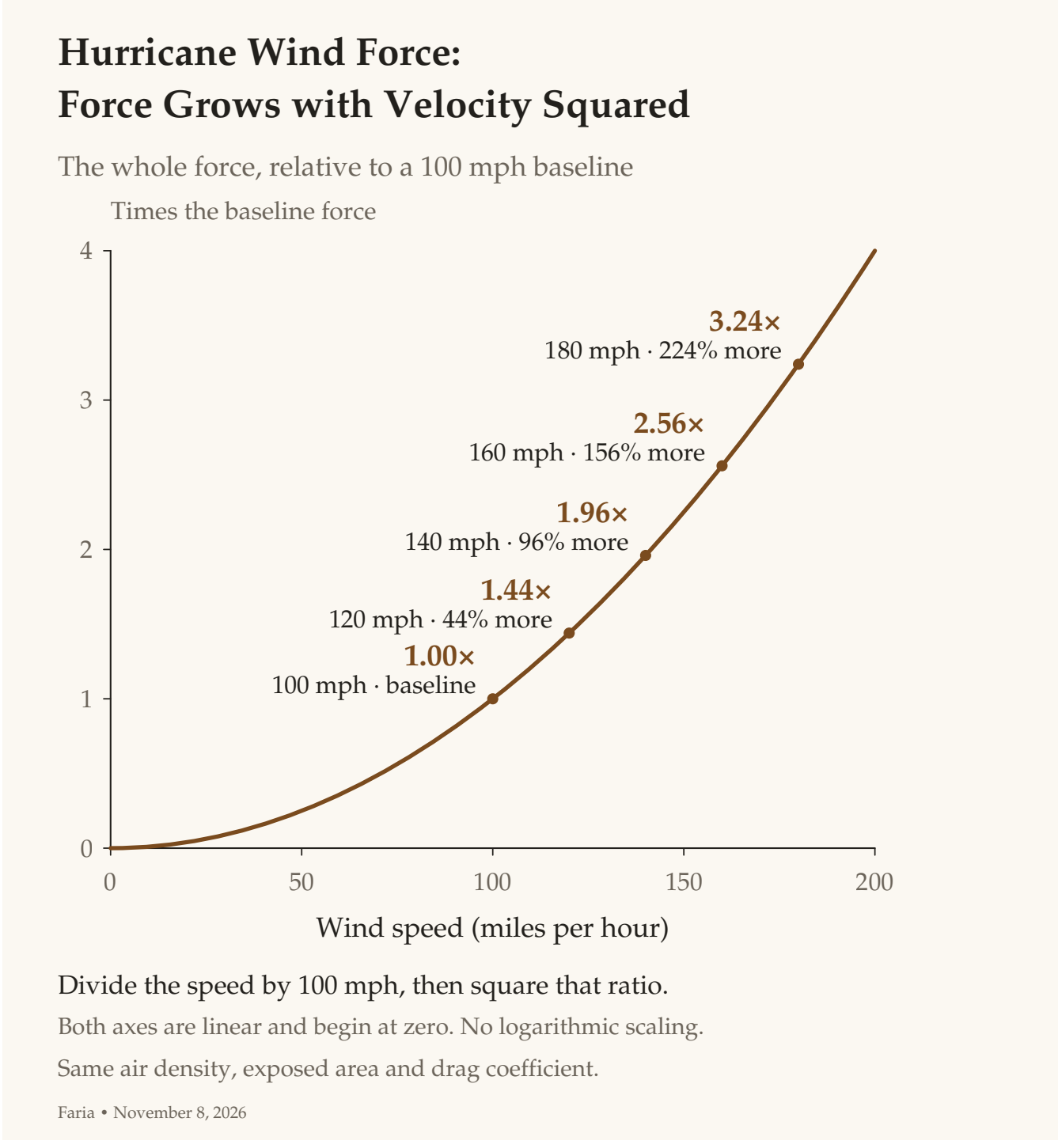
top and the bottom, so they cancel. What is left is the plain number 1.20, which means "1.20 times as fast." Square it: 1.20 times 1.20 is 1.44. The force is 1.44 times the force at 100 miles per hour.

"Times as much" and "percent more" are not two names for one careless phrase. 1.44 times as much means you still have the whole baseline, and then 0.44 of it besides. That extra is 44 percent of the baseline, so the phrase is 44 percent more. It is not 144 percent more. 144 percent more would be 2.44 times the baseline. Say "times" when you mean the whole new amount. Say "percent more" when you mean only the extra.

The same arithmetic, still against 100 miles per hour:

Wind	Speed divided by 100	That number squared: times the baseline force	Percent more than the baseline	Added by this 20 mph step	This force divided by the previous force
100 mph	1.00	1.00	the baseline		
120 mph	1.20	1.44	44%	0.44	1.44
140 mph	1.40	1.96	96%	0.52	1.361
160 mph	1.60	2.56	156%	0.60	1.306
180 mph	1.80	3.24	224%	0.68	1.266

Check the squares on a scrap of paper. 1.4 times 1.4 is 1.96. 1.6 times 1.6 is 2.56. 1.8 times 1.8 is 3.24. The percent-more column is the times column, minus 1, times 100. 2.56 times the baseline is one double, which would be 100 percent more, and then another 0.56, which is 56 percent more: 156 percent more. 3.24 times is a bit more than triple. Triple would be 200 percent more. The extra 0.24 makes 224 percent more.



The whole force follows the square of wind speed. Both axes are linear and begin at zero; air density, exposed area and drag coefficient are held fixed.

Read wind speed along the horizontal axis and the whole force up the vertical axis. Both axes begin at zero and use even spacing. At 100 mph the force ratio is 1.00; at 180 mph it is 3.24: 224 percent more than the baseline. The rising curve is quadratic, not exponential.

https://faria.jbf.com/topics/71-hurricane-wind-force-square-not-exponential/

Why the curve bends, and why that is not exponential

Look down the "added" column. Each equal step of 20 miles per hour brings a larger piece than the step before: 0.44, then 0.52, then 0.60, then 0.68 of the baseline force. The pieces themselves grow by about 0.08 of the baseline each time. That is the square announcing itself. When you multiply a larger speed by itself, the new product jumps farther than the last one did. The graph of force against speed bends upward.

Now look at the last column, the whole force divided by the force at the previous step. The factors are about 1.44, then 1.36, then 1.31, then 1.27. They shrink. Equal steps in miles per hour do not keep multiplying the force by one number.

Exponential is that other pattern, and it needs a clean definition. Take equal steps in whatever you are counting: years, generations, or, in a picture that is not the wind, equal steps of 20 miles per hour. Each new amount is the previous amount multiplied by the same factor. The factor does not change. A bad shorthand says "multiply each step by the last." That sentence does not say which number is the amount and which is the factor. Leave it. The amount becomes the old amount times one fixed number.

Compound interest is the household example. The U.S. Securities and Exchange Commission's Investor.gov glossary defines it as interest paid on principal and on accumulated interest. One hundred dollars at 5 percent a year is 105 dollars after a year. The next year pays 5 percent of 105, not only of the original 100, and the balance is 110.25. Every year the balance is 1.05 times what it was. That 1.05 does not wear out. A population under an ideal constant growth rate is the same arithmetic: each generation a fixed multiple of the one before, and nothing else changing. Real populations are not that obedient. The pattern is the ideal.

Set that pattern beside the wind, with made-up numbers labeled as made-up. Suppose, only as a picture, that every 20 miles per hour multiplied the force by 1.5. Start at 1 for 100 miles per hour. Then 1.50, 2.25, 3.375, 5.0625. Check the divisions: each entry is exactly 1.5 times the one above it. By 180 miles per hour the imaginary rule says a bit more than 5 times the baseline. The square law says 3.24 times. The gap is the definition. The square law's step-to-step factor was already falling. The imaginary rule refuses to fall.

Wind	Square of the speed ratio	Imaginary rule: times 1.5 at every 20 mph
100 mph	1.00	1.00
120 mph	1.44	1.50
140 mph	1.96	2.25
160 mph	2.56	3.375
180 mph	3.24	5.0625

The right-hand column is not a forecast and not a wind-tunnel result. It is there so the fixed factor can be seen next to the square.

The fist, kept honest

The old anchor is a good one if it is not asked to do too much. The same fist, the same mass, at 20 miles per hour instead of 10, has four times the kinetic energy. Kinetic energy is the energy of motion. NASA writes it as one half times the mass times the speed times the speed. Double the speed, and the speed-times-speed part becomes four times what it was. The mass did not change, so the energy is four times.

"Barely stings" is a feeling, and a feeling is not a force. We will not claim that 10 miles per hour barely stings, or that 20 miles per hour hurts four times as much. Pain is not on this page.

The push during the stop is a further sentence. The energy has to be spent over the distance in which the fist is brought to rest. If that distance is the same in the two cases, the average force is four times as well, because the same sort of stop is being asked to swallow four times the energy. If the stop is softer, or shorter, the average force changes even though the energy was fixed by the speed. Comparable stopping distance is the condition. It is not a promise about a particular impact, and this page is not a set of instructions for hitting anything.

Force is not power, and a category is not a force

Do not hand the square law to every phrase that contains the word wind. The rate at which energy arrives in the wind, which is what engineers mean by the power in the wind, follows the cube of the speed when density and area are held still. The U.S. Department of Energy writes that power as one half times density times area times speed times speed times speed. Doubling the speed multiplies that power by eight, not by four. The Government Accountability Office says the same thing in a sentence: the power available in the wind is proportional to the cube of its speed. A NOAA Air Resources Laboratory note on turbines uses that cube as well. Force and power are neighbors. They are not the same measurement. Damage on a street is a third thing again.

The National Hurricane Center's Saffir-Simpson scale is a 1 to 5 rating based only on maximum sustained wind speed. The Center says the scale does not take into account storm surge, rainfall flooding, or tornadoes. Its longer note adds that wind damage also depends on how long the wind lasts, whether the direction changes, how old the buildings are, and what code they were built to. A square-law comparison of two speeds on one wall is a piece of that story. It is not a damage forecast, and it is not a reason to treat every increase in a category as the same multiplication. How the basins name their scales is a separate lesson.

Hold the three sentences apart. The push on a comparable surface follows the square of the speed. The power flowing in the wind follows the cube. An exponential curve is a fixed factor, repeated, as in compound interest. The hurricane did not sign up for that third sentence.

Sources

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- NASA, "STEMonstrations: Kinetic and Potential Energy," kinetic energy as one half times mass times speed squared: https://www.nasa.gov/wp-content/uploads/2023/03/stemonstrations-kinetic-potential-energy.pdf
- U.S. Securities and Exchange Commission, Investor.gov, "Compound Interest," interest paid on principal and on accumulated interest: https://www.investor.gov/introduction-investing/investing-basics/glossary/compound-interest
- National Hurricane Center, "Saffir-Simpson Hurricane Wind Scale," a rating based only on maximum sustained wind: https://www.nhc.noaa.gov/aboutsshws.php
- National Hurricane Center, Saffir-Simpson description (updated May 2021), on duration, wind direction, age of structures, and building codes: https://www.nhc.noaa.gov/pdf/sshws.pdf

Key facts

- With density, area, and shape held still, the force of the wind follows the square of the speed. NASA Glenn states that doubling the speed quadruples that force. The dynamic pressure is one half times density times speed squared.
- Against a 100 mph baseline, 120 mph is 1.44 times the force (44 percent more), 140 mph is 1.96 times (96 percent more), 160 mph is 2.56 times (156 percent more), and 180 mph is 3.24 times (224 percent more). The miles per hour cancel in the ratio.
- "Times as much" is the whole new amount. "Percent more" is only the extra. 1.44 times is 44 percent more, not 144 percent more.
- Equal 20 mph steps add larger pieces of force (0.44, then 0.52, then 0.60, then 0.68 of the baseline), but the factor from one step to the next shrinks. That is quadratic. Exponential means each new amount is the previous amount times one fixed factor, as in compound interest.
- The power in the wind, energy per time, follows the cube of the speed in the standard wind-power equation. It is not the force. Damage also depends on surge, rain, duration, debris, and the building. The square law is not a universal damage rule.
- The same mass at twice the speed has four times the kinetic energy. Average impact force is four times only if the stopping distance is comparable. Pain is not a measurement of that force.

Sunday, November 8, 2026

A Feather and an Anvil Falling Through Air

A feather and an anvil, released from rest, start with the same acceleration, 9.81 metres per second squared. Air resistance then separates the two curves. The sheet uses chosen terminal speeds of 2 and 60 metres per second. Those speeds are an illustration, not measurements. Terminal speed means the acceleration has become zero. The object keeps falling.

Release a feather and an anvil from rest, in air, and follow the downward speed. At the first instant the two speeds rise together. The slope on the sheet is 9.81 metres per second squared for both. That is the familiar acceleration of a body near Earth's surface when the air has not yet begun to push back. The sheet states the consequence in one line: the anvil does not accelerate faster at release.

Acceleration is the change in velocity during a second. Velocity on this page is the downward speed, in metres per second. Time is seconds since the release. Both curves start at zero because both objects start at rest.

Air resistance is the push of the air against a body that is already moving. The push grows as the speed grows. Weight stays the weight. At the opening instant the speed is still zero, so the two objects share that first acceleration. Afterward the paths separate. The feather is light and spread out, and the air's push soon balances its weight. The anvil in this picture is heavy and compact, and it keeps gaining speed much longer.

Balance is the terminal speed. The air's push has matched the weight, so the acceleration is zero and the downward speed stops changing. The object is still moving downward at that speed. Terminal speed is the end of the speeding-up. It is the continued fall at a steady pace.

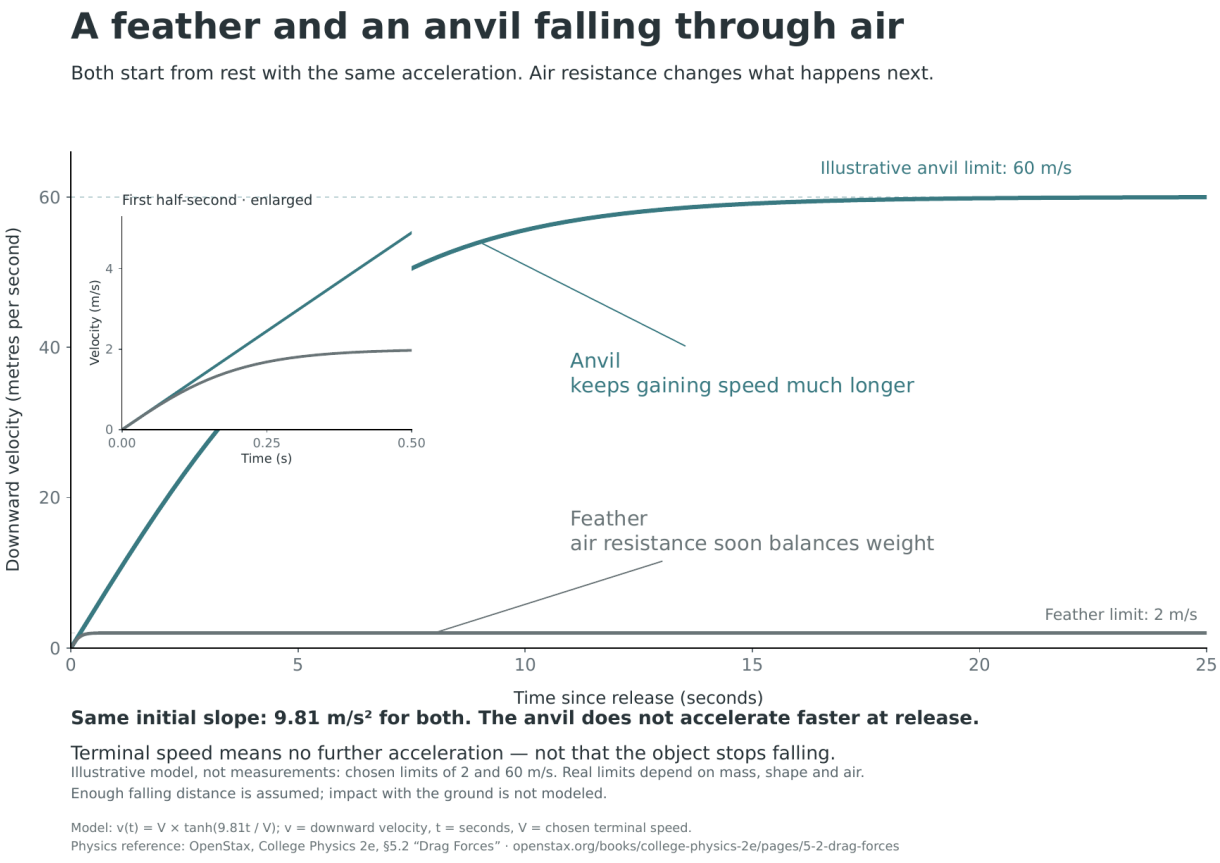
The sheet is an illustrative model. It is not a record of a measured feather or a measured anvil. The chosen limit for the feather is 2 metres per second. The chosen limit for the anvil is 60 metres per second. A real object takes its limit from its mass, its shape, and the air it falls through. The two numbers were picked so both curves fit on one graph.

The formula printed on the sheet is $v(t) = V \times \tanh(9.81\ t / V)$. Here v is the downward velocity, t is the time in seconds, and V is the chosen terminal speed. tanh is the hyperbolic tangent. At the instant of release, $t = 0$, and the formula gives $v = 0$. The opening slope is 9.81 metres per second squared for either choice of V. That is why the curves leave the origin together. As time goes on, v approaches V and then stays there. This is the formula for a drag force that grows with the square of the speed.

The small panel enlarges the first half-second, so the shared slope can be read before the feather's curve bends over. The main panel runs to 25 seconds. By then the feather

has long been at its chosen 2 metres per second, still falling, and the anvil is still climbing toward its chosen 60.

The sheet assumes enough distance to fall. Impact with the ground is not part of the model.



A feather and an anvil falling through air. Both start from rest with the same acceleration. The feather's chosen limit is 2 metres per second. The anvil's chosen limit is 60.

Same initial slope, 9.81 metres per second squared. The feather's air resistance soon balances its weight. The anvil keeps gaining speed much longer. The limits, 2 and 60 metres per second, are chosen for the drawing.

The file itself, unchanged, is A feather and an anvil falling through air (PDF).

Sources

- The velocity-time sheet: illustrative limits of 2 and 60 metres per second, a shared initial slope of 9.81 m/s², and the model v(t) = V × tanh(9.81 t / V).
- OpenStax, College Physics 2e, section 5.2, "Drag Forces," the reference named on the sheet: https://openstax.org/books/college-physics-2e/pages/5-2-drag-forces

Key facts

- Both objects start from rest. The initial slope is 9.81 m/s² for both. The anvil does not accelerate faster at release.
- The limits 2 m/s and 60 m/s are chosen for the illustration. They are not measurements. A real terminal speed depends on mass, shape, and the air.
- Terminal speed means the acceleration has become zero. The object keeps falling at that steady speed.
- The model assumes enough falling distance. Impact with the ground is outside the sheet.

5. Words, books, memory and history

The remaining public lessons follow their original topic order. No new lessons have been invented to bridge gaps in the source collection.

26 complete topics • begins on binder page 232

Saturday, October 31, 2026

Disaster means "bad star

The word disaster comes from Italian dis-astro, a bad star, and belongs to a big family of star words.

The word

Disaster comes from the Italian disastro: dis- (bad, ill, apart) plus astro (star). It reached English through French désastre in the late 1500s.

The idea behind it is astrology. People believed the position of the stars and planets ruled what happened on Earth. A catastrophe meant the stars had lined up against you, an "ill-starred" event. So every time a disaster response team deploys, it is, in the oldest sense of the word, answering a bad star.

The Greek and Latin root

The root goes back to Greek astēr / astron (star), which also gave Latin astrum. Words in the same family:

- **Astronomy**: astron + nomos (law, arrangement), the laws of the stars
- **Astrology**: astron + logos (word, study), reading the stars for meaning
- **Astronaut**: astron + nautēs (sailor), a star sailor (see the next topic)
- **Asterisk** (\): asteriskos*, "little star", the star-shaped mark on a page
- **Aster**: the flower named for its star-shaped bloom
- **Asteroid**: "star-like", because early telescopes showed them as points like stars
- **Disaster**: the bad star

Shakespeare used it this way

In King Lear, Edmund mocks people who "make guilty of our disasters the sun, the moon, and the stars." He means it literally: blaming the stars. See Edmund's star speech.

Key facts

- Disaster: Italian dis-astro, bad star
- Astro words share Greek astron (star): astronomy, astrology, astronaut, asterisk, aster, asteroid
- "Ill-starred" carries the same astrology idea in plain English

Saturday, October 31, 2026

Astronaut, nautical and the sailor root

Astronaut means star sailor. The naut- root runs through nautical, nausea and Nautilus.

Star sailor

Astronaut is Greek astron (star) + nautēs (sailor). So an astronaut is a "star sailor." The Soviet word **cosmonaut** is kosmos (the ordered universe) + nautēs, a "universe sailor."

The sailor family

Greek naus means ship, and nautēs is the person who sails it. Latin borrowed the same idea as navis (ship). Relatives:

- **Nautical**: to do with sailors and ships
- **Nautical mile**: the sailor's mile, one minute of latitude (about 1.15 ordinary miles)
- **Navy**, **naval**, **navigate**: from Latin navis
- **Nausea**: literally seasickness, from naus
- **Argonauts**: the sailors of the ship Argo in the Jason myth
- **Nautilus**: see below

Naut- yes, but what is -tilus?

Nautilus is the Greek nautilos, "sailor," built on the same naut- root. The -ilos ending just makes it a noun, so there is no separate "tilus" word. The Greeks gave the name to the sea creature with the coiled shell (the chambered nautilus, and the paper nautilus), because they imagined it sailing on the surface.

Robert Fulton borrowed the name for his 1800 submarine, and Jules Verne borrowed it again for Captain Nemo's submarine. See Fulton's Nautilus.

Key facts

- Astronaut: Greek astron (star) + nautēs (sailor) = star sailor
- Nausea literally means seasickness (Greek naus, ship)
- Nautilus: Greek nautilos, sailor; no separate "tilus" root

Saturday, October 31, 2026

King Lear: Edmund's star speech and the scientific method

Gloucester blames eclipses; his son Edmund mocks blaming the stars, a modern idea in a play about ancient Britain.

The setup (Act 1, Scene 2)

The Earl of Gloucester has just been fooled by his son Edmund into thinking his other son, Edgar, is plotting against him. Shaken, he blames the sky:

> These late eclipses in the sun and moon portend no good to us.

There were in fact real eclipses in 1605, just before Shakespeare wrote the play, and pamphlets were predicting doom from them. His audience would have heard it as current news.

Edmund's answer

Once his father leaves, Edmund, alone on stage, mocks the whole idea:

> This is the excellent foppery of the world, that when we are sick in fortune, often the surfeit of our own behaviour, we make guilty of our disasters the sun, the moon, and the stars; as if we were villains by necessity, fools by heavenly compulsion, knaves, thieves, and treachers by spherical predominance; drunkards, liars, and adulterers by an enforced obedience of planetary influence.

And then:

> I should have been that I am, had the maidenliest star in the firmament twinkled on my bastardizing.

In plain words: "We do bad things, then blame the planets. I'd be exactly who I am no matter which star was shining when I was conceived."

Why it matters

Notice the word "disasters" here. It means "bad stars" literally (see Disaster means bad star). Edmund rejects it.

The irony is that the villain has the modern, almost scientific view: look for causes in people's behaviour, not in signs in the sky. This was the moment when that view was being born. Francis Bacon, Shakespeare's contemporary, was arguing that knowledge should come from observation and testing rather than from authority and omens. His

Advancement of Learning came out in 1605, the same time as Lear, and his Novum Organum (1620) laid out what became the scientific method.

Shakespeare doesn't make it simple, though. The clear-eyed man is the liar and schemer, and the superstitious old father is the one who ends up blinded.

Key facts

- Gloucester: "These late eclipses in the sun and moon portend no good to us" (1.2)
- Edmund: we "make guilty of our disasters the sun, the moon, and the stars"
- Real eclipses in 1605 came just before Lear was written
- Francis Bacon, Shakespeare's contemporary, pushed observation over omens (Advancement of Learning, 1605)

Saturday, October 31, 2026

King Lear: the whole play

Two families, two foolish fathers. Lear has three daughters; Gloucester has two sons.

Two stories woven together

King Lear (written about 1605–06) has a main plot and a subplot that mirror each other.

- **Lear's family**: King Lear and his three daughters, **Goneril**, **Regan** and **Cordelia**.
- **Gloucester's family**: the Earl of Gloucester and his two sons, **Edgar** (legitimate, older) and **Edmund** (illegitimate, the "bastard").

That's where the sons come in. Each father misjudges his children, trusts the false ones and drives out the loyal one.

Act 1: the love test

Old Lear decides to divide Britain among his daughters, giving the most to whoever says she loves him most. Goneril and Regan flatter him extravagantly. Cordelia, the youngest and truest, won't play: she says she loves him "according to my bond, no more nor less." Lear disowns her. The King of France marries her anyway, without a dowry. The loyal Earl of Kent speaks up for her and is banished, but returns in disguise to keep serving Lear.

Meanwhile Edmund forges a letter to make Gloucester believe Edgar plans to kill him. Edgar flees and disguises himself as a mad beggar, **Poor Tom**.

Acts 2–3: the storm

Lear goes to stay with Goneril, then Regan. Both cut his knights and treat him with contempt. Lear rages out into a storm on the heath with his Fool and the disguised Kent. In the storm he begins to go mad, and also to see clearly for the first time how the poor live (see Lear's great lines). They meet Poor Tom (Edgar).

Gloucester secretly helps Lear. Edmund betrays him to Regan and her husband Cornwall, who gouge out Gloucester's eyes. Only then does Gloucester learn Edmund betrayed him and Edgar was innocent.

Act 4: the blind and the mad

Edgar, still disguised, leads his blind father toward Dover. Gloucester wants to throw himself off the cliffs; Edgar tricks him into a harmless fall to save him. Cordelia lands

with a French army to rescue her father. Lear, mad and crowned with wildflowers, meets blind Gloucester. Then Lear wakes, cared for by Cordelia, and asks her forgiveness: "I am a very foolish fond old man."

Act 5: the ending

The British forces win and Lear and Cordelia are captured. Goneril and Regan both lust after Edmund; Goneril poisons Regan, then kills herself. Edgar fights and mortally wounds Edmund in a duel. Dying, Edmund tries to cancel his order to have Cordelia killed, but it is too late. Lear enters carrying her dead body, and dies over it. Gloucester has already died, his heart bursting between joy and grief when Edgar revealed himself. Edgar, Kent and Albany (Goneril's decent husband) are left.

Key facts

- Lear: three daughters, Goneril, Regan, Cordelia
- Gloucester subplot: two sons, Edgar (loyal) and Edmund (illegitimate, the schemer)
- Edgar hides as the mad beggar Poor Tom
- Cordelia: loves him "according to my bond, no more nor less"

Saturday, October 31, 2026

King Lear: the great lines, and how to say Gloucester

Poor naked wretches, I have ta'en too little care of this, a very foolish fond old man, and GLOSS-ter.

Pronunciation: Gloucester

GLOSS-ter (rhymes with "foster"). Two syllables. The middle "-ces-" is silent, just as in Worcester (WUSS-ter) and Leicester (LESS-ter). It is an English county and city, and the earldom in the play is named for it.

"Poor naked wretches" (Act 3, Scene 4)

Out in the storm, just before he goes into the hovel and meets Poor Tom, Lear thinks of the homeless for the first time in his life:

> Poor naked wretches, wheresoe'er you are,
> That bide the pelting of this pitiless storm,
> How shall your houseless heads and unfed sides,
> Your looped and windowed raggedness, defend you
> From seasons such as these? O, I have ta'en
> Too little care of this! Take physic, pomp;
> Expose thyself to feel what wretches feel,
> That thou mayst shake the superflux to them,
> And show the heavens more just.

(The order matters: this speech comes **before** he meets Poor Tom, not after.)

In plain words: "I was king and never thought about people with no roof. Medicine for the powerful: go out and feel what they feel, then give them your surplus." Looped and windowed means full of holes; superflux means the extra you don't need.

For anyone in disaster work it is the line that hits hardest: **"I have ta'en too little care of this."**

"A very foolish fond old man" (Act 4, Scene 7)

Lear wakes from madness to find Cordelia caring for him:

> Pray, do not mock me:
> I am a very foolish fond old man,
> Fourscore and upward, not an hour more nor less;

> And, to deal plainly,
> I fear I am not in my perfect mind.

Fond here means foolish, doting, which was its older meaning. Fourscore is eighty.

Key facts

- Gloucester is pronounced GLOSS-ter
- "O, I have ta'en too little care of this!" (King Lear 3.4)
- "I am a very foolish fond old man" (King Lear 4.7)
- "Poor naked wretches" comes before Lear meets Poor Tom
- Fond once meant foolish; fourscore means eighty

Saturday, October 31, 2026

Fulton's Nautilus, a submarine for Napoleon

Robert Fulton built a working submarine for France in 1800, seven years before his famous steamboat.

The surprise

Robert Fulton (1765–1815), the American famous for the steamboat, built a **submarine first**. In 1800, living in France, he built the **Nautilus** and offered it to Napoleon's government as a weapon against the British navy, which was blockading France.

The boat

- About 21 feet long, copper over an iron frame, shaped like a stretched egg.
- Hand-cranked propeller underwater; a folding mast and sail for the surface.
- Tested on the Seine and off the French coast at Le Havre and Brest; Fulton and a small crew stayed submerged for hours using compressed air.
- Its weapon was a "torpedo" (what we would now call a mine), towed and attached to an enemy hull.

What happened

It worked, but it never sank a British ship: the British ships simply sailed away whenever it came near. Napoleon lost interest, and the French admirals thought it a dishonorable way to fight. Fulton then took the idea to the British, who also passed.

So he went home and turned to steam. His **Clermont** (officially North River Steamboat) began running on the Hudson from New York to Albany in **1807**, the first commercially successful steamboat. **The sub preceded the steamboat by seven years.**

The name

Nautilus is Greek nautilos, "sailor" (see Astronaut and nautical). Jules Verne named Captain Nemo's submarine after Fulton's boat, and has Nemo mention it. Later, the US Navy's first nuclear submarine, USS Nautilus (1954), carried the name on.

Key facts

- Fulton built the Nautilus submarine for Napoleon in 1800, seven years before his steamboat
- Clermont steamboat on the Hudson, 1807
- Verne's Nautilus was named after Fulton's

Saturday, October 31, 2026

The Count of Monte Cristo and Napoleon

Edmond Dantès is jailed for carrying a letter from Elba, and meets the Abbé Faria in the Château d'If.

The book

Alexandre Dumas published The Count of Monte Cristo as a newspaper serial in 1844–46.

The Napoleon connection

The story starts in Marseille in **February 1815**. Napoleon has been exiled to the island of **Elba** off Italy, and France has its king back. Young sailor **Edmond Dantès**, about to be made captain and to marry his fiancée **Mercédès**, has stopped at Elba on his dying captain's orders and is carrying a letter for Bonapartists in Paris.

Three men who envy him (Danglars, who wants his job; Fernand, who wants Mercédès; and the cowardly neighbor Caderousse) send an anonymous denunciation calling him a Bonapartist agent. He is arrested at his own betrothal feast. The prosecutor, **Villefort**, sees that the letter is addressed to his own father, a Bonapartist, and buries Dantès in the island fortress of the **Château d'If** to protect himself.

Days later Napoleon really does escape Elba and returns for the **Hundred Days**, ending at Waterloo in June 1815. In that political chaos, a prisoner like Dantès is forgotten.

The Abbé Faria

After years alone, Dantès hears scratching: another prisoner, the **Abbé Faria**, an Italian priest digging an escape tunnel, has dug into the wrong cell. Faria becomes his teacher. He teaches Dantès languages, history, science and mathematics, and he works out, by pure reasoning, who betrayed him. He also tells him of a hidden treasure on the island of **Monte Cristo**.

When Faria dies, Dantès takes his place in the burial sack, is thrown into the sea, escapes, finds the treasure, and returns as the Count of Monte Cristo to take revenge.

(This website is named for the Abbé: the teacher in the next cell.)

Key facts

- Monte Cristo opens in 1815 with Napoleon exiled on Elba
- Dantès is jailed in the Château d'If for carrying a letter from Elba

- The Abbé Faria teaches Dantès everything and reveals the treasure of Monte Cristo

Saturday, October 31, 2026

Dumas's father, the general who defied Napoleon

General Thomas-Alexandre Dumas, born enslaved in Saint-Domingue, fell out with Napoleon and was imprisoned, the seed of Monte Cristo.

The general

Alexandre Dumas's father was **Thomas-Alexandre Dumas** (1762–1806), born in the French colony of **Saint-Domingue** (now Haiti), the son of a French marquis and an enslaved African woman, Marie-Cessette Dumas. His father brought him to France, where he joined the army as a private under his mother's surname.

In the Revolution he rose spectacularly to become a general, the highest-ranking man of color in any European army of his time, famous for his size, strength and courage.

Napoleon

He served under Napoleon in the invasion of **Egypt** in 1798 and openly criticized the campaign. The two fell out. Sailing home in 1799, his ship put in at **Taranto** in the Kingdom of Naples, where he was taken prisoner and held for about two years in a fortress dungeon, falling ill (he believed he was being poisoned). Napoleon did nothing to help him and never gave him his back pay or pension after his release. He died in 1806, when his son Alexandre was three.

Into the novel

A man unjustly imprisoned in a fortress, abandoned, betrayed by those in power: Dumas's father's story echoes through Edmond Dantès. Tom Reiss told the full story in The Black Count (2012), which won the Pulitzer Prize.

Key facts

- Dumas's father, General Thomas-Alexandre Dumas, was born in Saint-Domingue to a French noble and an enslaved woman
- He fell out with Napoleon in Egypt and was imprisoned at Taranto, 1799 to 1801
- Tom Reiss, The Black Count (2012, Pulitzer)

Saturday, October 31, 2026

Jules Verne's A Floating City

Verne's 1871 novel of his own 1867 crossing on the Great Eastern. Good as a travelogue today, thin as a story.

Where it came from

In spring 1867 Jules Verne and his brother Paul sailed from Liverpool to New York on Brunel's **Great Eastern**, then went up to see **Niagara Falls** before returning. Verne turned it into Une ville flottante, **A Floating City** (1871). The title is the ship itself: a city afloat, with thousands of people aboard.

The storyline

The narrator (essentially Verne) boards the Great Eastern. He meets an old friend, **Captain Fabian MacElwin**, a British officer who is heartbroken: the woman he loved, **Ellen**, was forced to marry another man and has lost her reason from grief. It turns out that Ellen is aboard, with her cruel husband, **Harry Drake**, a gambler who married her for money.

Through the crossing, storms, shipboard life and gossip, tension builds between Fabian and Drake, until a duel is arranged in America. Before it can happen, Drake is killed by lightning during a storm at sea (a convenient act of heaven). At Niagara, Ellen's mind returns, and she and Fabian can be together.

Is it any good today?

Honestly, the romance is melodrama and the plot is thin. What makes it worth reading is the **ship**: Verne's eyewitness description of the Great Eastern, her engines, her paddle wheels, her crowds, the storm that smashed her wheel, and life aboard the largest ship in the world. It's a short read and best taken as a travel journal by a man who loved machines. For the ship, see the Great Eastern.

Key facts

- A Floating City (1871) is based on Verne's own 1867 Great Eastern crossing
- The story: Fabian, Ellen and her villainous husband Harry Drake; ends at Niagara Falls
- Worth reading today mainly for the eyewitness picture of the Great Eastern

Saturday, October 31, 2026

Twenty Thousand Leagues Under the Seas

Professor Aronnax, Conseil and Ned Land aboard the Nautilus with Captain Nemo, "no one." The leagues are distance traveled, not depth.

The book

Jules Verne serialized Vingt mille lieues sous les mers in 1869–70. The title means distance traveled underwater, **20,000 leagues**, about 50,000 miles (twice around the world), **not depth**. No sea is anywhere near that deep.

The characters

- **Professor Pierre Aronnax**, French naturalist and narrator
- **Conseil**, his devoted servant, obsessed with classifying every creature
- **Ned Land**, a hot-tempered Canadian harpooner who wants only to escape
- **Captain Nemo**, the mysterious commander. Nemo is Latin for "**no one**," the same trick Odysseus played on the Cyclops by calling himself "Nobody."

The story

In 1866 ships report a huge sea monster. The US frigate Abraham Lincoln hunts it with Aronnax, Conseil and Ned aboard. In the struggle the three are thrown overboard and land on the "monster," which is a submarine, the **Nautilus**, named after Fulton's (see Fulton's Nautilus).

Nemo has cut himself off from all nations. He runs the Nautilus on electricity and lives entirely from the sea. His guests are prisoners in luxury: they can never leave. Highlights of the voyage:

- An undersea hunting trip through a forest on the sea floor, in diving suits
- A burial in a coral cemetery for a crewman
- Pearl beds off Ceylon, where Nemo saves an Indian pearl diver from a shark
- Through the secret "Arabian Tunnel" from the Red Sea to the Mediterranean
- Walking among the ruins of **Atlantis**
- Reaching the **South Pole** (in 1868, decades before Amundsen), then being trapped under the ice and cutting free with air running out
- Passing the **transatlantic telegraph cable** on the sea floor
- The fight with the **giant squid** (poulpes in French), which kills a crewman

- Nemo rams and sinks a warship from the nation that wronged him, and weeps before the portraits of his wife and children

Finally, off Norway, the Nautilus is caught in the **Maelstrom**, the great whirlpool. The three prisoners escape in the dinghy and wake up on shore. The fate of Nemo and the Nautilus is unknown.

Who was Nemo?

Verne answers that in The Mysterious Island (1874–75). Nemo is **Prince Dakkar**, an Indian prince, son of a rajah, who fought the British in the 1857 rebellion, lost his family and his country, and vanished under the sea. (Verne had first imagined him as a Pole fighting Russia, but his publisher worried about the Russian market.)

Nemo and Dantès

Like Edmond Dantès in The Count of Monte Cristo, Nemo is a man destroyed by injustice who reinvents himself under a new name, possesses vast hidden wealth and knowledge, and uses it to take revenge. Both are rescued from despair by learning, and both find that revenge does not bring peace. See Monte Cristo and Napoleon.

Key facts

- Twenty Thousand Leagues (1869–70): 20,000 leagues is distance traveled, about 50,000 miles, not depth
- Nemo is Latin for "no one"
- Aronnax, Conseil and Ned Land escape the Nautilus in the Maelstrom off Norway
- In The Mysterious Island, Nemo is revealed as Prince Dakkar of India

Saturday, October 31, 2026

Mark Twain's The Innocents Abroad

Twain sailed on the Quaker City in 1867 and mailed newspaper letters home as he went; they became his best-selling book.

The voyage

In June 1867 Mark Twain, then a 31-year-old newspaper writer, joined one of the first organized American pleasure cruises, aboard the steamship **Quaker City**, for about five months through the Mediterranean: France, Italy, Greece, Constantinople, the Black Sea, the Holy Land and Egypt.

Did he send installments while traveling? Yes.

The San Francisco **Daily Alta California** paid his fare (about $1,250) in exchange for travel letters, about **$20 apiece**. He wrote around 50 letters along the way and mailed them from ports. They were printed as they arrived, weeks later. He also sent several to the New York Tribune and the New York Herald.

Back home, he reworked the letters into a book, The Innocents Abroad, or The New Pilgrims' Progress (**1869**), sold door to door by subscription. It was his best-selling book in his lifetime and one of the best-selling travel books ever.

Why it was fresh

Travel books of the time were reverent. Twain was not. He mocked the guides who showed him the "authentic" relics everywhere, his fellow pilgrims who chipped souvenirs off monuments, and himself. An example: in Italy, he and his friends tortured guides by asking, about every famous name, "Is he dead?"

It was the same summer, 1867, that Jules Verne crossed the Atlantic on the Great Eastern. See A Floating City.

Key facts

- Twain sailed on the Quaker City, June to November 1867
- He mailed about 50 letters to the Daily Alta California, about $20 each, as he traveled
- The Innocents Abroad (1869) was his best-selling book in his lifetime

Sunday, November 1, 2026

Clue, the Labyrinth and the Minotaur

A clue was a ball of thread, Ariadne's thread. The labyrinth may be the house of the double axe.

A ball of thread

Clue comes from **clew**, an old English word for a **ball of thread or yarn**. The story behind it is Greek myth.

Theseus, Ariadne and the Minotaur

King **Minos** of Crete kept the **Minotaur**, half man, half bull, in the **Labyrinth**, a maze built by Daedalus. Athens had to send young men and women every few years to be fed to it. The hero **Theseus** went as one of them. Minos's daughter **Ariadne** fell in love with him and gave him a **ball of thread**. He tied one end at the entrance, unwound it as he went in, killed the Minotaur, and followed the thread back out.

So a clew was the thing that leads you out of a maze, and by the 1600s, spelled **clue**, it meant anything that leads you to the answer to a puzzle.

A **clew** is also a sailing term: the **lower corner of a sail**, where the sheet line attaches.

Labyrinth: house of the double axe?

Labyrinth may come from **labrys**, the Minoan **double-headed axe**, a sacred symbol carved all over the great palace at **Knossos** on Crete. "Labyrinth" would then mean "**house of the double axe**," the vast, confusing palace itself. The link is likely but not proven.

Minotaur

Minotaur = **Minos** + **tauros**, **bull**: "Minos's bull." The same taur is in **Taurus**, the bull of the zodiac.

Key facts

- Clue: from clew, a ball of thread, Ariadne's thread for Theseus in the Labyrinth
- The spelling "clue" was in use by the 1600s; a clew is also a sail's lower corner
- Labyrinth may come from labrys, the Minoan double axe: house of the double axe at Knossos
- Minotaur: Minos + tauros, bull, as in Taurus

Sunday, November 1, 2026

Shakespeare's Sonnet 18: Shall I compare thee

The full sonnet, its turn in the final couplet, and the legal metaphor of summer's lease.

The poem

> Shall I compare thee to a summer's day?
> Thou art more lovely and more temperate:
> Rough winds do shake the darling buds of May,
> And summer's lease hath all too short a date;
> Sometime too hot the eye of heaven shines,
> And often is his gold complexion dimm'd;
> And every fair from fair sometime declines,
> By chance or nature's changing course untrimm'd;
> But thy eternal summer shall not fade,
> Nor lose possession of that fair thou ow'st;
> Nor shall Death brag thou wander'st in his shade,
> When in eternal lines to time thou grow'st:
> So long as men can breathe or eyes can see,
> So long lives this, and this gives life to thee.

The shape

A Shakespearean sonnet is 14 lines: three groups of four (quatrains) and a rhyming couplet. The first eight lines show how summer fails: too windy, too short, too hot, too cloudy, and every beautiful thing fades. Line 9 turns with "**But**": your summer won't fade.

The real **turn** comes in the final couplet. Why won't the beloved fade? Because of "**this**," **the poem itself**. As long as people read it, it keeps the beloved alive. The poem claims to beat Time, and four centuries later it is still being read, so it was right. (Compare Ozymandias, where only words survive too.)

Summer's lease: a legal metaphor

"**And summer's lease hath all too short a date**" is property law.

- Summer is a **tenant** renting the year.
- **Date** means the **end date of the lease**, when it expires.
- **Time** is the **landlord**, who always takes the property back.

"**Nor lose possession of that fair thou ow'st**": ow'st means **ownest**, "the beauty you own." "Possession" is a legal term too. The poem gives the beloved an **eternal lease**: possession that Time can't end.

Shakespeare the lawyer?

Legal language runs all through the sonnets. **Sonnet 87** ("Farewell! thou art too dear for my possessing") is built on contracts: "The **charter** of thy worth gives thee releasing; / My **bonds** in thee are all determinate." Because of this, some scholars have argued Shakespeare worked as a lawyer's clerk in his "lost years" before London. There's no record to prove it; legal talk was common among Elizabethans, who sued each other constantly.

Key facts

- Sonnet 18's final couplet turns on "this," the poem itself, which gives the beloved life
- "Summer's lease hath all too short a date": summer the tenant, date the lease's end, Time the landlord
- "Thou ow'st" means "you own"
- Sonnet 87 uses charter and bonds; the lawyer theory is unproven

Sunday, November 1, 2026

T. S. Eliot and The Waste Land

A St. Louis boy who became an English banker-poet, and the 1922 poem that opens "April is the cruellest month.

The man

Thomas Stearns Eliot (1888–1965)

- Born in **St. Louis**, Missouri, to an old New England family
- Studied at **Harvard**, then in Paris, Germany and Oxford
- Settled in **London in 1914**, just as World War I began
- Worked in the foreign department of **Lloyds Bank** (1917–1925), then as an editor at the publishers Faber
- Became a **British citizen** and joined the **Church of England** in **1927**
- **Nobel Prize in Literature, 1948**

The Waste Land (1922)

Eliot's most famous long poem, written after the war and during a breakdown. It is a collage of voices, languages and quotations about a broken civilization. It opens:

> April is the cruellest month, breeding
> Lilacs out of the dead land, mixing
> Memory and desire, stirring
> Dull roots with spring rain.

This **reverses Chaucer**. The Canterbury Tales opens with April as the sweet month when showers bring the world back to life ("Whan that Aprill with his shoures soote") and people set off on pilgrimage. For Eliot, spring is cruel because it forces life and memory back into people who would rather stay numb.

Other famous lines:

- "**I will show you fear in a handful of dust.**"
- "**Those are pearls that were his eyes**," borrowed from Ariel's song in Shakespeare's The Tempest ("Full fathom five thy father lies... Those are pearls that were his eyes"), about a man drowned and changed by the sea.

Cats

Eliot had a playful side. Old Possum's Book of Practical Cats (1939), light verse he wrote for his godchildren ("Old Possum" was his nickname from Ezra Pound), became Andrew Lloyd Webber's musical **Cats** (1981).

Key facts

- T. S. Eliot, 1888 to 1965: born St. Louis, Harvard, London 1914, Lloyds Bank
- British citizen and Church of England, 1927; Nobel Prize 1948
- The Waste Land (1922): "April is the cruellest month" reverses Chaucer's sweet April
- "I will show you fear in a handful of dust"
- "Those are pearls that were his eyes" comes from The Tempest
- Old Possum's Book of Practical Cats (1939) became the musical Cats

Sunday, November 1, 2026

The Love Song of J. Alfred Prufrock

Coffee spoons, a peach, an attendant lord and the mermaids. Eliot's 1915 poem of a man afraid of looking foolish.

The poem

Published in **1915** (written mostly in 1910–11, when Eliot was barely in his twenties), it is the inner voice of a timid, aging man going to a society tea, wanting to ask a woman an "overwhelming question" and never daring to.

The opening

> Let us go then, you and I,
> When the evening is spread out against the sky
> Like a patient etherized upon a table;

A shocking image in 1915: the sunset compared to a patient numbed on an operating table. It sets the mood: everything is anesthetized, waiting.

Coffee spoons

> I have measured out my life with coffee spoons;

A whole life counted in tiny, polite, repeated social moments.

Not Prince Hamlet

> No! I am not Prince Hamlet, nor was meant to be;
> Am an attendant lord, one that will do
> To swell a progress, start a scene or two,
> Advise the prince; no doubt, an easy tool,
> Deferential, glad to be of use,
> Politic, cautious, and meticulous;
> Full of high sentence, but a bit obtuse;
> At times, indeed, almost ridiculous—
> Almost, at times, the Fool.

He isn't even the hero of his own story, just a minor courtier like Polonius, or at worst the Fool.

The peach

> I grow old ... I grow old ...
> I shall wear the bottoms of my trousers rolled.
> Shall I part my hair behind? Do I dare to eat a peach?
> I shall wear white flannel trousers, and walk upon the beach.

Why a **peach**? It's soft, juicy, **messy and sensual**: you can't eat one gracefully. Daring to eat a peach in public means risking dripping juice and looking undignified. It stands for his **fear of looking foolish**, and of desire itself. The white flannel trousers are the fashion of a young man at the seaside, which he worries he is too old for.

The mermaids

> I have heard the mermaids singing, each to each.
> I do not think that they will sing to me.

> We have lingered in the chambers of the sea
> By sea-girls wreathed with seaweed red and brown
> Till human voices wake us, and we drown.

He can imagine beauty but believes it isn't for him, and when real life ("human voices") breaks in, the dream drowns.

The name

Prufrock almost certainly came from the **Prufrock-Littau Company**, furniture dealers in St. Louis, Eliot's hometown, which advertised there in the 1910s. Asked about it later, Eliot said he had no memory of taking the name that way, "but I think that it must be assumed that I did, and that the memory has been obliterated."

"**Prude in a frock**" is a **readers' interpretation**, not something Eliot ever said.

Key facts

- Prufrock (1915) opens with the evening "like a patient etherized upon a table"
- "I have measured out my life with coffee spoons"
- "I am not Prince Hamlet... an attendant lord... almost, at times, the Fool"
- "Do I dare to eat a peach?": the messy, sensual peach stands for fear of looking foolish
- The name came from the Prufrock-Littau furniture company of St. Louis
- "Prude in a frock" is a readers' reading, not Eliot's

Sunday, November 1, 2026

The method of loci and a memory palace

Simonides remembered where the dead guests sat. Place things in rooms, and walk through to remember them.

Pronunciation

Loci: **LOH-sigh** (Latin for "places"; singular locus).

Simonides and the collapsed banquet hall

The Greek poet **Simonides of Ceos** (around 556–468 BC) was performing at a banquet when he was called outside. While he was gone, the **roof of the hall collapsed**, crushing the guests so badly their families couldn't recognize them. Simonides found he could **identify every body by remembering where each guest had been sitting**.

He realized memory works best with **places**. To remember a list, picture a building you know, put a vivid image of each item in a spot along a route through it, then walk the route in your mind and pick them up in order.

Cicero and the orators

The Roman orator **Cicero** tells the Simonides story in De Oratore (55 BC). Roman speakers memorized long speeches by placing each point in a room of an imagined house. Our phrase "**in the first place**" comes from this: the first point was literally in the first place along the walk.

Topic comes from Greek **topos**, "**place**": a topic was the place where an argument was stored.

Our practice memory palace

Walk through the house, from the front door:

1. **Front door**: **Eratosthenes** stands on the doormat with his stick, measuring its shadow. See Eratosthenes.
2. **Couch**: **Galileo** sprawled with his telescope, muttering "and yet it moves." See Bruno and Galileo.
3. **Kitchen**: **Brunel** trying to squeeze the giant **Great Eastern** into the sink. See the Great Eastern.
4. **Bedroom**: the **Nautilus** surfacing out of the bed, with **Verne** on deck. See Twenty Thousand Leagues.

5. **Bathroom**: **Walter Reed** swatting mosquitoes over the tub. See Reed and Finlay.

The sillier and more vivid the image, the better it sticks.

Key facts

- Method of loci (LOH-sigh): memory by places
- Simonides of Ceos (around 500 BC) identified crushed banquet guests by where they sat
- Cicero tells the story; "in the first place" comes from it
- Topic comes from Greek topos, place
- Practice palace: door Eratosthenes, couch Galileo, kitchen Brunel, bedroom Nautilus, bathroom Reed

Sunday, November 1, 2026

Shelley's Ozymandias

A shattered statue in the desert, a king's boast, and the irony that only the words survive.

The poem (1818)

> I met a traveller from an antique land,
> Who said—"Two vast and trunkless legs of stone
> Stand in the desert. . . . Near them, on the sand,
> Half sunk a shattered visage lies, whose frown,
> And wrinkled lip, and sneer of cold command,
> Tell that its sculptor well those passions read
> Which yet survive, stamped on these lifeless things,
> The hand that mocked them, and the heart that fed;
> And on the pedestal, these words appear:
> My name is Ozymandias, King of Kings;
> Look on my Works, ye Mighty, and despair!
> Nothing beside remains. Round the decay
> Of that colossal Wreck, boundless and bare
> The lone and level sands stretch far away."

Who was Ozymandias?

Ozymandias is the Greek form of a name of **Ramesses II**, the great Egyptian pharaoh (reigned about 1279–1213 BC), from his throne name User-maat-re. The Greek historian Diodorus Siculus recorded the inscription on his statue: "King of Kings am I, Ozymandias. If anyone would know how great I am and where I lie, let him surpass one of my works."

In **1818** a huge granite head and chest of Ramesses II, called the **Younger Memnon**, arrived at the **British Museum** in London, where it still is. The news of it inspired the poem.

The irony

- "**Look on my Works, ye Mighty, and despair!**" The king meant: despair, because you can never equal me. Now the meaning **reverses**: despair, because this is what all your power comes to. Nothing beside remains.
- The sculptor "well those passions read": the artist saw through the king and carved his cruelty. The sneer survives; the empire doesn't.

- The final irony: the only things that last are **words**, the inscription, and the poem itself. That links it to Sonnet 18, where the poem promises to outlive Time.

A sonnet contest

Shelley wrote it in friendly competition with his friend **Horace Smith**; each wrote a sonnet on the same subject, and both were published in The Examiner in early 1818. Smith's poem is long forgotten. Shelley's is one of the most famous poems in English.

Shelley and tyrants

Percy Bysshe Shelley (1792–1822) was a radical who hated tyranny in every form. He wrote in the years just after **Napoleon**'s fall (1815), when Europe had just watched an emperor who conquered a continent end on a small island. The poem is about every ruler who believes his works will last forever.

Key facts

- Ozymandias (Shelley, 1818): "Look on my Works, ye Mighty, and despair!"
- Ozymandias is the Greek name for Ramesses II
- The British Museum's giant head of Ramesses II, the Younger Memnon, inspired it
- "Despair" reverses meaning: only the words survive, as in Sonnet 18
- Written in a sonnet contest with Horace Smith, whose poem is forgotten

Monday, November 2, 2026

Milton's Paradise Lost

The blind, defeated revolutionary who dictated England's great epic; Satan's magnetism, the cosmos of Heaven, Chaos and Hell, the churches' verdicts, and the quiet, beautiful ending.

Milton the man

In 1660 the English monarchy came back, and Charles II hunted down the people who had defended executing his father, Charles I. John Milton had written that defense. He hid, was arrested and was released, probably thanks to friends like the poet **Andrew Marvell**. By then he was completely blind, poor and on the losing side of a revolution.

Then he wrote the greatest epic in English. He composed verses in his head at night and dictated them each morning to helpers, saying he wanted to be "milked." The poem was published in **1667 in ten books**, and in **twelve from 1674**: almost **10,600 lines of blank verse** (unrhymed iambic pentameter, Shakespeare's da-DUM beat). Milton defended dropping rhyme as freedom from

> "the troublesome and modern bondage of rhyming."

His stated aim was to

> "justify the ways of God to men."

Yet the poem opens in Hell, and its most magnetic character is Satan, the rebel who lost a war against a king.

Satan in Book I

The rebel angels fall for nine days through Chaos onto a burning lake. Satan rises and speaks:

> "What though the field be lost? / All is not lost; the unconquerable will, / And study of revenge, immortal hate, / And courage never to submit or yield."

> **"The mind is its own place, and in itself / Can make a Heav'n of Hell, a Hell of Heav'n."** (I.254–55, Jeff's favorite)

> "Better to reign in Hell, than serve in Heav'n." (I.263)

William **Blake** said Milton was "of the Devil's party without knowing it." Percy **Shelley** thought Satan morally finer than the God he fights.

Milton's trick: the grandeur is undercut from the start. Satan's face is "intrenched" with scars, and he is "Vaunting aloud, but racked with deep despair." He shrinks across the poem: from archangel, to a cormorant perched on the Tree of Life, to a toad squatting at Eve's ear, to a serpent on his belly.

Why twelve books, why so long

Virgil's Aeneid has twelve books. The 1667 edition had ten, perhaps to stand apart from Virgil's imperial poem; in 1674 Milton split two books, placing himself in Virgil's line. **Homer's** epics have 24 books each, so twelve is half a Homer.

Epic was the highest genre, and Milton took the biggest subject there is: the war in Heaven, the Creation, the Fall, and history down to the Flood and beyond. He promised

> "Things unattempted yet in prose or rhyme."

At about 10,600 lines it is still shorter than the Iliad (about 15,700).

How big is it today; the audiobook

- The poem alone runs about **300–350 paperback pages**; annotated editions (Penguin, Norton Critical) run about **450–650 pages**.
- Read aloud it takes about **11–12 hours**. For scale, The Count of Monte Cristo is over 1,200 pages.
- **Why listen:** Milton was blind and composed by ear, so the sound is the design. His long sentences roll over the line endings.
- **Best approach:** listen while following the printed text.
- **Recordings:** Anton Lesser (Naxos, unabridged, about 12 hours). On Audible, Jeff chose **Simon Vance**.

Did Milton know his success?

He sold it in 1667 for **£5 down**, with £5 more after the first edition of 1,300 copies sold, which took about two years. His widow later settled the rights for **£8**. He lived to see the 1674 twelve-book edition and died in November 1674, aged 65.

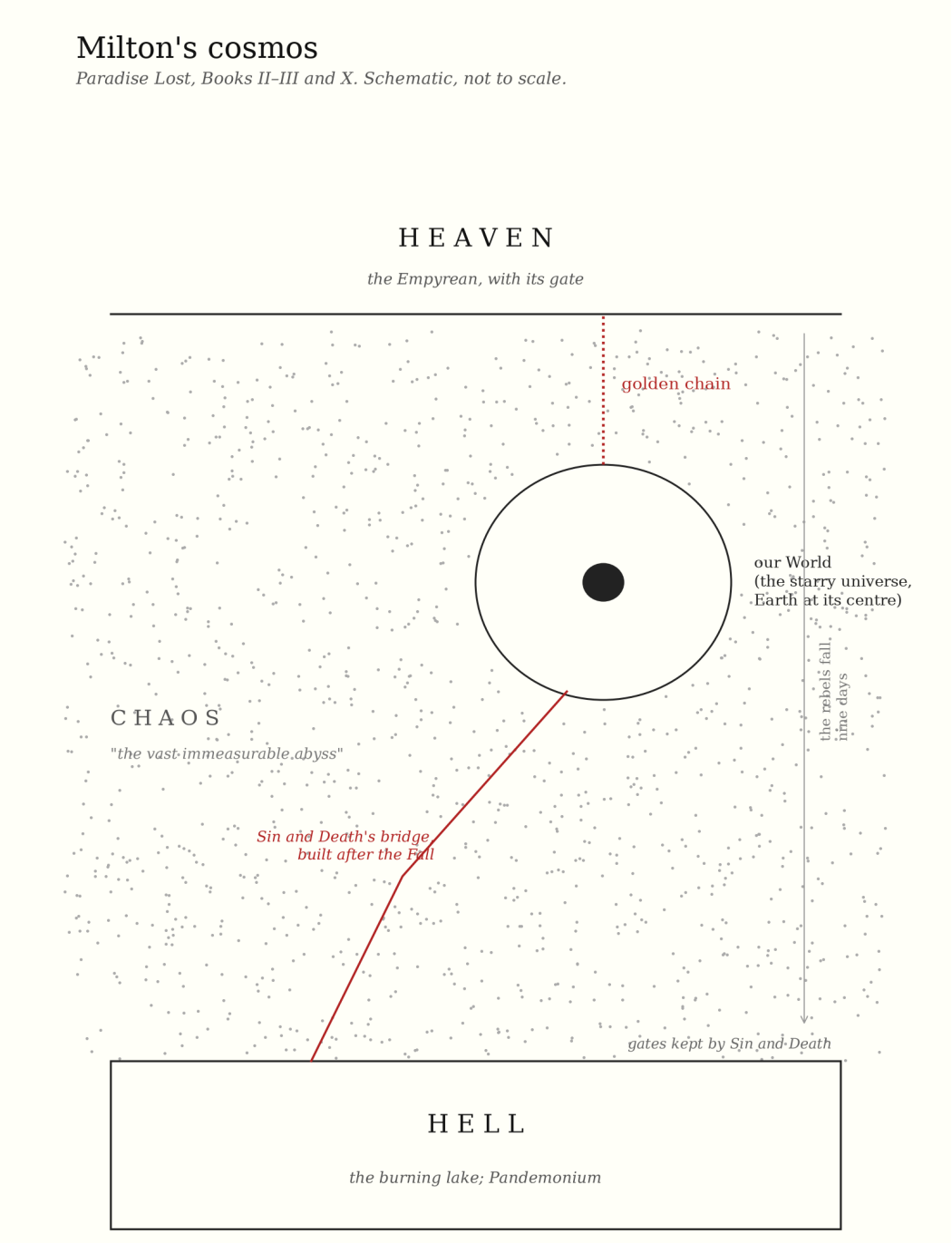
John **Dryden** reportedly said:

> "This man cuts us all out, and the ancients too."

Dryden asked to turn it into a rhymed opera; Milton reportedly replied, "Ay, you may tag my verses." Within a generation the poem was treated as England's national epic.

Milton's cosmos

Heaven (the Empyrean) is at the top. Our starry **World** hangs from it by a **golden chain** inside **Chaos**, "the vast immeasurable abyss," with **Hell** far below, its gates kept by **Sin and Death**. After the Fall, Sin and Death build a **bridge across Chaos** from Hell to our world.



Milton's cosmos: Heaven above, our World hanging by a golden chain in Chaos, Hell below with Sin and Death's bridge

What the churches say

- **No pope has ever ruled on it**, and no church has officially endorsed or banned it.
- Milton was fiercely **anti-Catholic**: in Book III's "Paradise of Fools," monks, relics and papal bulls are blown about by the wind.
- His **prose works** went onto the Vatican's Index of Forbidden Books. Paradise Lost itself is often reported as listed, but that is **uncertain**.
- Catholic critics today admire the art and reject the theology.
- For 200 years Protestant England treated it almost as a second Bible. Much of the popular picture of Satan's rebellion and the war in Heaven comes **from Milton, not Scripture**.
- **Heresies:** On Christian Doctrine (found in 1823) denies the Trinity (the Son is lesser than the Father), argues for divorce, and rejects creation from nothing.
- C. S. **Lewis**, A Preface to Paradise Lost (1942), argued Satan is meant to be absurd, and that readers cheering him are fooled as Eve was.

"God the Father turns a School-Divine"

The line is Alexander **Pope's**, from Imitations of Horace (1737). A school divine is a scholastic theologian who argues by dry logic. When God speaks in Book III, explaining free will and why the Fall isn't his fault, he sounds like a professor defending a thesis:

> "I made him just and right, / Sufficient to have stood, though free to fall."

Satan gets the thrilling speeches; God gets lecture notes. Milton set out to justify God's ways to men, and in Pope's view he made God sound like a man justifying himself.

How it ends

It ends quietly, not with thunder. In Books XI and XII the archangel Michael shows Adam the future: Cain killing Abel, the Flood, and finally the coming of Christ. Adam is consoled that good will come out of evil. Then Michael leads Adam and Eve out of Eden by the hand, while behind them the flaming sword of the cherubim guards the gate.

These are the last lines, among the most beautiful in English:

> **Some natural tears they dropped, but wiped them soon;**
> **The world was all before them, where to choose**
> **Their place of rest, and Providence their guide:**
> **They hand in hand with wandering steps and slow,**
> **Through Eden took their solitary way.**

Notice the mix. They are exiled and alone, "solitary," yet they walk "hand in hand." It's grief, but also freedom. "The world was all before them" sounds almost like a new beginning. The epic that opened in Hell closes with two people walking into our world, which is where every one of us begins.

Connection

This is the first of the creature's three books in Frankenstein (then Plutarch's Lives, then Goethe's Werther). The creature reads Paradise Lost as true history, comparing himself to both Adam and Satan. See Tambora, 1816 and Frankenstein.

Key facts

- **"The mind is its own place, and in itself / Can make a Heav'n of Hell, a Hell of Heav'n."** (I.254–55)
- "Better to reign in Hell, than serve in Heav'n." (I.263)
- Milton was blind, defeated and nearly executed after the 1660 Restoration; Andrew Marvell probably helped save him.
- Published 1667 in ten books, twelve from 1674; about 10,600 lines of blank verse.
- Aim: "justify the ways of God to men."
- Twelve books echo Virgil's Aeneid; Homer's epics have 24.
- Sold for £5 down plus £5 more; his widow settled the rights for £8.
- Audiobook about 12 hours: Anton Lesser (Naxos) or Simon Vance (Audible).
- Cosmos: Heaven above, our World on a golden chain in Chaos, Hell below; Sin and Death bridge Chaos.
- Blake: Milton was "of the Devil's party without knowing it."
- Pope: "God the Father turns a School-Divine."
- Ending: "The world was all before them... They hand in hand with wandering steps and slow, / Through Eden took their solitary way."

Monday, November 2, 2026

Plutarch and the Parallel Lives

The Greek priest of Delphi who paired famous Greeks with famous Romans to hold up mirrors to character; the second of the creature's three books, and Shakespeare's source.

Who he was

Plutarch (about **46 to 120 AD**) was a Greek from the small town of **Chaeronea**, living under Roman rule. He was also a **Roman citizen**. For about the last **thirty years of his life** he served at **Delphi** as one of the **two priests of Apollo**.

Portrait head of a philosopher in the Delphi Archaeological Museum, often identified as Plutarch (the identification is uncertain)

Portrait head of a philosopher in the Delphi Archaeological Museum, often identified as Plutarch (the identification is uncertain). Odysses, CC BY-SA 4.0, via Wikimedia Commons.

The Parallel Lives

He wrote the Lives in **pairs**: a famous Greek matched with a famous Roman.

- Theseus and Romulus
- Alexander and Caesar
- Demosthenes and Cicero

Most pairs end with a short **"comparison." Twenty-three pairs survive, plus four single lives.**

Not histories but lives

The key to reading him is his own statement at the start of Alexander:

> **"It is not histories I am writing, but lives."**

He says a small thing, "a phrase or a jest," often reveals a man's character better than battles in which thousands die. He isn't a historian. He's a **moralist who uses great men as mirrors**, so readers can look at them and fix their own character.

Why the creature reads him

That's why Mary Shelley gave the book to the creature in Frankenstein. From Plutarch he learns about public virtue. He says the book taught him "high thoughts":

> "I felt the greatest ardour for virtue rise within me."

He admires the **lawgivers Numa, Solon and Lycurgus** more than the conquerors. Paradise Lost gives him the cosmic story, of being made and cast out. Plutarch gives him the social one, a picture of how a good man lives among other people. He learns both and still has nobody to live among.

Plutarch and Shakespeare

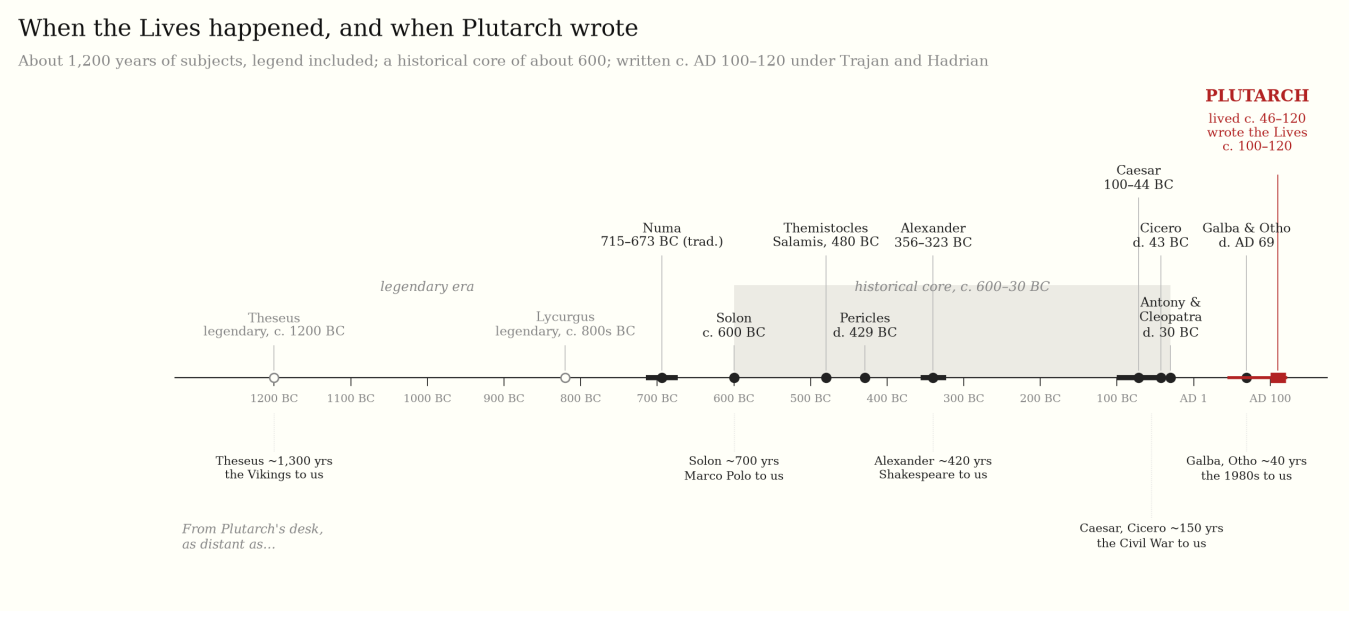
Thomas North's 1579 English translation was Shakespeare's source for **Julius Caesar**, **Antony and Cleopatra** and **Coriolanus**. Shakespeare's Cleopatra on her barge on the Cydnus follows North's prose almost line for line, set into verse.

Question to think about

Plutarch thinks a small, unguarded moment reveals character better than a great deed. In disaster work, do you learn more about a person from how they run the big operation, or from some small thing they do at 2 a.m. in a shelter?

When the Lives happened, and when Plutarch wrote

The Lives span about **1,200 years including legend**, from **Theseus (around 1200 BC)** to **Galba and Otho (AD 69)**. The **historical core** is about **600 years**, from **Solon (around 600 BC)** to **Antony and Cleopatra (30 BC)**. Plutarch wrote them around **AD 100–120, under Trajan and Hadrian**.



Timeline of the subjects of Plutarch's Lives from Theseus to Galba and Otho, with Plutarch's lifetime and writing years in red

Timeline drawn for this site. Hollow markers are legendary figures.

How far back he was looking, from his desk:

- **Theseus**: about **1,300 years**, as distant as **the Vikings** are to us.
- **Solon**: about **700 years**, as **Marco Polo** is to us.
- **Alexander**: about **420 years**, as **Shakespeare** is to us.
- **Caesar and Cicero**: about **150 years**, as **the Civil War** is to us.
- **Galba and Otho**: about **40 years**, when he was in his **twenties**, as **the 1980s** are to us.

It's like writing lives **from the Vikings to Reagan**: legend, old books, and living memory, all in one work.

Why the Lives mattered

It preserved lost sources. Much of what we know about **Lycurgus, Themistocles, Pericles, Pompey and Antony** survives mainly through Plutarch, who read books now lost. Many of history's most famous scenes, **Caesar's death, the Rubicon, Cleopatra's barge**, reach us in his telling.

It invented biography. Before Plutarch, lives were mostly praise or history. He made the individual character the subject, the model behind *Boswell's Life of Johnson*** and nearly every biography since.

The Renaissance. Jacques Amyot's French translation (1559) made Plutarch a European bestseller; **Thomas North's English (1579)** was made from Amyot. **Montaigne** loved him above all authors, and **Shakespeare** built three plays on North.

The revolutions. The American founders, **Adams, Hamilton and Madison**, were raised on Plutarch. The Federalist Papers were signed **"Publius,"** after **Publius Valerius Publicola**, a founder of the Roman Republic from Plutarch's Lives. **Washington's favorite play** was Addison's **Cato**, about Plutarch's Cato the Younger. **Rousseau** said Plutarch was his **first book**. **Charlotte Corday** read Plutarch **the day before she killed Marat**.

Leaders. Napoleon carried it on campaign. **Emerson** called it

> "a bible for heroes."

Truman read it as a boy.

Cincinnatus, the Roman called from his plow to save the state who then went back to his farm, was the model for **Washington**, who gave up command and later the presidency. (Cincinnati is named for the society of officers honoring him.)

The creature. In Frankenstein, Plutarch gives the creature **the education of a citizen with no city to live in**.

Cincinnatus

Lucius Quinctius Cincinnatus comes to us mainly from **Livy**; Plutarch wrote **no Life of him**, but he is the Roman ideal the Lives keep circling.

458 BC. A Roman army was **trapped by the Aequi in the Alban Hills**. The Senate named Cincinnatus **dictator**, an office of **emergency powers for up to six months**. The messengers found him **plowing his small farm across the Tiber**; his wife **Racilia** fetched his **toga** so he could receive them properly.

Cincinnatus Leaves the Plow to Dictate Laws to Rome (1806), Juan Antonio Ribera

Cincinnatus Leaves the Plow to Dictate Laws to Rome (1806), Juan Antonio Ribera. Juan Antonio de Ribera, Public domain, via Wikimedia Commons.

He **won in about 15–16 days**, then **resigned**, handing back about **5.5 months** of unused absolute power, and went home to his plow. He was called on **again around age 80, in 439 BC**, and again gave the power back.

He became the image of Roman **virtus**, **civic virtue**: power as a **duty**, not a prize. He is **the opposite of Caesar**, who crossed the Rubicon to take power and made himself dictator for life.

Washington. George Washington **resigned his commission in 1783** and **refused a third term in 1797**. **King George III reportedly said** that if Washington gave up power, he would be

> "the greatest man in the world." (reported; the exact wording is uncertain)

He was hailed as **"the Cincinnatus of the West."** Revolutionary officers founded the **Society of the Cincinnati**, with **Washington as its first president**, and

Cincinnati, Ohio is named after the Society. Its emblem shows **Cincinnatus taking up the sword and leaving the plow**.

Statue of Cincinnatus at Sawyer Point, Cincinnati, Ohio, holding the fasces and handing back power

Statue of Cincinnatus at Sawyer Point, Cincinnati, Ohio, holding the fasces and handing back power. Chris Light, CC BY-SA 4.0, via Wikimedia Commons.

Etymology: from Latin **cincinnus**, **"curl,"** so Cincinnatus means **"curly-haired."**

Question: in disaster work, who hands power back when the job is done?

How to read the Lives

Translations:

- **Penguin Classics thematic volumes**: The Rise and Fall of Athens, Makers of Rome, The Age of Alexander. Modern, readable, grouped by era.
- **Dryden's translation, revised by Clough**: free online, and available as an audiobook.
- **North, 1579**: beautiful, Shakespeare's own, but hard going.

A reading order:

1. Alexander
2. Julius Caesar
3. Antony (read alongside Shakespeare's Antony and Cleopatra)
4. Lycurgus and Numa
5. Themistocles
6. Pericles
7. Cato the Younger, Brutus, Cicero

Pace: about **1–2 hours per life**, so roughly **a month of evenings**. As you read, watch for **the small revealing moment Plutarch picks**, the phrase or jest he said shows character better than battles.

Key facts

- Plutarch: c. 46–120 AD, from Chaeronea; Greek, and a Roman citizen; priest of Apollo at Delphi for about his last 30 years.
- Parallel Lives: 23 surviving pairs plus 4 single lives, most pairs with a closing comparison.
- "It is not histories I am writing, but lives." (Alexander, opening)
- The creature in Frankenstein: "I felt the greatest ardour for virtue rise within me."
- North's 1579 translation was the source for Julius Caesar, Antony and Cleopatra and Coriolanus.
- Amyot's French 1559; North's English 1579; "Publius" of the Federalist after Publicola; Emerson: "a bible for heroes."
- The Lives span ~1,200 years with legend (Theseus c. 1200 BC to Galba/Otho AD 69); historical core ~600 years; written c. AD 100–120.
- Cincinnatus, dictator 458 BC, resigned after ~16 days; Washington, "the Cincinnatus of the West"; cincinnus = "curl."

Monday, November 2, 2026

Delphi and the Oracle

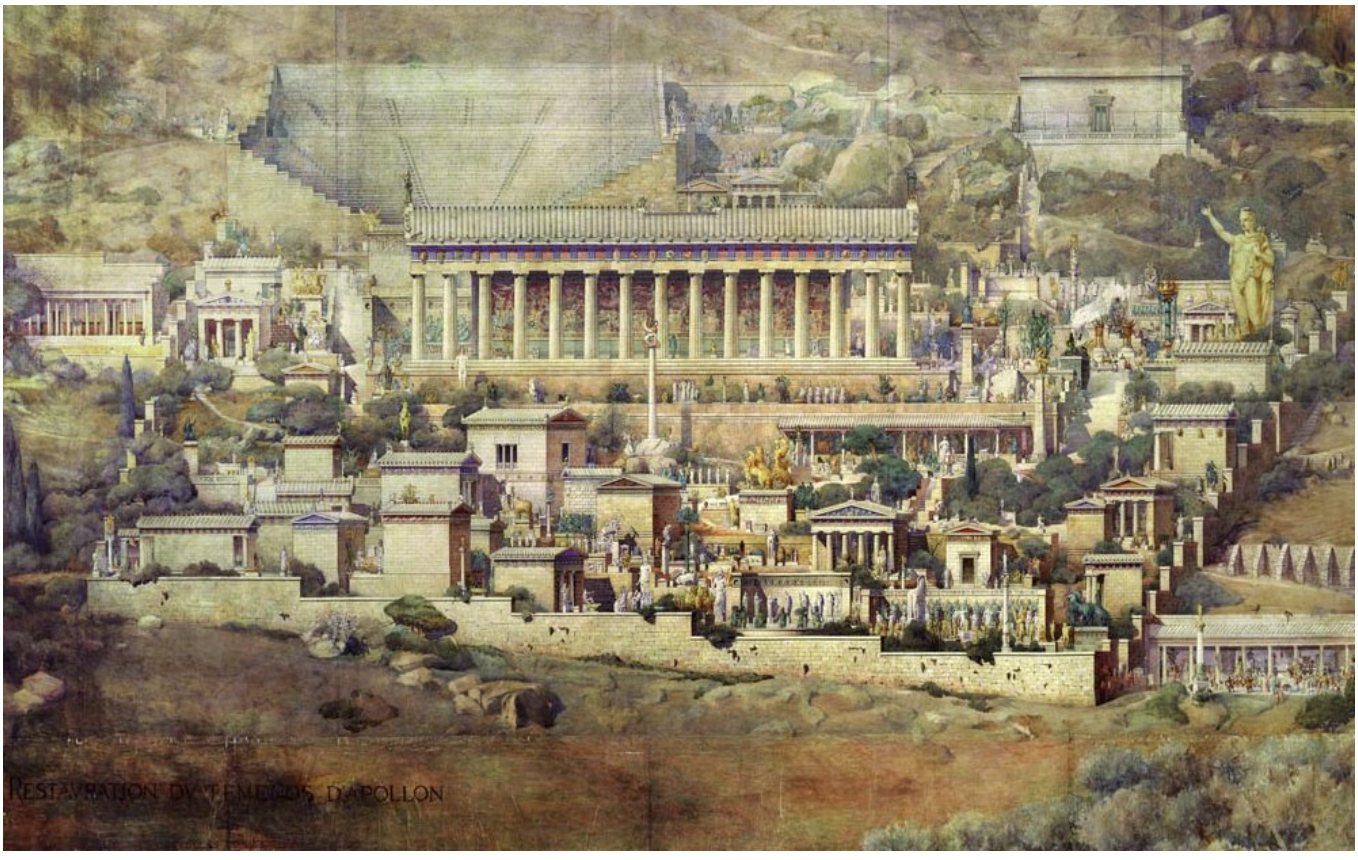
The navel of the Greek world: the Pythia on her tripod over a fault line, the riddling answers that shaped history, and how earthquakes both destroyed Delphi and preserved its treasures.

The navel of the world

Delphi was the holiest site in ancient Greece. It sits high on the slopes of **Mount Parnassus** and held the **temple of Apollo** and his oracle. The Greeks called it the **omphalos**, the "navel" of the world. In the myth, Zeus released two eagles from opposite ends of the earth, and they met there.

Where Delphi is: on Mount Parnassus above the Gulf of Corinth, with Itea, Chaeronea, Athens and Salamis

Map drawn for this site from Natural Earth data.



Reconstruction of the sanctuary of Delphi by the French architect Albert Tournaire (c. 1894), the temple of Apollo at center

Reconstruction of the sanctuary of Delphi by the French architect Albert Tournaire (c. 1894), the temple of Apollo at center. Albert Tournaire, Public domain, via Wikimedia Commons.

The temple carried three short sayings. The best known is the Delphic maxim:

> **"Know thyself"** (gnōthi seauton)

That suits Plutarch, who was a priest there: the Lives are one long exercise in knowing yourself by studying other people.

Etymology: the name comes from delphys, "womb," and is related to delphis, "dolphin." In one myth Apollo took the form of a dolphin to bring his first priests there.

The Pythia

The oracle was not one woman who lived forever. **"The Pythia" was a job title**, held by a succession of real women over **about 1,000 years**; when one died, another was chosen. Plutarch, who worked beside them, says that in his day she was a **local peasant woman of good character**, not educated or high-born. At first the Pythia was young; later she was a **woman over fifty dressed as a girl**. At Delphi's busiest, **three Pythias** took turns.

Priestess of Delphi (1891), John Collier's imagining of the Pythia on her tripod over the vapor

Priestess of Delphi (1891), John Collier's imagining of the Pythia on her tripod over the vapor. John Collier, Public domain, via Wikimedia Commons.jpg).

A visit to the oracle

- She answered only on certain days: at first **once a year**, later **once a month**, and **never in winter**, when Apollo was said to be away.

- You paid a **fee** and offered a sacrifice, usually a **goat**. The priests sprinkled it with **cold water**; if it didn't **shiver**, the god was unwilling and you went home.
- The Pythia purified herself in the **Castalian spring**, went down into the **adyton** (the "do-not-enter" room) under the temple, and sat on a **bronze tripod** over a **crack in the rock**, holding **laurel**, Apollo's plant.
- You asked your question **through a priest**. She went into a trance and answered, and the priests often **put her words into verse**.

King Aegeus consulting the oracle: the goddess Themis sits on the tripod holding laurel. Attic red-figure kylix by the Kodros Painter, c. 440–430 BC, Antikensammlung Berlin F2538. The only surviving ancient image of the oracle at work

King Aegeus consulting the oracle: the goddess Themis sits on the tripod holding laurel. Attic red-figure kylix by the Kodros Painter, c. 440–430 BC, Antikensammlung Berlin F2538. The only surviving ancient image of the oracle at work. Kodros Painter, Public domain, via Wikimedia Commons.

The vapor and the faults

Ancient writers said a sweet-smelling vapor, the **pneuma**, rose from the crack and brought on the trance. Plutarch himself thought the vapor was **real and weakening**, which he gave as one reason the oracle was declining.

In **2001**, geologists found **two faults crossing right under the temple**, and traces of **ethylene**, a sweet-smelling gas that can cause a trance-like state, in a nearby spring. The ethylene is still debated; the faults are real. The most famous holy place in Greece may have been set by a fault line.

Question: if the priests shaped her words into verse, who really spoke for Apollo, the woman in the trance or the men writing it down?

The great stories

Croesus tests the oracles. Croesus, king of Lydia ("rich as Croesus"), tested several oracles by having messengers ask, on the **hundredth day**, what the king was doing at that moment. Only Delphi got it right: he was **boiling a tortoise and a lamb together in a bronze pot**. So he trusted Delphi, sent fabulous gold, and was told that if he attacked Persia **"a great empire would fall."** One did: **his own**. Herodotus tells this story.

The wooden wall. In **480 BC** Xerxes invaded Greece. Athens got a terrifying answer, then a second: trust in the "wooden wall," and

> "divine Salamis, you will destroy the children of women."

The old men said it meant the wooden fence around the Acropolis. **Themistocles** argued it meant **ships**. Athens abandoned the city, put everything on the fleet, and crushed the Persian navy at **Salamis**. One reading of one sentence may have saved Western history. Plutarch tells it in his Life of Themistocles.

Socrates, the wisest man. A friend asked whether anyone was wiser than Socrates. The Pythia answered no one. Socrates, who knew he knew nothing, questioned people who claimed to be wise and found they didn't know either. He decided his only wisdom was knowing that he didn't know. That questioning is the **Socratic method**, and it may have started with this answer.

Oedipus. The oracle tells Oedipus he will kill his father and marry his mother. He runs away to avoid it, and running away is exactly how it comes true. That is the Greek idea of fate: trying to escape it brings it about.

The treasuries. Cities built small marble treasuries along the Sacred Way to show off their gifts, a kind of competitive generosity. The **Athenian treasury** still stands, rebuilt, and holds the **earliest notated music** we have: **two hymns to Apollo** carved in stone with notes above the words.

The pattern: the oracle rarely told you what to do. It handed you a riddle and made you reason, which is why it lasted. Question: if its answers were so ambiguous, why did the smartest Greeks keep coming back for a thousand years?

The Sacred Way today, climbing past the treasuries toward the temple

The Sacred Way today, climbing past the treasuries toward the temple. Ronny Siegel, CC BY 3.0, via Wikimedia Commons.

The decline

- Rome sacked Delphi more than once: **Sulla** plundered its treasures in **86 BC**, and **Nero** is said to have carried off **500 bronze statues**.
- Plutarch's essay **"On the Obsolescence of Oracles"** asks why the shrines had grown so quiet. In it he tells of the ship's pilot **Thamus**, who sailing past the island of Paxi heard a voice telling him to announce that **"Great Pan is dead."** When he did, a great wailing rose from the shore. Later Christians read it as the old gods dying at Christ's coming.
- **Theodosius** banned pagan worship in **391–392 AD**, and the oracle fell silent. A late legend gives its last answer:

> "Tell the king the fair-wrought hall has fallen... the speaking water is quenched."

- **Earthquakes**, from the same faults, brought the temples down, and **landslides** from Parnassus buried them.

The village on top

A Greek village, **Kastri**, grew over the ruins, its houses built from the **old marble**; ancient inscriptions ended up in garden walls and goat pens. Travelers in the 1600s and 1700s came looking for Delphi and found peasants living on it. **Lord Byron** visited in **1809** and **carved his name** on a column in the gymnasium.

The village of Castri (Kastri) on the site of Delphi, from Christopher Wordsworth's Greece (1882 edition)

The village of Castri (Kastri) on the site of Delphi, from Christopher Wordsworth's Greece (1882 edition). Christopher Wordsworth, Public domain, via Wikimedia Commons.

The rediscovery

In **1891 France won the right to dig**, partly by agreeing to **buy Greek currants**. Greece moved the **entire village about 1 km west**; the new Kastri is the modern town of Delphi. The **French School at Athens** dug from **1892 to 1903**, the **"Great Excavation,"** finding:

- the Athenian treasury (rebuilt) and the hymns with musical notes
- the temple foundations and the theater
- the **Charioteer of Delphi**, a life-size bronze from about **470 BC**, buried by the **373 BC earthquake**. That burial is why it survived when nearly every other ancient bronze was melted down. His inlaid eyes still look at you.

The Charioteer of Delphi, bronze, c. 470 BC, Delphi Archaeological Museum

The Charioteer of Delphi, bronze, c. 470 BC, Delphi Archaeological Museum. Odysses, CC BY-SA 4.0, via Wikimedia Commons.

The earthquakes that destroyed Delphi also preserved its greatest treasure, a pattern any disaster person will recognize.

Delphi today

The ruins climb the mountainside in terraces; below them, the Sanctuary of Athena Pronaia holds the round Tholos; and the modern town, the relocated Kastri, sits about a kilometer west along the ridge.

The temple of Apollo today; the adyton where the Pythia sat lay beneath it

The temple of Apollo today; the adyton where the Pythia sat lay beneath it. Bernard Gagnon, CC BY-SA 4.0, via Wikimedia Commons.

The theater above the temple, with the valley below

The theater above the temple, with the valley below. Bernard Gagnon, CC BY-SA 4.0, via Wikimedia Commons.

The Tholos in the Sanctuary of Athena Pronaia, c. 380 BC

The Tholos in the Sanctuary of Athena Pronaia, c. 380 BC. Bernard Gagnon, CC BY-SA 4.0, via Wikimedia Commons.

The modern town of Delphi, the relocated village of Kastri

The modern town of Delphi, the relocated village of Kastri. Angela Monika Arnold, Berlin, CC BY-SA 2.5, via Wikimedia Commons.

Key facts

- Delphi: Mount Parnassus; temple of Apollo; the omphalos, "navel" of the world.
- "Know thyself" (gnōthi seauton).
- The Pythia: a title held by many women over ~1,000 years; up to three at once.
- 2001: two faults under the temple; ethylene traces in a spring (debated).
- Salamis, 480 BC: the "wooden wall" read as ships.
- Silenced by Theodosius, 391–392 AD.
- Kastri moved ~1 km west; Great Excavation 1892–1903; Charioteer c. 470 BC, buried 373 BC.

Monday, November 2, 2026

Ancient Greek Religion

A religion with no name, no book and no church; where "pagan" and "myth" come from, how Greece and Rome shared one set of gods, and whether every religion eventually becomes a myth.

A religion with no name

It had **no name**. The Greeks didn't treat "religion" as something separate from life, so they had no word for their own. The closest were **ta theia**, "the divine things," and **eusebeia**, "piety," proper reverence for the gods. There was **no holy book, no creed and no single church**: only the rituals, festivals and stories each city kept, with **Homer and Hesiod** as the shared reference.

Scholars today call it **ancient Greek religion** or **Hellenic polytheism** (Hellēn "Greek," poly "many," theos "god"). The Roman version is called Roman religion.

"Pagan," "heathen" and "myth"

Pagan comes from Latin **paganus**, "country person, villager," from **pagus**, "rural district." Christianity spread first in the cities, so those who kept to the old gods were the country folk. In effect, "pagan" first meant "hick." **Heathen** may have the same story in English: people of the heath, the open countryside. The only name the old religion ever got was given by its rivals, and it meant "the people out in the sticks."

Myth comes from Greek **mythos**, which first meant simply "word" or "story." Only later did it come to mean "a story that isn't true," largely because Christians used it against the old religion.

How old was it?

- **Before the Greeks, around 1400 BC:** Mycenaean traces at Delphi. Myth says Delphi first belonged to **Gaia**, the Earth, guarded by the serpent **Python**; **Apollo killed Python** and took over, which is the origin of the name **Pythia**. Many scholars read this as an older earth cult being replaced.
- **Linear B** clay tablets from Mycenaean palaces already name **Zeus, Hera, Poseidon and Dionysus**.
- **The 700s BC to about 390 AD:** the oracle of Apollo at its height, about **1,100 years**.
- The religion overall lasted about **1,800 years**, close to the time from Jesus to today.

Greek or Roman? Both

It was Greek first and became both. Delphi was always a Greek sanctuary, run in Greek, but for its last five centuries Rome ruled and honored it.

- From about 500 BC Rome absorbed the Greek gods: **Zeus as Jupiter, Hera as Juno, Aphrodite as Venus**. **Apollo** is the exception: Rome kept his Greek name.
- Rome sent delegations to Delphi as early as the **500s and 400s BC**.
- After Rome conquered Greece in **146 BC**, emperors like **Augustus** and **Hadrian** paid for repairs and visited.
- **Plutarch** was both a Greek priest at Delphi and a Roman citizen: the "both" in one man. One religion seen through two languages, which is the idea behind the Parallel Lives.

Plutarch's own beliefs

Delphi is far older than Christianity, but Plutarch lived in the first century AD, while **Paul was preaching in Greece** and the **Gospels were being written**. He **never mentions Christianity** once. For him Apollo and Zeus were the living religion of his world. He was a **Platonist** who believed in **one supreme, perfect divine being**, and saw the stories of the gods as partly poetic veils over it. He said outright that some myths were **crude** and not to be taken literally.

Question: Plutarch believed in one perfect God behind a crowd of stories. How far is that from the Christianity that was about to replace his world?

Will Christianity be called a myth?

Jeff's point: **"In a thousand years some might consider Christianity a myth."**

History gives that weight. Every religion of the ancient Mediterranean that once seemed permanent is now called "mythology." **Egypt's gods lasted about 3,000 years**, longer than Christianity has existed, and are now in museums and cartoons.

But religions are **absorbed as well as dying**. Christianity took over Greek philosophy, Roman festivals and even sacred sites; Delphi's ruins have Christian buildings among them. A thousand years from now Christianity might be called a myth, or live on inside whatever comes next, the way Plato lives on inside it.

> **The word "myth" usually just marks whose story lost.**

Question: which lasts longer, what a religion teaches about how to live, or its story about the gods? Plutarch would say the first: his moral essays are still read; nobody prays to his Apollo.

Does every religion become a myth?

Not every one, but most. **Those tied to a fallen state did:** Egypt, Mesopotamia, Greece and Rome, the Norse, the Aztecs. **The survivors:**

- **Hinduism**: the Vedas from about 1500 BC; about a billion people.
- **Judaism**: about 3,000 years; it moved its center from the Temple to a book and a way of life.
- **Zoroastrianism**: about 100,000–200,000 followers, still alive after about 3,000 years.

The pattern: text plus community plus daily practice outlasts temples and kings. Christianity, Islam and Buddhism have those survival traits, but the future is unknown. (Continued on the Nov 3 session.)

Key facts

- No native name; ta theia ("divine things"), eusebeia ("piety"); no book, creed or church.
- "Pagan" < Latin paganus, "villager"; "myth" < mythos, "word, story."
- Linear B names Zeus, Hera, Poseidon, Dionysus; ~1,800 years overall.
- Rome: Zeus → Jupiter, Hera → Juno, Aphrodite → Venus; Apollo stayed Apollo.

Monday, November 2, 2026

Alexander and Caesar

Plutarch's most famous pair, told through small moments; the ambition that made and ruined both men, and what the Rubicon really was.

Alexander the Great (356–323 BC)

Plutarch opens with his line on character: "It is not histories I am writing, but lives." He shows Alexander through small moments:

- As a boy he tamed the horse **Bucephalus**, which no one could ride, by noticing it was **afraid of its own shadow** and **turning it toward the sun**. His father Philip said:

> "My son, look for a kingdom equal to yourself; Macedonia is too small for you."

- His tutor was **Aristotle**.
- Meeting the philosopher **Diogenes**, he offered him anything he wanted. Diogenes:

> "Stand a little out of my sun."

Alexander:

> "If I were not Alexander, I would be Diogenes."

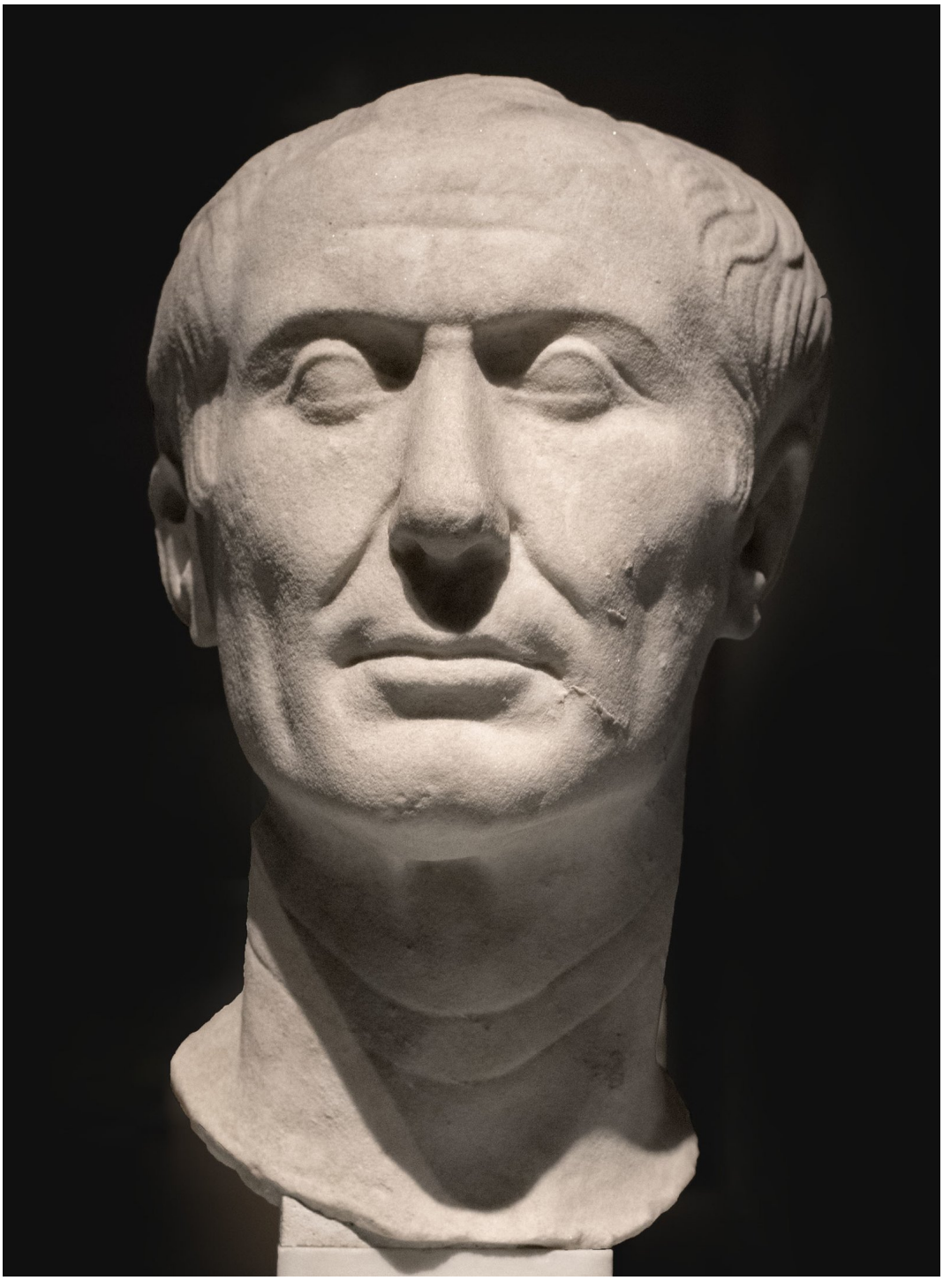
- His rage: drunk at a banquet, he **killed his friend Cleitus** with a spear, then grieved for days.
- His soldiers **refused to go beyond India**. He **died in Babylon at 32**, worn out and drinking heavily, still planning new conquests.

Alexander at the Battle of Issus, detail of the Alexander Mosaic from the House of the Faun, Pompeii (c. 100 BC), Naples

Alexander at the Battle of Issus, detail of the Alexander Mosaic from the House of the Faun, Pompeii (c. 100 BC), Naples. Unknown artist, Public domain, via Wikimedia Commons.

Julius Caesar (100–44 BC)

- Kidnapped by **pirates** as a young man, he told them to demand a **higher ransom**, joked he'd come back and **crucify** them, and later did.
- In Spain he **wept before a statue of Alexander**: at his age Alexander had conquered the world, and Caesar had done nothing memorable. This is the hinge of the pair: Caesar measured himself against Alexander, and Plutarch measures them against each other.
- He crossed the **Rubicon**: "Let the die be cast."
- He let himself be named **dictator for life**; senators who feared it stabbed him **23 times**, and he fell **at the foot of the statue of Pompey**, the rival he had beaten.
- He ignored the warnings about the **Ides of March**, which Shakespeare took straight from Plutarch.



Marble portrait of Julius Caesar

Marble portrait of Julius Caesar. Ángel M. Felicísimo from Mérida, España, Public domain, via Wikimedia Commons_(cropped).jpg).

The lesson: ambition

Both men were ruined by what made them great: **ambition**, an appetite for glory with no limit. The Greek word for its dark side is **pleonexia**, **"always wanting more."** The quality that lifts a man is often the one that ruins him, like Milton's Satan:

> "Better to reign in Hell, than serve in Heav'n."

What is the Rubicon?

A **small river in northeast Italy** flowing into the **Adriatic near Rimini**. In Caesar's day it marked the **border between his province, Cisalpine Gaul, and Italy proper**. Roman law forbade a general to bring his army across: doing so was **treason** and meant civil war.

In **January 49 BC** Caesar stopped at the bank with **one legion**. Plutarch says he hesitated a long time, weighing the disaster he was about to cause, then crossed saying, **in Greek**:

> "Let the die be cast."

The civil war that followed ended the Republic. **"Crossing the Rubicon"** now means passing a point of no return. The name likely comes from Latin **rubeus**, **"red,"** for its reddish, muddy water. Nobody is sure which modern stream it was: in **1933** Italy officially named the **Fiumicino** the **Rubicone**, but historians still argue. A good mapping puzzle.

Key facts

- Alexander 356–323 BC; tutor Aristotle; horse Bucephalus; died in Babylon at 32.
- Caesar 100–44 BC; crossed the Rubicon January 49 BC; 23 stab wounds; fell at Pompey's statue.
- Pleonexia: "always wanting more."
- Rubicon: Cisalpine Gaul / Italy border near Rimini; rubeus "red"; Fiumicino renamed Rubicone in 1933.

Tuesday, November 3, 2026

Lycurgus and Numa

Plutarch's pair of lawgivers, the men the creature in Frankenstein admired most: Sparta's founder, who built a city for war, and Rome's philosopher-king, who built one for peace.

"Nothing can be said that is not disputed"

Plutarch opens the Life of Lycurgus with an honest warning:

> "Nothing can be said that is not disputed."

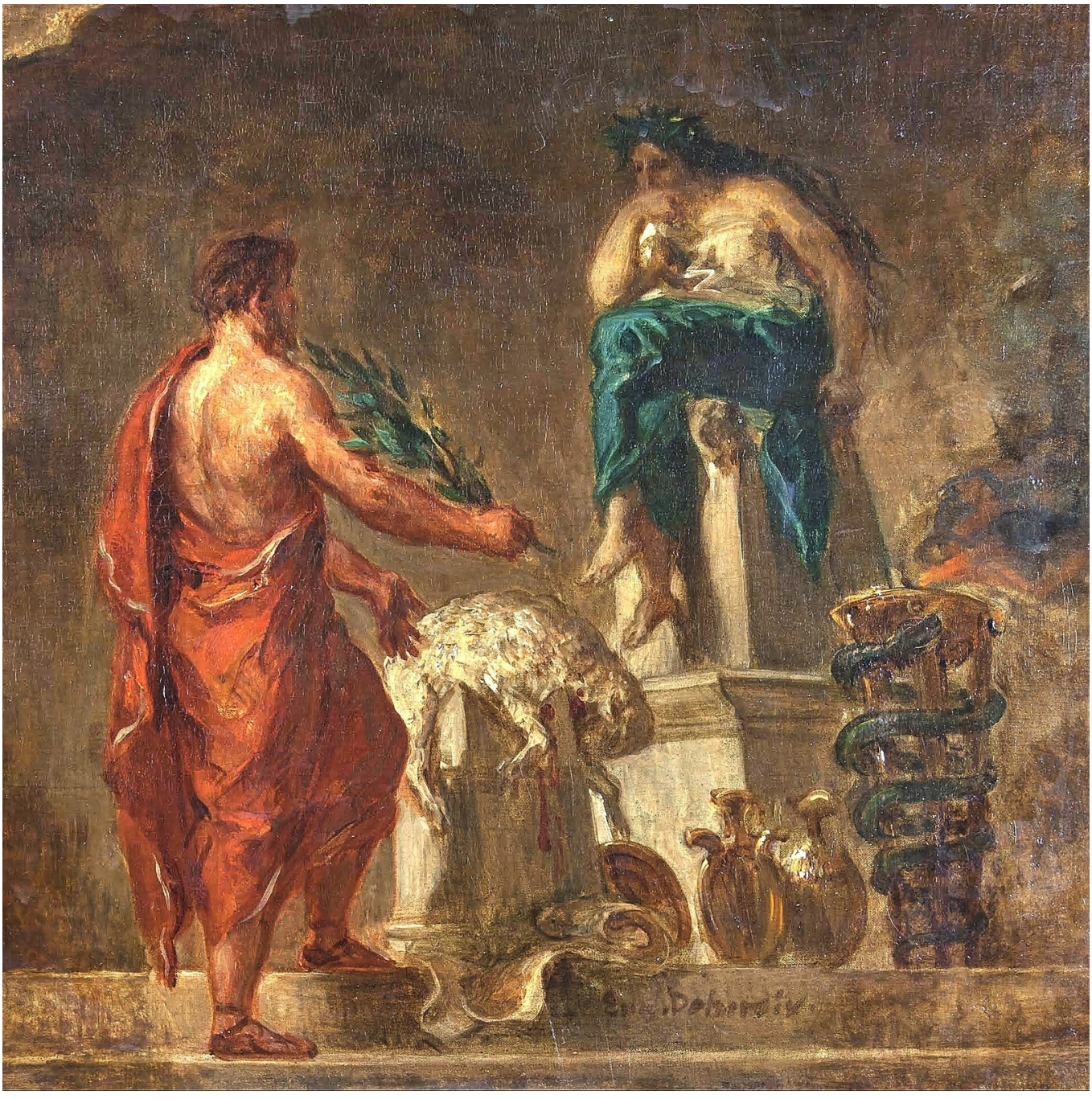
His birth, travels and death are all contested, and some scholars doubt he existed at all.

Lycurgus of Sparta

Lycurgus lived **around the 800s BC**, if he lived at all; he may be **legendary**.

Laws from Delphi. He got his laws from the **Delphic oracle**, and the Pythia greeted him as

> "more god than man."



Lycurgus Consulting the Pythia (detail), Eugène Delacroix

Lycurgus Consulting the Pythia (detail), Eugène Delacroix. Eugène Delacroix (1798–1863), Public domain, via Wikimedia Commons.jpg).

What he built:

- **Land divided equally** among the citizens.
- **Gold and silver banned**, replaced with **heavy iron bars**. A small fortune needed an **ox cart** to move it, so nobody bothered to steal, bribe or hoard.
- **Common mess halls**, where every citizen ate together, and the infamous **black broth**. A visitor who tasted it said:

> "Now I know why Spartans don't fear death."

- The **agoge**, the state training every boy entered **from age seven**.
- The story of the **boy who hid a stolen fox under his cloak** and let it **chew into him** rather than be caught.

His exit. He made the Spartans swear an **oath to keep his laws until he returned**. Then he went to Delphi and **starved himself**, so he **never returned** and the oath could never expire.

Numa Pompilius of Rome

Numa (traditionally **715–673 BC**) was **Rome's second king, after Romulus**. He was a **Sabine philosopher** who **refused the crown at first** and had to be persuaded.

- He founded the **priesthoods**, the **Vestal Virgins** and the **calendar**, adding **January and February**.
- He claimed counsel from the **nymph Egeria**, whom he met in a **sacred grove**. Plutarch suspected she was a **useful fiction**, a way to give laws divine weight.
- The doors of the **temple of Janus**, opened in war and shut in peace, **stayed shut for all 43 years of his reign**.

The Nymph Egeria Dictating to Numa, Ulpiano Checa

The Nymph Egeria Dictating to Numa, Ulpiano Checa. Ulpiano Checa, CC BY-SA 3.0, via Wikimedia Commons.

The comparison

Lycurgus built a city for war; Numa built one for peace. Sparta's laws lasted **about 500 years**. Rome **dropped Numa's peace after his death**. Plutarch's verdict leans on endurance: the harsher system lasted, the gentler one didn't.

Words from Sparta

- **Laconic**, "using few words," from **Laconia**, Sparta's region. Philip of Macedon threatened:

> "If I enter Laconia, I will raze Sparta."

The Spartans' whole reply:

> "If."

- **Spartan**, "austere, without comfort," from the mess halls, iron money and agoge.

The Frankenstein link

These are the lawgivers the creature admired most when he read Plutarch: not conquerors, but men who shaped how people live together. He studied founders of cities while having no city of his own.

Question to think about

Lycurgus said his laws came from Apollo; Numa said his came from a nymph. **Why would a lawgiver claim his laws came from a god?**

Key facts

- Plutarch: "Nothing can be said that is not disputed."
- Lycurgus, c. 800s BC (possibly legendary); the Pythia: "more god than man"; iron money, mess halls, black broth, agoge from age 7.
- Numa, trad. 715–673 BC; Vestals, priesthoods, calendar (January, February); Egeria; Janus closed 43 years.
- Sparta's laws ~500 years; Rome abandoned Numa's peace after him.
- "Laconic" < Laconia; "If."

Wednesday, November 4, 2026

The Real Abbé Faria

The Goan priest, revolutionary and hypnotist behind Dumas's prisoner-tutor, who argued that the trance comes from the subject's own mind, not from a magnetic fluid.

Birth in Goa

José Custódio de Faria was born on **31 May 1756** (a few sources say 30 May) in **Candolim, Goa**, then **Portuguese India**. He came from a **Goan Catholic family of Brahmin descent**: his father's line went back to **Anantha Shenoy**, a Goud Saraswat Brahmin village clerk who converted in 1583. His father was **Caetano Vitorino de Faria**; his mother **Rosa Maria de Sousa**. The marriage was unhappy; the couple separated with the Church's permission, his **father became a priest** and his **mother a nun** at the **Convent of St. Monica** in Old Goa.

Rome and the "vegetables"

Father and son reached **Lisbon in December 1771**, when José was 15, and King Joseph I sent them to **Rome**. José trained for the priesthood and earned a **doctorate in theology**, dedicating his thesis to **Queen Maria I of Portugal**. He was reportedly invited to preach in the **Sistine Chapel** before the Pope.

Back in Lisbon he was invited to preach before **Queen Maria I**. The famous **anecdote** (from family tradition, not a documented record): climbing the pulpit, he froze at the sight of the court, until his father whispered in **Konkani**:

> "Hi sogli baji; kator re baji." "They're all vegetables; cut the vegetables!"

Jolted, he lost his fear and preached fluently. The story goes that he spent the rest of his life wondering how a single phrase could change a mind's state so completely: the seed of his work on suggestion.

The Pinto Conspiracy and flight to France

In **1787** Goan priests and others plotted the **Conspiracy of the Pintos**, the first organized attempt to overthrow Portuguese rule in Goa, named for the Pinto family of Candolim in whose house it was planned. It was betrayed and crushed. Faria was **implicated**, though how deeply is unclear, and he left for **France in 1788**.

Revolution and prison

In Paris he is reported to have **led one of the sectional battalions in the uprising of 13 Vendémiaire (5 October 1795)**, the royalist-leaning revolt against the National Convention, which the young **Napoleon** broke with his "whiff of grapeshot." (Wikipedia's English article says "10 Vendémiaire" and has him helping cause the Convention's fall; the uprising was in fact on 13 Vendémiaire, and it failed.) He knew **Chateaubriand** and was a friend of the **Marquis de Puységur**, Mesmer's leading disciple.

He was **imprisoned at some point**, but the details are thin. **Whether he was ever held in the Château d'If**, as in Dumas's novel, is **doubtful and probably legendary**; no reliable record puts him there.

In **1811** he became **professor of philosophy at Nîmes** and a member of the **Société Médicale de Marseille**.

Mesmer and animal magnetism

Franz Anton Mesmer (1734–1815) claimed that an invisible **magnetic fluid** filled the universe and living bodies, and that a "magnetizer" could channel it to cure illness. A royal commission including **Benjamin Franklin** and **Lavoisier** investigated in 1784 and concluded the effects came from imagination. His name gave us **"mesmerize."**

Faria's idea: it's in the subject's mind

From his first public sessions in **1814** in Paris, Faria broke with the fluid theory:

- **Nothing comes from the magnetizer; everything comes from the subject**, from imagination and expectation working inside the subject's own mind. This is what we now call **suggestion**, and he also pointed to **autosuggestion**.
- He called the trance **"sommeil lucide"** (**lucid sleep**), the subject a **"concentrated"** person (concentré), and himself the **"concentrator"** (concentrateur). The state, he said, is part of the **natural order**, a form of sleep.
- His method was **command, following expectancy**: he seated the subject, asked them to close their eyes and focus on sleep, then ordered in a firm voice:

> **"Dormez!"** ("Sleep!")

Fixed gaze and mental fatigue helped, but the cause was moral, inside the person.

His book. In **1819** he published **De la cause du sommeil lucide, ou Étude de la nature de l'homme** (On the Cause of Lucid Sleep, or Study of the Nature of Man). It was planned in several volumes; **only volume 1 appeared**. Laurent Carrer published the first English translation in 2004.

Mockery and an obscure death

Paris turned on him. He was accused of being a **charlatan**, and he was mocked on stage, including in the comedy **La Magnétismomanie** (1816), after a public demonstration reportedly went wrong when an actor pretended to be entranced. He retired as **chaplain to an obscure religious house** and **died of a stroke in Paris on 20 September 1819** (one passage in the English Wikipedia says 30 September; most sources give the 20th). He left no address, and his **grave is unmarked**, somewhere in **Montmartre**.

Pioneer of hypnosis

Later historians restored him as **a pioneer of the scientific study of hypnosis**, the first to put **suggestion** at its center. The theory is sometimes called **Fariism**. The line runs through **Ambroise-Auguste Liébeault** and **Hippolyte Bernheim** of the **Nancy school**, and **Émile Coué**'s autosuggestion. The Scottish surgeon **James Braid**, who coined "hypnotism" in 1841, reached similar conclusions; how directly Faria influenced him is debated. Faria is also seen as a **forerunner of the placebo idea**: belief and expectation themselves produce real effects.

Dumas's Faria

Alexandre Dumas took the name for the imprisoned priest in **The Count of Monte Cristo (1844)**: the "mad" abbé in the Château d'If who tunnels into Edmond Dantès's cell, educates him in languages, science and history, and reveals the treasure of Monte Cristo. Dumas's Faria is a** fictional composite**. Apart from the name, the priesthood and a certain mental self-mastery, he bears little resemblance to the real man.

The statue in Panaji

The bronze statue of Abbé Faria hypnotizing a woman, Panaji, Goa

Statue of Abbé Faria, Panaji, Goa. Vn nilesh at English Wikipedia, CC BY-SA 3.0, via Wikimedia Commons.

In central **Panaji (Panjim)**, next to the **Old Secretariat** (the Adil Shah or Idalcão Palace), stands a **bronze statue of Faria, arms outstretched over a woman falling into a trance**. It dates from **1945** and is usually credited to the Goan sculptor **Ramchandra Pandurang Kamat**; some sources credit **Constâncio Fernandes** instead. Portugal marked his 250th birthday in **2006** with a stamp showing the statue, and Lisbon and Margao have streets named for him.

(No freely licensed portrait of Faria turned up; the Bibliothèque nationale in Paris holds a contemporary caricature.)

Closing reflection

Dumas's Faria **educated Dantès's mind in prison**. The real Faria showed that **the mind can put itself into a trance**: the hypnotist only gives the cue, and the subject does the work. That's a question for any tutor:

> **Does the tutor teach, or does the student learn?**

Key facts

- José Custódio de Faria: born 31 May 1756, Candolim, Goa; died 20 September 1819, Paris.
- Doctorate in theology in Rome; preached before Queen Maria I ("cut the vegetables," an anecdote).
- Implicated in the Pinto Conspiracy (1787); to France 1788; reportedly in the 13 Vendémiaire uprising (1795).
- Château d'If imprisonment: doubtful, probably legend.
- Rejected Mesmer's fluid: the trance comes from the subject (suggestion); "sommeil lucide," "Dormez!"
- De la cause du sommeil lucide (1819), only volume 1 published.
- Forerunner of the Nancy school (Liébeault, Bernheim) and the placebo idea.
- Panaji statue, 1945, usually credited to Ramchandra Pandurang Kamat.

Saturday, November 7, 2026

Jos. A. Bank: an American clothier, not Joseph Banks

The Baltimore menswear firm is a twentieth-century family business, later sold to Quaker Oats, then to Men's Wearhouse in 2014. It is not the British botanist, and it was not bought by Brooks Brothers.

The name looks like the botanist and is not him. Jos. is an abbreviation of Joseph. Bank, on the shop signs, has no final s. Sir Joseph Banks sailed with Cook and is the subject of another lesson on this day. This page is a clothing company in Baltimore.

The usual founding date is 1905, with a family workshop behind it. Charles Bank, a Lithuanian immigrant, was a tailor. His grandson Joseph worked young, as a cloth cutter and a salesman. A family trade, a formal company, and a brand name are three dates. Do not collapse them into one man who did everything in one year. Sources even disagree about the grandson's middle name, so this lesson uses Joseph and does not pretend to settle it.

Joseph married Anna Hartz, of a Baltimore clothing family. The partnership called L. Hartz & Bank belongs to those early years. Published accounts put it in 1912 or in 1922. Both years are in print, so this page will not pick one it cannot defend. In 1945 Joseph and his son Howard bought the Hartz share. The brand Jos. A. Bank is the name from that change of ownership.

Shops of their own came in the 1950s, a catalog around 1960. There were 11 stores when Quaker Oats bought the firm in 1981. A leveraged buyout in 1986 left a heavy debt and a near failure, then a recovery and a public sale of shares in 1994.

On 18 June 2014 Men's Wearhouse completed the purchase at 65 dollars a share, about 1.8 billion dollars, after the two companies had tried to buy each other in 2013. It was not a sale to Brooks Brothers in 2011. The parent company took the name Tailored Brands in 2016. In 2020 that parent went through Chapter 11 and emerged as a private company on 1 December. The brand still sells clothes. A current store count is easy to invent and is left out.

No British chain named for the botanist turned up in the records checked for this lesson. Not finding a chain is not a proof that none ever existed. It is a reason not to tell that story. The Chicago firm Hart Schaffner & Marx, next, is another similar name with no documented tie to these Hartzes.

The illustrated lesson is download the PDF.

Sources

- Encyclopedia.com, Jos. A. Bank Clothiers: https://www.encyclopedia.com/books/politics-and-business-magazines/jos-bank-clothiers-inc
- Tailored Brands, 18 June 2014, on the Men's Wearhouse acquisition: https://www.tailoredbrands.com/2014/06/18/mens-wearhouse-completes-acquisition-of-jos-a-bank/
- Tailored Brands, 1 December 2020, on emerging from restructuring: https://www.tailoredbrands.com/2020/12/01/tailored-brands-successfully-completes-financial-restructuring/

Key facts

- Jos. A. Bank is a Baltimore clothier. Joseph Banks the botanist is a different person. The spellings differ.
- The firm is dated from 1905, with earlier family work. Charles Bank was the immigrant tailor. Joseph, his grandson, was the young cutter. Do not force one sole founder.
- In 1945 Joseph and Howard bought the Hartz interest. The 1912 and 1922 partnership dates conflict, so neither is asserted here.
- Men's Wearhouse bought the company on 18 June 2014 for about 1.8 billion dollars. Not Brooks Brothers in 2011. Tailored Brands emerged from Chapter 11 on 1 December 2020.

Saturday, November 7, 2026

Hartz and Hart Schaffner and Marx: names are not a pedigree

Baltimore's Hartz family, tied to Jos. A. Bank, is not the Chicago firm Hart Schaffner and Marx. No document found links them. Absence of a document does not prove the families were unrelated in Europe.

Lena Hartz and her daughter Anna belong to a Baltimore clothing family. Anna married Joseph Bank, and that story is the clothier lesson. Harry Hart and Max Hart belong to Chicago. The trade is the same. The name is similar. A pedigree is something else.

In Chicago the roots are a Hart brothers' shop in 1872. In 1879 Levi Abt and Marcus Marx joined them. Joseph Schaffner joined in 1887, and the familiar name Hart Schaffner & Marx is that company. Do not say the firm was formed after 1945. That year is the Baltimore buyout, in another city, of a Hartz share in a different business.

A careful history of Schaffner's migration, from the German Historical Institute's work, shows a network of people from near Worms. It shows how a young man entered a Chicago house. It does not show a contract with Baltimore. Some accounts say Abt and Marx left when Schaffner arrived. Others say Marx stayed. Those details conflict, they are not needed for this question, and they are not a bridge to the Hartzes. They are left out on purpose.

No partnership paper, no share ledger, and no family record found for this lesson ties the Baltimore Hartz firm to the Chicago company. Read that as a limit, not as a triumph. Not finding a document does not prove the families were strangers back in Europe. It does forbid the sentence "they were one house."

What would count? For a business, a partnership agreement, articles of incorporation, or a record of ownership. For kinship, a birth, a marriage, a ship's list, a census, or a will that names people on both sides. A German-sounding surname in the garment trade is the start of a search. It is not the result.

If a "Hart" in Chicago is needed for context, there is already another one. Cousins Abraham and Henry Hart had a different firm, also spoken of as Hart brothers, and it was not this company. Similar names inside one city are a warning, not a family tree. The Bank lesson has the Baltimore side. This page only refuses the false pedigree.

The illustrated lesson is download the PDF.

Sources

- Encyclopedia of Chicago, Hart, Schaffner & Marx: https://encyclopedia.chicagohistory.org/pages/2695.html

- Immigrant Entrepreneurship, "Joseph Schaffner": https://www.immigrantentrepreneurship.org/entries/joseph-schaffner/
- Encyclopedia.com, Hartmarx: https://www.encyclopedia.com/social-sciences-and-law/economics-business-and-labor/businesses-and-occupations/hartmarx-corp

Key facts

- Baltimore's Hartz family is part of the Jos. A. Bank story. Chicago's Harry and Max Hart are part of Hart Schaffner & Marx. They are not the same firm.
- The Chicago company grows from 1872, with Marx and Abt in 1879 and Schaffner in 1887. It was not formed after 1945.
- No business record or kinship record found here links the two. That does not prove there was never a remote family tie.
- A real link would be a partnership, an ownership record, or a civil document naming both sides. The same trade and a similar name are not that link.

Appendix A. Complete companion papers

These are the complete standalone lesson papers and maps, not summaries. Twenty-four public downloads match papers in this collection byte for byte; each matching PDF is included once. HTML lessons may still overlap with their fuller or illustrated paper editions.

30 complete papers • begins on binder page 308

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Aristarchus: a moving Earth and a measurable Moon

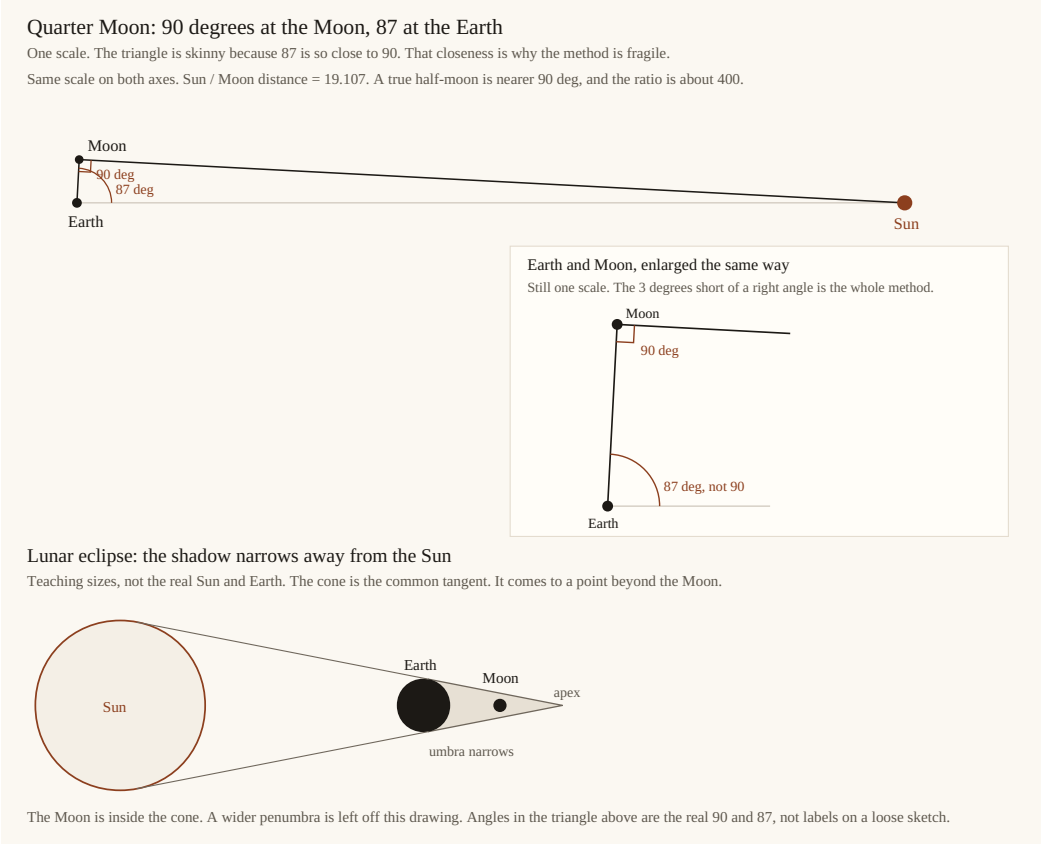
Aristarchus of Samos, about 310 to 230 BCE, is the first Greek we can name who set the Earth moving around the Sun, and turning once a day, with the stars so distant that the year shows no shift. We know that proposal because Archimedes reports it. The book itself is lost. The book we still have, On the Sizes and Distances of the Sun and Moon, does geometry. It is not the heliocentric treatise, and it should not be asked to prove one.

Three claims get fused in retellings. They are different.

First, the heliocentric idea, attested, not surviving in his own diagrams.

Second, the quarter Moon. At half phase, the angle at the Moon, between Sun, Moon, and Earth, is a right angle. Measure how far the Moon stands from the Sun in the sky, and the triangle gives the Sun's distance in units of the Moon's. Aristarchus took that elongation as 87 degrees and found a Sun about 18 to 20 times farther than the Moon. The real ratio is about 400. Half-moon is so close to 90 degrees from the Sun that a small mistake in "exactly half" wrecks the answer. The diagram draws 87 degrees so you can see the right angle. The sky is less obliging.

Third, a lunar eclipse. The Moon crosses Earth's shadow. That shadow is a cone that tapers, because the Sun is a wide disk, not a point of light. You need the Moon's angular size and the Sun's geometry. The width of the dark patch, alone, is not a distance.



The quarter-moon triangle and a tapering eclipse shadow

The triangle is one scale: 90 degrees at the Moon, 87 at the Earth, Sun 19.107 times the Moon's distance. The eclipse uses teaching sizes. The true half-moon is nearer 90 degrees, and the ratio is about 400.

Do not hand Aristarchus the modern "about 60 Earth radii" as if his book printed it and got it right. Reconstructions of the angles he adopted are shaky. What is solid is the distinction he forces on us: a physical width, an angle, and a distance are three different things, and a triangle can connect them. No calculus is required. Hipparchus, next, uses a baseline on the Earth and other people's eclipse reports. A thumb at arm's length, much later in this day's lessons, is the same triangle with worse numbers and an assumed distance.

The illustrated lesson is download the PDF.

Sources

- MacTutor, biography of Aristarchus: https://mathshistory.st-andrews.ac.uk/Biographies/Aristarchus/
- MacTutor, on the sizes and distances: https://mathshistory.st-andrews.ac.uk/BEA/aristachus_bea.pdf

Key facts

- The heliocentric proposal is known from Archimedes. It is not the surviving Sizes and Distances.
- The quarter-moon triangle, with an assumed 87-degree elongation, gave a Sun only about 18 to 20 times farther than the Moon. The real ratio is about 400.
- A lunar eclipse needs the taper of Earth's shadow and the Moon's angular size. Shadow width alone is not the distance.
- The modern Moon, near 60 Earth radii, is not a figure to attribute to Aristarchus as a successful original result.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Bounty: mutiny, survival, and the people of Pitcairn

On 28 April 1789 Fletcher Christian took HMS Bounty from William Bligh. Bligh and eighteen others were put in the launch. Not everyone left aboard had chosen the mutiny in the same way, and not every man in the boat was a loyal volunteer in a simple story. The ship's jobs, nursery and boat, are the ships lesson. This page is what people did.

Bligh could navigate. He was also harsh to his company. A court later acquitted him of losing the ship. An acquittal says the Navy did not convict him. It does not wash the temper out of the record.

At Tofua on 2 May his own narrative is the one we have. He had told his men to agree on a cover: the ship had sunk on 1 May. Islanders tried to haul the boat ashore, and Bligh brandished a cutlass. When he refused to stay the night, a chief threatened them. John Norton ran up the beach to cast off the stern line, though he was called back. As the crew boarded, the attack began. Norton was struck down. Bligh cut the rope with a pocket knife. Bligh inferred that the visitors meant to take the boat and the stores, and that they had learned the party had no firearms. That is his inference. There is no account here from Tofua, and there is no proven single provocation that rests on Norton alone.

They reached Timor. Not every man of the launch lived to see England again.

In January 1790 nine British mutineers, six Polynesian men, twelve women, and an infant came to Pitcairn. Infant counts vary, and this page will not pretend they do not. People had lived on the island before. It was empty then. The Bounty was burned on 23 January. The Pitcairn government describes the Polynesian men as treated more like slaves than like partners. Land, weapons, and the women's choices were not shared. In September 1793 five of the mutineers were killed: Christian, Mills, Martin, Williams, and Brown. By 1794 all six Polynesian men were dead. Later tellings disagree about who struck whom. McCoy killed himself in 1798. Young and Adams killed Quintal in 1799. Young died in 1800. Adams lived until 1829, the last of the mutineers, not the author of a peaceful idyll. The women and the children are the center of what continued.

The American ship Topaz, under Mayhew Folger, found the settlement in 1808. In 1831 a removal to Tahiti brought disease and deaths, then a return. In 1856 the community went to Norfolk Island. Some families came back to Pitcairn later. Head counts differ by source, so this lesson will not pick a false exact number. Descendants live on both islands. The breadfruit those ships were built to move is the next lesson in the chain.

The illustrated lesson is download the PDF.

Sources

- Royal Museums Greenwich, the mutiny and Bligh: https://www.rmg.co.uk/stories/maritime-history/mutiny-on-bounty and https://www.rmg.co.uk/stories/maritime-history/william-bligh
- William Bligh, A Voyage to the South Sea (Project Gutenberg): https://www.gutenberg.org/cache/epub/15411/pg15411.txt
- Government of Pitcairn, history: https://www.government.pn/history
- Maike Gauger and others, on Pitcairn's early years, ANU Press: https://press-files.anu.edu.au/downloads/press/n2159/pdf/ch06.pdf

Key facts

- The mutiny was 28 April 1789. Bligh and 18 others took the launch. Not everyone remaining aboard was a willing mutineer in the same sense.
- At Tofua on 2 May, Norton was killed while casting off the stern line. Bligh's book is the account we have. It infers motive. It is not a Tongan testimony.
- Pitcairn, January 1790: nine British mutineers, six Polynesian men, twelve women, and an infant. The ship was burned on 23 January.
- By 1794 the Polynesian men were dead and five mutineers had been killed. Adams was the last surviving mutineer, not the whole community.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Breadfruit and empire: moving a living crop

Breadfruit, Artocarpus altilis, is a starchy staple of the Pacific. People there cultivated it long before a British ship asked for cuttings. Many of the kinds grown for food are seedless. You do not ship them as a bag of seed. You ship a living plant: stem, root, and the water and shade those roots need. That is why a warship's cabin was rebuilt as a nursery. The botany is real. It is also the reason the voyage was so easy to ruin.

A seedless fruit has no seed to dry and pocket. Gardeners cut a shoot, keep the cut damp, and wait for roots. At sea the same shoot needs fresh water, air that is not a hold's fug, and space so one mold does not take the whole crate. Those are practical facts. They are also why the first cargo was fragile, and why a mutiny did not merely delay a sack of grain. It ended a greenhouse.

Joseph Banks organized the plan. He did not captain it. The stated use was cheap food for enslaved workers on Caribbean plantations, not a clean gift from one people to another. Scientific care and an oppressive purpose traveled in the same pots. The Banks lesson has the organizer. The Bounty lesson has the mutiny that threw the first cargo away in 1789.

A second try worked. HMS Providence and Assistant, under Bligh again, sailed in 1791 and in 1793 landed breadfruit at St Vincent and Jamaica, first into botanical gardens and then out toward plantations. The trees that grow in the Caribbean now are part of the food of those islands. Genetic studies tie modern cultivars back to Pacific stock moved in that era. A plant can become ordinary in a kitchen and still have a harsh reason for the crossing.

Nothing in that success rewrites the first plan. Ingenuity in keeping a tree alive at sea is not the same thing as a kind motive. The gardeners who received the pots in the Caribbean were doing botany. The people the fruit was meant to feed were not free to refuse the plantation. If you want the ship that failed, read the mutiny. If you want the man who ordered the trees, read Banks. This page is only the crop, and the fact that it was sent as provisions for people who were not free.

The illustrated lesson is download the PDF.

Sources

- Natural History Museum, on Banks: https://www.nhm.ac.uk/discover/joseph-banks-scientist-explorer-botanist.html
- Current Biology (2023), on the genetic history of breadfruit introductions: https://www.cell.com/current-biology/fulltext/S0960-9822(22)01901-7

- Royal Museums Greenwich, William Bligh: https://www.rmg.co.uk/stories/maritime-history/william-bligh
- Tropical Agriculture, on breadfruit in the Caribbean: https://journals.sta.uwi.edu/ojs/index.php/ta/article/view/6507

Key facts

- Artocarpus altilis was cultivated in the Pacific long before European voyages. Seedless kinds travel as living plants, not as seed.
- Banks's plan for the Bounty was cheap food for enslaved Caribbean workers, not a purely humanitarian gift.
- The Bounty never delivered. Providence and Assistant brought plants to St Vincent and Jamaica in 1793.
- Modern Caribbean trees are tied by genetics and by history to that transfer. The meal does not erase the purpose.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Captain James Cook: precision, encounter, and empire

James Cook was born in Yorkshire in 1728. He learned the sea in the coal trade, then joined the Royal Navy in 1755. Before any Pacific voyage he surveyed Newfoundland. The Navy already trusted him to draw a coast so that the next ship could use the drawing.

In 1768 the Admiralty gave him Endeavour. The announced task was the transit of Venus, 3 June 1769, at Tahiti, with the astronomer Charles Green. Joseph Banks and Daniel Solander sailed to collect and classify. Cook commanded, and he surveyed. After Tahiti he opened sealed instructions to sail farther and to chart what he found. In 1769 and 1770 he charted New Zealand and the east coast of Australia. Those shores were not empty, and they were not unknown to the people who lived there. A European chart is a new record for London. It is not the first knowledge of a place.

Cook's three Pacific voyages on a timeline

Three commissions, drawn for this lesson. The first includes the Tahiti transit and the charts of New Zealand and eastern Australia. The second crosses the Antarctic Circle. The third runs from 1776 to 1780. Cook died in 1779. The ships came home the next year.

The second voyage, 1772 to 1775, in Resolution and Adventure, crossed the Antarctic Circle. Cook put an end to the hope of a green southern continent sitting in the temperate ocean. He did not claim to have seen the mainland of Antarctica.

The third voyage, 1776 to 1780, in Resolution and Discovery, searched for a passage in the north. In 1778 the ships reached Hawaii. On 14 February 1779, at Kealakekua Bay, Cook was killed as a relationship that had already turned violent broke apart. His death did not end the expedition. Charles Clerke took command, and the ships reached England in 1780. The charts stayed in the Navy's hands. So did the harder consequence: disease, weapons, and later colonial power traveled along the same routes.

Hold both parts of the man. The angles and the coastlines were careful. The encounters were not a blank page, and they were not a moral slogan in either direction. The transit itself, and the risk of

getting the observers there, is the next lesson in this day's chain. Banks, who collected beside him and did not sail the second voyage, is the lesson after that.

The illustrated lesson is download the PDF.

Sources

- Royal Museums Greenwich, "Captain James Cook: timeline," https://www.rmg.co.uk/stories/maritime-history/captain-james-cook-timeline
- Whipple Museum, Cambridge, on the quadrant stolen at Tahiti and recovered: https://www.whipplemuseum.cam.ac.uk/explore-whipple-collections/astronomy-and-empire/stolen-quadrant

Key facts

- Cook, born 1728 in Yorkshire, joined the Royal Navy in 1755 and had already surveyed Newfoundland.
- Endeavour, 1768-1771: Tahiti transit on 3 June 1769 with Green; then charts of New Zealand and Australia's east coast, lands already inhabited.
- Second voyage, 1772-1775, crossed the Antarctic Circle and did not claim a sighting of the Antarctic mainland.
- Third voyage, 1776-1780, in Resolution and Discovery. Cook was killed at Kealakekua Bay on 14 February 1779. The ships came home in 1780.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Charles Mason: an astronomer on difficult ground

Charles Mason was born in Gloucestershire in 1728, a baker's or miller's son. In 1756 James Bradley brought him to Greenwich as an assistant. Not in 1750. The five years sometimes counted backward from 1761 are a way of saying he had already been there a while. The appointment year is 1756.

In September 1760 the Royal Society named him for the transit expedition toward Bencoolen, in Sumatra, with Jeremiah Dixon as his assistant. John Bird, the instrument maker, is the man usually credited with putting Dixon forward. That recommendation is the sturdy tradition. A document of their first handshake is not. This page will not invent the meeting.

The plan did not survive the voyage. After the attack on the Seahorse, told in the transit lesson, they observed from the Cape of Good Hope on 6 June 1761, and only the ending of the transit. Mason's times then entered other people's reductions. James Short compared the Cape with a set of European stations. Mason was an observer in a network, not the owner of the solar parallax.

Pingre later worried that the Cape longitude might be off, and Short had to answer him in print. The argument is a reminder, not a verdict against Mason. A clock time from the Cape is only as good as the place you assign to the Cape. Observers in Europe could not repair a bad longitude by wishing. They could only compare, reject, and say so.

In 1763 the same pair took the Penn-Calvert boundary. A zenith sector watches stars nearly overhead and yields a latitude. Chains and offsets turn that latitude into a mark on the ground. The legal line is geometry. It is useless if you do not know where the instrument stood. That is the transit rule brought ashore: the observer's place is part of the observation.

The survey lesson tells how the tangent was tried, found 22.51 chains off an older point, and corrected without a theatrical restart. Dixon's lesson tells the partnership from his side. Mason's job here is the simpler moral of the instruments. A number in the sky, or a post in the woods, means something only after you can say where you were standing when you took it.

The illustrated lesson is download the PDF.

Sources

- Edwin Danson, on Mason and Dixon's astronomy, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf

- Todd Babcock, on the boundary survey, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Mason and Dixon's 1767 letter on the line, from the survey record: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

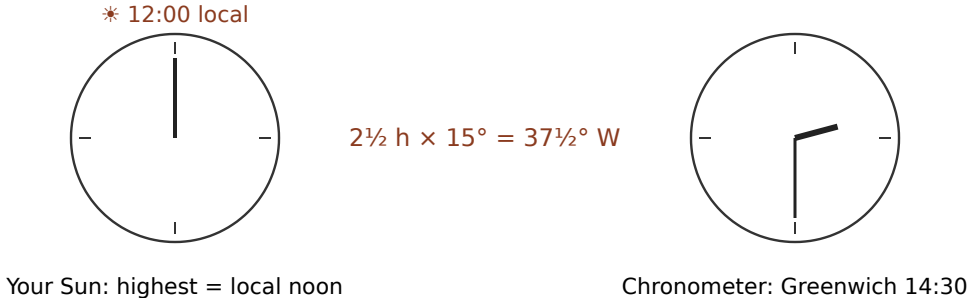
- Mason was born in 1728 and became Bradley's assistant at Greenwich in 1756, not 1750.
- He led the 1761 transit party, with Dixon as assistant. Bird is generally credited with recommending Dixon. Their first meeting is not documented.
- After the Seahorse attack they timed the end of the transit from the Cape. Short and others reduced those times with European stations.
- On the boundary, latitude from a zenith sector still required a known place on the ground.

FARIA · 7 NOVEMBER 2026

Longitude with a clock, in plain language

Before you sail, set an accurate chronometer to Greenwich time. Never reset it at sea. It is a piece of Greenwich you carry with you.

At sea, watch the Sun. Local noon is the moment it stands highest in your sky, not when an ordinary clock says 12. At that instant, read the chronometer.



Same instant, two times. Greenwich is ahead, so you are west.

Say it reads 14:30. Greenwich is 2 hours 30 minutes ahead of you. Earth turns 360° in 24 hours, so every hour is 15°. Two and a half hours is 37½° west. That number is your longitude. It is not how long you sailed. If Greenwich read 10:00 instead, you would be 30° east.

The 15° per hour holds at every latitude. What changes is the ground underneath: the circles you sail along (lines of latitude) shrink toward the poles, so the same 15° covers fewer miles. The clock still works everywhere.

What gets hard near the poles is the Sun, not the arithmetic. Its highest point is flatter and harder to pin down, and in polar winter it may not rise for weeks. At the pole itself every meridian meets, so longitude has no meaning.

What if we had started with tens?

We could divide a day into 10 hours, each hour into 100 minutes, and each minute into 100 seconds. A circle could have 400 divisions, called gradians, rather than 360 ordinary degrees. Revolutionary France actually tried decimal time. Tens fit our ten fingers and the way we write numbers.

Sixty has a useful advantage: it divides evenly by 2, 3, 4, 5, 6, 10 and 12. Ten does not. Decimal notation and modern calculation make that bookkeeping advantage less important, though one-third still cannot be written as a finite decimal.

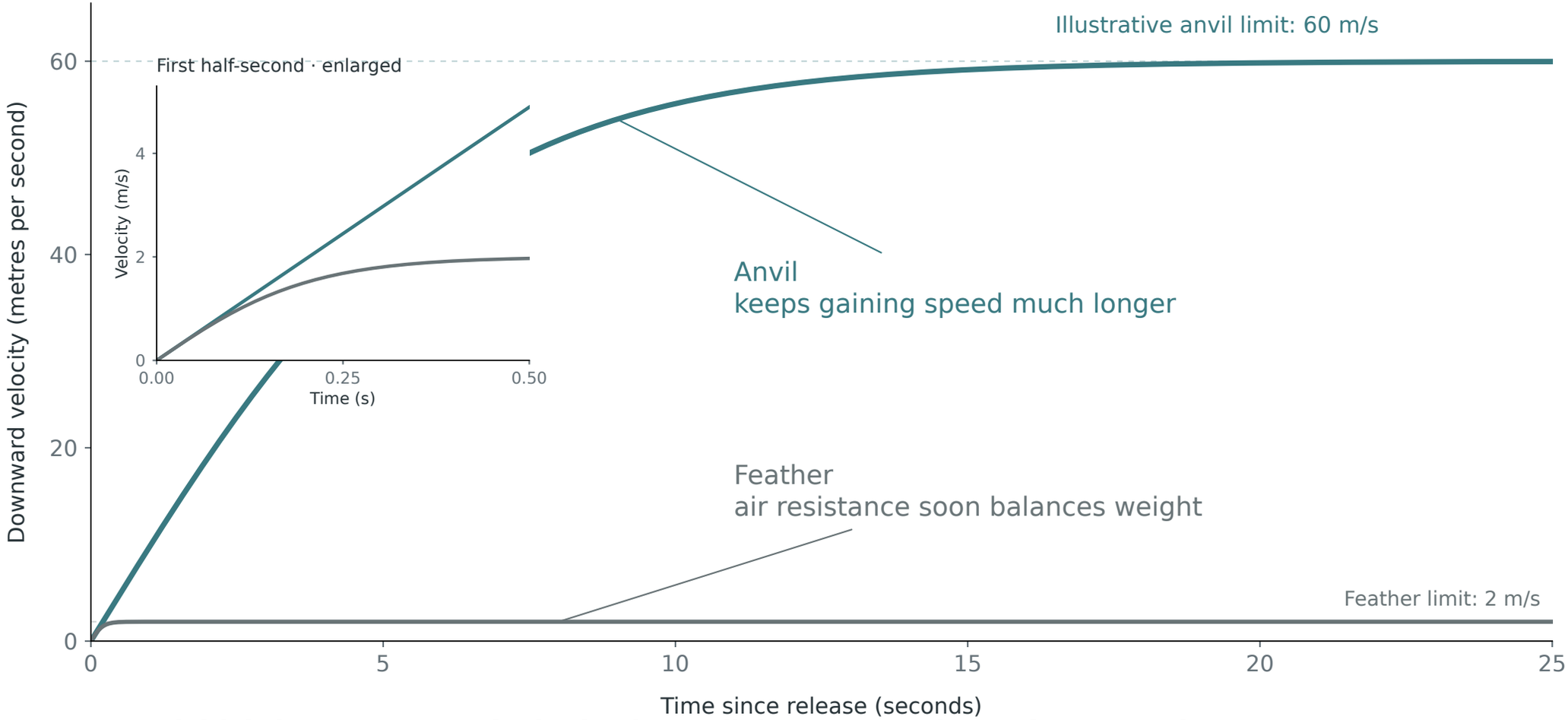
A 400-day year would look tidy beside a 400-part circle, but nature does not oblige: the seasonal year is about 365.24 ordinary days. Changing clock units changes neither the day nor the year, and would not remove or reduce the need for leap-day corrections.

This simple example ignores the small seasonal difference between sundial time and clock time (the equation of time, up to about 16 minutes). Navigators correct for it with an almanac.

Royal Museums Greenwich, “What is longitude?” rmg.co.uk/stories/time/what-longitude · U.S. Naval Observatory, “The Equation of Time” aa.usno.navy.mil/faq/eqtime · Decimal time: IMCCE, “The decimal clock” · Calendar: USNO, “Leap Years”

A feather and an anvil falling through air

Both start from rest with the same acceleration. Air resistance changes what happens next.



Same initial slope: 9.81 m/s² for both. The anvil does not accelerate faster at release.

Terminal speed means no further acceleration — not that the object stops falling.

Illustrative model, not measurements: chosen limits of 2 and 60 m/s. Real limits depend on mass, shape and air.

Enough falling distance is assumed; impact with the ground is not modeled.

Model: v(t) = V × tanh(9.81t / V); v = downward velocity, t = seconds, V = chosen terminal speed.

Physics reference: OpenStax, College Physics 2e, §5.2 “Drag Forces” · openstax.org/books/college-physics-2e/pages/5-2-drag-forces

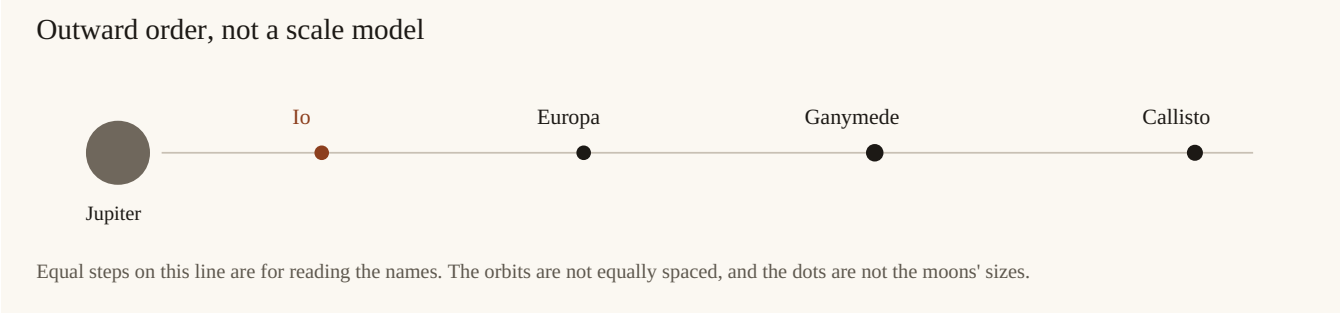
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Galileo's moons: new centers and lasting names

On 7 January 1610 Galileo saw points of light beside Jupiter, lined up east and west and changing place from night to night. He came to understand four bodies going around Jupiter. Sidereus Nuncius, in March 1610, told Europe. The fuller story of his telescope and his trial is the Galileo page. This page is only the moons, and the names.

They matter because not every moving body has to orbit the Earth. One old objection to Copernicus was that a traveling Earth would leave its Moon behind. Jupiter travels, in any of the systems then argued, and it keeps four companions. That weakens the objection. It does not finish the case. Tycho Brahe's system, Sun around the Earth and the other planets around the Sun, can hold moons around Jupiter just as well. New centers of motion are real. They are not, by themselves, a proof of a moving Earth.

Galileo named the four, together, the Medicean Stars, for the Medici who paid the patronage. He did not give them the names on a modern chart. Those names, Io, Europa, Ganymede, and Callisto, in that order outward, are Simon Marius's, in Mundus Iovialis of 1614. Marius said Kepler suggested them.



The four moons in outward order, not to scale

A reading order, drawn for this lesson. The gaps are equal so the names can be read. The orbits are not equally spaced, and the dots are not the moons' sizes.

Marius also said he first saw the moons on 29 December 1609. That date is Julian. On the Gregorian calendar it is 8 January 1610, the day after Galileo's 7 January. Do not treat the two calendars as if they were the same day, and do not call Marius a fraud because the numbers sit close. Who published a dated book first is one question. Who looked, independently, is another. The quarrel is real. A collapsed calendar is not evidence.

Io, the inner one, later became a clock. When it enters Jupiter's shadow the reflected sunlight stops, and the timetable of those eclipses taught Rømer that light takes time. That is the next lesson.

The illustrated lesson is download the PDF.

Sources

- Museo Galileo, on Jupiter's moons: https://catalogue.museogalileo.it/indepth/JupitersMoonsSatellites.html
- Museo Galileo, Sidereus Nuncius (1610): https://www2.museogalileo.it/it/biblioteca-e-istituto-di-ricerca/progetti/incontra-galileo/31-opere/454-sidereus-nuncius-1610.html
- Pasachoff, "Simon Marius's Mundus Iovialis and the discovery of the moons of Jupiter," IAU proceedings, 2018: https://ui.adsabs.harvard.edu/abs/2018iau3.book...38P/abstract

Key facts

- Galileo recorded moons of Jupiter from 7 January 1610 and published them in March 1610 as the Medicean Stars.
- Moons around Jupiter answer one objection to a moving Earth. Tycho's system can include them too. They are not a complete proof of Copernicus.
- Io, Europa, Ganymede, and Callisto are Marius's 1614 names, which he credited to a suggestion from Kepler.
- Marius's 29 December 1609 is Julian. Gregorian, it is 8 January 1610. Do not equate the calendars or assume fraud from the numbers.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Hartz and Hart Schaffner and Marx: names are not a pedigree

Lena Hartz and her daughter Anna belong to a Baltimore clothing family. Anna married Joseph Bank, and that story is the clothier lesson. Harry Hart and Max Hart belong to Chicago. The trade is the same. The name is similar. A pedigree is something else.

In Chicago the roots are a Hart brothers' shop in 1872. In 1879 Levi Abt and Marcus Marx joined them. Joseph Schaffner joined in 1887, and the familiar name Hart Schaffner & Marx is that company. Do not say the firm was formed after 1945. That year is the Baltimore buyout, in another city, of a Hartz share in a different business.

A careful history of Schaffner's migration, from the German Historical Institute's work, shows a network of people from near Worms. It shows how a young man entered a Chicago house. It does not show a contract with Baltimore. Some accounts say Abt and Marx left when Schaffner arrived. Others say Marx stayed. Those details conflict, they are not needed for this question, and they are not a bridge to the Hartzes. They are left out on purpose.

No partnership paper, no share ledger, and no family record found for this lesson ties the Baltimore Hartz firm to the Chicago company. Read that as a limit, not as a triumph. Not finding a document does not prove the families were strangers back in Europe. It does forbid the sentence "they were one house."

What would count? For a business, a partnership agreement, articles of incorporation, or a record of ownership. For kinship, a birth, a marriage, a ship's list, a census, or a will that names people on both sides. A German-sounding surname in the garment trade is the start of a search. It is not the result.

If a "Hart" in Chicago is needed for context, there is already another one. Cousins Abraham and Henry Hart had a different firm, also spoken of as Hart brothers, and it was not this company. Similar names inside one city are a warning, not a family tree. The Bank lesson has the Baltimore side. This page only refuses the false pedigree.

The illustrated lesson is download the PDF.

Sources

- Encyclopedia of Chicago, Hart, Schaffner & Marx: https://encyclopedia.chicagohistory.org/pages/2695.html

- Immigrant Entrepreneurship, "Joseph Schaffner": https://www.immigrantentrepreneurship.org/entries/joseph-schaffner/
- Encyclopedia.com, Hartmarx: https://www.encyclopedia.com/social-sciences-and-law/economics-business-and-labor/businesses-and-occupations/hartmarx-corp

Key facts

- Baltimore's Hartz family is part of the Jos. A. Bank story. Chicago's Harry and Max Hart are part of Hart Schaffner & Marx. They are not the same firm.
- The Chicago company grows from 1872, with Marx and Abt in 1879 and Schaffner in 1887. It was not formed after 1945.
- No business record or kinship record found here links the two. That does not prove there was never a remote family tie.
- A real link would be a partnership, an ownership record, or a civil document naming both sides. The same trade and a similar name are not that link.

The Harvard Computers

Women who measured the stars — before computer meant a machine

Abbé Faria lesson for Jeff Franzen · 6 November 2026 · faria.jbf.com/topics/51-harvard-computers/

Jeff, reading about Cepheids on the Polaris page, stopped at "Henrietta Swan Leavitt (1868–1921), a Harvard 'computer'" and said he could not be left hanging on the word. This lesson is the full answer.

Women astronomical computers at Harvard College Observatory, c. 1891 — "Pickering's Harem." Williamina Fleming near center; Antonia Maury (with glass) third from left. Leavitt and Cannon are not in this photo. Harvard College Observatory / Public domain.

1. The word: computer meant a person

Computer is compute + -er: a person who computes.

Earliest recorded English use: Richard Brathwaite (also spelled Braithwaite), The Yong Mans Gleanings (London, 1613):[1]

> *"I haue read the truest computer of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number."*

The OED's historical sense: a person employed to make calculations in an observatory, in surveying, etc. The electronic-machine sense clusters in the 1940s. Until then, computer almost always meant a human being.

Gloss. In Brathwaite the "computer" is God, the reckoner of our days ("The daies of Man are threescore and ten"); the word then moved from a reckoner of time, to human calculators like the Harvard women, to machines.

[1] Footnote — Brathwaite 1613. Exact wording as cited by the Oxford English Dictionary (earliest quotation for computer): Richard Brathwaite ("R. B."), The Yong Mans Gleanings (London, 1613), p. 1: "I haue read the truest computer of Times, and the best Arithmetician that euer breathed, and he reduceth thy dayes into a short number." Original spelling preserved (haue, euer, dayes). The passage continues: "The daies of Man are threescore and ten" (Psalm 90:10) — so the "computer of Times" is God counting out a human life, not a clerk with an abacus. Same quotation in Wikipedia's Human computer etymology note.

Human computers before Harvard. The British Nautical Almanac hired dozens of human computers for lunar and solar tables. From 1765–1811 about 35 worked on it; Mary Edwards was a woman who computed for a living. That world is the cousin of Nathaniel Bowditch's world of checked tables a navigator trusted with a ship.

2. Edward Pickering and the factory observatory

Edward Charles Pickering (1846–1919) became director of the Harvard College Observatory in 1877. Photography and spectroscopy flooded astronomy with glass plates. Starting in the early 1880s (Fleming moved from household staff into the computing room in 1881), he hired women to measure positions, magnitudes, and spectra by day. Over the decades about 80 women served. They were called computers. Observing itself stayed mostly closed to them.

Pay was about 25 cents an hour (Fleming as curator ~50¢; Leavitt reportedly ~30¢). Men doing similar work earned substantially more. The mocking nickname was "Pickering's Harem." Modern writers say the Harvard Computers.

The Observatory (Cambridge, plus Arequipa, Peru) amassed on the order of 500,000 glass plates. Funding came largely from Anna Palmer Draper, widow of Henry Draper — the Henry Draper Memorial that paid for the spectral survey and the Henry Draper Catalogue (HD).

Computers at work over astronomical photographs, c. 1890. Public domain.

3. Williamina Fleming: from housemaid to curator

Williamina Paton Stevens Fleming, c. 1890s. Public domain.

Williamina Paton Stevens Fleming (1857–1911), a Scottish schoolteacher, emigrated to Boston, was abandoned while pregnant, and worked as Pickering's maid/housekeeper (~1879). In 1881 he put her on the Observatory payroll. She helped invent the early Harvard spectral letters; discovered the Horsehead Nebula on a plate taken 1888 Feb 6 by W. H. Pickering (she noted it 1888 May 19); found hundreds of variables, novae, and the first recognized white dwarfs; and in 1899 became Harvard's first Curator of Astronomical Photographs.

4. Antonia Maury: a finer classification

Antonia Maury (1866–1952), a niece of Henry Draper, built a finer scheme that distinguished giants from dwarfs by line width. Pickering preferred Fleming's simpler letters for the big catalogue; Maury's subtler distinctions were vindicated when Hertzsprung and Russell made luminosity classes central to stellar evolution.

Oh Be A Fine Girl, Kiss Me

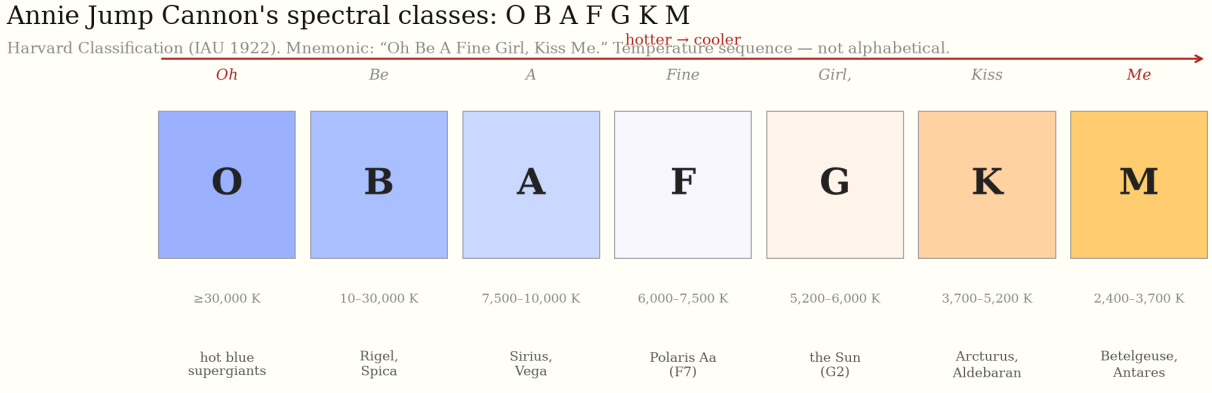
How Annie Jump Cannon sorted the stars.

Annie Jump Cannon, 1922. Public domain.

Annie Jump Cannon (1863–1941) studied at Wellesley, lost much of her hearing to scarlet fever, and joined in 1896. She reordered Fleming's letters into the temperature sequence still used worldwide:

O – B – A – F – G – K – M

That mnemonic — the heading of this section — looks alphabetical; it is a temperature sequence (O hottest, M coolest). The Sun is G2. Polaris Aa is F7Ib.



Cannon's spectral sequence with approximate temperatures and familiar examples. Diagram for this lesson.

For the nine volumes of the Henry Draper Catalogue (1918–24) she classified about 225,300 spectra; across her lifetime about 350,000 (some tallies near 400,000). The IAU adopted her system in 1922. She received the Henry Draper Medal (1931) but was not made a Harvard professor until 1938.

6. Henrietta Swan Leavitt: the period–luminosity law

Henrietta Swan Leavitt (1868–1921). Public domain.

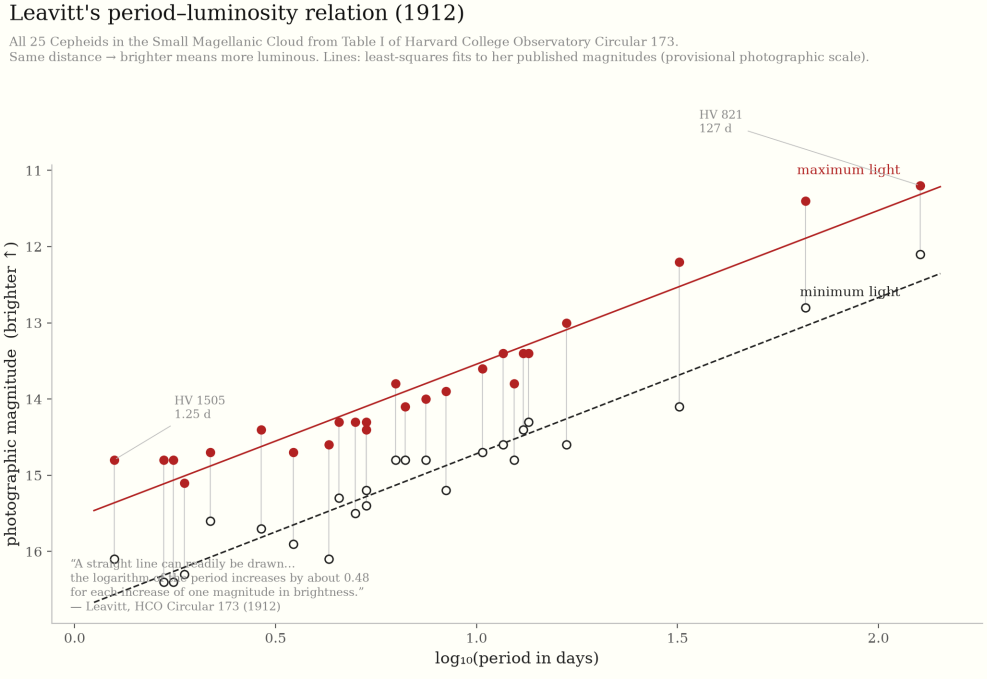
Henrietta Swan Leavitt (4 Jul 1868 – 12 Dec 1921) graduated from what became Radcliffe, volunteered in 1895, took a permanent post in the early 1900s, and was nearly deaf. Her assignment: variable stars on Magellanic Cloud plates.

1908. In 1777 Variables in the Magellanic Clouds she wrote:

> *"It is worthy of notice that the brighter variables have the longer periods."*

1912. HCO Circular 173 (3 Mar 1912), prepared by Miss Leavitt under Pickering's signature, plotted 25 Cepheids:

> *"A straight line can readily be drawn among each of the two series of points corresponding to maxima and minima… The logarithm of the period increases by about 0.48 for each increase of one magnitude in brightness."*



All 25 stars from Table I of Circular 173 (1912), replotted from her published periods and provisional magnitudes. Brighter is upward.

If period tells true brightness, comparing true with apparent brightness yields distance. Cepheids become standard candles. Hertzsprung (1913) and Shapley calibrated the zero point; Edwin Hubble (1923–25) used Cepheids in

Andromeda to show it lies outside the Milky Way. The relation is now often called Leavitt's Law.

The Nobel letter that came too late. On 23 Feb 1925, Gösta Mittag-Leffler wrote, unaware she had died in 1921:

> *"Honoured Miss Leavitt… I feel seriously inclined to nominate you to the Nobel prize in physics for 1926…"*

Nobels are not awarded posthumously. She never received the nomination.

Annie Jump Cannon (left) and Henrietta Swan Leavitt (right), 1913. Public domain.

7. Cecilia Payne: stars are mostly hydrogen

Cecilia Payne (later Payne-Gaposchkin, 1900–1979) came from Cambridge to Harvard in 1923. Her 1925 Ph.D. thesis used Cannon's classes and Saha ionization theory to show stars are overwhelmingly hydrogen and helium. A senior reviewer urged her to soft-pedal; she did in print, then lived to see it become orthodoxy. In 1956 she became Harvard's first woman professor of astronomy.

8. Later recognition

- Dava Sobel, The Glass Universe (2016).
- DASCH and Project PHaEDRA: digitizing ~500,000 plates and the computers' notebooks.
- Lunar crater Leavitt; others honor Cannon and Fleming. AAS (2008) urged calling the relation Leavitt's Law.

Review questions

1. What did computer mean in 1613, and who used it?
2. Rough pay per hour, and the mocking nickname?
3. Who found the Horsehead Nebula, on whose plate, what year?
4. Recite O B A F G K M and the mnemonic. Sun's letter? Polaris Aa?
5. What did Leavitt notice in 1908, and what did Circular 173 prove?

6. Why does a shared distance make a period–luminosity relation possible?
7. What did Mittag-Leffler write in 1925, and why no Nobel?
8. What did Payne's 1925 thesis conclude?

Answers

1. A person who calculates; Richard Brathwaite, Yong Mans Gleanings (1613), "the truest computer of Times" — first of God, later of human calculators, then of machines.
2. ~25¢/hour; "Pickering's Harem."
3. Williamina Fleming, on W. H. Pickering's plate, 1888.
4. O B A F G K M — "Oh Be A Fine Girl, Kiss Me." Sun = G (G2). Polaris Aa = F (F7Ib).
5. Brighter Magellanic variables have longer periods; 1912: straight-line log(period)–brightness relation for 25 Cepheids.
6. Same distance → apparent brightness differences are true luminosity differences.
7. He was inclined to nominate her for the 1926 Physics Nobel; she had died in 1921.
8. Stars are mostly hydrogen and helium.

Companion page: https://faria.jbf.com/topics/51-harvard-computers/ · Related: Polaris (48), Bowditch (49), Sidereal day (50).

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

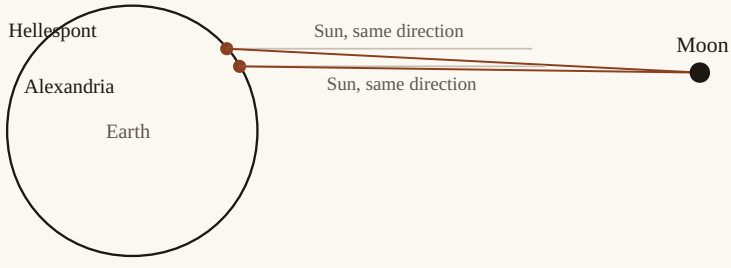
Hipparchus: the Moon changes place when we move

Hold a finger at arm's length and look with one eye, then the other. The finger jumps against the far wall. The gap between your eyes is the baseline. The angle of the jump, with that baseline, is a distance. That is parallax. Hipparchus, in the second century BCE, used it on the Moon. Most of his books are lost. Later astronomers rebuilt the argument from what they still had.

One famous piece uses a solar eclipse. Near the Hellespont the Sun's disk was totally obscured. At Alexandria four-fifths of that disk was obscured, so only about one-fifth of the Sun's diameter stayed uncovered. The whole shift between the two places was that fifth of the diameter. It is a fraction of the diameter, not of the area: four-fifths of the area would be a different, larger bite. A date often proposed is 14 March 190 BCE. The identification is not secure, and this lesson will not pretend it is. Hipparchus used reports. He did not stand in both cities.

Two places on the Earth, one Moon, parallel sunlight

Stations are on the circle. Grey lines are parallel. The Moon is drawn much too close.



Both dots are on the circle. Hellespont is north of Alexandria. The grey lines are parallel. The Moon is drawn much too close, so the angle is enlarged. Total at the Hellespont. At Alexandria four-fifths of the diameter was covered, so about one-fifth stayed bright.

Two stations, one Moon, a shared solar direction

The angle is drawn far too large. The Sun's own parallax is treated as negligible beside the Moon's. Drawn for this lesson.

The baseline is the north-south distance on the Earth between those latitudes. The sightlines to the Moon differ. The Sun is so far that its direction can be treated as the shared wall. G. J. Toomer's reading of the reconstruction gives one lower bound near 71 Earth radii. Other ways of treating the shadow and the Sun's distance give a band roughly 59 to 67 Earth radii. That band is not a modern confidence interval. The Moon really averages near 60 Earth radii, and the distance changes a little through the month. An ancient model can land close to the truth and still be a model with wide edges. Nearness is not the same thing as a tight measurement.

Compare the sticks. The Venus stations of 1761 and 1769 stand on the Earth's surface, a baseline of hundreds or thousands of miles. Kepler's Mars triangles, later in this day's lessons, use a baseline the size of Earth's orbit. Hipparchus is the same idea with the cities of two reports. Aristarchus, just before this page, was comparing sizes inside one triangle. Here the observer moves, and the Moon moves on the sky.

The illustrated lesson is download the PDF.

Sources

- MacTutor, biography of Hipparchus: https://mathshistory.st-andrews.ac.uk/Biographies/Hipparchus/
- MacTutor, Dictionary of Scientific Biography excerpt: https://mathshistory.st-andrews.ac.uk/DSB/Hipparchus.pdf

Key facts

- Parallax is a known baseline and an angle. Hipparchus's own books are mostly lost and were rebuilt by later writers.
- The eclipse was total at the Hellespont. At Alexandria four-fifths of the Sun's diameter was covered, so about one-fifth of the diameter stayed bright. That fifth is the parallax shift. It is not four-fifths of the area. He used reports. The 190 BCE date is uncertain.
- Reconstructions give a lower bound near 71 Earth radii in one reading, and roughly 59 to 67 in others. That is not one confidence interval.
- The modern mean is near 60 Earth radii and varies. Venus transits use a surface baseline. Kepler's Mars work uses the orbit.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Jeremiah Dixon: the surveyor in the partnership

Jeremiah Dixon was born in 1733 at Cockfield, in County Durham, into a Quaker family with means. He trained as a surveyor and worked as an amateur astronomer. When the Royal Society needed a second man for the 1761 transit, John Bird's recommendation is the account to trust, and to trust only as far as it goes. Bird made instruments and knew both worlds, the workshop and the observatory. A family story about an examination at Woolwich is thinner: the Ordnance minutes that would confirm it have not been produced. Say "recommended through Bird's network," not "we watched them meet."

A Quaker surveyor in the north of England already knew how a chain lies, how a slope cheats a distance, and how a neighbor disputes a hedge. That craft is not astronomy, and astronomy is not that craft. The Royal Society needed both on one road, which is why the pairing matters more than which man stood an inch to the left.

Dixon was not a spare hand hired to hold Mason's chain. At the Cape he observed. On the boundary, from 1763, surveying was his profession and astronomy was Mason's daily habit. The work needed both. A chain of standard length, a slope corrected, a trial line checked by a second measurement: those are the surveyor's corrections, and they are legitimate. They are not a confession that the first number was a scandal.

In October 1767 the Six Nations escort stopped at the war path. Their leader told the party that this was as far as his commission from the chiefs ran, and that he would not go one step farther west. That sentence is about a limit set by the people who sent him. It is not a license to invent a speech about fear, and it is not a permission slip written by the geometry. Some of the hired men, separately, had already refused to cross a river for fear of the Shawnee and the Delaware. Two facts. Do not melt them into one slogan.

Dixon died in Britain in 1779. The line he helped lay is the next lesson. Mason's lesson has the Greenwich years and the rule that a place must be known before a sight is interpreted. The pair only works if you can see both men.

Norway in 1769: Dixon and Chappe compared

Who observed the transit of 3 June 1769 better, Jeremiah Dixon in Norway or Jean-Baptiste Chappe d'Auteroche in Baja California? Weather, instruments, timing, and later praise all belong in the question. They do not add up to a winner.

Dixon was not at Vardo with Maximilian Hell, and he was not beside William Bayly. Nils Voje Johansen's account puts Bayly at Honningsvag, near the North Cape, and Dixon at Rypeklubben, just outside Hammerfest. Cloud held Dixon until 20 May, his first clear observations. On 3 June the weather was cloud and gales. Johansen writes that both men "failed to observe the transit satisfactorily." That sentence is about the transit. Dixon and Bayly still published the positions of the observatories, altitudes of the Sun, meridian transits, the solar eclipse that followed, and notes on the weather. A missed contact is not an empty expedition.

Dixon's own instrument list is not known. Johansen reconstructs Bayly's from the papers they printed: a quadrant of one-foot radius by Bird, a two-foot reflector by Dollond, a four-foot transit instrument by Bird with a Dollond lens, and clocks by Shelton, including an astronomical clock with a gridiron pendulum. Bayly said Dixon's instruments were similar, except that Dixon had no dipping needle. That is a reconstruction, not a case opened and listed. A contact time to the second, for Dixon, was not established here. Bayly wrote down a hazy contact of his own. Those seconds stay his.

Chappe's list is his own, quoted by Espinoza Meléndez: a three-foot-radius quadrant by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. He had checked the clock and the telescopes, with repeated observations of the Sun, before 3 June. The landing, the illness, and the death on 1 August are on the Venus lesson. So is the longitude. Cassini's adopted figure in the 1772 volume comes from the transit, not from an improvement measured by the eclipse of 18 June. A seconds-level accuracy for Chappe himself was not established. Cassini's 20 or 30 seconds of time is a caution about the Jupiter-moon method, not a personal error bar.

Later pages praise the Baja work. A Smithsonian exhibition describes the conditions as ideal. An American Scientist article calls the observations among the best made of either transit, and the position work unusually accurate for hard fieldwork. That is later praise, from those pages, not a jury sitting in 1769. No formal ranking of Dixon against Chappe was found, and nothing found here says Dixon was the less skillful man. The sky at Hammerfest failed him. No weighting by Encke that picks a winner was found.

The illustrated lesson is download the PDF.

Sources

- Edwin Danson, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf

- Todd Babcock, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- The 1767 letter, Swarthmore transcription: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html
- Nils Voje Johansen, "The Expeditions of William Bayly and Jeremiah Dixon to Honningsvåg and Hammerfest, 1769," Journal of Astronomical Data 19, 1 (2013): https://journalofastronomicaldata.be/JAD19/jad19_1/jad19_1f.pdf
- Pedro Espinoza Meléndez, "Medir el sistema solar," Meyibo (UABC). Year and pages are not shown on the copy used here: http://meyibo.tij.uabc.mx/index.php/meyibo/en/article/download/118/110/195
- Chappe d'Auteroche, *Voyage en Californie*, Cassini's reductions, Paris, 1772, pp. 88-92: https://archive.org/download/voyageencaliforn00chap/voyageencaliforn00chap_djvu.txt
- Smithsonian Libraries, Chasing Venus, "Measuring the Universe": https://www.sil.si.edu/exhibitions/chasing-venus/measuring.htm
- American Scientist, "Transits, Travels and Tribulations IV": https://www.americanscientist.org/article/transits-travels-and-tribulations-iv

Key facts

- Dixon was born in 1733 at Cockfield, County Durham, and worked as a surveyor and amateur astronomer.
- Bird's recommendation brought him to Mason. The literal first meeting is unknown, and the Woolwich exam story is unverified.
- He observed at the Cape in 1761 and surveyed the boundary from 1763. Planned corrections are part of the method.
- In October 1767 the escort stopped at the commission's geographic limit. Dixon died in Britain in 1779.
- In 1769 Dixon was at Rypeklubben, near Hammerfest, and Bayly at Honningsvag. Hell at Vardo was a separate station. Cloud until 20 May, and cloud and gales on 3 June. Johansen's words are that both "failed to observe the transit satisfactorily." They still published the place and other observations.
- Dixon's exact instruments are not known. Johansen's list reconstructs Bayly's kit, which Bayly called similar except for the dipping needle. Chappe's own list names the Canivet and English

quadrants, a transit instrument, a parallactic machine, two Dollond telescopes, and a Berthoud clock.

- Later histories praise Chappe's Baja observations. No contemporary ranking, and no Encke weighting, was found. The adopted longitude in the 1772 volume is the transit's, not an eclipse improvement. The weather, not a judgment of skill, is what failed Dixon.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Jos. A. Bank: an American clothier, not Joseph Banks

The name looks like the botanist and is not him. Jos. is an abbreviation of Joseph. Bank, on the shop signs, has no final s. Sir Joseph Banks sailed with Cook and is the subject of another lesson on this day. This page is a clothing company in Baltimore.

The usual founding date is 1905, with a family workshop behind it. Charles Bank, a Lithuanian immigrant, was a tailor. His grandson Joseph worked young, as a cloth cutter and a salesman. A family trade, a formal company, and a brand name are three dates. Do not collapse them into one man who did everything in one year. Sources even disagree about the grandson's middle name, so this lesson uses Joseph and does not pretend to settle it.

Joseph married Anna Hartz, of a Baltimore clothing family. The partnership called L. Hartz & Bank belongs to those early years. Published accounts put it in 1912 or in 1922. Both years are in print, so this page will not pick one it cannot defend. In 1945 Joseph and his son Howard bought the Hartz share. The brand Jos. A. Bank is the name from that change of ownership.

Shops of their own came in the 1950s, a catalog around 1960. There were 11 stores when Quaker Oats bought the firm in 1981. A leveraged buyout in 1986 left a heavy debt and a near failure, then a recovery and a public sale of shares in 1994.

On 18 June 2014 Men's Wearhouse completed the purchase at 65 dollars a share, about 1.8 billion dollars, after the two companies had tried to buy each other in 2013. It was not a sale to Brooks Brothers in 2011. The parent company took the name Tailored Brands in 2016. In 2020 that parent went through Chapter 11 and emerged as a private company on 1 December. The brand still sells clothes. A current store count is easy to invent and is left out.

No British chain named for the botanist turned up in the records checked for this lesson. Not finding a chain is not a proof that none ever existed. It is a reason not to tell that story. The Chicago firm Hart Schaffner & Marx, next, is another similar name with no documented tie to these Hartzes.

The illustrated lesson is download the PDF.

Sources

- Encyclopedia.com, Jos. A. Bank Clothiers: https://www.encyclopedia.com/books/politics-and-business-magazines/jos-bank-clothiers-inc
- Tailored Brands, 18 June 2014, on the Men's Wearhouse acquisition: https://www.tailoredbrands.com/2014/06/18/mens-wearhouse-completes-acquisition-of-jos-a-bank/

- Tailored Brands, 1 December 2020, on emerging from restructuring: https://www.tailoredbrands.com/2020/12/01/tailored-brands-successfully-completes-financial-restructuring/

Key facts

- Jos. A. Bank is a Baltimore clothier. Joseph Banks the botanist is a different person. The spellings differ.
- The firm is dated from 1905, with earlier family work. Charles Bank was the immigrant tailor. Joseph, his grandson, was the young cutter. Do not force one sole founder.
- In 1945 Joseph and Howard bought the Hartz interest. The 1912 and 1922 partnership dates conflict, so neither is asserted here.
- Men's Wearhouse bought the company on 18 June 2014 for about 1.8 billion dollars. Not Brooks Brothers in 2011. Tailored Brands emerged from Chapter 11 on 1 December 2020.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Joseph Banks: the botanist who built a network

Joseph Banks (1743-1820) was a British naturalist with money enough to turn curiosity into cargo. In 1766 he collected in Newfoundland and Labrador. In 1768 he joined Cook on Endeavour. Daniel Solander classified. Sydney Parkinson drew. A dried plant, a label, and a drawing are a scientific record. They are a European record. People who already grew or gathered those plants had not been waiting for a London name.

What traveled home was not a conquest of the plant. A pressed specimen keeps a shape. Parkinson's drawing keeps a color the dried sheet loses. Solander's name puts the plant in a European cabinet. None of those three objects moves the people who already cooked the leaves or fruit. The cabinet is a record of a visit. It is not a discovery in the sense of finding an empty world.

Banks came home in 1771. He did not go on Cook's second voyage. In 1772 he sailed to Iceland instead. From 1778 until he died in 1820 he was president of the Royal Society. He advised King George III and had a long influence on the gardens at Kew. Influence is the right word. He was not the formal director of Kew, and this lesson will not give him that title.

He is also the organizer of the breadfruit voyages, not their captain. The plan was to move a living Pacific crop to the Caribbean as cheap food for enslaved plantation workers. The packing of living trees was ingenious. The purpose sat inside an empire that ran on forced labor. Both sentences belong in the account. The voyage itself, and the mutiny that stopped the first attempt, are told with the Bounty. The plant is told in the breadfruit lesson.

One warning about the name. Jos. A. Bank, the Baltimore clothier, is a different man and a different century. The botanist has a final s. The firm usually does not. Nothing in Banks's career is a clothing chain, and this page will not invent one.

The link back to Cook is the ship they shared in 1768. The link forward is the crate of breadfruit trees Banks wanted sent, and the American firm that only looks like his name. A label in a cabinet is not a deed to the plant, and a presidency is not a captain's log.

The illustrated lesson is download the PDF.

Sources

- Natural History Museum, "Joseph Banks: scientist, explorer, botanist," https://www.nhm.ac.uk/discover/joseph-banks-scientist-explorer-botanist.html

- Royal Botanic Gardens, Kew, history of the gardens: https://www.kew.org/about-us/history-of-kew
- Royal Society note on Banks: https://royalsociety.org/science-events-and-lectures/2017/09/joseph-banks/

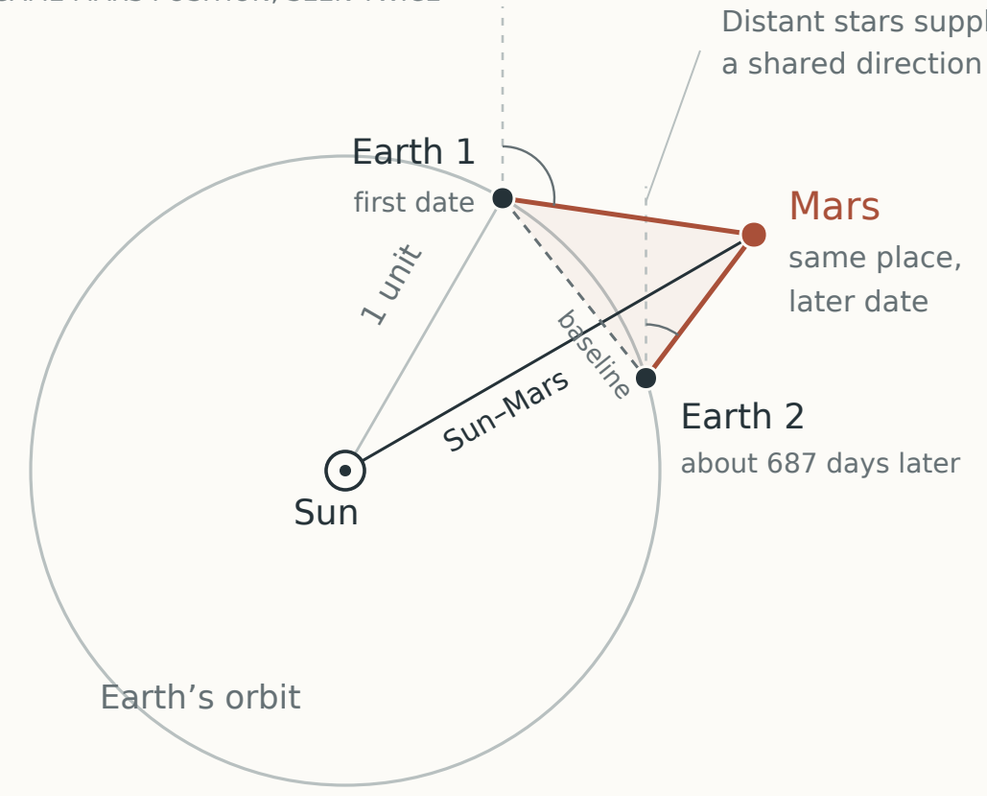
Key facts

- Banks (1743-1820) collected in Newfoundland and Labrador in 1766 and sailed on Endeavour in 1768 with Solander and Parkinson.
- He returned in 1771, did not join Cook's second voyage, and went to Iceland in 1772.
- President of the Royal Society 1778-1820. He advised the king and influenced Kew. He was not Kew's formal director.
- He organized the breadfruit missions. He was not captain of the Bounty. He is not the clothier Jos. A. Bank.

How Kepler found Mars without going there

One returning planet. Two observing places. A triangle with a scale.

THE SAME MARS POSITION, SEEN TWICE



1 unit · mean Earth–Sun distance

Teaching circle only. Kepler also reconstructed and refined Earth’s orbit.

1 Wait for Mars to return.

Tycho’s records supplied paired sightings about 687 days apart: one Mars orbit, not the interval between successive oppositions. Earth is now elsewhere.

2 Cross the two sightlines.

Place both Earth locations on a map of Earth’s orbit. Draw each measured direction to Mars. Their intersection recovers this one Mars position.

3 Read a relative distance.

The Earth–Earth chord sets the scale. Measure Sun to Mars on that map: about 1.5 Earth–Sun units here. That distance varies around the orbit.

A star direction alone is not a distance.

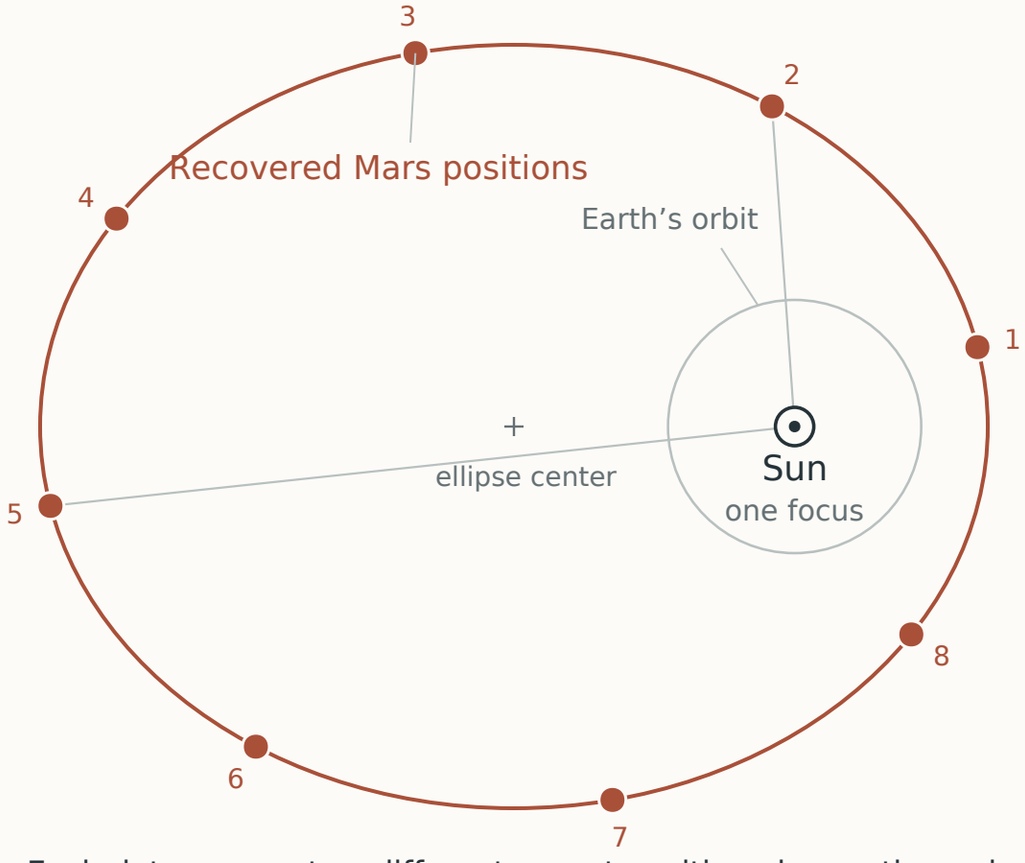
Use a true triangle, not three points in a line.

Sources: “Using Kepler’s Triangulation Technique,” Univ. of Central Arkansas teaching handout, pp. 2–3 (faculty.uca.edu/saustin/3110/mars.pdf).
History: O. Gingerich, “The great Martian catastrophe and how Kepler fixed it,” Physics Today 64(9), 50–54 (2011). • Prepared 7 Nov 2026.

Many triangles reveal an ellipse

Faria’s lesson: locate the points first; let the evidence choose the curve.

REPEAT AT OTHER PLACES IN MARS’S ORBIT

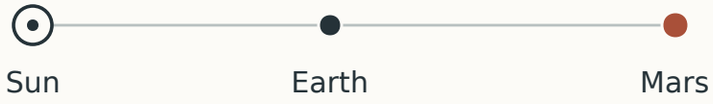


Each dot represents a different repeat-position observation pair.

Ellipse flattening and focus offset exaggerated; this panel is not to scale.

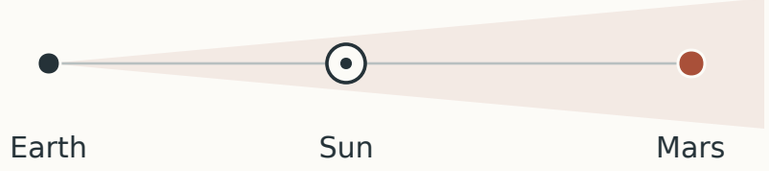
TWO ALIGNMENTS — NOT TWO DISTANCE METHODS

Opposition: Earth is in the middle.



Mars is opposite the Sun in Earth’s sky.
The straight alignment alone does not fix distance.

Conjunction: the Sun is in the middle.



Mars lies in the Sun’s direction: solar glare prevents a usable ordinary sighting near conjunction.

No permanently hidden half-orbit.

Visibility changes with Earth’s viewpoint.
Mars can be seen in many orbital places over time.

Sources: NASA Science, “Orbits and Kepler’s Laws”; “Close Encounter: Mars at Opposition”; “Geometry of Mars Solar Conjunction.”
science.nasa.gov • Geometry simplified to one plane. Dots, bodies and glare cone are illustrative; no historical observations are plotted.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

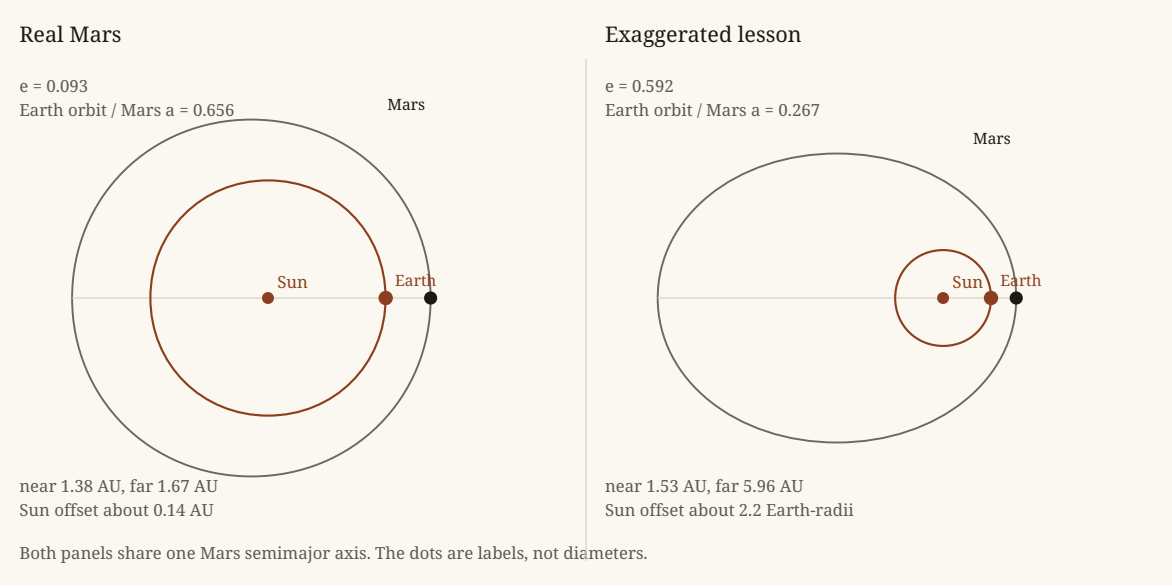
Kepler's Mars triangles, and an honest scale check

Kepler did not guess Mars's orbit from one night. Mars returns to the same place in its path about every 687 days. Earth has moved. If you know where Earth is, two lines of sight to that same Martian place cross, and you have a point in space. Angles measured against the stars give the directions. One angle plus a guessed radius is not enough. Kepler also had to repair Earth's own orbit before the crossings would sit still. Many pairs, not one clever triangle, drew the curve. It is an ellipse, with the Sun at a focus.

Opposition and conjunction are easy to mix up with a measurement. At opposition the order is Sun, Earth, Mars, with Earth in the middle. Mars is up all night and near its brightest. At conjunction the order is Earth, Sun, Mars. Mars is lost in the glare. Nothing is eclipsed. A straight alignment is not, by itself, a distance. Hipparchus's lesson uses a short baseline on the Earth. Here the baseline is Earth's whole orbit, the same stick that, once it has a length in miles, carries the transit lesson outward.

An early illustrated sheet of this ellipse, made to be obvious in a small picture, stretched the truth and said so in a caption. It was not a printer's accident. On that sheet Mars's semiaxes were 247 and 199 drawing units, and Earth's orbital radius was 66. The ratio is about 3.74, against the real ratio of semimajor axes, about 1.524. The drawing is about 2.46 times too stretched. Its eccentricity is about 0.59, against Mars's real 0.093, roughly six times too oval. The Sun sits about 2.2 of those Earth-radii from the center. In the sky the offset is about 0.14 AU. If you treat the drawn Earth radius as 1 AU, the sheet's nearest and farthest distances are about 1.53 and 5.96 AU. The real ones are about 1.38 and 1.67 AU.

Earth's orbital radius ought to be about 66 percent of Mars's semimajor axis, not about 27 percent. That is why Earth looks tiny on the old panel. The tininess is the exaggeration. It is the wrong reason for the eye to think Earth is small. The planet dots on that sheet were 14 and 12 units across, a ratio near 1.17. Mars's diameter is about 6,792 kilometers and Earth's about 12,756, a ratio near 0.53. Dots are names, not worlds.



Real Mars beside the exaggerated teaching ellipse

Both panels use the same Mars semimajor axis, and both put the Sun at the focus. The right-hand ellipse is the exaggerated lesson, labeled as such. The dots are not the planets' physical sizes.

The illustrated lesson is download the PDF.

Sources

- Steven Austin, notes on Kepler's Mars orbit: https://faculty.uca.edu/saustin/3110/mars.pdf
- Physics Today, on Kepler and Mars: https://physicstoday.aip.org/features/the-great-martian-catastrophe-and-how-kepler-fixed-it
- NASA Mars fact sheet: https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html
- NASA, Mars at opposition: https://science.nasa.gov/asset/hubble/close-encounter-mars-at-opposition/
- NASA, geometry of a Mars conjunction: https://science.nasa.gov/resource/geometry-of-mars-solar-conjunction/

Key facts

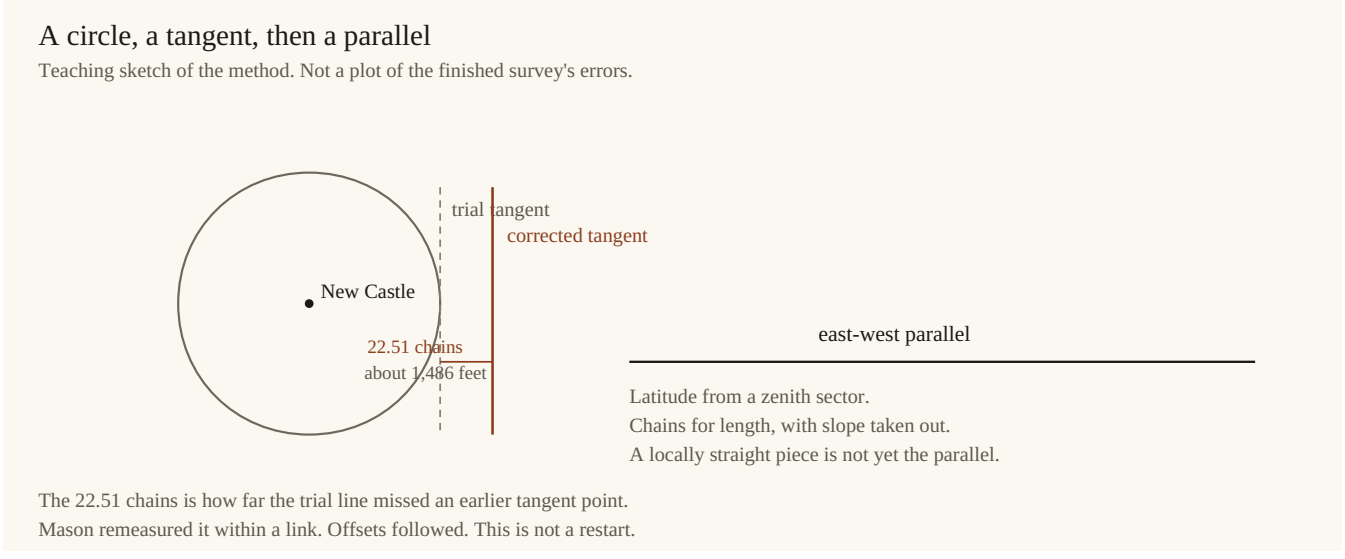
- A repeated heliocentric place of Mars, seen from two known positions of Earth, fixes a point. Many points make the ellipse. One angle is not enough.
- Opposition puts Earth between the Sun and Mars. Conjunction hides Mars in the Sun's glare. Neither alignment is a distance by itself.
- The exaggerated sheet has orbital ratio about 3.74 instead of 1.52, and eccentricity about 0.59 instead of 0.093. Earth's orbit should be about 66 percent of Mars's semimajor axis, not 27 percent. The dots are not sizes: Mars is about half Earth's diameter.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

The Mason-Dixon survey: how a boundary became real

The Mason-Dixon line is not one endless straight stroke. The Penn and Calvert grant needed a circle around New Castle, a tangent across the Delmarva peninsula, and then an east-west parallel. Each piece answers a clause in a charter. None of them is "draw west until the paper runs out."

Latitude came from a zenith sector, a telescope pointed near the zenith so the star's height gives the parallel. Length came from chains of a known size, with slope leveled out. On a round Earth a short piece that looks straight on the ground is not yet a parallel of latitude. You compute the parallel, then you step the offsets that keep you on it.



The New Castle circle, a trial tangent, and the parallel

A sketch of the method, drawn for this lesson. It is not a graph of leftover error along the finished line.

In August 1764 a trial tangent about 81 miles long missed an earlier colonial tangent point by 22.51 chains, about 1,486 feet. Read that carefully. It is the offset of a planned trial from an older mark, not the collapse of a finished survey. Mason took a man and remeasured the gap himself. He came within a link, about eight inches. Through the autumn they laid proportional offsets and declared a true tangent. They did not burn the work and begin again. No document found shows one single gross mistake that forced a full restart. That sentence does not prove the party never erred. Later resurveys, in the nineteenth and twentieth centuries, are different instruments and different legal questions.

On 17 September 1767 a falling tree killed the pack-horse drivers William Baker and John Carpenter. The names come from a careful later account of the survey; they are not written in Mason's own journal, so the lesson keeps that distinction. On 9 October the leader of the escort, who had joined in July with Mohawk and Onondaga men, said the war path was as far as the chiefs

of the Six Nations had commissioned him to go. He would not proceed one step farther west. Earlier, twenty-six hired men had refused a river for fear of the Shawnee and the Delaware. The journal states the commission's limit. It does not, in its own words, explain why the chiefs drew it there. Later markers that say hostile nations "ordered them to halt" are an interpretation, not the 1767 sentence.

The line later became a symbol of North and South. That symbol is a nineteenth-century use of an eighteenth-century colonial boundary. Dixon's lesson has the man who helped lay it. The geometry here is only the geometry.

The illustrated lesson is download the PDF.

Sources

- Todd Babcock, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_babcock.pdf
- Edwin Danson, FIG 2002: https://www.fig.net/resources/proceedings/fig_proceedings/fig_2002/HS1/HS1_danson.pdf
- Mason and Dixon, 1767 letter: https://peterschmidt.domains.swarthmore.edu/engl52b/m-d.letter.html

Key facts

- The boundary is a New Castle circle, a Delmarva tangent, and an east-west parallel, not one long line.
- The 1764 trial tangent missed an older point by 22.51 chains. Mason remeasured within a link, and offsets corrected the line. It was not a documented full restart.
- On 9 October 1767 the escort stopped at the war path, the limit of the Six Nations commission.
- Baker and Carpenter were killed by a falling tree on 17 September 1767 in later accounts of the work, not in Mason's journal itself.

A 2,100-Mile Circle Across the United States

Chicago, Boston, Miami and Denver · Fixed-Radius Best Fit

Approximate Moon-Diameter Comparison

Best fit, not a circle through all four cities. Chicago is 272 miles inside; Boston is 327 miles outside.

The 2,100-mile span is the lesson's rounded estimate (the Moon is about 2,160 miles wide), not a literal lunar footprint.

Circle radius: 1,050 statute miles. Center minimizes squared WGS84 radial errors across all four unchanged city points.

Natural Earth (Public Domain) · NASA Moon Facts · WGS84 Geodesics

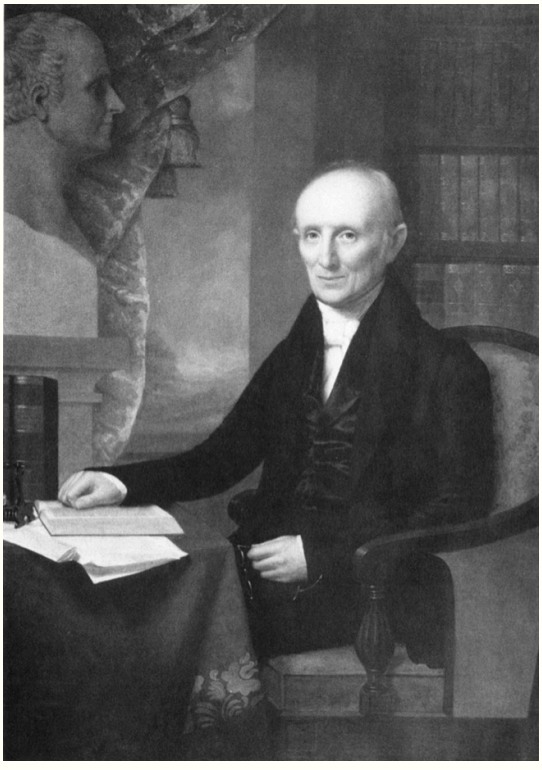
Azimuthal Equidistant · Center 31.0393°N, 91.6444°W · Distances from center are exact; scale elsewhere varies.

7 Nov 2026 · Public

Nathaniel Bowditch: the self-taught navigator

Faria lesson · 2026-11-06 · faria.jbf.com/topics/49-bowditch/

Jeff asked who Bowditch was after we talked about Polaris correction tables. This page is the full answer. It pairs with Polaris (the star whose tiny offset Bowditch's tables correct), The sidereal day and latitude sailing (why the correction is needed, and Jeff's "oscilloscope wave" insight), Galileo (longitude as time, before chronometers), and Map Surprises.



Portrait by Charles Osgood. Public domain (Peabody Essex Museum), via Wikimedia Commons.

1. Salem childhood: school ends at ten

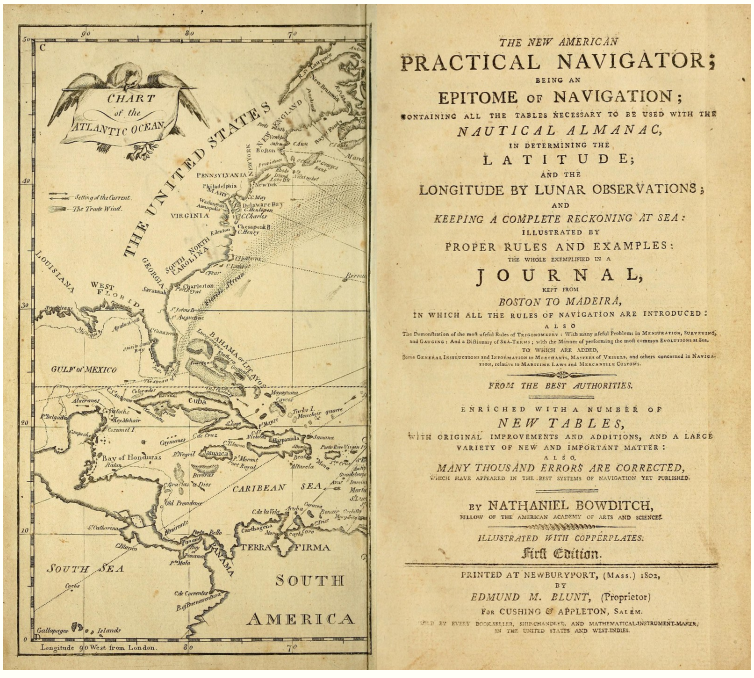
Nathaniel Bowditch was born in Salem, Massachusetts, on 26 March 1773, the fourth of seven children of Habakkuk Bowditch, a cooper (and sometime sailor who quit the sea after his ship went aground in 1775), and Mary Ingersoll Bowditch.

Family money ran short. At about age ten his formal schooling stopped and he went to work in his father's cooperage. At twelve he was indentured for nine years as a bookkeeping apprentice to a ship chandler — a dealer in rope, canvas, tar, and the other stores a ship needs. He lived and worked in the chandlery and learned accounts.

That "left school at ten" is the fact Jeff asked about. It is confirmed in the standard biographies (MacTutor / Biographical Encyclopedia of Astronomers; Wikipedia following Henry Ingersoll Bowditch's memoir of his father).

2. Self-taught: algebra, calculus, Latin, French, Newton's Principia

In the chandlery Bowditch taught himself.



1802 first edition. Public domain (U.S. Naval Observatory), via Wikimedia Commons.

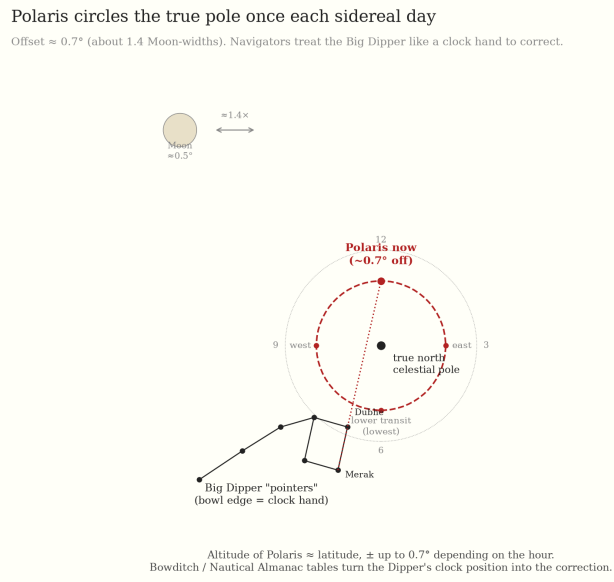
- 1786 (age 14): algebra.
- 1788: calculus.
- 1790: Latin, specifically so he could read *Isaac Newton's Philosophiae Naturalis Principia Mathematica.
- 1792: French, for the European mathematical literature.

At eighteen he copied out all the mathematical papers in the Philosophical Transactions of the Royal Society. In 1793 he found an error in Newton's Principia.

Serendipity helped. In 1781 the Beverly privateer Pilgrim captured a ship carrying the scientific library of the Irish chemist Richard Kirwan — Newton, the Bernoullis, Chambers's Cyclopedia, and more. The books were bought for Salem and eventually housed with the Rev. John Prince, who in 1791 let the eighteen-year-old Bowditch use what had become the best scientific library north of Philadelphia. Without that captured library, the self-education would have been far harder.

3. Five voyages, 1795–1803

Bowditch went to sea:



Polaris's ~0.7° circle and the Big Dipper clock. Drawn for faria.jbf.com.

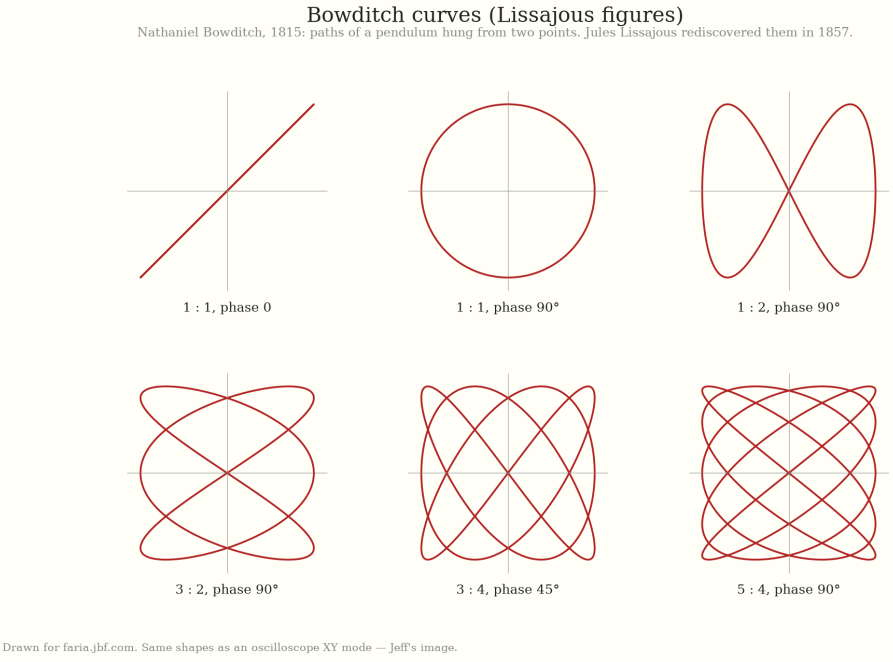
- Voyage — Role — Notes
- 1st — Ship's clerk / second mate — Began finding errors in Moore's tables; conceived a clearer method of lunars
- 2nd–4th — Supercargo — Continued corrections; taught crew navigation
- 5th — Master and part owner — Last voyage; returned to Salem 1803

Between the first and second voyages, in March 1798, he married Elizabeth Boardman; she died seven months later while he was at sea. In 1800 he married his cousin Mary "Polly" Ingersoll; they had eight children.

After 1803 he left the sea for mathematics, insurance, and the book that made his name.

4. ~8,000 errors in Moore's Practical Navigator

The standard English handbook was John Hamilton Moore's The Practical Navigator (London). Its tables were known to be faulty. On his first voyage Bowditch began checking them. By the time he was done he had compiled a list of more than 8,000 errors.



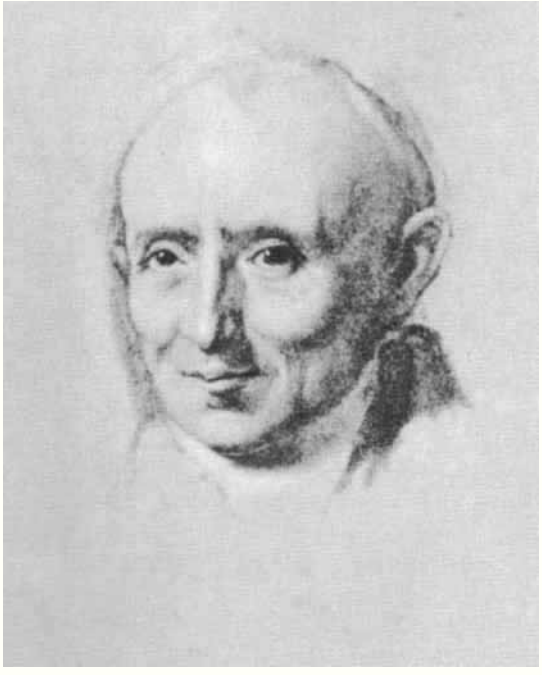
Bowditch curves (Lissajous figures), 1815 — oscilloscope XY shapes. Drawn for faria.jbf.com.

A careful caveat (from a modern study of Moore and Bowditch in the Journal of Navigation): most of those errors were small — last-decimal-place mistakes in traverse tables, proportional logarithms, and log-trig functions. Saying "8,000 errors" can make Moore sound like a fraud; he was not. But at sea a wrong last digit can still put you on a reef. Bowditch recomputed everything.

He first sent corrections to the American publisher Edmund March Blunt. Blunt issued corrected American editions of Moore in 1799 and 1800. By the third round the book had changed so much that Blunt published it under Bowditch's own name.

5. The New American Practical Navigator (1802)

In 1802 Blunt published Nathaniel Bowditch, The New American Practical Navigator, printed at Newburyport, Massachusetts. Bowditch's rule for what went in:



Unfinished Gilbert Stuart portrait, 1828. Public domain, via Wikimedia Commons.

> *"put down in the book nothing I can't teach the crew."*

The book rearranged Moore, added tables, and included Bowditch's clearer solutions for the spherical triangle and for lunar distances — finding longitude by measuring the Moon's angular distance from the Sun or a star, then comparing with an almanac. Accurate marine chronometers (Harrison's) existed but were still too expensive for most merchant captains; lunars were the poor man's clock in the sky. (See Galileo for the older dream of a Jupiter-moon clock.)

Still in print. In 1867–68 the U.S. Hydrographic Office bought the copyright. The book has been in continuous government publication ever since. Today it is NGA Pub. No. 9, The American Practical Navigator, issued free by the National Geospatial-Intelligence Agency in two volumes (2024 edition as of this writing). Every commissioned U.S. Navy ship still carries it.

Sailors simply call it "Bowditch" — "the bible of the sea." Midshipmen before the Naval Academy existed called it "the immaculate Bowditch."

6. Teaching the crew — including the cook

Bowditch's teaching story is the one Jeff heard.

On the fourth voyage, to Manila in the ship Astræa, Captain Henry Prince was asked in Manila how an American ship had found its way so well against the northeast monsoon. Prince answered (as recorded in Henry Ingersoll Bowditch's memoir of his father) that he had a crew of twelve men, every one of whom could

> *"take and work a lunar observation as well, for all practical purposes, as Sir Isaac Newton himself were he alive."*

Bowditch sat through the conversation "as modest as a maid," slate pencil in his mouth. Another merchant added that there was "more knowledge of navigation on board that American ship than there has been in all the ships that ever came into Manila Bay."

Later tellings (including Baron Franz von Zach and many navy histories) add that even the ship's cook became proficient in lunar distances. The cook detail is traditional and widely repeated; the core claim — an entire twelve-man crew able to work lunars — is anchored in Captain Prince's contemporary boast. Bowditch's own standard for the book was the same spirit: write nothing you cannot teach the forecastle.

7. How Bowditch corrects Polaris

This is the bridge to Polaris and to Jeff's insight on the sidereal day page.

Polaris is not exactly on the north celestial pole. It sits about 0.7° off (roughly 1.4 Moon-widths), and therefore circles the true pole once each sidereal day. If you take the raw altitude of Polaris as your latitude, you can be wrong by up to 0.7°, about 42 nautical miles (since 1° of latitude = 60 nmi).

Navigators fix that with a Polaris correction. The classic rule of thumb treats the Big Dipper's pointer stars (Dubhe and Merak, the outer edge of the bowl) like a clock hand around the pole. From the "hour" on that clock you look up a correction in the tables — in Bowditch historically, and today in the Nautical Almanac (corrections a0, a1, a2) — and add or subtract it. The wave flattens. The altitude of Polaris, corrected, is your latitude.

8. Laplace's Mécanique céleste

Bowditch taught himself French partly to read Pierre-Simon Laplace's monumental Traité de mécanique céleste. He finished translating the first four volumes by about 1818, then spent years adding commentaries that roughly doubled the length: supplying steps Laplace had skipped, inserting later results, and giving credits Laplace had omitted. Publication (at Bowditch's own expense) ran 1829–1839. The translation became a foundation of American physical astronomy in the nineteenth century. Benjamin Peirce helped with the later volumes.

Harvard offered him the chair of mathematics and physics in 1806; West Point and the University of Virginia also offered chairs. He turned them all down — the insurance salary was higher, and he preferred the work.

9. Actuary and "money manager"

In 1804 Bowditch became president of the Essex Fire and Marine Insurance Company in Salem — often called America's first insurance actuary in the modern sense. The company prospered through the War of 1812. In 1823 he moved to Boston as actuary for the Massachusetts Hospital Life Insurance Company, investing the fortunes of sea-made wealth into manufacturing towns such as Lowell. When he moved, he took more than 2,500 books, 100 maps, and 29 volumes of his own manuscripts.

He was elected to the American Academy of Arts and Sciences (1799), the American Philosophical Society (1809), and later the Royal Societies of London and Edinburgh and the Royal Irish Academy. In 1829 he became the first American foreign associate of the Royal Astronomical Society. Harvard gave him an honorary degree in 1802 and an LL.D. in 1816.

10. Bowditch curves = Lissajous figures = Jeff's oscilloscope

In 1815 Bowditch published "On the Motion of a Pendulum Suspended from Two Points" in the Memoirs of the American Academy of Arts and Sciences. A pendulum hung from two points (a Y-shaped or Blackburn suspension) can swing at different frequencies in two perpendicular directions. The bob traces the curves

$$x = a\,\sin(n\,t + c), \quad y = b\,\sin(t)$$

Jules Antoine Lissajous studied them independently and more famously in 1857, using light reflected from tuning forks; his name stuck. Because Bowditch was first, mathematicians also call them Bowditch curves.

Jeff's connection. When Jeff pictured an uncorrected Polaris track as "cycles on an oscilloscope," he reached for exactly this family of shapes. An oscilloscope in XY mode draws Lissajous / Bowditch figures. The navigator who wrote the bible of the sea also drew the curves on the screen Jeff was imagining. See The sidereal day and latitude sailing for the wavy track itself.

11. Death, eulogy, and legacy

Bowditch died in Boston on 16 March 1838 of stomach cancer. He is buried in Mount Auburn Cemetery, Cambridge; the memorial statue by Robert Ball Hughes was the first life-size bronze cast in America.

The Salem Marine Society eulogized him:

> *"In his death a public, a national, a human benefactor has departed. Not this community nor our country only, but the whole world has reason to do honor to his memory. When the voice of eulogy shall cease to flow, no monument will be needed to keep alive his memory among men; but as long as ships shall sail, the needle point to the north, and the stars go through their wonted courses in the heavens, the name of Dr. Bowditch will be revered as of one who has helped his fellowmen in time of need, who was and is a guide to them over the pathless oceans, and one who forwarded the great interests of mankind."*

Named for him: the oceanographic survey ship USNS Bowditch; a lunar crater; a Salem–Boston ferry; a Maine windjammer; schools; the Department of Marine Transportation building at the U.S. Merchant Marine Academy; and Jean Lee Latham's 1955 Newbery Medal novel Carry On, Mr. Bowditch.

Review questions

1. Why did Bowditch teach himself Latin?

2. What was his rule for what belonged in the Navigator?

3. Why does Polaris need a correction table, and what "clock" do sailors use?

4. What do Bowditch curves have to do with an oscilloscope?

Key facts

- Nathaniel Bowditch, Salem, 26 Mar 1773 – Boston, 16 Mar 1838.
- Left school ~age 10; indentured at 12 to a ship chandler for nine years; self-taught algebra, calculus, Latin (for Newton), French (for Laplace).
- Five voyages 1795–1803; found more than 8,000 errors in J. H. Moore's Practical Navigator (mostly small tabular mistakes).
- The New American Practical Navigator, 1802; still published as NGA Pub. No. 9; sailors call it "Bowditch."
- Rule: "put down in the book nothing I can't teach the crew"; on the Manila voyage the whole twelve-man crew (cook included, by tradition) could work lunars.
- Polaris correction: star is ~0.7° off the pole; Big Dipper pointers as clock; tables remove up to ~42 nmi of latitude error.
- Translated and annotated Laplace's Mécanique céleste (vols. I–IV, pub. 1829–39); declined Harvard / West Point / Virginia chairs.
- America's early insurance actuary (Essex Fire & Marine, then Mass. Hospital Life).
- Bowditch curves (1815) = Lissajous figures (1857) = oscilloscope XY shapes.
- USNS Bowditch; Mount Auburn bronze; Salem Marine Society eulogy: "as long as ships shall sail…"

Answers

1. So he could read Newton's Principia in the original Latin.

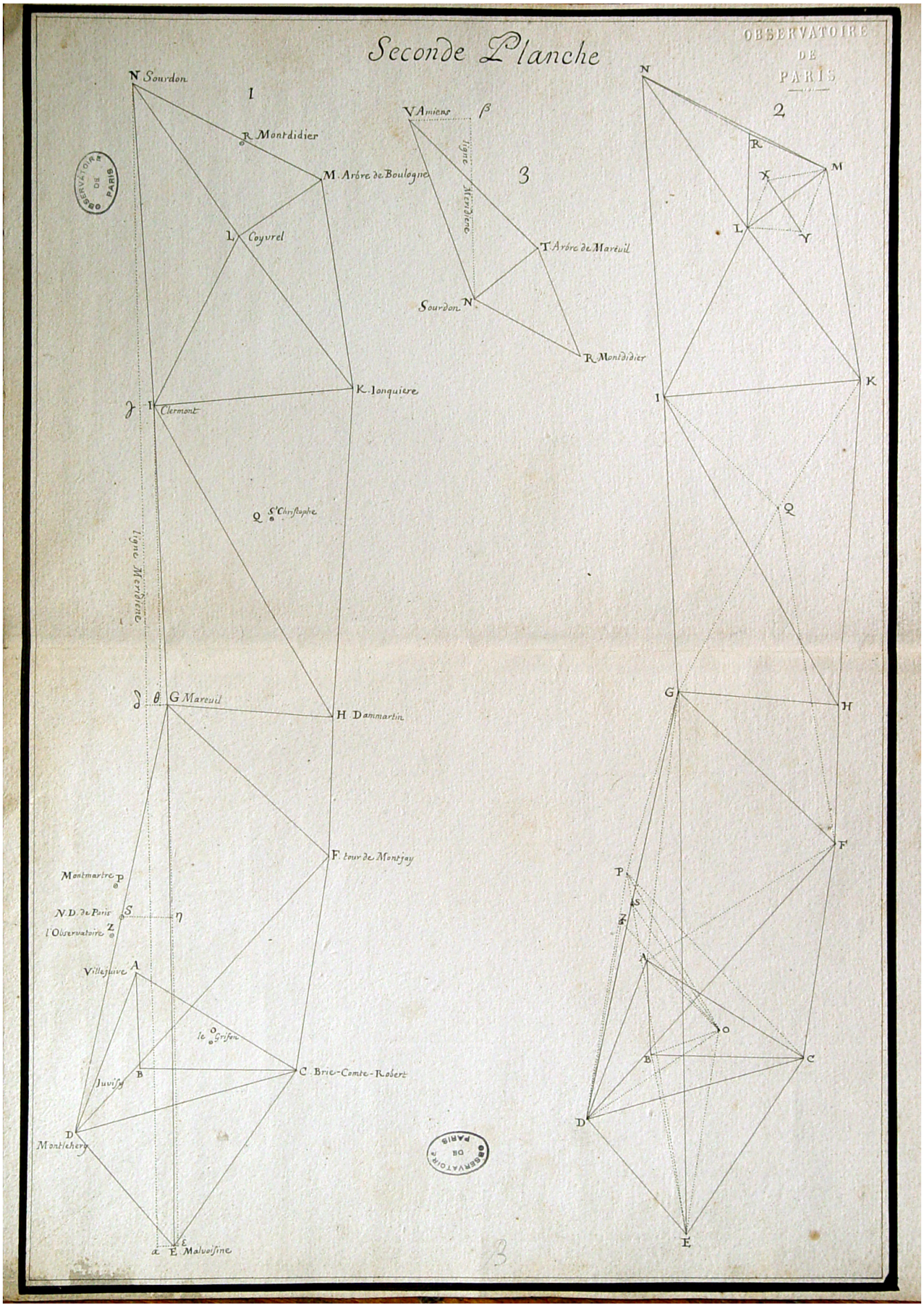
2. "Put down in the book nothing I can't teach the crew."

3. Polaris is ~0.7° from the true pole and circles it once per sidereal day; raw altitude can be wrong by up to ~42 nmi. Sailors use the Big Dipper pointers as a clock hand and look up the correction in Bowditch / the Almanac.

4. They are the same family of curves: Bowditch studied them in 1815 as pendulum paths; an oscilloscope in XY mode draws them live.

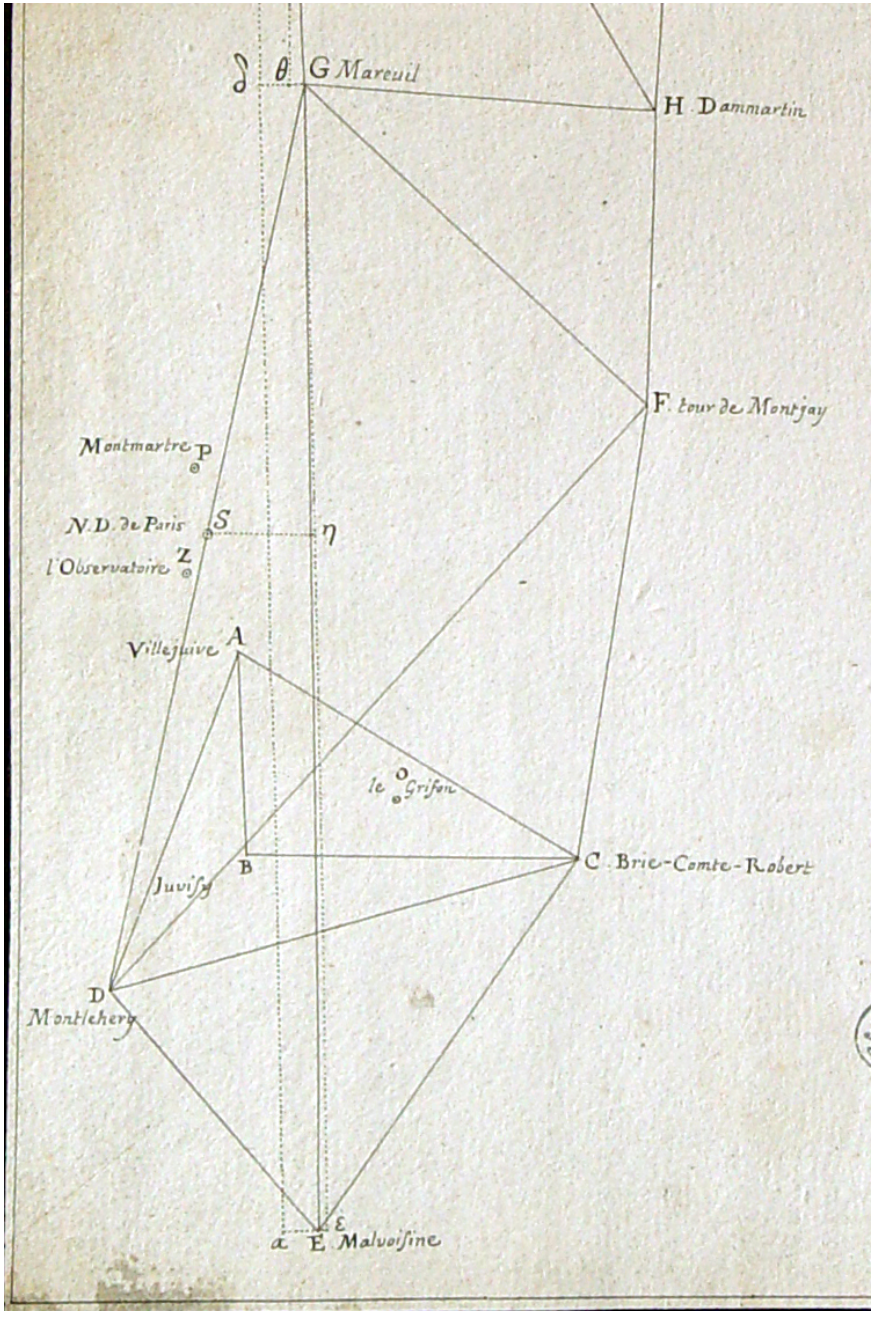
Picard's triangulation, 1669-1670

Historical preparatory plate, 1671. Original geometry and lettering preserved.



Source: Bibliotheque de l'Observatoire de Paris, D5/42 (14). Creator unknown.
Seconde planche preparatoire de la Mesure de la Terre, 1671.
Archive credit: Bibliotheque de l'Observatoire de Paris. No commercial use.
https://bibnum.obspm.fr/ark:/11287/2hkG3

Read the southern baseline and its triangles



Enlargement of the same archival plate; no synthetic map elements added.
https://bibnum.obspm.fr/ark:/11287/2hkG3

The physically measured baseline

A: Villejuif windmill

B: Juvisy pavilion

AB = 5,663 toises of Paris

About 11.04 km / 6.86 miles

Other points visible in this enlargement

C: Brie-Comte-Robert

D: Montlhery

E: Malvoisine

F: ruins of Montjay

G: Mareuil (Mareil)

H: Dammartin

What the complete plate shows

Left: named stations in the principal network.

Right: a second, lettered triangle diagram.

Top centre: an inset extending to Amiens.

Clermont is labelled on the full plate;
the label does not itself say Saint-Samson.

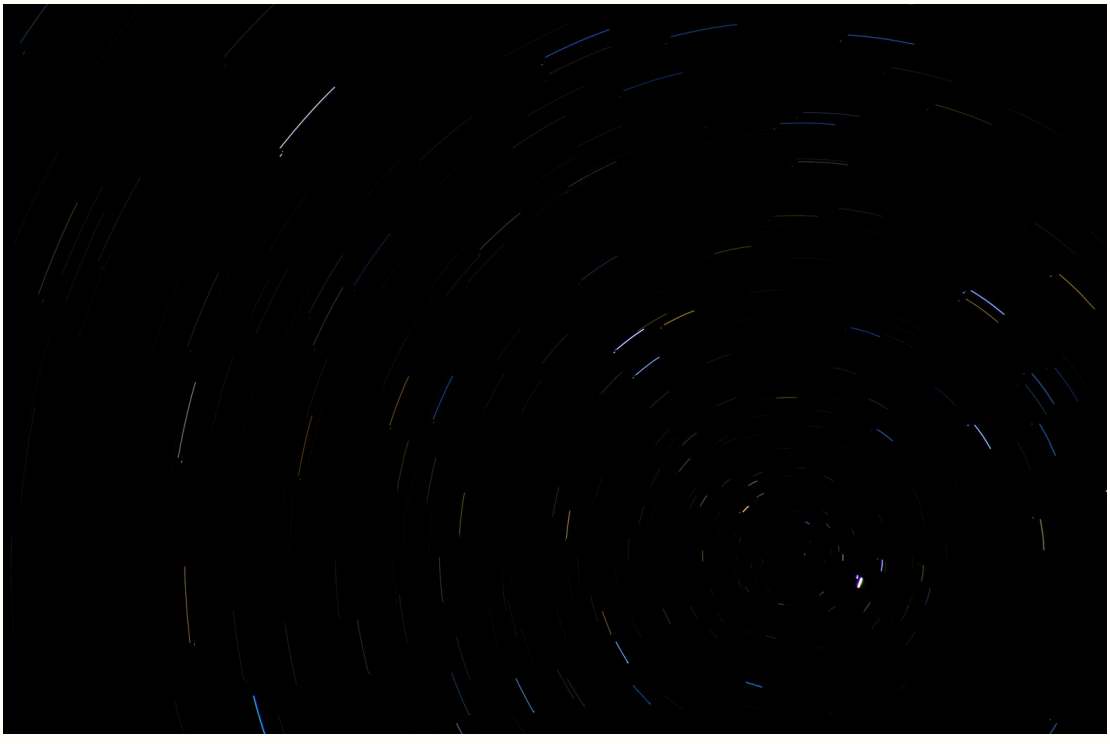
Not Orleans

The main chain runs from Malvoisine north
to Sourdon, with an Amiens extension.
Orleans is not an endpoint of this network.

Polaris: the North Star, explained

A Faria lesson for Jeff Franzen · 6 November 2026 (America/New_York)

The whole Wikipedia entry on Polaris, made clear: a triple star, a Cepheid yardstick, the precession of the pole, every name from cynosure to Dhruva, and Shakespeare's Sonnet 116 line by line. Live page: faria.jbf.com/topics/48-polaris/



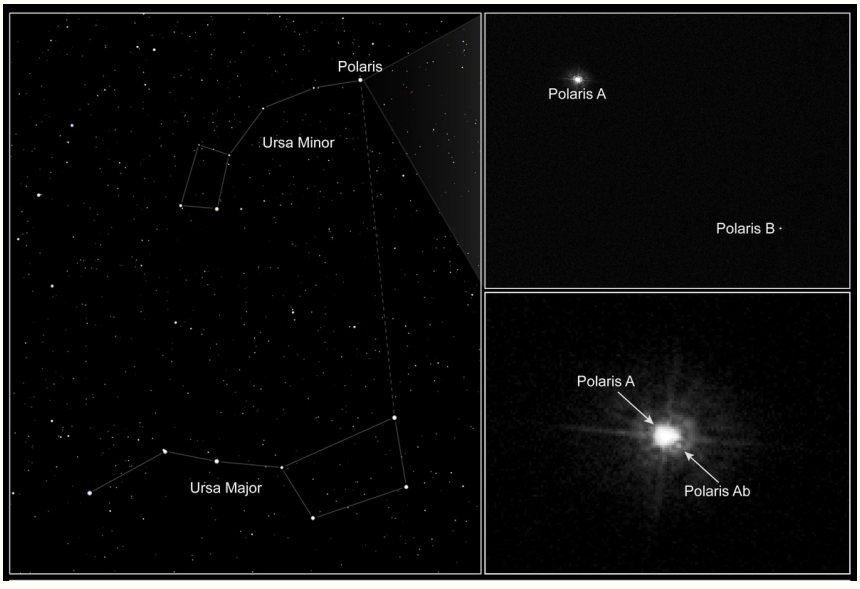
Circumpolar star trails; Polaris near the still center. Daniel Seßer, CC BY-SA 3.0, Wikimedia Commons.

1. Not one star: a multiple system

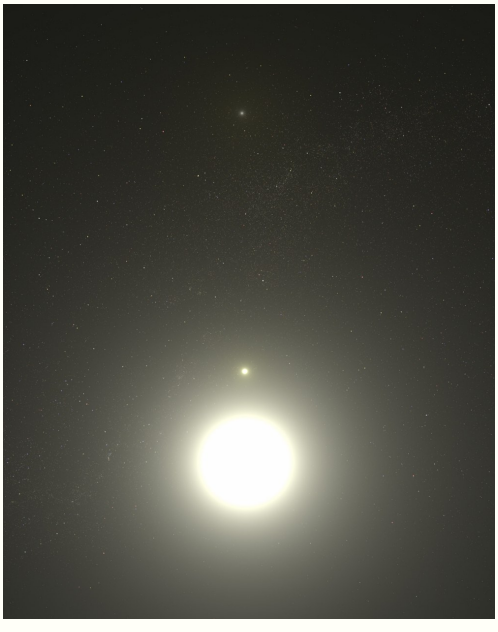
To the naked eye Polaris is a single point of light. It is not. α Ursae Minoris (Alpha Ursae Minoris, α UMi) means "the brightest star of Ursa Minor," the Little Bear. Greek letters were assigned by Johann Bayer in 1603.

- Polaris Aa — yellow supergiant, spectral type F7Ib, mass about 5.13 ± 0.28 solar masses (Evans et al. 2024), radius ~46 solar radii. The star you see; a classical Cepheid.
- Polaris Ab — close dwarf (~F6V), orbits Aa in ~29.4 years. Hubble imaged it in January 2006. Closest approach ~6 AU at the Gaia distance.
- Polaris B — F3V dwarf at ~2,400 AU. William Herschel found it in August 1779 with a telescope he built himself.
- Polaris C and D were once listed as companions but are not physically associated with the system.

Spectral classes F7 / F3. The sequence O B A F G K M runs hot to cool. F stars are yellow-white, a little hotter than our G-type Sun. Ib = luminous supergiant; V = main-sequence dwarf.



Hubble image of the Polaris system. NASA/ESA/HST, public domain.



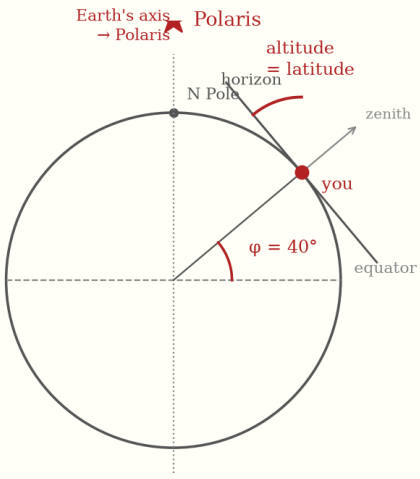
Artist's concept of Aa, Ab, and B. NASA/ESA/G. Bacon (STScI), public domain.

2. Why it stands still

Earth's spin axis points nearly at Polaris, so northern stars wheel around it while Polaris barely moves. In 2018 it was about 0.66° (~40′, about 1.4 Moon-widths) from the true pole, tracing a tiny daily circle ~1.3° across.

The Big Dipper pointers Dubhe and Merak aim at Polaris. Nathaniel Bowditch's American Practical Navigator (1802) lists Polaris among the navigational stars and teaches the clock-face correction from the Dipper.

Altitude of Polaris = latitude. At the North Pole it is overhead; at the equator on the horizon; in Florida (~28°N) about 28° up. That is why latitude was easy for sailors while longitude needed a clock (see the Galileo lesson on faria.jbf.com).



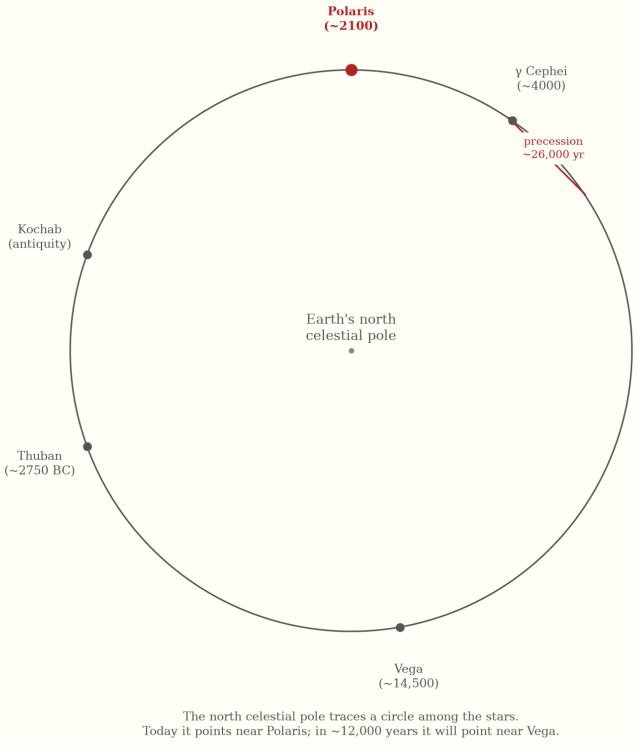
Because Polaris lies almost on Earth's axis, its height above your northern horizon equals your latitude. At the North Pole it is overhead; at the equator it sits on the horizon; in Florida (~28°N) it stands ~28° up.

Altitude of Polaris equals the observer's latitude. Diagram for this lesson.

3. Precession: it will not always be the pole star

Earth's axis wobbles once every ~26,000 years (precession of the equinoxes). The north celestial pole drifts among the stars:

- ~2750 BC — Thuban (α Draconis), near the pole as the pyramids rose
- Classical antiquity — closer to Kochab (β UMi) than to Polaris (~10° from either)
- ~320 BC — Pytheas described the pole as devoid of stars
- 5th c. AD — Stobaeus: ἀεί φανής, "always visible"
- ~2100 AD — Polaris closest (~0.45°)
- ~4000 — Gamma Cephei (Errai); ~14,500 — near Vega; then back after one full cycle

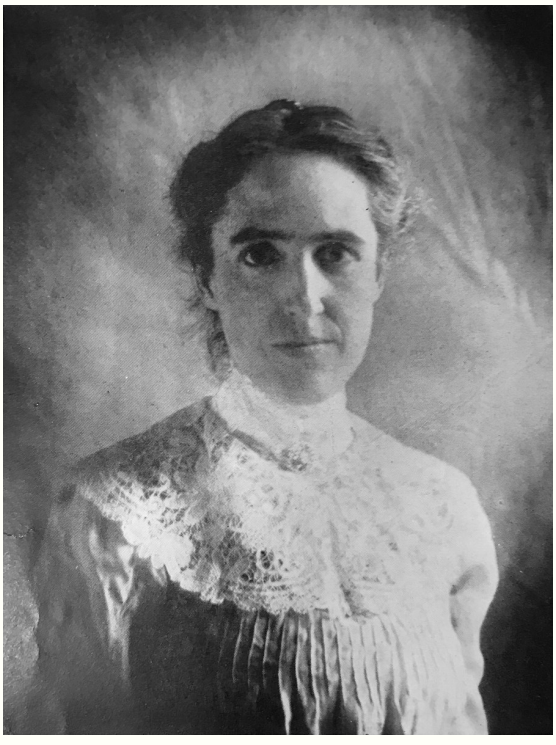


The ~26,000-year precession circle of pole stars. Diagram for this lesson.

4. Cepheids: a pulsing yardstick

Pronounce it SEF-ee-id. A Cepheid swells and shrinks on a steady beat. Named for Delta Cephei.

Henrietta Swan Leavitt (1868–1921), a Harvard "computer," found in 1912 that a Cepheid's period tells its true brightness — the slower the pulse, the more luminous the star. Compare true and apparent brightness and you know the distance. Edwin Hubble used her law on Andromeda Cepheids to show that Andromeda is a separate galaxy and that the universe is expanding.



Henrietta Swan Leavitt. Public domain, Wikimedia Commons.

Polaris is the nearest classical Cepheid (period ~3.97 days). Its amplitude shrank sharply after 1966; the period lengthened ~4.5 s/yr for much of the modern record (Torres 2023 finds a reversal since ~2010). Research in Science (2004) suggests it may be ~2.5× brighter than in Ptolemy's day; Edward Guinan: if real, "100 times larger" than theory predicts.

5. Distance and parallax

Finger-blink analogy: hold a finger at arm's length and blink each eye; it jumps against the background. Astronomers do the same with Earth's orbit (January vs July). That angle is stellar parallax.

- Hipparcos (1989–93, revised 2007): ~432 light-years
- Turner et al. spectral/photometric "short" distance: ~323 light-years
- Gaia DR3 (via companion B): ~446.5 light-years — current best geometric estimate

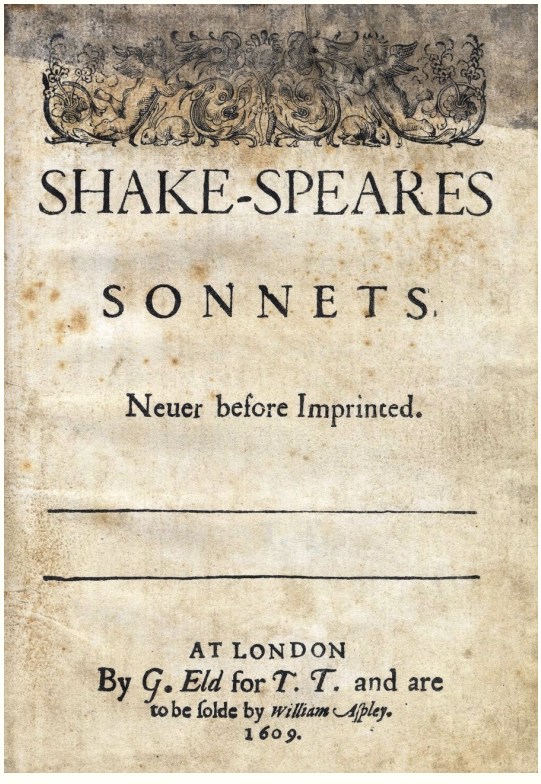
Close companion Ab tugs Aa on a 29-year orbit, mixing orbital wobble with parallax and Cepheid pulsation — binaries spoil clean measurements. Polaris calibrates the cosmic distance ladder; an error here feeds every rung above (galaxies, supernovae, Hubble's constant).

6. Names

- Cynosure (SY-nuh-shoor) — Greek kynosoura, "dog's tail"; thence "center of attention."
- Lodestar — guiding star; kin to lodestone.
- Spenser's "steadfast star" (Faerie Queene I.ii.1): "the northern wagoner had set / His sevenfold team behind the steadfast star."
- Stella polaris / Polaris — Neo-Latin; Alfonsine Tables 1492; IAU proper name 2016.
- Stella Maris / Ave Maris Stella — "star of the sea," Marian title.
- Sanskrit Dhruva — "fixed, constant, immovable."
- Arabic al-Jady ("the kid"); also al-qutb al-shamali ("northern axle") and mismār ("needle").

• Old English scip-steorra ("ship-star"); Rune Poem Tir: "ever on its course over the mists of night and never fails."

7. Shakespeare, Sonnet 116 — the centerpiece


SHAKE-SPEARES

SONNETS.

Neuer before Imprinted.

AT LONDON
By G. Eld for T. T. and are
to be solde by William Aspley.
1609.

Title page, Shake-speares Sonnets, 1609 Quarto. Public domain.

Full text (public domain; 1609 Quarto, modernized spelling):

Let me not to the marriage of true minds
Admit impediments. Love is not love
Which alters when it alteration finds,
Or bends with the remover to remove:
O, no! it is an ever-fixed mark
That looks on tempests and is never shaken;
It is the star to every wandering bark,
Whose worth's unknown, although his height be taken.
Love's not Time's fool, though rosy lips and cheeks
Within his bending sickle's compass come;
Love alters not with his brief hours and weeks,
But bears it out even to the edge of doom.
If this be error and upon me proved,
I never writ, nor no man ever loved.

Close reading of the GIS couplet

"It is the star to every wandering bark, / Whose worth's unknown, although his height be taken."

- Bark = ship (Old French barque).
- The star = Polaris, the sailor's guide.
- Height be taken = measuring the star's altitude with astrolabe, quadrant, or cross-staff — which is the ship's latitude. Exactly the GIS Month rule: altitude of Polaris = latitude.
- Worth's unknown = you can measure and depend on the guide without knowing its true nature. In 1609 no one knew Polaris was a triple Cepheid ~446 light-years away. Love, likewise, can be relied on without being fathomed.

"Ever-fixed mark" = a sea-mark / beacon that looks on tempests and is never shaken — the same steadfastness Spenser gave the pole star. "Edge of doom" = Judgment Day. The closing couplet stakes Shakespeare's entire life's work on the claim.

Pronunciation: Polaris (puh-LAIR-iss); Cepheid (SEF-ee-id); cynosure (SY-nuh-shoor); precession (pree-SESH-un); parallax (PAIR-uh-lacks).

8. Coordinates: sky latitude and longitude

Declination +89° 15′ 50.8″ (sky latitude; +90° = north celestial pole). Right ascension 02h 31m 49.09s (sky longitude, in hours because the sky turns 15° per hour — the same arithmetic as terrestrial longitude).

Review questions

1. At the North Pole, where is Polaris, and what latitude does that report?
2. What did Henrietta Leavitt discover in 1912, and how did that let Hubble prove Andromeda is a separate galaxy?
3. In Sonnet 116, what is a "wandering bark," and what does it mean to "take the height" of the star?
4. Why will Polaris not be the North Star in the year 14,500?
5. Why does a wrong distance to Polaris ripple into measurements of the whole universe?

Answers

1. Straight overhead (zenith); latitude 90°N.
2. A Cepheid's period reveals its true brightness → distance. Hubble found Cepheids in Andromeda far outside the Milky Way.
3. Bark = ship. Taking the height = measuring altitude for latitude.
4. Precession will point Earth's axis near Vega.
5. Polaris is the nearest classical Cepheid; Cepheids calibrate the cosmic distance ladder.

Sources checked against Wikipedia "Polaris" (retrieved 2026-11-06), the 1609 Quarto text of Sonnet 116 (public domain), Spenser Faerie Queene I.ii.1, and the Old English Rune Poem (Tir). Live lesson: https://faria.jbf.com/topics/48-polaris/

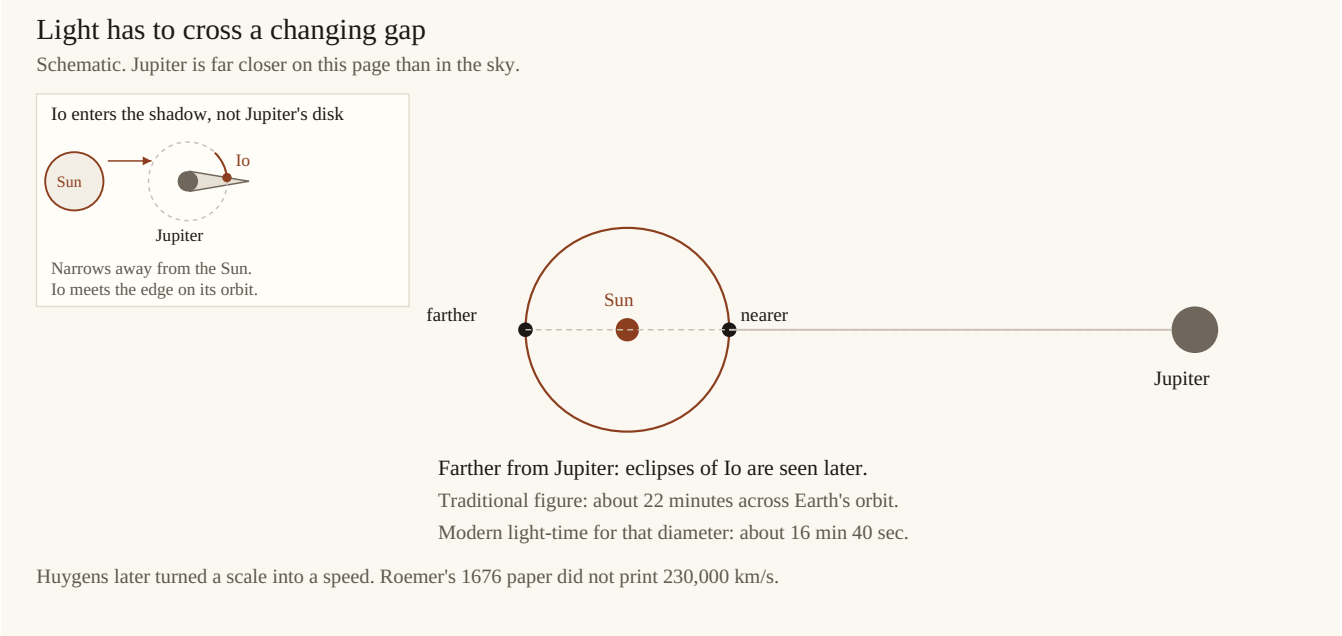
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Romer and Io: light needs time

Io goes around Jupiter in about 42.5 hours. We see it by sunlight reflected from its surface. When Io enters Jupiter's shadow, that sunlight is cut off. The moon has not switched off a lamp of its own. The disappearances and returns are eclipses, and they can be timed. The names and the order of the four moons are the previous lesson.

Giovanni Cassini had been building tables of those times. Ole Romer, working with that program in Paris, argued in 1676 that the tables made sense if light travels at a finite speed. Earth moves around the Sun once a year. When Earth is traveling away from Jupiter, each eclipse is seen a little later, because the light has a longer trip. When Earth is traveling toward Jupiter, the events come early. The extra minutes are the light-time of the extra distance.

Jupiter moves too, and Io's orbit is not a perfect clock. Many nights are required before the leftover can be blamed on light rather than on the tables. The traditional round number is about 22 minutes for light to cross the diameter of Earth's orbit. The modern light-time for that same diameter is about 16 minutes 40 seconds.



Io's shadow eclipse, and Earth near and far from Jupiter

A schematic, drawn for this lesson. Jupiter is much farther, and it moves. The inset is the local event: reflected light cut off by the shadow.

Romer did not publish a speed of 230,000 kilometers a second. Christiaan Huygens, in the Treatise on Light of 1690, converted an orbital scale into a speed that comes out near that figure in modern

units. The defined speed of light is now 299,792.458 kilometers a second. A conversion is not a sentence Romer printed.

You also cannot pick one opposition and one conjunction and call them a clean pair. Near conjunction, Jupiter sits in the Sun's glare and Io's eclipses cannot be watched. The argument is a stack of dates. The drawing shows the idea, not the observing calendar.

The illustrated lesson is download the PDF.

Sources

- Paris Observatory, "340 years ago, Romer showed that light has a finite speed": https://observatoiredeparis.psl.eu/340-years-ago-romer-showed.html
- Observatoire de Paris, CLEA / CLIC notes on Romer: https://clic.obspm.fr/

Key facts

- Io's eclipses are the cutting off of reflected sunlight in Jupiter's shadow, on a clock of about 42.5 hours. Io does not emit the light.
- Romer in 1676 explained early and late eclipses by a finite speed of light as Earth moved nearer or farther.
- The traditional 22 minutes across Earth's orbit compares with a modern 16 minutes 40 seconds. Jupiter's motion and Io's orbit also affect the times.
- Huygens, not Romer, produced the speed later quoted near 230,000 km/s. The defined value is 299,792.458 km/s.

FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

Ships of exploration: four craft, four different jobs

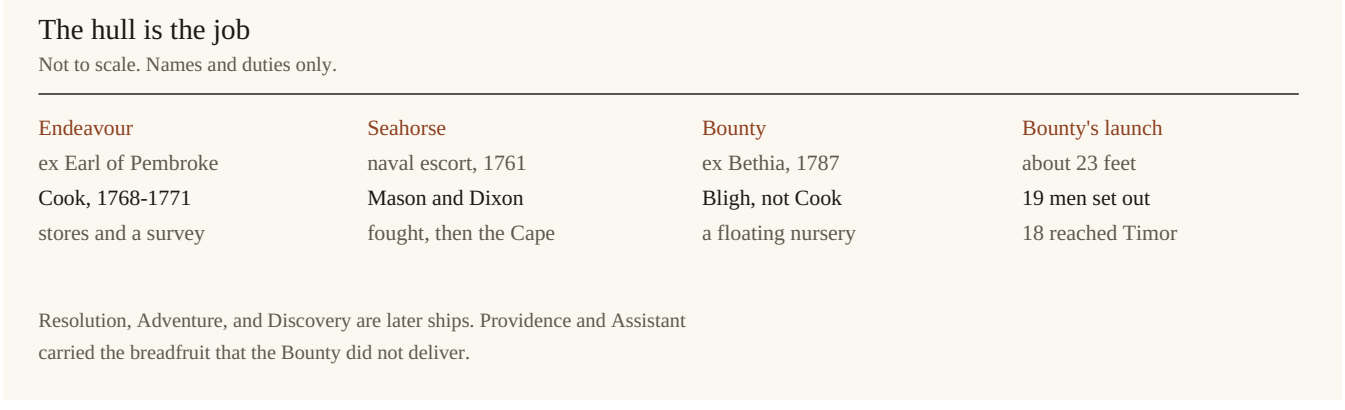
A ship's name is a job description if you let it be. Four vessels from this day's stories are easy to tangle, and they should stay apart.

Endeavour had been the collier Earl of Pembroke. The Navy bought a cargo ship because Cook knew that kind of hull and because a survey voyage needs room for stores, not a frigate's gun deck. She carried the first voyage, 1768 to 1771: Tahiti, New Zealand, the Australian east coast. Resolution, Adventure, and Discovery came later, for the second and third voyages. They are not nicknames for Endeavour.

Seahorse, in 1761, was the naval ship that took Charles Mason and Jeremiah Dixon toward Sumatra. In January she fought a French frigate. Eleven men were killed and thirty-seven wounded, by Mason's letter. The transit party went on to the Cape. Seahorse did not draw the American boundary, and she had nothing to do with the Bounty.

Bounty had been the merchant Bethia. The Navy took her up in 1787 and rebuilt her to carry living breadfruit, a nursery where a great cabin had been. William Bligh commanded. Cook had been dead for eight years. Joseph Banks organized the plan and stayed in England. Bounty is a breadfruit ship that became a mutiny. She is not Cook's surveyor.

Her launch was a working boat about 23 feet long, carried on board like any ship's boat. After the mutiny, nineteen men were in it. At Tofua, John Norton was killed. The remaining eighteen reached Timor. The distance usually given is about 3,618 nautical miles, not statute miles. How you count the days depends on which day's entry you start from, so this page will not pretend one duration is the only honest one. A later pair, Providence and Assistant, in 1791 to 1793, did deliver breadfruit. The boat's story is the mutiny lesson. The plant's story is the breadfruit lesson. Cook's collier is the Cook lesson.



Four vessels and the job each one actually did

Roles, not a scale model. Drawn for this lesson.

The illustrated lesson is download the PDF.

Sources

- Royal Museums Greenwich, Cook timeline: https://www.rmg.co.uk/stories/maritime-history/captain-james-cook-timeline
- Royal Museums Greenwich, mutiny on the Bounty: https://www.rmg.co.uk/stories/maritime-history/mutiny-on-bounty
- Royal Museums Greenwich, William Bligh: https://www.rmg.co.uk/stories/maritime-history/william-bligh
- Royal Museums Greenwich, the Bounty launch model: https://www.rmg.co.uk/collections/objects/rmgc-object-68956

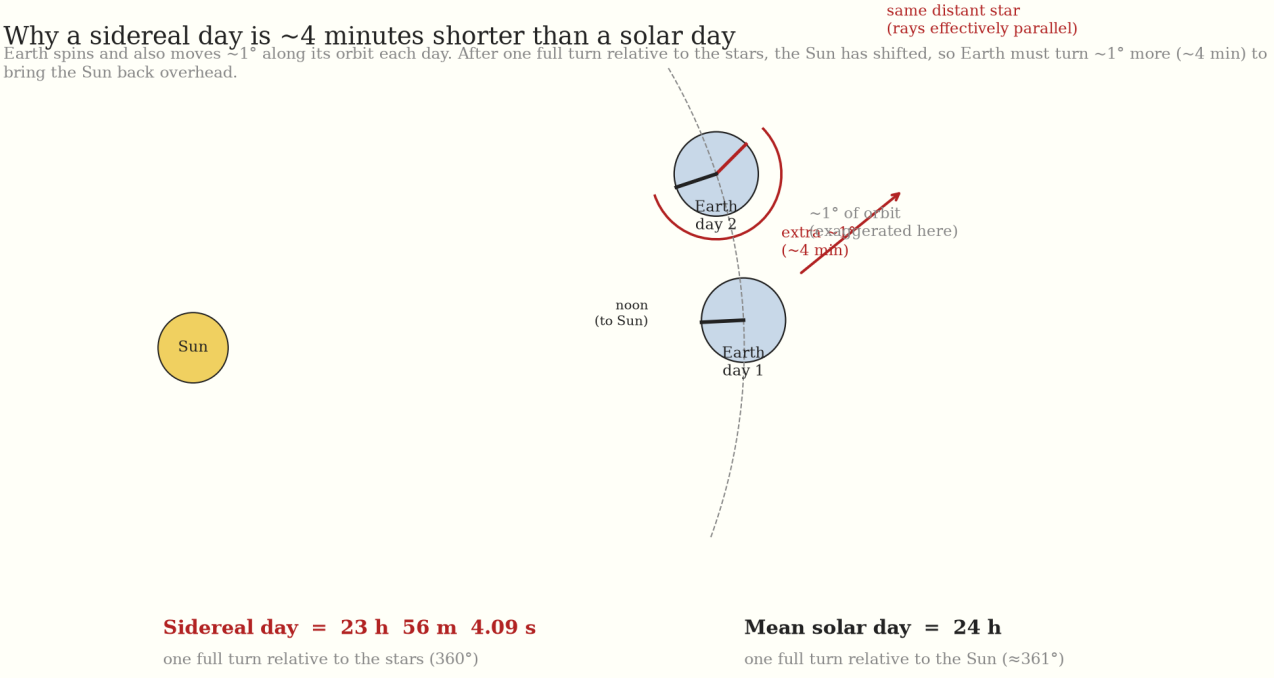
Key facts

- Endeavour, ex Earl of Pembroke, was Cook's first-voyage collier. Later voyages used other ships.
- Seahorse carried Mason and Dixon in 1761 and was attacked. She is not the boundary survey and not the Bounty.
- Bounty, ex Bethia, was fitted in 1787 as a breadfruit nursery under Bligh, not under Cook or Banks.
- The launch of about 23 feet took 19 men. After Norton's death at Tofua, 18 reached Timor, some 3,618 nautical miles, not statute miles.

The sidereal day and latitude sailing

Faria lesson · 2026-11-06 · faria.jbf.com/topics/50-sidereal-day/

Jeff asked what sidereal means and why a day measured by the stars is only 23 hours 56 minutes. Then he pictured what happens if you trust Polaris raw while sailing west: the track would "wiggle… like cycles on an oscilloscope." This page is both answers, plus the older sailor's trick of latitude sailing. It grows straight out of Polaris and Bowditch, and sits beside Galileo (longitude as time) and Map Surprises.



Why a sidereal day is ~4 minutes shorter than a solar day. Drawn for faria.jbf.com.

1. The word: sidereal

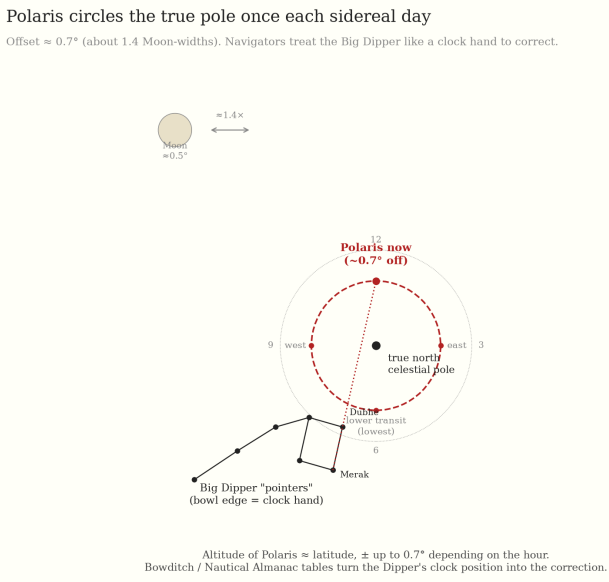
Sidereal — pronounce sy-DEER-ee-ul.

From Latin sidus, genitive sideris, "star" (also "constellation, heavenly body"). Related English words: sidereal, consider (literally "with the stars," from an old sense of consulting the heavens), desire (from de-sidus, "down from a star," via a folk etymology of longing), and the astronomical sidereal time, sidereal year.

A sidereal day is a day measured against the stars, not against the Sun.

2. 23 h 56 m 4.09 s versus 24 h — and why

Two different "days":



Polaris circles the true pole once each sidereal day. Drawn for faria.jbf.com.

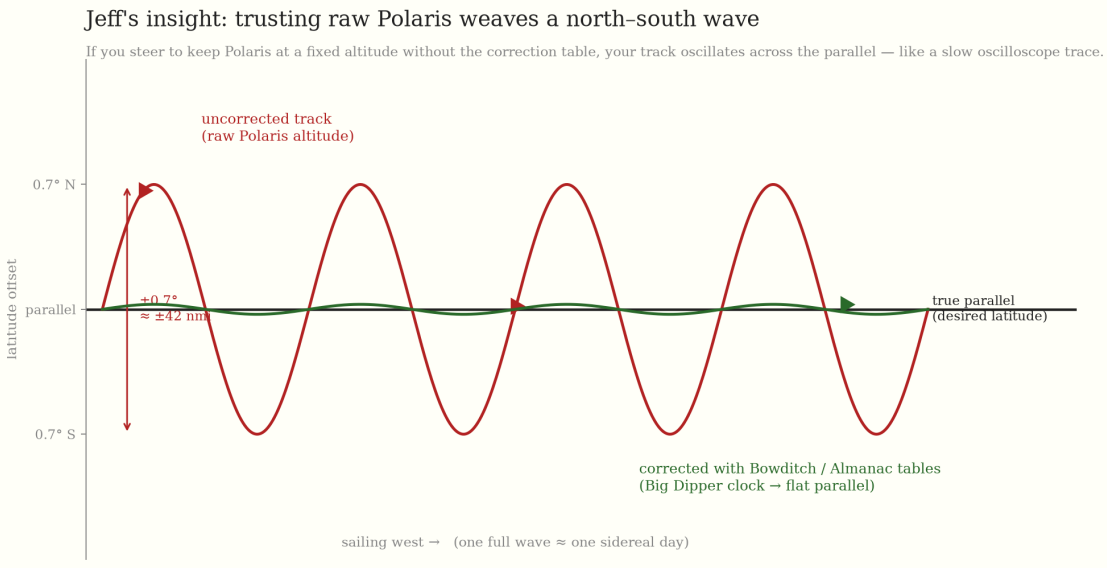
- Kind of day — Length — What returns overhead
- Sidereal day — 23 h 56 m 4.091 s of mean solar time — A distant star (really: the vernal equinox)
- Mean solar day — 24 h exactly (by definition of civil time) — The mean Sun

Why the difference? While Earth spins, it also moves along its orbit around the Sun — about 360° / 365.25 ≈ 0.986° per day, "about a degree." After Earth has turned a full 360° relative to the stars, the Sun has shifted by that ~1° in the sky. Earth must turn about one more degree to bring the Sun back to the same meridian. One degree of rotation takes 4 minutes (because 360° × 4 min = 1440 min = 24 h). So the solar day is about four minutes longer than the sidereal day.

- "Why only 56 minutes?" — because 24 h minus ~4 minutes is 23 h 56 m. The precise figure is 23 h 56 m 4.091 s.
- "366 sidereal days per year?" — yes, almost: in one mean solar year (~365.2422 mean solar days) Earth turns once extra relative to the stars, so there are about 366.2422 sidereal days in a year. Roughly: 365 Sun-days, 366 star-days.

3. Stars rise ~4 minutes earlier each night

Because the sidereal day is shorter, the starry sky runs about 4 minutes ahead of civil clocks each night. Consequences Jeff already uses without naming them:

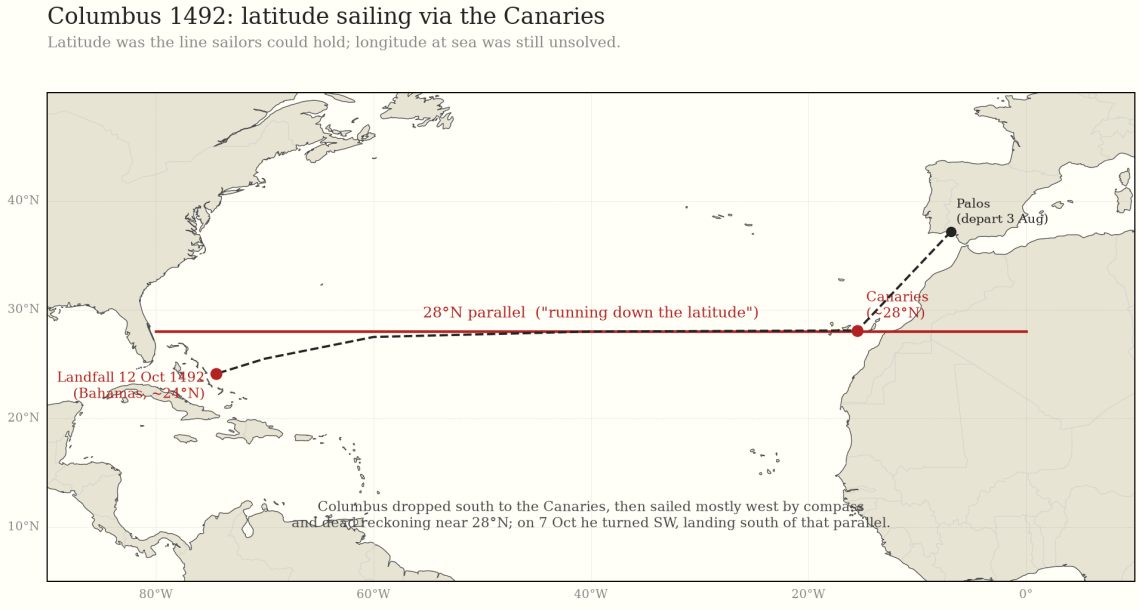


Jeff's insight: uncorrected track weaves north–south like an oscilloscope. Drawn for faria.jbf.com.

- ~2 hours per month. 4 min × 30 ≈ 2 hours. The same constellation rises two hours earlier after a month.
- Seasonal constellations. Over a year those daily 4-minute slips add up to a full 24 hours: winter evenings show Orion; summer evenings show Scorpius. Same sky, different clock time.
- Right ascension in hours. Astronomers measure sky-longitude in hours precisely because the sky turns 15° per hour — the same arithmetic as terrestrial longitude (Galileo, Polaris §8).

4. Polaris's tiny circle — once per sidereal day

Polaris is about 0.7° from the true north celestial pole (epoch ~2010: ~0.69°; ~0.66° in more recent figures; closest approach ~0.45° around 2100). It therefore traces a small circle around the pole once each sidereal day (23 h 56 m), not once each solar day.



Columbus 1492 via the Canaries along ~28°N. Drawn for faria.jbf.com.

Only at upper and lower transit (when Polaris is due above or below the pole on the meridian) is its azimuth true north. The rest of the time its altitude is a little high or a little low compared with the observer's true latitude. Maximum error ≈ 0.7°.

How far is 0.7° on Earth? One degree of latitude is 60 nautical miles (that is the definition that makes the nautical mile useful). So 0.7 × 60 = 42 nmi — about 48 statute miles, or roughly the distance from Washington, D.C. to Baltimore. Trust raw Polaris and you can be almost a Baltimore-run off your parallel.

5. Jeff's insight: the oscilloscope wave

Jeff, thinking about sailing west while keeping Polaris at a fixed altitude:

> *"You would wiggle your line… like cycles on an oscilloscope. You just kind of go a little left and right."*

Almost perfect. The wiggle is north–south, not left–right (east–west), because altitude error is a latitude error. As Polaris rides its daily circle — high, then low — a helmsman who steers to "correct" the raw altitude weaves above and below the parallel. One full wave takes one sidereal day. Amplitude up to ±0.7° ≈ ±42 nmi.

The fix is exactly what Bowditch teaches: read the Big Dipper like a clock, look up the Polaris correction, apply it. The wave flattens into a straight parallel. Modern Nautical Almanac tables (a0, a1, a2) do the same job.

And the oscilloscope image is not a stretch: the curves an oscilloscope draws in XY mode are Lissajous figures, also called Bowditch curves, first studied by Nathaniel Bowditch in 1815. Jeff reached for the navigator's own mathematics without knowing it.

6. Latitude sailing: "running down the latitude"

Before sailors could find longitude at sea, they could still find latitude — from the Sun at noon, or from Polaris at night. So a standard tactic was:

1. Sail north or south until you reach the latitude of your destination.

2. Turn east or west and hold that latitude (keep Polaris, or the noon Sun, at the right height) until land appears.

This is called latitude sailing, parallel sailing, or "running down the latitude." You follow a rhumb line along a parallel: constant compass heading, constant latitude. It is not the shortest path on a globe (that is a great circle), but it is the path you can find again tomorrow with only a quadrant and a compass.

Why it mattered. Longitude at sea was the unsolved problem until John Harrison's marine chronometers in the 1760s (and, for the poor, workable lunar-distance methods that Bowditch helped teach). See Galileo for the earlier dream of timing Io's eclipses. Latitude was the only line you could trust. So you aimed for the right parallel and ran along it.

7. Columbus 1492: mostly that idea, with a caveat

Christopher Columbus, first voyage, 1492:

- Left Palos, Spain, 3 August.
- Dropped south to the Canary Islands at about 28°N — a deliberate move into the easterly trade winds and onto a useful parallel.
- From the Canaries he sailed mostly west, by compass and dead reckoning, near that latitude. He was not yet a Polaris-table navigator in the Bowditch sense; the star method became routine later. But the logic is latitude sailing: get on the parallel, hold west.
- On 7 October, seeing flocks of birds, he turned southwest.
- Landfall 12 October in the Bahamas, around 24°N — south of the 28°N parallel he had been roughly holding. (Which island is still argued: traditionally San Salvador / Watling.)

So: Columbus is the textbook illustration of "drop to the latitude, then run west," with the honest footnote that he steered more by compass and log than by nightly Polaris altitudes, and that his final days left the parallel on purpose.

8. Rhumb line versus great circle

A quick GIS distinction for the talk:

- Path — What it is — On a Mercator chart
- Rhumb line (loxodrome) — Constant compass bearing — A straight line
- Great circle — Shortest path on the sphere — A curve (unless it is the equator or a meridian)

Latitude sailing along a parallel is a special rhumb line (bearing due east or due west). Airlines and great-circle sailing prefer the shorter curved path; sixteenth- and seventeenth-century ocean sailors preferred the rhumb they could hold with a compass. That is why Mercator charts were invented: so a rhumb looks straight.

Review questions

1. Pronounce sidereal and give its Latin root.
2. Why must Earth turn ~1° extra each solar day?
3. If Polaris is 0.7° off the pole, what is the maximum latitude error in nautical miles, and why?
4. Jeff said the uncorrected track would wiggle "left and right." What is the one-word correction to that picture?
5. What is "running down the latitude," and why did sailors use it before chronometers?

Key facts

- Sidereal (sy-DEER-ee-ul) ← Latin sidus, sideris "star."
- Sidereal day = 23 h 56 m 4.091 s; mean solar day = 24 h; difference ≈ 4 min because of ~1°/day orbital motion.
- ≈ 366.24 sidereal days per mean solar year (one extra turn relative to the stars).
- Stars rise ~4 min earlier each night → ~2 h/month → seasonal constellations.
- Polaris ~0.7° from the pole; circles it once per sidereal day; raw altitude error up to ±42 nmi.
- Jeff's insight: uncorrected track = north–south oscilloscope wave; Bowditch / Almanac tables flatten it. Those curves are themselves Bowditch/Lissajous figures.
- Latitude sailing / parallel sailing / "running down the latitude": reach the destination's latitude, then hold east/west on that parallel (a rhumb). Used because longitude at sea was unsolved before chronometers (and practical lunars).
- Columbus 1492: Canaries ~28°N, then mostly west; turned SW 7 Oct; Bahamas landfall ~24°N.
- Rhumb (constant bearing, straight on Mercator) ≠ great circle (shortest, curved on Mercator).

Answers

1. sy-DEER-ee-ul; Latin sidus / sideris, "star."

2. Because Earth also advanced ~1° along its orbit; after a 360° spin relative to the stars, the Sun has moved, so Earth needs ~1° more rotation (~4 min) to bring the Sun back overhead.

3. 42 nmi, because 1° of latitude = 60 nmi, and 0.7 × 60 = 42.

4. North–south (a latitude error), not east–west.

5. Sail to the destination's latitude, then hold east or west along that parallel until land appears. Latitude could be measured from the sky; longitude at sea could not (until chronometers / lunars).

Thumb to Moon

Calculating the Moon’s diameter

A little window on a large Moon

Let us put the arithmetic aside for a moment. Imagine the Moon as a distant coin. It has a real width, measured in miles, and an apparent width: how much of your view it occupies. A nearby thumb can hide that enormous object because apparent width depends on distance as well as size.

Close one eye and hold your thumb at arm’s length. For a rough picture, think of a thumb about two centimeters—roughly three-quarters of an inch—wide, held about sixty centimeters, or two feet, from your eye. It covers about two degrees. Your own thumb and arm may differ; this is a picture, not a precision instrument. Never use this exercise to look at the Sun.

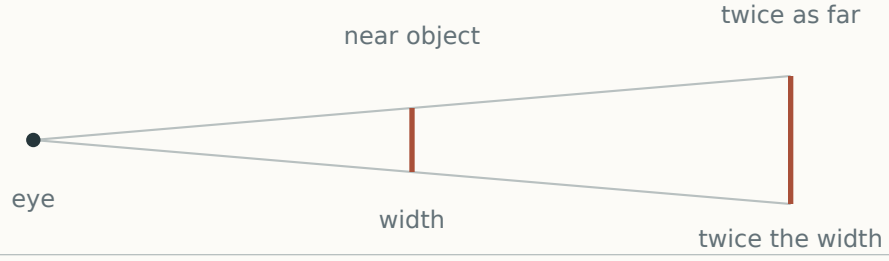
The Moon spans about half a degree. That is roughly one-quarter of our two-degree thumb width. So a small object covering the Moon at a distance of two feet would be only about a fifth of an inch wide: approximately 0.21 inches, or 5.3 millimeters, at 24 inches.

A correction worth keeping

A TWO-INCH-wide thumb held 24 inches away would cover about 4.77 degrees, not two degrees. Centimeters and inches cannot be exchanged without changing the answer. The rough two-degree example above uses two centimeters, not two inches.

The rule behind the picture

For the SAME apparent width, an object twice as far away must be twice as wide. Imagine sliding a small coin farther from your eye while enlarging it just enough to keep its edges on the same two sightlines. The triangle grows; its shape stays the same. Moving your thumb does not change the Moon. It changes only how much of your view the thumb covers.



Same sightlines.
Same apparent width.

Objects are schematic.

Source: NASA Science, Moon Facts • science.nasa.gov/moon/facts/ • Prepared 7 Nov 2026
Geometry is illustrative. Thumb/arm estimates are approximate. Never look at the Sun for this exercise.

Now give the picture a scale

From your arm to the Moon

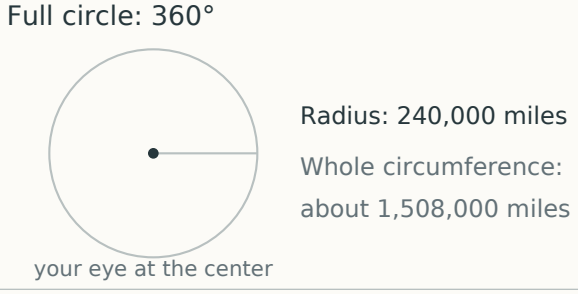
We must already know the distance before we can calculate the Moon's physical width. Take a rounded distance of 240,000 miles. A mile contains 5,280 feet, so that is 1,267,200,000 feet. Divide by our two-foot arm length: the Moon is about 633,600,000 arm lengths away.

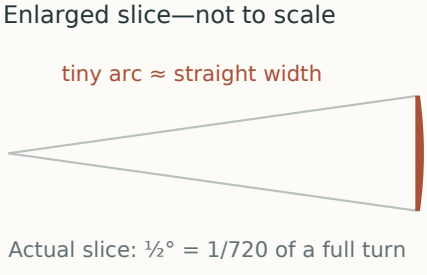
Enlarge the 0.21-inch Moon-covering width by that same factor. The units look cumbersome, but the thought is simple: the same narrow view reaches a much wider object farther away. Converting the enlarged width back into miles gives about 2,100 miles.

An easier route: go around a circle

Imagine an enormous circle centered on your eye, with a radius reaching to the Moon: 240,000 miles. This is an imaginary measuring circle, not a claim about the Moon's orbit. A full turn contains 360 degrees; half a degree is one part in 720.

We must divide the WHOLE circumference by 720—not the radius. The distance around a circle is about 6.28 times its radius. That factor is twice pi; pi is approximately 3.14. Our circle is therefore about 1,508,000 miles around.





Diameter ≈ 240,000 × 6.28 ÷ 720 ≈ 2,100 miles.

The little arc covering half a degree is almost the same length as the straight width across it. At this tiny angle, the difference is negligible for our rough inputs. Dividing the distance alone by 720 would give about 333 miles and omit the essential distance-around-the-circle step.

Does the answer make sense?

Yes. The Moon's known mean diameter is about 2,160 miles, or 3,475 kilometers. Our rounded calculation gives about 2,100 miles, roughly 3,370 kilometers: a good approximation, not an exact measurement. Apparent size changes with lunar distance, and our thumb and arm are rough guides.

We have not discovered the distance with a thumb. We assumed it. Angular width PLUS distance gives physical width. Ancient eclipse geometry offers a different route to size and distance relationships; that is a separate lesson.

Source: NASA Science, Moon Facts • science.nasa.gov/moon/facts/ • Prepared 7 Nov 2026

Geometry is illustrative. Thumb/arm estimates are approximate. Never look at the Sun for this exercise.

Time to Longitude

Converting Hours, Minutes and Seconds to Degrees

Why time becomes angle

Earth turns once relative to the mean Sun in twenty-four hours of solar time. That full turn is three hundred sixty degrees of longitude. Divide evenly and each hour of solar-time difference corresponds to fifteen degrees of longitude: two hours, thirty degrees. We work with solar time here, not with the slightly shorter sidereal day of a pure spin against the stars.

Keep the two unit systems apart. Time uses hours, minutes, and seconds (h, m, s). Angle uses degrees, arcminutes, and arcseconds (° , ′ , ″). One degree equals sixty arcminutes; one arcminute equals sixty arcseconds. One hour of time equals sixty minutes; one minute equals sixty seconds. The bridge: one hour of time = 15° of angle; one minute of time = 15′ of arc; one second of time = 15″ of arc.

The formula

Longitude difference in degrees = 15 × (hours + minutes/60 + seconds/3600)

Worked example: 2 h 35 m west

Suppose local solar time is two hours thirty-five minutes behind the time at Greenwich, compared at the same instant. Work row by row.

- 2 h of time → 2 × 15° = 30°
- 35 m of time → 35 × 15′ = 525′ of arc. Convert: 525 ÷ 60 = 8° with remainder 45′, so 8°45′.
- Add: 30° + 8°45′ = 38°45′00″ west of Greenwich.
- As a decimal: 38.75° W. Signed convention (east positive, west negative): −38.75°.

Which side of Greenwich?

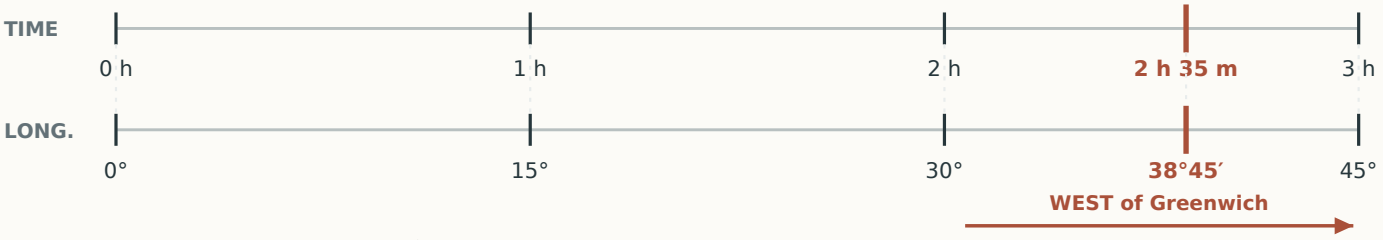
If local solar time is behind Greenwich at the same instant, you are west. If local solar time is ahead, you are east. Picture Greenwich at 12:00 while a western place still shows 9:25: that is 2 h 35 m behind, hence 38°45′ west. Always name the reference meridian—today Greenwich is usual; Chappe's published result, for example, used Paris.

This is a solar-time conversion, not a way to read exact longitude from today's time-zone label. Civil and political zones, and daylight-saving shifts, are administrative clocks, not precise meridians.

Coordinates may be written both ways. Computers commonly work with decimal degrees; people often read degrees, minutes and seconds. GPS and mapping apps can display either.

Paired scales — solar-time difference ↔ longitude west of Greenwich

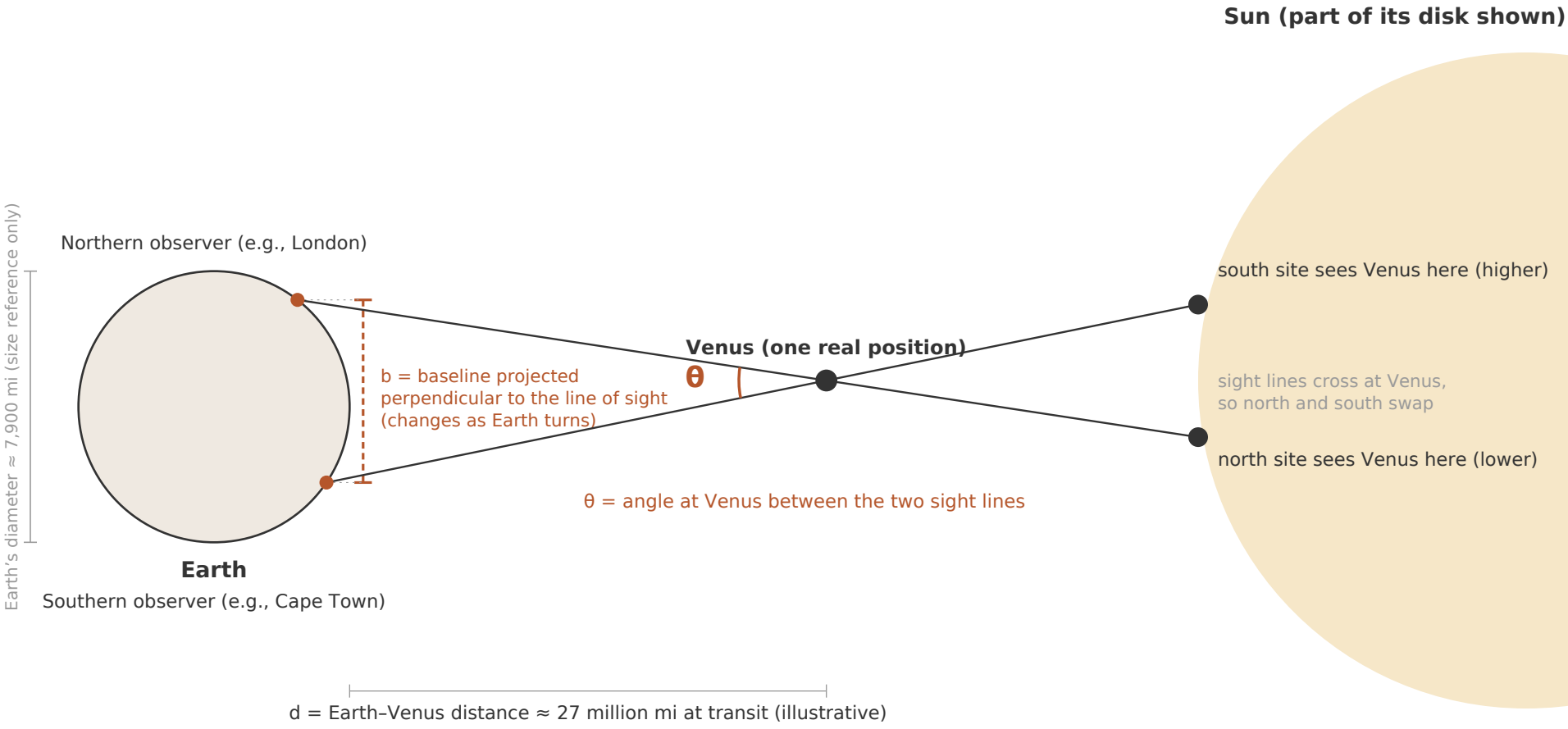
West increases to the right. Endpoint at 2 h 35 m = 38°45′ (38.75°) in true proportion.



Recall — If local solar time is one hour behind Greenwich, what longitude does that suggest? Fifteen degrees west of Greenwich.

Measuring the Sun’s distance with the 1761 transit of Venus

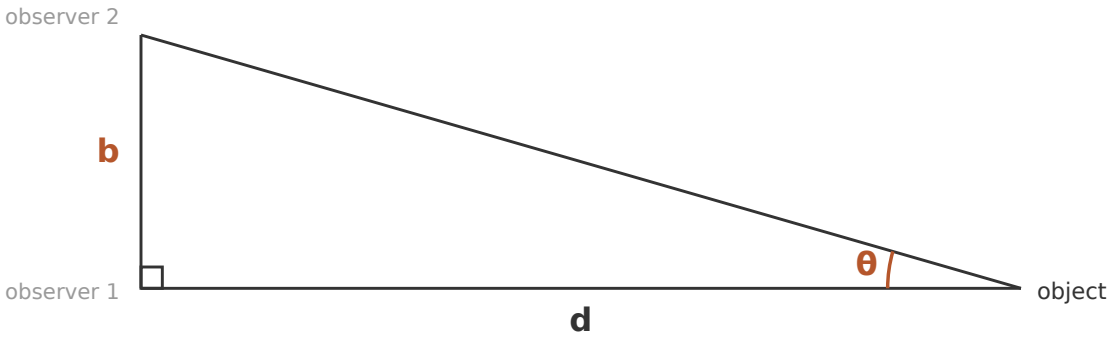
Panel 1 — the geometry (schematic side view; not to scale; angles hugely exaggerated)



How to read it: both observers look at the same Venus. Because they stand in different places, their sight lines cross at Venus and land on different spots on the Sun. The Sun is also at a finite distance, so it shifts a little too; what observers actually measure is Venus’s shift RELATIVE to the Sun (Venus’s parallax minus the Sun’s own parallax). In 1761, that shift showed up as differences in contact timings. Schematic note: London and Cape Town are NOT opposite ends of a diameter and are not placed from a real map here; the useful baseline is only the part of their separation perpendicular to the line of sight, and it changes hour by hour as Earth rotates.

Panel 2 — turning an angle into a distance

Basic trigonometry; round, historical-style numbers for illustration (not the 1761 London–Cape Town measurement)



Right triangle, b perpendicular to d, θ at the far object:

d = b / tan θ (exact for this one-sided triangle)
Symmetric case (b split evenly about the sight line):
d = b / [2 tan(θ/2)] ≈ b / θ (θ in radians)

Solar inset — the classic textbook triangle

Solar horizontal parallax p☉ = the angle one Earth RADIUS (≈ 3,959 mi, ~4,000 rounded) spans as seen from the Sun.

Using p☉ = 8.5″ = 8.5 / 206,265 rad ≈ 0.0000412 rad:
d☉ = R / tan p☉ ≈ 3,959 / 0.0000412 ≈ 96 million mi
(≈ 97 million mi with R rounded to 4,000)

Historical illustration only. 8.5″ was an assumed working value (e.g., James Short, 1761) — NOT Halley's 1716 figure (he assumed 12.5″). Modern value ≈ 8.794″ → 1 AU ≈ 93.0 million mi.

From the Sun to Venus — and everything else (Kepler's third law)

$P^2 \propto a^3$. From the known orbital periods (Venus 224.7 days, Earth 365.25 days): a_Venus / a_Earth = (224.7/365.25)^(2/3) ≈ 0.723.
At inferior conjunction, with circular orbits as an approximation: Earth–Venus ≈ (1 − 0.723) × Earth–Sun ≈ 0.277 × 96 M mi ≈ 27 million mi.
At that distance one Earth radius spans ≈ 30.7″ from Venus; Venus's shift relative to the Sun ≈ 30.7″ − 8.5″ ≈ 22.2″ (double for a full diameter).
Kepler gives ratios only; one measured distance sets the scale for all planets. Real 1761 analyses used full orbital geometry, not circles.

Historical footnote

In 1761 Cape Town and Greenwich could compare only the egress (end), not a whole transit. Short found 435 s observed vs 439 s predicted for 8.5″, implying ≈ 8.42″. Illustrative numbers above are not those observations.

Sources

Halley (1716), English tr.: eclipse.gsfc.nasa.gov/transit/HalleyParallax.html · Short (1761), Phil. Trans.: doi.org/10.1098/rstl.1761.0102
NASA Venus fact sheet (orbit, period): nssdc.gsfc.nasa.gov/planetary/factsheet/venusfact.html

FARIA · 6 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

From a Venus transit to the Cepheids

How a clock in the planets, a clock on Earth, a baseline the size of our orbit, and a clock inside a star came to measure distance.

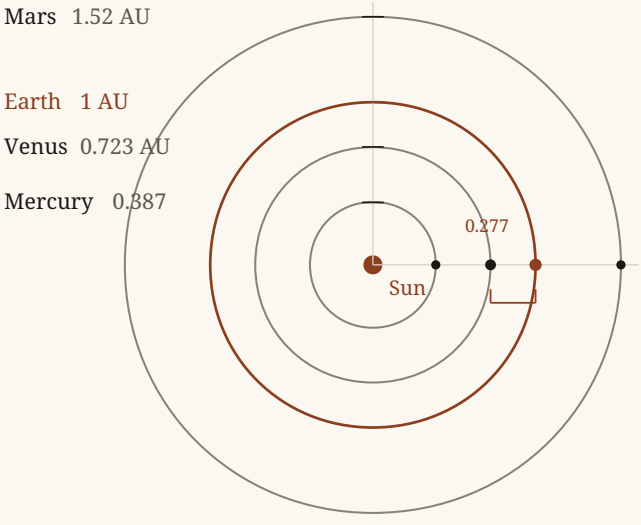
Jeff, the Polaris page called a Cepheid a yardstick, and the Harvard Computers page showed where Henrietta Leavitt's straight line came from. This page is the road under that yardstick. Four rungs, in order:

1. **Kepler's third law** gives planetary distances *relative to Earth's orbit*. It does not give miles.
2. **Transits of Venus in 1761 and 1769**, timed from widely separated places, turn that relative map into a length. The thing measured is a *difference* of parallaxes, read from durations or from contact times, not one angle taken at one station and not a latitude by itself.
3. Once the astronomical unit is a known length, **Earth's orbit is a baseline**. A nearby star's annual shift against the background is its parallax. January and July are a teaching pair, not a rule for every star. The first success, Bessel in 1838, used a heliometer, not a photograph.
4. **Leavitt's period-luminosity relation** tells which Cepheid is intrinsically brighter than which other one. It does not, by itself, say how many watts a ten-day Cepheid emits. Later calibrations, anchored on parallax and on other rungs, pin that zero point. Then Cepheids measure distances across the Milky Way and out into other galaxies.

A printed copy of this whole lesson, with the same four diagrams, is Venus transit to Cepheid distances (PDF).

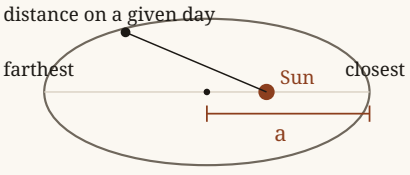
Kepler's third law: periods fix relative distances

a^3 / T^2 is the same number for every planet. a is the semimajor axis. T is the period.



a is not today's distance

Eccentricity exaggerated so the focus is obvious.



a is half the long axis: the average of closest and farthest. The line drawn to one day has a different length.

From the clock: a = T^(2/3)

Planet	Period T	a, in AU
Mercury	0.241 year	0.387
Venus	0.615 year	0.723
Earth	1 year	1
Mars	1.881 years	1.524
Jupiter	11.86 years	5.20
Saturn	29.46 years	9.54

Rounded sidereal periods, Earth set to 1. Modern semimajor axes agree here. Venus shifts a^3/T^2 by parts per million. Jupiter shifts it by about 0.1 percent. Not one rule for all.

Dots: planets on the near side. The bracket is the 0.277 AU gap, Earth minus Venus, in the circular model. Jupiter (5.2 AU) and Saturn (9.5 AU) would fall off the page.

Kepler's third law drawn for this lesson. The orbits are circles of the right semimajor axes. The ellipse at right has its eccentricity exaggerated so the focus is obvious. Jupiter and Saturn are in the table and off the drawing.

1. The clock, and what it measures

In 1619 Johannes Kepler published *Harmonices Mundi* (*The Harmony of the World*). In Book 5 he stated what we still call his third law. In his words, the ratio of the periodic times of any two planets is the **sesquialterate** proportion of their mean distances: the 3/2 power. If Earth is the unit,

> **a cubed, over T squared, is the same number for every planet.**

T is the sidereal period in Earth years. a is the **semimajor axis** in astronomical units. Set Earth's year and Earth's semimajor axis both to 1, and the shared number is 1. Then

> **a = T to the power 2/3.**

"Mean distance," in Kepler's language, is this a. It is half the long axis of the ellipse. It equals the average of the closest distance and the farthest distance. It is **not** the distance on a Tuesday.

Look at the ellipse on the diagram. The Sun sits at a focus, not at the center. The long axis is 2a. The segment marked a runs from the center to either end. The line from the Sun to "distance on a given day" has a different length. For Venus the difference is small, because the orbit is nearly round (eccentricity about 0.0068). For a comet it is the whole story. The law never said "the distance right now."

A deeper step, if you want it. Newton showed why the number is shared. For a small planet around a heavy sun,

> a^3 / T^2 = G (M + m) / (4 pi squared),

where M is the Sun's mass and m is the planet's. The constant grows by the factor (M + m) / M, which is 1 + m/M. That factor is not the same for every planet. Venus is a few parts per million of the solar mass, so for Venus the shift in a^3/T^2 really does sit in the sixth decimal. Jupiter is about 1/1047 of the Sun, and the constant is larger by about **0.1 percent**. Saturn is about 0.03 percent. The rounded table on the diagram agrees at the printed decimals because the periods were rounded. It does not mean every planet's mass is negligible.

Worked arithmetic for Venus. NASA's Venus fact sheet gives a sidereal period of **224.701 days**. Earth's sidereal year is **365.256 days**.

- 224.701 / 365.256 = **0.61519** year.
- Square it: 0.61519 squared = **0.3785**.
- The semimajor axis is the cube root of 0.3785. Try 0.72: 0.72 cubed = 0.373. Try 0.723: 0.723 squared = 0.5227, times 0.723 = **0.3779**. Close enough: **a = 0.723 AU**.

The fact sheet's own semimajor axis is **0.723332 AU**. The clock and the modern orbit agree. The same rule, with rounded sidereal periods, fills the table on the diagram: Mercury 0.387, Mars 1.524, Jupiter 5.20, Saturn 9.54.

The law was not the only road to a ratio. Venus never strays much past about **46 degrees** from the Sun. In a circular picture, the sine of that greatest elongation *is* the size of Venus's orbit in AU. sin 46 degrees = **0.719**, and the modern a is 0.723. Copernicus already had that triangle. What the third law adds is a single rule that ties every planet's *period* to the same scale, including Mars, Jupiter, and Saturn, for which there is no "greatest elongation" of the Venus kind. The transit calculators of the 1760s used that relative model. They were not inventing the shape. They were trying to put a mile on it.

2. The simplified gap, and why it is simplified

Put the orbits in a flat circle and line the planets up on the near side of the Sun, as the dots on the diagram do. Earth's orbit is 1 AU. Venus's is 0.723 AU. The gap is

> **1 - 0.723 = 0.277 AU.**

That is the Earth-Venus separation at inferior conjunction **in this model**: circular, coplanar, and using the semimajor axes as if they were the distances that day.

Three real complications, none of which wreck the lesson:

- The orbits are ellipses. Venus's Sun-distance runs from about 0.718 to 0.728 AU, Earth's from about 0.983 to 1.017. The radial gap at a real conjunction is near 0.277 AU and is not identical to it.

- Venus's orbit is tilted about **3.4 degrees** to Earth's. The true separation is a little larger than the difference of the two solar distances, except near the nodes, which is exactly where a transit can happen.
- "Conjunction" means the same ecliptic longitude, not "the smallest possible 3-D gap on an average Tuesday."

So 0.277 AU is the right number to *think* with. A reduction of a real transit uses the orbital elements for that week. Nobody in 1769 subtracted 0.723 from 1.000 on a slate and called the result the measured distance.

3. A ratio is not a mile

An astronomical unit, in the eighteenth century, meant "Earth's semimajor axis, whatever that is in feet." Kepler hands you Venus at 0.723 of that unit. He does not hand you the unit.

The transit measures an *angle*: the **solar parallax**. The horizontal solar parallax is the angle subtended by Earth's equatorial radius as seen from a distance of 1 AU. Once you have that angle, and once you know Earth's radius in miles from a survey,

> **AU = Earth's radius / the parallax, in radians.**

The survey is the old problem on the Eratosthenes page, done much more carefully by 1769. Earth's size was already known far better than the Sun's distance. The blur in the transit era was the angle, not the radius. Halley, in 1716, still had to remind his readers how bad the old guesses were: Ptolemy at 1,200 Earth radii, Kepler at almost 3,500, and his own working hypothesis of 12.5 arcseconds, which is about 16,500 Earth radii. The modern figure is about **23,500** Earth radii. The expeditions were trying to settle a number that learned men still had wrong by a large factor.

4. Three different parallaxes

This is the step that gets flattened in a hurry, so it is worth going slowly.

Solar parallax, written here as pi-sun, is Earth's radius seen from the Sun. The modern value is about **8.794 arcseconds**. It is no longer an independent mystery: the astronomical unit is a defined length (below), Earth's equatorial radius is about **6,378.137 km**, and the angle follows.

> 6,378.137 / 149,597,870.7 = 0.000042635, in radians. Times 206,265 arcseconds in a radian = **8.794 arcseconds**.

Venus's parallax is the same baseline seen from Venus. In the circular model Venus is 0.277 AU away at inferior conjunction, so

> pi-Venus is about pi-sun / 0.277, which is about **3.61 times** the solar parallax.

The differential parallax is what two observers actually see as a disagreement in where Venus sits *on the Sun*. Both bodies shift when you move your eye. The nearer one shifts more. The relative shift is the difference:

> pi-Venus minus pi-sun is about **(3.61 - 1) times pi-sun = 2.61 times the solar parallax**.

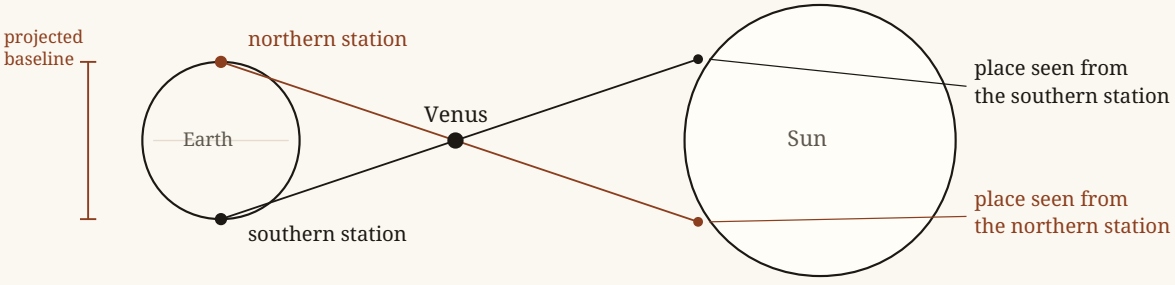
Check the arithmetic the long way. 1 / 0.277 = 3.610. Subtract 1. That is 2.610. If you insert the modern solar parallax only to see the size of the effect, 2.61 times 8.79 arcseconds is about **23 arcseconds**.

Read that 23 carefully. It is the track separation for one stated baseline: a projected separation of **one Earth radius**, square across the line of sight. That is the same length hidden in the definition of horizontal parallax. It is an illustration, not a number from a logbook, and it is **not a ceiling**. The straight-line distance between two observers can be longer than one radius. It can be as much as about an Earth **diameter**, two radii, when the stations stand near opposite ends of the sunlit hemisphere and can both still see the Sun. A side-on diameter would about double the shift, to something near 46 arcseconds in this same circular model. Real stations fall somewhere in that range, and the baseline is tilted, so the separation on the disk is whatever that projected length produces. The diagram labels 23 arcseconds as the one-radius case.

Halley himself, in 1716, did the same kind of arithmetic with a *trial* parallax, and he said so. He supposed the solar parallax was 12.5 arcseconds "till its precise quantity be made to appear more certain by the trial I propose." On that supposition he put Venus's parallax at 43 arcseconds, "almost 4 times" the Sun's (43 / 12.5 = 3.44, so he was using an Earth-Venus distance near 0.29 AU, close to our 0.277 and not identical). Subtract, and "there will remain half a minute at least" for the parallax of Venus *from* the Sun. Half a minute is 30 arcseconds. That is his hypothesis. It is not a measurement, and it is not the modern 8.8 arcseconds.

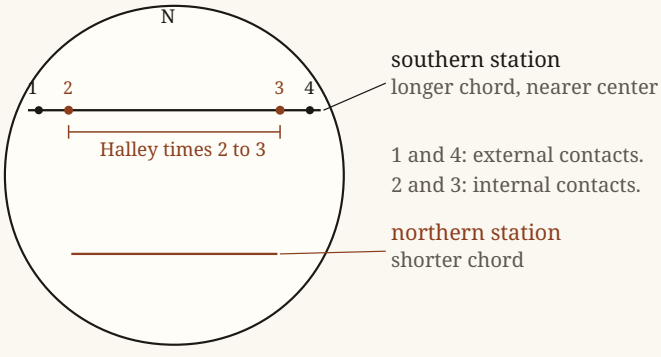
What two stations actually compare

Schematic. Sizes and angles are exaggerated so the shift can be seen. Nothing here is traced from a 1769 logbook.

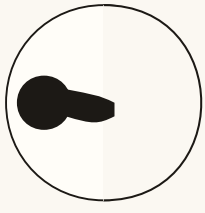


Raise your head and a near finger jumps down against a far wall.
A northern station sees Venus shifted south on the Sun. The Sun shifts too.
Venus, being nearer, shifts more. The measurement is that difference.

Over the hours, each station sees a chord across the Sun



Why contact 2 smears



Black drop. A dark neck seems to tie Venus to the outside sky, so "contact" is a judgment, not a click.

Circular model: Earth to Venus at inferior conjunction is about 0.277 AU.
Differential parallax is about (1/0.277 − 1) times the solar parallax, or 2.61 times it.

Illustrative, not a measurement. One Earth radius, side-on, separates the tracks by about 23 arcseconds if the solar parallax is the modern 8.79. That is one radius, not a ceiling.
Stations can be farther apart, up to about a diameter, and the baseline is usually tilted.

Delisle times one contact in absolute time and needs longitude. Halley's duration uses one local clock. Latitude by itself is not the baseline.

Mercury shows a black drop too. The neck is mostly blur and the Sun's dim edge, not only Venus's air.

Schematic, drawn for this lesson. Earth, Venus, and the angles are much too large. The 23 arcseconds in the note is the one-radius illustration above, not a ceiling and not a 1769 observation. North is up on the solar disk: a northern station sees the nearer planet shifted south.

5. Halley's clock, and Delisle's clock

Edmond Halley watched a transit of Mercury from St Helena in 1677. Mercury is too far, at a transit, for its parallax to stand well clear of the Sun's. Venus at inferior conjunction is the useful one: nearer, larger differential, rare.

He published the method in the *Philosophical Transactions* in **1716** (*Methodus singularis*, volume 29, number 348, page 454). He would be dead before 1761. The paper is a set of instructions to the next generation.

What he asked them to time. Not a tiny angle on a divided circle. He had already said that single seconds of arc, even ten seconds, defeat the instruments of the day. He asked for the **duration**: the local time from internal ingress (contact 2, when Venus has just come fully onto the disk) to internal egress (contact 3, when it is about to leave). A clock that runs truly between those two moments is enough for the duration. An error in the absolute longitude mostly cancels, because you subtract two readings on the same clock.

Why a duration is a parallax: the stations see Venus travel different **chords** of the Sun. A chord nearer the center is longer, so the transit lasts longer. The difference of durations, the known speed of Venus across the disk, and the known direction of the path turn back into an angle: the separation of the chords. That separation is the **differential** parallax for the **projected baseline** between the stations. Divide by the geometric factor (about 2.61 in the simple model; the real factor uses that month's distances) and you have the solar parallax. Then the AU, in Earth radii, is the reciprocal of that angle in radians.

Halley's own trial numbers show the proportion, and only the proportion. He reckoned that if the solar parallax were 12.5 arcseconds, the duration at Hudson Bay would exceed the duration in the East Indies by about **17 minutes**. A larger observed difference would mean a larger parallax, in nearly the same ratio. He thought two seconds of time would pin the parallax to a small fraction of an arcsecond, and the distance to one part in five hundred. Hold that hope lightly. He also assumed a path that Venus did not take. In 1761 she passed much nearer the Sun's center than his 4 minutes of latitude, his favored stations were wrong for the whole duration, and a later note on the translation records an error in the angle of the path. The 17 minutes is a worked example under stated assumptions. It is not the 1769 result, and it is not a number to paste onto Cook's journal.

Delisle's method is the other half of the campaign. Joseph-Nicolas Delisle, who had visited Halley, proposed timing **one** internal contact at two stations in absolute time. You no longer need both ingress and egress, so a station that sees only the entrance, or only the exit, still counts. The price is the longitude. Both clocks have to be reduced to one meridian, and the east-west part of the baseline matters as much as the north-south part. Halley's duration is forgiving of a bad absolute clock and hungry for a complete transit. Delisle's instant is forgiving of a cloud in the middle of the afternoon and hungry for a known longitude. Eighteenth-century longitude was the harder measurement. Neither method is "stand on a latitude and measure one angle."

The projected baseline is the component that actually separates the chords or shifts the contact time: the station-to-station line as seen from the Sun, square to the line of sight, and then taken along the direction that matters for the path. Two towns at the same latitude, far apart in longitude, are useless for a pure north-south chord shift and very useful for a timing method. Two towns on the same meridian, far apart in latitude, are Halley's classic pair. The map coordinate you print is not the baseline. The geometry of that day is.

6. Who went, and what "Siberia" means

Transits of Venus come in pairs eight years apart, then not again for more than a century. Jeremiah Horrocks and William Crabtree saw the transit of **1639** from England. There was no distant second station, so Halley's comparison could not be made. The coordinated campaigns are **1761** and **1769**.

1761, 6 June. The French Academy sent Jean-Baptiste Chappe d'Auteroche overland to **Tobolsk**, in western Siberia. He got there, timed a lunar eclipse to fix the longitude, and, in clear weather and with armed protection (local feeling blamed the strange instruments for a flood), watched the whole transit. He published the observations; the journey became *Voyage en Sibérie* (1768). Charles Mason and Jeremiah Dixon, sent toward Bencoolen in Sumatra and diverted by war and weather, observed from the **Cape of Good Hope**. Dozens of other stations in Europe took part. The published solar parallaxes from this transit scattered by several tenths of an arcsecond.

Mikhail Lomonosov, watching from St Petersburg, reported a luminous ring at the limb and argued that Venus has an atmosphere. The atmosphere is real. How much of what he saw was the atmosphere, and how much was the optics of a bright edge, is still argued. Keep that ring separate from the black drop below. One is a bright aureole. The other is a dark neck that spoils a clock reading.

1769, 3 June at Tahiti and 4 June at stations far to the east, the same transit on two local dates. The Royal Society sent James Cook in the *Endeavour*, with the astronomer Charles Green, formerly Maskelyne's assistant at Greenwich. They anchored in Matavai Bay on **13 April 1769**, built Fort Venus on the point still called Point Venus, and had a clear day. Cook, Green, and Daniel Solander observed. Cook's journal records the gift and the trouble in one breath: not a cloud, and yet

> "we very distinctly saw an Atmosphere or dusky shade round the body of the Planet which very much disturbed the times of the Contacts particularly the two internal ones."

He adds that he, Green, and Solander "differ'd from one another in observeing the times of the Contacts much more than could be expected." The times and the drawings of that ligament were published in the *Philosophical Transactions* for **1771** (volume 61, pages 397-421), Green named with Cook. Green did not live to finish the voyage home.

The black drop. Near internal contact, Venus seems to stay tied to the Sun's limb by a dark neck, so the moment of "internal contact" is a judgment spread over many seconds. It is tempting to blame Venus's air, and Cook's "dusky shade" points that way. Mercury, which has almost no atmosphere, shows a black drop too. The neck is mostly the telescope's blur acting on the Sun's darkened limb. The atmosphere produces a different, real, bright rim. The practical result in 1769 was the same either way: Halley's hoped-for single second of time was not what the observers could agree on.

Russia in 1769 is a network, not one Siberian dot. Stepan Rumovsky, director of the Academy observatory, organized it under Catherine II, with observers from Russia, Germany, Sweden, and Geneva. The stations, from the IMCCE chronological table and the Siberian local-history accounts, were:

Station	Who	Where it actually is
Yakutsk	Ivan Isleniev; the Siberian encyclopedia also names F. Chernyi, while the IMCCE table says an unnamed companion	On the Lena, deep in Siberia. The Siberian station.
Orsk	Christoph Euler	Southern Urals, the edge of Europe and Asia, not deep Siberia
Orenburg	Wolfgang Ludwig Krafft	Same edge
Guriev	Petr Inokhodtsev and Georg Moritz Lowitz	Caspian shore, now Atyrau, Kazakhstan
Kola	Rumovsky	Kola Peninsula, European Arctic
Ponoi	Jacques Andre Mallet	Kola Peninsula
Umba	Jean-Louis Pictet	Kola Peninsula
St Petersburg	Anders Lexell and colleagues	The capital

Isleniev's party was in the Yakutsk region from June 1768 to August 1769. The published work includes Venus leaving the Sun, Jupiter's satellites for the longitude, and coordinates that later improved the Academy's maps. Do not fold every Russian station into the word Siberia, and do not move Chappe's **1761** Tobolsk into 1769.

Other 1769 stations that matter for the geometry, and that are not Siberia: Maximilian Hell and Janos Sajnovics at **Vardo** in northern Norway; William Wales and Joseph Dymond at Prince of Wales Fort on **Hudson Bay**, the northern station Halley had named in 1716; and Chappe again, this time at **San Jose del Cabo** in Baja California. He observed the transit on 3 June and died in the epidemic there on 1 August 1769. The observations were carried back.

No single station, and no single transit, printed the modern astronomical unit. The 1761 reductions disagreed among themselves. The 1769 campaign was better and still left a spread. Johann Franz Encke's careful combination of both transits, published in **1824**, gave **8.5776 arcseconds**, with an uncertainty he put near 0.037. That is about 0.22 arcsecond below the modern 8.794, which makes the Sun about **2.5 percent too far** (8.794 / 8.5776 is about 1.025). A revision in 1835, after a dispute over Hell's times, moved it only to about 8.571. The transits of **1874 and 1882** brought photography and another generation of reductions. Simon Newcomb's later synthesis sat near **8.79 to 8.80 arcseconds**, close to what we use, and still not a definition frozen on transit day. Radar echoes from Venus, in the twentieth century, measured a distance in kilometres by the travel time of a signal. The transit had become history. The geometry it taught had not.

7. The number we actually use

On 31 August 2012 the International Astronomical Union, in Resolution B2, redefined the astronomical unit as a conventional length:

> **exactly 149,597,870,700 metres.**

The symbol they asked for is **au**. The old definition, tied to the Gaussian gravitational constant and to the Earth's orbit, was retired. The Earth's real Sun-distance still changes through the year, from about 0.983 au at perihelion to about 1.017 au at aphelion. "One au" is no longer a synonym for "however far the Sun is today."

In international miles, 1,609.344 metres to the mile:

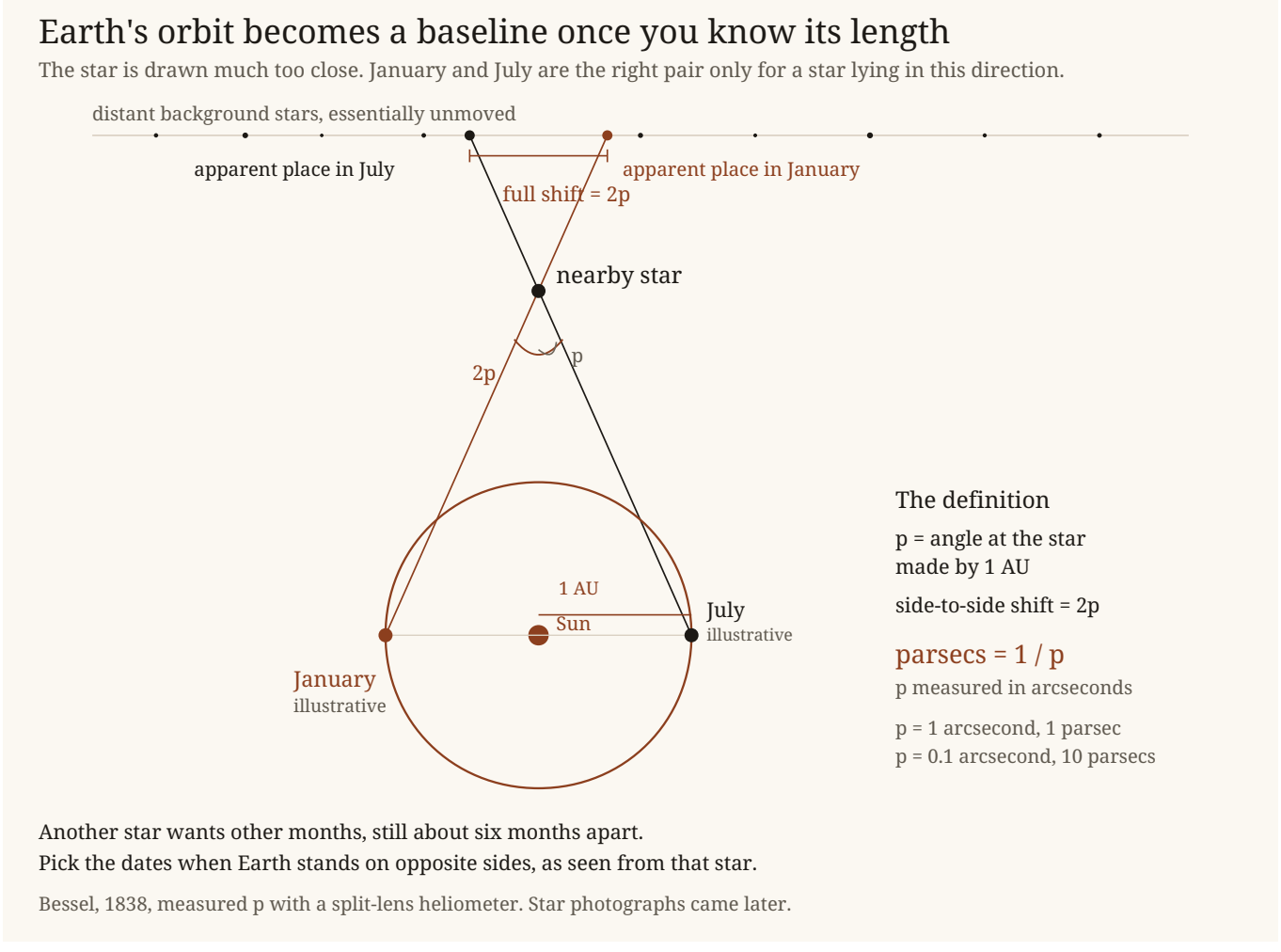
> 149,597,870,700 / 1,609.344 = **92,955,807 miles**,

which is **92.96 million miles** if you round to the hundredth of a million. That is the figure to remember. It is a definition, inherited from the best radar-era estimates, not a stopwatch reading from Tahiti.

8. The orbit as a baseline

You now have a stick. It is one au long, and you know that length in metres. Look at a nearby star in two seasons, from opposite sides of the stick, and the star shifts against the distant background. That shift is the **annual parallax**.

Hold a finger at arm's length and blink each eye. The finger jumps. The nearer it is, the larger the jump. The distant wall barely moves. Earth's orbit is the gap between the two eyes. The star is the finger.



Drawn much too close, or the angle would be invisible. January and July are the right pair for a star in this direction. Another star wants other months. Bessel's 1838 measurement was a heliometer angle, not a pair of photographs.

p and 2p. The parallax angle p is the angle at the star subtended by **1 AU**, not by the full width of the orbit. The star's apparent jump from one side of the orbit to the other is **2p**. Textbooks that say "the parallax is the shift from January to July" are describing 2p and naming p. The definition that makes the distance formula clean is the half-shift:

> **distance in parsecs = 1 / p**, with p in arcseconds.

One parsec is the distance at which 1 AU subtends one arcsecond. A tenth of an arcsecond means ten parsecs. A parsec is about 3.26 light-years, or about 206,265 AU.

January and July are a drawing, not a calendar rule. They are opposite, and they are the right pair when the star lies square to the January-July line, as on the diagram. A star somewhere else on the ecliptic wants the two dates when Earth is on opposite sides of the Sun *as seen from that star*. Those dates slide around the year with the star's ecliptic longitude. A star near the ecliptic pole traces a small circle, and any opposite pair will do. What you must not do is observe in January and February and call the tiny piece of the orbit a baseline of 2 AU.

The first success was not a photograph. Friedrich Wilhelm Bessel, at Konigsberg, used a Fraunhofer **heliometer**: a telescope whose objective lens is cut in half, so the two halves can be slid until a nearby star and a comparison star coincide. He chose **61 Cygni** for its large proper motion, a hint that it is near, and because the pair can be measured cleanly. In 1838, in *Astronomische Nachrichten*, he published

> **p = 0.3136 arcsecond, give or take 0.0202.**

He turned that into 657,700 sun-distances and a light-time of **10.3 years**. The modern parallax is about **0.286 arc-second**, near **11.4 light-years**. He was close, and he was first to publish a result that the errors could not swallow.

Two others were on the same problem. Thomas Henderson had measured Alpha Centauri from the Cape in 1832-33 and published in 1839, after Bessel. Wilhelm Struve measured Vega; a preliminary notice is 1837 and the fuller result 1840, and it was less accurate. "Bessel, 1838" is the right first *success*. It is not a claim that nobody else was looking.

Photography cannot be the method, because it was not ready. Daguerre's process was announced in 1839. The first successful photograph of a star, Vega, was made at Harvard by John Adams Whipple and William Cranch Bond on 17 July 1850. A pair of plates, January against July, is a clear way to *explain* the geometry, and it is how later parallax programs worked. It is not what Bessel did. The Polaris page uses the January/July picture for the finger-blink. That picture is fair teaching. The instrument in 1838 was the split lens.

Polaris itself is a warning about easy parallaxes. It is the nearest classical Cepheid, and the distance fight on that page (Hipparcos near 432 light-years, a spectroscopic "short" distance near 323, Gaia near 446) exists because a companion and a pulsation sit on top of a tiny angle. The nearest famous star is not the easiest calibrator.

9. Brightness is not distance

A star can look faint because it is dim, or because it is far, or because dust is in the way. **Apparent** brightness is what the plate records. **Intrinsic** brightness, the luminosity, is what the star emits.

Light spreads over a sphere. Double the distance and the same light covers four times the area, so the flux falls to a quarter. That is the **inverse-square law**. In magnitudes, with distance d in parsecs and no dust,

> **m - M = 5 log10(d) - 5.**

m is the apparent magnitude. M is the absolute magnitude, the apparent magnitude the star would have at 10 parsecs. If dust dims the star by A magnitudes, add A to the right-hand side. Ignore the dust and you will set the star too far.

A purely illustrative sum, not a real star. Suppose a calibrated relation says some Cepheid has absolute magnitude -4.0, and you measure its apparent magnitude as 16.0, and you ignore dust. The difference m - M is 20. Then

> d = 10 to the power of (20 + 5) / 5 = 10 to the 5th = **100,000 parsecs**.

The arithmetic is the whole point. The -4.0 was *supplied*. Nothing in the period, by itself, wrote that -4.0 down.

10. What Leavitt measured, and what she did not

The Harvard Computers page has her life, the plates, and the plot of all 25 stars. The facts that matter for the ladder are these.

The Small Magellanic Cloud is a cloud of stars at essentially **one** distance. (The depth of the cloud is real and small beside the distance.) Stars in that cloud can be faint because they are faint, not because one of them is vastly farther. Difference in apparent brightness is difference in luminosity.

In 1908, in *Annals of the Harvard College Observatory*, volume 60, Leavitt wrote that it was worthy of notice that **the brighter variables have the longer periods**, and that the sample was still small. In **Harvard College Observatory Circular 173**, 3 March 1912, "prepared by Miss Leavitt" and issued over Pickering's name, she had 25 periods. On a plot of magnitude against the logarithm of the period, "a straight line can readily be drawn." She gave the slope in her own words: the logarithm of the period increases by about **0.48** for each increase of one magnitude in brightness. The average swing from maximum to minimum was about 1.2 magnitudes. And, because they share a distance, "their periods are apparently associated with their actual emission of light."

That is the **slope**. A longer period means a more luminous star, by a definite step on that line.

It is not the **zero point**. The circular is careful. The magnitudes are on a provisional photographic scale. When she says "absolute scale of magnitudes," she means tying those faint comparison stars to a standard sequence, the North Polar Sequence. She does not mean the modern M, which already requires a distance. The closing hope is explicit:

> "It is to be hoped, also, that the parallaxes of some variables of this type may be measured."

In 1912 those parallaxes were out of reach. A classical Cepheid's parallax is a few thousandths of an arcsecond. The period told her which star was brighter than which other star in the Cloud. It did not tell her how far the Cloud was, in parsecs, until someone pinned the line to a luminosity.

The slope is not yet a distance

Distance grows upward. The plot is a sketch of the idea. Leavitt's twenty-five stars are on the Harvard page.

Farther still: Type Ia supernovae
the next rung, not this lesson

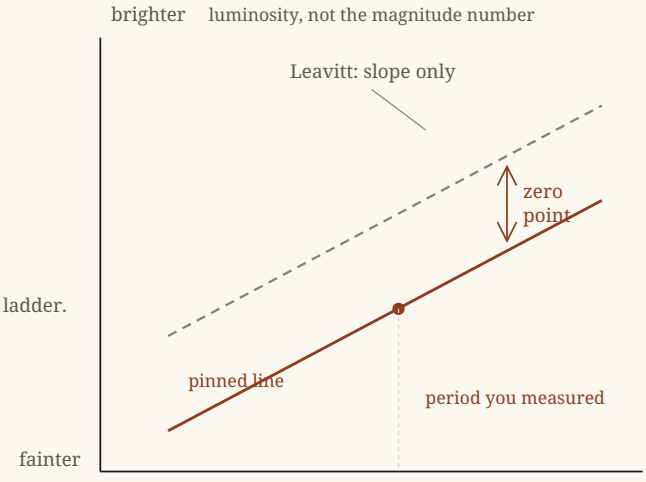
4. Cepheids, after calibration
Leavitt's slope, 1908 and 1912
Zero point from rung 3, clusters, later parallaxes
Across the Milky Way, and out into other galaxies

3. Nearby stars
Parallax. The baseline is the length from rung 2.
parsecs = 1 / p, with p in arcseconds
Reaches nearby Cepheids. One star checks; it does not pin the ladder.

2. A length, not only a ratio
Transit timings give the solar parallax.
AU = Earth's radius divided by that angle.
A few percent in 1769. Radar, then a definition, came later.

1. Relative orbits
Kepler, 1619. a^3 / T^2 is shared.
Venus 0.723 AU. Conjunction gap about 0.277 AU.
The shape of the system. No miles yet.

Same slope. The zero point can still slide.



Read luminosity off the pinned line.
Compare it with how bright the star looks.
Light falls as 1 over distance squared.

The period alone does not say how many watts. Dust makes a star look too far. Another Cepheid family lies on another line. The arrow is only a slide.

Which way it slides is settled by measurement, not by this sketch.

The ladder drawn for this lesson. Distance increases upward. On the graph, brighter means more luminous, upward, so a longer period stands higher. That axis is not a magnitude scale, on which the numbers run the other way. The graph is a sketch of a sliding zero point, not a replot of Leavitt's 25 stars; those are on the Harvard Computers page. The arrow can slide either way. Measurement decides which way.

11. Pinning the line

Ejnar Hertzsprung, in 1913, made the first rough zero point from the motions of galactic Cepheids: a statistical parallax, not a clean trigonometric one. **Harlow Shapley, around 1918**, revised that calibration and used it to size the Milky Way and to put the Sun off center. The revision was bold and partly wrong. He mixed classical Cepheids, the bright young pulsators of Leavitt's Cloud, with fainter stars of similar light curves in globular clusters (Type II Cepheids, and the RR Lyrae stars then called cluster variables). Interstellar dust was not yet in the sum. The zero point in use came out too faint by about **1.5 magnitudes**.

Edwin Hubble, on plates of 1923, found a Cepheid in the Andromeda "nebula." The result read to the astronomers in 1925 placed Andromeda far outside the Milky Way, on Shapley's scale. The island-universe question was answered. The miles were still short.

Walter Baade, in 1952, separated the two families. A classical, Population I Cepheid is about 1.5 magnitudes brighter than the Population II star of the same period that had been sitting on the same line. A magnitude differ-

ence of 1.5 is a distance factor of 10 to the power 1.5/5, which is 10 to the 0.3, which is **about 2**. The galaxies jumped outward. That was not a new slope from the Cloud. It was a new pin.

Dust, if you want the extra care: grains dim a star and redden it. Leave them out and the inverse-square distance is too large. A color excess estimates the dimming. Metallicity is a smaller, real shift: Cloud Cepheids are poorer in heavy elements than Cepheids near the Sun, and the luminosity at a given period depends slightly on that composition. Overtone pulsators, which complete a cycle in a shorter period than the fundamental tone, must not be dropped onto the fundamental line. Polaris is a nearby check and an awkward one. The pin is a crowd of Cepheids with geometric distances, not one star.

The modern pin is geometric. Parallaxes of Milky Way Cepheids, and of Cepheids that belong to open clusters whose distances are also parallaxes, put luminosities on the line directly. Gaia's data releases are the current basis of that work; one 2023 cluster study (Cruz Reyes and Anderson, *Astronomy & Astrophysics*) is an example of a zero point at the one-percent level, and of the remaining argument over a tiny parallax offset. Hubble Space Telescope parallaxes of Galactic Cepheids did the same job a generation earlier, with fewer stars. The top of the ladder, where Cepheids calibrate Type Ia supernovae and those supernovae measure the expansion, is still debated. The debate is about a few percent and about the next rung. It is not a return to "we have no idea how far."

Cepheids are bright: thousands of times the Sun, and the slow ones brighter still. They are visible across the Milky Way, in the Magellanic Clouds where Leavitt found them, in Andromeda, and, with a telescope in orbit, in galaxies tens of millions of light-years away that also hosted a supernova. The rung is not "inside our galaxy and then a wall." The wall is the next method.

12. The four rungs, said once

Rung	What you use	What you get	What you still do not have
Kepler, 1619	Periods, and a^3 / T^2	a in AU. Venus 0.723. Gap about 0.277 in the circular model	Miles
Transits, 1761 and 1769	Durations or contact times, a known baseline, Earth's radius	The AU as a length, to a few percent; refined for two centuries after	A single transit's exact modern metre
Parallax	The AU as a baseline	parsecs = 1 / p. Nearby stars, including nearby Cepheids	A luminosity for a Cepheid you cannot resolve
Cepheids, calibrated	Leavitt's slope, plus a zero point from parallax and later rungs	Distances across the Milky Way and into other galaxies	The supernova rung beyond them

Today the second rung is a definition plus radar history. The shape of the argument is unchanged. A ratio, then a length, then an angle against the stars, then a clock inside a star whose brightness that angle has taught you.

Try the arithmetic again

1. Venus takes 0.615 year. Square it. You should be near 0.378. What cube root, to three decimals, brings you back?
2. In the circular model, what is 1 minus that cube root?
3. 1 / 0.277 is about 3.61. What is the differential factor, and what angle is 2.61 times 8.79 arcseconds for a side-on baseline of one Earth radius? Is that a ceiling?

4. A star has p = 0.25 arcsecond. How many parsecs? About how many light-years, at 3.26 light-years to the parsec?
5. Two Cepheids have the same apparent brightness. One has twice the period of the other. What else do you need before you can say which is farther?

Review questions

1. What does a shared a^3 / T^2 give you, and what does it refuse to give you?
2. Why is the semimajor axis not the distance today?
3. Where does the 0.277 AU come from, and what three facts make it a model rather than a transit-day measurement?
4. Name the solar parallax, Venus's parallax, and the differential parallax. Which one moves the black spot differently for two observers?
5. What did Halley tell observers to time, and why is that not one angle at one place?
6. Why does Delisle's method need longitude, while a duration mostly cancels a bad absolute clock?
7. Who observed at Tahiti on 3 June 1769, and which 1769 station is the one deep in Siberia? Where was Chappe in 1761, and where was he in 1769?
8. What did Encke publish in 1824, and why is "the transit gave us 92.96 million miles" false?
9. In the parallax drawing, what is p, what is 2p, and why are January and July only examples?
10. What instrument did Bessel use in 1838, and why is a photograph the wrong picture of that measurement?
11. What did the Small Magellanic Cloud do for Leavitt that a scatter of Milky Way Cepheids could not, and what did she still ask for?
12. Why did Baade's 1.5 magnitudes roughly double the distances to galaxies?

Glossary

- **Semimajor axis (a).** Half the long axis of an elliptical orbit. The average of the nearest and farthest distances from the focus. Kepler's "mean distance."
- **Astronomical unit (au).** Since 2012, exactly 149,597,870,700 metres. Historically, the Earth's semimajor axis.
- **Inferior conjunction.** An inner planet between the Earth and the Sun, on the near side. A transit is an inferior conjunction that also crosses the Sun's disk.
- **Solar parallax.** The angle subtended by Earth's equatorial radius at a distance of 1 au. About 8.794 arcseconds.
- **Differential parallax.** Venus's parallax minus the Sun's, the shift of Venus relative to the solar disk. About 2.61 times the solar parallax in the circular model.
- **Projected baseline.** The station-to-station separation as it appears from the Sun, in the direction that changes the chord or the contact time. Not the latitude printed on a map.
- **Internal contacts.** Contact 2, Venus fully onto the disk, and contact 3, about to leave. Halley's duration runs from 2 to 3.
- **Black drop.** The dark neck that makes internal contact uncertain. Mostly optical. Not the same thing as Venus's bright atmospheric rim.

- **Parallax angle p.** The angle at a star subtended by 1 au. The full side-to-side shift is 2p. Distance in parsecs is 1/p if p is in arcseconds.
- **Parsec.** The distance at which 1 au subtends 1 arcsecond. About 3.26 light-years.
- **Heliometer.** A telescope with a split objective, used to measure small angles. Bessel's instrument.
- **Apparent magnitude m.** How bright a star looks.
- **Absolute magnitude M.** How bright it would look from 10 parsecs. A measure of luminosity.
- **Period-luminosity relation.** Longer period, higher luminosity, for a given family of Cepheids. Leavitt found the slope. The zero point is the pin that turns the slope into a distance.
- **Zero point.** Where the line sits vertically: the actual luminosity at a chosen period. Sliding it changes every distance and leaves the slope alone.
- **Standard candle.** An object whose luminosity you know, so apparent brightness gives distance by the inverse-square law.

Sources

These are the works the numbers and the names come from. Expedition lists disagree about assistants; where they disagree, the lesson says so.

- Kepler, *Harmonices Mundi* (Linz, 1619), Book 5. The sesquialterate proportion of the periodic times to the mean distances.
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- Halley, "Methodus singularis," *Philosophical Transactions* 29 (1716), no. 348, p. 454. English translation in the 1809 abridgement, reprinted by Fred Espenak at NASA Goddard. The 12.5 arcsecond trial, the 43 arcsecond Venus parallax, the half-minute differential, and the 17-minute example are all his hypothesis.
- Suzanne Debarbat on Delisle's one-contact method, in the *Journal of Astronomical Data* 19 (2013), and the account of Halley's durations against Delisle's longitudes in the transit-campaign literature (the 1760 *mappemonde* and the later syntheses).
- Cook and Green, "Observations made, by appointment of the Royal Society, at King George's Island in the South Sea," *Philosophical Transactions* 61 (1771), 397-421. Cook's journal for 3 June 1769, in J. C. Beaglehole (ed.), *The Journals of Captain James Cook*, vol. 1 (1968), for the "dusky shade" and the disagreement on contacts.
- IMCCE, chronological table of the Venus transit expeditions (the 2004 transit-history project), for the station list: Tobolsk, the Cape, Tahiti, Yakutsk, Orsk, Orenburg, Guriev, Kola, Ponoi, Umba, St Petersburg, Vardo, Hudson Bay, San Jose del Cabo.
- Library of Siberian Local History, entries on the Yakutsk astronomical expedition of 1768-69 and on Isleniev: Rumovsky's direction, Yakutsk as the Siberian station, and F. Chernyi named beside him.
- Chappe d'Auteroche, observations at Tobolsk, 6 June 1761, in the *Histoire de l'Academie royale des sciences* for 1761, and *Voyage en Siberie* (1768). His 1769 Baja California expedition and death on 1 August 1769 are in the standard biographies and in the French expedition accounts.

- Encke's 1824 solar parallax 8.5776 arcseconds, and the 1835 revision near 8.571, as summarized by Andreas Verdun, *Archives des Sciences* 57 (2004), and by the historical survey arXiv:1402.1945. Newcomb's later value near 8.79-8.80 arcseconds is from his re-reductions for the American Ephemeris papers.
- IAU Resolution B2 (2012), Beijing: the astronomical unit is exactly 149,597,870,700 metres.
- Bessel, "Bestimmung der Entfernung des 61sten Sterns des Schwans," *Astronomische Nachrichten* 16 (1838); English in *Monthly Notices of the Royal Astronomical Society* 4 (1838), 152-161. The 0.3136 and 0.0202 arcseconds, and the 10.3 years, are his. The heliometer is described in the AAS "This Month in Astronomical History," November 2021.
- Leavitt, *Annals of the Harvard College Observatory* 60 (1908), and Harvard College Observatory Circular 173 (3 March 1912). Quotations from the circular as transcribed by the AIP Center for History of Physics. The 25-star figure is on the Harvard Computers page.
- Walter Baade's 1952 separation of the two Cepheid populations, as told by Allan Sandage, "The First 50 Years at Palomar," and the standard factor: 1.5 magnitudes, about twice the distance. Hubble's Andromeda Cepheid is the 1923-25 result.
- Cruz Reyes and Anderson, "A 0.9% calibration of the Galactic Cepheid luminosity scale based on Gaia DR3," *Astronomy & Astrophysics* (2023), as one modern geometric pin, not as the last word.

Key facts

- Kepler, 1619: a^3 / T^2 is shared. a is the semimajor axis, the average of nearest and farthest, not the distance today.
- Venus: period 224.701 days = 0.615 year, so a = 0.723 AU. Circular coplanar gap at inferior conjunction: 1 - 0.723 = 0.277 AU. Real conjunctions are near that, not identical to it.
- Greatest elongation near 46 degrees already gave sin 46 = 0.72 by geometry. The third law extends one scale to every planet.
- The transit measures a differential parallax, about 2.61 times the solar parallax in the simple model, using a projected baseline. Halley compared durations of internal contacts. Delisle compared one contact in absolute time and needed longitude. Not one angle, and not latitude alone.
- Halley's 12.5 arcseconds, 43 arcseconds, and 17 minutes are his 1716 trial, not results. About 23 arcseconds is the illustrative shift for a side-on baseline of one Earth radius and the modern 8.79 arcsecond solar parallax. A baseline can be longer, up to about a diameter. It is not a logbook entry and not a ceiling.
- 1761: Chappe at Tobolsk, Siberia, 6 June. Mason and Dixon at the Cape. 1769: Cook, Green, and Solander at Point Venus, Tahiti, 3 June; Isleniev at Yakutsk, the deep-Siberian station, in Rumovsky's wider Russian network. Hell was at Vardo, not Siberia. Chappe's 1769 station was Baja California.
- The black drop spoiled contact times. It is mostly telescope blur and the Sun's dim limb. Venus's atmosphere is a separate bright rim. Mercury shows a black drop too.
- Encke 1824: 8.5776 arcseconds, about 2.5 percent too small, so the Sun too far. No one transit delivered the modern au.
- Since IAU 2012 Resolution B2, 1 au = 149,597,870,700 m exactly, about 92.96 million miles. Earth's actual distance still varies from about 0.983 to 1.017 au.

- Stellar parallax: full opposite-orbit shift is 2p; the defined p is the angle of 1 AU; parsecs = 1/p in arcseconds. January and July are illustrative. Bessel, 1838, 61 Cygni, heliometer, 0.3136 arcsecond. Photography of stars begins later (Vega, 1850).
- Leavitt 1908 and Circular 173 (1912) fixed the slope from SMC stars at one distance. She asked for parallaxes. She did not set the zero point. Hertzsprung and Shapley tried; Baade in 1952 separated two Cepheid families and roughly doubled the extragalactic distances. Gaia-era parallaxes pin the luminosity now. Cepheids reach across the Milky Way and into other galaxies.

Answers

Arithmetic. (1) 0.723 AU. (2) 0.277 AU. (3) The factor is 2.61. 2.61 times 8.79 is about 23 arcseconds for a side-on baseline of one Earth radius. That is an illustration, not a measurement, and not a ceiling: stations can be up to about a diameter apart. (4) 4 parsecs, about 13 light-years. (5) The calibrated period-luminosity relation, so you know which period is the more luminous. Equal apparent brightness plus a known luminosity ratio gives the distance ratio. The period ratio alone does not.

Review. (1) Relative distances in AU. Not miles, and not the distance on a given day. (2) It is half the long axis, the average of closest and farthest. A point on the ellipse lies at some other distance from the focus. (3) From 1 minus 0.723 in a circular, flat model. Real orbits are ellipses, tilted by about 3.4 degrees, and a conjunction is a shared longitude, so the gap that week is computed from the elements. (4) Solar parallax: Earth's radius at 1 au, about 8.794 arcseconds. Venus's parallax: the same baseline at Venus's distance, about 3.61 times larger in the model. Differential: the difference, about 2.61 times the solar parallax. The differential is what shifts Venus on the disk. (5) The duration from internal contact 2 to internal contact 3. The difference of durations is a difference of chords, which is a differential parallax across a known projected baseline. (6) One contact has to be timed on a shared absolute clock, so the longitudes must be known. A duration subtracts two times on one local clock, so an overall clock error mostly cancels. The places still have to be known. (7) Tahiti: Cook, Green, and Solander, 3 June 1769. Deep Siberia in 1769: Isleniev at Yakutsk. Chappe was at Tobolsk in 1761 and at San Jose del Cabo in 1769. (8) 8.5776 arcseconds. It is about 2.5 percent smaller than the modern solar parallax, and the modern mile-length is a 2012 definition, 149,597,870,700 metres, after radar. The expeditions got the scale to a few percent and did not finish it. (9) p is the angle subtended by 1 AU. The full shift from one side of the orbit to the other is 2p. January and July fit the star on the diagram. Another ecliptic longitude wants another pair, still about six months apart. (10) A Fraunhofer heliometer at Konigsberg. The first stellar photograph, Vega, is 1850. Plates are a later way to explain the same angle. (11) A common distance turns apparent-brightness differences into luminosity differences, so period can be matched to luminosity without already knowing the miles. She still asked that parallaxes of brighter variables of the type be measured. That request is the missing zero point. (12) 1.5 magnitudes is a factor of 10 to the power 0.3, about 2, in distance. The galaxies were about twice as far as the old pin had said.

Faria learning log, faria.jbf.com. Lesson dated 6 November 2026. Diagrams drawn for this lesson. The 23 arcseconds is the one-radius illustration, not a ceiling and not a 1769 measurement.

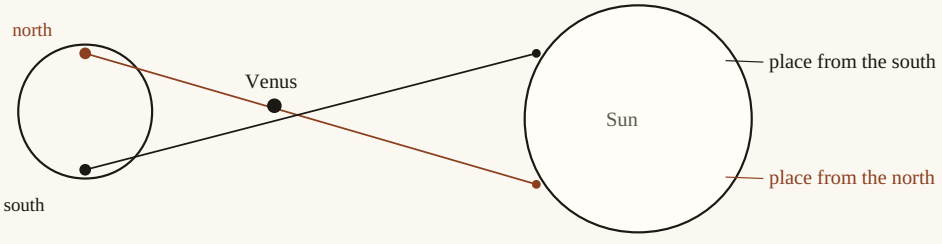
FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

The Venus transits: a worldwide ruler

Kepler's clock gives the planets in units of Earth's orbit. A transit of Venus can turn that ratio into a length, if two stations far apart time the same event. What they compare is a differential parallax: Venus, being nearer, shifts on the Sun more than the Sun itself shifts. The full story of the ladder, including why 23 arcseconds is a one-radius illustration and not a ceiling, is the Venus-to-Cepheid lesson. This page is the human campaign.

What the stations compare

Schematic. Angles exaggerated. Not a traced 1769 track.



Differential parallax is about (1/0.277 - 1) times the solar parallax, about 2.61 times it.

One Earth radius, side-on, is about 23 arcseconds if the solar parallax is the modern 8.79.

That length is an illustration, not a ceiling. Stations can stand up to about a diameter apart.

Halley compared full internal-contact durations. One contact, timed in absolute time, needs longitude.

The black drop is mostly blur and the Sun's dim limb, not Venus's air by itself.

Two stations and the differential shift of Venus on the Sun

Teaching geometry, not a 1769 tracing. Halley compared durations of the internal contacts. A single contact timed on an absolute clock needs the longitude. The black drop is mostly blur and limb darkening.

Jeremiah Horrocks and William Crabtree, observing separately, saw the transit of 1639. Horrocks argued from Venus's small apparent disk, and from a model of sizes, that the solar parallax was not above 14 arcseconds, and the Sun at least about 15,000 Earth radii. That is not a two-station parallax. The book came out after his death, in 1662. Converted with modern constants, 14 arcseconds is about 58.4 million miles and 15,000 Earth radii about 59.4 million miles. Those mile figures are our conversion, not his words.

In 1761 Mason and Dixon were sent toward Bencoolen. Their ship, the Seahorse, was attacked in January. Mason's letter counts eleven killed and thirty-seven wounded. After repairs they observed from the Cape, and only the end of the transit, after sunrise on 6 June. Cloud early, then a clearing for the contacts. Do not promote that into a cloudless day. James Short compared the Cape with fifteen European stations. He was not their exclusive partner. His 8.47 and 8.52 arcseconds are for the day of the transit, not the mean distance. The mean-distance figure near 8.65 converts to about

94.5 million miles, roughly 1.7 percent high. A later Short value, 8.69, converts to about 94.1 million miles. Again, the miles are a modern conversion.

There was no one official answer. Pingre's 1765 reduction from Rodrigues, 10.25 arcseconds, converts to about 79.8 million miles, about 14 percent low. Hornsby, using 1769 and publishing in 1771-72, got a mean near 8.78 arcseconds, about 93.1 million miles, roughly 0.16 percent above today's 92.956 million miles. The astronomical unit is now defined as exactly 149,597,870,700 metres. Encke in 1824 combined both transits and got 8.5776 arcseconds, about 95.3 million miles, 2.5 percent high. That combination is not the 1769 result alone.

Some observers got little or nothing. Le Gentil, kept from Pondicherry by war in 1761, watched from a moving ship and the record was unusable. In 1769 a cloud hid Pondicherry again. Maskelyne at St Helena could not send a complete observation. Pingre at Rodrigues saw the Sun rise in thick cloud. At Greenwich, Bliss and Green missed the start for cloud.

In 1769 Cook and Green had a clear 3 June at Tahiti, after the quadrant was stolen on 2 May and recovered damaged but repairable. Green died of dysentery on the way home, on 29 January 1771. Wales and Dymond wintered at Hudson Bay and did observe. Their story is not a stopped clock. Mason's share of the Cape work is the next lesson. Dixon's sky in Norway, and whether anyone "won" against Chappe, is Dixon's lesson. This page will not rank them.

Chappe in Baja California, 1769

Pedro Espinoza Meléndez, writing in Meyibo, follows Chappe to San Jose del Cabo. The copy used here does not show the article's year or its pages. It says the party reached California on 19 May 1769, two weeks before the transit, after a month crossing the gulf. Chappe's diary, quoted there, has them camped on the shore and determined to reach San Jose the next day. Nineteen May is the landing the article dates. The walk into the town may be the day after. The article does not decide.

He set up in a granary, half the roof removed on the south side. Before 3 June he set the pendulum clock, checked the telescopes, and made repeated observations of the Sun. The transit was 3 June. On 5 June the Spaniards with the party fell ill, and so did the people of San Jose. Chappe nursed the sick and then became ill himself. The article does not date each of those days of care. On 18 June, fifteen days after the transit, he observed a lunar eclipse while very weak. Cassini, printing the register in 1772, says Chappe was then too weak to take the Moon at the meridian. That night was not the last line in the book. The same tables still enter later emergences of Jupiter's moons as Chappe's, including 20 June and 16 July. He did not leave Baja. He died on 1 August 1769. Counted on the calendar, that is fifty-nine days after the transit and seventy-four days after the 19 May landing. Those counts are arithmetic, not a sentence in either source. No date was found here for

when the survivors sailed. The Meyibo article says it is not clear whether the disease was typhus or yellow fever. This page says epidemic.

The article quotes Chappe's own instruments: a quadrant of three-foot radius by Canivet, a smaller English quadrant of one-and-a-half-foot radius, a transit instrument, a parallactic machine, Dollond telescopes of ten feet and three feet, and a pendulum clock by Berthoud. Dixon's weather, and the comparison of the two kits, is his lesson.

What the 18 June eclipse did not settle

The eclipse was a check on the longitude of San Jose. It did not become the longitude the book adopts, and it did not leave a measured improvement.

In *Voyage en Californie* (Paris, 1772) Cassini works the place in the astronomical observations, pages 88 to 92. Five emergences of Jupiter's moons, which he attributes to Chappe, average 7 hours 28 minutes 53 seconds west of Paris. The five include 20 June and 16 July, after the eclipse, so the average is not a baseline from before it. Cassini gives more weight to the transit. From the transit he has 7 hours 28 minutes 6 seconds, and he adopts 7 hours 28 minutes 10 seconds, which is 112 degrees 2 minutes 30 seconds west of Paris. Turning those hours into degrees is the time-to-longitude lesson. The worked example there uses Greenwich. This one uses Paris.

The eclipse, set only against a time calculated for Paris, gives 7 hours 29 minutes 7 seconds at the start and 7 hours 29 minutes 40 seconds at the end. Cassini says he knew of no observation of that eclipse at another known place. The two results stand 57 seconds and 90 seconds of time away from the figure he adopted. In angle, at fifteen seconds of arc to one second of time, that is 14 minutes 15 seconds and 22 minutes 30 seconds of longitude. The conversion is arithmetic done here. It is a disagreement. It is not a measured error, and it is not a measured improvement. No distance in miles is taken from it.

His remark that Jupiter's moons, without enough matched observations, should not be trusted closer than about 20 or 30 seconds of time is a caution about the method. It is not a personal error for Chappe. Nothing on these pages shows the eclipse tightening the longitude. The figure he asks the reader to keep is the one from the transit. A later ESO note says he stayed in order to fix the longitude. Staying made the check possible. It did not write the adopted value.

The illustrated lesson is download the PDF.

Sources

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- Nicholas Kollerstrom, on Horrocks, in the IAU transit volume: https://www.dioi.org/kn/IAUVenus-Transit.pdf
- James Short, Philosophical Transactions, 1761: https://royalsocietypublishing.org/rstl/article-pdf/doi/10.1098/rstl.1761.0102/1458661/rstl.1761.0102.pdf
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- Jean-Baptiste Chappe d'Auteroche, *Voyage en Californie*, Cassini's reductions, Paris, 1772, pp. 88-92: https://archive.org/download/voyageencaliforn00chap/voyageencaliforn00chap_djvu.txt
- ESO, on Chappe remaining for the 18 June 1769 lunar eclipse: https://www.eso.org/public/outreach/eduoff/vt-2004/Background/Infol2/EIS-F3_pf.html

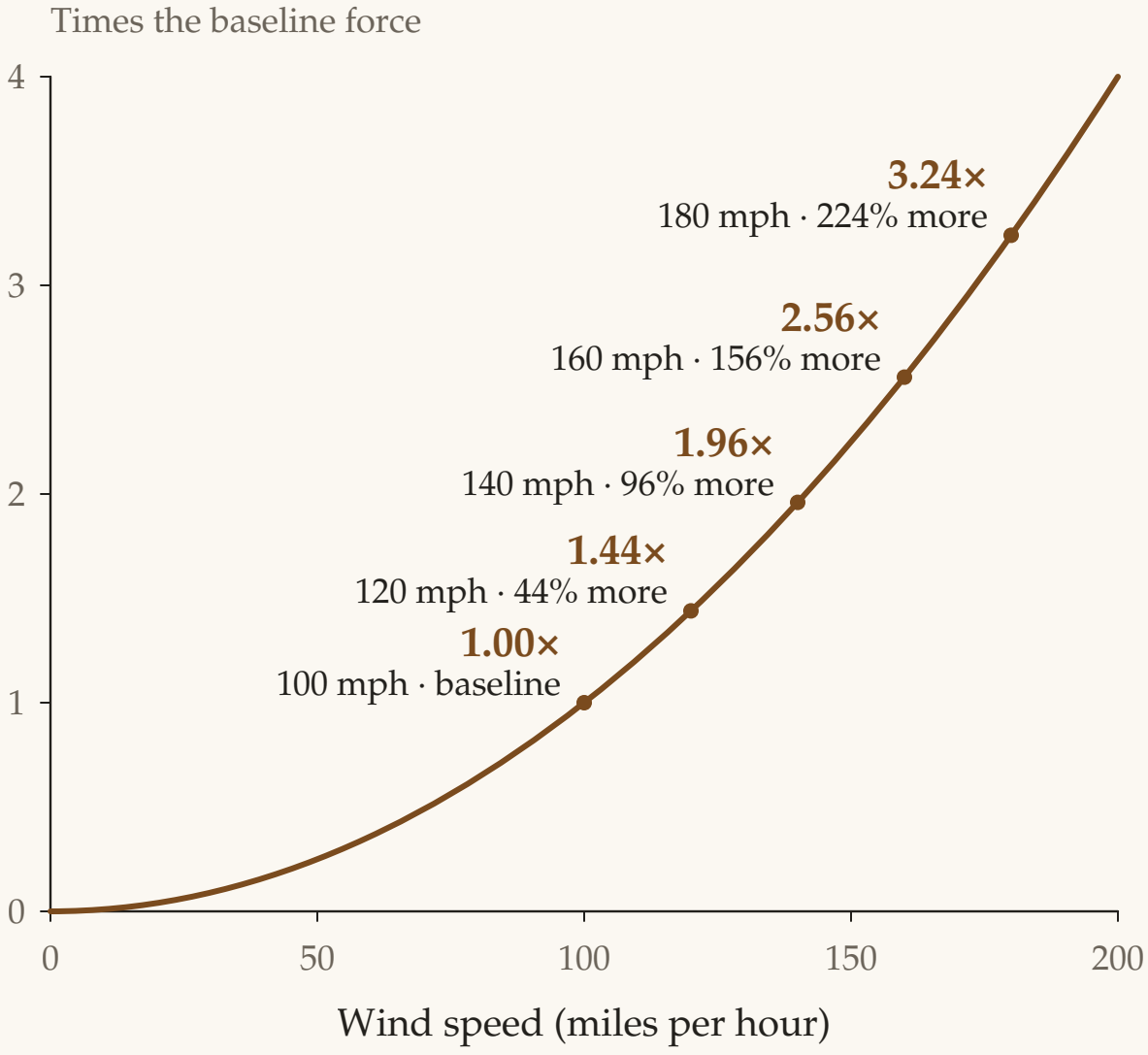
Key facts

- Horrocks in 1639 set an upper bound near 14 arcseconds from Venus's apparent size, not from a second station. Publication was posthumous, 1662.
- Mason and Dixon reached the Cape after the Seahorse fight (11 dead, 37 wounded) and timed the end of the 6 June 1761 transit.
- Short, Hornsby, Pingre, and Encke published different parallaxes. Converted miles on this page are modern arithmetic, not their sentences.
- The modern au is exactly 149,597,870,700 metres, about 92.956 million miles. No 1761 or 1769 paper printed that definition.

- Chappe's party reached the Baja shore on 19 May 1769. The diary leaves open whether the town was the next day. Transit on 3 June, lunar eclipse on 18 June while he was very weak, death there on 1 August, of an epidemic. Later Jupiter-moon timings are still entered as his.
- Cassini, *Voyage en Californie* (1772), pp. 88-92, adopts 7 hours 28 minutes 10 seconds west of Paris from the transit. The eclipse was a check against a calculated Paris time. No measured improvement is given.

Hurricane Wind Force:
Force Grows with Velocity Squared

The whole force, relative to a 100 mph baseline



Divide the speed by 100 mph, then square that ratio.

Both axes are linear and begin at zero. No logarithmic scaling.

Same air density, exposed area and drag coefficient.

Faria • November 8, 2026

Appendix B. Earlier mathematics editions

Archival versions are preserved for completeness, not as recommended starting points. The current 22-page edition is in Section 1. These three earlier editions are unchanged source facsimiles.

3 complete papers • begins on binder page 405

A MASTER PAPER / BEGIN WHEREVER YOU FEEL COMFORTABLE

Mathematics, one idea at a time

Jeff, this is not a test. Think of it as a walk from a shelter room to a ladder, then out to the Moon and the moving planets. We will name each idea before asking it to do any work.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: **say what you know, say what you want, and connect the two.**

CONTENTS

1. Algebra 2

Solving an unknown • shelter rooms and a truck

2. Geometry 4

Euclid • right triangles • Pythagoras • equal sides

3. Trigonometry 7

One ladder • sine, cosine and tangent • Galileo

4. Calculus 13

Newton and Leibniz • change • accumulation • orbits

Sources & further reading 17

The history and science behind the stories

How to read: stop whenever an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation.

The contents and colored page references are clickable. Click any page footer to return here. Examples simplify real situations; they are not operating limits or engineering advice.

1.1 / ALGEBRA STARTS WITH AN EMPTY PLACE

How many people in the shelter?

Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An **unknown** is simply the answer we have not found yet. We can leave an empty box for it, or give it a letter as a label.

Here, let **C** stand for the total number of people. The letter is not a special object. It is just the name on the empty place we want to fill.

The sign **×** means “times,” or “multiplied by.” The sign **=** means “has the same value as.” It is not an instruction to guess.

THREE GROUPS OF FORTY PEOPLE

C = 3 × 40 = 120 people

Check the answer: 40 people in the first room, 40 in the second and 40 in the third make 120. In symbols, **+** means “add”: **40 + 40 + 40 = 120**.

We did not make the problem harder by writing C. We made the question visible: “What total belongs here?”

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements and local rules.

1.2 / WORK BACKWARD, THEN CHECK

How many pallets in the truck?

Now a truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let **P** be the number of pallets. Before we know P, we can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

NAME THE UNKNOWN

P × 500 pounds = 3,000 pounds

“How many groups of 500 fit into 3,000?” is a division question. The sign ÷ means “divided by.” Division undoes multiplication.

FILL THE EMPTY PLACE

P = 3,000 ÷ 500 = 6 pallets

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting our answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves **P = 6**.

Name the unknown, use the relationship, then check. We will do it again with the ladder in Section 3 (page 7).

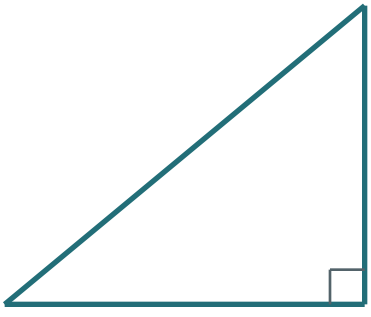
“lb” on the boxes means pounds. This is weight arithmetic only—not a loading plan. Real loads must meet vehicle, axle, space, balance and securement limits.

2.1 / GEOMETRY GIVES THE PROBLEM A SHAPE

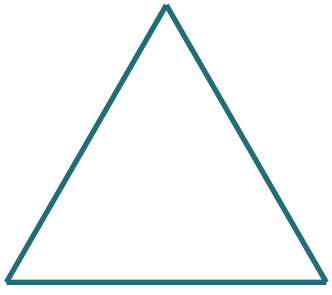
Euclid’s world of lines and corners

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE—about 2,300 years ago—he organized much earlier knowledge into a book called Elements.

Geometry studies shapes, sizes and positions. Euclid’s lasting example was to begin with clear assumptions and build an explanation, step by step. A **proof** is reasoning that shows why a claim must be true.



Right triangle



Equilateral triangle

An **angle** describes how widely two lines open apart. We measure angles in **degrees**, written with the little sign °. A full turn is 360 degrees; a square corner is 90 degrees.

A **right triangle** has one square corner. “Right” here means 90 degrees, not the right-hand side of the page. Its **hypotenuse** is the side across from that corner, and is always the longest side.

An **equilateral triangle** has three equal sides. On flat paper, each of its three angles is 60 degrees. It does **not** have a square corner.

These are different triangles. We can split the equal-sided one to make a right triangle; that connection will explain the ladder’s one-half relationship on the next diagram (page 6).

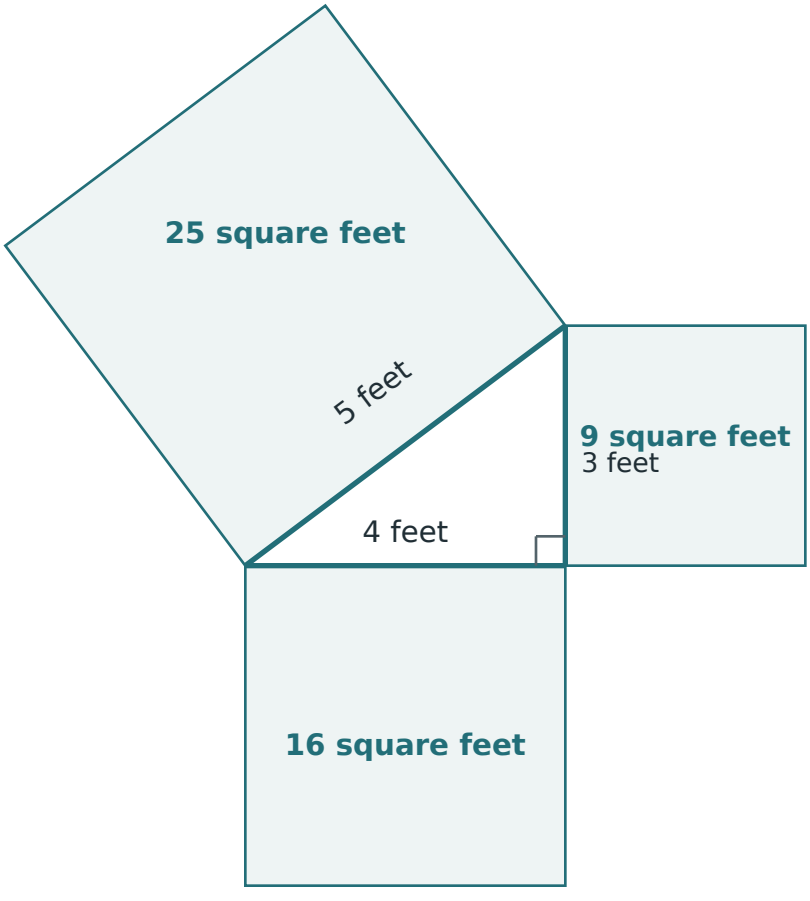
BCE means “before the common era.” Geometry is much older than any one mathematician. History sources are collected on page 17.

2.2 / PYTHAGORAS CONNECTS THREE LENGTHS

The 3–4–5 triangle

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A **theorem** is a statement established by proof.



Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A **square foot** is a square one foot on each side.

So our picture says: **3 × 3 + 4 × 4 = 5 × 5**, or **9 + 16 = 25**. If the long side were unknown, we would ask: "What length times itself gives 25?" The answer is 5.

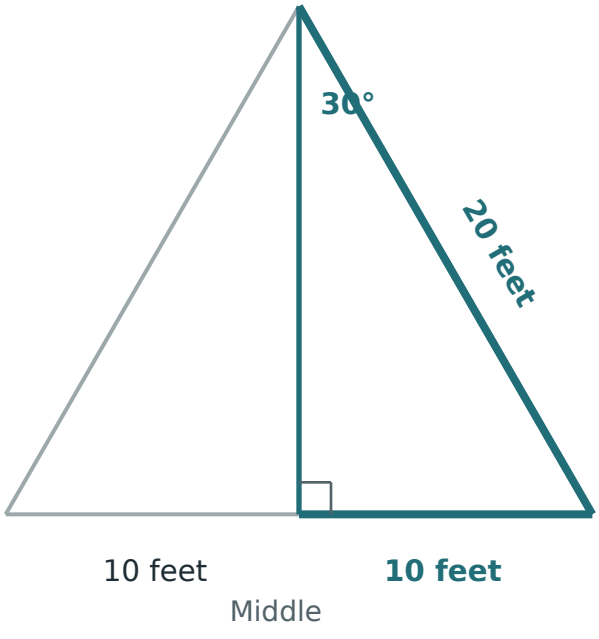
The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning (page 12).

2.3 / A BRIDGE FROM GEOMETRY TO TRIGONOMETRY

Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.



In either half, the side **opposite the 30-degree angle** is 10 feet long. The sloping side—the hypotenuse—is 20 feet long. So the opposite side is **half** the hypotenuse.

Turn that half-triangle around on the page: the lengths and angles do not change. It is the same shape as our 20-foot ladder reaching 10 feet up the wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not a lucky measurement or a number invented by a calculator.

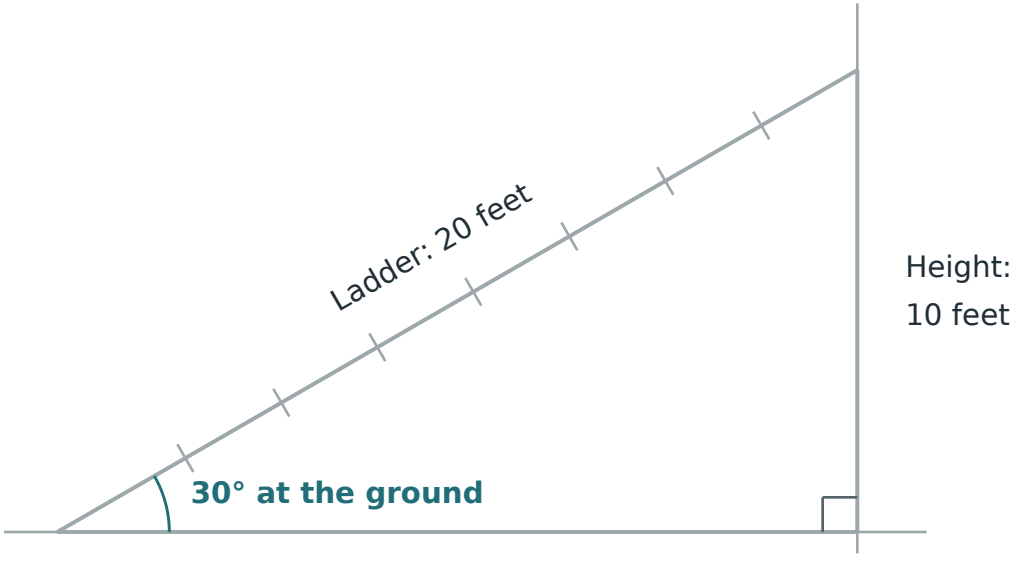
This ratio is called **sine** (Section 3, page 8). Geometry supplies the shape; trigonometry uses its proportions.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Return to the right triangle from Section 2 (page 4). A ladder leaning against a wall makes that shape. We will keep the **same ladder** throughout this section.

Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown (Section 1, page 2). You do not need calculus yet.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means "degrees." A square corner is 90 degrees.

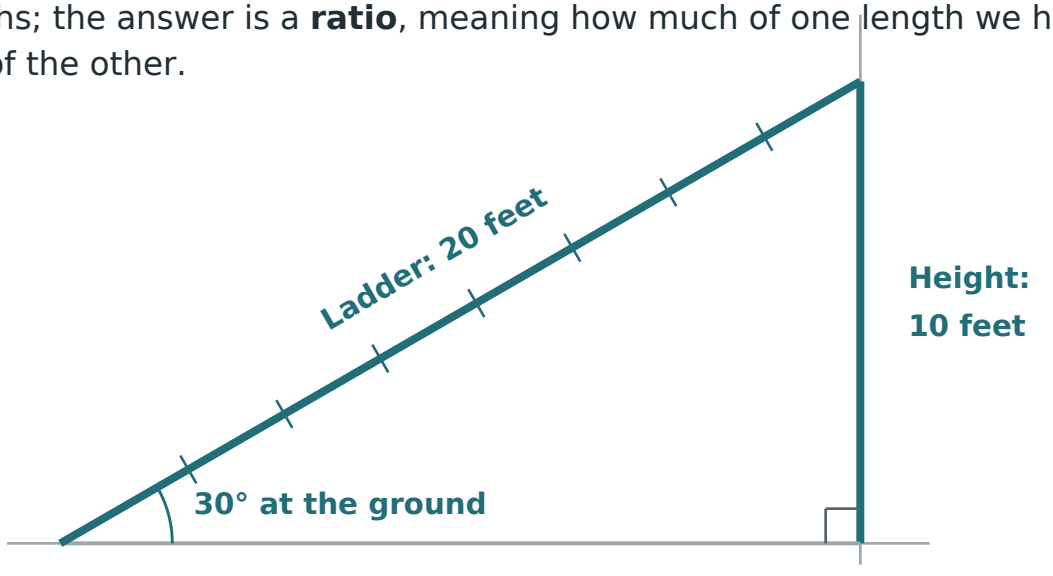
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. The equal-sided triangle on page 6 explains why; a calculator or table supplies the same ratio.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

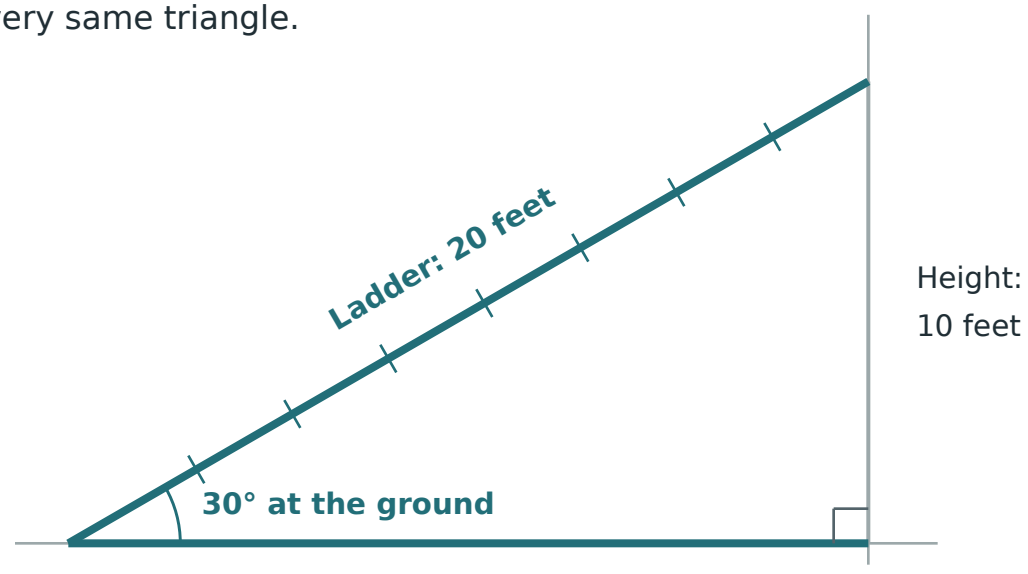
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



Base: about 17.32 feet

For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

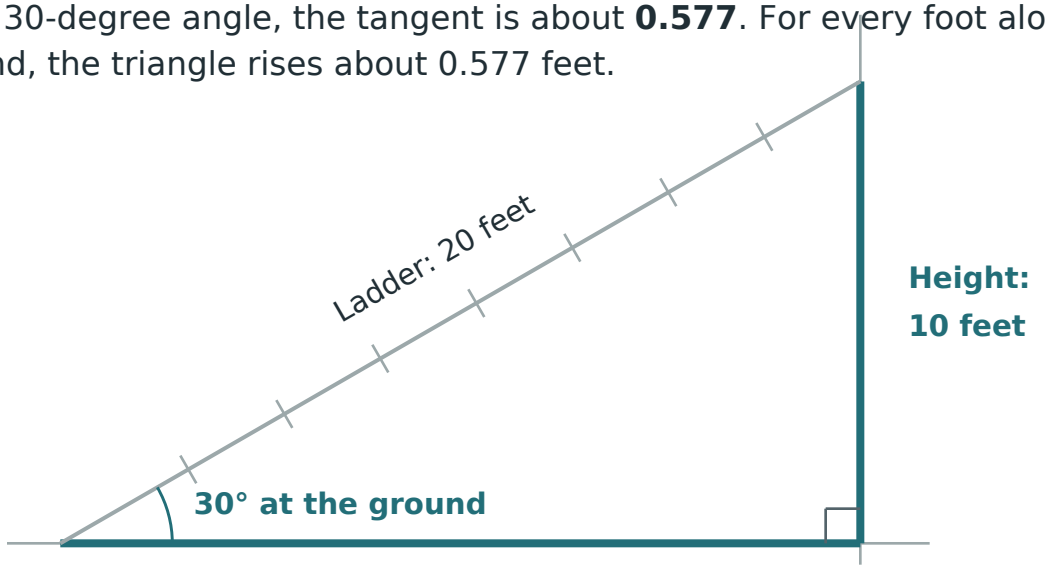
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

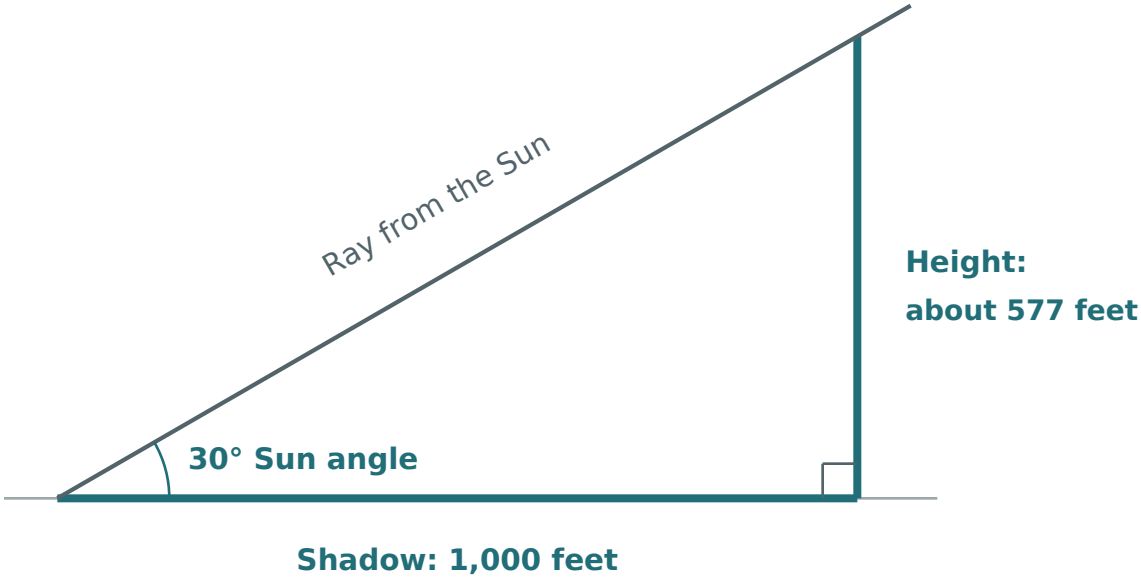


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule** (Section 2, page 5), the relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Where the story goes next

Geometry gives us a shape; trigonometry compares its sides. What if the shape or position changes from moment to moment? That is our next subject: Section 4, calculus (page 13).

Galileo's original text and the related shadow method are linked in the sources on page 17. The shadow example on page 11 is invented, not Galileo's data.

4.1 / NEWTON, LEIBNIZ AND THE MOVING WORLD

Now let something change

The shelter total, triangle and ladder all held still while we examined them. But a truck speeds up, a planet changes direction, and a tide rises. **Calculus** helps us describe continuous change and what builds up from it.

In the late 1600s, Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this new mathematics closely with motion. It was not one person's invention out of nothing.

How fast, right now?

A rate of change

How much altogether?

Accumulation

A **rate of change** compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If we know the truck's speed throughout a journey, we can add up the small distances it covers to find the whole trip.

These two questions are partners: from the journey's distance record, find the changing speed; from the changing speed, recover the distance traveled.

The earlier sections are foundations, not a claim that Pythagoras invented calculus. Geometry describes positions and lengths; algebra expresses relationships. Calculus asks how those relationships behave as things change.

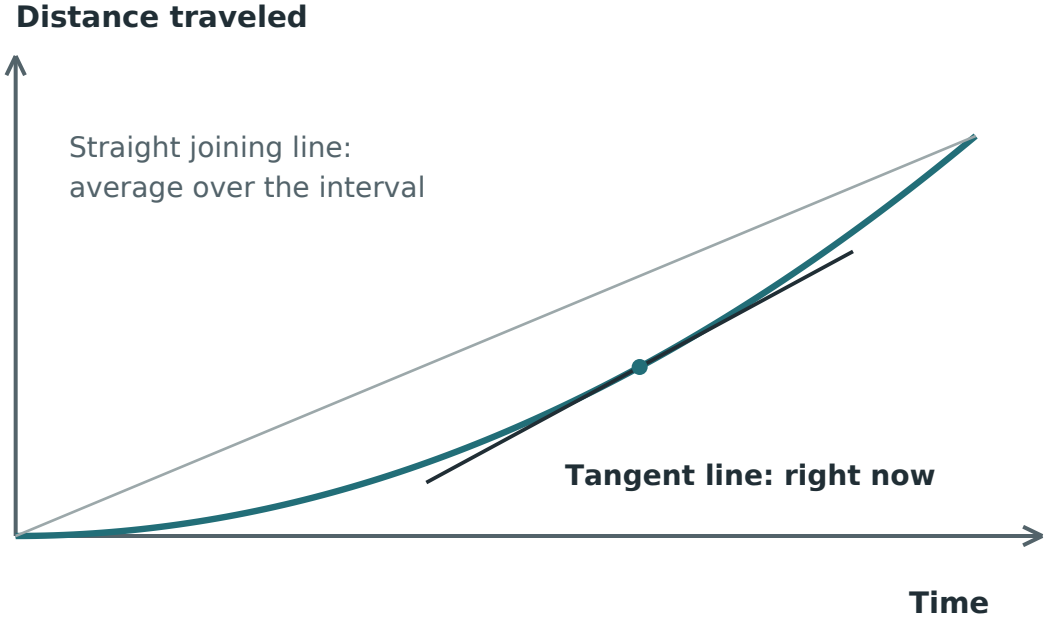
You can understand the central ideas without learning the formal symbols first. We will stay with the truck before returning to the sky.

4.2 / AVERAGE IS NOT THE SAME AS RIGHT NOW

A speedometer and a curve

Suppose the truck travels 60 miles in two hours. Its **average speed** is the total distance divided by the total time: **60 ÷ 2 = 30 miles per hour**. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment—the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.



This picture is a **graph**: time runs across; distance runs upward. The curve records a journey. Its **slope** means rise compared with horizontal run—here, distance change compared with time change.

A **tangent line** follows the curve’s direction at one point. Its slope gives the rate there. Finding that local rate is called **differentiation**; the rate we find is called a **derivative**.

This is not quite the same use of “tangent” as on page 10. There, **tan** is a triangle’s height-to-base ratio. Here, a **tangent line** is a line beside a curve. Their connection is slope: rise divided by run.

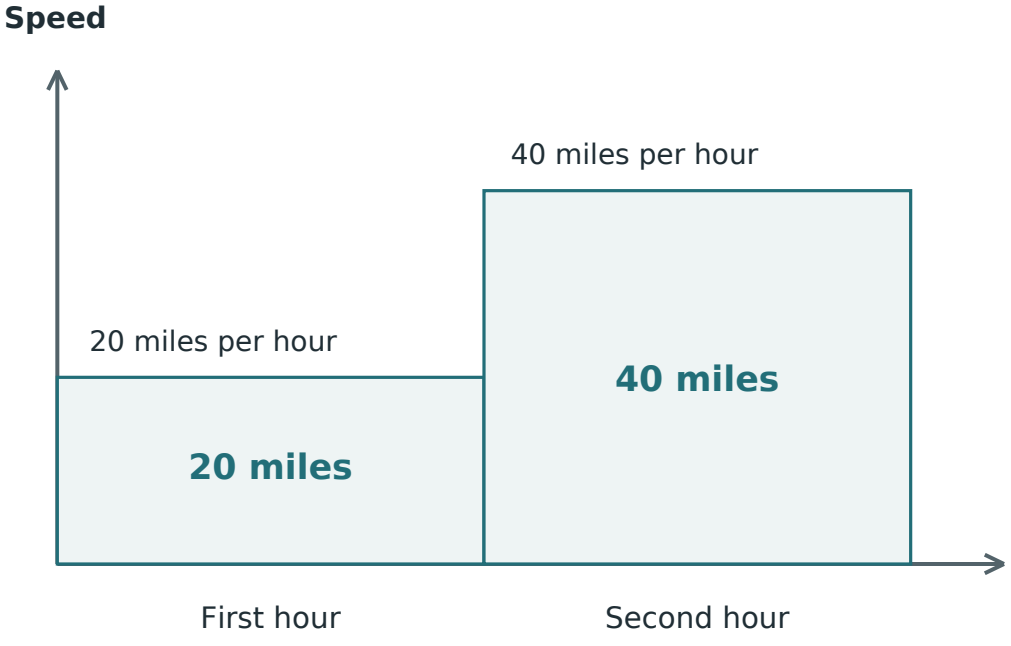
The graph is conceptual, not the numerical record of the 60-mile trip above. No formula for the curve is needed to read the idea.

4.3 / ADD THE LITTLE PIECES

From speed back to distance

Now turn the question around. Suppose we know how fast the truck moved. If it held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

So the whole distance is **20 + 40 = 60 miles**. We found a total by adding the contributions of different parts of the journey.



In a speed-versus-time graph, a rectangle’s height is the speed and its width is the time. Multiplying them gives the distance during that time. The **area under the speed graph** therefore represents distance.

When speed changes smoothly, divide the journey into short time pieces. Use each piece’s speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called **integration**; the accumulated result is an **integral**.

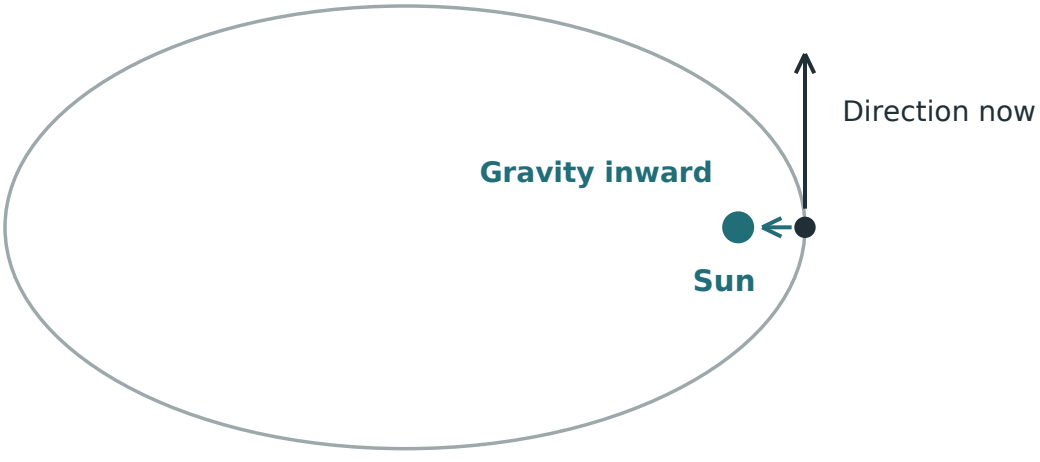
Differentiation asks for the local rate; integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 / FROM MOVING TRUCKS TO MOVING PLANETS

Orbits, tides and changing direction

For Newton, motion raised a deeper question: how does a pull change an object's journey? **Gravity** is the attraction between masses—between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. **Acceleration** means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity and acceleration from moment to moment.



Schematic ellipse; sizes and distances are not to scale.

Forward motion and the Sun's inward pull make the planet's path curve. Without gravity it would continue straight in its current direction. Orbits need not be circles; many are **ellipses**—oval-shaped paths.

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Newton's gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape and other effects. Calculus describes change; it does not make geography disappear.

Geometry describes shapes; calculus describes how they change.

READ ONLY IF YOU WANT TO FOLLOW THE THREADS

Sources, and what to remember

The historical stories are real; the shelter, truck and shadow numbers are teaching examples. The master paper reworks the standalone ladder lesson into Section 3. Original PDFs are left unchanged.

Geometry and its history
University of St Andrews, MacTutor: Euclid and Pythagoras. Euclid organized earlier knowledge; the right-triangle relationship predates Pythagoras.

[1] Galileo's actual mountain calculation
Galileo, Sidereus Nuncius (1610), E. S. Carlos translation (1880), pp. 28–30: Project Gutenberg ebook 46036. Sunlit peaks beyond the terminator; right-triangle calculation; "more than 4 Italian miles."

[2] The related shadow method
Mary McIntyre, BBC Sky at Night Magazine: "How to measure craters on the Moon". Height equals shadow length times the tangent of the local Sun angle, with simplifying assumptions.

Calculus and its development
University of St Andrews, MacTutor: "A history of the calculus". Earlier work, Newton, Leibniz, rates and accumulation. Orbits: NASA, "Gravity & Mechanics".

Why tides happen
NOAA: "Tides" and "What Causes Tides?". Uneven gravitational pull, the Moon and Sun, and the influence of local geography.

Four questions, one growing toolkit
Algebra: What number is missing?
Geometry: What shape and lengths do we have?
Trigonometry: What do the angles tell us about the sides?
Calculus: How is it changing, and what does it add up to?

You do not have to memorize every word. Recognizing the question is already a beginning.

A MASTER PAPER / BEGIN WHEREVER YOU FEEL COMFORTABLE

Mathematics, one idea at a time

Jeff, this is not a test. Think of it as a walk from a shelter room to a ladder, then out to the Moon and the moving planets. We will name each idea before asking it to do any work.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: **say what you know, say what you want, and connect the two.**

CONTENTS

1. Algebra 2

Solving an unknown • shelter rooms and a truck

2. Geometry 4

Euclid • right triangles • Pythagoras • equal sides

3. Trigonometry 7

One ladder • sine, cosine and tangent • Galileo

4. Calculus 13

Newton and Leibniz • change • accumulation • orbits

4.5 Differential equations — coffee cooling 17

Sources & further reading 18

The history and science behind the stories

How to read: stop whenever an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation.

The contents and colored page references are clickable. Click any page footer to return here. Examples simplify real situations; they are not operating limits or engineering advice.

1.1 / ALGEBRA STARTS WITH AN EMPTY PLACE

How many people in the shelter?

Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An **unknown** is simply the answer we have not found yet. We can leave an empty box for it, or give it a letter as a label.

Here, let **C** stand for the total number of people. The letter is not a special object. It is just the name on the empty place we want to fill.

The sign **×** means “times,” or “multiplied by.” The sign **=** means “has the same value as.” It is not an instruction to guess.

THREE GROUPS OF FORTY PEOPLE

C = 3 × 40 = 120 people

Check the answer: 40 people in the first room, 40 in the second and 40 in the third make 120. In symbols, **+** means “add”: **40 + 40 + 40 = 120**.

We did not make the problem harder by writing C. We made the question visible: “What total belongs here?”

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements and local rules.

1.2 / WORK BACKWARD, THEN CHECK

How many pallets in the truck?

Now a truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let **P** be the number of pallets. Before we know P, we can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

NAME THE UNKNOWN

P × 500 pounds = 3,000 pounds

“How many groups of 500 fit into 3,000?” is a division question. The sign ÷ means “divided by.” Division undoes multiplication.

FILL THE EMPTY PLACE

P = 3,000 ÷ 500 = 6 pallets

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting our answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves **P = 6**.

Name the unknown, use the relationship, then check. We will do it again with the ladder in Section 3 (page 7).

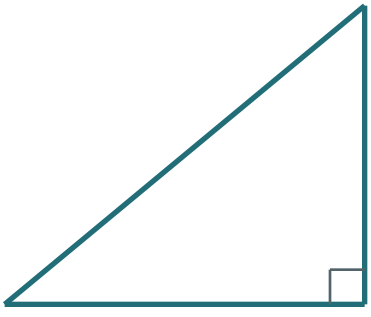
“lb” on the boxes means pounds. This is weight arithmetic only—not a loading plan. Real loads must meet vehicle, axle, space, balance and securement limits.

2.1 / GEOMETRY GIVES THE PROBLEM A SHAPE

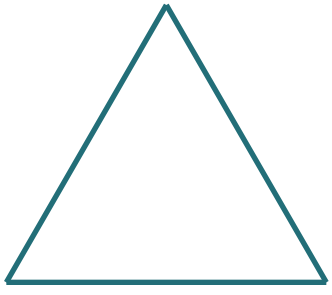
Euclid’s world of lines and corners

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE—about 2,300 years ago—he organized much earlier knowledge into a book called Elements.

Geometry studies shapes, sizes and positions. Euclid’s lasting example was to begin with clear assumptions and build an explanation, step by step. A **proof** is reasoning that shows why a claim must be true.



Right triangle



Equilateral triangle

An **angle** describes how widely two lines open apart. We measure angles in **degrees**, written with the little sign °. A full turn is 360 degrees; a square corner is 90 degrees.

A **right triangle** has one square corner. “Right” here means 90 degrees, not the right-hand side of the page. Its **hypotenuse** is the side across from that corner, and is always the longest side.

An **equilateral triangle** has three equal sides. On flat paper, each of its three angles is 60 degrees. It does **not** have a square corner.

These are different triangles. We can split the equal-sided one to make a right triangle; that connection will explain the ladder’s one-half relationship on the next diagram (page 6).

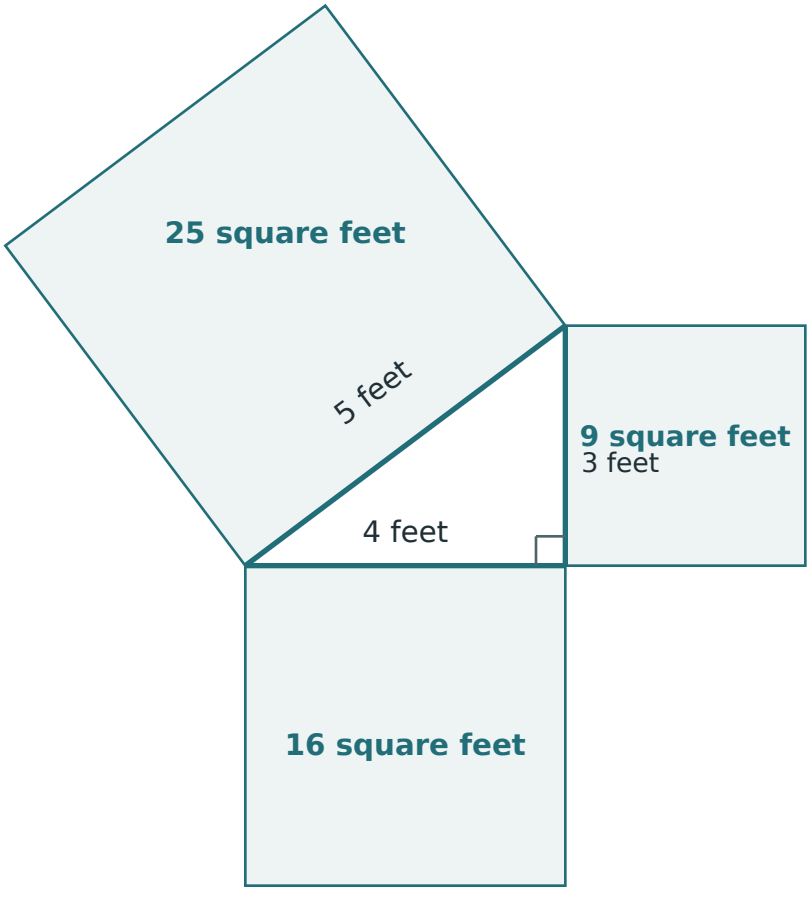
BCE means “before the common era.” Geometry is much older than any one mathematician. History sources are collected on page 18.

2.2 / PYTHAGORAS CONNECTS THREE LENGTHS

The 3–4–5 triangle

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A **theorem** is a statement established by proof.



Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A **square foot** is a square one foot on each side.

So our picture says: **3 × 3 + 4 × 4 = 5 × 5**, or **9 + 16 = 25**. If the long side were unknown, we would ask: "What length times itself gives 25?" The answer is 5.

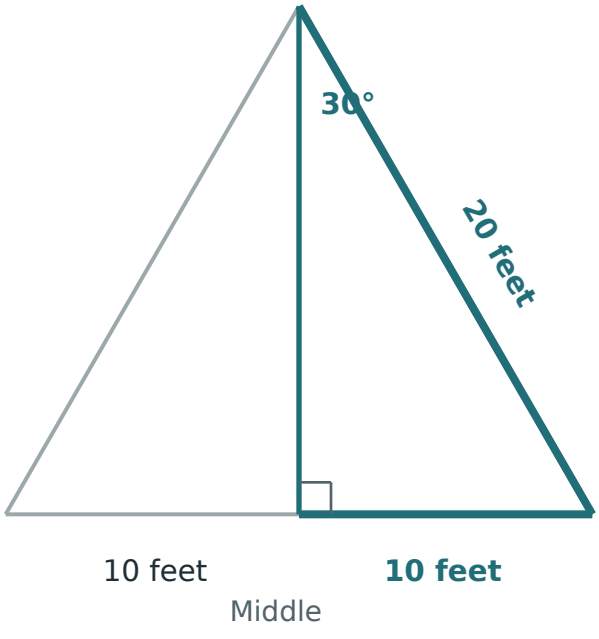
The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning (page 12).

2.3 / A BRIDGE FROM GEOMETRY TO TRIGONOMETRY

Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.



In either half, the side **opposite the 30-degree angle** is 10 feet long. The sloping side—the hypotenuse—is 20 feet long. So the opposite side is **half** the hypotenuse.

Turn that half-triangle around on the page: the lengths and angles do not change. It is the same shape as our 20-foot ladder reaching 10 feet up the wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not a lucky measurement or a number invented by a calculator.

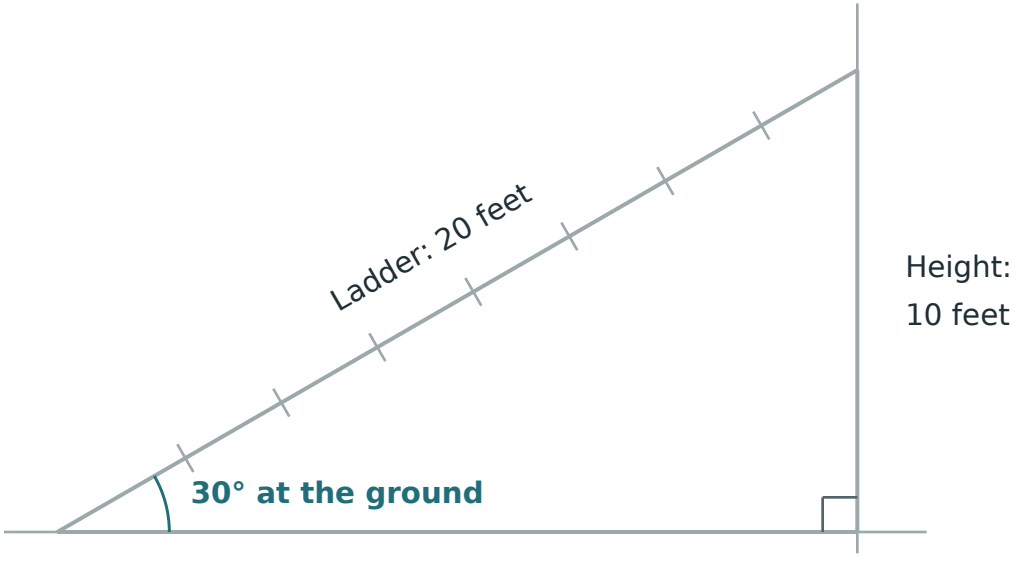
This ratio is called **sine** (Section 3, page 8). Geometry supplies the shape; trigonometry uses its proportions.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Return to the right triangle from Section 2 (page 4). A ladder leaning against a wall makes that shape. We will keep the **same ladder** throughout this section.

Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown (Section 1, page 2). You do not need calculus yet.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means "degrees." A square corner is 90 degrees.

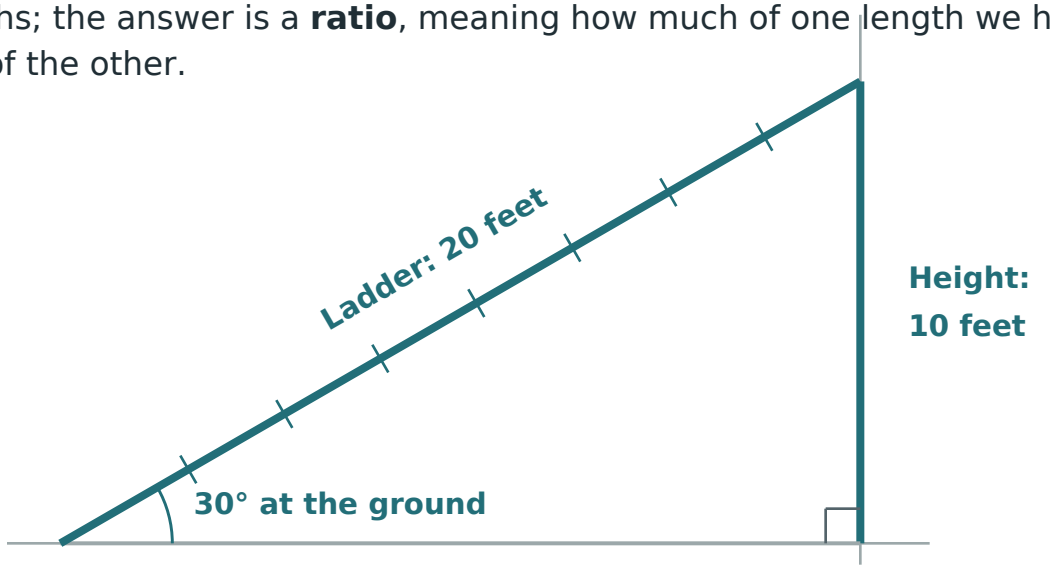
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. The equal-sided triangle on page 6 explains why; a calculator or table supplies the same ratio.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

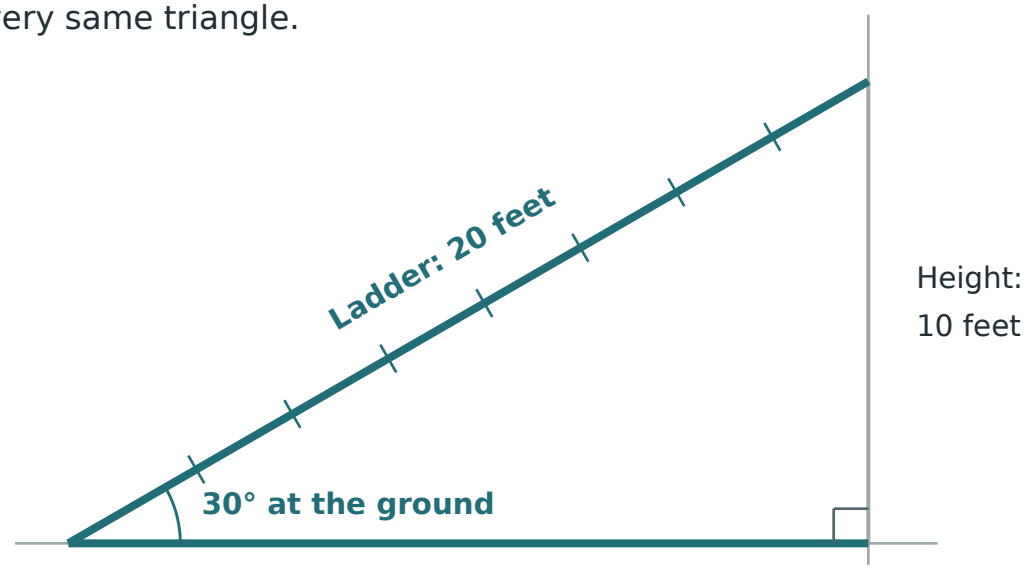
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



Base: about 17.32 feet

For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

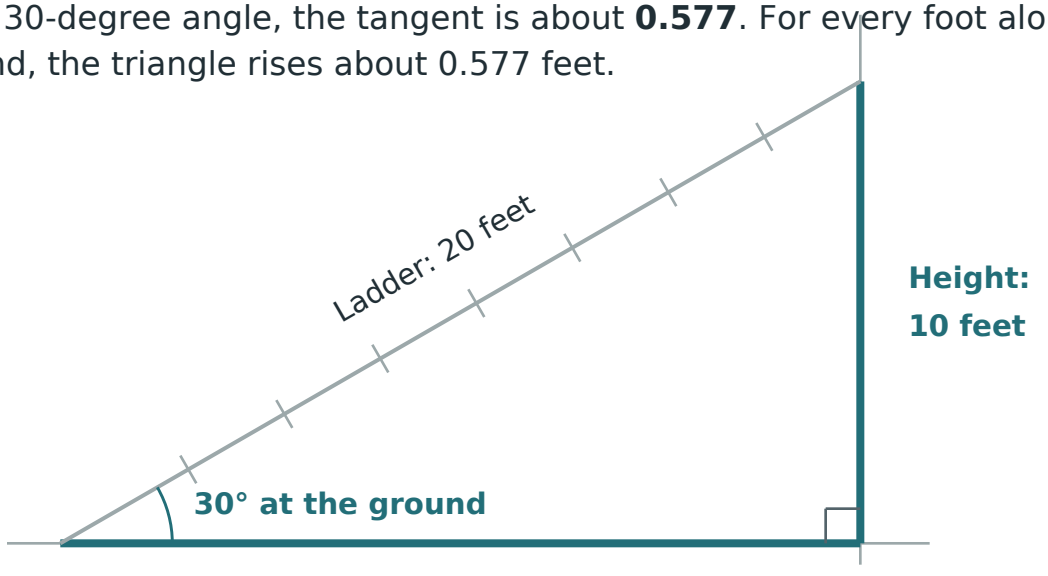
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

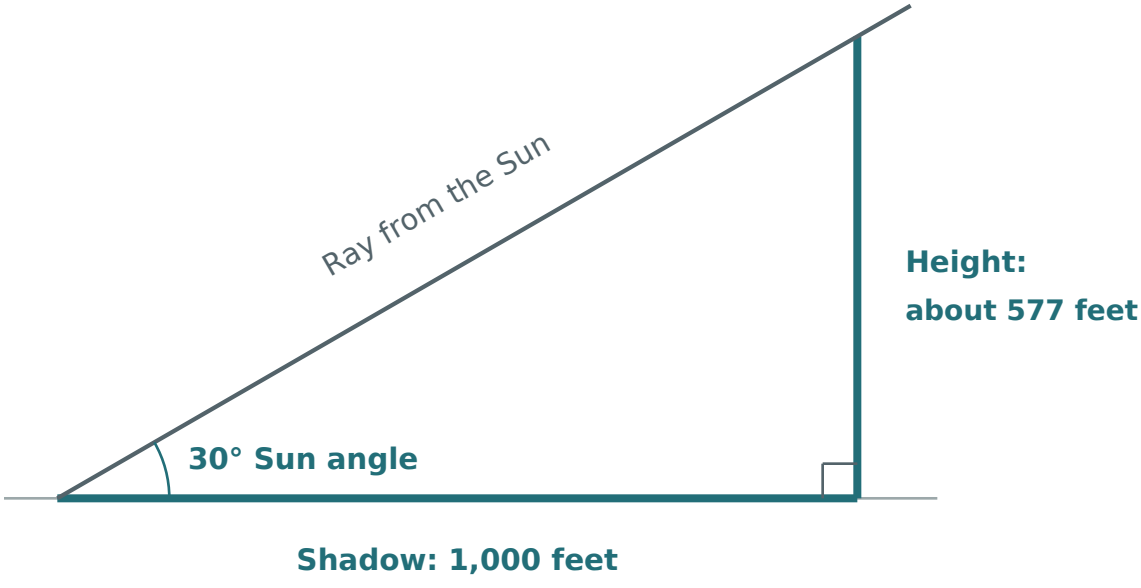


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule** (Section 2, page 5), the relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Where the story goes next

Geometry gives us a shape; trigonometry compares its sides. What if the shape or position changes from moment to moment? That is our next subject: Section 4, calculus (page 13).

Galileo's original text and the related shadow method are linked in the sources on page 18. The shadow example on page 11 is invented, not Galileo's data.

4.1 / NEWTON, LEIBNIZ AND THE MOVING WORLD

Now let something change

The shelter total, triangle and ladder all held still while we examined them. But a truck speeds up, a planet changes direction, and a tide rises. **Calculus** helps us describe continuous change and what builds up from it.

In the late 1600s, Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this new mathematics closely with motion. It was not one person's invention out of nothing.

How fast, right now?

A rate of change

How much altogether?

Accumulation

A **rate of change** compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If we know the truck's speed throughout a journey, we can add up the small distances it covers to find the whole trip.

These two questions are partners: from the journey's distance record, find the changing speed; from the changing speed, recover the distance traveled.

The earlier sections are foundations, not a claim that Pythagoras invented calculus. Geometry describes positions and lengths; algebra expresses relationships. Calculus asks how those relationships behave as things change.

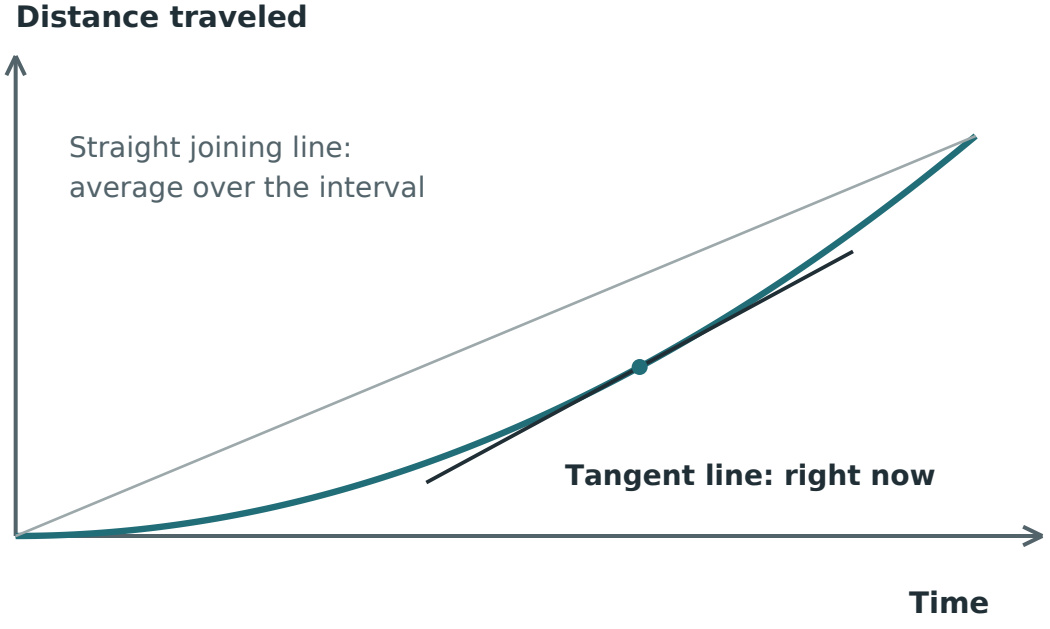
You can understand the central ideas without learning the formal symbols first. We will stay with the truck before returning to the sky.

4.2 / AVERAGE IS NOT THE SAME AS RIGHT NOW

A speedometer and a curve

Suppose the truck travels 60 miles in two hours. Its **average speed** is the total distance divided by the total time: **60 ÷ 2 = 30 miles per hour**. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment—the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.



This picture is a **graph**: time runs across; distance runs upward. The curve records a journey. Its **slope** means rise compared with horizontal run—here, distance change compared with time change.

A **tangent line** follows the curve’s direction at one point. Its slope gives the rate there. Finding that local rate is called **differentiation**; the rate we find is called a **derivative**.

This is not quite the same use of “tangent” as on page 10. There, **tan** is a triangle’s height-to-base ratio. Here, a **tangent line** is a line beside a curve. Their connection is slope: rise divided by run.

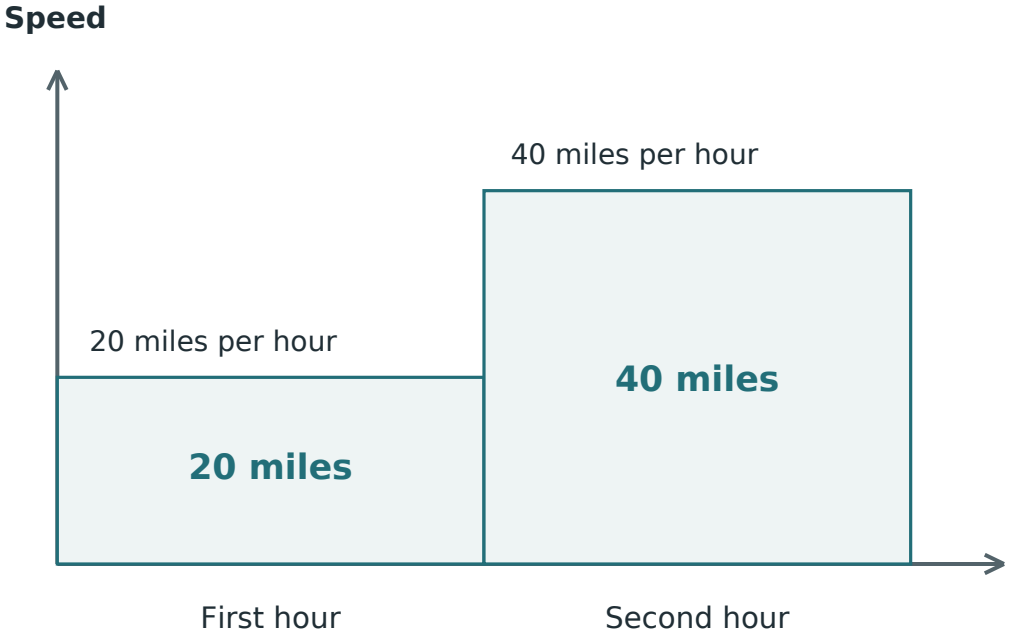
The graph is conceptual, not the numerical record of the 60-mile trip above. No formula for the curve is needed to read the idea.

4.3 / ADD THE LITTLE PIECES

From speed back to distance

Now turn the question around. Suppose we know how fast the truck moved. If it held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

So the whole distance is **20 + 40 = 60 miles**. We found a total by adding the contributions of different parts of the journey.



In a speed-versus-time graph, a rectangle’s height is the speed and its width is the time. Multiplying them gives the distance during that time. The **area under the speed graph** therefore represents distance.

When speed changes smoothly, divide the journey into short time pieces. Use each piece’s speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called **integration**; the accumulated result is an **integral**.

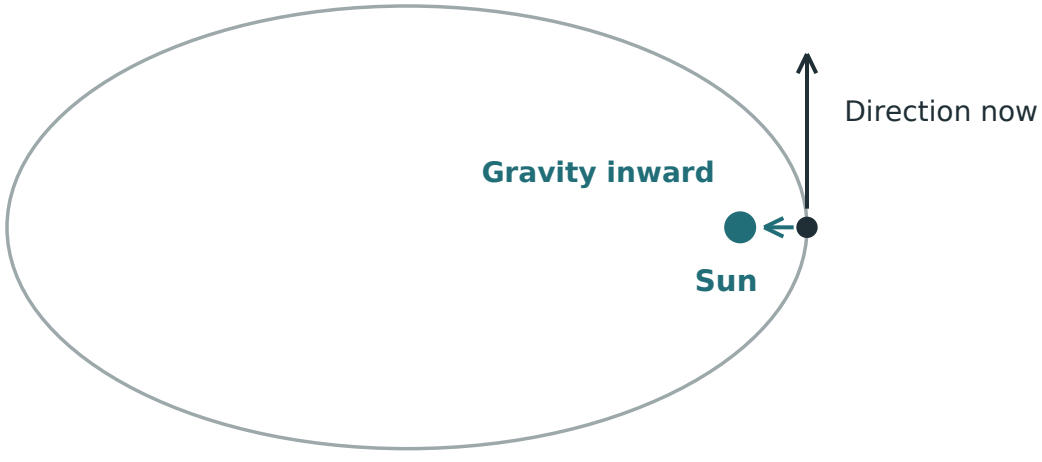
Differentiation asks for the local rate; integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 / FROM MOVING TRUCKS TO MOVING PLANETS

Orbits, tides and changing direction

For Newton, motion raised a deeper question: how does a pull change an object’s journey? **Gravity** is the attraction between masses—between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. **Acceleration** means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity and acceleration from moment to moment.



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Newton’s gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape and other effects. Calculus describes change; it does not make geography disappear.

Geometry describes shapes; calculus describes how they change.

4.5 / DIFFERENTIAL EQUATIONS, IN EVERYDAY LIFE

Why hot coffee cools more slowly

A fresh cup of coffee cools quickly at first, then more slowly as it approaches room temperature. A **differential equation** connects a changing quantity with its **rate of change**—how quickly it changes. Here is a simple model, called Newton's law of cooling.

First, name the pieces. **T** is the coffee's temperature, in degrees Fahrenheit. **t** is elapsed time, in minutes. **T_room** is the steady room temperature; "room" is a label, not multiplication. The **gap** is coffee temperature minus room temperature: **T − T_room**. The sign **−** means subtract.

dT/dt reads "rate of temperature change": degrees Fahrenheit per minute. Think of **dT** as a tiny temperature change and **dt** as a tiny time change; the derivative describes the limiting rate. **k** is a positive **cooling coefficient**—a multiplier telling how quickly the gap shrinks. Its units are **per minute**, or inverse time.

The equals sign means both sides have the same value. Parentheses group the gap. Writing k beside parentheses means multiply. Now we can read the equation:

dT/dt = −k(T − T_room)

In words: **the size of the cooling rate is k times the gap**. The minus sign before k makes the temperature-change rate negative: hot coffee's temperature is falling. A smaller gap gives slower cooling.

The **time constant** is one characteristic cooling time. Its name is **tau**, written **τ**. The rule is **τ = 1/k**: one divided by k. Because k is per minute, tau is in minutes. The actual time depends on the cup and conditions; we have not chosen a number of minutes.

The sign **%** means "out of a hundred." After **one** time constant, about **37% of the original gap remains**: 63% has closed. After **three**, about **5% remains**: 95% has closed. These percentages describe the **gap**, not the room temperature.

Example: coffee starts at 170°F in a 70°F room: a 100-degree gap. "°F" means degrees Fahrenheit. After one time constant it is about **107°F**; after three, about **75°F**.

The model assumes steady room temperature and approximately constant k. It only approximates real coffee. It approaches room temperature without reaching it exactly in finite time: page 14's local-rate idea at work.

READ ONLY IF YOU WANT TO FOLLOW THE THREADS

Sources, and what to remember

The historical stories are real; the shelter, truck and shadow numbers are teaching examples. The master paper reworks the standalone ladder lesson into Section 3. Original PDFs are left unchanged.

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University of St Andrews, MacTutor: Euclid and Pythagoras. Euclid organized earlier knowledge; the right-triangle relationship predates Pythagoras.

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Mathematics, one idea at a time

Jeff, this is not a test. Think of it as a walk from a shelter room to a ladder, then out to the Moon and the moving planets. We will name each idea before asking it to do any work.

The four subjects are not four locked doors. Each adds a new kind of question to the same habit: **say what you know, say what you want, and connect the two.**

CONTENTS

1. Algebra 2

Solving an unknown • shelter rooms and a truck

2. Geometry 4

Euclid • right triangles • Pythagoras • equal sides

3. Trigonometry 7

One ladder • sine, cosine and tangent • Galileo

4. Calculus 13

Newton and Leibniz • change • accumulation • orbits

4.5–4.6 Coffee cooling and its derivative 17–18

Sources & further reading 19

The history and science behind the stories

How to read: stop whenever an idea feels new. Read the words, look at the picture, then try the numbers. Nothing here requires formal calculus notation.

The contents and colored page references are clickable. Click any page footer to return here. Examples simplify real situations; they are not operating limits or engineering advice.

1.1 / ALGEBRA STARTS WITH AN EMPTY PLACE

How many people in the shelter?

Picture a shelter with three rooms. For this arithmetic example, imagine each room holds 40 people. How many people is that altogether?

Algebra is a way of reasoning about an unknown number. An **unknown** is simply the answer we have not found yet. We can leave an empty box for it, or give it a letter as a label.

Here, let **C** stand for the total number of people. The letter is not a special object. It is just the name on the empty place we want to fill.

The sign **×** means “times,” or “multiplied by.” The sign **=** means “has the same value as.” It is not an instruction to guess.

THREE GROUPS OF FORTY PEOPLE

C = 3 × 40 = 120 people

Check the answer: 40 people in the first room, 40 in the second and 40 in the third make 120. In symbols, **+** means “add”: **40 + 40 + 40 = 120**.

We did not make the problem harder by writing C. We made the question visible: “What total belongs here?”

The room capacity is invented for teaching. Real shelter capacity depends on space, access, staffing, safety requirements and local rules.

1.2 / WORK BACKWARD, THEN CHECK

How many pallets in the truck?

Now a truck needs to carry a 3,000-pound load. Suppose each loaded pallet weighs 500 pounds. How many pallets make that load?

Let **P** be the number of pallets. Before we know P, we can still describe the relationship: the number of pallets times the weight of each pallet gives the total weight.

NAME THE UNKNOWN

P × 500 pounds = 3,000 pounds

“How many groups of 500 fit into 3,000?” is a division question. The sign ÷ means “divided by.” Division undoes multiplication.

FILL THE EMPTY PLACE

P = 3,000 ÷ 500 = 6 pallets

Check: six pallets times 500 pounds per pallet gives 3,000 pounds. Putting our answer back into the original relationship confirms that it fits.

Dividing both sides of an equality by the same number other than zero keeps them equal. Here, dividing each side by 500 leaves **P = 6**.

Name the unknown, use the relationship, then check. We will do it again with the ladder in Section 3 (page 7).

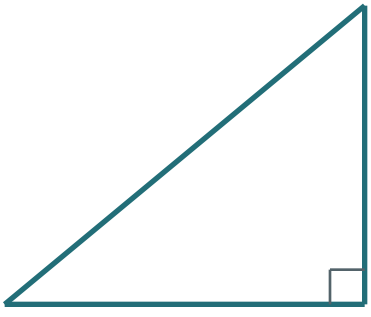
“lb” on the boxes means pounds. This is weight arithmetic only—not a loading plan. Real loads must meet vehicle, axle, space, balance and securement limits.

2.1 / GEOMETRY GIVES THE PROBLEM A SHAPE

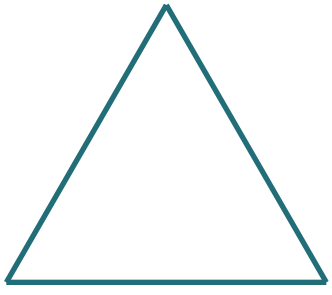
Euclid’s world of lines and corners

People measured fields and built structures long before the Greek mathematician Euclid. Around 300 BCE—about 2,300 years ago—he organized much earlier knowledge into a book called Elements.

Geometry studies shapes, sizes and positions. Euclid’s lasting example was to begin with clear assumptions and build an explanation, step by step. A **proof** is reasoning that shows why a claim must be true.



Right triangle



Equilateral triangle

An **angle** describes how widely two lines open apart. We measure angles in **degrees**, written with the little sign °. A full turn is 360 degrees; a square corner is 90 degrees.

A **right triangle** has one square corner. “Right” here means 90 degrees, not the right-hand side of the page. Its **hypotenuse** is the side across from that corner, and is always the longest side.

An **equilateral triangle** has three equal sides. On flat paper, each of its three angles is 60 degrees. It does **not** have a square corner.

These are different triangles. We can split the equal-sided one to make a right triangle; that connection will explain the ladder’s one-half relationship on the next diagram (page 6).

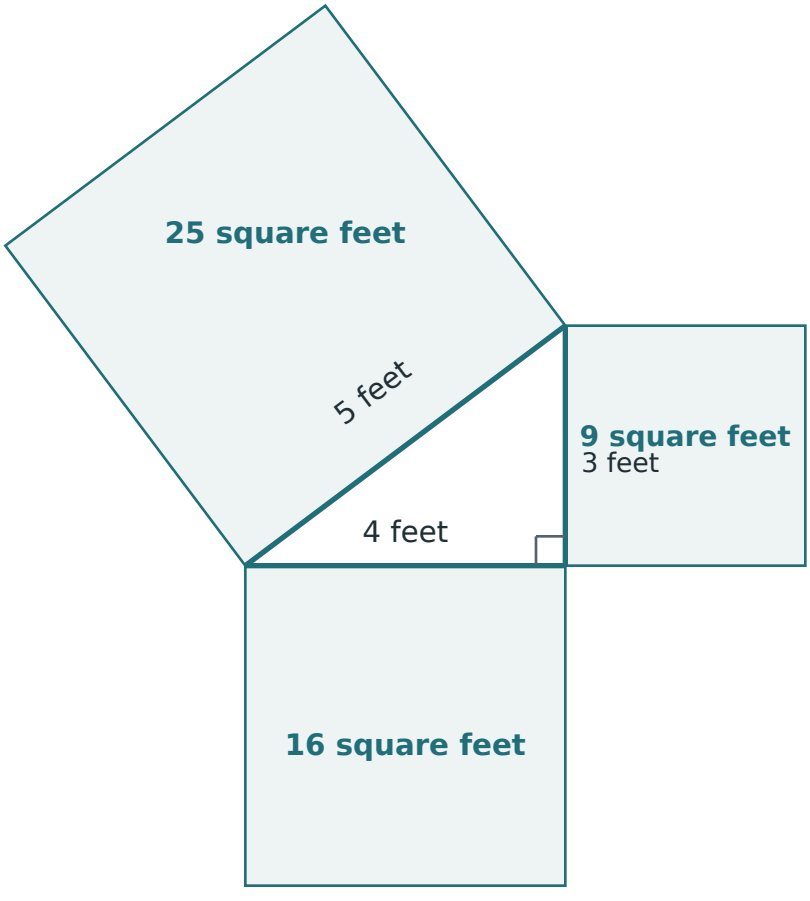
BCE means “before the common era.” Geometry is much older than any one mathematician. History sources are collected on page 19.

2.2 / PYTHAGORAS CONNECTS THREE LENGTHS

The 3–4–5 triangle

Imagine a right triangle whose two square-corner sides measure 3 feet and 4 feet. Its hypotenuse measures 5 feet. Why do these three lengths fit?

Pythagoras's theorem says that the area of the square built on the hypotenuse equals the areas of the other two squares added together. A **theorem** is a statement established by proof.



Area means how much surface is covered. A square 3 feet on each side covers 3 times 3, or 9 square feet. A **square foot** is a square one foot on each side.

So our picture says: **3 × 3 + 4 × 4 = 5 × 5**, or **9 + 16 = 25**. If the long side were unknown, we would ask: "What length times itself gives 25?" The answer is 5.

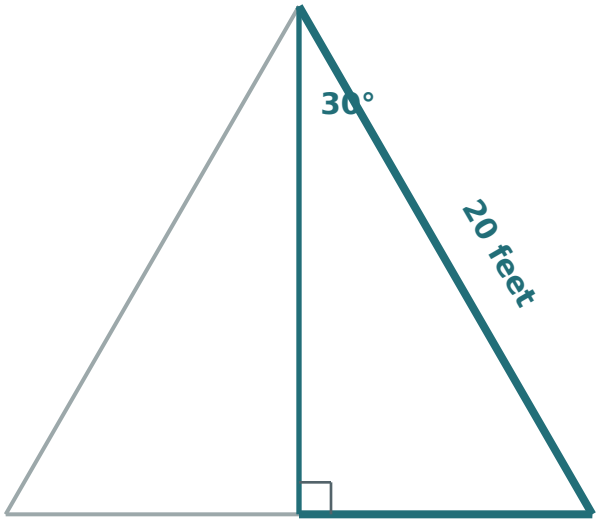
The theorem bears Pythagoras's name, but the relationship was known in Babylonian mathematics much earlier. Galileo used this kind of right-triangle reasoning (page 12).

2.3 / A BRIDGE FROM GEOMETRY TO TRIGONOMETRY

Why thirty degrees gives one half

Begin with an equilateral triangle: three equal sides, each 20 feet long. Every corner is 60 degrees. Draw a line from the top corner straight down to the middle of the bottom side.

The line splits the picture into two matching right triangles. It cuts the top angle into two 30-degree angles and the bottom side into two 10-foot pieces.



In either half, the side **opposite the 30-degree angle** is 10 feet long. The sloping side—the hypotenuse—is 20 feet long. So the opposite side is **half** the hypotenuse.

Turn that half-triangle around on the page: the lengths and angles do not change. It is the same shape as our 20-foot ladder reaching 10 feet up the wall.

A ratio compares two lengths by division. Here, 10 ÷ 20 is one half. That comparison comes from the shape, not a lucky measurement or a number invented by a calculator.

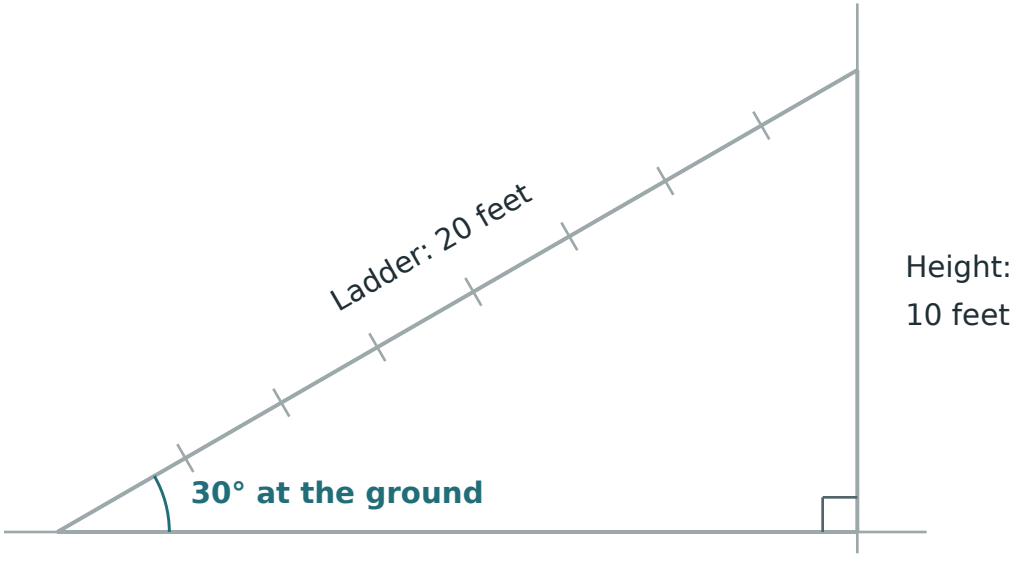
This ratio is called **sine** (Section 3, page 8). Geometry supplies the shape; trigonometry uses its proportions.

BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Return to the right triangle from Section 2 (page 4). A ladder leaning against a wall makes that shape. We will keep the **same ladder** throughout this section.

Trigonometry connects angles with side lengths. It adds known ratios to algebra's search for an unknown (Section 1, page 2). You do not need calculus yet.



First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means "degrees." A square corner is 90 degrees.

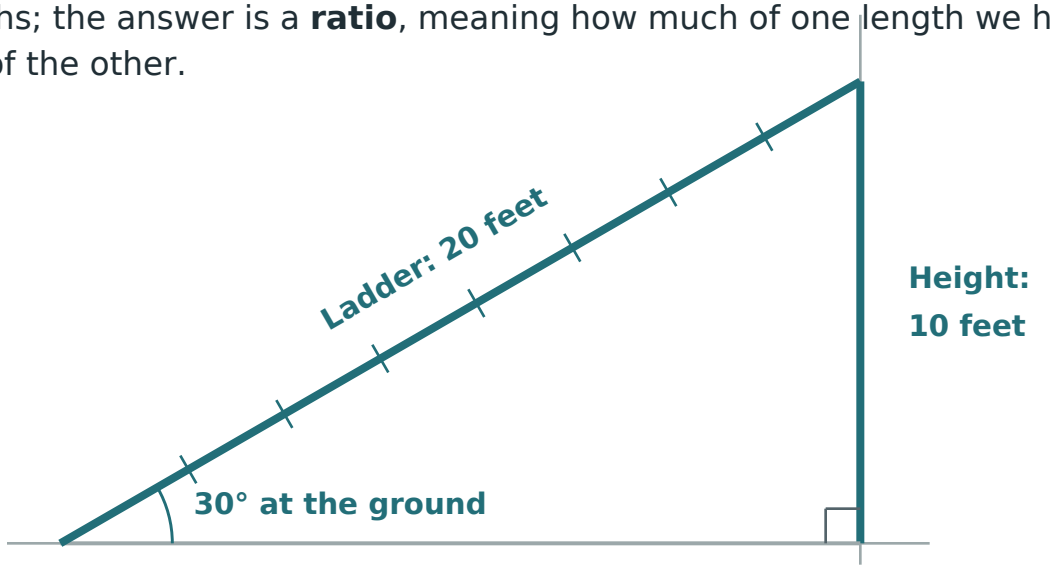
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. The equal-sided triangle on page 6 explains why; a calculator or table supplies the same ratio.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

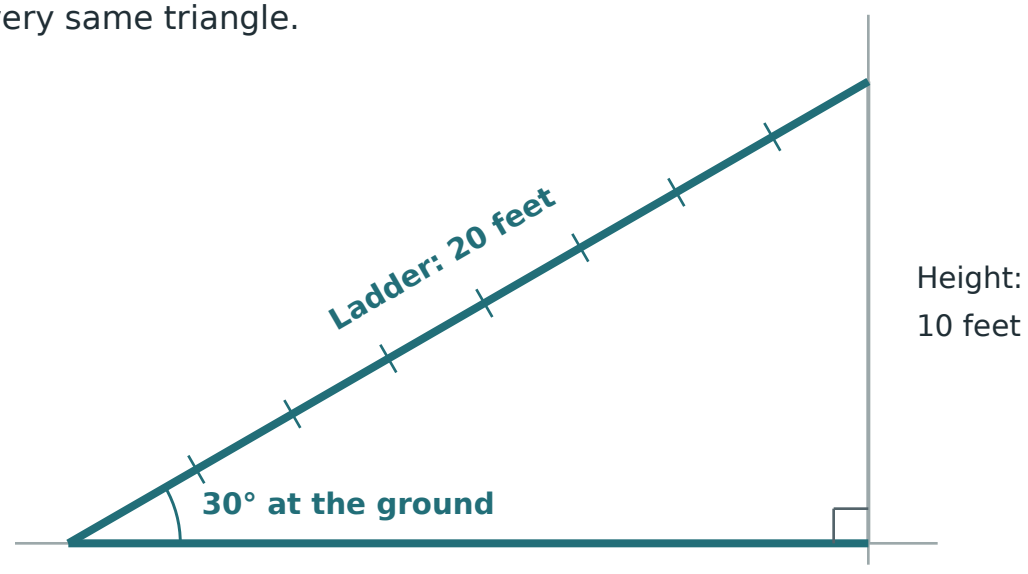
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



Base: about 17.32 feet

For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

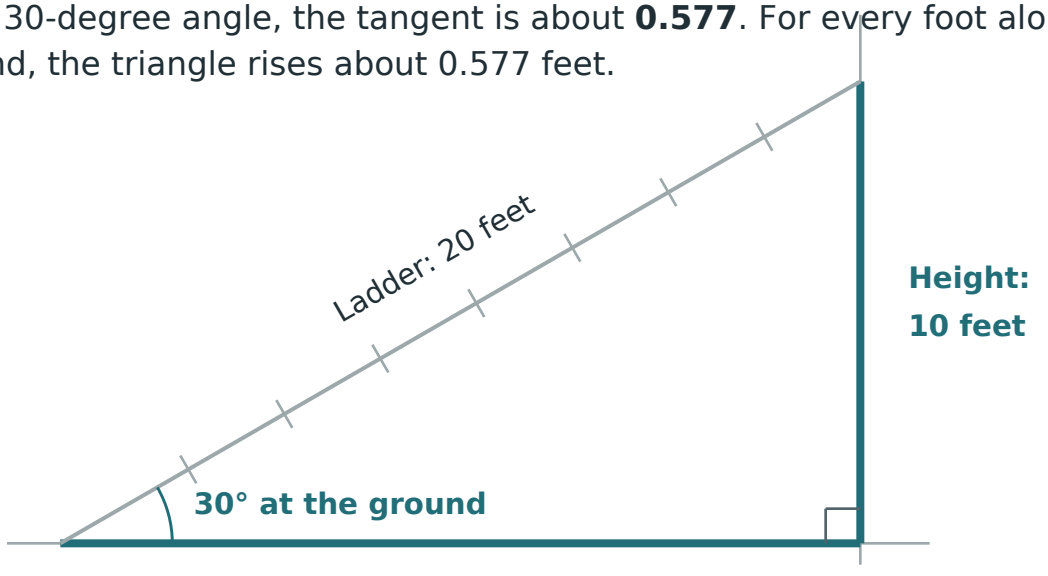
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

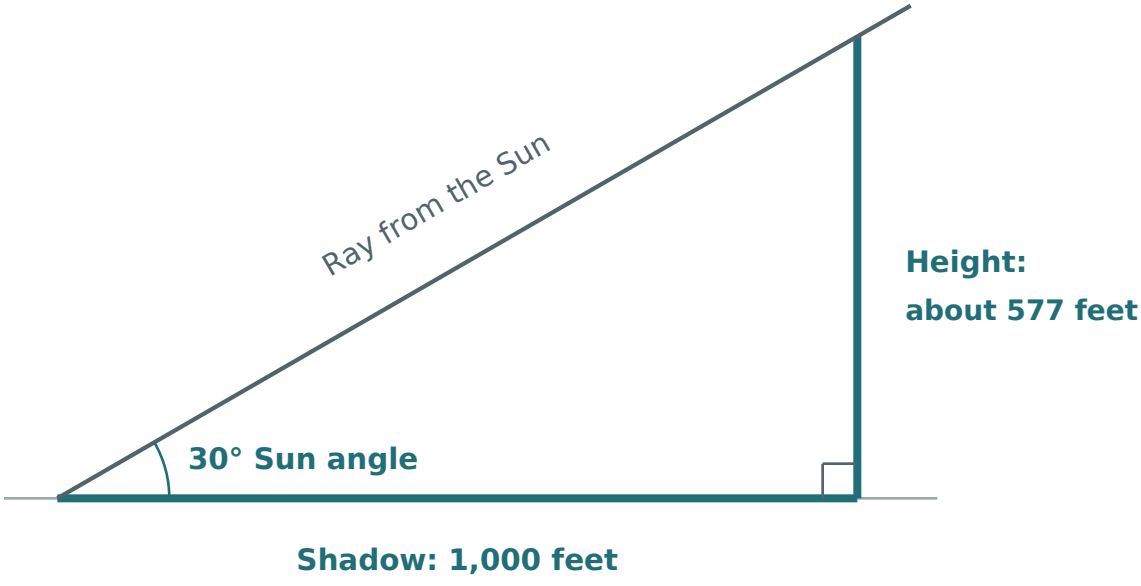


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule** (Section 2, page 5), the relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Where the story goes next

Geometry gives us a shape; trigonometry compares its sides. What if the shape or position changes from moment to moment? That is our next subject: Section 4, calculus (page 13).

Galileo's original text and the related shadow method are linked in the sources on page 19. The shadow example on page 11 is invented, not Galileo's data.

4.1 / NEWTON, LEIBNIZ AND THE MOVING WORLD

Now let something change

The shelter total, triangle and ladder all held still while we examined them. But a truck speeds up, a planet changes direction, and a tide rises. **Calculus** helps us describe continuous change and what builds up from it.

In the late 1600s, Isaac Newton and Gottfried Wilhelm Leibniz developed calculus independently, building on earlier work. Newton connected this new mathematics closely with motion. It was not one person's invention out of nothing.

How fast, right now?

A rate of change

How much altogether?

Accumulation

A **rate of change** compares a change in one quantity with a change in another. Speed is a familiar example: how much the distance changes as time passes.

Accumulation means adding up what happens over an interval. If we know the truck's speed throughout a journey, we can add up the small distances it covers to find the whole trip.

These two questions are partners: from the journey's distance record, find the changing speed; from the changing speed, recover the distance traveled.

The earlier sections are foundations, not a claim that Pythagoras invented calculus. Geometry describes positions and lengths; algebra expresses relationships. Calculus asks how those relationships behave as things change.

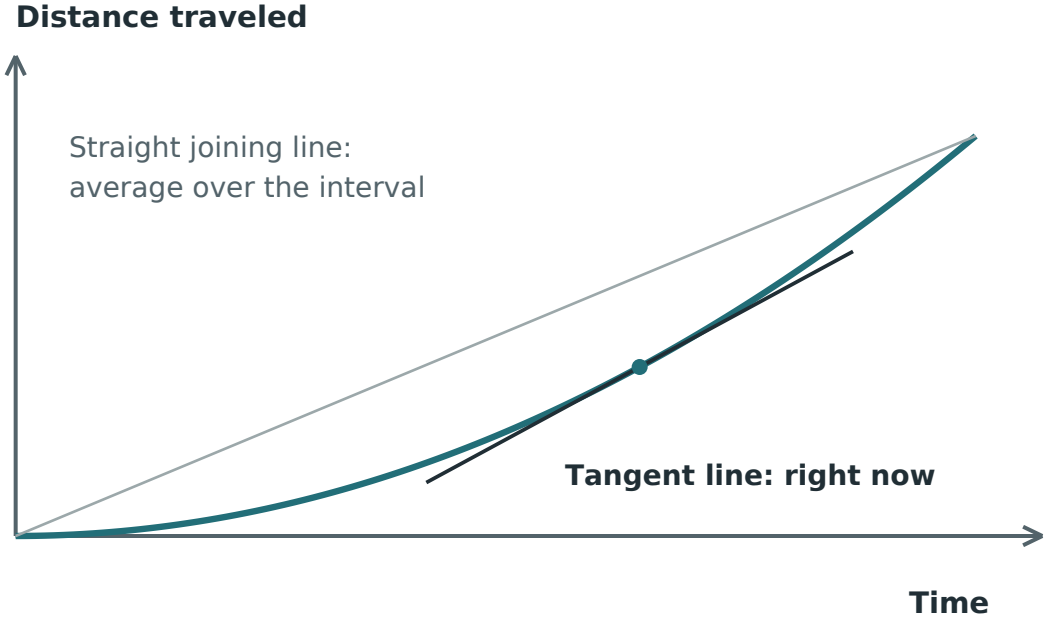
You can understand the central ideas without learning the formal symbols first. We will stay with the truck before returning to the sky.

4.2 / AVERAGE IS NOT THE SAME AS RIGHT NOW

A speedometer and a curve

Suppose the truck travels 60 miles in two hours. Its **average speed** is the total distance divided by the total time: **60 ÷ 2 = 30 miles per hour**. That does not mean it moved at 30 the whole way.

Instantaneous speed means speed at a particular moment—the kind of reading a speedometer gives. To estimate it, measure distance over a very short time. Calculus asks what happens as that time interval gets smaller and smaller.



This picture is a **graph**: time runs across; distance runs upward. The curve records a journey. Its **slope** means rise compared with horizontal run—here, distance change compared with time change.

A **tangent line** follows the curve’s direction at one point. Its slope gives the rate there. Finding that local rate is called **differentiation**; the rate we find is called a **derivative**.

This is not quite the same use of “tangent” as on page 10. There, **tan** is a triangle’s height-to-base ratio. Here, a **tangent line** is a line beside a curve. Their connection is slope: rise divided by run.

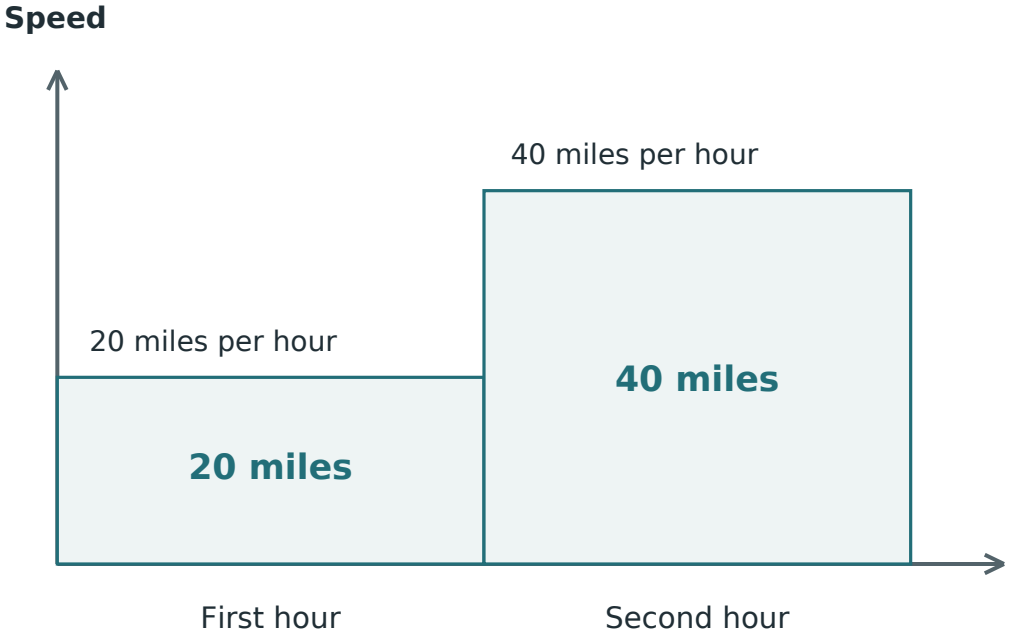
The graph is conceptual, not the numerical record of the 60-mile trip above. No formula for the curve is needed to read the idea.

4.3 / ADD THE LITTLE PIECES

From speed back to distance

Now turn the question around. Suppose we know how fast the truck moved. If it held 20 miles per hour for one hour, it covered 20 miles. If it then held 40 for another hour, it covered another 40.

So the whole distance is **20 + 40 = 60 miles**. We found a total by adding the contributions of different parts of the journey.



In a speed-versus-time graph, a rectangle’s height is the speed and its width is the time. Multiplying them gives the distance during that time. The **area under the speed graph** therefore represents distance.

When speed changes smoothly, divide the journey into short time pieces. Use each piece’s speed to estimate its distance, then add the distances.

Calculus makes this idea precise by taking smaller and smaller pieces. Adding a continuously changing quantity this way is called **integration**; the accumulated result is an **integral**.

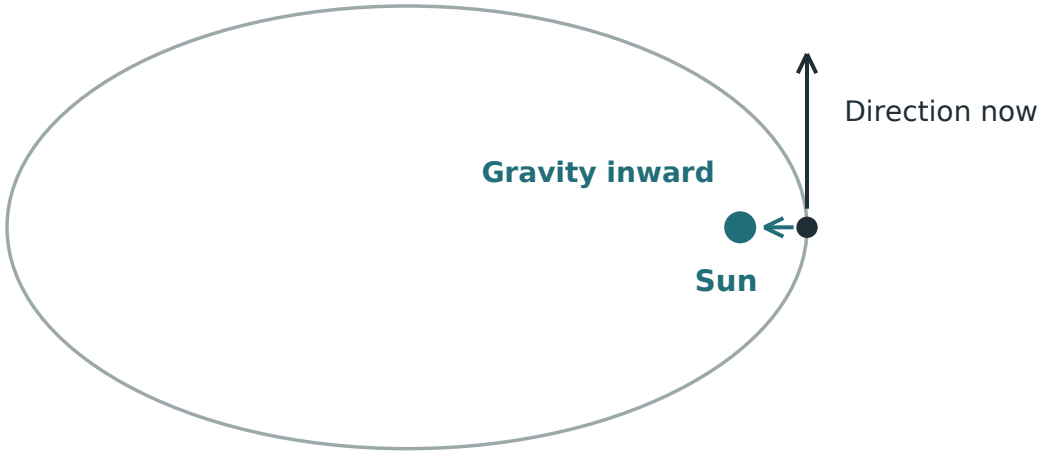
Differentiation asks for the local rate; integration adds the pieces. Under suitable conditions, these are reverse operations. That partnership is the central link in calculus.

4.4 / FROM MOVING TRUCKS TO MOVING PLANETS

Orbits, tides and changing direction

For Newton, motion raised a deeper question: how does a pull change an object’s journey? **Gravity** is the attraction between masses—between the Sun and a planet, or between Earth and the Moon.

Velocity means speed with a direction. **Acceleration** means a change in velocity, so turning counts even when speed stays the same. Calculus relates position, velocity and acceleration from moment to moment.



Schematic ellipse; sizes and distances are not to scale.

Forward motion and the Sun’s inward pull make the planet’s path curve. Without gravity it would continue straight in its current direction. Orbits need not be circles; many are **ellipses**—oval-shaped paths.

The Moon pulls harder on Earth’s near side than on its center, and less on the far side. This **difference across Earth** helps stretch the oceans and creates tides. The Sun contributes too.

Newton’s gravity explains the basic tide-raising forces. Local water-level predictions also need ocean depth, coastline shape and other effects. Calculus describes change; it does not make geography disappear.

Geometry describes shapes; calculus describes how they change.

4.5 / DIFFERENTIAL EQUATIONS, IN EVERYDAY LIFE

Why hot coffee cools more slowly

A fresh cup of coffee cools quickly at first, then more slowly as it approaches room temperature. A **differential equation** connects a changing quantity with its **rate of change**—how quickly it changes. Here is a simple model, called Newton's law of cooling.

First, name the pieces. **T** is the coffee's temperature, in degrees Fahrenheit. **t** is elapsed time, in minutes. **T_room** is the steady room temperature; "room" is a label, not multiplication. The **gap** is coffee temperature minus room temperature: **T − T_room**. The sign **−** means subtract.

dT/dt reads "rate of temperature change": degrees Fahrenheit per minute. Think of **dT** as a tiny temperature change and **dt** as a tiny time change; the derivative describes the limiting rate. **k** is a positive **cooling coefficient**—a multiplier telling how quickly the gap shrinks. Its units are **per minute**, or inverse time.

The equals sign means both sides have the same value. Parentheses group the gap. Writing k beside parentheses means multiply. Now we can read the equation:

dT/dt = −k(T − T_room)

In words: **the size of the cooling rate is k times the gap**. The minus sign before k makes the temperature-change rate negative: hot coffee's temperature is falling. A smaller gap gives slower cooling.

The **time constant** is one characteristic cooling time. Its name is **tau**, written **τ**. The rule is **τ = 1/k**: one divided by k. Because k is per minute, tau is in minutes. The actual time depends on the cup and conditions; we have not chosen a number of minutes.

The sign **%** means "out of a hundred." After **one** time constant, about **37% of the original gap remains**: 63% has closed. After **three**, about **5% remains**: 95% has closed. These percentages describe the **gap**, not the room temperature.

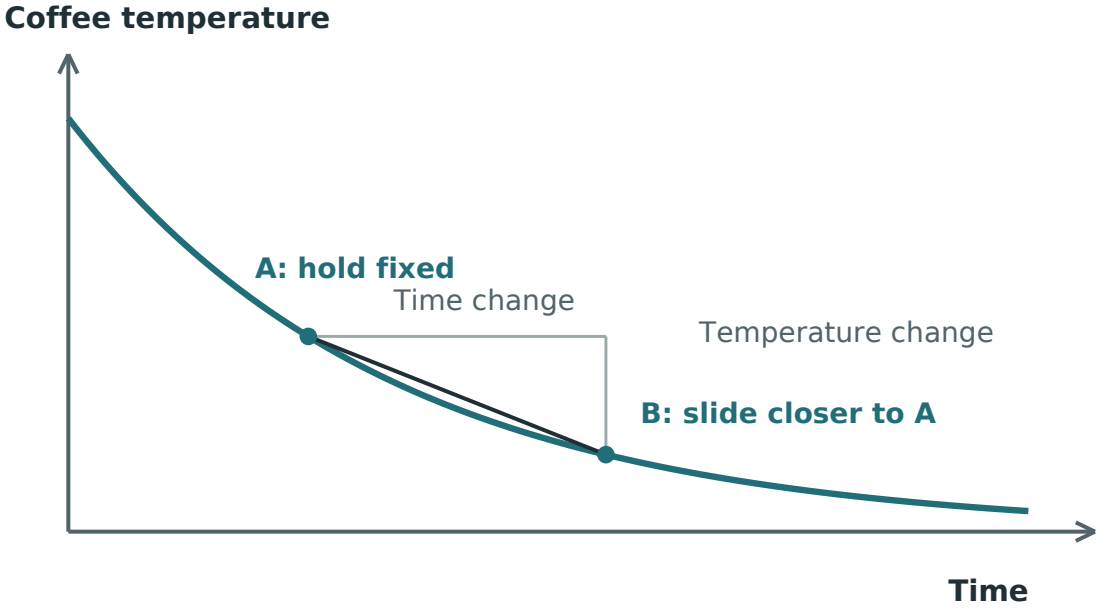
Example: coffee starts at 170°F in a 70°F room: a 100-degree gap. "°F" means degrees Fahrenheit. After one time constant it is about **107°F**; after three, about **75°F**.

The model assumes steady room temperature and approximately constant k. It only approximates real coffee. It approaches room temperature without reaching it exactly in finite time: page 14's local-rate idea at work.

4.6 / THE DERIVATIVE, ONE INSTANT AT A TIME

Two points, then one instant

On a **coffee-temperature graph**, time runs horizontally and temperature runs vertically. Choose an instant and mark its temperature as **point A**. Choose a nearby later instant and mark its temperature as **point B**. The letters are only labels.



Schematic curve, not observed measurements. The dark line joins A and B.

Join the points. The line’s slope is the **vertical temperature change divided by the horizontal time change**—rise divided by run. The run is positive; the rise is negative because coffee cools. The ratio is degrees Fahrenheit per minute.

Hold A fixed and slide B closer along the curve. Each two-point ratio estimates the slope at A. The value these ratios approach as the time gap shrinks is the **limit**: the derivative at A. We never divide by zero.

Every instant has its own slope. In this cooling model the slopes become less negative as the curve flattens. The limit is exact for the ideal smooth model; real measurements only estimate it.

READ ONLY IF YOU WANT TO FOLLOW THE THREADS

Sources, and what to remember

The historical stories are real; the shelter, truck and shadow numbers are teaching examples. The master paper reworks the standalone ladder lesson into Section 3. Original PDFs are left unchanged.

Geometry and its history
University of St Andrews, MacTutor: Euclid and Pythagoras. Euclid organized earlier knowledge; the right-triangle relationship predates Pythagoras.

[1] Galileo's actual mountain calculation
Galileo, Sidereus Nuncius (1610), E. S. Carlos translation (1880), pp. 28–30: Project Gutenberg ebook 46036. Sunlit peaks beyond the terminator; right-triangle calculation; "more than 4 Italian miles."

[2] The related shadow method
Mary McIntyre, BBC Sky at Night Magazine: "How to measure craters on the Moon". Height equals shadow length times the tangent of the local Sun angle, with simplifying assumptions.

Calculus and its development
University of St Andrews, MacTutor: "A history of the calculus". Earlier work, Newton, Leibniz, rates and accumulation. Orbits: NASA, "Gravity & Mechanics".

Why tides happen
NOAA: "Tides" and "What Causes Tides?". Uneven gravitational pull, the Moon and Sun, and the influence of local geography.

Four questions, one growing toolkit
Algebra: What number is missing?
Geometry: What shape and lengths do we have?
Trigonometry: What do the angles tell us about the sides?
Calculus: How is it changing, and what does it add up to?

You do not have to memorize every word. Recognizing the question is already a beginning.